

OCEAN DATA, IN THREE DIMENSIONS

Depth Atlas

Main documentation



01

Locate

Choose a dataset

02

Explore

Look beneath the surface

03

Compare

Inspect the evidence

METHODS / PRODUCT / ENGINEERING

Verification, deployment and phase history



Geographic context: Natural Earth, 1:110m

TEAM SEIFUKU

Browser-native ocean science workspace

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RELEASE 0.17.1
27 September 2026

depth-atlas-seifuku.vercel.app

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Current release first. Detailed records follow. Select an entry or use the PDF bookmarks to jump directly to a chapter.

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01 Project overview and current release

Depth Atlas is a browser application for exploring the ocean below its surface. It places three-dimensional ocean model fields beside real instrument measurements, with their original positions, depths, timestamps, units and quality information. Users can inspect a supported historical case, compare eligible measurements with its model and preserve an investigation so that the calculation can be checked again.

Project: Depth Atlas. **Team:** Seifuku, ID 150123. **Document edition:** 27 September 2026. **Application release:** 0.17.1.

Website: depth-atlas-seifuku.vercel.app. The earlier project addresses remain available and serve the same production release. Browser saves belong to their website origin: export an investigation from the old address and import it at the new address when moving saved work.

Team and responsibility

Person or detail	Confirmed role or value
Adwita Kurle	Team leader
Gaurav Ranade	Website creator and operator
Anushree Dixit	Team member
Spruha Kurle	Team member
Hrishikesh Tarade	Team member
Tanisha Natrajan	Team member
Public contact	ranadegaurav30@gmail.com
Location	Pune, Maharashtra, India

The project is independent. No university is affiliated with its development. Additional member responsibilities, mentors and endorsements are not claimed.

How to read this document

This opening handbook describes the current release. It is followed by the scientific contracts, capability matrix, delivery and security reports, phase records, method reviews, development instructions, design decisions and prepared report text. The contents page and PDF bookmarks provide direct navigation.

Earlier phase records describe the implementation and checks at their stated dates. Statements such as a feature being planned, a smaller catalogue or a later phase not having started are historical records, not the current status. The current release summary and final capability audit take precedence when describing today's delivery. Repeated test counts from different reports overlap and must not be summed into a single unique-test total.

This edition consolidates the maintained project documentation rather than reproducing raw logs, every test trace or third-party publications. Original evidence paths, scientific references and material limitations remain available. The final appendix inventories the included source documents. It is a project handbook, not proof of submission to the organizer.

02 Problem statement and delivery

The project brief asks for a web-based, browser-native 3D ocean visualization system that integrates ocean model outputs and observational data. The main priority is an interactive scientific workspace: volumetric temperature, salinity and current views, instrument profiles, supported file ingestion, configurable display controls and an extensible backend. Open standards and science outreach are also part of the stated scope.

The intended operational audience is INCOIS forecasters and ocean-data teams. Learners, teachers, outreach visitors and policymakers are additional audiences. The purpose is to make depth-resolved information and its evidence easier to inspect. Faster operational advisories are an intended benefit in the problem statement; this prototype has not measured forecast improvement or received institutional acceptance.

PS expectation	Current delivery	Remaining boundary
Browser-native 3D water column	Volume rendering, optional cutaway, native inspection, depth slices, sections and isosurfaces	Supported case bounds, missing cells and display resolution are explicit; Basic 2D is a labelled fallback

PS expectation	Current delivery	Remaining boundary
Temperature, salinity and current fields	Indian cases include all three; selected Pacific cases include monthly potential temperature	Quantities are source-specific; unsupported variables are not fabricated
Time-step animation	Playback follows actual available snapshots	No invented intermediate source dates
Argo, glider, CTD and BGC overlays	Genuine source profiles, positions, timestamps and QC-aware readings	Some glider and CTD examples have no matching loaded model case
Model/profile co-display	Linked scientific views and eligible numerical comparison	Spatial overlap alone does not establish a valid match
NetCDF and delimited text	Supported observation parsers, preview, mapping and bounded batch review; fresh model preparation through adapters	Arbitrary files and unlimited model uploads are not supported
Colour and variable controls	Palettes, range, log/linear scale, opacity and vertical exaggeration	Display settings do not change native calculations
Modular extension	Declared source, variable and instrument contracts	New extensions still require metadata, unit and verification work
REST and open standards	Public REST/downloads; actual local WMS, WCS and OPeNDAP checks	Persistent public THREDDS is not configured
CF-compatible data	Scoped metadata and original-value checks	Full independent conformance certification remains open
Portable institutional deployment	Source package and isolated runtime setup checked on Windows	Docker and actual INCOIS installation are unverified
Public outreach	Default Basic guide, optional In-depth guide and feature tours	Real unfamiliar-user comprehension evidence remains open

These are scoped delivery states, not an official scoring rubric or a percentage of SIH completion. Source: [official SIH problem statements](#), project brief, checked 27 September 2026.

03 Using the application

A complete investigation

1. Open the website. Choose the default Basic guide, explicitly choose In-depth, or select Skip tutorial.
2. Use the data globe to locate a supported model case or a genuine instrument profile. Read its dates and area, then explicitly open the model or profile.
3. Explore the water column. Choose a variable, open the optional cutaway, change depth and select a source time. Read the colour key and native value.
4. Inspect a measurement. Its date, position, quality and depth remain distinct from the model context.
5. Compare compatible data. Apply the time, distance, quality and vertical-gap rules. Inspect accepted pairs, excluded readings and the model-minus-observation residual.
6. Save deliberately. Download a portable investigation or supported offline viewer, and use recalculation to check saved source and method identities.
7. Open Tools for the other scientific investigations. Additional tutorials demonstrate those tools through real app actions.

The globe remains a deliberate geographic view. Loading model metadata does not automatically replace it with a cube. Geographic dots show actual profile locations; they do not represent continuous concentration fields, bloom extents or confidence.

LOCATE THE DATA

Where shall we look?

Find a model area or a measured profile. Then open it at depth.

Model areas

Instrument locations

Chlorophyll observations

Globe

Map

Available model case

Bay of Bengal · 7–10 January 2024

MODEL DOMAIN

Bay of Bengal · 7–10 January 2024

HYCOM GOFS 3.1 + Argo / INCOIS DAC

Data dates

2024-01-07 00:00:00 UTC
2024-01-10 00:00:00 UTC

Water column

0 to 5,000 m

Fields

Temperature, Salinity, Eastward current, Northward current

Open this 3D ocean →

The outline is the grid extent. Land, seabed and missing cells stay masked in the scientific view.

► Data publication and updates

Looking outside these cases?

Explore wider source regions →

Actual data globe in the deployed release

Reading a 3D ocean view

Colours encode the selected quantity and its units. The coloured water column represents model values, not the natural appearance of seawater. A depth slice colours one selected depth. A volume includes information through the supported column; an optional cutaway reveals its interior. Vertical exaggeration stretches the display to make depth visible without changing the physical coordinate or source value.

Masked land and missing depths remain missing. Isosurfaces and rendering may use bounded display data. Numerical inspection and the declared analytical operations retain their native-source contracts.

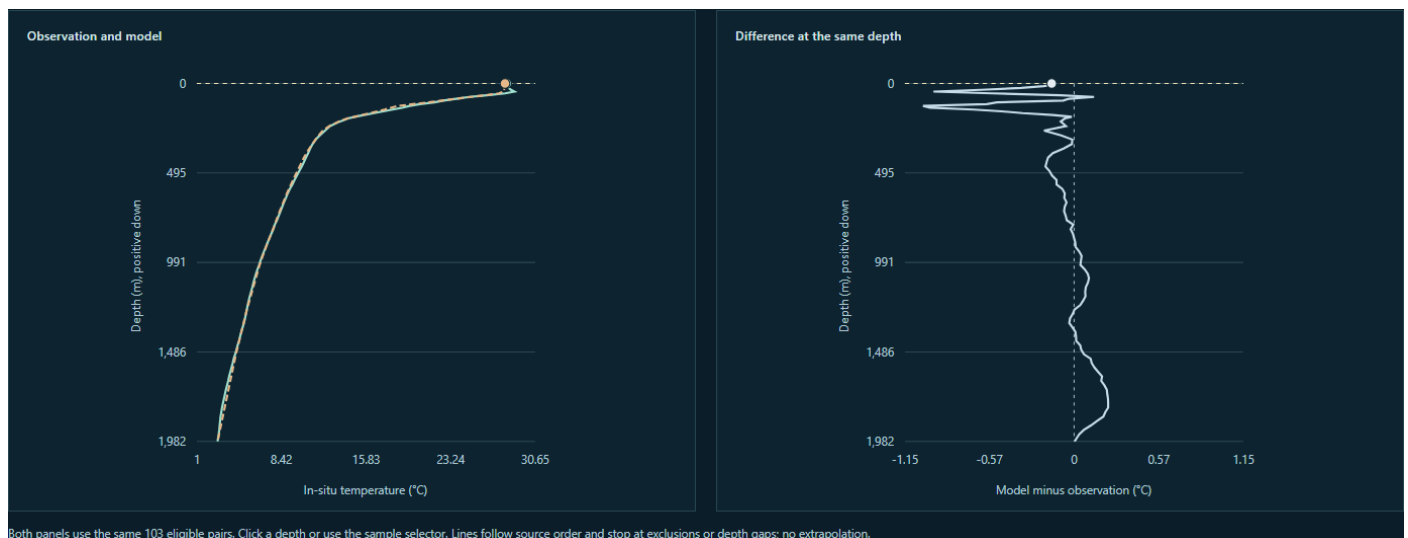


Actual cutaway and source value

Measurements, differences and saved evidence

An observation is a measured sample. A model value is a calculated source field. A residual is the model value minus the observation at a compatible matched depth. If a rule fails, the application explains the exclusion instead of producing a plausible difference.

In the checked January Bay example, Argo platform 1902669 and the 7 January 2024, 12:00 UTC snapshot produce 103 eligible pairs under the demonstrated settings. The selected residual is -0.152 degrees Celsius. These numbers describe that exact case; they do not establish the model's accuracy everywhere or independent validation of assimilated observations.



Actual matched profiles and residuals

Saves use local browser storage only when requested. Portable files retain source identities, methods, settings and supported results. Imported investigations include the original observation bytes and mapping. A replay checks those identities and recalculates. An offline viewer displays saved evidence; it does not run every scientific laboratory without a service.

04 Data, coverage and scientific interpretation

The six core cases comprise January Bay of Bengal, January Arabian Sea, a fresh March Bay of Bengal case and three selected tropical Pacific seasons. Indian model fields are bounded historical HYCOM cases. Pacific comparisons use selected GODAS seasons from 2013, 2015 and 2022. Source dates remain historical even though this project edition is dated 2026.

Measured sources include Argo, glider, ship CTD and biogeochemical profiles. Supported chlorophyll readings are observations. The bundled core ocean model does not contain a chlorophyll field, so the app does not claim a model chlorophyll volume, model chlorophyll residual or bloom map.

Additional regional exploration has source-specific geographic and variable limits. Main-model laboratory support is not automatically available for every wider source. A source catalogue is not a claim of live or uniform global coverage.

The implementation preserves physical coordinates, irregular native levels, variable definitions, masks, quality flags and source fingerprints. Unsupported data returns a reason. Quality categories belong to the original source scheme. The scientific-contract

and method-review chapters document interpolation, matching, derived quantities and independent numerical references.

Additional investigations

Tool	Purpose	Interpretation boundary
Find structures	Find connected native model cells satisfying a threshold and inspect their shape, depth and volume	A threshold component is not automatically an identified eddy
Observation support	Locate eligible samples and residuals within a selected structure, and inspect gaps	Coverage is not calibrated confidence
Matching sensitivity	Compare outcomes under different eligibility rules	Sensitivity is not a real-world probability
Drift	Release passive particles into historical horizontal currents and inspect their trajectories	Fixed-depth tracers are not sinking pebbles or a rescue forecast
Heat & Depth	Detect supported historical surface heat events and inspect overlapping depth evidence	A surface event does not prove a subsurface heatwave
Climate Event Lab	Compare selected El Nino, La Nina and reference seasons using declared baselines	Historical comparison is not adjustable climate prediction or local causal attribution
Virtual Expedition	Propose sampling locations and compare bounded reconstruction experiments	Model sampling is not new observed data or operational survey improvement
Feature evolution	Compare native threshold regions across source frames, including branches and gaps	Shape correspondence is not proof that the same water travelled
Observation blackout	Exclude selected evidence and inspect retained comparisons	The ocean model and its assimilation are not rerun
Wider regions	Explore supported bounded regions and source variables	Coverage and available tools differ by product

05 Architecture, data lifecycle and operation

The browser uses React, TypeScript, Vite, Three.js and chart components. FastAPI provides the scientific API. NumPy, netCDF4 and GSW support runtime parsing and calculations; xarray belongs to offline model preparation. The project uses a shared workspace and data contracts rather than disconnected demonstrations.

The flow is: **original source files > preparation adapters > checked case packs > bounded API > linked geographic, volume and profile views > explicit saved investigation > export and verified replay.**

Prepared read-only packs and the frontend/API release are published together. Runtime visitors do not trigger arbitrary provider downloads. A maintainer validates a new candidate publication before activating it. If validation fails, the earlier usable publication remains available. Sources and the publication status retain distinct acquisition, coverage and release dates.

The public app runs on Vercel. A separate THREDDS service has a different deployment lifecycle. The portable package includes ordinary server setup and a container recipe, but successful Windows installation does not establish actual institutional installation.

Current resource and storage limits

Area	Current boundary
Original observation request	2,000,000 bytes
Ordinary scientific JSON request	65,536 bytes before parsing
Imported comparison/support or investigation requests	3,000,000 bytes
Regional replay request	4,500,000 bytes
Upload body completion	15 seconds after admission
File review queue	Up to eight sequential files; 20,000-sample workspace ceiling plus source/profile/decoded-byte limits
API rate budget	360 requests/minute per client per worker, burst 360

Area	Current boundary
POST rate budget	60 calculations/minute per client per worker, burst 60
Concurrency	24 total API requests and four POST calculations per worker
Saved investigations	Explicit local IndexedDB storage with declared record/byte limits

These limits describe application controls. They are not distributed quotas, a performance promise or unlimited upload support. People behind one network may share its client budget. Accepted server work may finish after a browser cancels its request.

06 Verification, security and remaining work

The final audit checked every project requirement, feature and original differentiating idea against implementation and evidence. Calculations retain their source definitions. Display changes and security hardening did not change the scientific arrays or methods.

Recorded verification	Result and scope
Complete integrated backend suite	963 passed
Client calculation checks	82 passed
Public release, API and asset checks	24 passed, no retries
Exact public replay checks	61 checks over 17 requests passed
Complete public tutorial	39 lessons passed on desktop and mobile emulation
Selected WebKit and mobile interactions	126 passed
Public security checks	14 passed over 13 bounded requests
Portable installation	35 source/API checks, 61 replay checks and nine browser checks passed on the same Windows host

The detailed reports preserve initial failures and targeted corrections. The real Sources focus defect was corrected. An obsolete case count and stale navigation expectations were fixed in tests. A Firefox pointer-rounding test was checked against the actual source cell and corrected without changing numerical tolerances. Counts above overlap and are

not a count of unique requirements.

In three fresh browser contexts per condition, the median automated route to a checked comparison took 6.837 seconds on desktop and 9.333 seconds under the stated mobile network/CPU emulation. All six runs retained interactive 3D and exact values. Two four-client regional-request rounds succeeded. These are small technical samples without human reading time, not a test of 100 simultaneous users.

The interface was inspected across 28 feature captures plus focused follow-ups. Detailed mobile charts sometimes require internal scrolling or Try it myself to remove the tutorial card. Actual novice comprehension, physical-device support and domain acceptance still require real participants.

Security controls include bounded input, finite upload time, same-origin browser POST checks, per-worker admission controls, constrained script policies and response headers. Exact-pin dependency advisory checks and a listed secret-signature scan found no findings in their recorded scope. Those checks do not certify complete security, audit every native library or sandbox scientific parsers.

Phase progression

Phase	Delivered work or acceptance state
P00-P02	Planning, application foundation and checked historical data
P03-P06	3D rendering, observations, numerical comparison, adapters and local standards services
P07-P08	Native connected structures and saved, replayable investigations
P09	Guided comparison cases implemented; actual unfamiliar-user pilot remains open
P10-P14	Expedition, Drift, Heat & Depth, Climate, evolution and blackout
P15	Bounded wider-source exploration and supported offline handling
P16	Basic/In-depth real-action tutorials and feature discovery
P16A-P16E	Connected imports, fresh data, imported replay, linked workflow, batch limits, retained publication and offline continuity
P17	Capability audit, security, technical verification, measured performance, portable setup and demonstration materials; external acceptance remains open

Next work should gather unfamiliar-user observations and oceanographer feedback, test another physical device and verify the applicable private submission instructions. Persistent public THREDDS, full independent CF checking, compatible glider/CTD matchup data and actual INCOIS deployment remain separate dependencies. No competition placement, institutional endorsement or operational forecast benefit is claimed.

Demonstration and submission

Use one connected investigation: locate the January Bay case, explore the water column, inspect a genuine profile, show an eligible comparison and save/recalculate the evidence. Present additional tools when relevant. A 43.12-second silent recording preserves actual public app behavior; separate narration and judge questions are included in the reference chapters.

A six-slide PDF/PPTX draft follows the official idea template. Source, documentation and the recording are prepared assets, not an organizer submission. Public SIH sources were checked on 27 September 2026 and the problem page displayed 30 September 2026. The user's applicable private round instructions, upload fields, file-size limits and nomination status still need confirmation.

Document and evidence boundary

The remainder of this handbook consolidates the maintained engineering and scientific documents, with repeated internal handoff material replaced by this current summary. Original records remain in the private development archive. The source inventory at the end records the included files and their edition. Earlier reports remain historical so their original failures, corrections and limitations are not rewritten into unsupported success claims.

Depth Atlas release update

Release 0.17.1 changes the app name, public address, interface labels and documentation cover. The full name remains visible on small screens; a narrow header overflow was corrected. The deployed update passed 24 release/API/asset checks, 61 exact replay checks, and the real desktop, mobile and 320-pixel branding journey. The numerical methods and case-pack bytes are unchanged. These are internal technical checks, not new participant or oceanographer results.

The public source is [GauravR17A/depth-atlas](#). The first public snapshot includes the runnable app, checked datasets, tests and method notes, with earlier private

planning/history excluded. Anonymous repository access was verified. Its full backend run passed 854 tests and skipped 109 tests requiring original raw reference archives. The 145 affected checks pass where those originals are available. No numerical tolerance or production rate limit was weakened for publication. The draft presentation and older backup recording remain historical artifacts pending any final branding refresh.

07 Public project identity

Current release reference | Scope-edited excerpt | Source: docs/TEAM.md

Confirmed by the user on 26 September 2026. This is the public identity used by About, Privacy and Terms, not a claim of institutional endorsement or completion of the SIH portal submission.

Field	Confirmed value
Project	Ocean Navigator
Problem statement	project brief
Team	Seifuku
Team ID	150123
Website creator and operator	Gaurav Ranade
Public email	ranadegaurav30@gmail.com
Location	Pune, Maharashtra, India
Team leader	Adwita Kurle

Members: Adwita Kurle, Gaurav Ranade, Anushree Dixit, Spruha Kurle, Hrishikesh Tarade and Tanisha Natrajan. Leadership, website-creation and website-operation roles are confirmed; do not invent engineering roles for the other members. Preserve the supplied spelling of Tanisha Natrajan.

The user requested no social, repository or other public team links. The website retains necessary source citations, software attribution and hosting-policy references. No mentor information was supplied. The user clarified that the project has no university affiliation or university involvement in its development. Do not attribute its creation, development or operation to a university. No personal phone number or home address is needed for the

current public pages.

When identity changes, update all three `web/*.html` pages and this record together. Review Privacy and Terms whenever account, storage, upload, analytics, provider or security-related data handling changes.

08 Scientific contracts

Current release reference | Scope-edited excerpt | Source: `docs/SCIENTIFIC_CONTRACTS.md`

Current scope reviewed 27 September 2026 during P17. The P02-P06 sections below identify the original contracts, which are retained for reproducible replay. Later adapters do not silently change those earlier definitions. The capability audit in `P17_CAPABILITY_AUDIT.md` distinguishes implemented behavior, internal numerical evidence and external validation still needed.

Model/legacy observation source: `science/contracts.py`, schema v1. P04 instruments: `science/instruments.py`, schema v2. Public API models derive from these Pydantic contracts; frontend runtime readers reject incompatible responses. The model catalogue is also schema v2; its version is separate from scientific-record versions.

Contract	Meaning and required distinctions
Variable	Stable ID, original variable name/units, standard name, physical definition, decoded units and original packing/missing metadata.
Coordinates	Actual UTC timestamps, calendar and source time units; irregular positive-down metre depths; sorted latitude and longitude in <code>[-180,180)</code> . Original coordinate values remain in source metadata.
Provenance	Provider, version, model/observation/derived/simulation kind, source/licence links, citation, retrieval time, file fingerprints, transformations and limitations.
CaseManifest	Verified subset bounds/time/depth coverage, analytical/display coordinates and methods, actual profile summaries, source records, samples and derived-file hashes.
Observation	Platform/cycle/time/location, mode and QC, source file fingerprint, raw/adjusted values/errors, selected fields, pressure and derived depth, source-order indices and explicit eligibility.

Contract	Meaning and required distinctions
OperationResult	Source/case/variable, actual timestamp, shape, coordinates, units, representation, processing and manifest fingerprint. Flattened values are ordered depth then latitude then longitude; longitude changes fastest.

Model arrays are scalar fields for temperature, salinity, eastward velocity and northward velocity. "Currents" in the workspace refers to the two source horizontal components. No vertical current or derived speed is silently substituted.

Supported input

The P02 HYCOM-style NetCDF adapter supports independent one-dimensional time, depth, lat, lon coordinates and four compatible scalar fields on that shared grid. An explicit variable-name map can bind different source names with the same declared definitions. Dimension order may differ; the adapter reorders it. Coordinate sorting always reorders the data as well.

Packed data are masked for fill/missing/declared valid ranges **before** decoding as packed $\times \text{scale_factor} + \text{add_offset}$ in float64. Source attributes are preserved. Existing in-range longitude floats are not altered unnecessarily. Source precision is not improved by the larger decoded dtype.

The legacy P02 Argo adapter supports bounded core-profile NetCDF. It remains unchanged for the pinned model manifest. The P04 observation adapter below adds per-parameter modes and genuine sensor examples through a separate contract.

Instrument contract v2

InstrumentSummary carries stable source-derived identity, sensor type, platform, source time range, actual location, track, sample count, variable metadata and provenance. InstrumentProfile adds ordered Measurement rows, methods, source metadata and warnings. Each Parameter records its definition, units, original field, processing mode, QC scheme and accepted count. Each Reading retains selected/raw/adjusted values, their flags, supplied adjustment error and an eligibility reason. Missing readings remain null. Original pressure, derived depth, per-sample position/time and structured coordinate flags stay separate.

Argo R selects raw; A/D selects adjusted, with no raw fallback. BGC uses each parameter's processing mode and raw R BGC values are inspect-only. Argo 1/2, QARTOD 1, WOCE 2 and declared CSV 1/2 are distinct eligibility rules. Coordinate checks apply separately; unavailable source flags are blank, not invented. Depth uses GSW 3.6.20 `-z_from_p` with

sea pressure in dbar and source latitude. P04 preserves rather than collocates sensor records.

Supported runtime formats are Argo/BGC NetCDF, documented IOOS glider row-NetCDF, original CTD Exchange and Ocean Navigator CSV v1. The parser bounds each input to 2 MB/24 profiles/5,000 samples and output to 3.5 MB. Metadata/units errors return named 422 responses; resource limits return 413. The server reparses original bytes per request and does not persist user-uploaded content. Temporary imports remain in browser memory until clear/reload. Since P16C, explicit investigation saves retain the original bytes, mappings, parsed profiles, settings and results in IndexedDB on the user's device; portable JSON/ZIP exports contain those records. Replay sends the included original bytes back to the service for verification. P16E allows up to eight sequentially reviewed files, retaining the per-file limits and a 20,000-sample workspace ceiling plus profile/source/decoded-byte budgets. This is not an unlimited batch or server archive. See PHASE_16C_IMPORTED_REPLAY.md, PHASE_16E_CONTINUITY.md and the public observation import guide.

Errors and resource limits

Unsupported science raises a named `UnsupportedData` error. API errors use `{error: {code, message, request_id}}`. Invalid inputs never return plausible substitute values.

- Unsupported time, depth, variable semantics, incomplete/outside bounds: HTTP 422.
- Unknown case/profile: HTTP 404.
- Core case-subset requests above 100,000 scalar values: HTTP 413 with smaller-request guidance. Wider-source, structure, import and replay workflows have their own explicit limits; this is not a universal limit for every operation.
- Failed packaged-data integrity: HTTP 503.
- Missing cell: valid result with JSON `null`, not an HTTP error and not zero.
- Offline model preparation: at most 12 million scalar values per input file; supported Argo files at most 500 profiles and 10,000 levels per profile.

No endpoint fetches a caller-supplied URL or opens a caller-supplied path. IDs resolve through the packaged catalogue. Cache capacity is explicit and does not grow with arbitrary requests.

The model/profile overlap record describes availability only. It must not be reused as an independent validation score, confidence estimate or a completed collocation result. Numerical comparisons separately retain their declared method, inputs, tolerances and evidence.

P05 comparison and coverage contracts

`science/evidence_contracts.py` defines schema 1 comparisons and coverage, method `p05-native-column-v1`. Both carry identical matching settings and model/library fingerprints. Each comparison retains all original rows with source index, timestamp/coordinates, QC/mode, selected value, signed time offset, model column/distance, native bracket and interpolation weight, eligibility and one first-failed reason. Model and residual are null when a sample is excluded. Aggregate counts and sample metrics must agree with accepted rows.

Coverage includes exact accepted depths and accepted-sample time/distance ranges. For a zero-pair profile, source time offsets are diagnostic only and distances remain null. Minimum absolute accepted time offsets and suggested alternate offsets/distances support truthful ranking even for moving casts. The browser checks requested identities, arithmetic, units, timestamps, limits, metrics and count consistency before displaying results. Scientific definitions and gate order are specified in [PHASE_05_EVIDENCE.md](#). The P02 model manifest remains unchanged.

P16C applies these same physical/QC gates to supported imported temperature and practical-salinity observations by reparsing the original file. Imported coverage describes that file only. Chlorophyll is inspectable and saveable as an observation, with no supplied model chlorophyll field or chlorophyll residual. P16D adds exact native-cell observation-support association and sensitivity, without turning sample coverage into a probability or confidence estimate. Historical saved results retain their original method/caveat text; obsolete capability wording in a frozen record does not define the current import workflow.

P06 exchange and derived products

The CF-1.10 exchange uses native float64 fields and coordinates, positive-down metre depths, and proleptic Gregorian time. Practical salinity is explicitly dimensionless with the original psu metadata retained. Horizontal kinetic energy is $(u*u + v*v)/2$, in square metres per square second, calculated before display reduction. It uses the joint finite mask, excludes vertical velocity, and is not eddy kinetic energy. Operation results identify it as `kind: derived` and include method `horizontal-ke-v1`.

`/api/products` lists installed product metadata. `/api/data-access` describes the exchange artifact, registered adapters, supported protocols and actual availability.

`/data/bay-bengal-2024-01.nc` and `/data/manifest.json` are public CDN assets on Vercel and file responses locally. The existing source case manifest is unchanged.

THREDDS WMS/WCS responses have a tested 0.0001-degree horizontal-coordinate tolerance; WCS maximum tested longitude regularization was 0.0000826928 degree. Exact-coordinate consumers should use DAP or the native file. Source/protocol numerical checks use 1e-12 absolute tolerance and identical missing masks. This is scoped interpretation and interoperability, not universal CF/OGC certification. See PHASE_06_STANDARDS.md and deploy/standards/README.md.

Later historical-source and laboratory contracts

- The three HYCOM workspace cases preserve 40 native depth levels from 0 to 5,000 m, with missing wet/land/deep cells retained. They cover bounded January 2024 Bay of Bengal/Arabian Sea and March 2024 Bay of Bengal windows. Source depths do not guarantee a value at every location or depth.
- The three Pacific GODAS workspace cases use SON 2013, 2015 and 2022 monthly potential temperature, with 28 native levels from 5 to 459 m. Climate anomalies use a complete 1991-2020 matching-calendar-month baseline and explicitly equal-weight monthly SON means. This is not instantaneous in-situ temperature or a new climate simulation.
- Drift uses historical horizontal currents at fixed selected native depths for passive tracers. It does not model sinking pebbles, wind-driven objects or search-and-rescue probabilities. Heat & Depth separates historical surface-event detection from linked native profiles and explicitly named density, stratification and column potential-enthalpy diagnostics.
- Feature evolution measures correspondence between threshold-defined native regions. It does not establish the identity or path of water parcels. Blackout excludes observations from evidence while leaving the original model and assimilation unchanged. The expedition experiment evaluates historical model sampling/reconstruction, not forecast improvement.

09 P17 capability and scientific acceptance audit

Current release reference | Scope-edited excerpt | Source: docs/P17_CAPABILITY_AUDIT.md

Reviewed 27 September 2026. Baseline runtime source: 20aaa2c6c4412809d5d7262ad6c8863c8d02e2bc, release 0.16.12. This review checks current source and the scientific test suite, rather than treating earlier phase checkmarks as fresh evidence. Subsequent P17 security, browser, deployment and delivery results belong to their own reports and the final project-state handoff.

What the evidence means

- **Implemented and internally checked** means the named supported behavior exists and has source, analytical, numerical or interaction evidence. It does not mean arbitrary inputs, every ocean, every device or production-scale use works.
- **Partial** means useful implementation exists but the wider claim still has a listed missing part.
- **Open** means the acceptance requirement needs evidence that has not been produced.

The requirement interpretation uses the preserved official [project brief statement](reference/official-2026/project brief-statement.txt). The statement prioritizes 3D model fields, colocated instrument profiles, supported ingestion, controls, extensibility and standards. It also calls for outreach. It does not publish feature weights, demand a photorealistic globe or supply a numeric completion score. The globe helps locate actual case bounds and source instruments; it is not itself the volumetric ocean model.

Fresh scientific checks

Executed check	Result	Evidence
Full existing backend suite, python -m pytest tests -q with JUnit output	931 passed, 1 failed, zero skipped; 276.55 seconds. The sole failure expected five cases after a sixth real March case had been added.	Initial log, JUnit

Executed check	Result	Evidence
New independent P17 boundary suite	18 passed; 0.42 seconds. Unit-vector great-circle oracle in both hemispheres, Decimal interpolation/residual checks with irregular depths and masked brackets, and whole-domain spherical-shell volume integration.	Log, JUnit, tests/test_p17_scientific_audit.py
Corrected wider-source suite plus all new P17 checks	68 passed; 5.66 seconds. The obsolete catalogue test now checks all six exact case IDs and the health count.	Log, JUnit

PS requirement matrix

The IDs are the project's tracking scheme, not an official SIH marking rubric. Every row remains subject to the external-review limitation above.

ID	Present implementation and internal evidence	Acceptance boundary
R01	Implemented and internally checked. Three.js WebGL2 volume ray marching, explicit cutaway and native inspection: ocean/scene.ts, ocean/grid.ts, test_case_pack.py; previous renderer/browser evidence retained.	Native HYCOM depths span 0-5,000 m in three bounded cases. Missing deep/land cells remain missing. Pacific cases supply only 5-459 m. Basic 2D fallback is labelled and does not claim volume rendering.
R02	Implemented and internally checked. In-situ temperature, practical salinity and east/north current components; horizontal kinetic energy extension. test_science_adapters.py, test_standards_extensions.py, test_fresh_model.py.	No supplied vertical current, chlorophyll model field or automatic support for arbitrary variables. Pacific source is potential temperature.
R03	Implemented and internally checked. Depth slices, scalar isosurfaces, native sections and actual source-step playback. ocean/grid.ts, ocean/scene.ts; grid and browser regressions.	Isosurfaces are rendered from declared display data; native analyses and picks retain their separate source precision. No inferred times between snapshots.

ID	Present implementation and internal evidence	Acceptance boundary
R04	Implemented and internally checked. Genuine Argo, glider, CTD and BGC profiles, geographic markers and source/QC handling. test_instruments.py, test_import_preview.py.	Genuine glider/CTD examples have no matching main model case at their supplied times/locations. Showing the marker is not an eligible residual.
R05	Implemented and internally checked. Clickable timestamped profiles, model context, imported comparisons and support links. test_evidence.py, test_imported_investigations.py, test_structure_support.py; linked frontend workspaces.	Separate observation/model dates remain visible. Matching uses declared tolerances and compatible quantities. Agreement with assimilative analysis is not independent validation.
R06	Implemented and internally checked. Supported native NetCDF, delimited CSV, CTD Exchange, preview, mapping and bounded batch import. Fresh March model preparation works. Adapter/import/fresh-model tests pass.	Browser imports are supported observations. Fresh model ingestion is a maintainer preparation workflow, not arbitrary large model uploads or a live stream subscription.
R07	Implemented and internally checked. Variable, palette, range, log/linear, opacity and exaggeration controls. Grid/contracts tests and retained interaction evidence.	Invalid logarithmic cases are rejected. Display controls do not alter native scientific calculations.
R08	Implemented and internally checked. React/TypeScript frontend, FastAPI REST and reproducible OPeNDAP path. api/app.py, deploy/standards, test_standards_extensions.py.	Public Vercel service exposes REST/downloads. Recorded WMS/WCS/DAP2 checks used local THREDDS; persistent public THREDDS is unconfigured.
R09	Partial deployment proof. Browser-only client plus portable server package; P16B archived-source isolated installation passed API and browser checks.	This is portable-package evidence on the same Windows host, not an actual INCOIS installation or institutional acceptance. P17 clean-setup evidence is recorded separately.

ID	Present implementation and internal evidence	Acceptance boundary
R11	Partial standards assurance. Scoped CF-1.10 metadata/value checks and actual local WMS/WCS/DAP2 values, including March input. <code>test_standards_extensions.py</code> , P16B protocol reports.	Not universal CF/OGC certification. An independent full CF checker was unavailable on the recorded Windows setup; missing public THREDDS remains explicit.
R12	Partial; human evidence open. Scientific workspace, Basic/In-depth real-action tutorials, source explanations and responsive/keyboard paths exist.	P09.5 actual unfamiliar-user task/comprehension evidence and representative oceanographer feedback remain required. Automation cannot close this row.

Additional feature register, all rows

File names below resolve in `science`, `api`, `tests` or `web/src` as named. Phase method reports hold the retained earlier source/browser evidence; the fresh backend results above cover the current Python implementation.

ID	Implementation and evidence reviewed	Remaining limit or acceptance status
X01	Acquisition journals, immutable source/pack hashes, publication pinning; <code>test_case_pack.py</code> , <code>test_publication.py</code> .	Internally checked; a hash proves byte identity, not scientific truth.
X02	Physical-coordinate sections/native picking; <code>features.py</code> , <code>test_features_reference.py</code> , grid tests.	Internally checked; unsupported or missing source support stays explicit.
X03	Readiness and compatibility explanations in imports/lab gates; <code>test_fresh_model.py</code> , import tests.	Internally checked; case-specific supported-tool gates retained.
X04	Native profile matcher, exclusions, bias/RMSE/MAE; <code>test_evidence.py</code> , new P17 Decimal checks.	Internally checked; equal sample weights, no independence or forecast skill claim.
X05	Eligible-depth coverage and compatible-comparison discovery; evidence/support tests.	Internally checked; no ocean-area confidence field.

ID	Implementation and evidence reviewed	Remaining limit or acceptance status
X06	Matching/QC/reference settings and threshold sensitivity; evidence/investigation/evolution/support tests.	Internally checked; sensitivity is not a real-world probability distribution.
X07	CF exchange adapter and actual local DAP subset evidence; standards/fresh-model tests.	Internally checked local capability; public persistent DAP endpoint still unavailable.
X08	Six-face native connected components, masks, membership/volume; features/reference and new P17 shell-volume checks.	Internally checked; threshold regions are not automatically identified eddies.
X09	Drawn nearest-native-column sections and eligible source associations; <code>test_features_reference.py</code> .	Internally checked; inferred cell bounds/volume are declared estimates.
X10	IndexedDB investigations, imported original bytes, original source/method/settings and exact replay; investigation/imported tests.	Internally checked; no hosted shared-account database.
X11	Readable reports, JSON/ZIP, numerical CSV and original provenance; investigation/imported tests.	Internally checked; exports retain licence and source constraints.
X12	Real January Bay/Arabian catalogue plus fresh March Bay case; case/fresh-model/guided-case tests.	Internally checked; historical bounded regions, not live whole-ocean coverage.
X13	Same-surface/different-depth source-verified guided pair; <code>test_guided_cases.py</code> .	Internally checked; similarity is not a causal explanation.
X14	Task-grouped tools, adaptive views, accessible controls and guided actions; frontend/browser evidence.	Open actual-user gate , despite implemented interface and automated interaction checks.
X15	Bounded station suggestions with explicit objective, spacing and budget; <code>test_expedition.py</code> , independent tests.	Internally checked; proposed model sampling, not an operational approved cruise plan.
X16	Native model transect/survey sampling; expedition independent-source tests.	Internally checked simulated measurements, not new observations.

ID	Implementation and evidence reviewed	Remaining limit or acceptance status
X17	Held-out historical reconstruction with equal budgets/baselines; test_expedition_independent.py.	Internally checked experimental design; no real observing-system or forecast-improvement claim.
X18	Fixed-depth passive RK4 tracer trajectories; test_drift_independent.py, drift API tests.	Analytical/upstream/source checks pass; no matched real drifter validation established.
X19	Release/depth/date sensitivity and target-entry accounting; drift independent/API tests.	Internally checked; arrival fraction is a simulation result, not rescue probability.
X20	Six fixed-cell OISST heatwave series and complete 1982-2011 baseline; test_heat_independent.py.	Pinned upstream event comparison checked; not a global live heatwave service.
X21	Native depth links, density/temperature MLD estimates and gradients; heat independent/API tests.	Internally checked; unavailable event/model overlap is not fabricated.
X22	GSW density, signed stratification and named potential-enthalpy integral; heat independent tests.	Numerical/reference checked; no external domain sign-off and no unsupported subsurface anomaly.
X23	Historical ENSO/IOD labels, matching-month 1991-2020 baseline and anomalies; climate/index/independent tests.	Internally checked selected SON 2013/2015/2022; no adjustable-strength climate simulation.
X24	Bounded tropical Pacific and separate Indian context; climate source tests.	Internally checked; no ENSO-to-local-weather causal inference.
X25	Native-volume overlap, branches/splits/merges/gaps, deterministic v2; evolution/reference tests.	Internally checked region correspondence, not water-parcel identity.
X26	Exact source-profile exclusions and retained-pair metrics; blackout/P14 tests.	Internally checked; model assimilation is unchanged.

ID	Implementation and evidence reviewed	Remaining limit or acceptance status
X28	Progressive displays, budgets, cancellation, retained publication and standalone offline evidence; wider/publication tests and P16E reports.	Internally checked declared bounds; final P17 load/device conclusions remain workload-specific. Offline viewer does not recompute every lab.
X30	Basic and explicit In-depth lessons over real operations, direct feature tours and Tools handoff; learning/globe frontend/browser tests.	Implemented and internally checked. Human comprehension remains open under R12/X14.
X32	This scientific audit plus P17 security, performance, browser and user-evidence package.	Open combined gate: independent human/usability evidence cannot be supplied by a coding test.
X33	Existing public release and portable artifacts; P17 freeze/demo/submission work is reported separately.	Delivered at the final 0.17.0 release checkpoint. P17_DELIVERY and p17-acceptance.json record setup, recording and draft assets. Actual organizer submission and institutional installation are separate actions.

Original USP index cross-check

Every named row in USP_INDEX.md is retained. The implementation paths are:

Original or later-discussed USP	Current path and relevant feature IDs	Scientific boundary
El Nino / La Nina; IOD	Climate, X23-X24	Selected historical seasons and separately labelled indices.
Moving pebbles / particles	Drift, X18-X19	Passive fixed-depth tracers, not sinking objects.
Marine heatwaves	Heat & Depth, X20	Six complete fixed-cell SST series.
Look below a warm surface	Heat & Depth, X21-X22	Source profiles and explicitly derived diagnostics.
Evidence Inspector	Compare and Instruments, X04-X05	Eligible comparisons, source QC and assimilation limits.

Original or later-discussed USP	Current path and relevant feature IDs	Scientific boundary
Change assumptions	Compare, Drift, Structures/Evolution, X06/X19/X25	Recomputed settings; no calibrated probability.
Save whole investigation	Saved work and portable export, X10-X11	Declared supported recipes and original-source replay.
Virtual survey planner	Expedition, X15-X17	Model-based proposals and synthetic sampling.
Feature tracking	Evolution, X25	Native-region correspondence with explicit gaps.
Global expansion / broader inputs	Regions and adapters, X27-X29	Pinned sources and coverage limits.
Search connected underwater structures	Structures, X08	Threshold components, not automatic feature classification.
Draw section through a structure	Structures section, X09	Native source coordinates and eligible observations.
Evidence coverage and useful comparisons	Compare/support, X05	Sample support and missing coverage are distinct.
Same surface, different ocean	Guided case, X13	Real retained source pair.
Suggest additional sampling locations	Expedition, X15	Declared objectives, constraints and candidate grid.
Fair sampling/reconstruction sandbox	Expedition, X17	Historical reconstruction rather than operational impact.
Observation blackout	Blackout, X26	Exclusion rehearsal leaves the ocean model unchanged.
Data-readiness explanations	Imports, source controls and tool gates, X03	Explicit unsupported quantities, dates and operations.

Corrections and outstanding evidence

Outstanding evidence is specific: an unfamiliar-user task study, representative oceanographer review, actual INCOIS-host installation/approval, public persistent standards hosting and independent full CF-checker results. Original drifter validation remains unavailable for the checked cases. Genuine glider/CTD examples need compatible model cases before numerical matchup can be demonstrated. Adding a UI label or passing a build cannot close these dependencies.

This audit supports a source-backed historical scientific prototype with broad connected functionality. It does not establish operational forecast skill, institutional acceptance, universal input/device support or competition placement. P17.1's matrix can be accepted; the human/external parts of P17.2/P17.4 and the combined X32 gate remain open until actual evidence is recorded.

Final release addendum

The 0.17.0 integrated backend suite passed all 963 tests after the documented test-isolation corrections. Public release, replay, full tours, browser engines and security checks are recorded in PROJECT_STATE and evidence/p17-acceptance.json. The scientific arrays and methods reviewed above remain unchanged. P17_PERFORMANCE supplies stated-environment measurements, P17_DELIVERY records the isolated portable setup, and the actual backup recording and six-slide draft establish X33's prepared-asset gate. These results do not close P09.5, R12, X14 or X32's human evidence requirement. The baseline failures above remain part of the record.

10 P17 security review and request safeguards

Current release reference | Scope-edited excerpt | Source: docs/P17_SECURITY.md

Reviewed 27 September 2026. This is an application security review with targeted regression tests, not an independent penetration test, a compliance certification or a promise that the service cannot be abused.

Implemented changes

The public API now limits requests before decoding JSON or running scientific work. A pure ASGI guard counts bytes even when Content-Length is absent. It rejects invalid or oversized lengths, compressed request bodies, excessive header/query data and browser POST requests from another origin. A body deadline stops incomplete uploads. Errors retain the existing JSON envelope, request ID, app version and no-store response policy.

Boundary	Current limit
Ordinary scientific JSON	65,536 bytes before parsing
Original observation upload	2,000,000 bytes
Imported comparison/support and investigation replay/export	3,000,000 bytes
Regional pack replay	4,500,000 bytes
Request headers / query string	32,768 / 8,192 bytes
Body completion	15 seconds after admission
API requests per client, per process	360 per minute, burst 360
POST calculations per client, per process	60 per minute, burst 60
Concurrent API requests per process	24
Concurrent POST calculations per process	4
Counter identities per process	2,048, idle expiry after 10 minutes

Admission is immediate. A full calculation slot or exhausted token bucket returns HTTP 429 with Retry-After instead of growing an unbounded queue. Slots are released after a response, a disconnect, timeout or exception. Counters use a process-specific salted HMAC of the client address; raw addresses and request payloads are not retained by this guard. The Privacy Policy describes these counters. It does not make a provider-wide log deletion promise.

Vercel deployments use the platform-supplied `x-vercel-forwarded-for` only when the server's VERCEL environment flag is set. Direct installations use the ASGI peer address and ignore raw forwarded headers. Institutions using a reverse proxy must configure its trusted proxy addresses correctly and apply their own public edge limits. The generic X-Forwarded-For header is not trusted by application code. [Vercel request header documentation](#).

These are process-local controls. Independent serverless workers have separate counters; cold starts reset them. They are not a distributed quota, bot detector or DDoS protection claim. The deployment uses the existing Vercel project. A shared edge policy or external atomic store would be needed for one globally consistent quota. No new paid service, account or secret was introduced.

The four-slot budget applies specifically to POST requests. Native subset, evidence comparison/coverage and drift-context GET requests can also perform substantial calculations; they share the 24-request total and the 360-per-minute client budget. This is a finite bound, not a claim that 24 cold scientific requests fit every machine's memory or response target. An institutional proxy must tune its workload limits against measured datasets and worker resources. No security setting is exposed as a public API or user-controlled environment input.

Browser and deployment controls

The viewer uses a Content Security Policy with same-origin API connections and scripts, the exact SHA-256 of its existing inline startup recovery guard, no arbitrary inline JavaScript, no eval, no objects, no base URL changes and no form submission. Inline styles remain allowed because React places the tutorial and scientific overlays dynamically. Framing is denied, and camera, microphone, device location, payment and USB permissions are disabled. Vercel responses declare HTTPS Strict-Transport-Security.

The separate `/api/docs` page retains its pre-existing Swagger CDN dependency with an explicit CDN allowlist and the hash of its fixed startup script. This exception applies to API documentation, not the application workspace. The docs page still needs internet access for its reference UI; the OpenAPI JSON is served locally. Existing downloaded offline investigation files have their own hash-based CSP and no network permissions. Viewer controls do not replace those file policies.

Both Python and Vercel header definitions are checked against the source startup script. If that script changes, its allowed hash must change in both places before deployment. Source maps remain disabled in the frontend build. `.env*`, `.vercel`, local runtime directories, tests, logs and documentation remain excluded from the public deployment.

Existing boundaries checked

- Public source data is packaged by maintainer tools. Runtime requests do not fetch an arbitrary user URL. Case IDs, example names and profiles resolve through checked collections, not arbitrary file paths.
- Observation parsing rejects unsupported formats and nested NetCDF groups. Limits include 160 variables, 32 dimensions, 5,000 source samples and 24 Argo profiles; Argo total decoded elements and individual selected NetCDF variables have bounds before decoding. Imported responses are capped at 3.5 MB. Native NetCDF access is serialized.
- Imported original files are reparsed from bounded bytes and retained only for the request. Uploaded files do not enter the public immutable result cache. File checksums and parser/method versions remain checked during replay.
- Existing scientific schemas cap coordinate ranges, sample counts, iteration settings and output sizes. The broader new transport guard does not replace these scientific checks or change calculations.
- Investigation HTML escapes source/user text. Offline data escapes the < character before embedding JSON and uses `textContent` for display. CSV exports protect formula-like text while retaining negative numeric values. Portable original filenames reject directory separators and control characters.
- The publication switch remains a maintainer-only local operation; there is no public publication mutation endpoint. New publication/source identities are validated before startup.

Evidence and reproduction

- `evidence/p17-security-regression.log`: 84 passing guard, API, original/import-preview, imported-replay and structure-support tests. This includes three parallel sessions, a 120-request read burst, rejection/recovery, timeout, disconnect, oversized chunked input, malformed headers and exact script hashes.
- `evidence/p17-security-tests.log`: initial 82 passing tests before the extra parallel-session and disconnect cases were added.
- `evidence/p17-security-deadline-tests.log`: all 13 guard tests pass after the independent code reviewer identified and corrected an absolute-deadline edge case. The regression advances a fake clock between immediately available chunks and requires a timeout before another chunk is read.
- `evidence/p17-python-secret-audit.json`: all 19 exact runtime pins queried successfully using the primary PyPI release JSON advisory records. Zero active advisories were

returned at the recorded time. The scan found no listed secret signatures or tracked environment/credential paths among 1,270 tracked/build files.

- `evidence/p17-npm-audit.json`: npm reported zero known advisories in 158 frontend/build dependencies at the recorded audit.
- `evidence/p17-python-secret-audit-final.json`: the exact-pin/signature audit was repeated against the final 0.17.0 build. All 19 PyPI queries succeeded, with zero active advisories, candidate secret signatures or tracked sensitive paths in the same 1,270-file scan scope.
- `evidence/p17-security-built-headers.json`: 8/8 checks pass on the final local 0.17.0 worker at port 8038. The built startup script hash and headers on the workspace, three legal pages, API reference and health endpoint agree with the policy. This verifies local responses; the separate final public evidence is recorded below.
- `evidence/p17-drift-security-isolation.log` and `.xml`: 81/81 drift API and security checks passed after fixing scientific test isolation. The initial full backend run had 952 passes and 11 drift-test failures because one module-wide simulated client spent its 60-request POST burst across unrelated parameterized cases. Each scientific test now has a distinct simulated peer on the same shared app, preserving source caches, production rate defaults and stable identity within each test. No production limit was raised or bypassed. The original full-run failures remain in `p17-backend-final.log`.
- `evidence/p17-backend-accepted.log` and `.xml`: the complete integrated backend rerun passed all 963 tests in 295.24 seconds after that fixture correction. The three warnings concern deprecated test-client APIs and an existing JUnit property compatibility warning, not failed checks. Runtime source remained `5a5e6160a7165427d6f5a174b58083f5a2db7750`.
- `evidence/p17-security-public.json`: all 14 public security checks passed on READY deployment `dp1_6REGcXUreo8pXinNuhQp331W2ANe`, release 0.17.0, source checkpoint `0f5c417`. Thirteen bounded requests checked HTTPS/HSTS, exact workspace/legal/API-docs CSP, denied framing/permissions, unavailable private paths, malformed JSON and cross-origin rejection. Error envelopes retained app version, request IDs and no-store. A preceding single GET encountered a local report-script attribute error; no application defect was indicated and the corrected run is recorded.
- `evidence/p17-swagger-public.json` and `.png`: one actual public Chromium page rendered the Swagger API reference with 45 operations and version 0.17.0. Both existing CDN assets and local OpenAPI JSON returned 200. No page, console or request errors occurred. This checks that the API-docs CSP exception works in a browser, not only in response headers.

Run from the repository root:

```
.venv/Scripts/python.exe -m pytest tests/test_security.py tests/test_api.py
tests/test_instruments.py tests/test_import_preview.py
tests/test_imported_investigations.py tests/test_structure_support.py -q
.venv/Scripts/python.exe deploy/security_audit.py --report
docs/evidence/p17-python-secret-audit.json
npm --prefix web audit --json
```

The Python advisory check uses the same primary PyPI JSON records that are documented by [pip-audit](#). It is a direct exact-pin query, not a claim that pip-audit was installed or run. The scanner prints only the file, line and signature type if it finds a candidate; it does not print a matched credential value. Ignored personal credential stores and env files are not read. Advisory absence and signature scans cannot prove there are no vulnerabilities or secrets.

Remaining limits and deployment checks

The final public response-header, bounded-error and API-docs checks are recorded above. Complete tutorial, import, saved-export and numerical replay verification remains in the root release evidence, separate from these security requests. No public flood test was performed.

Native scientific dependencies include netCDF-C 4.9.3 and HDF5 1.14.6 in the checked Windows environment. The version-specific PyPI scan does not comprehensively audit those C libraries, the operating system or Vercel infrastructure. File limits and serialization reduce resource exposure but do not sandbox native parser vulnerabilities. A future institutional deployment accepting untrusted uploads at scale should isolate parsers in constrained workers and maintain its own OS/library patch process.

Already accepted server computations can finish after browser cancellation; this is stated in the Privacy Policy. Application bounds and Vercel's 30-second function duration remain relevant. A slow connection can exceed the 15-second upload admission window, and people sharing a network share its per-process client counter. Test automation is not evidence that every device or network will succeed.

Prepared explanation: The release adds limits around public requests and uploads, keeps credentials out of the browser bundle and saved investigations on the user's device, and restricts which browser resources can execute. The tests exercise those boundaries without changing scientific values. This is a hardened public prototype, with process-local rate limits and external infrastructure responsibilities stated explicitly.

11 P17 measured performance

Current release reference | Scope-edited excerpt | Source: docs/P17_PERFORMANCE.md

Measured 27 September 2026 against public release 0.17.0. These are technical measurements on the development computer, not participant timings, a capacity forecast or a claim about every evaluator device.

Browser workflow

The harness used fresh headless Chromium 153.0.8010.12 contexts on Windows 10.0.26200, an Intel Core i7-14650HX with 24 logical CPUs and 16,873,545,728 bytes of system memory. Browser caching was disabled. Server and operating-system caches could be warm. Three independent contexts were run per condition, without other browser suites or server benchmarks running concurrently.

Desktop used a 1440 by 900 viewport with no added CPU/network throttle. Mobile emulation used 390 by 844, touch, a 4x CPU slowdown, 150 ms added latency, 200,000 bytes/second download and 93,750 bytes/second upload. This is emulation on the same computer, not a physical phone or an unrelated GPU.

Measured event	Desktop median, seconds	Throttled mobile median, seconds
Navigation until welcome controls are visible	3.181	4.670
Navigation until globe case control is enabled, after Skip	3.572	4.988
Click Open 3D ocean until native model reading	2.514	2.483
Select depth slice until native reading	0.120	0.239
Open Compare and select source pair until eligible count	0.561	1.224

Measured event	Desktop median, seconds	Throttled mobile median, seconds
Entire automated route through checked comparison	6.837	9.333

The first two rows share the navigation start. Later rows time separate actions. Medians must not be added to estimate a total. The complete route ranged from 6.158 to 8.920 seconds on desktop and 8.075 to 9.859 seconds under mobile emulation. Automation does not spend human reading or decision time.

All six runs retained Interactive 3D, no uncaught page errors, no failed API response status and no document horizontal overflow. Each checked the unchanged native 22.303 degree Celsius reading and the January Argo comparison's 103 eligible pairs and selected -0.152 degree Celsius model-minus-observation residual. These exact figures verify a declared source pair, not general model accuracy. The report also retains resource timings, sizes and JavaScript heap measurements; heap size is not total browser memory.

Evidence: full measurements, harness. Three samples are too few to establish reliable tail latency, a speedup claim or service-level guarantees. The desktop and mobile conditions ran sequentially, so their medians are not a controlled hardware comparison.

Regional source and bounded concurrent requests

Two public rounds each sent four concurrent supported subset requests. All eight returned 200 with valid source-backed results; individual response times ranged from 0.795 to 1.369 seconds. Responses were gzip encoded. This is a bounded sample, not a stress test or evidence for 100 simultaneous users. The app's per-worker request limits and the regional calculation gate remain separate controls.

Evidence: source/cache/request report. Reproduction:

```
$env:OCEAN_TEST_URL='https://depth-atlas-seifuku.vercel.app'
$env:OCEAN_PERF_DIR='docs/evidence/p17-performance-public'
node web/benchmarks/p17-performance.mjs
.venv/Scripts/python.exe -m science.verify_p15_performance --base-url
https://depth-atlas-seifuku.vercel.app --output docs/evidence/p17-public-
capacity.json --source-audit
```

Use a new output path for later measurements so the release evidence is retained. The source audit needs the original wider-data files in the development reference environment.

Remaining acceptance

Chromium, Firefox and WebKit checks, fallback tests and the visual review are linked from the final acceptance index. They do not establish universal device support. The known large JavaScript chunks remain a loading-cost limitation. A fair unfamiliar-user comparison and an unrelated physical-device check are still required by P17.4. The prepared [participant protocol](#) contains no invented participant results.

12 P17 delivery and technical handover

Current release reference | Source: docs/P17_DELIVERY.md

Prepared 27 September 2026. The exact final source commit, public deployment and acceptance reports belong to PROJECT_STATE and the P17 evidence index. A release is a tested prototype of the declared scope; it is not scientific certification or institutional acceptance.

Verified public release

Release 0.17.0 is deployed Ready as `dp1_6REGcXUreo8pXinNuhQp331W2ANe` on the existing Vercel project. [Open the current app; immutable deployment.](#)

Runtime source: `5a5e6160a7165427d6f5a174b58083f5a2db7750`. Deployment packaging source: `0f5c417f607f04aa586ae94f16fbe3aec5b35da7`. The latter only excludes the submission materials from the website/function upload; application and scientific bytes are unchanged. The canonical URL can move with a later deployment, so retain the immutable URL and source identities for this release.

Public verification passed 24 release checks without retries, 61 exact replay checks, and both complete 39-lesson tours on desktop/mobile, using normal transport. These are bounded tests of this release, not proof of universal network reliability or participant comprehension.

The actual public-browser recording is a **silent technical backup**, 43.12 seconds at 1440 by 900 pixels, 3,707,695 bytes. Its nine real steps show the globe, model, cutaway, native depth reading, eligible comparison, explicit save, checked reopen and Tools. Capture completed without page errors; video seeking/decoding passed, and selected decoded frames were visually inspected. Recording metadata and separate narration cues bind it to this release, source and deployment. The exported demonstration investigation passed exact saved replay. This local file is not a narrated pitch, external video upload or portal

submission.

Architecture

flowchart LR

S[Original public NetCDF and observations] --> P[Offline adapters and preparation]

P --> C[Checked case packs and source identities]

C --> A[Bounded FastAPI scientific service]

C --> T[Separate persistent THREDDS]

A --> U[React and Three.js browser workspace]

T --> O[WMS WCS OPeNDAP clients]

F[User-selected supported file] --> A

U --> B[Explicit browser-local investigation]

B --> E[Portable evidence and offline viewer]

E --> A

The geographic view locates available cases and measured profiles. Display arrays make rendering manageable; analytical queries use declared native grids and methods. The model, observation and derived result remain separate contracts with source dates, definitions, QC and fingerprints. UI controls and tutorials invoke the same real commands. Changing the presentation does not manufacture scientific data.

The public app serves a compatible frontend, API and read-only publication together on Vercel. Runtime requests read included prepared packs; visitors do not trigger provider downloads. Imports are bounded request processing, and imported investigations reparse original bytes before comparison or replay. Library caches are separate from temporary user files. Browser saves and downloadable files are explicit; there is no application account or cloud database. The Privacy page and security audit describe the deployed metadata/security behavior.

THREDDS is a separate Java service with a different lifecycle. Its local standards evidence does not mean the Vercel app hosts a persistent THREDDS server. A receiving institution must select an approved host, storage, network/access policy and operating procedure.

Package inventory

Item	Location	Status and use
Public prototype	https://depth-atlas-seifuku.vercel.app/	Verify against the current release evidence before judging.
Frontend/backend/source	web/, api/, science/, adapters/	Source in this workspace; no public repository was created by P17.
Runtime data	casepacks/, api/publication.json	Bounded checked historical products, examples, source references and checksums.
Dependency locks	requirements*.txt, web/package-lock.json	Runtime-only install differs from reference/test environments.
Setup	README, DEVELOPMENT, this file, deploy/	Actual isolated setup evidence below. Docker recipe remains separately unexecuted when Docker is unavailable.
Model and observation methods	SCIENTIFIC_CONTRACTS and phase method reports	Coordinate/variable definitions, QC, matching, interpolation, scientific labs and declared limitations.
Coverage	P17_CAPABILITY_AUDIT and FEATURE_REGISTER	Distinguishes working scope, internal evidence and outstanding external acceptance.
Reproducible examples	Retained docs/evidence/*records*.json and replay verifiers	Select source-compatible current-method records; older incompatible versions deliberately report a mismatch.
User instructions	In-app Basic/In-depth tutorials, public import guide, P17_DEMO	Demonstrate the core model/profile/evidence workflow first.
Shortlisting draft	submission/project_brief_OceanNavigator_Idea_DRAFT.pdf and .pptx	Six slides, official template, reviewed local PDF. Not submitted. Portal metadata and limits need user confirmation.
Portal/deck support	submission/PORTAL_TEXT.md, submission/LINKS.md, submission/build_deck.py	Editable text and repeatable document source. No invented repository or social link.

Item	Location	Status and use
Backup recording	Public-browser WebM, capture details	Verified silent 43.12-second technical backup from release 0.17.0; no narration or external upload.
Attribution	web/public/third-party-notices.txt, each dataset's manifest/licence, science/vendor/PARCELS_LICENSE.txt	Preserve existing library and provider notices in distributions and derived outputs.

No project-wide open-source licence has been selected. This package does not grant an invented licence for original project code. Review redistribution terms and source visibility before any public repository or institutional transfer. Publicly available data is not automatically public domain; retain each provider's actual terms and citations.

Clean installation procedure

Extract a source archive into an empty directory, without copying the development virtual environment, node_modules, credentials or built assets. Use the runtime-only pinned requirements. From that directory in PowerShell:

```
python -m venv .venv
.venv/Scripts/python.exe -m pip install -r requirements.txt
.venv/Scripts/python.exe -m pip check
npm --prefix web ci --no-fund
npm --prefix web run build
.venv/Scripts/python.exe -m science.publication --candidate casepacks --manifest api/publication.json
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8000
--no-access-log
```

Check /api/health, the catalogue, a native source reading, an instrument, saved replay and anonymous static pages. The source-probe verifier runs with runtime dependencies:

```
.venv/Scripts/python.exe -m science.verify_p16b --url http://127.0.0.1:8000
--report portable-http.json
```

Run `science.verify_p16c --url http://127.0.0.1:8000 --report portable-replay.json` from a separate review checkout with `requirements-dev.txt` installed. It needs the HTTP test client and compares the serving runtime against newly calculated references and retained archives. Do not silently install reference/test dependencies into a runtime-only server and still describe it as runtime-only.

The frontend build typechecks before emitting assets. `science.publication` validates prepared files and creates a release-compatible manifest. Starting with a stale release manifest is an error, not a reason to bypass the identity check. Use the same method in a container image; the Dockerfile now includes this validation after copying its checked

packs. A changed build must be tested before replacing a running installation.

The final P17 isolated installation uses runtime commit

5a5e6160a7165427d6f5a174b58083f5a2db7750, release 0.17.0, in `.runtime/p17-portable`. The local archive `.runtime/p17-portable-source-0.17.0.zip` is 178,526,632 bytes with SHA-256 5ea0fe6739e42f11866e454e9da8d0c2b7fdb2901be0651198a6d57a02e34ad9. The subsequent 0f5c417 commit only excludes submission files from Vercel packaging; scientific/frontend/API runtime bytes are unchanged.

The environment began with a clean 0.16.12 source archive, fresh runtime-only Python dependencies and a fresh `npm ci`. The exact 0.17.0 archive was then overlaid; no tracked files were deleted between these source commits. Pinned `pip` installation, `npm ci`, `typecheck/build` and publication validation were repeated. `pip check` found no broken requirements; the known large-chunk/build-plugin warnings remain. The 529 compared `runtime/data/source` files match the final archive exactly. The regenerated publication has the correct application version and unchanged scientific publication identity. The runtime-only environment contains no `httpx` test client.

The isolated server passed **35 source/API checks over 33 requests, 61 exact imported/legacy replay checks over 17 requests** and **nine browser checks** across Chromium desktop, Chromium mobile emulation and WebKit. Browser checks cover native model readings, genuine chlorophyll import beside the correctly dated model and supported-case fallback. There were zero browser skips, unexpected failures or retries. A separate development environment ran the replay verifier without changing runtime dependencies. These counts describe different checks and are not combined into a unique-test total. Evidence: `p17-portable-summary.json`, source checks, replay checks and browser report. The verified worker on port 8043 was stopped after testing.

Host scope: Windows on the same development computer, Python 3.10, Node 22.17.1. `Get-Command docker` returned no executable on 27 September 2026. No Linux, Docker, INCOIS server or unrelated physical-device result is inferred from this run.

Institutional deployment boundary

The ordinary browser requires no Python, Java or plugin. Those are server/preparation dependencies only. A receiving institution should first run the included checked case and repeat its exact source/replay checks. Next it can prepare a compatible new model through a documented adapter, verify its coordinates/quantities and establish its own publication process.

Before exposure on an institutional network, configure TLS, trusted proxy handling, allowed origins, access policy, process and request budgets, host monitoring and

recovery. Application process-local protection does not replace distributed edge/WAF controls. See [P17_SECURITY.md](#) for implemented mechanisms and limits. Direct hosts use the ASGI peer and ignore forwarded client-IP headers; the Vercel platform flag selects its platform-owned client header. Proxy installations need a deliberate verified trust policy. No provider keys or new security environment settings are required for the supplied defaults. No paid service, provider account or operational security certification is bundled.

Use the separate [standards runbook](#) for THREDDS and its resource requirements. The Windows launcher verifies pinned vendor archives. Docker/Linux instructions are reproducible recipes but require execution on their target host. Current scoped CF checks and WMS/WCS/DAP source-value comparisons do not substitute for full independent CF certification.

Submission boundary and remaining decisions

Official guideline and template files were downloaded again on 27 September 2026 and match the 21 September copies byte for byte. The current PS entry matches the implemented priority and shows a 30 September 2026 submission deadline. See [evidence/p17-official-source-check.json](#) and [SUBMISSION_CHECKLIST](#).

The signed-in team dashboard was not inspected. Nomination, registered details, authorization letter, current per-field limits, PDF size cap, round-specific instructions and actual submitted status remain unverified. The deck is a review draft, not a submission. University non-affiliation in the product is distinct from any organizer-required student registration or institutional nomination; do not invent either.

Independent oceanography review, unfamiliar-user evidence, target-institution installation, persistent public THREDDS and full external CF validation need external access or participants. Keep those acceptance gaps explicit. Security and internal tests can support a bounded engineering release without falsely closing those external gates.

13 P17 demonstration and judge questions

Current release reference | Scope-edited excerpt | Source: docs/P17_DEMO.md

Prepared 27 September 2026. This is a rehearsal script, not a submission receipt or an account of a participant study. Use the release and public verification recorded in PROJECT_STATE. The script starts with the PS's core model/observation workflow; optional labs remain available for questions.

Thirty-second explanation

Ocean Navigator puts a three-dimensional ocean model beside real instrument measurements in a browser. A user chooses an available place and date, looks down through the water, inspects a sensor profile, and checks which measurements can be compared with the model. Saved investigations keep the sources and calculation settings so the result can be checked again. It is intended for oceanographers, data teams and learners. The current prototype uses historical data and does not issue operational advisories.

Main demonstration: about four minutes

The timings below are a presentation budget, not measured task-completion results. Use a supported January Bay of Bengal case for this route. Confirm the selected case/time in the interface rather than changing source dates to the current year.

Time budget	Action	What to say and show
0:00-0:30	Open the app. Show Basic tutorial and Skip. On the globe choose the January Bay case and explicitly open its model.	"This outline shows the area of the available dataset. The label gives its historical dates. The globe helps us choose evidence; it is not a live coverage map."
0:30-1:15	Rotate the water column, open the optional cutaway, choose a depth slice and inspect a native reading. Briefly show salinity or currents.	"Colour represents a named quantity and unit. Depth is exaggerated for visibility. The readout uses the original model grid; missing cells are not filled with invented values."

Time budget	Action	What to say and show
1:15-2:10	Open Instruments and a supported Argo profile. Keep the model beside it. Select a reading and open the eligible comparison.	"This is an observation with its own position, timestamp and QC. Comparison accepts only samples that meet the declared quantity, time, distance, depth and quality rules. Here are residuals and excluded samples."
2:10-2:55	Open Find structures, run the supported warm-region example, then Show observation support.	"This shape is a connected set of model cells meeting the threshold. These readings support a comparison in some locations; blank areas are gaps in eligible evidence, not low-confidence measurements. A stricter rule can change support."
2:55-3:35	Save the comparison or structure explicitly. Reopen from Saved, then download the evidence ZIP or offline viewer where supported.	"Reopening recalculates and verifies the source and method identities. The file carries settings and provenance. Browser storage is local, so a downloaded backup is useful."
3:35-4:00	Open Tools and show the extra feature tutorials.	"The same workspace includes Drift, Heat & Depth, Climate, Evolution and other experiments. Each declares the data and limits that apply to it. We can explore one if useful."

Do not race through all tools. If the judging slot is shorter, keep depth, one real profile, one comparison and saved evidence. The globe, rendering and extra labs do not substitute for this core proof.

Optional branches

Judge's interest	Open	State the boundary
New sensor data	Instruments > import preview, then the supplied CSV/NetCDF example	Read detected variables, units and QC before Add. Supported layouts and upload limits are explicit. The demo does not access a judge's local files automatically.

Judge's interest	Open	State the boundary
Chlorophyll	Globe > measured chlorophyll example > open profile beside March case	Genuine BGC readings, raw/adjusted provenance and QC. The bundled model has no chlorophyll field; no model chlorophyll residual or bloom map is claimed.
Currents and particles	Drift on a checked January Indian case	Passive fixed-depth tracers in historical horizontal currents. They are not sinking stones, a search-and-rescue forecast or a validated real-drifter prediction.
El Nino and La Nina	Climate Event Lab	Selected historical seasons and a declared anomaly baseline. Monthly potential temperature is distinct from instantaneous in-situ temperature.
Heat below the surface	Heat & Depth	Surface-event detection and available depth context. A surface event alone does not establish a marine heatwave at depth.
Feature changes	Evolution, then Blackout	Evolution links overlapping model-cell regions. Blackout removes eligible observation evidence; it does not rerun the model or measure forecast impact.
Sensor placement	Expedition	Sampling/reconstruction experiment under declared assumptions. A comparison is not proof of operational observing-system improvement.
Interoperability	Data access, native NetCDF; separately run the standards package	REST and bounded NetCDF are public. Local WMS/WCS/OPeNDAP have actual protocol evidence; persistent public THREDDS remains a deployment dependency.

Judge questions

What does the PS ask for? A browser-native 3D ocean data system that co-displays model fields and instruments, supports NetCDF/text, provides depth/time/variable/colour controls, and can grow through modular sources and open standards. It also needs an understandable outreach experience.

What is the main contribution? The connected path from source-backed 3D exploration to QC-aware model/profile evidence and reproducible investigations. Structure support, sensitivity and guided actions make the extra analysis inspectable instead of showing isolated animations.

Is this live ocean data? No. These are checked historical cases and bounded source products. Every view keeps the source date. A maintainer validates and publishes updates; the demo does not run a live ingest scheduler.

Are matching measurements an independent validation? Not automatically. A model may assimilate related observations. The app shows the comparison and provenance, while independent validation requires an appropriate held-out design.

Can any NetCDF file be uploaded? No. The supported observation formats, units, mappings and budgets are documented. Model preparation uses separate adapters. Unsupported metadata and physical definitions are rejected rather than silently guessed.

Can INCOIS run it? The frontend and API have a portable setup and checked packs; an isolated installation has been exercised on the development Windows host. Actual INCOIS infrastructure, access controls, persistent standards hosting and institutional approval need their own deployment test.

How much faster or more accurate is it? We report only measured technical timings from the stated audit environment. Human task-time improvement and operational forecast accuracy have not been demonstrated. The usability and domain-review protocols remain open until real participants provide evidence.

Where is user data stored? Temporary imports are processed for the current request/workspace. Explicit investigations are stored in this browser and may include original imported bytes. Exported files are controlled by the downloader. There is no account or application cloud database. See the current Privacy page for hosting and security metadata handling.

What still needs work? Use the current capability audit. Key external gaps include unfamiliar-user evidence, independent oceanographic review, institutional deployment and persistent public THREDDS. Full CF certification is not established.

Who built and operates it? Team Seifuku. Adwita Kurle is team leader; Gaurav Ranade is the website creator and operator. The project has no university affiliation or university development involvement.

Rehearsal and fallback

Before presenting, open a new browser context on the canonical HTTPS link, verify the source case loads, and confirm the release/publication identity. Prepare a known portable investigation and its readable offline viewer. Check downloads without developer authentication. Close unrelated tabs and notifications before recording.

The verified technical backup records actual actions on the public release 0.17.0. It is **silent**, 43.12 seconds long, 1440 by 900 pixels and 3,707,695 bytes. This is a compact technical demonstration, not the four-minute spoken pitch budget above. Recording metadata supplies nine timestamped narration cues, the Bay of Bengal source case, runtime/packaging commits and deployment ID. Capture passed without page errors; seeking and decoding were checked and selected video frames were visually reviewed. The exported saved investigation reproduced exactly when replayed.

The recording shows the default guide choice, data globe, selected historical model, optional cutaway, 100 m reading, eligible Argo comparison, explicit save, checked reopening and Tools. Narration cues remain a separate text track in the metadata. No voiceover or external video upload has been completed, and no portal submission is claimed. Keep the original capture and source identity if editing a future presentation copy.

If the live service is unavailable, explain the interruption once and switch to the real recording or offline saved evidence. Offline viewers retain saved values and provenance; they do not run new scientific calculations or the full 3D app. Keep the portable server as a separate technical fallback, with no claim that it runs on an evaluator's device automatically.

14 Final participant and scientific review protocol

Current release reference | Scope-edited excerpt | Source: docs/P17_USER_VALIDATION.md

Prepared 27 September 2026. No participant sessions or external oceanographer review have been performed by this document. Automated browser checks are separate engineering evidence.

Small usability comparison

Recruit 4-6 people who have not used Ocean Navigator if practical. Two or three can still reveal issues, but cannot support a reliable comparative claim. Include an ocean-data user separately. Ask consent for anonymous task notes and any recording. Use participant codes, not names or contact details.

Compare two entry paths to the same released build: Basic tutorial available, and tutorial skipped with normal tool labels. Alternate which path participants try first. Use equivalent January Bay and Arabian tasks, with task order reversed across participants to reduce learning effects. Do not compare the app with an imaginary competitor or a deliberately obstructed interface.

Tasks:

1. Identify the selected source and historical date. Open the ocean and show water below the surface.
2. Explain the colour, units, depth and whether vertical exaggeration changes a measurement.
3. Open an instrument profile. Identify its time and a QC exclusion.
4. Find an eligible model comparison, read the signed residual, and explain why agreement is not independent validation.
5. Save the investigation, reopen it and explain where the saved data lives.
6. Find Tools and one extra feature without help. Distinguish its measured data from a simulation.

Use the same task wording in both conditions. Let the participant think aloud. Record task completion, elapsed time, wrong turns, requested help and their actual words. Give help

only when requested or after the agreed task limit; mark assisted completion explicitly. An incorrect explanation is not a success just because a button was clicked. Record device/browser/screen and network context without identifying the participant.

Use a blank table per participant:

Task	Entry path	Completed in dependently	Time	Help/errors	Participant explanation
1-6					

Report raw counts and individual observations. With this small sample, avoid claims of statistical significance or universal ease of use. Fix repeated confusion and repeat the affected tasks, keeping first-run failures in the record.

Independent domain review

Ask a qualified ocean-data reviewer to inspect representative original-source and result pairs: native model sample, pressure-derived observation depth/QC, vertical interpolation/residual, heat diagnostic, passive tracer, monthly climate comparison and support/blackout limitations. Supply source hashes, units, dates, methods and numerical fixtures. Record the reviewer's scope, comments and unresolved objections.

The reviewer should check unsupported and missing-data cases as well as successful results. The project does not claim operational advisories, trajectory probabilities, bloom maps from single chlorophyll profiles, independent validation of assimilated observations, or INCOIS endorsement.

Available-device check

Have a participant open the canonical HTTPS link in their normal browser with no installation. Record whether the welcome, persistent globe, volume or labelled 2D fallback, comparison and save/reopen work. Test one actual phone and one non-developer laptop if available. Emulators do not replace these records.

Send findings back as observations and screenshots without personal data. P09.5, R12/X14 and the participant/domain portions of P17 remain open until this evidence exists.

15 Complete feature register

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt |
Source: docs/FEATURE_REGISTER.md

All implementation checkboxes start unchecked. Check an item only when its declared supported behavior has verifiable evidence, and link that evidence in PROJECT_STATE.md. A feature can be scoped to documented inputs; do not imply support for arbitrary data.

P01-P16 and the P16A-P16E completion checkpoints provide the connected historical-data workspace, scientific labs, bounded wider sources, learning and replay. P17 has audited the present implementation and delivered the 0.17.0 release evidence. Actual institutional deployment and human validation remain separate open dependencies. Historical notes below record earlier milestones; the table and P17_CAPABILITY_AUDIT.md define the current scoped status. The geographic globe alone does not establish volumetric rendering.

R01-R12 retain the guide's tracking labels. They are a compact requirement register, not an official scoring scheme. The official PS controls requirement interpretation. X01-X33 cover the guide's software vision and subsequent discussion. Phase completion alone does not excuse an unverified feature.

27 September globe/tutorial follow-up: the explicit data globe extends discovery for R04/R05 and X03, while PS-ordered core lessons and the expanded feature tours extend X30. GLOBE_AND_TUTORIAL_REVIEW.md records the requirement-to-interface review. Globe markers are actual source locations, not a concentration surface or validated coverage estimate. No new core requirement checkbox is closed by this presentation change; final release evidence is in PROJECT_STATE.md.

The subsequent 0.16.12 reliability follow-up refines R01/R03/R05/R08 and X30: bounded slower-read recovery, next-frame-only preloading and cancellation of unused speculative work. Scientific methods and source bytes are unchanged. RELIABILITY_FOLLOWUP.md and PROJECT_STATE.md record reproduced failures and verification; no new requirement or P17 checkbox is closed.

Official requirements

Verified	ID	Capability	Implementation phase(s)	Required evidence
[x]	R01	Browser-native full-water-column 3D model visualization	P03	Real depth-resolved volume; correct positions/depths and numerical picking
[x]	R02	Temperature, salinity, current vectors and extensible variables	P02, P03, P06	Real supported fields, units/vector orientation, and a working extension example
[x]	R03	Depth slices, isosurfaces and actual time-step animation	P03	All three operate on scientific values and real timestamps
[x]	R04	Geolocated Argo, glider, CTD and BGC observations	P04, P16A	Genuine sample for each claimed type; correct coordinates, time and QC
[x]	R05	Clickable timestamped profiles alongside model fields	P04, P05, P16A	Profile selection and linked charts with eligible matching behavior
[x]	R06	Automated supported NetCDF and delimited-text ingestion	P02, P04, P16A	Additional compatible input works; malformed/unsupported data handled
[x]	R07	Palettes, ranges, linear/log scale, opacity and vertical exaggeration	P03	Valid controls function; invalid logarithmic cases rejected; analysis unaffected
[x]	R08	Modern JS frontend and lightweight REST/OPeNDAP backend path	P01, P06	Runnable browser client, real documented API/protocol requests
[x]	R09	Browser-only user experience and INCOIS-deployable server package	P01, P06, P16B	Reproducible server package and browser client; actual institutional installation separately recorded
[x]	R11	CF-NetCDF interpretation and OGC WMS/WCS interoperability	P02, P06	Metadata checks and actual map/coverage requests independently verified
[]	R12	Scientific investigation and public-outreach usability	P03, P09, P16	Usable investigation and basic public/student guided path at P09; expanded learning mode at P16; actual user feedback

Additional product capabilities

Verified	ID	Capability	Implementation phase(s)	Required evidence
[x]	X01	Versioned acquisition/cache and data provenance	P02	Original/subset fingerprints, retrieval recipe, licence and transformation record
[x]	X02	Vertical sections and linked analytical inspection	P03, P07	Physical coordinates/depths agree with source values
[x]	X03	Data-readiness explanations	P04	Actual supported operations, missing variables and compatible coverage explained
[x]	X04	Evidence Inspector with residuals and optional sample metrics	P05	Validated matches, physical conversions, exclusions and assimilation caveat
[x]	X05	Evidence coverage view and useful-comparison discovery	P05	Accepted observations/depths/ages, explicit ranking and correct empty cases
[x]	X06	Matching/QC sensitivity and comparison of assumptions	P05, P08	Settings change results and sample counts consistently; original can replay
[x]	X07	Tested OPeNDAP path and documented adapter contract	P06	Actual subset request plus extension example
[x]	X08	Structured 3D feature search and connected-region inspection	P07	Numerical region membership, volume/extents and threshold behavior verified
[x]	X09	Draw-a-section interaction linked to a selected feature	P07	Drawn transect produces correct section and eligible observation links
[x]	X10	Saved/reopened/shareable investigation	P08	Numerical replay in a fresh session and version-mismatch handling
[x]	X11	Readable report, JSON settings and numerical CSV evidence export	P08	Exports agree with analysis and retain provenance/licensing constraints
[x]	X12	Bay of Bengal and Arabian Sea case catalogue	P02, P09	Two real cases use shared pipelines; dates/bounds/coverage declared

Verified	ID	Capability	Implementation phase(s)	Required evidence
[x]	X13	Same-surface/different-depth guided example	P09	Actual source-verified pair; no fabricated contrast or causal claim
[]	X14	Simple guided interface and responsive/accessible interaction	P03, P09	Unfamiliar-user task evidence, usable keyboard/error states and mobile fallback
[x]	X15	Virtual Expedition station suggestions	P10	Explicit objectives/reasons; budget/spacing constraints verified
[x]	X16	Virtual survey/transect sampling	P10	Synthetic samples match supplied model and remain labelled as simulated
[x]	X17	Fair sampling/reconstruction experiment	P10	Hidden evaluation, equal-budget baselines, multiple cases and reproducible results
[x]	X18	Virtual-particle/"pebbles" simulation: bounded historical Drift Lab	P11	Analytical/convergence/boundary checks, true current forcing and reproducible paths
[x]	X19	Drift sensitivity, arrivals and travel-time reporting	P11	Correct denominators/assumptions; no inferred real-world probabilities
[x]	X20	Surface marine-heatwave detection	P12	Six complete fixed-cell SST series; declared 1982-2011 climatology and event rules; all six agree with pinned reference code; see PHASE_12_HEAT_DEPTH.md
[x]	X21	Linked surface event, profiles and mixed-layer/gradient views	P12	Actual event/model-date overlap, original source/QC links and native-depth gradients; two declared derived MLD criteria; unsupported Arabian event dates remain unavailable
[x]	X22	Reviewed density, stratification and heat-content diagnostics	P12	Independent GSW/source/analytical checks; signed N-squared and complete column potential-enthalpy integral relative to CT=0; no subsurface anomaly claim

Verifi ed	ID	Capability	Implementation phase(s)	Required evidence
[x]	X23	El Nino, La Nina, neutral ENSO and IOD historical comparisons	P13	Pinned indices/baselines, same-season alignment and verified arithmetic
[x]	X24	Selected tropical Pacific coverage linked to Indian Ocean context	P13	Real compatible cases; dateline handling; no unsupported causal attribution
[x]	X25	Feature evolution with splits/merges and threshold sensitivity	P14	Native overlap fixtures, source oracle, explicit branches/gaps and exact replay; p14-native-overlap-v2
[x]	X26	Observation blackout rehearsal	P14	Exact exclusions update evidence/comparisons, with original/modified saves; unchanged model and QC; p14-blackout-v1
[x]	X27	Wider/global bounded access	P15	New supported regions work; coverage and unsupported areas visible
[x]	X28	Progressive loading, resource limits and cached/offline cases	P01, P11, P15	Measured resource behavior, cancellation and labelled cached fallback
[x]	X30	Teacher/public learning mode	P09, P16	Guided scientific explanations and comprehension checks
[]	X32	Scientific, usability and performance evidence package	P17	Actual commands/results, representative independent checks, stated hardware and fair comparisons
[x]	X33	Public release, portable package and national-round demo assets	P01, P17	Verified URL/setup, real-behavior video, attribution, limitations and requirement matrix

Cross-cutting acceptance rules

The 26 September frontend rebuild adds technical evidence toward X14 and the existing outreach/navigation requirements. Its paged introduction, grouped tool access, keyboard focus, responsive layouts and preserved scientific workflows are described in `FRONTEND_REBUILD.md`. X14 remains unchecked until actual unfamiliar-user task evidence exists; implementation and browser tests are not a substitute for that pilot.

- Display source/time/units and distinguish observed, modelled, derived and simulated values across all features.
- Preserve no-data, rejected-data, unsupported-input, running, failed and cancelled states.
- Inspect scientific definitions, coordinate conversions, masks and QC before matching or calculating.
- Share the same selected region/time/variable and investigation manifest across tools where meaningful; explicitly show when a lab uses a different source or baseline.
- Scientific computations use declared analytical data, not pixels or unlabelled low-resolution display values.
- Every input-dependent cache includes relevant data and method versions/settings.
- Never present a planned capability as working or a synthetic demonstration as a real record.

External dependencies that remain separately visible

- Public hosting account/access and any spending authorization.
- Data account access, licences and real source availability.
- Independent scientific review and suitable real drifter validation tracks.
- Actual INCOIS infrastructure installation and institutional acceptance.
- Optional explanation-model credentials and any paid API use.

These dependencies are not resolved by adding a UI button. Record their actual state in `PROJECT_STATE.md`.

P14 implementation evidence (24 September 2026)

X25-X26 are complete for their declared supported cases in release 0.14.1. Evolution links full native memberships through an inspectable volume-overlap rule, with splits, merges, ambiguity, selected-depth limits, skipped-frame gaps and nearby-threshold comparisons. Blackout excludes the union of selected profiles/groups and identifies exact original source sample IDs, unchanged eligibility gates and before/after evidence. The model is not rerun. Nine new investigations reproduce exactly in both directions between Windows and Vercel, including numerical exports; all 36 retained older records replay. Final evidence includes 807 backend tests, 34 frontend numerical/parser tests, 87 public browser checks, 67 public HTTP checks and 43 public visual captures. See [Phase 14 acceptance](#). Evolution is restricted to the two checked January 2024 Indian cases; monthly Pacific potential temperature cannot produce instantaneous in-situ residuals. P09.5 and P15-P17 remain open.

P02 implementation evidence (21 September 2026)

X01 is complete for the pinned first case: source/subset fingerprints, exact download recipe, licence/provenance records, independent source-value checks, separate analytical/display assets, bounded cached APIs and a deterministic pack rebuild. See `PHASE_02_DATA.md`, `SCIENTIFIC_CONTRACTS.md` and `evidence/phase-02-*`.

R02, R06 and R11 have partial foundations from P02 but remain unchecked: their renderer/extension example, delimited-text/additional input workflows and actual OGC interoperability gates belong to later phases. Viewing real Argo records in the P02 inspector does not complete P04 map overlays or P05 model/observation comparison.

P03 implementation evidence (22 September 2026)

R01, R03 and R07 are complete for the pinned historical case: real scalar volume, native analytical picking, physical depths/masks, slices, isosurfaces, vectors, actual timestamp playback and validated display controls. X02 is complete for fixed-latitude east-west sections with independently checked source picking. P07.4 still adds arbitrary line-drawn sections and linked observations.

See `PHASE_03_RENDERING.md` and `PROJECT_STATE.md` for the 12 grid checks, 21 backend checks, independent NetCDF fixtures, 50 passing local/public browser checks per suite and measured graphics limitations. R02 awaits the P06 variable-extension example; R12 awaits guided outreach and user feedback. No later lab or extra region was marked complete.

The 0.3.1 P03 clarity revision adds independently checked cut-face picking, explicit depth windows and missing-cell hatching. It passes 16 grid checks, 21 backend checks and 53 public browser checks, with one intentional skip. This improves the existing completed features; R12 and X14 still need real unfamiliar-user evidence. See the current `PROJECT_STATE.md` handoff.

The 0.3.2 continuity/motion revision further refines the existing P03 features. See `evidence/motion-plan-check.json` for alignment to P03.1-P03.5 and `PROJECT_STATE.md` for source, numerical, browser and runtime results, including initial public timeouts and their rechecks. No future-phase or unfamiliar-user acceptance gate was silently marked complete.

P04 implementation evidence (22 September 2026)

R04, R06 and X03 are complete for the documented inputs and checked examples. Thirteen genuine profiles cover core Argo, glider, ship CTD and BGC; each retains source coordinates/time, per-variable QC, units and processing modes. Both additional native NetCDF files and documented delimited text import through the public service. Malformed files, unsupported layouts, incompatible units and file limits have verified error paths. Missing variables, excluded readings and geographic compatibility are explained in the workspace and import guide.

R05 remains unchecked because P05 still owes eligible model-observation matching. P04's map/3D selection, timestamped depth chart and depth-navigation link are implemented, but they do not calculate residuals or claim simultaneous measurements. No future lab, region expansion or standards gate was marked complete. See `PHASE_04_INSTRUMENTS.md` and the P04 project-state handoff.

P05 evidence update

R05, X04 and X05 are complete for the checked historical model and observation library. Comparisons use explicit native-grid space/time/depth gates, compatible temperature/salinity definitions, preserved source QC and a source-review hold. Charts, metrics, actual-depth coverage and discovery share accepted counts. Local/public verification and independent source references are recorded in `PHASE_05_EVIDENCE.md` and the P05 project-state handoff.

At the P05 handoff, X06 remained unchecked as a full feature. P05 implements controlled matching/QC sensitivity and an in-memory reference with restoration; durable reproducible replay belongs to P08. No later lab, standards service or new model region is marked complete.

P06 verifies R02, R08, R11 and X07 for the bounded case. Native CF exchange, actual local THREDDS WMS/WCS/DAP2 requests, a second adapter and horizontal kinetic energy are implemented. Public REST/downloads are on Vercel; persistent public THREDDS hosting is not configured. R09 and R10 remain unchecked for later packaging/integration verification, and R12 retains outreach/user-feedback gates. See PHASE_06_STANDARDS.md for Checkpoint B.

P08 investigation evidence (22 September 2026)

X06, X10 and X11 are complete for the supported source-backed investigation modes. Applied matching policies and a reference comparison survive file/link replay; full results are recalculated and checked before reopening. Explicit browser saves, portable records, stateless replay links, readable HTML, full-precision numerical CSV and evidence ZIPs retain source and method identities. Uploaded observations remain temporary. No cloud database, permanent-link availability or identity signature is claimed.

Final release 0.8.1 passed 307 backend tests, 27 public P08 browser checks, 19 public replay/export checks and 49 existing HTTP/source checks. The full regression, responsive screenshot evidence, failures and fixes are retained in PHASE_08_INVESTIGATIONS.md and PROJECT_STATE.md. Later regions, labs, public persistent standards hosting and unfamiliar-user gates are unchanged.

P09 technical implementation

X12 and X13 now have real two-case and independent source evidence. Arabian Sea uses the shared tools, and the disclosed surface/depth contrast is independently verified. The basic guided path for R12 and X14 is implemented, but those rows remain unchecked because real unfamiliar-user findings are pending; P16 expanded learning is now implemented, as recorded below. P09.5 is open. See PHASE_09_GUIDED_CASES.md for tests, limitations, source selection and Checkpoint C.

P10 implementation evidence (22 September 2026)

X15-X17 are complete for the checked Bay and Arabian cases, temperature/salinity and native depths from 0 to 1000 m. Prior-only plans, native simulated surveys, fixed reconstruction comparisons and replay/export are verified. Final acceptance includes 77 backend checks, 51 public browser checks, 116 public HTTP checks, 18 public visual captures and exact replay of eight records in each direction between Windows and Vercel. See [Phase 10 methods and acceptance](#). Initial replay failures and their correction are retained. These are historical model reconstruction experiments, not operational observing-system validation. P09.5 and X14 remain open for real unfamiliar-user evidence.

P11 implementation evidence (23 September 2026)

X18-X19 are complete for passive tracers in the two checked historical cases at native depths from 0 to 1000 m. Source current interpretation, adapted RK4 integration, masks, boundaries, seeded releases, target-entry accounting, comparison, request-scoped jobs, progress, cancellation and saved replay/export are verified. Final release 0.11.1 passes 139 focused backend checks, including 45 independent scientific checks, 54 public browser checks and 100 public HTTP checks. Eight records replay exactly in each direction between Windows and Vercel. Twenty public captures and representative visual inspection cover paths, results, errors and responsive layouts. See [Phase 11 methods and acceptance](#).

The initial platform-dependent mean-distance summary failure and its versioned correction remain documented. No overlapping NOAA GDP tracks were retrieved in the queried bounds/dates, so independent drifter validation is unavailable. The feature models passive horizontal tracers, not sinking pebbles, oil or search-and-rescue probabilities. X28 was incomplete at P11; the broader P15 scope is now accepted as recorded below, and X14/P09.5 still require actual user evidence. No later lab or official submission requirement was marked complete.

P13 implementation evidence (23 September 2026)

X23-X24 are complete for selected SON 2013, 2015 and 2022 cases. Real monthly GODAS potential-temperature fields, a declared 1991-2020 baseline, native depth profiles, matched calendar windows, separately versioned RONI/ONI/DMI context and original Argo QC are verified. Pacific exploration and Indian Ocean historical context share the checked source product without causal attribution. Exact saves and numerical exports work in both directions between Windows and Vercel. Release 0.13.0 passes 683 backend tests, 111 public browser checks, 133 public HTTP checks and a 24-view final public visual audit. See [Phase 13 methods and acceptance](#). The original missing-variable, monthly-versus-instantaneous and geographic limitations remain explicit. P09.5, P14-P17 and their unfulfilled feature rows remain open.

P15 acceptance, 26 September 2026

P16 learning acceptance

X30 and the scope of X31 are implemented in release 0.16.0. Basic and explicitly chosen In-depth tutorials execute the same 27 real demonstrations with concepts, methods and optional questions. Source-linked explanations use current parsed facts, and bounded requests expose validated parameters before applying them. Unsupported questions and unavailable results do not become fabricated measurements. See [PHASE_16_LEARNING.md](#) and [PROJECT_STATE.md](#) for actual acceptance reports and retained corrections.

R12 and X14 remain unchecked because P09.5 requires actual unfamiliar-user feedback. This release does not close that participant gate or start P17.

P16 usability follow-up, release 0.16.1: X30 now uses callouts attached to visible controls or charts, automatic destination scrolling, a pointer/highlight, responsive layout and in-form save guidance. Hidden retained workspaces are excluded. This strengthens the tutorial interaction without changing X31's scientific contract or closing the R12/X14 participant gate. Evidence is in [PROJECT_STATE.md](#).

P16 discovery follow-up, release 0.16.2: X30 adds eight teaser cards and direct topic tours using the ten existing extra-lab demonstrations. Entry points appear in the welcome, active guide, core completion and feature completion. Topic tours retain Basic/In-depth, real action/result checks and explicit next-feature choices. A delayed catalogue cannot lose the first destination, and Skip cancels the pending tour. Local feature acceptance passes 39/39 and final visual acceptance covers 29 captures. Public release results are recorded separately in [PROJECT_STATE.md](#). Scientific methods, data coverage, X31

provider status and the open R12/X14 participant gate are unchanged.

P16A integration acceptance, 27 September 2026

Release 0.16.5 extends R04/R05/R06 and X03 with temporary imported markers, co-visible model/profile inspection, pre-use metadata/QC review and explicitly supported CSV column/unit mapping. The real BGC chlorophyll example retains original source identity, adjustment and flags. Its March date does not overlap the January Bay model, and no chlorophyll model field is claimed. All original packed source bytes and saved-library replay identities are preserved. PHASE_16A_IMPORTS.md and PROJECT_STATE.md contain source, numerical, browser, visual and public-deployment evidence.

Glider and CTD source examples are importable/inspectable but have no overlapping loaded main model case; the exact data dependency is documented. Imported numerical comparison/save/replay remains P16C. R12/X14 remain unchecked because automated layouts and tutorial checks do not close the actual-participant gate. P16B is accepted below; P16C-P16E and P17 remain unstated.

P16B acceptance, 27 September 2026

Release 0.16.7 proves a fresh compatible historical model acquisition through the existing scientific viewer and native API. Source checks cover all 25 decoded/derived arrays. The March chlorophyll import overlaps this case in area and time without implying model chlorophyll or a numerical imported comparison. Every pre-existing scientific pack file is byte-identical. Catalogue tool availability follows the retained scientific case gates.

R09 is checked for the reproducible browser/server package: a committed source archive was installed with fresh runtime-only Python and frontend dependencies, then passed 35 API/source/replay checks and 24 final browser checks outside Vercel. This is an isolated installation on the same Windows host. Docker, actual institutional installation and INCOIS environment approval are not established.

R11 retains its bounded meaning: January/March local WMS/WCS/DAP2 source-value checks pass, along with scoped CF and adversarial metadata checks. Full independent IOOS CF checking could not install due to missing UDUNITS; public THREDDS remains unavailable. R12/X14 still require actual unfamiliar-user feedback. P17's final scientific/security/performance audit remains open. Evidence: PHASE_16B_FRESH_DATA.md and evidence/p16b-acceptance.json.

P16C acceptance, 27 September 2026

Release 0.16.8 extends R05/R06, X03, X10 and X11 to supported imported observations. Original-file reparsing supplies eligible temperature/salinity comparisons and explicit file-specific coverage. Local saves include original bytes, mappings, profiles, settings and results; JSON/ZIP exports and exact replay preserve parser/model/method/source identities. Chlorophyll can be saved/reopened as an observation, with no invented model field. Original bundled libraries and archived replay identities remain unchanged.

All P16C.1-P16C.4 gates pass for the bounded paths in PHASE_16C_IMPORTED_REPLAY.md and evidence/p16c-acceptance.json. Backend/source checks, fresh-process/served replay, local Chromium/WebKit and public desktop/mobile flows, and visual review provide the evidence. R12/X14 actual-user gates, full independent CF verification, matched glider/CTD model cases, public THREDDS/actual INCOIS deployment and the P17 security audit remain open. No unrelated feature checkbox changes.

P16D connected investigation extension

P16D extends R05, X03-X06 and X10-X11 with source-sample identity links, co-visible model/comparison panels, eligible structure support from one declared observation collection, matching sensitivity and a five-step Investigation Builder. It owns follow-ups 19-22 and 29. Existing thresholds, observation gates, model source bytes and prior replay identities remain. See PHASE_16D_CONNECTED_WORKFLOW.md and the final PROJECT_STATE.md handoff for accepted evidence. Support snapshots do not add a new replay format. R12 and X14 remain unchecked until actual unfamiliar-user/domain-user task evidence exists; the prepared P16D_USER_TASK.md is not completed testing.

P16E extension evidence, 27 September 2026

Release 0.16.10 extends R05/R06, R09, X03, X10/X11 and X28 for bounded batch review, retained publication and portable offline evidence. Up to eight files are reviewed sequentially, still limited to 2 MB each; the imported workspace is limited to 20,000 samples plus profile/source/decoded-byte budgets. Partial failure, cancellation and a late model response preserve earlier usable imports. Maintainer-only publication validates prepared scientific bytes, pins readers and retains the last usable release after a failed update. Vercel serves a compatible bundled deployment; it does not supply a persistent refresh scheduler.

Saved observation, comparison and native-column viewers retain exact values, QC, source dates and original investigation JSON without network dependencies. They display saved

results rather than performing offline lab recomputation. PHASE_16E_CONTINUITY.md and evidence/p16e-acceptance.json record the accepted scope, measurements, affected-path safeguards and failed/repeated checks. The public transfer issue is unresolved at transport level; explicit Retry and retained evidence are tested. P17's final security/scalability/release audit and R12/X14 participant evidence remain open.

P17 release checkpoint, 27 September 2026: X33 is accepted for the verified 0.17.0 URL, source-compatible portable setup, actual browser backup recording, six-slide draft, attribution, methods and limitations. P17_DELIVERY.md and evidence/p17-acceptance.json link actual artifacts. This does not mean an idea was submitted or INCOIS installed the system. X32, R12 and X14 retain their real-human evidence gates. P17_CAPABILITY_AUDIT.md covers all requirement/feature/USP rows; no synthetic completion percentage or official scoring weight is assigned.

16 Original USP features: where each one is built

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/USP_INDEX.md

Updated 27 September 2026. The learning and all P16 completion checkpoints are implemented. P17 has audited the release and deployed 0.17.0; use PROJECT_STATE.md for the evidence and open acceptance gates.

The user asked on 2026-09-21 to make the original PDF's USP features explicit. They were already assigned to phases, but "Climate Event Lab" and "Drift Lab" obscured the familiar names. This index preserves those names and maps them to implementation work.

Here "USP" means a proposed differentiating product capability, not a verified world-first invention. This index preserves the original scope. Several capabilities are now implemented; use [the feature register](#) and project state for current acceptance status. Heat & Depth remains available with method p12-heat-v1, Drift Lab with p11-drift-v2 and Climate with p13-climate-v1. P14 adds native feature correspondence with p14-native-overlap-v2 and observation exclusion with p14-blackout-v1. Its final numerical, public browser, exact replay and visual checks pass. The first public aggregate mismatch is retained alongside the correction and successful verification. P15 wider source access and bounded offline/scaling evidence are recorded; P16 learning is delivered and P17 audits every row in P17_CAPABILITY_AUDIT.md. The P09.5 real unfamiliar-user pilot remains open.

Original PDF capabilities

Feature the user remembers	What it means in the app	Phase / feature IDs	Original guide pages
El Nino and La Nina	Implemented for SON 2013 neutral, 2015 El Nino and 2022 La Nina: compare matching monthly/seasonal potential-temperature fields, depth profiles and 1991-2020 anomalies using pinned RONI labels	P13 complete , X23-X24	16
Indian Ocean Dipole (IOD)	Matching-date Indian Ocean GODAS sections, separately labelled CPC DMI values and documented BOM episode assessments; no inferred causal link	P13 complete , X23-X24	16
Moving pebbles / particles	Working in both checked Indian Ocean cases: release passive tracers, calculate fixed-depth trajectories, compare starting places/dates/depths, inspect simulated arrivals and save/replay both runs	P11 complete , X18-X19	11-13
Marine heatwaves	Working at six fixed source cells: inspect historical surface events against the 1982-2011 seasonal baseline, duration/intensity and complete event boundaries	P12 complete , X20	14

Feature the user remembers	What it means in the app	Phase / feature IDs	Original guide pages
Look below a warm surface	Inspect available native profiles, two derived mixed-layer estimates, gradients, density, stratification and a declared column potential-enthalpy integral; dates without depth coverage stay unavailable	P12 complete , X21-X22	15, 19
Evidence Inspector	Compare compatible model and observed profiles with quality flags, matching distances/times, residuals and explicit exclusions	P05 , X04-X05	9-10
Change assumptions / sensitivity	Compare accepted time windows/QC choices and particle releases; the P14 build reruns native region correspondence at nearby thresholds and shows changed branches	P05, P08, P11, P14 complete , X06, X19, X25	17
Save the whole investigation	Reopen sources/settings/methods and reproduce calculations; export readable and machine-readable evidence	P08 , X10-X11	18
Virtual survey planner	Draw a transect and sample the supplied model at declared positions/depths/times, labelled as simulated measurements	P10 , X16	19
Feature the user remembers	What it means in the app	Phase / feature IDs	Original guide pages
Feature tracking	Compare native-cell overlap across	P14 complete , X25	19

Feature the user remembers	What it means in the app	Phase / feature IDs	Original guide pages
	consecutive Indian Ocean snapshots; inspect branches, splits, merges, unresolved support, explicit gaps and threshold sensitivity, with each selected region's footprint and depth range		

Additions from our later discussion

Capability	Phase / feature IDs
Search connected underwater structures using temperature/depth conditions	P07, X08
Draw a section through a selected structure	P07, X09
Evidence coverage view and discovery of useful real comparisons	P05, X05
Same surface, different ocean: verified historical demonstration	P09, X13
Virtual Expedition: suggest additional sampling locations with reasons	P10, X15
Fair sampling/reconstruction sandbox	P10, X17
Capability	Phase / feature IDs
Observation blackout: exclude profiles or instrument/platform/collection groups, compare original	P14 complete, X26

Capability	Phase / feature IDs
and remaining evidence, and retain the exact eligible source sample IDs that disappear	
Data-readiness explanations before an unsupported operation	P04, X03

What the names do and do not promise

- Drift Lab's particles represent passive tracers carried by historical currents at a fixed depth. Actual sinking stones, oil chemistry or object-specific drift require additional physics and are not silently implied. Numerical checks and exact replay do not establish an independently validated real-drifter forecast.
- Climate compares three selected historical Pacific seasons and their matching Indian Ocean context. It has no event-strength control that generates a new climate scenario.
- The implemented build contains six workspace cases: a fresh bounded March 2024 Bay HYCOM case, two January 2024 Indian Ocean cases with temperature, salinity and currents, plus three Pacific SON cases with monthly potential temperature only, from 5 to 459 m. Pacific fields use a 1991-2020 matching-month baseline with complete 30-year support. These cases do not imply equal data support across every region or lab.
- Pacific Ocean Explorer, Find structures and original instrument inspection are supported. Direct measurement comparison, Expedition, Drift, Heat and P14 Evolution remain specific to the checked Indian Ocean cases. Blackout permits Pacific source exclusions, but monthly potential temperature stays incompatible with instantaneous in-situ Argo residuals.
- Evolution describes correspondence between threshold-defined regions, not the trajectory or identity of a water parcel. It uses the supplied 12-hour Indian Ocean snapshots. Skipped frames have no inferred links; missing source support can leave endpoints unresolved.
- Evolution v2 changes the summation of P14 aggregates for reproducibility. Source data, region memberships and P07 structure-search calculations are unchanged. Older v1 archives remain readable with explicit replay method mismatch. All nine new records reproduce exactly in both directions between Windows and Vercel; all 36 retained earlier records replay.

- Blackout changes available evidence, not the model or its original assimilation. Reduced coverage is not a calibrated loss of confidence, and a smaller residual after removing observations is not model improvement.
- A surface heatwave detection and a depth inspection remain separate analytical claims.
- The complete PDF/USP vision includes P11-P13. A P09 prototype checkpoint must not be presented as the finished full product.
- The submission deck should feature the connected lab vision prominently, with each capability labelled honestly as working or planned. New feature-search ideas do not replace the original labs.

Release audit

Before describing the full product as complete, check every row against the feature register and actual evidence. Keep any unfinished row visible. Build details and acceptance checks remain in `ROADMAP.md`; actual status remains in `PROJECT_STATE.md`.

The completed particle implementation and its scientific limits are documented in [Phase 11](#), and the surface/depth methods in [Phase 12](#). The implemented climate capability, source checks and completed public acceptance are recorded in [Phase 13](#). [Phase 14 methods and acceptance](#) records the completed evolution and blackout workflows, final release checks and retained initial failures. [Phase 15](#) adds wider historical source access, external oxygen/uncertainty and offline regional packs. [Phase 16](#) adds Basic/In-depth real-action tutorials and source-grounded explanations. P16B adds fresh March model input and portable setup proof. P16C release 0.16.8 adds imported comparisons, explicit local saves and original-file portable replay. P16D release 0.16.9 connects source sample selection, model/comparison panels, structure support and guided investigation. P09.5 still requires real unfamiliar-user feedback.

17 Full phased build plan

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: `docs/ROADMAP.md`

This roadmap preserves the user's entire software vision. It specifies proposed work, not completed functionality. The authoritative progress record is `PROJECT_STATE.md`; individual capability evidence is tracked in `FEATURE_REGISTER.md`.

For the features named in the original PDF, see `USP_INDEX.md`: **particle/"pebbles" simulation is P11; marine heatwaves and depth analysis are P12; El Nino, La Nina and IOD are P13**. These remain part of the full build, not deleted ideas. Submission materials and their verified official status are tracked in `SUBMISSION_CHECKLIST.md`.

How phases work

- Each phase has small step IDs, defined inputs and acceptance checks. Stop/resume at those step IDs rather than reconstructing work from conversation memory.
- Complete the requested phase, update the handoff and report the next phase. Continue across phases only when the user authorizes that.
- Dependencies mean the stated implementation inputs must work. An unrelated external blocker, such as unavailable hosting credentials, need not prevent local scientific development in a subsequently requested phase.
- Tests, truthful scientific labels, accessibility basics and incremental deployment belong throughout the build. P17 is a final audit, not the first verification pass.
- Keep report, slide and demo wording ready in `PHASE_WRITEUPS.md` as each phase progresses. Update it with verified results, limits and significant fixes before closing the phase. Use direct language without em dashes or promotional filler.
- Follow `STACK_AND_EXPERIENCE.md`: evaluators use unknown devices during online round 1; ease of use, impressive data-faithful visuals and adaptive quality are explicit product priorities.
- When a feature cannot pass its gate, leave it incomplete with a specific reason. Do not replace it with a decorative imitation or remove it from the plan.

Overview

These are working recommendations, not required settings. For reasons and optional switches at specific steps, see `PHASE_REASONING_GUIDE.md`.

P01 : Foundation and first deployment path

Outcome: a small runnable app with a coherent shell, a backend health check, repeatable setup and a concrete hosting path.

Steps:

- [x] **P01.1** Inspect installed runtimes and repository state. Establish compatible frontend/backend environments and the initial `web/`, `api/`, `science/`, `adapters/`, `tests/`,

deploy/ and docs/ structure where needed.

- [x] **P01.2** Build the workspace shell: region/date/variable area, main viewport placeholder clearly labelled as such, evidence drawer, responsive navigation and basic loading/error states. Establish shared types and API error shapes.
- [x] **P01.3** Add a health endpoint, production build, documented local start commands and a portable deployment package. Keep configuration and secrets separate.
- [x] **P01.4** Establish available hosting access and deploy a thin build when possible. Record a real verified URL, or the exact external blocker; do not invent successful deployment.

Done when: the app starts from documented commands, the production build serves successfully, the frontend reaches the API, and the deployment step has verifiable evidence. An externally blocked deployment stays open even if local steps are complete. Do not spend money without authorization.

User verification: open the actual app and health endpoint; see clear application states.

P01 evidence: public Vercel deployment and API verified without authentication; evidence/phase-01-public-http.json, evidence/phase-01-public-browser.json, browser report and screenshots. Build and five API checks passed. Twelve browser checks passed; four Firefox cases could not launch due to a Windows browser-binary configuration error, also after reinstalling the test browser. Docker recipe is prepared but unexecuted because Docker is unavailable. No scientific capability is marked complete by this foundation milestone.

P02 : Real data and scientific contracts

Outcome: one real model subset and eligible observations are acquired and interpreted correctly, with a shared contract used by all later features.

Steps:

- [x] **P02.1** Inspect candidate data availability. Select a bounded Bay of Bengal historical case with usable model/profile overlap. Record licence/access route, dataset version, bounds, dates, checksums and download instructions.
- [x] **P02.2** Define model, observation, provenance, operation-result and variable contracts. Include real coordinates, depth/pressure semantics, time calendar, units, masks, fill values, scale/offset and variable definitions. Define unsupported-input errors.
- [x] **P02.3** Implement a supported NetCDF model adapter using scientific libraries. Normalize only declared semantics; preserve originals. Create separate analytical and display representations with documented transformations.

- [x] **P02.4** Fetch and parse a real Argo sample as an early compatibility check, preserving QC/adjusted metadata. Establish a small cache and bounded subset endpoints for supported variables and timesteps.
- [x] **P02.5** Independently inspect representative source values and tricky fixtures, including reversed depth, longitude conventions, irregular levels, missing cells and incompatible variables. Record expected values and processing decisions.

Done when: real regional temperature, salinity and horizontal velocity data are usable; at least one real profile has usable overlap; coordinate/value checks pass; raw and derived data are distinguishable. Chlorophyll requires its own suitable source and must not be invented from physics-only files.

User verification: a dataset summary identifies exactly what exists, where and when, with sample numerical values.

P02 evidence: genuine HYCOM and INCOIS-DAC Argo case prepared and deployed on 21 September 2026. Historical coverage is 7-10 January 2024, explicitly labelled. See PHASE_02_DATA.md, SCIENTIFIC_CONTRACTS.md, the versioned casepacks/bay-bengal-2024-01/manifest.json, 21 API/science checks, 60 independent source-value checks, deterministic rebuild verification and 18 passing local/public supported-browser tests. The initial WebKit focus issue is fixed; Firefox remains an environment launch blocker. P03 rendering has not started.

P03 : 3D Ocean Explorer

Outcome: a genuine scientific volume can be inspected across depth and time in the browser.

Steps:

- [x] **P03.1** Render the real scalar field using a bounded display grid. Preserve geographic placement and physical depth; label any resampling and vertical exaggeration.
- [x] **P03.2** Add temperature/salinity selection, depth slicing, vertical cross-sections, a simple isosurface and current-vector display. Keep picking linked to analytical values.
- [x] **P03.3** Add playback over actual timestamps, legends/units, palettes, explicit ranges, valid linear/log mapping, opacity, camera reset and vertical-stretch controls.
- [x] **P03.4** Implement loading, no-data, capability/context-loss handling and a useful reduced-detail/2D fallback. Start in an adaptive Auto mode for unknown evaluator hardware; do not require WebGPU or a discrete GPU. Add keyboard and readable control basics.

- [x] **P03.5** Compare selected displayed positions and values with the source read; benchmark bounded real data on declared reference setups. Assess transfer functions, depth cues and cinematic shading against responsiveness; confirm appearance/quality controls never alter analysis. Follow `STACK_AND_EXPERIENCE.md` and record measurements rather than claiming universal performance.

Done when: volumes, slices, isosurfaces, vectors and time controls function on real data; missing cells remain missing; depth geometry is correct; required controls are usable. A globe or stacked decorative planes alone do not satisfy volumetric rendering.

Checkpoint A completed, 22 September 2026: v0.3.2 includes the verified clarity, loading-continuity and optional animated-cutaway revisions and is deployed on Vercel. See `PHASE_03_RENDERING.md` and `PROJECT_STATE.md` for source, numerical, browser and performance evidence and the remaining limits.

P04 : Instruments and supported ingestion

Outcome: real instrument records appear at their actual locations with usable depth profiles and quality metadata.

Steps:

- [x] **P04.1** Complete the Argo adapter: parameter-specific raw/adjusted selection, QC policy, timestamp/location handling and documented pressure-to-depth conversion.
- [x] **P04.2** Add genuine glider, ship CTD and BGC examples through a common contract. Preserve glider movement and per-sample position/time. Separate demonstration cases are acceptable where one region lacks all types; disclose coverage.
- [x] **P04.3** Implement one documented delimited-text/CSV adapter plus supported NetCDF observation ingestion. Provide automatic metadata inspection, missing-field explanations, parser/file limits and explicit unsupported-format errors.
- [x] **P04.4** Add selectable instrument markers/tracks, timestamped profiles, units, data mode, source and QC inspection; link the selected depth to the 3D view.
- [x] **P04.5** Verify each claimed instrument type using a genuine record; exercise malformed input, missing variables, bad flags, moving samples and incompatible units.

Done when: each claimed sensor type has a real verified example, NetCDF and the documented text format work, and imported metadata remains inspectable. Missing required sensor support stays a visible requirement gap.

User verification: select different instrument types, inspect their profiles and import an additional compatible file.

P04 completed, 22 September 2026: release 0.4.0 includes 13 genuine profiles, four instrument types, supported CSV/NetCDF imports and linked map/chart/depth selection. Source/QC, local/public browser and import evidence is recorded in PHASE_04_INSTRUMENTS.md and PROJECT_STATE.md. Model matching remains P05.

P05 : Evidence Inspector and coverage view

Outcome: users can understand whether a comparison is eligible and exactly what observations support it.

Steps:

- [x] **P05.1** Implement explicit spatial, temporal and vertical matching rules, domain masks and extrapolation policy. Convert compatible physical quantities correctly; reject incompatible definitions.
- [x] **P05.2** Show observed/model/residual profiles with a synchronized depth cursor. Include matched coordinates/time, separation, QC, method, sample counts, exclusions and source type.
- [x] **P05.3** Add a coverage view showing accepted profile locations, sampled depths and age under declared filters. Preserve the model as separate context. Do not fill unsampled areas with invented observations or confidence.
- [x] **P05.4** Add "Find a useful comparison" using eligible records. Offer explicit sorting such as nearest eligible match or largest residual among accepted pairs; show sample coverage and definitions.
- [x] **P05.5** Add controlled sensitivity to matching windows and accepted QC categories. Test known matches, deliberate mismatches, no-valid-pair states and count/plot consistency. Record assimilation caveats.

Done when: comparison values pass independent checks, incompatible/no-match cases are truthful, and users can explain both the result and the exclusions. Bias/RMSE summaries describe the selected sample only.

User verification: change a time window and see both the curves and their accepted sample counts change consistently.

P05 completed, 22 September 2026: release 0.5.0 adds checked temperature/salinity comparisons, actual observation coverage, explicit discovery and temporary matching sensitivity. Methods, numerical references, public verification and limitations are in PHASE_05_EVIDENCE.md. Durable replay remains P08.

P06 : Standards and extension contract

Outcome: supported data can be exchanged through real scientific protocols and new compatible adapters can be registered without rewriting the viewer.

Steps:

- [x] **P06.1** Configure a mature service such as THREDDS for the supported grid. Test actual WMS capabilities/map requests and WCS coverage requests; verify geolocation and numerical values with an independent client.
- [x] **P06.2** Demonstrate a supported OPeNDAP subset path alongside the app's documented REST endpoints. The PS uses REST/OPeNDAP wording; this project's explicit interoperability target includes a tested OPeNDAP path.
- [x] **P06.3** Validate relevant CF interpretation and publish supported conventions, axes/calendars and known exclusions. Opening a .nc file is not a complete CF check.
- [x] **P06.5** Package the service stack for a reproducible server deployment, document resource limits and institutional deployment needs, and update public/local builds as access permits.

Done when: actual protocol requests pass, the extension example works, and deployment instructions are reproducible. Do not claim universal standards certification or installation on INCOIS infrastructure.

Checkpoint B: audit R01-R12 against the supported core. Record every outstanding requirement rather than treating it as optional.

P06 completed, 22 September 2026: release 0.6.0 provides native NetCDF access and a derived-variable extension. Actual THREDDS WMS/WCS/DAP2 requests passed against the local package. Public persistent THREDDS hosting remains unconfigured. Checkpoint B and acceptance evidence are in PHASE_06_STANDARDS.md.

P07 : Search underwater structures

Outcome: users can query and inspect connected 3D regions rather than manually browse every slice.

Steps:

- [x] **P07.1** Implement visible structured queries for variable, threshold/range, depth and available time. Validate units, masks and supported combinations.
- [x] **P07.2** Detect connected qualifying regions with a documented adjacency rule. Compute physical extent and volume with actual cell geometry; keep display

resampling distinct from analysis. For this point-coordinate source without supplied bounds, volume is explicitly estimated from clipped midpoint bins; see D028 and PHASE_07_FEATURES.md.

- [x] **P07.3** Render selectable region meshes and an inspector showing query, extent, statistics and eligible observations intersecting/near the feature under declared rules.
- [x] **P07.4** Add "Draw a question": a line-selected vertical section with linked values and observations. Provide explicit ordering such as largest qualifying region or most observation coverage.
- [x] **P07.5** Test known synthetic geometries and real source selections; check missing cells, unequal depth spacing, threshold equality and empty results.

Done when: a changed query produces numerically correct regions, exported values agree with source selections, and the app never presents threshold regions as automatically identified eddies or transported water parcels.

User verification: enter a new supported threshold/depth query, select a result and draw a section through it.

P07 completed, 22 September 2026: release 0.7.1 is deployed on Vercel. The full backend suite passed 281 tests, including independent native-region references; all 99 independent public P07 API checks and all 35 public visual checks passed. The full public browser suite passed 144 initially, with eight loading failures passing unchanged rechecks and one deliberate skip. Estimated geometry, network limitations and the component-wide audit are recorded in PHASE_07_FEATURES.md and P07_COMPONENT_AUDIT.md. P08 had not started at that handoff.

P08 : Save, replay and export investigations

Outcome: another user can reconstruct the actual analysis, not only its camera position.

Steps:

- [x] **P08.1** Define a versioned investigation manifest containing source versions/subset fingerprints, region/time/depth, variables, QC/matching rules, queries, methods and software versions.
- [x] **P08.2** Save and reopen state plus calculated results through stable records/share links as supported by the deployment. Keep mutable upstream data and cache invalidation explicit.
- [x] **P08.3** Export a readable report, settings JSON and paired numerical CSV with appropriate source references. Bundle data only when permitted; otherwise include authorized retrieval instructions and checksums.

- [x] **P08.4** Establish a shared result/experiment manifest so subsequent labs add parameters, random seeds and outputs consistently.
- [x] **P08.5** Reproduce the same numerical results from a saved case in a fresh process/session. Exercise missing source data and version mismatch handling.

Done when: the investigation replays, numerical exports match the UI, and a source/settings change cannot silently reuse an incompatible result.

User verification: save, close and reopen an investigation; inspect the exported evidence and repeat the numbers.

P08 completed, 22 September 2026: release 0.8.1 is deployed on Vercel. Four investigation modes reproduce in fresh processes/sessions. Final checks passed 307 backend tests, 27 public P08 browser checks, 19 public replay/export checks and 49 existing public HTTP/source checks. The 0.8.0 full browser run and 30 public visual checks, its unchanged loading rechecks, the corrected ZIP attribution packaging and storage limits are recorded in PHASE_08_INVESTIGATIONS.md. P09 progress is recorded below.

P09 : Guided Indian Ocean cases and usability

Outcome: an unfamiliar user can complete a meaningful investigation, including a second regional example.

Steps:

- [x] **P09.1** Add an Arabian Sea case using the existing pipeline and document any required adapter correction. Avoid hidden hardcoded fixes.
- [x] **P09.2** Find and verify suitable real "same surface, different ocean" examples. Display similar surface conditions and different depth profiles with real dates/sources; do not invent a dramatic pair.
- [x] **P09.3** Build a guided journey: explore below the surface, compare a measurement, search a structure and save the result. Include a basic public/student path with plain-language explanations of variables, depth and model-versus-observation meaning; do not defer all outreach to P16. Offer available dates/variables, clear next actions, undo/reset and expandable expert controls.
- [x] **P09.4** Check responsive layout, keyboard access, contrast, reduced motion, clear units, progress/cancellation and useful failure states. Open directly into a useful case with task-based controls; keep read-only public demonstration access simple where hosting permits. Confirm the experience works without assumptions about the evaluator's device or screen size.

- [] **P09.5** Run a small unfamiliar-user task pilot. Record participant count, task time, mistakes and help needed; fix observed problems and rehearse the actual demonstration.

26 September usability follow-up: the user requested a simpler frontend before P16. The rebuild introduces a short paged visual introduction, one grouped Tools menu, a scene-first explorer and on-demand source/settings panels. Existing scientific workflows are retained and their browser tests now use the new navigation. See [FRONTEND_REBUILD.md](#) and [PROJECT_STATE.md](#) for actual verification and deployment. This follow-up does not close P09.5.

Done when: both regional cases work through shared tools, the guided example is source-verified, a basic public/student path is usable, and usability findings are recorded. A small pilot is not a universal impact claim.

Checkpoint C: first competition-focused public-prototype candidate. Audit official requirements and record which later phases are still planned. This is not the full PDF/USP release: particle simulation, Heat & Depth, and El Nino/La Nina/IOD remain unfinished until P11-P13 pass their checks. The ten-day target is assessed against actual progress here, not guaranteed in advance.

P09 technical handoff, 22 September 2026, release 0.9.1: P09.1-P09.4 are implemented with two real cases, a source-verified teaching pair, a basic public/student guide and technical usability checks. P09.5 remains open for a real unfamiliar-user pilot. See [PHASE_09_GUIDED_CASES.md](#) and [P09_USER_PILOT.md](#). P10 was not started at that handoff; its subsequent acceptance is recorded below.

P10 : Virtual Expedition and sampling sandbox

Outcome: users can explore where additional measurements could be useful and compare explicit sampling strategies within a bounded experiment.

Steps:

- [x] **P10.1** Define one supported sampling objective, finite candidate locations, measurement budget, minimum spacing and geographic/depth constraints. Keep coverage, gradients and observed mismatch as distinct explanations.
- [x] **P10.2** Suggest candidate stations with transparent rationale. Let budget/objective changes update the plan; explain unsatisfied constraints. This is a scientific planning aid, not an operational ship/gliders route.
- [x] **P10.3** Add a virtual transect/survey: draw a supported route and sample model values at declared positions/depths/times. Label every sample as simulated and retain

the source and interpolation method.

- [x] **P10.4** Implement a bounded sampling/reconstruction sandbox with frozen prior information, hidden evaluation data, a declared reconstruction method and equal-budget uniform/random baselines. Prevent evaluation-data leakage; use multiple held-out cases and report failures as well as successes.
- [x] **P10.5** Save/replay plans and experiments. Verify budget/spacing constraints and numerical samples; present benchmark results as reconstruction results only. Do not call this a validated OSSE or claim forecast gains from reanalysis experiments.

Done when: suggestions have inspectable reasons, virtual samples are correct, and sandbox evaluation is reproducible and fairly separated from planning information. If sandbox verification fails, keep X17 incomplete even if station suggestions are usable.

User verification: change a station budget and compare saved strategies under identical experimental conditions.

P10 acceptance, 22 September 2026: release 0.10.1, method p10-sampling-v2, deployed to the existing Vercel project. All five steps pass within the two checked cases and declared experiment: 77 final backend checks, 51 public browser checks, 116 public HTTP checks, 18 public visual captures and exact replay of eight records in each direction between Windows and Vercel. Selection is isolated from later fields and reserved evaluation points in both cases; their correlated snapshots are not independent event validation. See [methods and acceptance](#) and [independent scientific review](#). P09.5 real-user feedback remains open. P11 is not started.

P11 : Particle simulation ("pebbles") / Drift Lab

Outcome: users can run and compare bounded historical passive-tracer experiments using supplied currents.

This is the PDF's virtual-particle experiment: the user releases visible markers and follows their calculated movement through the supplied current field. The particles can look like small pebbles/dots; the calculation represents passive tracers, not sinking stones with unimplemented physics.

Steps:

- [x] **P11.1** Integrate a suitable tested particle framework. Verify current units, component orientation, grid staggering/masks, spherical coordinates and interpolation conventions.
- [x] **P11.2** Support release points/region, start time, fixed horizontal depth, available duration, particle count and optional target region. Declare numerical integration and

boundary policies.

- [x] **P11.3** Add jobs, progress, cancellation, resource limits, caching and trajectory rendering. Distinguish instantaneous current visualization from calculated time-dependent paths.
- [x] **P11.4** Compare releases/dates and report simulated arrivals/travel times with denominators and assumptions. Integrate reproducible seeds/settings and exports.
- [x] **P11.5** Test zero/constant currents, physical displacement, timestep convergence, coast/missing-data boundaries and exhaustion of forcing data. Seek independent drifter tracks; record unavailable validation rather than implying it occurred.

Done when: analytical/reference checks pass, unsupported conditions fail visibly, and a saved run reproduces. Initial scope excludes windage, oil chemistry, swimming, operational rescue probabilities and invented vertical motion.

User verification: change a release and compare trajectories; inspect what happens when the selected duration exceeds available forcing.

P11 handoff, 23 September 2026, release 0.11.1: P11.1-P11.5 and X18-X19 are complete within the documented historical cases. An adapted pinned OceanParcels RK4 kernel runs in bounded request-scoped jobs, with genuine progress, cancellation, fixed-depth trajectories, target accounting, comparisons and source-checked saves/exports. Independent verification uses the complete framework separately. The final release passes 139 focused backend checks, 54 public browser checks, 100 public HTTP checks and exact replay of eight records in each direction across Windows and Vercel. Twenty public visual captures and representative visual review are retained. No matching real drifter tracks were retrieved in the queried bounds/dates, so independent drifter validation remains unavailable. See [Phase 11 methods and acceptance](#), D037-D039 and the retained initial summation-mismatch evidence. P09.5 stays open. P12 requires a separate user instruction.

P12 : Heat & Depth Lab

Outcome: users can inspect a checked surface marine-heatwave event and related depth structure, with reviewed layer diagnostics.

Steps:

- [x] **P12.1** Acquire a suitable long daily surface-temperature series. Pin baseline period, seasonal threshold method, duration/event-joining rules, leap-day and missing-data policies.

- [x] **P12.2** Implement and compare surface event detection with a documented reference method. Show temperature, baseline, threshold, duration/intensity and supported event locations.
- [x] **P12.3** Link a selected surface event to available depth fields and observations. Retain separate labels: surface event detection does not prove a heatwave throughout the water column.
- [x] **P12.4** Add source-defined mixed-layer depth and actual-depth temperature gradients. Extend to reviewed density, stratification and heat-content calculations with TEOS-10/GSW as appropriate, explicit definitions and integration limits.
- [x] **P12.5** Verify reference-series events and independently calculated profiles/integrals. Export the complete methods and baselines. Omit unsupported diagnostic outputs while keeping their feature status open.

Completed 23 September 2026: release 0.12.0, method p12-heat-v1. Six fixed OISST cells have complete 1982-2025 series and a 1982-2011 baseline. All 590 backend tests pass, including 76 independent scientific tests; 54 public browser checks, 105 public HTTP checks and 20 public visual captures pass. Eight saved records replay and export exactly in each direction between Windows and Vercel. No provider MLD variable is supplied, so the two mixed-layer outputs are explicitly derived threshold estimates, as defined in D041. The integral is potential-enthalpy content relative to $CT=0$, not a subsurface anomaly. Arabian surface events have no overlap with the available model dates and correctly show unavailable depth evidence. See [Phase 12 acceptance](#). P09.5 actual human feedback remains open; P13 has not started.

Done when: event dates match the declared method, diagnostics agree with independent expected results, and physical definitions/coverage are visible. No cyclone-intensity or ecosystem-damage prediction is inferred from a profile.

User verification: inspect one event, open its depth view and see the separate evidence for surface and subsurface statements.

P13 : El Nino, La Nina, IOD / Climate Event Lab and tropical Pacific

Outcome: users can select documented **El Nino, La Nina and neutral ENSO periods**, inspect tropical Pacific depth structures, and compare the separately defined **Indian Ocean Dipole (IOD)** and Indian Ocean historical context using the same workspace.

Steps:

- [x] **P13.1** Acquire and validate a bounded tropical Pacific model/observation case through existing contracts. Check longitude/dateline behavior and source availability.

- [x] **P13.2** Build a small historical event catalogue explicitly covering El Nino, La Nina, neutral ENSO conditions and supported IOD cases, with pinned index definitions/releases. Keep ENSO and IOD distinct; do not silently mix ONI/RONI or later revisions.
- [x] **P13.3** Implement same-season comparisons, compatible seasonal anomalies and synchronized sections/profiles across cases. Show baseline periods and dataset versions.
- [x] **P13.4** Link Pacific exploration to corresponding Indian Ocean historical context without claiming causal attribution from two selected years. Save/export all event settings.
- [x] **P13.5** Verify date alignment, event labels, baseline arithmetic and missing coverage against independent source checks.

Completed 23 September 2026: release 0.13.0, method p13-climate-v1. Three checked Pacific GODAS cases cover SON 2013, 2015 and 2022 with a complete-source 1991-2020 anomaly baseline, separate official ENSO/IOD definitions, nine original Argo profiles and shared 3D/structure tools. All 683 backend tests, 111 public browser checks and 133 public HTTP checks pass. Eight investigations replay and export exactly in each direction, and all 28 older records remain reproducible. The final public visual audit captures 24 checked views, including corrected sidebar source/date metadata; nine focused public browser checks also pass after the final copy correction. See [Phase 13 acceptance](#). P09.5 remains open and P14 is unstarted.

Done when: selected Pacific and Indian Ocean cases work with shared tools and versioned event definitions. No strength slider pretends to produce a new coupled climate or local-weather forecast.

Checkpoint D: all principal labs have an implemented, tested supported path, with exact limitations recorded.

P14 : Feature evolution and observation blackout

Outcome: users can follow defined structures across available times and see how conclusions depend on an observation group.

Steps:

- [x] **P14.1** Implement feature correspondence across frames using an explicit, inspectable rule. Represent splits, merges, disappearance and ambiguous matches rather than forcing a single continuous identity.

- [x] **P14.2** Display feature evolution with linked thresholds, extents and available observations. Compare nearby threshold choices and preserve time-gap information.
- [x] **P14.3** Add observation blackout: exclude an instrument/group and recompute evidence coverage, eligible comparisons and affected statements. Clearly state that the underlying ocean model has not been rerun.
- [x] **P14.4** Verify known feature evolution fixtures, ambiguous events and exclusion effects. Save both original and modified investigations.

Completed 24 September 2026: release 0.14.1, methods `p14-native-overlap-v2` and `p14-blackout-v1`. Native overlap graphs retain branches, ambiguity, nearby thresholds and explicit gaps. Blackout preserves the model and source QC while identifying the exact eligible evidence removed. All 807 backend tests, 34 frontend numerical/parser tests, 87 final public browser scenarios and 67 public HTTP checks pass. Nine new records replay and export exactly between Windows and Vercel in both directions; 36 earlier records still replay. The final public visual run has 43 captures, with nine manually inspected. See [Phase 14 methods and acceptance](#). P09.5 remains open.

Done when: tracking and exclusions are reproducible, ambiguity is visible, and no region trajectory is mislabelled as the transport path of the same water.

User verification: remove an instrument and see exactly which evidence disappears; inspect a feature split or disappearance.

P15 : Wider geographic coverage and scale

Outcome: the architecture works beyond the curated initial cases with declared geographic/data limits and measured resource use.

Steps:

- [x] **P15.1** Add authorized bounded-region requests over wider/global source coverage. Show supported dates, depths, variables and instrument availability before loading.
- [x] **P15.2** Implement progressive display loading, bounded frame/variable caches and appropriate detail levels. Preserve analytical precision and label display resolution.
- [x] **P15.4** Test dateline crossings, longitude conventions, land/ice/missing data, supported grid types and resource ceilings. Reject unsupported arbitrary grids clearly.
- [x] **P15.5** Benchmark cold/cached loads, transfer size, memory and representative concurrent evaluator sessions; verify cancellation, quotas and reproducible cached/offline case packs. Test slow connections and adaptive display quality separately from scientific calculation settings.

Done when: new supported regions work through the shared pipeline and measured scaling behavior is documented. Global navigation does not imply validated analysis in every ocean or live data everywhere.

User verification: open a new supported region and a constrained case; inspect resolution/source differences and the cached fallback.

Completion checkpoints before P17

Planning revised on 26 September 2026 after the user objected that the two-checkpoint proposal appeared to reduce the 31-item scope. Use **P16A-P16E**, followed by **P17**, to account for the entire list. These are groups of implementation and verification tasks, not individual features. Earlier phase numbers, historical acceptance records and open P09.5/P16.3 gates are unchanged. The full mapping is in `CORE_COMPLETENESS_REVIEW.md`. Product improvements retain their non-mandatory PS status while receiving a concrete proposed delivery phase. None of these checkpoints starts through this planning discussion; phase-by-phase authorization still applies.

P16A: imports and one connected scientific view

Purpose: complete the core observation/model workflow without overwhelming the workspace. Owns review items **1, 4, 5, 6, 11, 12, 13**. Related requirements: R04-R06 and the usability work toward R12/X14; existing checkmarks do not establish these added paths.

- [x] **P16A.1** Feed successfully parsed temporary observations into the shared instrument layer with real coordinates, per-sample times, provenance and QC. Make out-of-domain/time support explicit.
- [x] **P16A.2** Inspect an original or imported profile while retaining useful model context. Use a compact inspector, responsive layout and meaningful depth/position links; preserve independent observation/model dates.
- [x] **P16A.3** Add a concise import summary of detected variables, units, coordinates, time/depth coverage, accepted/excluded samples and unsupported fields. Preserve strict definitions and existing safety bounds; arbitrary user-defined unit conversion is not included.
- [x] **P16A.4** Demonstrate the existing chlorophyll parser with a genuine source and QC-aware display. Distinguish observational chlorophyll from a separately sourced volumetric model field. Verify each instrument type's available model overlap; acquire compatible data when needed or record the exact source dependency.

- [x] **P16A.5** Run relevant parser, numerical-picking and browser checks, inspect responsive layouts and the full import-to-profile path, update prepared text and deploy to the existing Vercel project.
- [x] **P16A.6** Before final import, let the user resolve supported variable-name and unit mappings. Show the interpreted quantity and any allowed conversion; reject incompatible dimensions and unsupported conversions. Verify ambiguous names, unit mismatches and unchanged already-compatible inputs. Implement this before the final P16A.5 acceptance run.

Done when: a new supported observation file appears in the shared scene at its real location and opens an understandable profile without losing model context. Import preview, supported mapping and QC feedback work together. Genuine chlorophyll behavior and each claimed instrument overlay have source-backed evidence. Unsupported comparisons stay explicit. Imported numerical comparison and persistence belong to P16C and are not claimed by this checkpoint.

Accepted 27 September 2026: release 0.16.5 delivers the supported import/preview/mapping/QC path and connected model/profile inspection. Real chlorophyll evidence is supplied as a previewable import, preserving frozen-library replay. The glider/CTD examples have no matching main model case; PHASE_16A_IMPORTS.md records their exact source dependency. No imported comparison/save, chlorophyll model volume or simultaneous four-instrument match is claimed. Local science/browser/visual and public API/browser gates pass; see PROJECT_STATE.md.

P16B: fresh data, extensibility and portability proof

Purpose: demonstrate that the architecture works beyond preselected examples. Owns review items **2, 3, 7, 8, 9**. Related requirements: R02, R06, R08-R11. This can require several sessions and external data/runtime access.

- [x] **P16B.1** Process one unfamiliar compatible model dataset through acquisition/preparation, registration, API and scientific viewer. Preserve source identity, units, native coordinates/masks and historical dates. A new file need not be a live data feed or a public arbitrary-upload endpoint.
- [x] **P16B.2** Record actual changes needed for a source/variable/sensor extension. Reuse registered contracts; identify remaining hard-coded coupling and fix what the demonstration requires. Reuse the P16A chlorophyll example where appropriate without claiming model-variable support from an observation alone.
- [x] **P16B.3** Reproduce WMS/WCS/OPeNDAP and scoped CF evidence using a documented setup. Report independent checker findings where available; keep missing public THREDDS hosting explicit.

- [x] **P16B.4** Demonstrate a clean app installation outside the current Vercel deployment and record startup instructions and measured representative workloads. Do not label this an actual INCOIS installation. Establish the versioned repository/setup baseline required for reproducibility.
- [x] **P16B.5** Check new-data values against source references, export/replay behavior where supported, update the coverage matrix and deploy any application changes. Carry final cross-feature security/performance/regression gates forward to P17.

Done when: a reviewer can see a genuinely new compatible input reach the application and reproduce the declared extension/deployment path. Unavailable data, services or infrastructure remain explicit dependencies, not completed checkboxes.

P16B accepted 27 September 2026: release 0.16.7 delivers a newly acquired compatible March HYCOM case, exact source-value checks, measured registry/preparation changes, January/March local WMS/WCS/DAP2 proof, a clean source-archive installation and public browser/API/visual checks. The external CF checker could not install because native UDUNITS is unavailable; scoped CF and source/protocol checks passed. Docker and actual INCOIS installation were not supplied. Persistent public THREDDS remains unconfigured. The initial menu exposed methods outside their checked case sets; the final catalogue-driven routing fixes that without loosening science gates. See PHASE_16B_FRESH_DATA.md and PROJECT_STATE.md. P16C started subsequently on the user's instruction.

P16C: imported comparisons, saving and replay

Purpose: give supported imported observations a complete investigation lifecycle. Owns review items **15, 16, 17, 18**. Depends on the P16A import contract and compatible model inputs. This is substantial implementation work, even though saved bundled investigations already exist.

- [x] **P16C.1** Pass validated imported profiles through eligible model matching, QC, physical-unit checks and numerical comparisons. Include source identity, exclusions and the assimilation limitation; verify results against independent fixtures.
- [x] **P16C.2** Save and reopen supported imported investigations with explicit storage location, quotas, deletion and parser/method versions. Start with the existing local storage architecture where practical; do not introduce server retention or accounts implicitly. Review policy wording against the implemented behavior.
- [x] **P16C.3** Extend portable bundles to include the permitted imported data or sufficient resolvable references, settings, results and provenance. Check source hashes and parser/calculation versions on replay. Reject corrupt or incompatible bundles and explain unavailable external references.

- [x] **P16C.4** Check save/reopen/export/reimport and numerical replay in a fresh session, including tampered and mismatched inputs. Update documentation and deploy.

Done when: an eligible imported profile can be compared, saved and reopened with reproducible results, and a supported portable bundle can be reopened with its dependencies and limits made explicit. Existing bundled investigations still replay correctly.

P16C accepted 27 September 2026: release 0.16.8 adds request-scoped imported comparisons, explicit IndexedDB saves, original-file JSON/ZIP bundles and parser/source/method/result-checked replay. Independent original-observation and native-model checks, 203 backend tests, 60 client checks, local Chromium/WebKit interactions, 40 public desktop/mobile browser checks and 61 public lifecycle checks pass. Frozen source data and old replay identities remain unchanged. Scope, quotas and limits: PHASE_16C_IMPORTED_REPLAY.md.

P16D: connected investigation workflow

Purpose: make existing analysis and the new import lifecycle understandable in one task flow. Owns review items **19, 20, 21, 22, 29**. Depends on P16A and P16C for the full imported-data journey. Reuse existing scientific tools and tutorials rather than duplicating their calculations.

- [x] **P16D.1** Synchronize eligible depth/time/position selections across scene, profile and comparison. Show distinct source timestamps and unmatched selections; never align unrelated measurements by relabelling dates.
- [x] **P16D.2** Make observation support, accepted residuals and coverage gaps visible for a selected structure. Expose sensitivity settings and their effects with units, counts and limits. Reuse checked methods and add numerical checks for any changed calculation.
- [x] **P16D.3** Connect region/data selection, visualization, observation inspection, analysis and saving in a Guided Investigation Builder. Keep a simple first action, contextual advanced controls and room for the scientific views. Preserve direct access to existing labs.
- [x] **P16D.4** Verify the complete task through keyboard, responsive and browser interactions, including empty/unsupported states. Prepare a participant task script, document actual behavior and deploy. Actual participant evidence remains P09.5/P17.4.

Done when: a visitor can follow one connected investigation from selecting data to saving its result, with meaningful linked views, discoverable support/sensitivity and recoverable interruptions. Automated interaction does not establish real-user comprehension.

P16D accepted 27 September 2026: release 0.16.9 connects source selections and eligible model context, adds native-cell observation support/residuals/gaps and matching sensitivity, and guides an explicit saved investigation. All four steps have evidence in PHASE_16D_CONNECTED_WORKFLOW.md and evidence/p16d-acceptance.json. Public release, exact scientific/replay, 34 browser checks and 12 responsive captures pass. Actual participant evidence is still open; the prepared task script does not close it. Initial transport and test failures remain recorded.

P16E: larger workloads, refresh and offline continuity

Purpose: extend practical data handling and preserve usable results through long operations or unavailable services. Owns review items **14, 23, 24, 30**. Depends on the P16B data pipeline and P16C storage/provenance contracts. This is substantial architecture and resource work; limits come from measurements.

- [x] **P16E.1** Support bounded batch/larger imports with per-file feedback, memory/request limits and validated partial-failure behavior. Measure realistic inputs before raising existing caps; do not claim unlimited uploads.
- [x] **P16E.2** Add an explicit dataset refresh pipeline with source timestamps, update status, failure reporting and retained last usable publication. Publish compatible frontend/backend data contracts together. Configure only available authorized infrastructure and record any external scheduler/hosting dependency.
- [x] **P16E.3** Extend offline investigations beyond the existing regional pack for declared supported workflows. Inventory required data/assets, honour source licences and quotas, expose cache age and removal, and distinguish unavailable analysis from cached results.
- [x] **P16E.4** Provide useful progress, cancellation and retained results across changed import/refresh/analysis paths. Verify cancellation, timeout, offline, quota and restart scenarios without misleading completion states. Document measured limits, recheck relevant security/data-handling controls, and deploy.

Done when: the declared larger import workload, refresh failure/recovery and expanded offline workflow pass real checks, and users can cancel long work while retaining earlier usable results. Missing infrastructure remains an open dependency. Final cross-feature scalability and security evidence belongs to P17.

P16E accepted with declared limits, 27 September 2026: release 0.16.10 supplies an eight-file sequential queue, a measured 20,000-sample workspace, maintainer-only retained publication and self-contained offline profile/comparison/native-column viewers. Cancellation, timeout, quota, restart and source-value checks have evidence in PHASE_16E_CONTINUITY.md and evidence/p16e-acceptance.json. The latest public phase

suite had 20 passes and two failures; all six targeted recovery checks then passed. The Chromium HTTP/2 transfer failure remains an explicit P17 network follow-up, not a silently cleared defect. Twelve final public responsive captures passed overflow/page-error checks and representative screens were manually inspected. No automatic external scheduler, full offline laboratory engine or actual institutional installation is claimed.

All 31 items have an owner: P16A (1, 4, 5, 6, 11, 12, 13); P16B (2, 3, 7, 8, 9); P16C (15, 16, 17, 18); P16D (19, 20, 21, 22, 29); P16E (14, 23, 24, 30); P17 (10, 25, 26, 27, 28, 31), plus the already requested security audit. Documentation, relevant checks and deployment happen throughout. Recruit actual participants early; do not wait until the release freeze to discover that no participants are available. No item is marked complete by assigning it a phase.

P17 : Independent checks and final release

An authorized globe/tutorial follow-up precedes P17: G1 explicit source-backed globe, G2 PS-ordered tours and visible Tools handoff, G3 responsive/scientific regression checks and Vercel deployment. Its acceptance is recorded in `PROJECT_STATE.md` and `GLOBE_AND_TUTORIAL_REVIEW.md`. This does not start or complete the P17 audit.

Outcome: a deployable release with an honest capability matrix, reproducible evidence and a rehearsed national-round demonstration.

Steps:

- [x] **P17.1** Audit every feature/requirement checkbox and every row in `USP_INDEX.md` against actual code and evidence. Separate implemented, internally tested, externally reviewed and incomplete capabilities. Do not close unfinished earlier phases to meet a date.
- [] **P17.2** Run the relevant end-to-end and numerical regression suite, clean setup, an unfamiliar compatible input and saved-investigation replay. Seek an independent scientific review of representative outputs and record its actual availability/scope.
- [] **P17.4** Record performance on stated devices/networks and a fair small usability comparison. Include Chromium/Firefox/WebKit engine checks, small viewports, network/CPU throttling and available real-device GPU/fallback tests; automation alone does not establish universal hardware support. Publish measured numbers only; disclose cached/precomputed work and study limits.
- [] **P17.5** Use `SUBMISSION_CHECKLIST.md` to prepare a short live demonstration, actual-behavior backup video, `README/setup`, architecture and methods notes, attribution/licences, limitations, coverage matrix and judge questions. Prepare the

shortlisting PDF before its deadline even if later build phases remain open. Reverify dashboard/organizer instructions for the actual round; supporting assets are not automatically mandatory portal uploads.

27 September 2026 release checkpoint: 0.17.0 is publicly deployed. P17.1 and P17.3 are accepted for their stated audit/security/portable-package scope. P17.2's numerical, browser, unfamiliar-input, clean-setup and replay work passed; independent oceanographic review is still unavailable. P17.4's stated-environment browser/network/resource measurements are recorded, while actual participant and unrelated-device evidence remain open. P17.5's six-slide draft, real silent backup recording, methods, attribution and demo materials are ready; private round/dashboard verification and any actual submission still require the applicable organizer information. Those combined checkboxes remain open rather than silently weakening their acceptance criteria. X33's prepared release/demo assets are delivered. See PROJECT_STATE.md and evidence/p17-acceptance.json.

Done when: the declared release is reproducible and deployable, evidence supports its claims, and open limitations are explicit. Public prototype readiness does not mean operational oceanographic certification or guaranteed competition placement.

First ten-day checkpoint

Do not assign a guaranteed day to each phase before the first data/graphics/deployment checks.

- First 1-2 days: establish the environment and hosting path; acquire real data and prove the first correct model/profile display. Begin the standards-service compatibility check early even though complete integration is P06.
- Midpoint: review actual core progress and resolve scientific/standards blockers before adding experiments.
- Last 2-3 days before the first demonstration: reserve time for usability, fresh setup, deployment checks and rehearsal. Assess a candidate release with the relevant P17 checks even if later roadmap phases remain unfinished.
- The desired first candidate is P01-P09. If that is not achieved, report the exact supported subset and remaining official requirements; retain P10-P17 and all incomplete work in the roadmap.
- The full roadmap continues after that checkpoint under the user's phase-by-phase instructions. Additional phases are not automatically completed merely because a public prototype has shipped.

18 Phase 2: historical Bay of Bengal case

Historical or maintained reference: read alongside the current release overview | Source: docs/PHASE_02_DATA.md

Project/access-review date: **21 September 2026**. The user explicitly reminded us that the current year is 2026. Scientific acquisition dates, observation dates, source update dates and project dates are separate fields. No January 2024 data is described as current or live.

Selected case

- Case ID: bay-benga1-2024-01.
- Model: HYCOM GOFS 3.1 + NCODA, GLBy0.08/expt_93.0, an assimilative **model analysis**, not an observation or a newly computed forecast.
- Seven instantaneous snapshots: 2024-01-07 00:00 UTC through 2024-01-10 00:00 UTC, every 12 hours. The archive has 3-hourly output; the intervening timestamps are deliberately not bundled.
- Requested box: 85-90° E, 12-15° N. Selected native coordinates: approximately 85.04-90° E, 12-15° N. The source's exact first longitude is **85.0400390625° E**, confirmed independently through its ASCII endpoint; retain that value for exact point requests rather than the rounded label.
- Native subset: 63 longitudes × 76 latitudes × 40 irregular output depths, from 0 to 5000 m. The source's 41 hybrid model layers are distinct from its 40 fixed output depths.
- Four scalar arrays: in-situ temperature, practical salinity, eastward velocity and northward velocity. The velocity components are geographic, not grid-relative. No chlorophyll or vertical velocity is supplied.
- Seven **delayed-mode Argo profiles** distributed through the INCOIS DAC at Ifremer GDAC, cycles 012 of floats 1902669, 2903891, 4903775, 4903776, 5907083, 7901125 and 7901126.

This box was chosen after inspecting the January 2024 Argo index and reading actual profile files. It is a small compatibility case, not a verified ocean event or basin-wide coverage claim. The geographic globe's larger navigation rectangle remains illustrative and is labelled separately.

Source-access audit

Candidate	Evidence on 21 September 2026	Decision
Copernicus GLORYS12	Product catalogue read successfully; official download documentation describes account credentials. No authenticated subset download was established in this session.	Retain as a later adapter/source candidate; do not block this phase or imply acquisition succeeded.
INCOIS services	ESSDP homepage returned HTTP 200. An older portal/datainfo.jsp URL returned 404. This is not evidence of access to an INCOIS numerical model download.	Preserve the institutional integration requirement. The selected Argo originals do come from the public INCOIS DAC directory.
HYCOM GOFS 3.1 archive	DDS, DAS and coordinate metadata returned HTTP 200. NCSS metadata timed out. Native netCDF4 remote access failed locally with a curl IO error; the same OPeNDAP archive worked through pydap. One later field request timed out and was resumed from verified snapshots.	Use serial pydap OPeNDAP subset requests, at most one HYCOM connection at a time, with bounded retries.
Ifremer Argo GDAC	Indian Ocean daily file and global profile index retrieved. The initially inspected 15 January daily file had no profiles in the tested Bay box. Index filtering found 57 January profiles in the broader candidate region; seven matching INCOIS profiles were downloaded for the final box/time span.	Use original individual delayed-mode NetCDF records with QC and metadata.

The raw audit is in `evidence/phase-02-access-audit.json`. Selected files, exact remote indices, source URLs, retrieval timestamp and SHA-256 fingerprints are recorded in the case manifest and source-metadata file. The profile index is a discovery aid, not a substitute for inspecting the selected original records.

Reproduction and stored representations

From the repository root:

```
.venv/Scripts/python.exe -m pip install -r requirements-science.txt
.venv/Scripts/python.exe -m science.acquire_case
.venv/Scripts/python.exe -m science.prepare_case
.venv/Scripts/python.exe -m pytest tests -q
```

The pinned offline environment was tested with Python 3.10.11. The web service uses the much smaller `requirements.txt`; `xarray`, `netCDF4`, `pydap` and `GSW` do not run in evaluator requests.

`data/raw/` is ignored and not uploaded to Vercel. Acquisition stores:

1. Original Argo profile NetCDF files, unchanged.
2. HYCOM **packed source subsets reconstructed as NetCDF** from OPeNDAP, with source attributes, original coordinates and exact index selections. These are not the original whole global archive files. Packing metadata retains the source Float32 scale/offset and Int16 fill values.
3. An acquisition journal with hashes. Verified cached snapshots can be reused after interrupted network requests. A changed cached fingerprint is rejected rather than silently accepted.

`casepacks/bay-bengal-2024-01/` is included in `source/deployment`:

- `manifest.json`: versioned contracts, coverage, provenance, limitations and file fingerprints.
- `source-metadata.json`: original coordinate/variable metadata, acquisition records and value statistics.
- `analytical/*.bin.gz`: lossless gzip of little-endian float64 values on the native subset grid. Source packing is decoded once; no interpolation or hole filling.
- `display/*.bin.gz`: every second latitude/longitude, plus the final coordinate, retaining all 40 depths, converted to float32. The resulting shape is $40 \times 39 \times 32$. The manifest records source indices and the measured maximum rounding error per variable.
- `profiles/*.json`: selected and original readings, adjusted errors, QC, calibration metadata, source indices and pressure-derived depths.

The analytical and display arrays share timestamps and variable definitions. Scientific calculations must use the analytical representation. Display decimation is not an analytical regridding operation. Missing values remain NaN in binary and JSON null over the API.

The committed case pack allows a fresh checkout and Vercel deployment to run without downloading external data or configuring a marine-data account. Upstream files may be revised; the current Argo file hashes and update metadata identify this acquisition. We do not claim the files belong to a frozen monthly DOI snapshot. A later source revision must produce a reviewed case version, not silently replace this pack.

Observational eligibility

- DATA_MODE=D or A selects only *_ADJUSTED. R selects raw values. An absent adjusted value is never backfilled from raw data.
- A sample is eligible for this early compatibility check only if selected pressure, temperature and salinity are finite with QC 1 or 2, pressure is nonnegative, and position/time QC are 1 or 2.
- All original source levels are retained, including excluded levels and raw/adjusted fields. The source order is preserved; regular vertical spacing is not assumed.
- Pressure in dbar is converted to positive-down depth in metres with `-gsw.z_from_p(pressure, latitude)`. Dynamic height and sea-surface geopotential use the documented zero defaults. Pressure itself is retained.
- JULD is decoded from its CF epoch and rounded to the source's second resolution, avoiding floating-day truncation by one second. Original numeric JULD, units and decoded timestamp remain in metadata.
- Profile eligibility additionally requires actual spatial, temporal and vertical overlap. The manifest records the nearest bundled model timestamp and signed offset. This is **not** collocation/interpolation, a mismatch metric, a confidence score or an independent validation result.

Full model/profile comparison methods belong to P05. HYCOM assimilates observations, so future agreement cannot automatically be described as independent validation.

API and caching

Endpoint	Output
/api/catalog	Schema v2, one real historical case and two planned regions
/api/cases/bay-bengal-2024-01	Full scientific manifest
/api/cases/bay-bengal-2024-01/subset?variable=temperature&time_index=0&depth_index=0	Native analytical surface slice
Endpoint	Output
/api/cases/bay-bengal-2024-01/subset?variable=temperature&time_index=0&...	Bounded display volume

Endpoint	Output
peration=volume&representation=display	
/api/cases/bay-bengal-2024-01/profiles/argo-1902669-012-0	Genuine Argo record, selected/raw/adjusted values and QC

Subset requests accept one variable and timestamp, an optional geographic box, and either a depth slice or volume. All four box limits are required together. Coordinates must lie within the selected case; unsupported dates, depths, variables and oversized requests produce explicit errors. A maximum of 100,000 scalar values keeps responses bounded. The larger complete analytical volume needs smaller bounds or slices.

A process-local LRU holds at most three decompressed arrays plus one manifest. Files are immutable deployment assets; the filesystem is not used as durable user storage. Array/profile fingerprints are checked before serving. Browser/API responses retain no-store; no external data request is triggered by opening the workspace.

Verification scope

Numerical tests use independently specified packed integers and coordinates, reversed axes, 0-360 longitudes, irregular depth, fill/range masks, invalid coordinates, incompatible temperature/current definitions and unsupported calendars. Argo fixtures verify adjusted-value selection, rejection of bad QC and missing-adjusted behavior. Pressure conversion is checked against the published TEOS-10 example, not against the same implementation recomputed as an expected result.

Case-pack tests check hashes, actual dates/coverage, source samples, display/analytical equality within declared float32 rounding, missing deep cells and bounded API failure paths. Additional verification reads raw packed source NetCDF values independently with netCDF4 and records direct source-service checks. Browser tests cover historical labels, sample selection, missing values, Argo QC records, keyboard focus, region switching and error recovery. Evidence files are listed in PROJECT_STATE.md.

Not implemented by this phase: volumetric rendering, map instrument overlays, profile comparison charts, arbitrary uploads, live feeds, chlorophyll, gliders, CTD/BGC adapters, WMS/WCS interoperability certification, simulations or operational advisories. These remain on their original phases.

Primary references

- HYCOM dataset and archive
 - HYCOM public-release terms and service limits
 - HYCOM source metadata and service links
 - Argo raw/adjusted selection and quality guidance
 - Argo attribution and DOI guidance
 - TEOS-10 pressure-to-height example and definition
 - CF Conventions 1.12, used for interpretation; source metadata declares CF-1.6/NAVO, not full CF-1.12 certification.
 - Copernicus GLORYS12 product
 - Copernicus credential documentation
-

19 Phase 3: scientific ocean rendering

Historical or maintained reference: read alongside the current release overview | Source: docs/PHASE_03_RENDERING.md

Implementation date: 22 September 2026, Asia/Calcutta. Release: 0.3.2 (continuity and motion revision). This phase uses the unchanged historical Bay of Bengal package from P02. It does not add a new region, observation source or simulated result.

Rendering method

Three.js 0.186.0 manages the WebGL2 scene. A custom ray marcher samples a real three-dimensional texture, with front-to-back opacity accumulation. This is volume rendering of model values, not a decorative ocean surface or a stack of screenshots. The independent geographic overview remains available through Geographic context.

The full display grid has 32 longitudes, 39 latitudes and all 40 physical depth levels, for 49,920 samples per scalar snapshot. The initial view selects the original 33 levels from 0 to 1,000 metres. The Full column option restores all 40 levels to 5,000 metres. Neither window invents or redistributes depths. It selects every second horizontal source coordinate plus the final edge. It does not add depth levels or change analytical data. Float32 display values and float64 analytical values remain separate. The case manifest remains SHA-256 9598b33dc9eef89cb909f2811e1590ca580a143c1dbcc25d97e51a84faf7d28d.

View	Method and limits
Volume	The initial view is a whole volume. Open cutaway beside the scene removes the southeast quarter for viewing and reveals two source-aligned coloured cross-sections. The button closes it again. 160 samples per settled viewing ray (64 during the short cut transition), physical-axis lookup and trilinear interpolation over the irregular rectilinear grid. Finite sampling can miss thin features; use a depth slice and native inspection for numerical examination.
Depth slice	Triangles through an actual supplied depth level; scalar colour interpolation is a display operation. A quad with a missing corner is excluded from scientific colour.
Section	East-west plane at one of the supplied display latitudes, using actual unequal depth intervals. Missing cells are shown with gray hatching. Arbitrary drawn transects belong to later work.
Isosurface	Each complete cell is split into six consistently ordered tetrahedra. Edge crossings are linearly interpolated. A cell with any missing corner is excluded. This is an approximation, not a new analytical feature-volume calculation.
Currents	Horizontal speed is derived from the supplied eastward and northward components. Arrows point east and north correctly; no vertical velocity or particle integration is inferred. Direction arrows have equal lengths: 0.35 scene units in 3D and 14 CSS pixels in Basic. Colour and native numeric values encode speed, rather than arrow length. Zero vectors have no direction arrow.
Basic 2D	Canvas depth slice or east-west section on physical coordinate axes. Cell colour is the average of four valid corners. Current vectors remain available. Volume and isosurface modes explicitly show a vertical-section fallback. Missing cells are hatched; Basic never claims to extract a 3D isosurface.

Coordinates use a local equirectangular projection around this small domain's centre latitude, with radius 6,371,008.8 metres. East is positive scene x, north is negative scene z and depth is negative scene y. This local display projection is not the method for future geodesic distance, area or volume calculations. It is not yet suitable for arbitrary global, polar or antimeridian cases.

Vertical exaggeration ranges from 1 to 250 and is labelled. Initial upper-ocean exaggeration is 200; selecting the full column uses 50. Both are display settings, not physical aspect ratios. Depth remains in metres. Changing exaggeration changes geometry only. The default camera and reset fit the domain and depth labels to the viewport. Users can orbit, zoom or use keyboard-accessible camera buttons.

Colour, masks and inspection

Temperature, practical salinity and horizontal current speed have labelled units. Palettes, minimum/maximum, linear/log mapping and opacity are editable. Invalid ranges show an explanation and use a stated default linear range until corrected. Log mapping requires a positive minimum and hides non-positive values. Scientific false-colour gradients are used only for the data legend and rendering.

The volume opacity transfer function is $1 - \exp(-\text{opacity} * \text{rayStep} * 0.8)$. Opacity is independent of the scalar value, so colder water no longer receives systematically weaker opacity. This replaces the higher-value weighting used in v0.3.0. Brighter lower-range colours and opaque cut faces make the vertical structure easier to see. Opacity and simple depth/surface shading aid inspection; neither is underwater illumination, confidence, probability or a physical water property. Colours in a translucent volume are composited along the view, so they should not be read as an exact point measurement.

Missing source values stay masked. Volume interpolation rejects a location if a contributing corner is missing, allowing only a small floating-point weight tolerance. This can conservatively omit some valid water near a masked boundary. Hatching is the conservative missing-cell boundary, not measured bathymetry. There is no invented seafloor surface or gap filling. The source grid extends to 5,000 metres, but deep cells are missing; the empty 5,000-metre slice is a deliberate verification case.

Clicking a slice, section or isosurface maps back to the nearest native grid point. A cutaway-face click uses that face's physical longitude, latitude and depth. Other volume clicks use the explicitly selected horizontal probe depth. Clicking also updates the inspection-depth control. At overview scale, some native rows project to less than one pointer pixel; zoom or use coordinate selectors when a specific source row is required. The analytical endpoint supplies all 40 native depth values at the selected exact longitude and latitude. The selected original depth index reads directly from this checked column, so changing depth alone does not require another request. Columns must match the displayed field's dataset fingerprint, variable, units, time and coordinates. It is not a colour readback, interpolated display value or estimate at an arbitrary continuous position. Longitude and latitude selectors provide an accessible alternative. Current speed is calculated from the two native component values and both components are shown.

Seven real timestamps from 7 to 10 January 2024 are available. Playback waits for each requested snapshot, advances by the actual 12-hour source interval and stops at the end. It does not invent intermediate frames. While a new timestamp or variable loads, the last verified field stays visible with its original variable, date, colour range, view and isosurface threshold. A small progress notice identifies the pending selection and the field still shown. Scene picking is paused during this interval; a failed request preserves the labelled

previous view and offers retry. The new field and its identifying labels switch together, without blending scalar values or timestamps. Reduced-motion preference disables automatic playback while retaining manual time selection.

Device and failure handling

Auto is the initial setting. WebGL2 failure, shader compilation failure or lost graphics context switches to a real 2D view with an explanation and retry. Failed scientific requests produce a recoverable error, without substituting illustrative data. The existing version-change reload path is preserved.

The renderer caps device pixel ratio at 1.25. Auto observes the browser frame following each draw, excludes hidden-tab and idle intervals, and allows five initial draws for warm-up. Four net slow observations above 55 ms reduce pixel ratio to 0.7. After another warm-up, four net observations above 100 ms switch to Basic. A fast observation reduces the slow count. These are practical responsiveness thresholds, not a measured hardware classification. Balanced 3D lets the user retain 3D without this automatic reduction. Analytical arrays, probe values and source timestamps do not change with graphics quality.

Only dirty frames are drawn. The renderer is dynamically loaded after the case is available. Switching to geographic context retains the explorer and camera while suspending its rendering and playback. Resources are disposed on actual unmount or a switch to Basic. Cutaway, volume-probe and graphics-preference updates reuse the existing texture and geometry where the field is unchanged. The client rejects mismatched variable, unit, timestamp and display-coordinate responses. Current renderer contracts bound each display axis to 64 coordinates; larger datasets need the later subsetting/scale work.

The API compresses eligible responses with gzip. The first temperature display response measured 882,764 decoded bytes and 131,927 compressed bytes locally. This is a real response-size observation, not a general network guarantee. The Three.js scene is a separate dynamically loaded bundle of approximately 580 kB minified and 145 kB gzipped. Vite reports its 500 kB chunk advisory; it is recorded rather than suppressed.

Verification and reproducibility

```
npm --prefix web run build
npm --prefix web run test:science
.venv/Scripts/python.exe -m pytest tests -q
$env:OCEAN_TEST_REPORT='../docs/evidence/phase-03-local-browser-final.json'
node web/node_modules/@playwright/test/cli.js test
--config=web/playwright.config.ts --project=chromium-desktop --project=webkit-
desktop --project=chromium-mobile
$env:OCEAN_BENCH_NAME='clarity-verification'
node web/benchmarks/continuity.cjs
```

The 17 pure numerical checks use an independently specified affine field and irregular depths. They cover physical interpolation, masks, metre geometry, depth ratios, sections, isosurface crossings, vector magnitude, log mapping and nearest-coordinate selection. Four checks cover depth windows, coverage counts, source-aligned cutaway faces and the separation of missing-cell hatching from scalar data. One additional check bounds the display-only easing and its reversal/reduced-motion endpoints. The existing 21 backend checks cover source handling and API contracts.

Browser picking checks use an independent perspective calculation and expected values read directly from packed NetCDF integers, with the original scale and offset applied without application adapters. One 100-metre point lies between display nodes but on the native grid; another section point is 1,000 metres deep. A new cut-face fixture at 500 metres is 9.832999517093413 degrees Celsius, read from the packed source independently. A 4,000-metre hatched-face click must return no value. The fixture records the source-file hash in `web/e2e/fixtures/phase-03-source.json`. Both 3D and Basic clicks must recover the expected source coordinates and values.

Browser checks also exercise every view, missing deep water, an empty isosurface, real timestamps, colour/exaggeration invariance, reduced motion, API recovery and missing WebGL. Context loss is injected in Chromium. An explicitly scaled frame clock checks Auto fallback without modifying scientific responses. That test is a failure-path check, not a performance benchmark.

Performance is measured separately without a concurrent browser suite.

`web/benchmarks/phase-03.cjs` records the browser, viewport, reported graphics renderer, first-volume latency, request sizes and RAF intervals during repeated camera rotation. Headless Chromium here uses SwiftShader software rendering. WebKit on Windows reports an Apple GPU string, which does not establish physical Apple-hardware coverage. Screenshots cover desktop, 390-pixel mobile and 200% text at 768 pixels. The clarity audit also checks 320-pixel width. Current results are under `docs/evidence/motion-*`; the previous clarity review is under `clarity-*`; original-release evidence remains under `phase-03-*`. The new audit measures RAF intervals in Auto mode, which may switch to

Basic during the interval and must not be reported as sustained 3D FPS.

Firefox automation still cannot launch on this Windows installation because of the previously recorded side-by-side binary error. Docker has not been run. Neither is reported as passing. Public browser checks and final release measurements are recorded in the project state.

Next boundary

P04 adds geospatial instrument overlays, depth-profile charts and supported ingestion workflows. The seven existing Argo records remain a list/record inspector in this release. P05 adds scientific model-observation comparisons. P11 particles, P12 heat/depth and P13 ENSO/IOD remain in the roadmap. This phase completes no claim of live forecasting, independent observational validation, arbitrary global coverage or operational INCOIS deployment.

Continuity and motion details

Verified historical fields and native columns are cached in browser memory for ten minutes of inactivity. Cache keys separate the application release, case, variable, timestamp and native location; columns additionally use the display field's manifest fingerprint. The case pack is immutable within a release. Data are not written to browser storage. After the selected field is ready, a small sequential queue preloads the next timestamp and other variables at the current timestamp, along with native columns at the current point. Data-saving and 2G connections skip this optional work. No all-ocean or indefinite prefetch runs.

The cutaway uses a 460 ms smoothstep reveal from the surface downward. It changes clipping thresholds only, leaving cross-section vertices, masks and scalar attributes at their original coordinates. Picking pauses during the transition. The moving preview uses 64 ray samples and caps DPR at 0.75, then restores 160 samples and the prior pixel ratio. This reduces drawing cost during motion without changing numerical inspection. An interrupted toggle reverses from its current display progress. Reduced motion applies the endpoint immediately.

A new field/view has a short 180 ms opacity arrival; datasets are not cross-blended. The first request still needs network time and has a loading explanation. Subsequent uncached requests retain the verified scene. No fabricated loading percentage or completion counter is shown. Cold loads, slow software graphics and remote service errors remain possible.

The browser suite verifies no extra source requests for depth changes, constant texture/rebuild counts for cutaway changes, correct labels and thresholds during slow updates, cached revisits, failure/retry, cancellation of an abandoned timestamp, reduced motion, geographic-context preservation and rejection of mismatched native coordinates. The original independent NetCDF checks remain in place. `web/benchmarks/interaction.cjs` records the request counts, individual action timings and transition frames separately from the test suite. A mixed animation/idle RAF interval is not a sustained 3D frame-rate claim.

20 Phase 4: observations and supported imports

Historical or maintained reference: read alongside the current release overview | Source: `docs/PHASE_04_INSTRUMENTS.md`

Completed: 22 September 2026. Release: 0.4.0, verified on Vercel. `PROJECT_STATE.md` records the deployment and next phase.

Implemented contract

The new observation contract is version 2, defined in `science/instruments.py` and checked in the browser. The P02 model manifest and its legacy profile endpoints remain unchanged. P04 reads the original Argo files again into a separate observation library using `adapters/instruments.py`. Each variable has its own source field, units, processing mode, QC scheme and accepted count. Every original sample retains its index, pressure, derived depth, position, timestamp and selected/raw/adjusted values, flags and supplied adjusted error. Structured `coordinate_qc` records pressure, position and time flags, leaving unavailable source flags blank; `coordinate_eligible` states the adapter's rule without inventing good flags.

Pressure is converted with `-gsw.z_from_p(pressure, latitude)` using GSW 3.6.20 and its zero dynamic-height/surface-geopotential defaults. Pressure is sea pressure in dbar; it is never relabelled as metres. Source order and duplicate depths are retained. Negative pressure has no usable depth. See the [TEOS-10 function and independent example](#).

Core Argo R selects the raw parameter; A/D selects its adjustment. BGC selects each parameter through `PARAMETER_DATA_MODE`. Missing adjusted data never backfill from raw data. The accepted curve requires Argo flags 1/2 for its variable, eligible pressure and position/time. Raw BGC R values remain inspect-only. These choices follow the [Argo data-use guide](#); acceptance by this filter is not independent validation.

QARTOD 1 is accepted; QARTOD 2 is not evaluated and excluded. WOCE CTD flag 2 is accepted; WOCE 1 is not promoted to good. CSV 1/2 are uploader declarations, not verified provenance. The source schemes remain visible. References: [IOOS temperature/salinity QC manual](#), [WOCE hydrographic data formats](#).

Genuine observation library

Collection	Records	Source and limit
Bay of Bengal, January 2024	Seven Argo profiles already acquired for P02	INCOIS DAC originals via Argo GDAC; inside the model's geographic domain, but no match is inferred
Tropical Atlantic, September 2023	Argo 6903091 cycle 100, including adjusted oxygen	Original GDAC S-profile ; S-file sensor alignment belongs to the provider, not this app
Caribbean Sea, April 2024	Four RU29 glider profiles, 3,787 source rows	IOOS/Rutgers delayed deployment ; bounded two-hour profile-time selection, full original samples within those profiles
Southwest Indian Ocean, February 1996	CIVA2 station 1, cast 1, 48 samples	CCHDO cruise 35MF103_1 ; original CTD Exchange member <code>i06sb_00001_00001_ct1.csv</code>

Glider per-sample `precise_time`, `precise_lat` and `precise_lon` are retained. Source positions can be interpolated between GPS fixes, which is disclosed. The source salinity declares `valid_min` 40 and `valid_max` 0. The adapter records both, preserves finite source readings for inspection and excludes salinity from the accepted curve. It does not silently swap the bounds or invent a valid range. Temperature remains eligible under its own QARTOD/pressure checks. A source-quality problem should not become a plausible green curve.

The CTD source has missing oxygen at this station. The variable remains inspectable with no accepted curve. The temperature scale must be confirmed from the cruise report before P05 quantitative comparison. Neither the ship cast nor the glider/BGC collections extend the Bay of Bengal model domain.

`science/acquire_instruments.py` records source URLs, sizes and hashes;
`science/prepare_instruments.py` creates the versioned profile JSON and checksummed index in `casepacks/instruments`. Raw downloads remain under `ignored data/raw/instruments`. Five downloadable example files include original core/BGC/glider

NetCDF, original CTD Exchange, and a documented CSV export of 20 genuine Argo levels. The CSV export retains selected readings and flags; its simpler schema does not preserve the full adjustment history.

Import boundary

`POST /api/instruments/import?filename=...` accepts a raw file body. Application code processes bytes in memory, does not fetch source URLs and does not persist uploads. The file name is display metadata only, not a filesystem path. The API serializes netCDF C-library access. The endpoint and browser reject excessive sizes; malformed metadata, calendars, unsupported variables, incompatible units and unsupported schemas get explicit errors. File limits: 2 MB, 24 profiles, 5,000 source samples, bounded NetCDF fields/elements and a 3.5 MB decoded response. Browser imports are capped at 24 profiles or 12,000 samples per open session.

Supported layouts are Argo/BGC profiles, documented IOOS glider row-NetCDF subsets, CTD Exchange casts and Ocean Navigator CSV v1. This is not arbitrary NetCDF/Excel support. The public import guide defines the exact columns and requirements. Existing model ingestion still uses the offline xarray adapter. Runtime observation ingestion adds netCDF4, NumPy and GSW, without pandas/xarray in the serverless runtime.

Profiles from uploads live in browser memory until cleared or reloaded. Application code does not log body content; filenames can appear in request URLs and hosting logs. The Privacy Policy and Terms now describe this behavior and retain the user's deferred operator/contact details.

Interaction and scientific limits

Numbered locations on the 3D model open the corresponding historical Argo profile. These are geographic surface markers, not simultaneous instrument fixes at the selected model timestamp. Markers can be hidden. Instruments opens a connected workspace with a collection selector, location map, selectable tracks and a variable-versus-depth chart. The map and 3D markers use actual coordinates. Glider Track detail zooms into the retained source track; the ordinary basemap is Natural Earth geographic context.

Charts use source samples in source order, with a linear physical depth axis. Lines break at missing or excluded samples. Isolated eligible samples appear as dots. Optional excluded crosses never join the accepted curve. Pointer selection chooses an actual sample; keyboard users have previous/next and sample controls. Each selected sample displays its own timestamp, coordinates, units and quality reason. Raw/adjusted fields and source metadata are inspectable.

View nearest model depth selects the nearest native model grid location/depth only, keeps the existing model timestamp and states the observation time. The button is unavailable for locations outside the current model domain. This navigation is not collocation, residual analysis, validation or a forecast. Those scientific matching rules belong to P05.

The desktop shows map and chart together. Narrow layouts prioritize the profile chart and move geographic detail below it. Opening instruments preserves the 3D camera and pauses the hidden renderer. The model controls and model-only case card are hidden in the observation workspace to avoid confusing dates/coverage across collections.

Verification

- Backend: 51 tests pass, comprising 30 observation/import cases and the 21 existing checks. Source-value tests read original NetCDF/Exchange examples independently. Report: `evidence/p04-science-api-final.xml`.
- Renderer: 17 numerical checks passed; P04 does not alter the checked model field values or geometry methods.
- Local browsers: 98 passed, one intentional WebKit context-loss skip. Report: `evidence/p04-local-browser-final.json`. Isolated eligible-sample dots were then added and passed all three focused engine checks in `evidence/p04-isolated-samples-recheck.json`.
- Public browsers: 94 passed, four page/data-loading timeouts, one intentional skip, across Chromium desktop/mobile emulation and Windows WebKit. Three failures passed the first unchanged recheck. The mobile fallback needed a second unchanged recheck, which passed. Reports: `evidence/p04-public-browser-final.json`, `evidence/p04-public-browser-recheck.json`, `evidence/p04-public-fallback-recheck.json`. The traces show several API requests aborted around the existing eight-second limit. The fallback trace includes 5.1 seconds for HTML and 7.9 seconds for the first volume response, leaving its dependent renderer too late for the assertion. The underlying service/network cause is not established. First-load latency remains a real limitation. Trace summaries: `evidence/p04-public-timeout-traces.json`, `evidence/p04-fallback-timeout-trace.json`.
- Public source/API checks: all 13 profiles agree with the checked local library. All five example files download with matching hashes and import successfully. Malformed and oversized uploads return 422/413, unknown IDs return 404, and the catalogue is unchanged after imports. The P02 manifest is unchanged and three independent model probes agree. Report: `evidence/p04-public-http.json`.

- Visual review: desktop, Windows WebKit, 390/320-pixel widths and 200% root text; screenshots cover the model, Argo/BGC/glider, excluded salinity, missing CTD oxygen and imports. No page errors or document overflow in the checked views. Reports: `evidence/p04-local-visual-review.json` and `evidence/p04-public-visual-review.json`. These are emulated layouts, not physical-device certification.
- Build/typecheck and `pip check` pass. The Three.js chunk remains about 581 kB minified, 146 kB gzip, exceeding Vite's advisory. Two existing Python test-client deprecation warnings remain. Firefox's Windows launch issue, Docker and actual institutional installation remain open.

Issues found and retained evidence

The initial local suite passed 91 and failed seven, with one skip. Existing tests needed explicit scope after adding Argo markers; field-picking tests now hide the marker layer before selecting the field. The deep-section test clicked exactly on the 1,000 m boundary, where subpixel rounding can fall outside the valid face. It now aims at 975 m inside that cell while still checking the independently sourced native 1,000 m value. A focused run passed eight of nine before that final targeting correction. No scientific values or picking tolerance were changed. Reports: `p04-local-browser-initial.json` and `p04-picking-recheck.json`.

The initial public HTTP check exposed a missing static route for bundled license files; `/licenses/` is now mounted. The initial public browser run passed 94 with four failures and one skip, but was contaminated by a concurrent local run deleting shared Playwright trace artifacts and a test edit targeting the not-yet-deployed isolated-dot feature. Those failures are retained in `p04-public-browser-initial.json`. The final suite ran alone against the final deployment.

Two initial Vercel builds failed during dependency configuration: universal resolution hit the Windows ARM64 NumPy constraint, then `uv` rejected a project-only setting in `uv.toml`. The successful build uses `pyproject.toml` with identical runtime pins and explicit supported server targets. See D023 and `evidence/p04-release-check.json`.

21 Phase 5: model comparisons and observation coverage

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt |
Source: docs/PHASE_05_EVIDENCE.md

Build date: 22 September 2026. Completed release: 0.5.0. Acceptance is recorded below.
The January 2024 model dates are historical source dates.

Implemented scope

The Compare workspace connects the checked observation library to the existing Bay of Bengal model. It provides observed/model profiles, model-minus-observation residuals, a shared depth cursor, native model-column navigation, exclusion explanations, actual observation coverage, eligible-record discovery and a temporary sensitivity reference. Comparisons support in-situ temperature and practical salinity. Imports remain inspectable in Instruments and are not included in quantitative comparisons.

Only this bounded model case is available. Glider, CTD and BGC examples keep their actual regions and dates. Their presence in the observation library does not extend model coverage.

Reproducible method

The versioned method is `p05-native-column-v1.science/evidence_contracts.py` defines settings, comparison rows and coverage records. `science/evidence.py` is the pure numerical matcher. `api/evidence_store.py` binds it to immutable, hash-checked arrays and library profiles with bounded process caches.

1. Choose one actual model timestamp. Each source sample retains its own timestamp. Signed offset is observation minus model time; the absolute value must be within the selected tolerance. The tolerance applies on either side of the selected snapshot, including collection endpoints. No model value is inferred at the observation time.
2. Require compatible physical definitions and units. Both temperature fields are in-situ Celsius, and salinity is practical salinity. Do not substitute potential or conservative temperature, or absolute salinity. CTD temperature needs a confirmed ITS-90 scale before quantitative use. The current cruise source does not supply that confirmation.

3. Preserve source variable and coordinate QC. Good-or-probably-good uses the existing P04 policy; good-only narrows Argo and declared flags to 1. QARTOD 1 and WOCE 2 retain their own meanings. A user setting cannot rescue a reading rejected by the source adapter.
4. Require a valid physical depth and coordinates inside the native rectangular subset. Search all native horizontal grid points by great-circle distance with Earth radius 6371.0088 km. Resolve an exact tie by the first native index. Require separation within the selected distance limit. Do not search for a replacement wet column.
5. Read the full-resolution analytical array, not the display grid. Interpolate linearly at the actual observation depth between the immediately adjacent finite native levels. An exact native level needs that cell only. Reject depth extrapolation, a masked endpoint or a bracket wider than the configured maximum. The existing GSW pressure-to-depth calculation remains unchanged.
6. Calculate residual as model minus observed. Bias is mean residual, RMSE is the square root of mean squared residual, and MAE is mean absolute residual. N counts accepted original samples, with equal sample weights. Source order is retained, including nonmonotonic casts and gaps.

Defaults are a 6-hour window on either side, 5 km separation, a 500 m maximum native bracket and good-or-probably-good QC. They are project choices, not universal scientific thresholds. Adjustable bounds are 0-72 hours, 0-50 km and 1-1,000 m. Each rejected sample receives one first-failed reason. The gate order in the matcher is authoritative, so reason counts add to the excluded count without double counting.

Coverage uses the same comparison results. Depth strokes are actual accepted sample depths. Time and distance ranges describe accepted samples when pairs exist; empty profiles retain source time offsets only to explain the mismatch. Dashed model bounds are separate context. There is no spatial confidence field, interpolation of observation coverage or inference about unsampled water.

Discovery ranks currently eligible profiles by absolute time separation followed by distance, or by their largest absolute residual. When the selected snapshot has no eligible pairs, it can suggest another bundled snapshot. Applying that suggestion is an explicit action. Ranking by residual is a navigation aid and must not be presented as an unbiased assessment of the model.

Source review hold

The original `D5907083_012.nc` calibration comment contains verification-run wording. A fresh HTTPS download is byte-identical to the cached source, with SHA-256 `17f3e33d95003693c24b7fc5737b1dbe9369334b6ffce679964288b5f96483dd`. Its meaning has not been established with the provider. This file is held out of quantitative comparisons under `source_review`, while its readings and unchanged source QC remain inspectable. The decision is based on provenance text, not residual size. It is not a claim that the source is fabricated or a provider QC rejection.

P05 also preserves original scientific calibration comments, equations and coefficients in the instrument metadata. Regenerating the 13-profile observation library changes its derived files and index, but not the original source hashes or P02 model manifest. Model manifest SHA-256 remains

`9598b33dc9eef89cb909f2811e1590ca580a143c1dbcc25d97e51a84faf7d28d`.

Independent numerical references

`science/verify_p05_reference.py` reads original NetCDF files without importing the application matcher. It manually decodes source packing into float64, independently scans the native grid and evaluates scalar depth interpolation.

`tests/fixtures/p05-source-reference.json` freezes seven profiles, 720 source rows, four snapshots, 576 count scenarios and 12 labelled synthetic edge probes. Synthetic probes are test data only and never appear as observations in the application.

The first Argo 1902669 sample is 0.3976914255 m deep. At the January 7 noon model snapshot, its interpolated model temperature is 28.0013865286 degrees Celsius, its observation is 28.1529998779 and its residual is -0.1516133493. Practical salinity is 33.7670006539 versus 32.8320007324, giving +0.9349999215. These values were also checked against a bounded live HYCOM ASCII response.

The default January 7 noon settings yield 206 pairs across Argo 1902669 and 4903775 after the provenance hold. A 2-hour window yields zero. A 100 m maximum native bracket yields 124 pairs; 25 m yields 50. January 8 noon yields 103 pairs and January 9 noon yields 307. These are acceptance references for this case, not a coverage or performance claim for the wider ocean.

See [P05_SCIENCE_REVIEW.md](#), source audit and the frozen fixture for original URLs, packing details and hashes.

API and browser behavior

- `GET /api/cases/{case_id}/evidence/profiles/{profile_id}` returns every original source row, its eligibility, original/paired values, source QC, time and distance, bracket/weight, residual, metrics, methods and caveats.
- `GET /api/cases/{case_id}/evidence/coverage` returns counts, actual accepted depths, accepted time/distance ranges, exclusions and optional snapshot suggestions for all checked profiles.
- Both use the same bounded variable, `time_index`, `time_window_hours`, `distance_km`, `max_vertical_gap_m` and `qc` settings.
- Responses include method, model-manifest and observation-library identities. The library identity hashes a canonical sorted compact JSON serialization of the index. It is not the byte hash of the pretty-printed index file.
- The browser validates contracts, counts, residual arithmetic, summary metrics and requested settings before showing results. Cache keys include release, case, profile and all settings. A pending selection does not relabel an earlier result as the new one.
- The model and comparison share one timestamp control state. Instruments' older nearest-depth action remains navigation only. The matched-column action explains that the 3D probe uses the nearest native depth while the numerical comparison uses physical-depth interpolation.
- The sensitivity reference stays in browser memory, including its original profile, variable, snapshot and filters. It is cleared on reload. Durable replay/export is P08 and is not claimed here.

Acceptance and deployment

Completed P05.1-P05.5 and feature rows R05, X04 and X05. X06 remains partial because its durable replay requirement belongs to P08.

- Build/typecheck passed. The frozen reference regenerated byte-identically with `--check`; its SHA-256 is 4773abc59e39596bc2d4cff951a40a805d6d3463f7dc173992b20ea1a7d574c9.
- `.venv/Scripts/python.exe -m pytest tests -q --junitxml=docs/evidence/p05-science-api-final.xml`: 137 passed, including 85 matching checks and a calibration-text fidelity regression. Two existing Starlette/AnyIO deprecations remain.
- `npm --prefix web run test:science`: 17 renderer numerical checks passed.

- Full local browser run: 116 passed and one deliberate WebKit context-loss skip in `evidence/p05-local-browser-final.json`.
- Full public browser run: 113 passed, three failed, one deliberate skip in `evidence/p05-public-browser-final.json`. All 18 new comparison checks passed. The failures were one native-value loading timeout, one page-load timeout and a test locator matching two alert panels. The service-update assertion was scoped to its own alert; the two loading cases were unchanged. All three passed in `evidence/p05-public-browser-recheck.json`. Original failure traces/screenshots are retained in `evidence/p05-public-initial-failures/`. No claim is made that intermittent network delays are eliminated.
- The initial new-feature browser run had 15 passes and three failures from an outdated expected message after the explicit snapshot action was added. The test wording was corrected; the full local and public comparison checks passed. Keep `evidence/p05-browser-new-initial.json`.
- Public HTTP audit: 66 checks across 70 GET requests passed in `evidence/p05-public-http-release.json`. It verifies every library profile in both variables, 12 coverage scenarios, independently frozen values/metrics, source identities, exclusions, release/API errors and assets. No audit request failed and the script does not automatically retry.
- Desktop Chromium, Windows WebKit, 390/320-pixel widths and 200% root text passed the final comparison/coverage/rules layout audit: `evidence/p05-public-final-visual-review.json`. There is no page overflow or captured page error. A Go to results link is checked, and a footnote-width assertion prevents an inherited sidebar width from returning. The earlier public screenshots exposed that width defect at enlarged text; it was corrected by giving the comparison footnote its own CSS class. These checks emulate device sizes and do not certify physical phones or Safari on Apple hardware.

Final Vercel deployment: `dp1_DDofskrZKtLVQkJmBVupBpnUybZZ`, READY. Public URL: <https://depth-atlas-seifuku.vercel.app>. Unique URL: <https://depth-atlas-seifuku.vercel.app>. The first 0.5.0 deployment was `dp1_4wGDN2zpQtTV8KJH5b4B5XJRTv16`; the second changes only the conflicting footnote CSS class and associated build assets. Scientific code and case data are unchanged between these deployments.

The first public HTTP run passed 63 checks and failed three overly strict Windows/Linux static-byte comparisons. Investigation found generated asset-name references, HTML newline differences and exactly one 21-byte CSS rule, `.table{display:table}`, emitted locally from a test-file token excluded from deployment. JavaScript matches after normalizing generated filenames. Final checks allow only these demonstrated differences

while keeping all other code, CSS and scientific data comparisons strict. Preserve `evidence/p05-public-http-initial.json`, `evidence/p05-public-asset-differences-final.json` and the earlier follow-up audit.

The final workspace bundle is about 514.74 kB minified, 166.01 kB gzip; the renderer is about 581.03 kB minified, 145.65 kB gzip. Both trigger Vite's 500 kB advisory. No performance guarantee is inferred. Broader loading/device improvements remain measurable work. Firefox cannot launch in the current Windows setup, Docker is unavailable and actual INCOIS installation remains unverified.

The roadmap audit preserves all 17 phases, 82 steps and 45 feature rows; authored P05 UI/method text contains no em dashes. See `evidence/p05-continuity-audit.json`.

Limits and next phase

The HYCOM analysis assimilates observations including Argo. Agreement is not independent validation, forecast skill or operational improvement. Profiles contain correlated samples, unequal depth spacing and no area weighting. This implementation does not interpolate in time or horizontally, compare currents/BGC variables, accept arbitrary uploads into matching, provide live feeds or establish observing-system impact.

P06 adds standards services and an extension contract. WMS/WCS and OPeNDAP require actual tested protocol responses; a REST endpoint with similar naming is not sufficient. The Vercel app remains the public host, while service placement and available infrastructure must be verified in that phase.

22 P06: standards and extension contract

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: `docs/PHASE_06_STANDARDS.md`

Implementation date: 22 September 2026. Historical case dates remain 7-10 January 2024.

What this phase provides

The app adds horizontal kinetic energy as a registered derived scalar, using the existing volume, slice, section, isosurface and native-point inspection paths. The formula is $(u^2 + v^2) / 2$, in m^2/s^2 . It is energy per unit mass from horizontal velocity, not total three-component or eddy kinetic energy. Both input masks are retained. Calculation precedes display decimation and float32 rounding.

The native exchange NetCDF contains all seven snapshots, forty irregular depths, 76 latitudes, 63 longitudes and five fields. It is generated from checked analytical arrays, not the display mesh. No interpolation or gap filling is applied. Its SHA-256 and original case identity are in `casepacks/standards/manifest.json`. The original model manifest remains unchanged.

The new Use the data page links the download, source identity, REST example, OpenAPI reference and service contract. It states public availability explicitly. Vercel serves the 27.6 MB file as a CDN asset, avoiding the serverless response size limit. Normal browsing still uses bounded, cached scientific requests.

Standards service and independent checks

The separate service is THREDDS 5.9, Apache Tomcat 10.1.60 and Eclipse Temurin Java 17.0.20.1. Download URLs and vendor checksums are locked in `deploy/standards-runtime-lock.json`. It binds to loopback port 8096. No system-wide Java change or public tunnel was made.

Actual requests were made with OWSLib 0.35.0 and pydap 3.5.9. The independent verifier decodes the original packed NetCDF files manually in float64, without importing the app's adapters, store or product calculator. It compares every field in every snapshot with xarray's reading of the exchange file, including missing masks.

The 143-check protocol/source run includes WMS 1.3.0 capabilities, PNG maps in CRS:84 and EPSG:4326, four geolocated GetFeatureInfo readings, WCS 1.0.0 capabilities and ten NetCDF3 coverages, and five strided DAP2 subsets. WCS requests check time, depth, coordinates, values and masks at 100 m and 5000 m. Unsupported WMS layers produce service exceptions. An oversized 1025-pixel map request was also rejected by the configured 1024-pixel limit.

An initial run exposed two issues in the verification assumptions. THREDDS regularizes horizontal axes for WCS; maximum tested longitude change was 0.0000826928 degree, approximately nine metres. A declared 0.0001-degree tolerance is used for WMS/WCS geolocation, with a separate 1e-12 absolute tolerance for values. Source arrays and values are not modified to accommodate this behavior. Use the download or OPeNDAP when

exact source coordinates matter. The pydap slice object drops attributes, so the checker must read `_FillValue` from its parent variable before masking the response. Both initial and corrected reports are retained.

The service was started again in a fresh runtime base with the packaged launcher and the final configuration. The protocol checks passed against that fresh instance. Docker remains unavailable on this machine, so the optional Compose image is prepared but unexecuted. Actual INCOIS installation, public THREDDS hosting, concurrency capacity and universal OGC certification have not been established.

CF interpretation

The exchange declares CF-1.10. Its scoped checker verifies 1D increasing numeric coordinates, explicit axis/standard-name/unit metadata, unfilled/unpacked coordinate variables, positive-down metre depths, Gregorian time round trips, four declared source-variable definitions, derived-field units, float64 fields with typed fill values, finite unmasked samples and source identity. Adversarial tests alter units, calendar, depth direction, fill metadata, variable definition, axis presence and coordinate order.

The original HYCOM practical-salinity field uses `sea_water_salinity` and `psu`. The exchange expresses the project's documented practical-salinity interpretation as `sea_water_practical_salinity`, unit 1, with unchanged numbers. Original standard name, units and variable name remain attached. This is a declared metadata normalization, not a numerical conversion to absolute salinity. The horizontal-only energy field has a descriptive `long_name` and units without an invented CF standard name.

Supported cases have independent rectilinear time/depth/latitude/longitude axes and declared in-situ temperature, practical salinity and earth-relative horizontal velocities. Curvilinear or staggered grids, rotated vectors, sigma or pressure vertical axes, arbitrary unit conversion and non-Gregorian calendars need distinct adapters. These checks do not certify every CF construct.

Extension contract

`adapters/registry.py` registers trusted, versioned Python entry points. The original HYCOM adapter and the new CF exchange adapter both return the same `ModelGrid/native-float64/v1` contract. The exchange adapter accepts latitude/longitude axis names and explicit dimensionless practical salinity, preserving values and physical coordinates. The existing preparation path now selects its HYCOM reader through the registry.

science/products.py registers metadata, dependency names and a native-array calculation. Duplicate registrations, missing dependencies, mismatched grids, changed output shapes and non-finite valid results are rejected. Output cannot fill an input's missing cells. Registrations are maintained code; uploading a file cannot install or execute a plugin.

The current public product allowlist, API schema and frontend presentation metadata still require explicit maintainer updates when adding another quantity; the scientific renderer does not.

Evidence

- evidence/p06-protocol-initial.json: retained initial protocol assumptions and failures.
- evidence/p06-protocol-verification.json: final 143 checks plus scoped CF results.
- evidence/p06-protocol/: actual capabilities, PNG maps and downloaded WCS coverages.
- evidence/p06-science-api.xml: 176 passing science/API tests, including 39 new P06 cases; two existing dependency deprecation warnings.
- Renderer numerical suite: 17 passing checks.
- Local browser full run: 122 passed, three new-test failures and one deliberate WebKit context-loss skip. Initial failures included a closed-details selector, a genuine frontend superscript-unit encoding defect, and a selector for a control hidden in Compare. After the unit correction and test fixes, all nine P06 checks passed. Initial, intermediate and final reports are retained.
- Public browser full run: 117 passed, eight failed, one deliberate skip. Failures involved incomplete loading, request recovery deadlines, one page-receive error and an import interaction timeout. All eight passed a focused recheck with the same code and three workers in 20.7 seconds. The precise transport/runtime cause is not established; these results do not promise uninterrupted service. All nine P06 checks passed in the initial public run.
- evidence/p06-public-http.json: 49 checks over 50 anonymous requests passed, including the complete 27,554,963-byte download and 27 independent derived-value/missing-data fixtures.
- evidence/p06-public-visual-review.json: fifteen views across desktop Chromium, Windows WebKit, 390 px, 320 px and 768 px with 200% text. No page overflow or page errors. Screenshots were retained and representative desktop, mobile and large-text views were inspected.

- `evidence/p06-public-assets.json`: public JavaScript matches after generated asset-name normalization. CSS differs only by the previously identified unused `.table{display:table}` rule emitted locally from test files. Both raw hashes are retained.
- `evidence/p06-deterministic-exchange.json`: a fresh export is byte-identical. `pip check` reports no dependency conflicts.
- `evidence/p06-release.json`: Vercel release 0.6.0, deployment `dp1_GnUzron9LPS3jh3TCVftUSjDjUTt`. The first publish attempt reported Not authorized; read-only identity/project inspection succeeded and an unchanged retry published successfully. No credentials, permissions or account plan were changed.

P06.1-P06.5 are complete for this documented scope. Public persistent THREDDS hosting, Docker execution, physical-device/Firefox coverage and actual INCOIS installation remain open. R09, R10 and R12 remain unchecked; R02, R08, R11 and X07 now have their required bounded evidence. P07 has not started.

Checkpoint B: official requirements

Requirement	Supported result	Remaining boundary
R01	Tested volume, slices, isosurfaces, native depth and time views	Device/GPU breadth and final release audit
R02	Temperature, salinity, current vectors and working derived scalar extension	New variables still require explicit definitions and registration
R03	Checked depth/time controls and deliberate display transitions	No temporal interpolation or live stream claim
R04	Genuine Argo, glider, CTD and BGC records, geographic markers	Only declared source collections; no global model coverage inferred
R05	Source/QC profile charts and eligible model comparison	One Argo provenance hold, CTD temperature scale unresolved
R06	Bounded NetCDF and documented delimited imports	Arbitrary NetCDF/Excel unsupported
R07	Palette, limits, log/linear, opacity and vertical exaggeration	Masks and analytical values are preserved
R08	Modern browser frontend, REST and tested DAP2 service path	Public Vercel REST; THREDDS public host not configured
R09	Browser-only client, portable app and tested local service package	Container execution and institutional deployment remain unverified

Requirement	Supported result	Remaining boundary
R11	Scoped CF interpretation and actual independently checked WMS/WCS	Listed coordinate tolerance; no universal certification
R12	Basic orientation and source explanations	Guided cases, outreach and user feedback remain P09/P16/P17

Primary references

- [Unidata installation requirements](#): Java 17 and Tomcat 10.1 for the current TDS.
- [Unidata service configuration](#) and [WCS reference](#): real service paths and WCS subset limits.
- [TDS release metadata](#) and [Tomcat downloads](#): actual runtime versions and checksums consulted for this implementation.
- [CF-1.10 conventions](#) and [CF standard-name table](#): coordinate, packing and practical-salinity interpretation.

23 Phase 7: connected structures and drawn sections

Historical or maintained reference: read alongside the current release overview | Source: docs/PHASE_07_FEATURES.md

Implementation date: 22 September 2026. Dataset dates: 7 to 10 January 2024.

Scope and scientific meaning

The Find structures workspace queries one actual model snapshot. It supports in-situ temperature, practical salinity, eastward velocity, northward velocity and derived horizontal kinetic energy. A query records its variable, exact units, timestamp index, inclusive threshold or closed range, and minimum/maximum depth. No natural-language interpretation or hidden threshold is involved.

Search uses every native float64 sample, never the decimated rendering grid. A sample qualifies only if it is finite, satisfies the numerical comparison, and its depth coordinate falls inside the closed requested interval. A qualifying sample must also have positive inferred cell thickness after clipping. Six-face neighbours connect through one native

depth, latitude or longitude index. Diagonal contact, missing samples and excluded samples do not connect.

The source supplies point coordinates without cell boundaries. Consequently the implementation cannot honestly claim a measured or provider-supplied water volume. It infers boundaries halfway between coordinates, truncates outer boundaries at the first/last sampled coordinates, and clips depth boundaries to the query interval. Each point value is treated as constant within that inferred bin. The displayed quantity is **estimated volume**. Irregular depth spacing and changing latitude area are retained. Seabed position between sampled depths remains unknown.

Volume integrates a spherical shell with radius 6,371,008.8 m. For positive-down depths a and b , the radial factor is $(b-a) * (3R^2 - 3R(a+b) + a^2 + ab + b^2) / 3$, multiplied by longitude width in radians and the sine difference of latitude boundaries. Division by 10^9 converts cubic metres to cubic kilometres. Both native centre extents and inferred cell extents are returned. Arithmetic sample mean and volume-weighted mean are separate fields.

The ID of a region uses its smallest global flat index in depth/latitude/longitude order. IDs are query-specific, not persistent identities of a water body. Ranking by estimated volume or eligible sample count does not change membership. The API returns the first 50 region summaries in that order and reports total regions/cells separately. The plan view shows the shallowest qualifying region per horizontal column, so deeper components can be selected through the list.

Geometry and sections

Each selected surface follows the exposed faces of inferred native cells. Coplanar rectangles merge without decimating or moving the boundary. The mesh is a visualization of the query's inferred bins. Depth exaggeration affects presentation only. The bounded fallback for more than 40,000 merged faces retains the footprint, numerical result and section instead of substituting a misleading simplified surface.

A drawn section is a straight segment in longitude/latitude. Its default 81 equally spaced stations include both endpoints. Each station selects the closest native column by great-circle distance, with deterministic first-index ties. Every selected native depth is retained. The section returns requested coordinates, sampled column coordinates, offset, cumulative station distance, source value and component membership. There is no horizontal or vertical interpolation, and missing values remain null. The station count is bounded to 2 through 201. Line geometry and sampling offsets remain explicit.

Observation context

Temperature and practical-salinity evidence reuse the tested P05 matcher and source holds. Defaults are a 6-hour absolute time window, 5-km native-column separation, maximum 500-m depth bracket and good/probably-good source quality. Eligible source samples are associated with the qualifying inferred depth bin in their matched native model column. Exact interior depth midpoints belong to the deeper bin; the final boundary is closed. Observation depth must also fall in the query interval. This is nearby model-column evidence, not proof that the observed value satisfies the threshold.

Section observations must pass the same P05 gates and lie within 5 km of a requested section station. This is a sampled proximity rule, not an exact continuous corridor. Changing station density can change those links. Velocity and kinetic energy have no comparable profiles in this library and do not receive invented observation matches. The provenance-held Argo source, incompatible definitions and out-of-time or out-of-domain examples stay excluded.

Interfaces and limits

- POST `/api/cases/{case_id}/features/search`: FeatureQuery.
- POST `/api/cases/{case_id}/features/region`: query plus query-specific region ID.
- POST `/api/cases/{case_id}/features/section`: query, two `[longitude, latitude]` endpoints and optional station count.
- Exact units, finite values, increasing query bounds, source timestamps, domain bounds and supported operators are validated.
- Native analysis is bounded to 250,000 cells and 5,000 components. The pinned case has 191,520 native cells.
- A small process cache holds immutable source-derived results. Ordering does not trigger fresh analysis. No query account, database or durable investigation is created.
- Numerical JSON exports contain query/result/source identity and section values. Durable saved investigations, fresh-session replay and share links remain P08.

Acceptance evidence

The local backend suite passed 281 tests, including 83 independent P07 tests and 22 request/geometry checks. The renderer suite passed 24 tests. The independent P07 verifier passed 99 local HTTP checks against original-source expectations. The existing standards verifier passed 143 checks again, and a separate audit matched all 56 native/display arrays across seven snapshots exactly, including missing masks.

The initial local 144-test browser run passed 140, failed three new pointer-test precision assertions and skipped one unavailable WebKit context-loss check. The pointer tests now use the browser's delivered event coordinates, rather than impossible subpixel placement assumptions. All 27 feature/scientific UI tests passed after correction. They passed again on the final 0.7.1 accessibility build. The standalone final visual audit passed 35 checks across Chromium, Windows WebKit, 390 px, 320 px and 768 px at 200% text. Actual screenshot inspection, rather than counters alone, identified and verified the small-screen fixes.

Release 0.7.0 was deployed and its full public suite passed 148 checks, with four loading timeouts and one deliberate skip. All four timeout checks passed unchanged on focused recheck. Traces show requests canceled around the existing eight-second client deadline; they do not establish the underlying network/server cause. Reports and traces are retained.

Release 0.7.1 adds bounded scrolling charts and readable enlarged-text controls. Its full public browser run passed 144 checks, failed eight during loading and skipped the same unavailable WebKit context-loss injection. All eight passed unchanged on focused recheck in 23.4 seconds. Independent review of every failed trace found 25 incomplete or failed transfers and no HTTP error response. Three tests stalled on the main static script before application code ran; API reads also hit the existing client deadline, and the WebKit feature POST reported a browser transport error. No returned numerical mismatch or application exception was demonstrated. The transport cause remains unresolved, so these passing rechecks do not establish uninterrupted availability.

Final public acceptance is complete for the declared scope. All 35 public visual checks passed across five configurations, with zero page errors and actual screenshot review. After that review, all 99 independent P07 HTTP checks passed against release 0.7.1, including original-source query memberships, sections, meshes, observation associations and errors. The existing HTTP verifier also passed all 49 checks over 50 requests, including the full native NetCDF checksum, source-derived kinetic energy, legal routes and static assets. Vercel inspection confirms the normal public alias points to production READY deployment `dp1_dfApBFR6zxN9wtK3Uo4ZXd74J3rR`.

Key records under evidence/: p07-science-api.xml, p07-independent-science.xml, p07-local-source-api.json, p07-protocol-verification.json, p07-audit-existing-source-arrays.json, p07-local-browser.json, p07-local-accessibility-browser.json, p07-local-071-features-visual.json, p07-public-browser.json, p07-public-browser-recheck.json, p07-public-timeouts.json, p07-public-071-browser.json, p07-public-071-browser-recheck.json, p07-audit-public-071-failures.json, p07-public-071-features-visual.json, p07-public-source-api.json, p07-public-existing-http.json and p07-release-check.json.

Defects found and corrected

- Basic current directions now account for plot aspect ratio and longitude convergence. Source speed and values are unchanged.
- Equal longitude widths are subtracted before conversion to radians, avoiding spurious unequal-volume ordering.
- Observation depth bins use unclipped midpoint edges for lookup, preserving exact inclusive upper-query boundaries before the clipped membership check.
- SVG picking uses the actual screen transform, including letterboxing and horizontal scrolling. Outer map boundaries no longer wrap to the first cell.
- Missing sections have no invented numeric colour scale. Constant sections identify their constant value. Exact depth-midpoint clicks follow the declared deeper-bin rule.
- Query thresholds retain their exact entered values; small positive quantities do not display as zero. Narrow depth intervals retain distinct tick labels, and returned section endpoints remain separate from draft inputs.
- The initial region camera exposes the side boundary and labels both the inferred depth extent and the selected exaggeration.
- The 320-pixel toolbar wraps, WebKit dropdown text is no longer clipped, and the full exaggeration value remains visible at 200% text. On narrow screens, charts retain readable axis labels inside keyboard-accessible horizontal scroll areas; the page itself does not overflow.

No current result is promoted to an eddy, a material trajectory, confidence score or validated forecast. Unknown-device performance, actual unfamiliar-user comprehension, Firefox and institutional deployment remain explicitly unverified.

Independent review: [P07_SCIENCE_REVIEW.md](#). Existing-component audit: [P07_COMPONENT_AUDIT.md](#).

24 Phase 8: saved investigations and numerical replay

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_08_INVESTIGATIONS.md

Completed: 22 September 2026. Deployed release: 0.8.1. P08.1-P08.5 and X06/X10/X11 are complete within the supported case and investigation modes.

What an investigation contains

An investigation saves one applied analysis from the ocean viewer, a checked observation profile, a model comparison or a structure query. It contains its recipe, case/region identity, source fingerprints, scientific method versions, software versions, calculated modules, source references and limitations. Comparison records retain an optional reference comparison with its own time, variable, profile and matching policy. Feature records retain the selected region, drawn section, selected native section sample and display choices.

Ocean records retain the selected native column and depth, variable/time, view, colour settings, depth window, cutaway, quality choice and exaggeration. Replay restores those settings; camera orbit and running animations reset. A saved investigation is one analysis, not a snapshot of every inactive tool. Unsubmitted form drafts and temporary observation imports are excluded. The interface disables capture while relevant requests or unapplied query changes are pending.

All numerical work still uses the existing P02/P04/P05/P07 methods. P08 does not add interpolation, change source arrays, relax quality gates or certify model skill. Native current profiles retain both horizontal components and their joint-mask speed. Missing values remain null in JSON and empty numeric CSV cells. Model and observation timestamps remain separate.

Storage and sharing

The public app remains on Vercel. There is no account, server database or saved file in the function filesystem. Saving writes an explicit record to IndexedDB in the current browser, limited to 25 records and 32 MB, with an 8 MB individual-file limit. Save failure leaves the newly calculated investigation available for download and does not claim a successful local save. Local copies can be removed independently of downloaded files and shared links.

Browser storage can be cleared or evicted, so the UI recommends a portable JSON backup. This follows the documented [IndexedDB storage model](#) and [browser storage/eviction rules](#). Vercel recommends external persistent storage for function writes; this implementation deliberately requires no paid storage service. See [Vercel file guidance](#).

A replay link stores a bounded recipe and expected source/result identities in a URL fragment, encoded as UTF-8 base64url, up to 16,000 encoded characters. It is self-contained settings, not a short server record ID. Opening it prepares the recipe. Recalculate and reopen sends it to the API for verification. Anyone with the link can read its title and settings; the UI and policies say so. Sharing does not transmit imported files. Long-term link replay still depends on the required sources and method implementations remaining available.

Replay and fingerprints

`POST /api/investigations/capture` recalculates a typed, bounded recipe against the current checked sources. Capture can require the fingerprint of the source visible when the user clicked Save. `POST /api/investigations/replay` checks the recipe fingerprint, method identity and source identity before calculating again. It compares the complete module results with the stored expected result fingerprint. A changed source, unsupported method, missing file or numerical mismatch stops reopening; it does not substitute another dataset or present a saved result as a successful recalculation.

The fingerprint format is `tagged-json-f64-sha256-v1`. Values carry explicit null, boolean, number, string, array and object tags. Numbers use their binary64 bytes in big-endian hexadecimal, with both zero signs normalized. Object keys sort by Unicode code point; arrays retain order. Tagged JSON uses ASCII escapes and compact separators before SHA-256. This avoids treating Python's `1.0` and JavaScript's `1` as different numerical values after a JSON round trip. Nonfinite values, unsafe integers and unsupported types are rejected. Browser file verification also checks the complete document and every module fingerprint.

Hashes establish consistency, not author identity or scientific truth. No digital signature or private signing key is claimed. Scientific result fingerprints exclude capture dates and the current app patch version. Software versions remain in the document, while explicit source/method identities decide compatibility. A UI-only patch can replay identical calculations; changed scientific methods require an explicit version and cannot silently migrate old records.

Each result module has `module`, `method_version`, `parameters`, `random_seed`, `output` and `output_sha256`. Current deterministic methods use a null seed. Later labs can add versioned typed recipes and modules using this envelope; unsupported modules are not accepted or executed in this release.

Exports

Investigation JSON preserves the complete portable record. A readable, script-free HTML report explains the analysis and displays summaries, methods, sources and limitations. Its detailed recipe and fingerprints are expandable. Browser printing can produce a PDF; no generated PDF layout is claimed.

Numerical CSV exports include original values, units, source times, positions and missingness. Comparison CSV contains eligible and excluded source rows, depth-bracket values/weights, residuals, QC and exclusion reasons. A reference comparison has its own CSV. Section CSV records requested stations, actual native columns, offsets, depths, values and region membership. Region CSV labels volume as estimated and records full membership fingerprints.

The evidence ZIP contains `investigation.json`, `settings.json`, `report.html`, the applicable CSV files, `SOURCES.json`, source credits and a README. Exports are recalculated and checked before download. Only supported public-source subsets and outputs are included, with source terms, original fingerprints and native-download instructions. Uploaded/private observations and complete global datasets are not bundled. The report escapes text; CSV protects formula-like text without altering negative numerical values.

Verification and fixes

The first P08 backend run passed 24 tests and exposed an incorrect fixture key in the new test. The corrected run passed all 25, including two separate fresh Python processes. The full suite then passed 306 tests and the existing renderer suite passed 24. Initial reports remain in `evidence/p08-initial-api.xml` and the subsequent records.

The first eight browser checks passed six and failed two text-comparison assertions: rendered `innerText` was compared with raw DOM text. Restored numeric selections were

correct. The assertions now consistently use rendered text; initial traces are retained. The broader browser run also found a real WebKit focus-return defect in the new dialog. A fallback to the persistent Saved button addresses mouse clicks that do not focus the triggering button; all 24 P08 browser checks passed on the final build across Chromium desktop, WebKit and Chromium mobile emulation.

All 30 local visual workflow checks passed at desktop, WebKit, 390 px, 320 px and 200% root text. Saved records, capture forms and the exported report were also inspected as actual screenshots. The enlarged-text run recorded 28 px dialog text, double its normal 14 px. Four complete investigation records replayed in a second fresh Python process. All 19 local HTTP capture/replay/export and mismatch checks passed.

The 0.8.0 public browser suite passed 172 checks initially, failed four during resource loading and skipped the intentionally unsupported WebKit context-loss injection. Every new P08 browser test passed in that full run. All four failures passed unchanged rechecks. Complete traces are retained in `evidence/p08-public-browser-failures/`; `p08-public-failure-review.json` records incomplete field/manifest transfers and a navigation timeout. The exact network/browser/CDN/backend origin is not established. An earlier `npm.ps1` launch consumed the project arguments and was interrupted; the corrected direct Playwright CLI ran the intended 177 checks. No test threshold was relaxed to obtain a pass.

All 30 separate public visual checks passed for 0.8.0. The subsequent 19-check archive audit found a real deployment defect: all four ZIPs omitted `source-credits.txt` because the backend bundle did not include the public attribution file. Numerical captures, replay and non-ZIP exports passed. Release 0.8.1 explicitly includes that file in the Vercel function and refuses to create a ZIP if required credits are missing. The regression test covers that failure path. All 307 backend tests and all 26 focused P08 backend tests pass on this revision.

All 19 public HTTP checks now pass against 0.8.1, using the earlier 0.8.0 records. Complete recalculated modules match the separately captured local records, and all four ZIPs include the required attribution, settings, report and numerical files. This also verifies that a packaging patch can replay unchanged scientific results without silently changing the method. Initial failing archive results remain in `evidence/p08-public-initial-replay.json`; final results are in `evidence/p08-public-replay.json`.

All 27 final public P08 browser checks pass on 0.8.1 across Chromium desktop, Windows WebKit and Chromium mobile emulation. The added deletion check confirms that removing a local copy persists after reload while its downloaded backup remains usable. All 49 existing public HTTP/source checks pass over 50 requests, including the complete native NetCDF checksum. Reports are `evidence/p08-public-081-browser.json` and

evidence/p08-public-existing-http.json. The full 177-check regression and 30-check screenshot audit were run on 0.8.0; 0.8.1 changes attribution packaging, explicit ZIP failure and release versions, with focused verification of those paths. No scientific method or source array changed.

Existing Firefox/physical-device/user-study, Docker, public THREDDS and intermittent public-network limitations remain separate from this phase's verification. Hash matching establishes reproducibility against these source/method versions, not independent validation of the ocean model.

25 Phase 9: guided Indian Ocean cases

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_09_GUIDED_CASES.md

Implementation date: 22 September 2026. Historical case dates: 7 to 10 January 2024.

What changed

Bay of Bengal and eastern Arabian Sea now use the same acquisition, preparation, visualization, comparison, structure search and investigation replay tools. The Arabian Sea pack has seven snapshots, 40 native depths, 76 latitudes and 63 longitudes. Its actual bounds are 66 to 70.9599609375 degrees east and 16 to 19 degrees north. It contains temperature, practical salinity and horizontal velocity components. Missing values are retained. Neither rectangle represents complete basin coverage.

The acquisition uses the original HYCOM GLBy0.08/expt_93.0 archive. Requests run serially, with a source hash and resumable checkpoint for each bounded snapshot. The existing model and Argo parsers required no scientific correction. Pipeline configuration and storage were generalized to accept an explicit case ID. Case IDs remain allowlisted; user strings cannot become file paths.

Source references: [HYCOM GOFS 3.1 analysis](#), [archive catalogue](#), [known archive gaps](#), [Argo attribution](#).

Five original INCOIS DAC files produce six profiles. R2902273_172 contains two profile records, so file count and profile count differ. The new observation index is separate from the original Bay index. Bay continues to expose its original 13-profile library. Arabian Sea exposes that library plus six new profiles, including out-of-domain examples with explicit comparison exclusions. This preserves the source identities of existing saved

investigations.

Calibration review found delayed-mode adjusted values for 1902671, 1902672 and 5907084, real-time raw values for 7901128, and adjusted real-time parameter modes for the two 2902273 records. Pressure-hysteresis comments are retained in the latter source families. The guide selects delayed-mode 1902671 for its Arabian Sea comparison. Source modes, QC, original values, adjusted values, error fields and calibration text remain inspectable. The original Bay 5907083 provenance hold remains in force. This review does not independently certify sensor calibration.

The teaching example

At 7 January 2024, 12:00 UTC, the selected Bay column is at 12.359999656677246 N, 87.760009765625 E. The Arabian Sea column is at 16 N, 70.56005859375 E. Their native surface temperatures differ by 0.007000000332482159 degrees Celsius. At 100 m they differ by 4.239000201341696 degrees Celsius.

Selection is explicit: consider every cross-case column pair at that snapshot with finite temperatures at 0 and 100 m and a surface difference no greater than 0.05 degrees Celsius. Select the greatest absolute difference at 100 m. Break ties by the first flattened Bay index, then Arabian index. This example is selected to illustrate contrast. It is not a representative estimate of either basin and does not establish a causal explanation, density contrast or forecast.

The guide displays a common-axis depth plot, a selectable native-depth value table, exact dates, coordinates and the selection rule. The lines connect native samples for readability. Missing values break lines. API values are checked against case fingerprints and native columns before being returned. The source verification script independently repeats the selection with a separate vectorized algorithm using packed source NetCDF files.

Guided journey

1. Select a study case and choose Start guided investigation.
2. Inspect the two-column example. Open this column selects the corresponding native point and opens the cutaway. Basic graphics remain available.
3. Compare a measurement loads a separate real Argo profile near a model column, at the declared snapshot. It does not imply that this float measured the teaching-example point. The initial Bay comparison has 103 eligible pairs; Arabian Sea has 101. These are sample counts, not accuracy scores.

4. Find warm water applies temperature at least 26 degrees Celsius over 0 to 300 m. It explains that connected cells are threshold regions, not automatically eddies or water trajectories.
5. Save the evidence points to the existing save/replay/export workflow. No step is marked scientifically successful merely because its button was pressed. Previous step, restart and exit allow navigation without a forced onboarding tour.

The study-case selector remains visible in comparison and structure views. Changing cases resets case-specific selections and result panels. Advanced colour and matching controls remain expandable. The new salinity colour default uses outward-rounded minima and maxima from all seven Arabian Sea snapshots. The original 30 to 36 range would saturate values above 36; the new case range includes those values without changing the science. Bay's original manifest and stored replay settings remain unchanged.

Pending calculations can be cancelled. Cancellation stops the client's request and keeps any previous result labelled with its original query. It does not promise to interrupt an already running server thread. Interrupted response-body downloads now remain retryable service errors. Malformed JSON remains a verification error. This fixes the classification, not every possible network or CDN failure.

Standards and deployment limits

Both cases have native CF-1.10 NetCDF downloads and REST access. The separately tested THREDDS configuration remains scoped to Bay of Bengal. No persistent public THREDDS address is configured, and Arabian Sea WMS/WCS/DAP2 service availability is not claimed. The new access page and API distinguish these scopes.

The model assimilates observations, including Argo. Agreement is not independent forecast validation. Derived structures use the existing P07 numerical method and midpoint-bin volume estimates. The profile pair and guided structure query are educational analyses of historical data, not operational advisories.

Verification status

- Independent source decoding: all 10,725,120 native scalar values across both cases, seven snapshots and four variables match their packed source files exactly. Display subsets match the declared float32 decimation, including masks. Both exchange files match their checked native temperature snapshot. The selected teaching pair passes an independent exhaustive selection check. Evidence: `evidence/p09-source-verification.json`.

- Original Argo audit: all six added profiles and 2,398 levels checked directly against the five original NetCDF files, including raw/adjusted/error values, flags, modes, source coordinates and depth conversion. Source JULD is preserved exactly; formatted timestamps are checked within their declared second-level presentation. This uses the declared GSW routine and is not an independent validation of GSW physics. Evidence: `evidence/p09-observation-source-check.json`.
- Backend regression: 317 passed. Includes two-case isolation, source checks, rejected mutations, capture/replay/export in all four Arabian modes and replay of original P08 records. Evidence: `evidence/p09-backend.xml`.
- Initial backend failures were a test double missing the new case argument and an incomplete new ocean-recipe test fixture. Both fixtures were corrected without weakening numerical assertions. Initial evidence is retained.
- The initial 198-case local browser run passed 190, failed seven and skipped one unavailable WebKit context-loss injection. Six failures were obsolete single-case copy assertions. One exposed WebKit response-parser classification, fixed by reading the body separately from local JSON parsing. The first guide run exposed an exact-label problem; the corrected guide suite passed all 18 cases. Corrected regression and deployment checks are recorded separately.
- Corrected local regression: all 78 browser cases passed across Chromium desktop, WebKit desktop and mobile Chromium emulation. New guide controls also pass narrow 320 px, enlarged text, reduced motion and keyboard-return checks. Evidence: `evidence/p09-local-corrected-regression.json`.
- Renderer/transport checks: all 26 passed, including numerical geometry and interrupted-body versus invalid-JSON recovery. Evidence: `evidence/p09-renderer-transport-final.txt`.
- Visual audit: 23 captured views across both cases, all five ocean modes, kinetic energy, comparison, search and 800/390/320 px guide widths, without detected horizontal page overflow, visible alerts or uncaught errors. Reviewed representative cutaway, guide, narrow-screen and structure screenshots. Mobile chart labels were enlarged after review. Evidence: `evidence/p09-visual-local-final/`. New explicit guide text/surface colour pairs exceed 4.5:1 contrast; this is not a full accessibility certification.
- Real unfamiliar-user pilot: pending. No participants or task outcomes have been invented. See `P09_USER_PILOT.md`.

Checkpoint C: official requirement review

The archived official statement is `reference/official-2026/project brief-statement.txt`. This is a review of that statement, not a claim that submission rules were reissued or rechecked today.

Requirement	Current implementation and remaining limit
Browser 3D volume, slices, isosurfaces, time	Working with both bounded historical cases; adaptive Basic graphics. Physical judge devices and Firefox remain unverified.
Temperature, salinity, currents	Real checked HYCOM fields. Kinetic energy is a labelled derived product.
Instrument markers and profile inspection	Argo, glider, CTD and BGC examples supported. Case overlap and numerical comparability are explicitly gated. Not every instrument overlaps both rectangles.
NetCDF and delimited ingestion	Supported, documented schemas and modular adapters. Arbitrary grids, variables and calendars are not automatically supported.
Palette, ranges, scale, opacity, vertical exaggeration	Implemented with unit labels, validation and saved settings.
REST/OPeNDAP and scalable deployment	Public Vercel REST/viewer/downloads. Separate local Bay THREDDS tested. Persistent institutional hosting, Docker execution and INCOIS installation are unverified.
CF and OGC interoperability	Native CF exchange for both cases. Local Bay WMS/WCS/DAP2 evidence retained with coordinate limitations.

This checkpoint does not complete those features, guarantee the ten-day deadline or establish a competition rank.

Investigation isolation correction

A direct API audit after the first 0.9.0 deployment found that observation-only capture could accept an Arabian-only profile under the Bay case. Numerical values were not fabricated, but the declared case-library identity did not include that record. Comparison already rejected this mismatch. The correction applies the same library membership gate before observation capture or replay. All 36 focused multi-case and investigation tests pass, including original P08 replay. This is a validation correction for an unsupported request, not a change to supported scientific calculations. The final patch is recorded in the release handoff.

Initial public run and retained rechecks

The 0.9.0 public browser run passed 189 cases, failed eight during loading and skipped one unavailable WebKit context-loss injection. All eight failed cases passed unchanged rechecks against the same deployment, without longer assertion timeouts or application changes. The original report, traces and transfer review are retained. They show pending transfers and several eight-second request aborts; the exact network/CDN/browser/backend contribution remains unestablished. This is not evidence of guaranteed availability.

Trace review also identified a separate unnecessary startup request: an observation query started under an undefined cache key, then was cancelled and restarted for the same default Bay case when the catalogue arrived. The final patch uses the actual default case ID from the beginning. This removes that redundant request; it is not claimed to explain or solve every public timeout. The final patch also enforces observation-only save library membership.

Final public release

Release 0.9.1 is deployed to the existing Vercel project as `dpl_CzrjqNDW5VceE7s3Eo9p73eTEqqk`. All 69 final public guide/instrument/investigation browser checks passed. The final case/source/isolation verifier passed 20 checks, the original-record replay/export verifier passed 19, and the release check verified eight public routes and their compiled assets. The scientific values and source fingerprints are unchanged by the final validation/startup patch. Evidence: `p09-final-public-browser.json`, `p09-final-public-http.json`, `p09-final-original-replay.json` and `p09-release-check.json` under `evidence/`.

P09.1-P09.4 are technically complete. P09.5 remains pending real participant evidence. This distinction is intentional.

Public screenshots cover 23 distinct views across the first capture run and the focused Arabian/mobile recheck. The first run stopped while waiting for Arabian current rendering after 30 seconds. Its precise cause was not captured by the original harness and remains unestablished. Both public current-component responses subsequently passed, and the unchanged focused screenshot run completed 13 captures with no uncaught errors, visible alerts or page overflow. The original partial run is retained separately. Representative deployed current, guide and narrow-screen images were visually reviewed. This is technical evidence, not a human usability study.

26 Phase 10: Virtual Expedition

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_10_EXPEDITION.md

Project date: 22 September 2026. Source dates: January 2024. Release: 0.10.1. P10.1-P10.5 and X15-X17 are complete within the bounded scope below. The release is deployed and verified on Vercel.

What this phase adds

Virtual Expedition connects three actions in the existing workspace: suggest stations, sample a line, and compare sampling strategies. It supports the checked Bay of Bengal and Arabian Sea cases, temperature or salinity, and exact native depths from 0 to 1000 m. It is an extra investigation tool built on the SIH viewer and evidence requirements.

The station budget counts one scalar measurement at one depth and snapshot. The user chooses 3 to 16 stations, minimum spacing from 0 to 150 km, and either prior gradient plus geographic spread or space-filling coverage. Every station exposes its coordinates, prior value, local gradient and selection reason. Changing the draft does not relabel previously calculated output.

The survey takes two endpoints and 2 to 81 positions along their straight longitude/latitude segment. Each position selects the nearest native column at the chosen depth and time. Requested coordinates, actual coordinates and offsets remain inspectable. Numerical ties within $1e-10$ km use source order. Missing values stay missing. The chart and export identify all values as simulated model samples.

Frozen experimental design

The first snapshot supplies all planning information. Candidate locations occupy every sixth native latitude/longitude index plus each final index, filtered only by the prior's finite mask. All remaining finite prior points are reserved for evaluation, disjoint from every possible station. Preview sampling is a display choice and never enters the calculation.

The gradient plan greedily balances normalized prior gradient (weight 0.65) with distance from selected stations (0.35). Physical differences use great-circle kilometres. The coverage strategy and uniform-coverage baseline use greedy farthest-point selection, starting nearest the candidate coordinate centre. It approximates uniform geographic

coverage rather than a regular lattice. Five fixed random-order plans use the same candidate pool, spacing and budget. All plans are finalized before later arrays are read. No retry is chosen because it scored better.

Snapshots 2, 4 and 6 supply simulated station measurements and evaluation truth. They are three correlated times in one case, not three independent events. Every sampled strategy reconstructs the field as the fixed prior plus inverse-distance-squared interpolation of station residuals. The power is fixed at 2, without tuning. Persistence retains the prior and uses no new measurements. All strategies are evaluated on the same target-finite reserved points. RMSE, mean absolute error and signed reconstructed-minus-target bias are point-weighted and retain units and denominators.

An incomplete greedy design or a missing target station is a failed trial. Missing evaluation truth is excluded identically across strategies. Failures and successful-run counts remain visible. Summary means describe successful runs only; unequal success sets must not be treated as directly comparable. Selected and uniform plans coincide by design when the coverage objective is chosen.

The future is hidden from selection algorithms, not secret from users who can inspect the same historical fields elsewhere. Repeatedly tuning settings after seeing errors is exploration, not external validation. There is no instrument noise, coastal barrier treatment, physical conservation, calibrated uncertainty, ship routing or demonstrated forecast benefit. These are model reconstruction experiments, not a validated operational OSSE.

Actual default results

Default: temperature, surface, eight stations, 30 km minimum spacing, gradient objective and seed 26067. Mean pointwise RMSE across the three times is shown below. Random averages pool five plans across those times, with no failures in these runs.

Case	Selected gradient plan	Space-filling baseline	Five random plans	Unchanged prior
Bay of Bengal	0.162475 degrees C	0.155987 degrees C	0.168171 degrees C	0.203111 degrees C
Arabian Sea	0.311115 degrees C	0.207471 degrees C	0.234092 degrees C	0.250760 degrees C

The gradient plan loses to space filling in both cases. It also loses to persistence in the Arabian case. We retain these results rather than presenting a suggested location as an optimal location. The methods were not changed after examining these scores.

Save, replay and export

Applied plan settings, a compatible survey and an optional experiment use the existing investigation system. The method identity is added only to expedition recipes, preserving earlier saved records. Reopening recalculates and checks source, recipe, output and document fingerprints. Records retain random seeds and failures. CSV files provide stations, simulated samples and all benchmark trials; ZIP includes these, the complete record, settings, a readable HTML report and source attribution. Browser saves remain local copies, not cloud backups.

Verification and limitations

Independent numerical tests decode the original packed NetCDF and calculate their own Cartesian great-circle distances and IDW errors. They verify both case benchmarks, surveys at 0/100/1000 m, gradient masks, exact and constant residuals, source-order ties, strict budgets, future-data isolation and retained missing-station failures. See `P10_INDEPENDENT_METHOD_REVIEW.md` and the evidence files for actual outcomes.

Local browser, public endpoint and visual acceptance results are recorded below. The earlier P09.5 unfamiliar-user pilot is still open. Automated interactions cannot prove that an unfamiliar judge understands the workflow. Existing Firefox runtime, physical-device, public standards hosting and institutional deployment limits remain in `PROJECT_STATE.md`.

Demonstration path

Open Expedition from the existing workspace. Inspect a numbered station and its reason. Change the budget and apply the plan. Draw a line or enter its coordinates, choose a source time, and inspect a simulated sample. Run Compare strategies and inspect all trial rows, including an example where the selected design loses. Increase spacing to 150 km with 16 stations and observe the explicit budget failure. Save the applied investigation, download the evidence, then recalculate and reopen it.

Next phase is P11 Drift Lab, the moving particle or pebbles experiment. It requires a separate user instruction.

Replay correction in 0.10.1

The initial public release passed all 87 targeted browser checks, but exact replay of eight Windows-generated expedition records failed on Vercel. The report retains 100 passes and 16 replay/export failures. Inspection found only tiny platform-dependent gradient-norm differences, with a maximum 6.94e-18 across the captured records. Source values, selected stations and benchmark errors were unchanged.

Method `p10-sampling-v2` computes the norm with explicit, separate binary64 multiplications, addition and square root. It keeps full source precision and strict fingerprints rather than rounding outputs or accepting approximate hashes. After this change, 77 focused backend checks passed, including 26 independent source/method checks, and all 24 local Expedition browser checks passed. These include reading an initial v1 record, retaining its downloaded JSON, and receiving an explicit method mismatch on replay. Existing P08/P09 records retain their original identities. Exact replay now passes in both directions between the checked Windows and Vercel environments. This does not establish bitwise portability on every possible future platform.

Final acceptance evidence

Check actually executed	Outcome	Evidence
Full backend suite before the targeted portability correction	364 passed	Backend report
Final method, original-source, API, export and investigation regression tests	77 passed, including 26 independent scientific checks	Final backend report
Existing renderer and transport checks	26 passed	Renderer record
Full local browser suite before the targeted correction	218 passed; one existing unsupported WebKit context-loss injection skipped	Full browser report

Check actually executed	Outcome	Evidence
Final local Expedition workflows, including archived v1 compatibility	24 passed	Local browser report
Final local HTTP, capture, replay and exports	100 checks passed across 103 requests	Local HTTP report
Final public HTTP, native samples, failures, guards, original records and exports	116 checks passed across 119 requests; no failed checks	Public HTTP report
Final public Expedition and existing investigation browser workflows	51 passed across desktop Chromium, Windows WebKit and Chromium mobile emulation	Public browser report
Exact record replay across the two checked platforms	Eight Windows records passed on Vercel, and eight Vercel records passed on Windows	Windows to Vercel, Vercel to Windows
Final public screenshot and layout audit	18 captures; no page errors, unexpected alerts or document overflow	Visual checks
Production build and deployment	TypeScript/Vite and Vercel Python bundle passed; deployment READY	Deployment record

Representative desktop, survey and 320-pixel benchmark screenshots were also visually inspected. Narrow tables intentionally scroll within their labelled container. These checks establish supported behavior on the tested configurations, not universal accessibility, device performance or first-time-user comprehension. Existing bundle-size and dependency warnings remain recorded.

Other verified corrections include support through the native 1000 m level, deterministic nearest-column ties, keeping draft routes separate from verified samples, matching the map and legend, accurate equal-error explanations, and retaining simulation status and actual sampling timestamps in standalone CSV files. Benchmark exports also identify their measurements as simulated.

The initial failed report p10-public-check.json and cross-platform diagnostic files remain intact. Original v1 Expedition archives remain readable and downloadable; create a fresh v2 plan to obtain a replayable current-method record. No original source values or previous P08/P09 record identities were rewritten.

Public URL: <https://depth-atlas-seifuku.vercel.app/>. Deployment:

dp1_F3GCPZiayJEEY8UKARqKDLpt6nz9. Choose **Expedition** in the navigation. The [simple project explanation](#) includes the SIH requirement comparison and a short spoken script.

27 Phase 11: Drift Lab

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_11_DRIFT.md

Project date: 23 September 2026. Sources: 7-10 January 2024. P11.1-P11.5 complete in release 0.11.1, method p11-drift-v2. Deployed on the existing [Vercel application](#). Scope and acceptance evidence are recorded below.

What users can do

Drift Lab releases passive tracers into either checked historical study area. A user chooses a point or a rectangular release area, one native depth from 0 to 1000 m, a source starting snapshot, 1 to 72 whole hours and 1 to 64 particles. The available source end bounds the duration. Unsupported requests fail clearly instead of silently shortening a trajectory. An optional target rectangle records first simulated entries.

The default uses 24 particles in a small central box over 24 hours. A point release has identical paths under identical forcing; no artificial jitter is added. Box releases use a pinned seeded generator, uniform in longitude and latitude rather than exactly uniform in area. Invalid release locations remain in the reported totals.

Run A and optional Run B retain their own applied settings. The user can change release location, date, depth, seed or duration and compare the results. A shared elapsed-time slider does not imply equal calendar timestamps. Each run retains its actual historical start and end. Changes to draft controls do not relabel completed paths or saved results.

The map has a study-area view and a fit-paths view, keyboard coordinate alternatives, an individual particle inspector and a complete results table. Start-snapshot current arrows provide context and remain explicitly static. Press Play to animate calculated output points. Rendering interpolates these points for visual continuity; it does not supply

additional numerical integration or physical detail. Reduced-motion settings pause automatic playback changes. There is no decorative random motion.

Numerical method

The production kernel adapts OceanParcels 3.1.4 AdvectionRK4, retaining the four-stage weighted displacement calculation. The array runner, data adapter, strict masks, progress stream and export logic belong to this application. We do not claim to ship the complete OceanParcels runtime on Vercel. Upstream MIT licensing and the source file identity are retained in `science/vendor/` and public third-party notices.

The full pinned framework is installed separately for offline reference checks using `requirements-drift-reference.txt`. This keeps its xarray, SciPy, Dask, Zarr and compiler-related dependencies out of the public function. The existing NumPy runtime is sufficient for the bounded runner. This replaces the initial separate-worker plan for P11's small checked cases; a durable worker/queue remains a possible later scaling choice, not an undeployed hidden dependency.

Both current components come from the same native rectilinear HYCOM z-level grid. Metadata declares eastward and northward m/s, so no additional grid rotation or staggering adjustment is applied. Interpolation is bilinear in native longitude/latitude and linear between available times. The depth is fixed at a selected native level. No vertical velocity, extrapolation, nearest-wet relocation or fill-to-zero conversion is introduced.

At every Runge-Kutta stage, velocity converts to angular motion using 111120 metres per latitude degree and the same factor times cosine(latitude) per longitude degree, matching the Parcels convention. A fixed-order binary64 cosine polynomial on the explicitly bounded latitude domain avoids platform-dependent trigonometric calls. Independent tests compare its accuracy across the supported domain and its trajectories against the full framework. Source values and numerical trajectories are not rounded to weaken replay checks.

Allowed integration steps are 300, 600 and 1200 seconds, default 600. Whole-hour duration and 12-hour source boundaries are exact multiples. Output cadence is `1cm(1800, timestep)`: 1800 seconds for 300/600 and 3600 seconds for 1200, with the last accepted point retained. Display cadence does not change the solver. Integration cannot reconstruct current variability absent from the source snapshots.

Boundaries, arrivals and distance

A complete finite U/V interpolation cell is required, including zero-weight spatial corners under the declared conservative rule. Exact-time point samples use that snapshot alone; movement through a time interval requires the source brackets supporting that interval. Every stage and endpoint must lie inside the source domain. The full rectangle of traversed interpolation cells must also remain supported. This deliberately conservative check prevents crossing an unsupported strip but can stop earlier than a geometric coastline intersection.

A failed step retains the last accepted position and elapsed time. Status distinguishes completed duration, leaving the study area, missing current support and invalid initial release. These are not observed beaching events or exact collision times. A stopped marker remains at its last supported point.

Target arrival is the first intersection with a closed target rectangle along an accepted integration-step segment. Time is interpolated linearly within that step, so this is an approximate numerical entry time. Valid initial particles already inside arrive at zero. Invalid releases never become arrivals. Arrival fraction uses all requested particles, including invalid releases; mean path distance uses valid releases only. Arrival can occur before a later early stop.

Distance accumulates the midpoint spherical-metric length of each accepted step. It is an approximation to the calculated path length, not endpoint separation or a navigable travel route. Neither arrivals nor comparisons form real-world probabilities or calibrated uncertainty.

Jobs, cancellation and reproducibility

Calculations are request-scoped jobs. `POST /drift/stream` emits actual completed integration-step counts and then a complete checked result. Work is chunked at no more than 12 solver steps between progress events. Cancelling disconnects the browser request; the server stops after the current bounded chunk. No partial result is saved or cached. A failed or cancelled request keeps the previous completed run visible. There is no durable background-job queue or database in this phase.

`POST /drift/run` produces the same result for numerical replay. Warm result caching is bounded to four results; current-plane caching is bounded to four case/depth combinations. Cached results retain the full source, method and query identity. A cold maximum-sized run measured about 0.67 seconds locally; that observation is not a public latency guarantee. The deployed function retains its 30-second bound.

Saved investigations use recipe mode `drift`, a required run A and an optional distinct run B. Modules `drift_run` and `reference_drift_run` retain seeds, methods, all paths and stopping reasons. Reopening recalculates and checks fingerprints. Original P08/P09/P10 method identities are unchanged. CSV exports retain actual timestamps, coordinates, native depth, seed, simulation status and source identity. The ZIP adds full JSON, settings, reports and attribution. Browser saves remain local copies.

Independent evidence and limits

The separate scientific review passed 45 checks with no skips locally. It decoded original packed provider files, checked analytical motion, manufactured-flow convergence, source interpolation, masks, step boundaries, forcing exhaustion, output cadence, target crossing and denominators. Twelve 24-hour paths across two cases and two depths matched full OceanParcels final coordinates exactly on the checked machine. The test tolerance and environment remain recorded separately from strict saved-record fingerprints.

The largest checked 600-versus-150-second endpoint difference was 0.002059 m. This measures numerical convergence for selected paths, not ocean accuracy. Independent NOAA GDP queries retrieved no overlapping tracks within either exact case and date range; a wider control query returned real records outside the cases. Independent drifter validation is therefore not established. See [independent review](#), reference comparisons and drifter search.

The tracer model excludes sinking, buoyancy, windage, swimming, added diffusion, oil chemistry and invented vertical movement. It does not establish operational rescue, forecast or ship-routing skill. Real unfamiliar-user feedback, physical-device coverage, the known Firefox runtime limitation, Docker execution and institutional/public standards hosting remain separately open.

Final acceptance

P11.1-P11.5 and feature rows X18-X19 pass their bounded acceptance gates. Real drifter validation remains unavailable, as permitted by P11.5's explicit requirement to seek and report it. No later feature or unfamiliar-user gate is marked complete.

The initial release passed 478 backend checks, 242 local browser checks with one inherited skip, 24 final local Drift checks and 51 public browser checks. Twenty local visual captures passed. Public HTTP verification retained 90 passes and eight failures involving replay/export of four Windows records. All trajectories and individual distances were exact across platforms; only the summary mean differed due to the changed floating-point summation in Python 3.12. See D039, initial HTTP evidence and exact

differences.

The 0.11.1 correction pins sequential addition as method p11-drift-v2. Old v1 archives remain readable and explicitly report a replay method mismatch. No fuzzy hashes or output rounding are introduced. Final acceptance verifies the new method in both directions, independently of the initial browser passes.

Check	Recorded outcome	Evidence
Full backend before summary correction	478 passed; no skips	Full report
Final backend scope	139 passed; 68 API, 45 independent scientific and 26 existing investigation checks; no skips	Final backend
Renderer and transport regression	26 passed	Renderer report
Full local browser before summary correction	242 passed; one inherited unsupported WebKit context-loss injection skipped	Full local browser
Final local Drift browser checks	27 passed across Chromium desktop, WebKit desktop and mobile Chromium emulation	Final local browser
Final public Drift and existing investigation workflows	54 passed across the same three projects	Public browser
Final public HTTP, streams, limits, old archives and exports	100 checks passed across 114 requests; no retries	Public API
Exact final-method replay across platforms	Eight Windows records replayed exactly on Vercel; eight Vercel records replayed exactly on Windows	Windows to Vercel, Vercel to Windows
Public responsive and scientific visual audit	20 captures; no page errors, document overflow or unexpected alerts; representative paths, arrival comparison and 320-pixel views visually inspected	Visual audit

Check	Recorded outcome	Evidence
Maximum record capacity	Two 64-particle, 72-hour, 300-second-step runs fit the existing import limit and retain complete exports; initial-method test remains preserved	Capacity evidence
Production build and deployment	TypeScript/Vite passed; existing large-chunk advisory remains; deployment READY with verified public release and assets	Final deployment

The public API check also verifies all four original P08 and eight P10 archives without changing their methods. Initial P11 failures, failed test-harness attempts and the first rejected CLI deployment request remain retained. The browser harness stopped depending on an unavailable DevTools response-body read for streaming requests; it now checks the visible result and separately checks the numerical endpoint. The initial unscoped Vercel CLI request returned an authorization error, while an explicit existing-team scoped retry succeeded. No account permissions or project configuration changed, and the initial request's cause is not established.

These checks establish implementation consistency for the supplied cases and tested platforms. They do not establish physical-device coverage, universal availability, unfamiliar-user understanding or observed ocean prediction skill.

Demonstration and next phase

Choose Drift Lab, inspect the release box, and calculate run A. Press Play paths, scrub time, inspect one particle and its source time, then add a target rectangle or move the release for run B. Compare the declared settings and denominators. Choose a late start with an excessive duration to see the explicit forcing limit. Save, export, then recalculate and reopen the experiment.

Next is P12 Heat & Depth Lab when the user authorizes it. P09.5 still needs actual unfamiliar-user task results.

28 Phase 12: Heat & Depth Lab

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_12_HEAT_DEPTH.md

Project date: 23 September 2026. P12.1-P12.5 are complete. Release 0.12.0 is deployed on the existing Vercel project, with scientific method p12-heat-v1. Source preparation, numerical acceptance, public API/export checks, exact replay in both platform directions, browser checks and visual review have passed within the supported scope.

Intended supported workflow

Choose one of three fixed SST locations in either the Bay of Bengal or Arabian Sea study rectangle. Inspect daily NOAA OISST analysed SST in 2023 or 2024 against the declared 1982-2011 seasonal baseline and 90th-percentile threshold. Select an actual detected event to inspect dates, duration and intensity. Original event boundaries remain visible even when they cross the displayed year.

For an event overlapping the available 7-10 January 2024 HYCOM case, choose an overlapping model snapshot and inspect the nearest native column. Temperature, practical salinity, density, actual-depth gradients and stratification remain separate quantities. Two mixed-layer criteria and a fully defined column potential-enthalpy integral expose their assumptions and missing-support states. Source profiles retain their own locations, dates, quality flags and provenance review status.

When an event has no supplied depth coverage, the app explains the gap. It does not relabel another January column as that event's subsurface conditions. Every surface point remains in the catalogue regardless of whether it has an event. These are historical source analyses and derived diagnostics, not forecasts or basin-wide heatwave maps.

Sources and method

See [surface source acquisition](#), [independent method review](#) and [API and export contract](#). D040-D043 record the substantive choices. OISST is a blended/interpolated surface analysis; HYCOM and the original instrument profiles are distinct sources. Data years remain separate from the 2026 project date.

Each of the six fixed cells has 16,071 daily values covering 1982-2025, with no missing temperature. All 10,957 baseline days are present at every cell. NOAA AOML supplies

14,966 dates, and retained NOAA PSL records supply 1,105 dates absent from AOML. This is a source merge, without temporal interpolation. The pack is 7,386,504 bytes. Its manifest SHA-256 is b2c1186439af2a66c982392961145300f0c015792327b6b96721216278d15fff.

The two distributions retain a small checked packing difference. Across 6,282 overlapping point-days, 973 differ by no more than one Float32 step, with a final maximum of 3.814697265625e-6 degrees Celsius; all remain in the same original centidegree bin. Original NCEI packed integers explain the tested decoding alternatives. The served values are preserved. Both preparations reproduced the manifest and all six location hashes exactly, and all 96,426 point timestamps reconstructed from their original per-file units. See `evidence/p12-source-rebuild.json`, `evidence/p12-source-packing-diagnostic.json`, `evidence/p12-distributor-comparison.json` and `evidence/p12-time-provenance-resume.json`.

The baseline uses all daily data from 1982 to 2011. For each calendar day, pool the five actual days before and after every occurrence inside the baseline. Clip at baseline edges. Calculate the mean and linear 90th percentile, interpolate the February 29 climatology before smoothing, then apply a 31-day circular average. Events need at least five consecutive days strictly above the threshold. Already-qualified events can merge across at most two finite intervening days. Unknown days are never filled or bridged.

Detection uses the supplied 2022-2025 context before selecting a display year. Boundary flags show where available context does not establish a complete start or end. Intensity means SST minus the seasonal mean. Merged duration and intensity accounting include observed intervening cool days.

The depth method uses GSW 3.6.20 with the original in-situ temperature, practical salinity, coordinates and native levels. It computes pressure, Absolute Salinity, Conservative Temperature, potential temperature, sigma0 and in-situ density. Adjacent valid levels supply signed temperature gradients and N-squared, retaining negative stratification values. Model temperature is not silently treated as potential temperature.

Derived mixed-layer depth uses a 10 m reference and the first crossing of either sigma0 increase 0.03 kg/m³ or absolute potential-temperature change 0.2 degrees Celsius. Each estimate reports crossing, no crossing, missing support or unavailable reference. It is not a supplied HYCOM MLD field. The no-crossing bound applies to the selected depth window.

Column potential-enthalpy content relative to CT=0 is the trapezoidal integral of rho times cp0 times CT from 0 m to a supported native limit of 100, 300, 700 or 1000 m. Its value is in J/m², displayed in GJ/m². Missing input prevents a complete integral. No subsurface climatological anomaly, heatwave excess, cyclone-energy prediction or causal attribution is claimed.

Integration and verification

The existing workspace, case selection and investigation system are reused. Saved records require the applied event selection and include surface-source identity, method, baseline curves, selected-year daily values, complete event boundaries, model diagnostics and source references. JSON, readable HTML and numerical CSV/ZIP exports retain these definitions. Existing recipe/source identities are preserved.

All 590 backend tests pass with zero skips in `evidence/p12-final-backend.xml`. This includes 76 independent scientific tests, separately recorded in `evidence/p12-independent-final.xml`, and 33 new heat API/integrity/export tests. The scientific checks cover six complete OISST series against the pinned Oliver implementation, exact event dates and durations, 18 exact primary NCEI comparisons, calendar/missing-data rules, eight original HYCOM columns, version-matched GSW check casts, separate quadrature and a closed-form integral. Initial checks remain in `evidence/p12-independent-initial.xml` and `evidence/p12-early-backend.xml`.

The older GSW 3.05 published Absolute Salinity example differs slightly from the pinned 3.6.20 implementation. The actual matching-version upstream check cast was used, and all 98 finite reference values matched exactly. Original discrepancy evidence is retained rather than hidden by widening a tolerance. See `evidence/p12-gsw-reference-selection.json`.

The 26 existing renderer tests pass. The existing browser suite passes 248 checks with one inherited WebKit context-loss injection skip. The first P12 browser run passed 23 and failed four: three tests incorrectly assumed that selecting display year 2023 precluded an event extending into January 2024, and one exposed a real 320 px WebKit select-overflow defect. The tests now choose an actual non-overlapping event, and the select contains long labels. The subsequent local P12 suite passes all 27 checks. Initial reports and failure artifacts remain in `evidence/p12-browser-local-initial.json` and its companion artifact directory. The final suite is `evidence/p12-browser-local-final.json`.

Final public browser acceptance passes all 54 Heat and saved-investigation scenarios across Chromium desktop, WebKit desktop and Chromium mobile emulation, with no failures, skips or flaky tests. The report is `evidence/p12-public-browser.json`. The final public visual run captures 20 views with no page errors, document overflow or unexpected alerts, recorded in `evidence/p12-visual-public/checks.json`. The frontend reviewer manually inspected the public Bay surface graph, temperature gradient, Arabian unavailable-depth panel and 320 px surface chart. A second review also inspected the public Bay surface and depth panels and the 320 px view. The source curves remain distinct, gradient labels retain meaningful precision, and narrow charts use clearly indicated horizontal scrolling. No public correction or retry was needed. Local refinement

results and the 20-view final local capture set remain retained separately.

The local HTTP verifier passes 88 checks across 85 requests with no retries, in `evidence/p12-local-http-initial.json`. It checks all six points and both years, actual Bay depth diagnostics at four limits, eight new captures/replays, numerical CSV/ZIP output, rejection paths and all 20 retained P08/P10/P11 records. The eight new records are retained in `evidence/p12-local-records.json`.

The public release passes 105 checks across 101 HTTP requests, with zero failures and no retries, in `evidence/p12-public-http-initial.json`. The same source, diagnostic, export and compatibility checks pass on Vercel. All eight Windows-produced heat investigations reproduce their complete scientific results and numerical ZIP contents exactly on the deployed service. The eight new Vercel records are retained in `evidence/p12-public-records.json`.

All eight Vercel-produced records also replay and export exactly on the Windows server. This reverse check passes 19 checks across 17 requests in `evidence/p12-public-to-local-replay.json`. No numerical tolerance was weakened, and no new platform-specific arithmetic correction was needed. The 76 independent tests are included in the 590 backend total; exact replay checks are included in the corresponding HTTP totals rather than additional unique tests.

Demonstration and remaining limits

Open the [public workspace](#), choose the Bay of Bengal study case, and open Heat & Depth through the toolbar or Heat on the navigation rail. Use the centre point and event year 2024, then select Analyse surface record if the controls contain unapplied changes. Identify the analysed SST, seasonal mean and threshold. The selected full event spans 11 October 2023 to 10 February 2024; its peak is in December, outside the displayed 2024 graph. Inspect a dated visible daily sample. Displayed values are rounded for reading; exports retain the numerical values.

The 7 January model snapshot supplies a separate native depth column. Change Depth variable, compare the two mixed-layer criteria, and change Column and integration limit. Select an event without January overlap to show the absence of depth evidence. Use Save heat investigation, give it a title and save it on the browser. Recalculate the saved record and download its evidence ZIP. Inspect the daily source series, seasonal curves, full event table and any supported depth tables.

Switch to the Arabian Sea case to inspect its genuine surface events and explicit absence of overlapping depth evidence. None of its detected events at the three selected cells overlaps the supplied model period. Changing display year alone does not guarantee

unavailable depth: a full event can begin in one year and extend into the next. The app retains the actual event boundaries and source dates.

Surface events do not establish a subsurface marine heatwave. Source and numerical agreement do not establish independent ocean validation. The real unfamiliar-user pilot, physical-device coverage, Firefox runtime, Docker execution and persistent public standards hosting remain separate project limitations.

Next planned phase is P13 Climate Event Lab, only after P12 acceptance and a separate user instruction.

29 Phase 13: Climate Event Lab and selected tropical Pacific cases

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_13_CLIMATE.md

Project date: 23 September 2026. Release 0.13.0, method `p13-climate-v1`, is deployed and verified on Vercel. P13.1-P13.5 and X23-X24 are complete for the supported historical cases. The separate P09.5 real unfamiliar-user pilot remains open.

Supported investigation

Compare September-November 2013, 2015 and 2022 at the same native Pacific coordinates. The pinned official RONI v6 snapshot identifies these as neutral, El Nino and La Nina respectively. Users can inspect SON or one of its three constituent monthly fields, compare potential-temperature anomalies against the same calendar window of the 1991-2020 GODAS baseline, and examine exact depth profiles and signed selected-minus-reference differences.

The Indian Ocean section uses the same source product, calendar window and baseline. Its context does not establish that ENSO caused the displayed Indian Ocean conditions. Official IOD episode assessments and numerical DMI values keep their separate definitions. A documented episode is not silently inferred by thresholding the displayed seasonal mean.

The three Pacific seasons are genuine shared-workspace cases, with native 3D, depth-slice and section access plus original instrument inspection. Potential temperature is the supported Pacific model variable. Find structures supports threshold queries on that field, with actual 5-459 m depth limits and no direct observation association. The existing

Indian Ocean HYCOM cases retain temperature, salinity and currents. The Pacific interface makes Compare, Expedition, Drift and Heat unavailable with a source-specific explanation. The climate views, Find structures and shared Ocean Explorer remain connected to the same native data. No unsupported current, heat-event or sampling-experiment path substitutes another field or dataset.

Sources and calendar definitions

See [GODAS and Argo preparation](#), [official index snapshots](#) and [API/export contract](#). D044-D046 record the substantive choices.

Thirty-one GODAS annual source subsets cover SON in 1991-2020 plus 2022. These are reconstructed native OPeNDAP subsets, not downloads of entire original annual NetCDF files. The combined source geometry has three monthly means, 28 native depths from 5 to 459 m, 30 native latitudes within 5 degrees of the equator, and 240 native longitudes from 40.5 to 279.5 degrees east. Each Pacific case retains the original 140.5-279.5 degree longitude subset. Original source values are potential temperature in kelvin. Preparation masks invalid values, promotes source Float32 values to Float64 and subtracts 273.15. It assigns the CF quantity name from NOAA's explicit variable description while recording that the original COARDS file lacked that `standard_name` attribute.

GODAS timestamps denote the first day of the averaging month. Each interval is start-inclusive and end-exclusive: September runs from 1 September to 1 October, and SON runs from 1 September to 1 December. The app retains that support interval and does not present a monthly mean as an instantaneous measurement. All twelve combinations of three years and four calendar windows are checked. The equatorial section uses the native latitude nearest zero, 0.16612949967384338 degrees north, without interpolation to the equator. Longitudes remain monotonic on the source 0-360 degree axis, while display labels can use east/west notation. Selected profiles and values use native longitude/depth indices, without interpolation or extrapolation beyond the supplied 5-459 m range.

The baseline requires all 30 source years at each month/cell. Its 535,279 complete month/cells retain their calculated means; 69,521 incomplete month/cells stay missing. SON is the equal-weight mean of the three complete monthly means, in fixed order. It is not a day-weighted mean. Anomaly means event field minus matching baseline. A between-event difference uses the same native coordinates and calendar window. No result uses an unsupported field value to fill a gap.

Later source review located the official NOAA 1991-2020 climatology under `godas/Derived/`. Its metadata describes a Float32 product that permits output with at least 3% of inputs present. P13 retains its declared Float64, 30-of-30 reconstruction from actual

annual fields. The earlier 404 from a guessed URL was not proof that no provider product existed. This corrected source explanation does not alter the chosen baseline or its hashes, and no numerical comparison with the provider climatology has been claimed. See the archived metadata check.

Seventeen acquired source documents support four separately named index definitions and three deployed event records. The official RONI v6 table is pinned separately from ONI v5 and ONI v6. The 2026 operational change is documented in the source review. The SON RONI values are -0.2, +2.0 and -1.1 degrees Celsius for 2013, 2015 and 2022. The qualifying warm/cold windows are checked against official provider classifications. Selecting September, October or November shows ASO, SON or OND index context respectively, with the actual three-month season name. It does not relabel a seasonal index as a monthly measurement, and it does not recompute RONI from the bounded GODAS fields.

CPC's specific downloaded ERSSTv6 DMI series declares a 1991-2020 baseline. The broader HTML page's different baseline wording is retained as a source metadata discrepancy. The published SON DMI values are -0.17, +0.39 and -0.48 degrees Celsius. BOM documents the 2015 positive IOD episode and 2022 negative episode. The 2015 SON value being below +0.4 does not erase that separate historical assessment or establish an identical time interval. No IOD episode label is invented for 2013.

Nine preselected original Argo profiles retain their original dates, coordinates, parameter-specific QC, raw/adjusted values and source hashes. September 2022 has 994 accepted temperature readings and no accepted salinity readings; joint core eligibility remains zero. The profile was not replaced to improve its appearance. Monthly potential temperature and instantaneous in-situ observations remain distinct, without a misleading direct residual or independent-validation claim.

Implementation and initial evidence

The shared case contracts now support a declared monotonic 0-360 longitude axis and temperature-only availability. Pacific defaults must respect the supplied 28 depth levels and three monthly timestamps. Case source labels come from their actual metadata. The climate manifest pins the field/index files plus all three model manifests and observation-library indexes, covering both the selected and reference cases. Its combined SHA-256 is 13b4c3aa2fc568272758f485d384044b5f98a54b87b9aa47d3d89743ebe9cd77. This supplies a separate optional saved-source identity; older investigation identities remain unchanged. Saved visual settings retain the display exaggeration without changing scientific coordinates or numerical results.

All 62 prepared source/case/instrument files reproduce byte-for-byte. Cached acquisition passes with networking disabled and the original journal unchanged. Independent NCSS requests match 1,008 source values exactly across six years, all SON months, 28 depths and both dateline-adjacent longitudes. The NCSS latitude differs from the original OPeNDAP coordinate by about $2.98\text{e-}7$ degrees; the original coordinate is retained and the cause is not asserted. See `evidence/p13-field-preparation.json`, `evidence/p13-field-rebuild.json`, `evidence/p13-field-cache-resume.json` and `evidence/p13-field-transport.json`.

Index acceptance passes 26 tests and 22 source audit checks. Independent field/method acceptance passes 29 tests, using original source arrays, scalar reference sums, native cell transcriptions and explicit missing-data fixtures. Source values use exact checks. Independent arithmetic comparisons use a stated $1\text{e-}12$ degree tolerance; saved-record replay still requires exact fingerprints. Evidence is in `evidence/p13-index-tests.xml`, `evidence/p13-index-audit.json` and `evidence/p13-independent-final.xml`.

The first API run had 27 passes and eight failed test expectations because the new tests addressed two existing endpoints incorrectly. All 35 passed after the test routes were corrected. The initial full backend regression then passed 680 tests with zero skips. Later API checks cover monthly source context, saved visual settings and the reference observation library's inclusion in the combined source identity. The latest focused report passes all 38 API tests in `evidence/p13-api-source-closure.xml`. The final full backend run passes 683 tests with zero failures or skips in 92.85 seconds; three existing dependency/report warnings remain. Evidence is `evidence/p13-final-backend.xml` and `evidence/p13-final-backend.log`. A retained intermediate context test used an unsupported 0 m lower depth; its corrected request respects the actual 5 m first level.

The local HTTP verifier passes 116 checks across 112 requests without retries, covering all twelve event/window combinations, nine original profiles, eight fresh exact replays and all 28 retained earlier investigations. Its initial run correctly received HTTP 413 for an oversized analytical volume while the test expected 422; the final run corrected only that expectation. Evidence is `evidence/p13-local-http-final.json`, with original failures preserved in their initial reports.

The first focused browser run recorded 24 passes and three failures from the legacy instantaneous-profile-overlap assumption. After the parser accepted the explicit monthly-support contract, the second focused run passed all 27 scenarios. Shared API review also corrected a Pacific HYCOM label and removed unrelated Bay-specific caveats from Pacific responses. Subsets and feature results now carry case-specific potential-temperature and monthly-support context. The completed browser, visual and public acceptance is recorded below; the earlier reports retain their original outcomes.

Final release acceptance

Release 0.13.0 is live at <https://depth-atlas-seifuku.vercel.app/>. Final Vercel deployment `dp1_4aVLAaqS82jcbV9iJKjHD3w43UAK` is READY. Its immutable URL is <https://depth-atlas-seifuku.vercel.app/>. The corrective production deploy command and result are retained in `evidence/p13-vercel-deploy-sidebar-final.log`. The first deployment and its full verification remain in the original reports.

The final build passes TypeScript and Vite. All 31 renderer/geographic checks pass. Final assets are `workspace-C3Ymu47W.js`, `scene-BgH9yDKL.js` and `workspace-0oAPrcy8.css`. The 13-path public audit matches every served build asset and static/legal page byte-for-byte, and verifies release 0.13.0 with five real cases. See `evidence/p13-public-release-sidebar-final.json`; the earlier 13-path audit is preserved.

The initial broad browser run recorded 260 passes, 15 failures and one inherited unsupported WebKit context-loss skip. The second broad run recorded 268 passes, seven failures and the same skip. They are not described as clean full runs. Most failures were old unavailable-Pacific expectations or unnecessarily changed Indian labels. Those labels were restored; revised availability tests verify actual Pacific support. A revised mobile test used a desktop-only button and was corrected to use the shared evidence panel. A separate final 69-case run passes all affected climate, feature, section-science and workspace scenarios, resolving all seven second-run failures. Reports and failed traces are retained under `evidence/p13-existing-browser-initial*`, `evidence/p13-existing-browser-second*` and `evidence/p13-browser-corrected-final.json`.

Actual screenshot review then caught a real longitude-boundary rounding defect in the Pacific instrument map. It placed nearby floats in different wrapped worlds and made the map unnecessarily wide. The helper now uses the exact next sorted coordinate; original source positions do not change. The regression includes the actual Argo longitudes and repeated stations. All 54 final climate/instrument checks pass, including existing Indian glider, CTD, BGC, imports and keyboard flows. See `evidence/p13-browser-map-final.json` and `evidence/p13-renderer-tests-map-final.log`.

Broad public browser acceptance on the initial deployment passes all 111 checks, with no failures, skips or flaky cases: 30 climate, 27 saved-investigation, 18 feature, 24 instrument and 12 workspace scenarios across Chromium desktop, WebKit desktop and Chromium mobile emulation. Evidence is `evidence/p13-browser-public.json`. The local final and public visual audits each capture 23 views with no page errors, unexpected alerts or document overflow. Manual inspection covers the actual climate sections, constant zero difference, native profiles, source/QC views, Pacific 3D/cutaway and narrow layouts. See `evidence/p13-visual-local-final/checks.json`, `evidence/p13-visual-public/checks.json` and `evidence/p13-visual-review.md`.

The final source-closure local verifier passes 116 checks across 112 requests. The public verifier passes 133 checks across 128 requests; the reverse run passes 19 across 17 requests. All eight final Windows records replay and export exactly on Vercel Python 3.12, and all eight Vercel records do so on Windows Python 3.10. All 28 retained P08/P10/P11/P12 records replay. There are zero verification retries or relaxed fingerprint comparisons. The final combined source identity is `13b4c3aa2fc568272758f485d384044b5f98a54b87b9aa47d3d89743ebe9cd77`. Reports are `evidence/p13-local-http-source-closure.json`, `evidence/p13-public-http-final.json` and `evidence/p13-public-to-local-replay.json`; corresponding portable files are `p13-local-records-source-closure.json` and `p13-public-records.json` in the same directory.

Visual refinements make the broad Pacific column legible using a labelled 6000x depth exaggeration, retain its real 5-459 m values, distinguish missing baselines from zero anomalies, explain single-month versus seasonal indices and keep rounded display values separate from exact exports. The public Privacy Policy, Terms, source credits and data-access page reflect climate requests and exports. Operator/contact details remain intentionally blank for P17.

Final sidebar source correction

A last desktop-sidebar audit found a surviving hardcoded HYCOM label on a Pacific case. The main climate and explorer labels were already correct, but that sidebar statement was wrong. The corrected sidebar derives its source label, coordinate notation, full monthly support interval and depth bounds from the actual manifest. The playback tooltip now says Play available times rather than claiming seven snapshots for a three-month case. No scientific method, source pack or API code changed.

The correction passed nine local source/date/sidebar checks and three focused playback-title checks, then nine public checks across all three supported browser projects. Final public evidence is `evidence/p13-sidebar-public-final.json`. The final visual audit captures 24 views, including the explicit corrected sidebar, with no page errors, alerts or document overflow: `evidence/p13-visual-public-final/checks.json`. The corrected public sidebar was manually inspected, alongside the earlier broad visual review. Initial 111-check and 23-view reports remain unchanged.

The final deployment also passes 13 public static/asset/health checks, with exact local/public bytes, and a 19-check replay/export smoke across 17 requests with zero retries. All eight final local records and all five numerical tables per ZIP still match exactly. See `evidence/p13-public-release-sidebar-final.json` and `evidence/p13-public-replay-sidebar-final.json`. The original full 133-check public run and 19-check reverse run remain valid evidence for the unchanged scientific backend; they were not falsely relabelled as rerun after this UI-only correction.

Demonstration and remaining limits

Open Climate Event Lab and compare 2015 with the 2013 neutral reference. Inspect the fixed SON event labels and their sources. Compare the two potential-temperature anomaly sections on a common scale. Select the same native longitude and depth in both years, then inspect the profiles and exact selected-minus-reference value. Switch to October to inspect actual monthly support and the separately labelled SON index context. Choose 2022 to examine the La Nina case. Expand Indian Ocean context and distinguish the DMI value from the BOM historical episode label.

Open the selected Pacific case in Ocean Explorer, inspect potential temperature and a genuine Argo profile, then return to the climate comparison. Save the applied settings, replay them and inspect the exported sections, baselines, profiles and index definitions. These actions pass the final public browser and exact replay/export checks.

Three selected seasons do not establish equal coverage of every ENSO or IOD event. No event-strength control produces a new forecast, and regional historical differences do not prove causality. P09.5 real unfamiliar-user testing, physical-device coverage and the project's existing external validation and hosting limits remain open.

30 Phase 14: feature evolution and observation blackout

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_14_EVOLUTION_BLACKOUT.md

P14.1-P14.4 and X25-X26 completed 24 September 2026. Release 0.14.1 is deployed and verified on the existing Vercel project. Implementation started 23 September. The initial 0.14.0 deployment and its failed acceptance remain recorded below.

User purpose

Follow a defined ocean structure across available historical frames, inspect competing branches, and see how nearby threshold choices affect the result. Separately remove selected observation profiles or groups from the available evidence and inspect exactly what disappears.

Method commitments

Evolution uses the full native three-dimensional memberships from the existing feature engine. Equal frame-local region IDs do not establish identity. Correspondence is an explicit shared-volume overlap rule, with both directional fractions retained. Splits, merges, competing matches, missing support and time gaps remain inspectable. No continuous water-parcel trajectory is inferred. Initial evolution cases are the two Indian Ocean January 2024 packs; the monthly Pacific cases are not silently treated as 12-hour snapshots.

Blackout exclusions are an additional user selection, separate from provider QC and the existing quantity/time/distance/depth gates. The original source fields and source observations remain intact. Original and excluded coverage, eligible comparisons, residual summaries and affected statements are computed under the same settings. Excluding an already ineligible profile may correctly have no effect on eligible evidence. This is an evidence-dependence experiment, not a model rerun or a forecast-impact test.

Both modes use versioned methods, original source identities and exact investigation replay. New recipe families must preserve prior saved records. Source and replay fingerprints remain exact; independent numerical checks do not relax that requirement.

Acceptance plan

- Hand-computable split, merge, ambiguous, disappearance, no-overlap, missing-support and skipped-frame fixtures.
- Native-volume weighting and threshold sensitivity, including depth/mask boundaries and explicit resource bounds.
- Remove-none, single profile, group union, remove-all, no-eligible-evidence and invalid-identity exclusion tests.
- Invariance of source arrays, QC and cached baseline results, plus exact original/modified replay and exported tables.
- Frontend-to-API workflows on supported desktop browsers and mobile emulation, error recovery, stale-result handling, keyboard controls and narrow layouts.
- Anonymous public API and browser checks after deployment, including cross-platform exact replay of new and retained earlier records.

Recovery checkpoint

A local source/data archive was created at `.runtime/p14-start-checkpoint.zip`; its SHA-256 and scope are in `evidence/p14-checkpoint.json`. It excludes ignored environment/credential/runtime files and the large `docs/evidence` tree. This is a local recovery copy, not a Git commit or remote backup.

Service and saved-record contract

Operation	Contract
POST <code>/api/cases/{case_id}/evolution/run</code>	EvolutionQuery: variable and exact units, inclusive threshold operator, depth interval, first/last source indices, frame step, minimum overlap and sensitivity delta. Returns every accepted frame region, native-column footprints, adjacency links, missing/gap events, eligibility context and sensitivity summaries.
POST <code>/api/cases/{case_id}/blackout/run</code>	BlackoutQuery: unchanged MatchSettings, excluded profile IDs, instrument types, platforms and collections. Returns original and retained coverage, eligible sample metrics, exact lost source sample indices and before/after evidence statements.
GET <code>/api/cases/{case_id}/blackout/catalog</code>	Source-derived profiles, exact selectable groups, model timestamps, available variables and default query.
Saved investigations	New evolution and blackout recipes. Evolution retains the selected node. Both pin model and observation-library hashes and their own methods. Original and modified blackout records are separately reproducible. Earlier methods and source identities remain unchanged.

Current evolution method: `p14-native-overlap-v2`. Blackout method: `p14-blackout-v1`. The first evolution method and its public replay failure are documented below.

Evolution is bounded to seven source frames, 512 regions per frame, 2,048 region instances per run, 8,192 overlapping pairs per transition and 200,000 footprint indices. The existing native engine also bounds cells and region construction. A zero sensitivity delta explicitly disables additional runs; a positive delta evaluates both shifted thresholds. A between interval shifts both endpoints. Work that exceeds bounds returns an explanation, not a truncated correspondence graph. The graph draws at most 12 connected nodes per frame, labels that presentation limit and leaves all calculated regions selectable.

Blackout accepts at most 128 profiles and 100,000 source samples per bounded library. Selections are sorted and deduplicated; their union removes each profile once. Unknown

selections are rejected. Source files are checked even when earlier caches are warm. The baseline includes source-backed alternate eligible timestamps, so exports retain references to all contributing model frames.

Exports provide evolution regions, overlap links, transitions and sensitivity tables, or blackout coverage, profile effects, removed eligible sample identifiers and changed statements. Numerical values are unrounded in CSV/JSON. Null residuals after complete exclusion remain missing, not zero. Depth contacts distinguish the selected depth interval from the original native domain boundary.

Demonstration

1. Choose Arabian Sea, then Evolution. The default 26 C, 0-300 m condition supplies seven native snapshots, 21 region instances and 26 overlap links. Two split groups and two merge groups occur under the declared rule. Select a graph node and inspect both overlap fractions, geographic extent and available observation context.
2. Compare the nearby threshold rows. Open the advanced controls and choose every second frame to demonstrate explicit 24-hour gaps. No links are inferred across those gaps.
3. Choose Bay of Bengal, then Blackout. The initial checked time has 206 eligible pairs. Select one contributing profile and apply the exclusion: 103 pairs remain. Removing an already ineligible glider changes the selected profile count without reducing the eligible-pair count. Removing all contributing profiles leaves no residual metric, rather than a zero error.
4. Save original evidence and modified evidence separately. Recalculate a saved record and download its evidence ZIP to inspect exact source sample indices, settings, source identities and numerical tables.
5. A Pacific blackout explicitly displays monthly potential temperature and its full averaging interval. It keeps the original observations inspectable but produces no incompatible instantaneous temperature residuals. Pacific monthly evolution remains unavailable.

Initial numerical and local acceptance

The initial complete backend suite passed 798 tests with zero skips and three existing warnings. The renderer/geographic suite passes 31 tests. These totals include the new API/save/replay cases and relevant earlier phase regressions; focused suites are not added to those totals again.

The evolution oracle reads the original packed HYCOM data and independently calculates segmentation, native geometry and overlap correspondence using separate SciPy, Decimal and set-based calculations. Its 56-test suite covers 42 frame/threshold combinations and 36 transition combinations. The generated fixture rebuilds byte-for-byte. The separate review checks all 26 emitted default Arabian overlap links exactly and both selected-depth boundary labels. See [independent method review](#) for what each check does and does not establish. Blackout's 46 tests include small arithmetic fixtures, original eligible sample identities, source invariance and unchanged cached evidence; the combined blackout/evidence run passes 131 tests.

Local HTTP acceptance passes 48 checks across 72 requests with no failures or retries. Eight new investigations match the local methods, capture, replay and export exactly. All 36 retained earlier investigations replay successfully. The archive check compares numerical CSV content, saved settings and source references with the checked record. It establishes transport consistency; its reused CSV helper is not an independent arithmetic oracle.

Recorded reports: `evidence/p14-backend-final.xml`, `evidence/p14-renderer.log`, `evidence/p14-evolution-fixtures-final.xml`, `evidence/p14-evolution-reference-rebuild.json`, `evidence/p14-independent-links.json`, `evidence/p14-local-http.json` and `evidence/p14-local-records.json`.

The subsequent [independent blackout review](#) identified two integration errors: original source sample identifiers can exceed a grouped profile's row count, and a monthly source file can list several timestamps rather than one. The browser parser now preserves sparse original identifiers, and blackout exports retain matching multi-timestamp model source files. Both errors were reproduced before correction. Their original reports are `evidence/p14-parser-sparse-initial.log` and `evidence/p14-monthly-source-initial.xml`.

The corrected API and existing saved-investigation suite passes 40 tests, with two existing warnings, in `evidence/p14-source-closure-tests.xml`. All 34 frontend numerical, geographic and parser tests pass in `evidence/p14-renderer-and-parser-final.log`; the three new parser tests are included in that total. Final local HTTP acceptance passes 49 checks across 76 requests, with zero failures or retries. Its nine records include a Pacific monthly blackout with zero incompatible residuals and the original GODAS source checksum. All 36 earlier investigations still replay. Reports are `evidence/p14-local-http-final.json` and `evidence/p14-local-records-final.json`. Scientific result identities and earlier phase reference behavior were not changed.

Corrections retained in the evidence

The review corrected missing labels for the requested depth interval, while keeping native source-boundary contacts separate. A blackout CSV column now says source or matching exclusions because those counts include the matching gates as well as source QC. Browser review corrected duplicate accessible labels, an empty-graph imaginary node, late responses overwriting newer control edits, and misleading Pacific instantaneous/in-situ labels. Residual-change validation now compares exact metrics even when rounded explanatory sentences happen to match.

Initial reports remain preserved. An early evolution test incorrectly expected a valid 5000 m source depth to be rejected. A blackout corruption test changed only JSON whitespace, which does not alter semantic source identity; it now changes actual source metadata. The first exploratory browser run had five passes and three failures: two duplicate-label failures and a request-wait error in a test. The subsequent 48-case run passed before the last three targeted scenarios were added. No source values, source QC, exact replay checks or numerical tolerances were weakened to obtain a pass.

Final interface and public acceptance

The initial broad browser regression passed 174 of 180 scenarios. Its six failures were two test assertions repeated across Chromium desktop, WebKit desktop and Chromium mobile: a fuzzy control label also matched a hidden Evolution control, and the generic disabled-state matcher misreported a disabled native option. Exact label selection and native `disabled` property checking corrected those assertions. The report and traces remain in `evidence/p14-local-browser-final.json` and `evidence/p14-browser-broad-initial-traces/`. Despite its filename, that JSON is the initial broad run, not a green final result.

The corrected Evolution, Blackout and structure-search suite passes all 75 checks, with zero failures, skips or flaky cases, in `evidence/p14-local-browser-corrections.json`. It includes the 57 new Phase 14 scenarios across three browser projects. The unchanged earlier navigation/evidence/climate/saved flows retain their successful checks from the broad run.

The local visual harness captured 43 views with zero page/console errors, visible alerts or document overflow. It exercised real saves and downloaded an actual evidence ZIP. Desktop, 320/390 px layouts and 200% root text are included. Nine actual captures were opened and inspected. Small pluralization corrections and a visible graph-scroll hint followed that inspection; the final public run below covers them. See `evidence/p14-visual-local-initial/checks.json` and visual review.

The initial 0.14.0 TypeScript/Vite build passed, retaining the existing large-chunk warning. `evidence/p14-build-release-final.log` identifies `workspace-BTy1L6-a.js`, `scene-DI_dmGZR.js`, `RegionMesh-DuSyiAHQ.js` and `workspace-BPdJCEkt.css`. Production deployment `dp1_5xCUoD13f6e7hSbrksvLNtEoqu8h` is READY on the existing project; its immutable URL is `https://depth-atlas-seifuku.vercel.app/`. The public alias remains `https://depth-atlas-seifuku.vercel.app/`. Deployment evidence is `evidence/p14-vercel-deploy.log`.

All 19 anonymous release checks pass in `evidence/p14-public-release.json`: `health/version`, exactly five cases, public pages, legal information, favicon, required notices and every built JS/CSS asset match the local production build. Five bounded `source/config/document/log` paths return 404. This is a release-content check, not an exhaustive security audit.

The first public scientific run did not pass: 61 checks passed and six failed across 88 requests, with no retries. Complete comparison isolated six Bay temperature differences and four salinity differences to copied NumPy region volumes and weighted means. Native memberships, frame totals, overlap links/fractions and sensitivity summaries matched exactly in the checked responses. Diagnostic requests, identities and every different leaf are retained in `evidence/p14-v1-cross-runtime-diagnosis.json`. A concurrent replay also reached the 30-second function limit. The first public browser run passed 81 of 84 scenarios; three WebKit evolution response waits timed out. Initial reports and traces remain in `evidence/p14-public-http.json`, `evidence/p14-public-browser.json` and `evidence/p14-public-browser-v1-traces/`.

The correction uses P14-only ordered `math.fsum` for region volumes, means and weighted means, under `p14-native-overlap-v2`. Source fields, native memberships, geometry, P07 methods and the independent oracle remain unchanged. An eight-result per-instance cache retains completed calculations, validates source integrity before every lookup, shares duplicate concurrent work and returns isolated copies. Eviction, failures and warmed-source corruption are tested. This corrects repeated work in the current P14 paths; it does not complete P15 scaling benchmarks. D049 records the choice.

All 64 focused v2 numerical/source/cache checks pass in `evidence/p14-evolution-v2-cache.xml`. The independent link checker again passes 28 checks in `evidence/p14-v2-independent-links.json`. All 34 frontend numerical/geographic/parser checks pass in `evidence/p14-v2-renderer-parser.log`. The corrected local HTTP run passes 49 checks/76 requests with no retries, including nine new records and 36 older investigations (`evidence/p14-v2-local-http.json`, `evidence/p14-v2-local-records.json`). All 30 v2 evolution and legacy-archive browser scenarios pass across the three supported projects in `evidence/p14-v2-local-browser.json`. V1 files remain readable and their original JSON can be downloaded unchanged, while

replay against v2 explicitly reports a different method.

Final 0.14.1 acceptance

All **807 backend tests** pass with zero failures or skips and three existing warnings in `evidence/p14-v2-backend-final.xml` and its log. The focused 64 evolution/source/cache checks are included in that total. Final TypeScript/Vite builds successfully with the retained bundle-size warning (`evidence/p14-v2-build-final.log`). Final assets include `workspace-fwVeVJXC.js`, `scene-CthBvSyi.js`, `RegionMesh-0_93-nOP.js` and `workspace-BPdJCEkt.css`.

Production deployment `dp1_B8ZWmGAqL3kunpzdwareaX41ZKDJ` is READY at [the existing public app](#), with [this immutable release URL](#). Its log is `evidence/p14-v2-vercel-deploy.log`. All **19 anonymous static-release checks** pass in `evidence/p14-v2-public-release.json`, including exact served assets, five cases and legal/source notices.

Final public HTTP acceptance passes **67 checks across 94 requests** with zero failures or retries in `evidence/p14-v2-public-http.json`. Nine Windows-generated investigations and their numerical exports reproduce exactly on Vercel, and all 36 earlier investigations replay. The separate reverse run passes **19 checks/19 requests**, reproducing all nine public records and exports on Windows (`evidence/p14-v2-public-to-local-replay.json`). The public and local record files are `p14-v2-public-records.json` and `p14-v2-local-records.json`. Original source fingerprints and full numerical precision remain intact.

The final public browser suite passes **87 scenarios**, with zero failures, skips or flaky cases, across Chromium desktop, WebKit desktop and Chromium mobile emulation (`evidence/p14-v2-public-browser.json`). It ran concurrently with public HTTP acceptance, including the previously timed-out evolution paths. This verifies the tested workload; it is not a wider load or performance benchmark.

The final public visual harness captures **43 views**, with zero page/console errors, visible alerts or document overflow. Original and modified saves and a 168,638-byte evidence ZIP work. The scroll hint and singular-count copy corrections are visible. See `evidence/p14-visual-public-final/checks.json` and the actual inspection record.

No blocking defect remains in the checked P14 paths. These checks do not establish water-parcel identity, operational forecast impact or full ocean validation. The existing P09.5 unfamiliar-user pilot remains open.

31 Phase 15: wider coverage and scale

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_15_WIDER_COVERAGE.md

Status: P15.1-P15.5 and X27-X29 complete, 26 September 2026. Release 0.15.2 is deployed and verified on the existing Vercel project. P16 has not started.

Scope and source contracts

The Regions workspace adds bounded requests over two pinned sources. It is part of the existing app and shares its renderer, native values, provenance and explicit-error conventions. Earlier scientific labs remain attached to their five verified case sources.

- NOAA GODAS monthly potential temperature and source salinity, September and October 2022. Native 28 x 418 x 360 grids cover 5-459 m and approximately 74.5 S to 64.499 N. Original 0-360 longitudes and irregular latitude/depth levels are retained. Potential temperature converts kelvin to Celsius in Float64. Salinity converts source kg/kg to g/kg, without claiming practical salinity or TEOS-10 Absolute Salinity.
- There is no arbitrary-region observation library. Existing Argo, glider, CTD and BGC support remains in the curated cases.

Method and resources

Method `p15-native-subset-v1` selects original source indices in an eastward-unwrapped bounded region. Finite ascending rectilinear latitude/vertical axes and one cyclic eastward longitude axis are required. Curvilinear grids, unsupported dates/variables, polar coverage, empty coordinate selections and oversized requests are explicit errors. Missing land/ice/source support is not filled, and no separate ice classification is invented.

Requests allow at most 90 longitude degrees, 40 latitude degrees and 120,000 native values. Query bodies are limited to 8 KB; imported/replayed envelopes to 4.5 MB and canonical numerical payloads to 4 MB. At most four regional operations execute per server instance; excess requests return 429. These are per-instance bounds, not distributed account quotas.

A 12 x 12 x 8 preview precedes a display using at most 40 x 40 horizontal points and all selected native vertical levels. Profiles, CSV and regional packs use native values. Server caches allow 40 MiB / three native arrays and 12 MiB / twelve responses. Source

checksums are rechecked before cache hits. Browser display cache is 8 MiB / six fields. Display settings do not change the scientific export.

Cancellation stops the browser from waiting and ignores late results. A bounded server calculation already started can finish. Adaptive/basic display is separate from native precision. Regional requests do not fetch user-provided URLs.

Portable evidence

A pack carries exact canonical JSON, source/packed-file hashes, native coordinate indices, method, query and values, with provider uncertainty where applicable. Local SHA-256 checks detect corruption; only online replay compares the pack with the currently pinned original sources. No signature or source authenticity is inferred from a self-contained checksum.

One explicit regional pack can be saved on the device, replaced or deleted. JSON import works without network calls. A separate self-contained HTML viewer opens with no app server or connection. This is not a promise that the entire website reloads offline. Privacy and Terms describe actual transmission and storage; operator identity/contact details remain blank as requested.

Initial verification and corrections

Initial full backend: 853 passed, three failures. One regression changed the missing-source HTTP status of older investigations and has been corrected without weakening the earlier contract. Adapter-count expectations were updated to the explicit four supported adapters. A unit-label encoding defect was fixed. Initial browser run: 36 passed across Chromium desktop, WebKit desktop and Chromium mobile emulation. Subsequent review corrected saved-pack replay cancellation, changed-selection comparison, single-level section availability and offline cell selection alignment. Final acceptance evidence is recorded below.

Demonstration

Limits and next work

This is historical, source-specific wider coverage, not live global analysis or independent ocean validation. Browser emulation does not establish physical-device support or actual unfamiliar-user comprehension; P09.5 remains open.

Source acquisition and reproducibility

`science.acquire_wider` retrieves GODAS in four-depth slabs because an initial larger provider response was truncated. Metadata pins units and native coordinates.

`science.acquire_gobai` reads bounded, explicitly acknowledged HTTP byte ranges from the original CDF-2 file, checks source ETag/Last-Modified identity during acquisition and retains the original range hashes. The 11 GB original is not downloaded in full and no full-file hash is claimed. Both commands are maintainers' preparation tools, not public arbitrary-fetch endpoints.

Measured resources

Measurements are small reproducible samples on this Windows computer, not capacity forecasts. The source preparation audit and fresh-process/cache benchmark are separate from public network measurements. Working-set memory is measured using the Windows process API; an initial measurement returned zero because its handle signature was missing, so those zero readings are invalid and superseded by `p15-local-performance-final.json`. OS disk cache was not flushed. A fresh store is used for the first preview; subsequent display/native stages reuse the source array.

The corrected local working-set samples are about 94-100 MiB, including the Python process. Source array and serialized response caches remain inside their 40/12 MiB ceilings. The browser cache has its independent 8 MiB/six-field bound. Four simultaneous HTTP clients completed both measured rounds, with deterministic saturation/recovery checked separately. Elapsed calculation durations are recorded; CPU utilization was not sampled, and there is no claim of a deployment-wide memory ceiling or sustained throughput guarantee.

A Chromium CDP regional-request throttle of 400 ms latency and 50,000 bytes/second download/upload produced a preview response in about 0.94 s and completion in about 3.43 s locally, with Save-Data selecting Basic view. The app shell had already loaded, so this is not an initial-page-load benchmark. The final hosted sample measured a 0.92 s preview response and 3.44 s completion under the same regional-request throttle. Cancellation ignores late responses and preserves completed data; the server can finish already-started bounded work.

React and interaction review

The Regions workspace is imported only when first opened. Shared scalar rendering does not require a different ocean calculation. Aborted requests and generation counters protect against late responses; source identities distinguish cached fields. Effects dispose renderers/listeners and stop background work when the workspace is hidden. Source/variable/month/quality controls have accessible labels, status/error regions identify progress and failure, and the inspected 320 px view has no document overflow. Display-only decimation keeps offline native packs below renderer axis limits without changing profiles or exports. These technical checks do not close the real-user pilot.

Final pressure-axis correction

Manual inspection of the first public screenshots found a clipped leading digit on a long pressure-axis tick label. Automated document-overflow checks had passed because this was text inside a canvas. Release 0.15.2 reserves a wider left margin for pressure sections and uses the same plotting geometry when interpreting clicks. A browser regression measures actual drawn text and independently checks that a click at the middle of the known longitude range requests 70 E. It passes in Chromium desktop, WebKit desktop and Chromium mobile. Existing metre-based plots keep their original geometry. No numerical source, method, export or cache setting changed in this visual patch.

The requested stop/restart command for the previous local server was blocked by automatic policy review. A new task-owned server was started non-destructively on port 8016 instead. The previous port 8015 is not the final acceptance server. No permission workaround or unrelated process termination was used.

Local transfer and cache samples

Source / resolution	First call, seconds	Cached call, seconds	JSON bytes	Gzip bytes
godas-2022 / preview	0.0746	0.0076	26078	7986
godas-2022 / display	0.0163	0.0113	430659	80422
godas-2022 / native	0.0171	0.0122	483972	89557

Gzip sizes above are measured compression of the JSON bytes, not a claim that every hosting response uses gzip. First-call timing is a fresh store only for preview; later

resolutions share the loaded native array. Browser cache reuse is separately checked by counting zero repeated subset requests.

Final acceptance record

- **Backend:** `.venv/Scripts/python.exe -m pytest tests -q --junitxml=docs/evidence/p15-v1-backend-final.xml` passed **857 tests**, zero failures/skips and three existing warnings. This includes 50 P15 tests. The final 0.15.2 patch changes pressure-axis drawing/click geometry and release identity only; scientific Python code is unchanged from that run.
- **Frontend/build:** `npm --prefix web run test:science` passes **39** checks in `evidence/p15-v2-frontend-science.log`. `npm --prefix web run build` passes TypeScript/Vite in `p15-v2-build-final.log`. The existing large-chunk warning remains.
- **Original sources:** `science.verify_p15_performance --source-audit` checks eight arrays, **28,964,160 original values**, all exact including missing values. The corrected memory/size/cache report is `evidence/p15-local-performance-final.json`. `science.acquire_wider` and `science.acquire_gobai` preserve all eleven prepared-file hashes on rebuild, in `evidence/p15-source-rebuild.json`.
- **Public transport:** `.venv/Scripts/python.exe -m science.verify_p15_public --base-url https://depth-atlas-seifuku.vercel.app --output docs/evidence/p15-v2-public-http.json --records-output docs/evidence/p15-v2-public-records.json --replay-records docs/evidence/p15-local-records.json` passes **23 checks across 79 requests**, without retries. Seven regional records match local native/display/profile/CSV calculations and exact pack identities. Reverse replay against local port 8016 passes **nine checks across sixteen requests** in `p15-v2-public-to-local-replay.json`.
- **Earlier phases:** retained replay of **45 earlier investigations**, with all nine P14 numerical export bundles, passes **55 checks/requests** on Vercel in `p15-v1-public-prior-replay.json`. That run precedes the pressure-axis-only UI patch. The same 55 checks pass locally. No older method or scientific fingerprint changed.
- **Browser regression:** the broader 0.15.1 public run passes **113 scenarios**, with one pre-existing WebKit graphics-context-loss skip and zero failures, in `p15-v1-public-browser.json`. After the pressure fix, direct Node Playwright runs over the wider suite and independent 3D/Basic coordinate test pass **45 scenarios**, zero failures/skips/flaky results, across Chromium desktop, WebKit desktop and Chromium mobile emulation. Final report: `evidence/p15-v2-public-browser.json`. The three focused local pressure-axis checks also pass.

- **Visual/offline:** `node web/benchmarks/p15-visual.mjs`
`https://depth-atlas-seifuku.vercel.app/docs/evidence/p15-v2-visual-public` captures **20 views**. Eighteen layout snapshots have no overflow/alerts; there are no recorded page/console errors. A 645,943-byte oxygen offline viewer opens with networking disabled and makes zero HTTP requests. Nine final screenshots were opened and inspected, including the corrected full pressure label at 320 px. See `evidence/p15-visual-review.md`.
- **Concurrency/performance:** `node web/benchmarks/p15-evaluators.mjs` passes four isolated simultaneous Chromium workflows through source selection, regional load, profile, device save and exact online replay. Region-load samples are **1.37-2.90 seconds**; sampled JS heap is about **9.9-24.4 MiB**. This is one computer and a small sample, not a 100-user guarantee. `science.verify_p15_performance` also passes two four-client HTTP rounds, **eight requests**, in `p15-v2-public-performance.json`, with sampled request durations **0.51-1.27 seconds**. Saturation, body limits, cache bounds, corrupt/missing files, cancellation and recovery have separate regression checks.
- **Deployment:** `npx --yes vercel@59.23.2 deploy --prod --yes --scope kai-vexen-studios` publishes to the existing project. Final deployment `dp1_4eG8oBzSgXxaL8XceTA767SUEmn` is READY at `https://depth-atlas-seifuku.vercel.app/`. The normal public URL serves the same release. `science.verify_p15_release --url https://depth-atlas-seifuku.vercel.app --report docs/evidence/p15-v2-public-release.json --deployment-id dp1_4eG8oBzSgXxaL8XceTA767SUEmn` passes **21 anonymous checks**, including API identity, all eight built JS/CSS assets, legal pages/favicon/notices and five bounded private-path 404 checks. This is not an exhaustive security audit.

All initial reports and failures are retained. The first full backend run was 853/856, not initially green. The initial source-hash and Windows memory verifier assumptions were corrected, and the canvas label defect found by actual visual inspection was fixed. The local focused pressure browser command reported a shell exit of 1 despite its JSON/list reporters confirming three passes, zero errors/failures/skips; final public runs independently pass those scenarios. No acceptance tolerance, original numerical field or older investigation fingerprint was weakened.

32 P16A: imported observations in a connected model view

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt |
Source: docs/PHASE_16A_IMPORTS.md

Started 26 September 2026; accepted 27 September 2026, Asia/Calcutta. Release 0.16.5 is deployed on the existing Vercel project. Final deployment evidence is recorded in PROJECT_STATE.md.

Delivered scope

P16A owns review items 1, 4, 5, 6, 11, 12 and 13. A supported observation file is reviewed before it enters the temporary workspace. Its eligible in-domain coordinates feed the shared instrument overlay. Opening a profile retains the model view and its selected timestamp. The observation sample has its own timestamp, position, source field, units and QC.

Desktop uses adjacent model and profile panels. Narrow screens stack them with a bounded, keyboard-accessible model panel. Location maps, tracks and detailed provenance expand on demand. Closing the profile restores the full model space. Basic graphics retains selectable instrument-location controls.

The preview lists detected variables and units, accepted, excluded and missing readings, coordinates, time/depth coverage, excluded-sample reasons and unsupported fields. Add commits to this tab; Discard or Cancel retains previously added profiles. Cancel aborts the browser request and ignores a late response; it does not guarantee interruption of parsing already running on the service.

CSV mapping supports explicit column selection for documented quantities. Kelvin converts to Celsius by subtracting 273.15; chlorophyll mg/L converts to mg/m³ by multiplying by 1000; ug/L has the same numerical value as mg/m³. Practical salinity accepts the declared psu/PSS-78 naming. Oxygen and nitrate require umol/kg. Coordinates, pressure, timezone-aware time and QC must already use the documented meanings. No density-dependent conversion, salinity redefinition, QC-scheme translation or arbitrary formula is offered. Canonical measurement columns cannot be relabelled as different quantities/units. Mapping changes require another review before Add becomes available.

Native NetCDF and CTD Exchange retain source metadata and processing rules. The original file hash survives CSV mapping; normalized-table and mapping hashes are recorded separately. Raw CSV inspection explicitly uses converted display units. Unknown source files do not inherit an official URL from their filename. Only an exact public-example byte hash receives the known original source URL.

Real examples and scientific limits

The machine-readable overlap report evaluates every example against all five main model cases using coordinate-eligible samples and the actual model date range. Intersection is not a scientific comparison or independent validation.

Example	Source date and location	Current model relationship
Argo 2903891 cycle 12, native and derived CSV	9 January 2024, 13.0667 N, 89.9833 E	Inside Bay rectangle/date range. Native file has 102 coordinate-eligible samples. The CSV contains 20 source levels from the same profile, not independent observations.
BGC 6903091 cycle 100	28 September 2023, 9.1326 N, 20.7304 W	Outside all five main model cases. Original oxygen adjustment and QC remain inspectable.
IOOS ru29 glider subset, four profiles	19 April 2024, near 17.7794 N, 67.0579 W	Outside all five main cases. Moving sample coordinates/times are preserved. Contradictory salinity range metadata retains its inspect-only policy.
CCHDO CTD station	21 February 1996, 33.1947 S, 28.0603 E	Outside all five main cases. Source WOCE flags and absent oxygen remain explicit.
New BGC 1902594 cycle 34	29 March 2024, 12.6794 N, 85.3031 E	Inside Bay rectangle, outside its January date range. Chlorophyll is an observed quantity, not a new model field.

The remaining dependency for simultaneous glider/CTD co-display is a supported model covering the actual example coordinates and dates, or an openly available, QC-documented observation subset inside the existing model domains/dates. No such matched source pair is supplied by this checkpoint. New model input is P16B; imported numerical matching is P16C. Source types are supported by the shared location contract, but this phase does not claim a real simultaneous four-instrument/model comparison.

Chlorophyll provenance

The source is [Argo GDAC/Coriolis SR1902594_034.nc](#), 518,900 bytes, SHA-256 1368b43fca3d535e3d4745cfcc0c477706ce5e121137ae18e0bf4ac9a613bc2c. It contains 1,475 original sample positions; 1,171 chlorophyll readings pass the implemented source/coordinate QC policy. CHLA_ADJUSTED is selected under the source's A mode. CHLA_FLUORESCENCE remains a distinct unsupported field. Source calibration metadata, raw values, adjusted values and flags remain available.

General source guidance: [Argo data FAQ](#) and [BGC-Argo chlorophyll reprocessing](#). These explain source processing; they do not independently validate this application. Future downloads can change after provider reprocessing. The preparation command rejects changed bytes instead of silently replacing the reviewed file.

The new example lives in `casepacks/instruments/import-examples.json`, separate from the frozen numerical library. An initial attempt to append it to the library changed the fingerprint required by saved investigations. The replay regression caught this. The original library bytes and fingerprints were restored; the new public example uses the normal preview/import workflow. Existing comparisons, source counts and older saved replay retain their original inputs.

Demonstration path

1. Skip the tutorial, open Tools, then Instruments. The model remains visible beside the profile.
2. Open Import observations and Formats and genuine example files. Preview D2903891_012.nc, inspect counts, then Add profiles to workspace.
3. Close the profile and select its Imported marker. Choose a reading and View nearest model depth. The model date does not change to the observation date, and no comparison is claimed.
4. Preview SR1902594_034.nc and Add. Chlorophyll is selected by default. Inspect the accepted curve, original source field and separate observation/model dates.
5. Use a supported CSV with alternate names to show explicit mapping, review and unit conversion. Test fixtures are analytical test data and must never be presented as real ocean observations.
6. Clear imported profiles. The checked library remains available. Reloading also removes temporary imports.

Verification

- `tests/test_import_preview.py`: native source comparisons, known chlorophyll count/hash, raw/adjusted values and missing samples; analytical Kelvin/chlorophyll conversion; malformed/ambiguous mappings; original identity; bounded request/response; no case-pack writes; all six examples; immutable library identity.
- Relevant existing parser, evidence, provenance, investigation and guided-case regression: 165 checks passed in `evidence/p16a-science-accepted.log`. This includes older saved investigation replay. Five API/release checks pass in `p16a-release-api.log`.
- `import-review.spec.ts`, `instruments.spec.ts` and `instrument-context.spec.ts`: review/Add/Discard/Cancel, mapping/QC, imported marker selection/clear, Basic fallback, original instrument types, chlorophyll values/dates, moving-position timestamps and dateline fixtures. The first settled set passed 45/45 across Chromium desktop/mobile and WebKit. Final-release and broader regression evidence is listed in the handoff.
- `science.verify_p16a` verifies every genuine example and its model overlap. With `--url`, it also checks that served example bytes and parsed JSON equal the local references.
- `web/benchmarks/p16a-visual.mjs` captures connected and chlorophyll views; manual findings and final public results are recorded separately.

Retained intermediate failures: the initial review-size check counted JSON spacing and wrongly rejected the glider response; compact wire-size accounting fixed it. The initial parser used an exception attribute that did not exist and accepted a malformed units type too late; both are corrected and covered. Adding chlorophyll to the frozen library broke replay and was replaced by the supplementary import catalogue. A first browser run accidentally included Firefox, whose launcher returned `spawn UNKNOWN`; no Firefox success is claimed. A mistyped Python test filename produced no tests; that failed command is retained separately from the accepted run. Existing build chunk-size and dependency deprecation warnings remain.

The first Vercel attempt rejected a 225.18 MB optimized function bundle against its 225 MB cap. The prior public release remained available. The packaging correction excludes developer browser benchmarks, client science-test files and offline `science/verify_*.py` commands from the server function. These files remain in the local source project. Runtime modules and all existing source-pack bytes are retained. Future dataset expansion must budget deployment size; this is not unlimited hosting capacity. The public verifier also initially compared Python coordinate tuples with JSON arrays; normalizing the local reference to its JSON wire form corrected that harness mismatch without changing numerical tolerances or product output.

The first public Python example check timed out while reading an uncompressed response. The verifier now requests and decodes the existing gzip response, as normal browsers do, and records per-example progress/failures. The final run verified all six served files and exact parsed JSON against the local references. The original timeout log is retained; this does not establish universal network reliability. No runtime API change or numerical tolerance change was made to get that result.

Remaining boundaries

Limits remain 2 MB input, 24 profiles and 5,000 samples per file, 3.5 MB decoded review and 24 profiles/12,000 samples per browser session. The application does not save uploads. The Privacy page still matches request-only processing, filenames in URLs and temporary in-tab profiles.

Imported numerical comparison/save/replay is P16C. Broader synchronization is P16D. Larger/batch imports and refresh are P16E. P17 security work remains deferred. Automated checks do not replace actual oceanography-user testing, independent scientific review or testing on every evaluator device.

33 P16B: fresh model input and portable setup

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_16B_FRESH_DATA.md

27 September 2026. Release 0.16.7 is accepted and deployed on the existing Vercel project.

What changed

A newly acquired HYCOM case covers 28-30 March 2024 in a bounded Bay of Bengal area. Five instantaneous snapshots retain all 40 native output depths and the 26 by 13 horizontal grid. Temperature, practical salinity and earth-relative currents pass through the existing adapter, preparation, API and scientific renderer. This is fresh acquisition of different historical files from the existing provider, not a new model family or live feed.

The real chlorophyll example SR1902594_034.nc now has a model case that includes its location and date. Import it manually to see both together. The model has no chlorophyll quantity. Imported numerical comparison and persistence remain P16C. No glider or CTD temporal/geographic overlap is claimed.

Reproduction

```
.venv/Scripts/python.exe -m science.acquire_case --case bay-bengal-2024-03
.venv/Scripts/python.exe -m science.prepare_case --case bay-bengal-2024-03
.venv/Scripts/python.exe -m science.prepare_standards --case bay-bengal-2024-03
.venv/Scripts/python.exe -m science.verify_p16b --reference
.venv/Scripts/python.exe -m pytest tests/test_fresh_model.py -q
.venv/Scripts/python.exe -m science.verify_p16b --url http://127.0.0.1:8026
--report docs/evidence/p16b-local-http.json
```

Source: https://tds.hycom.org/thredds/dodsC/GLBy0.08/expt_93.0 Provider description: <https://www.hycom.org/dataserver/gofs-3pt1/analysis> The acquisition journal in `casepacks/bay-bengal-2024-03/source-metadata.json` records request indices, exact timestamps, original variable metadata and SHA-256 fingerprints. Reconstructed NetCDF subsets retain original packed source values and float32 packing attributes. No spatial interpolation or missing-value filling is applied. A repeat acquisition verifies cached fingerprints instead of silently replacing them.

The prepared pack is 726,586 bytes. The native CF exchange is 1,595,051 bytes. These are file sizes, not runtime memory or total function-bundle size. All original case and observation-library files remain unchanged.

Extension work

Extension	Actual maintained changes	Boundary
Another compatible HYCOM case	One recipe declares bounds, exact dates, title, adapter and observation policy	Source coordinates and physical definitions must pass the registered adapter
Preparation and acquisition	Removed the fixed seven-frame assumption for recipes with explicit dates; optional observations produce no invented provenance	Original recipes retain their behavior
API registration	CaseStore includes maintained recipe IDs after the existing five IDs	User-provided paths never become registry keys
New case in browser	Existing case selector, scientific renderer, palettes and native-value inspection are reused	Tools with separate source requirements still need explicit availability
CF export and THREDDS	Date-aware export title/history; launcher registers integrity-checked exchanges	Not arbitrary NetCDF support or a public service

Extension	Actual maintained changes	Boundary
New variable/product/sensor	Existing trusted adapter, product and instrument contracts remain; CF exchange reimports every new native field; P16A proves chlorophyll observations	Public variable IDs, units and UI metadata remain maintained allowlists. No zero-code arbitrary sensor/variable claim

Heat & Depth requires its separately acquired surface time series. Expedition and Evolution also retain their explicit seven-frame January method gates. The catalogue now derives unavailable tools from the scientific registries and the case metadata, so the March case offers a supported January case before opening these tools. No method gate was loosened. Tool selection still contains those two checked fallback cases. This is recorded coupling, not universal automatic discovery. These availability rules also apply when changing cases while a tool is open. The wider connected workflow remains P16D.

Standards and CF

The existing January THREDDS service repeated 143 source/protocol checks. A fresh March service on loopback port 8097 passed 123 checks covering source-to-exchange values and masks, WMS 1.3.0 images and geolocated readings, WCS 1.0.0 coverages, and strided DAP2 requests. The differing count follows five versus seven source snapshots.

```
.venv/Scripts/python.exe -m deploy.run_standards --base .runtime/p06/p16b-march-server --port 8097
.venv/Scripts/python.exe -m science.verify_p06_standards --base
http://127.0.0.1:8097/thredds --prefix p16b-march-accepted --case bay-bengal-2024-03
```

Use the pinned runtime installer in `deploy/run_standards.py` when Java/Tomcat/TDS are not already installed. Port and runtime base are configurable; only loopback is bound. Native downloads and REST are public on Vercel after deployment. Persistent public THREDDS remains unconfigured.

The local scoped CF-1.10 checker and adversarial metadata tests pass. An independent IOOS compliance-checker 6.1.0 installation was attempted in an isolated environment. Its cf-units wheel build failed because UDUNITS2_XML_PATH/native UDUNITS support is absent on this Windows host. No full independent CF certification is claimed. Keep the installation failure log. THREDDS also logs a Windows EPSG-cache path error; the tested CRS:84/EPSG:4326 paths pass, but other projections are not verified.

Initial March protocol checking retained a fixed January image width/height in the verifier. This made the two image-shape assertions fail. Requests now use the new native horizontal dimensions. All original value and coordinate tolerances are unchanged; both the initial failure and final report remain.

Scientific and interaction evidence

- p16b-science.log: 117 passing tests, including complete native-array source hashes and CF reimport, retained replay and supported import boundaries.
- tests/fixtures/p16b-source-reference.json: 25 independently decoded arrays, five variables over five timestamps, with original file hashes and selected reference values.
- p16b-local-http.json: 35 checks over 33 serial requests, exact model replay/export and native download checksum. These timings are measurements, not a concurrency capacity claim.
- p16b-browser.json: 51 passing desktop Chromium, mobile emulation and Windows WebKit checks. Six are repeated pure marker-location fixtures. Browser checks cover the fresh case, imported chlorophyll, tool availability, old saved investigations and standards links.
- p16b-protocol-verification.json and p16b-march-accepted-protocol-verification.json: retained January and March protocol/source checks.

Visual review, clean installation and public checks are recorded at final acceptance. Docker is not present on this machine; its recipe is not claimed executed. No actual INCOIS installation, independent oceanographer review or physical-device testing is claimed.

Demo path

Skip the introductory tutorial, choose Bay of Bengal 28-30 March 2024, then choose 29 March 00:00 UTC. Inspect the native 100 m temperature value and switch to a depth slice. Open Instruments, Import observations, Formats and genuine example files, Preview SR1902594_034.nc, then Add profiles. Inspect the chlorophyll curve beside the correctly dated model. Explain that this is real observation/model co-display, with no model chlorophyll comparison. Use the data provides the native NetCDF and source identity.

Remaining coverage

Area	Current March case support
Explorer, slices, isosurfaces, currents, native picking and source metadata	Existing engine with fresh source fields; tested rendering and native values
Instrument co-display	March BGC import overlaps geographically and temporally; no new frozen-library profiles
Model save/export/replay	Exact verified model investigation
Imported comparison/save/replay	P16C
Surface heatwave series	Not supplied for March; choose a checked January case
Glider/CTD matched evidence	Still needs a suitable source-model pair
WMS/WCS/DAP2	Tested local January and March services; no persistent public endpoint
Expedition / Evolution	March is outside their checked case sets; choose January before opening
Docker / INCOIS installation	Not executed / not supplied

Additional boundary found during acceptance

The initial 0.16.6 candidate passed core viewer/import checks but exposed Expedition and Evolution for March because the old menu inferred all lab availability from the presence of currents. Their backends correctly refused this unverified case. The broader lab check caught both errors on desktop and mobile. Release 0.16.7 adds catalogue availability derived from the actual method registries. Users are offered a checked January case before either tool opens. Switching to March while already in an unsupported tool returns to Explorer with an explanation. The original method restrictions and source fingerprints remain unchanged.

Two older tests pinned exactly five cases and only the January protocol case. Those expectations were updated to six cases and the two verified Bay protocol routes. Their original failed runs remain. The final regression passes all 106 cases. No numerical tolerance was relaxed.

The clean source baseline also required `.gitattributes` to preserve scientific JSON bytes. All 304 staged case-pack files were compared byte-for-byte; all 260 pre-existing scientific pack files are identical to the P16A checkpoint. Large Functions is configured after its real fallback path passed. These fixes address packaging and availability; they are not the deferred P17 security audit.

Final acceptance

The final code baseline is local commit `3cd0f399bdce4f1bf195b7a50763753e7913929f`. A fresh source archive was extracted into `.runtime/p16b-portable-final`, installed using `requirements.txt` and `npm ci`, built and served on port 8028. Its 35 HTTP checks and 24 Chromium desktop/mobile and WebKit checks passed. Dependency consistency passed. This is a separate installation environment on the same host, not another operating system or institution.

Production release 0.16.7 is READY as `dpl_Ceg6EQpB7TpR3PpzSsWUaMBmZuqq` at <https://depth-atlas-seifuku.vercel.app/>. Public results: 24 release/asset checks, 35 scientific HTTP checks, 36 desktop/mobile browser checks and six visual captures without horizontal overflow/page errors. The final public serial HTTP sample had a 0.4885-second median and a 3.0495-second maximum over 33 requests, including a native download. Final isolated local requests had a 0.0279-second median and a 0.1083-second maximum. These include client/network conditions and do not establish concurrent scalability or guarantees.

Before the final availability change, 117 science/adaptor/import/replay tests, 106 core regression checks and 60 client science tests passed. The final capability patch passed 88 affected backend checks and 24 fresh-case browser checks. Counts overlap and must not be added into a unique-test total. The older 51-check browser suite and initial 26-check public suite are retained as candidate evidence, not a replacement for final checks.

Evidence index: `p16b-acceptance.json`, `p16b-final-public-release.json`, `p16b-final-public-http.json`, `p16b-final-public-browser.json`, `p16b-portable-final-summary.json`, `p16b-capabilities-browser.json` and `p16b-visual-review.md`. Initial failures, CF installation limitations and intermediate release evidence remain. All P16B.1-P16B.5 checks are complete within this documented scope.

34 P16C: imported comparison and reproducible saving

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_16C_IMPORTED_REPLAY.md

Work date: 27 September 2026. Release 0.16.8 is deployed. All four acceptance steps pass for the declared scope. Public and local evidence is recorded in PROJECT_STATE.md and evidence/p16c-acceptance.json. This checkpoint owns follow-ups 15-18, extending R05/R06, X03, X10 and X11.

What changed

A supported imported observation can use the existing native-column comparator. The original file is reparsed for every comparison, coverage, capture, replay and verified export request. The service does not accept client-supplied normalized profiles as scientific input. The selected file supplies its own temporary observation catalogue; the frozen bundled library and its source hashes remain unchanged.

Comparison retains the existing variable-definition, source-review, coordinate QC, reading QC, physical depth, geography, time-window, distance, mask and adjacent-depth interpolation gates. Temperature means in-situ Celsius and salinity means practical salinity. No potential-temperature substitution, temporal interpolation, masked-gap bridging or chlorophyll model value is introduced. Residuals are model minus observation; summary statistics use eligible source samples only. Coverage and discovery explicitly cover the selected imported file. A reference can compare settings within that file; changing between files or the bundled library clears the old reference.

The real BGC 1902594 cycle 34 file dated 29 March 2024 overlaps the March HYCOM case. At the 29 March 00:00 UTC snapshot and the default 6-hour, 5-km, 500-m limits, 245 temperature samples and 243 salinity samples qualify. These are observed counts for this file and configuration, not a performance or accuracy score. Chlorophyll remains an observed estimate with its original adjustment and QC; the model has no chlorophyll field. HYCOM assimilation prevents treating agreement as independent validation.

Source identity and replay

- Original bytes are embedded as bounded base64 with filename, SHA-256, parser version and explicit CSV mapping. Source URLs are never fetched during replay. Recognized public examples retain their checked attribution by byte hash.
- Parser: observation-preview-v1. Imported workflow: p16c-original-input-v1. Comparison mathematics: p05-native-column-v1. Pressure-to-depth and native source metadata remain those of the existing adapter. Parser changes require a new version and explicit migration policy.
- Source identity binds the model manifest, imported catalogue, original-file context and all reparsed profiles. The settings/result fingerprints use the existing tagged-json-f64-sha256-v1 cross-language representation.
- Reopening reparses the embedded original, verifies parser/method/source identity, recalculates outputs and checks exact result hashes before restoring the workspace. Corruption, unavailable models, changed mappings or unsupported versions stop replay. Hashes establish consistency, not authenticity, ownership or scientific truth.
- Saved imported profiles preserve selected variable, sample, excluded-value preference and model context snapshot. The new optional snapshot field is omitted when absent, preserving old instrument recipes. Comparisons retain snapshot, variable, matching policy, selected sample and optional reference policy.
- Existing archived records keep their original recipe, library and result identities. An old result's historical capability sentence about inspect-only imports remains inside the fingerprinted archive. Current UI and readable reports explain the current imported-comparison capability without rewriting historical numerical modules.

Storage, sharing and limits

Save on this browser is explicit. IndexedDB retains the original file, mapping, parsed profiles, settings and results. Existing limits remain 25 records, 32 MB total and 8 MB per local record. Removing a local copy does not delete downloads. Clearing imported profiles clears that active workspace and its imported comparison query cache, without deleting separately saved records. Browser cleanup or storage eviction can remove local records.

JSON downloads and evidence ZIPs include the original uploaded file. The ZIP also provides the unmodified file under original-observation/, full-precision numerical CSV, settings, source references and a readable report. Only share files that the uploader is authorized to distribute. Imported data is excluded from URL sharing; model data remains a pinned external dependency. HTML/CSV are readable evidence, not complete offline

replay engines. Reopening needs the app service and matching model pack.

The service processes uploads in request memory, does not write upload files or saved records, and bypasses the bundled comparator's process-wide caches for imported profiles. Comparison/replay/export send the embedded file back to the service. Privacy and Terms describe this behavior, with the existing operator and no-affiliation wording retained. This is not completion of the deferred P17 security audit.

Original file limit: 2 MB. Existing parser bounds: 24 profiles and 5,000 samples per file. Workspace: 24 profiles and 12,000 samples. New JSON request envelope: 3 MB. Imported investigation response budget: 4.2 MB; an oversized result returns an explicit smaller-input suggestion. Local records remain limited to 8 MB. Larger uploads, batch jobs and persistent server storage are not added by this checkpoint.

Demonstration path

1. Open the March Bay of Bengal study case and choose 29 March 2024 00:00 UTC.
2. Tools > Open instruments > Import observations > Formats and genuine example files. Preview SR1902594_034.nc, review its variables/QC and add it.
3. Inspect chlorophyll and its source flags. Choose Compare with model to compare temperature, or switch the comparison variable to salinity. Read eligible counts, residuals and exclusions.
4. Keep a reference. Narrow the time window to one hour and apply it. The temperature comparison becomes empty because the actual observation is at 02:30:44 UTC; the six-hour reference remains available.
5. Save investigation, name it and choose Save on this browser. Download JSON or the evidence ZIP. A replay link is intentionally unavailable for imported data.
6. Reload, or use another browser context. Open the record from Saved or import the JSON, then choose Recalculate and reopen. Matching source, settings and results are required before restoration.

Verification and boundaries

tests/test_imported_investigations.py checks original TEMP values/QC from the raw BGC NetCDF; independent great-circle column selection, physical-depth interpolation and metrics from the checked native CF exchange; mapping and missing/QC behavior; source/parser/method/result failures; request budgets; exact exports; and no mutation of the frozen library or import retention through its global caches. Existing investigation, import, evidence, source-provenance, instrument and fresh-model tests are retained.

web/e2e/imported-investigations.spec.ts exercises fresh-context portable replay, reload/deletion, mapped Kelvin CSV, rejected tampering, source switching and matching sensitivity. Existing saved investigations, import review and March model paths are regression checked. Browser automation is not a substitute for an oceanographer or unfamiliar-user study.

science.verify_p16c compares served results with fresh local calculations, checks original bytes in ZIPs, rejection paths and four archived P08 modes.

web/benchmarks/p16c-visual.mjs captures desktop/tablet/mobile controls, scientific charts and save disclosure. Actual accepted run counts, public URLs and reviewed images are in PROJECT_STATE.md and the acceptance report.

Glider/CTD examples still need appropriate co-located model cases for a real simultaneous demonstration. No universal model compatibility, independent scientific certification, offline exact server replay, cloud account storage or operational forecast skill is claimed.

35 P16D: connected scientific investigation

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_16D_CONNECTED_WORKFLOW.md

Started 27 September 2026 after the user's "lessgo". Accepted release: 0.16.9. This record owns follow-ups 19, 20, 21, 22 and 29. Final accepted deployment and evidence are recorded in PROJECT_STATE.md.

Implemented scope

Source sample identity and supported variable now travel between source profile and comparison. Both retain the shared model timestamp. Eligible comparisons position the native model probe in the matched column at its nearest native depth; the numerical residual still uses the existing native-depth interpolation. Unmatched samples, independent model picks, changed time and changed variable do not become invented observation matches. Profile saves retain the model context time for both bundled and imported data. Legacy records omit that optional field and retain their old replay identity.

The comparison has a co-visible model panel, with an explicit option to expand the comparison. Narrow screens stack the model and charts. Existing responsive charts, Basic graphics fallback, matching controls and tool access remain. The moving Basic/In-depth tutorials and the prepared guided case remain separate from the new five-step

Investigation Builder. The builder walks through choosing data, exploring depth, inspecting a reading, comparing and explicitly saving. It performs no automatic save, preserves its step during a lab detour, resets when the case changes and states unsupported monthly comparisons.

Find structures adds an optional observation-support panel for the selected region. Its plan view shows that region's native footprint, columns with eligible pairs and coverage gaps. Rings make supported columns visible without enlarging their native cell area. Each accepted sample exposes the source date, physical depth, observed/model values and residual. Opening a pair restores its matching rules, model snapshot, profile and exact source sample in comparison. Matching sensitivity preserves a reference for the same structure and observation source. A source or structure change clears that support reference.

Scientific contract

POST `/api/cases/{case_id}/features/support` takes a P07 query and region ID, compatible P05 matching settings and an optional bounded P16C original-file envelope. The query and matching variable/time must agree. Monthly potential temperature and unsupported observed quantities are rejected. Source/QC/time/distance/native-depth/mask gates are unchanged. Imported bytes are reparsed with the same request-scoped stores, without global file/result caching.

Eligible pairs are associated with the selected region by their matched native column and source depth's midpoint bin. The depth centre and source depth must both lie within the queried interval, following the existing P07 rule. Distinct native cells, distinct columns and profiles are counted once even when several samples land there. Residuals are model minus observed at the source depth, not a bin-centre difference. Observations need not satisfy the model's threshold. Exclusions apply to the selected collection, not an inferred count inside the structure.

Coverage counts are neither confidence nor volume fractions. A supported plan-view column can retain unsampled depths. Gaps describe the chosen collection and matching rules, not absence of every observation. Model agreement may involve assimilated observations and is not independent validation. No source dates, model arrays or existing P05/P07 responses are rewritten.

Support method: `p16d-native-support-v1`. Request body remains at most 3 MB; imported originals remain 2 MB. Support collections are bounded at 128 profiles/100,000 samples and decoded JSON output at 4.2 MB. Oversized output rejects rather than silently truncates. No new upload, hosting or concurrency capacity is claimed.

Save and export boundary

The existing comparison/structure records still provide recalculated replay. The support panel exports its complete accepted rows, applied rules, source/model identities and optional sensitivity reference as a JSON result snapshot. Imported snapshots include source, parser, parsed-profile and mapping-context identities but do not embed original bytes. A support snapshot is not a replay bundle. Save the structure or linked comparison separately for checked replay. Builder progress and support references remain in the tab until reload. Privacy and Terms reflect these operations.

Demonstration

Open Build investigation and follow the five steps with the January Bay case. Select a source reading, compare it and inspect the distinct dates. Use Find a useful comparison when the selected snapshot has no pairs. Save explicitly, download and reopen.

For structure support, choose the January Bay case at 7 January 12:00 UTC, default temperature threshold 26 degrees C and 0-300 m. The largest region has 34 eligible pairs in two profiles, touching 23 of 80,203 native cells; 80,180 cells have no eligible pair under those rules. These are source-backed counts, not validation percentages. Retain the support reference and set a zero-hour time window to see the eligible count become zero without changing the structure or source dates.

For the imported path, choose March Bay at 29 March 00:00 UTC. Preview/add SR1902594_034.nc in Instruments. Find structures at temperature ≥ 20 degrees C over 0-300 m, then choose that imported file in Observation support. Open an eligible pair in comparison. The full profile has 245 eligible temperature pairs under default rules, while the structure panel contains only the associated subset. The observation keeps 29 March 02:30:44 UTC. Chlorophyll remains observation-only.

Acceptance and remaining evidence

Numerical acceptance uses independent irregular-bin boundary/duplicate fixtures, real P05 row equality, January association counts, March original-file identity and stricter-window empty cases. Client contracts reject changed residuals, counts, dates, native coordinates, settings and import identities. Browser checks cover linked selections, explicit saves, restored source samples, imported support, failed requests, monthly incompatibility, keyboard and responsive paths. Final local/public test logs and visual review are linked from PROJECT_STATE.md.

The participant script is P16D_USER_TASK.md. Automated interaction and manual screenshot review do not close the actual unfamiliar-user/domain-user gates. P17 security

and release audit remain deferred until authorized.

Accepted 27 September 2026: all four steps passed their declared gates. Backend regression: 163 passes; client science: 73; final public release: 24; support: 24; imported and legacy replay: 61; public browser: 34; public visual: 12 captures at three sizes. Counts overlap. Exact commands, timings, initial failures and corrections are retained in PROJECT_STATE.md and evidence/p16d-acceptance.json. The runtime was not changed to make the public HTTP/2 transport failure disappear; a separate unchanged-build repeat passed.

36 P16E: import workload, publication and offline continuity

Historical or maintained reference: read alongside the current release overview | Source: docs/PHASE_16E_CONTINUITY.md

27 September 2026. Release 0.16.10. The scoped acceptance record is evidence/p16e-acceptance.json; public connection limitations and failed checks remain part of that record.

Delivered scope

P16E owns follow-ups 14, 23, 24 and 30. Batch review remains mounted while model metadata arrives or the model context changes. It reads up to eight original observations sequentially, shows each result and lets the user add ready files explicitly. A failed file, cancelled queue or workspace quota failure preserves earlier successful additions. CSV mapping and QC review remain available per file. Duplicate profile IDs replace their earlier copy rather than consuming another sample quota. Original input, parser identity, units and scientific matching methods are unchanged.

Sources > Data publication and updates shows the serving dataset identity, historical coverage, source retrieval dates and publication check time. A status check preserves its previous information when offline, cancelled or timed out. The public site publishes checked case packs with an application deployment. A maintainer-only local pipeline supports isolated candidates, integrity checks, atomic activation, durable status and retained previous releases. There is no public dataset-write endpoint, live-data claim or scheduled provider acquisition.

Saved investigations > Offline viewer downloads a self-contained HTML for observation profiles, model comparisons, comparison references and native model columns. It preserves exact saved values, irregular depths, missing samples, QC and source identity. Observation dates and model dates are shown separately above the chart. It includes variable/sample controls, source context, a plot and the original investigation JSON. The UI states the snapshot age and unavailable computations. Other laboratory modes keep existing JSON, CSV and report exports. The existing regional offline pack also remains available.

Save, replay and verified online exports expose cancellable request stages. Cancellation ignores late client results and keeps earlier records. A server calculation that has already started may finish within its existing bounds. The short IndexedDB commit stage cannot be cancelled midway; it is labelled Writing local record. Quota/storage failure leaves verified portable evidence available without claiming a successful local save.

Measured and enforced budgets

Boundary	Limit
Each original observation	2 MB, 24 profiles, 5,000 source samples
Each parser review response	Existing 3.5 MB decoded JSON limit
Batch review queue	8 files, 16 MB originals, 24 MB decoded reviews
Imported workspace	32 profiles, 20,000 samples, 16 MB originals, 24 MB decoded profiles
Imported replay/comparison request	Existing 3 MB JSON envelope
Imported investigation response	Existing 4.2 MB limit
Local investigation record/library	Existing 8 MB per record, 25 records / 32 MB total
Offline viewer data envelope	16 MB, including original record and prepared series; HTML escaping and viewer code add bytes
Prepared publication	250 MB, 5,000 files, 100 MB per file
Maintainer publication store	1 GB including retained releases; no automatic deletion
Import/status/investigation request deadline	30 / 15 / 45 seconds per client request

MB means decimal bytes. Queue and workspace can coexist, so their budgets add. Base64, object overhead, temporary JSON serialization, React and scientific scenes require additional memory. Limits bound accepted data; they do not certify a fixed browser RSS on every device.

`science.verify_p16e_workloads` parses six genuine source examples and eight generated resource fixtures. The generated fixtures are clearly marked tests and are never bundled as ocean observations. Their aggregate 20,000 samples occupy 1,693,480 original bytes and 13,611,208 normalized JSON bytes. Serial parser time was 2.339 seconds on this Windows machine. Tracemalloc measures Python allocations rather than all native memory. Browser measurements and deployment checks are separate evidence, with no concurrency or universal device claim.

Dataset publication procedure

1. Acquire and prepare supported sources in a separate checkout or staging tree using the existing source recipes and adapters. Retain provider timestamps, attribution, masks and original hashes. New case IDs or layouts still need their registered compatibility changes and scientific verification.
2. Use the compatible application version to run:

```
.venv/Scripts/python.exe -m science.publication --candidate C:/prepared/casepacks  
--store C:/ocean-publications --timeout 120
```

The candidate and store must be separate directories. The pipeline validates registered model manifests, declared file checksums, decoded array dimensions, observation libraries, example files and standards downloads. A portable, case-sensitive ordering of relative path strings makes publication IDs identical on Windows and Linux. An inventory binds every prepared file, including other laboratory packs; their existing module-specific gates remain unchanged. Validation is a compatibility/integrity check, not a new independent scientific certification.

1. Each success writes a new immutable-by-convention release under `releases/<attempt>/casepacks`, its full inventory and an atomic `active.json` pointer. Running readers keep their startup-pinned release. Failure, timeout, Ctrl+C or exhausted storage leaves the previous pointer untouched. Before starting an institutional/local service, set `OCEAN_PUBLICATION_ROOT` to the store. Startup rechecks the selected inventory and application version. Restart to activate a new pointer. There is no in-request swapping of scientific arrays.
2. For Vercel, do not set a writable publication store. `deploy/prepare_vercel.py` validates bundled case packs and generates `api/publication.json` before CDN packaging. Deploy the compatible frontend/API/data together on the existing project, then run `release`

and numerical checks. Vercel's previously deployed release remains the fallback if the candidate build fails. Runtime status exposes no server paths or uploaded files.

The job lock blocks concurrent writers. After an abrupt kill, the last stage can remain recorded and the lock stays in place. Confirm its recorded PID is no longer running before removing that exact lock and any unselected incomplete staging directory. Do not delete the selected release or any release still used by a running process. Storage reclamation and scheduler configuration require a maintainer. A status journal is not a job heartbeat. Power loss, clustered writers and distributed filesystem guarantees are not certified by these tests.

No persistent external scheduler, public THREDDS service or institutional deployment is available. The phase supplies a working bounded manual publication pipeline and records those infrastructure dependencies. A refreshed publication does not change historical measurements to 2026 observations. Changed data/method identities stop old replay; keeping an old directory does not automatically route replay to it.

Offline inventory and data rights

The downloaded HTML contains its viewer script/styles, original saved investigation, result arrays, QC, settings, references and any original import already embedded in the saved record. No fonts, tiles, textures, libraries or network services are required. A SHA-256 envelope is checked before displaying data; this checks file integrity, not authenticity. A restrictive CSP blocks network access, forms and external content. Source-derived text is inserted as text, not executable markup.

Full model grids, server calculations, new observations, interactive 3D navigation and unlisted lab computation are omitted. Model comparison plots use existing saved pair/residual values. They do not recompute metrics or interpolate new samples. Native current components remain separate quantities. Missing values are omitted, excluded values are optional hollow markers and depth remains positive down. Chlorophyll remains an observed quantity with no model residual.

Save dates and snapshot age refer to the record, while source timestamps remain attached to data. Delete downloaded HTML/JSON files to remove those copies. Remove local copy in Saved deletes only IndexedDB records; clearing site data does not delete downloads. The original providers' licences and uploader's sharing permissions still apply. Privacy Policy and Terms describe these flows. No new account, analytics, external provider or upload storage was added.

Demonstration

Open Instruments > Import observations. Select genuine Argo and BGC files together. Review both, then choose Add all reviewed files. Select the BGC collection and inspect chlorophyll QC. Save the profile and download Offline viewer. Open that HTML with networking unavailable, switch variables and inspect an original sample. The file still shows its source time and exclusions. Recalculation remains an explicit online operation.

For publication status, return to Sources and open Data publication and updates. Explain the difference between observation/model dates, provider retrieval and publication checking. The maintainer demonstration publishes an unchanged genuine historical pack, adds another checked historical case, then rejects a corrupted candidate. A running process keeps its old catalogue until restarted, and the failed attempt preserves the last usable pointer.

Limits and next work

The final full public phase run had 20 passes and two failures: a Chromium HTTP/2 body-transfer failure, and a comparison assertion that ended while the result was still loading. All six targeted desktop/mobile recovery checks subsequently passed, including a deliberately interrupted transfer followed by explicit Retry. The original failures are retained. This establishes the tested recovery path, not universal network reliability. The public visual report is separate from the API/source and browser reports.

P17 retains final cross-feature performance/security audit, participant/domain-user testing, release freeze, capability wording review and submission preparation. P16E does not establish universal hardware support, scientific endorsement, operational forecast improvement, equal coverage everywhere or actual INCOIS installation. Full offline laboratory recomputation and live automatic provider updates are outside this declared implementation.

37 Phase 16: learning through real actions

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt |
Source: docs/PHASE_16_LEARNING.md

Completed implementation on 26 September 2026 and deployed as release 0.16.0.

Problem-statement priority

The official SIH2026 problem page was rechecked on 26 September:

<https://www.sih.gov.in/sih2026PS>, statement project brief. Its central request is a browser workspace combining depth-resolved model fields with instrument observations. The current statement retains the core rendering, profiles, ingestion, configurable display, lightweight backend, extensibility and open-standards requirements already recorded in docs/reference/official-2026/project-brief-statement.txt. It also explicitly includes outreach to students and the public. There is no published feature-weight table in this statement.

Our demonstration order is a product decision, not an official scoring scheme:

1. See a real model in 3D, open a cutaway, inspect depth, switch variables and time, and read the colour key.
2. Co-display measurements, inspect timestamped profiles, compare eligible readings, and demonstrate supported file ingestion.
3. Show source access and preservation of evidence. Explain browser access, backend and interoperability without making infrastructure the visitor's first task.
4. After the core tour, offer investigations: structures, Drift, Heat & Depth, Climate, Expedition, evolution, blackout and wider regions.

User-requested interaction

Basic is selected by default. It performs one real app action per step and briefly explains the resulting output. In-depth is deliberately selected and adds the concepts, method and limitations behind each feature, plus optional comprehension questions. Visitors control Next, Back, replay, practice and Skip. There is no forced continuous autoplay. The initial choice fits the viewport without scrolling. Shared investigation links retain direct replay access.

Tutorial actions call existing component handlers and use parsed result snapshots. Missing or failed requests cannot produce a success explanation. Source dates remain historical. No account or external model service will be invented to mark that integration complete.

Acceptance to run

- Both modes, every core and additional demonstration, Next/Back/Skip/replay/practice, keyboard, narrow/landscape layouts and reduced motion.
- Actual backend values and query parameters agree with tutorial outputs, source references and explanations.
- Unknown questions, forged values/source attribution, absent sources, delayed/stale requests and resource bounds.
- Existing scientific fixtures and representative prior workflows, then local/public visual checks and the deployed release verifier.
- Record exact outcomes and corrected initial failures. Real unfamiliar-user feedback remains the separate P09.5 gate.

Implemented design

There are 27 demonstrations: 17 essential steps followed by 10 optional lab steps. Model depth, co-located instruments and eligible model-observation comparison appear before the additional experiments. Visitors can choose any lesson, pause to experiment, resume, replay, or skip. The first choice is a short illustrated panel, not a scrolling document. Release 0.16.1 replaces the fixed coach with anchored callouts. On a small screen the highlighted control or chart stays beside or above the callout; Next stays available. Completed source values take priority over a repeated action sentence on short screens.

The same lesson action contracts are used in both modes. In-depth adds separate Watch, Understand, Method and optional question views. The question feedback teaches distinctions such as model versus observation, missing versus zero, and a display stretch versus a changed physical depth. No score or answer history is stored. Shared investigation links retain direct replay rather than being blocked by a new tutorial.

`web/src/learning/bridge.ts` connects each existing workspace handler to a current, parsed result snapshot. A lesson waits for its own command ID and an available result. Actual imports use the bundled public CSV/NetCDF files and the ordinary parser. Numerical simulations call the existing solver. Save opens the real form without automatically persisting, downloading or sharing a record.

Prepared short explanation

The tutorial operates Ocean Navigator one step at a time. First it opens real model fields, then checks source measurements and explains the output. Basic mode shows how to use each feature. In-depth adds the science, the method and the limits. Visitors can stop, try the controls and resume whenever they want. Explain reads the current result and its sources, so missing evidence stays missing.

Demonstration path

Open the app with Basic selected. Start, open the cutaway, inspect 100 m, and proceed to sensor markers, the Argo profile and eligible comparison. Show the actual count and explain why agreement is not automatically independent validation. Continue to time, salinity, currents, isosurface, section, display controls and genuine imports. Open the save form and close it without saving if no record is needed. At the core completion screen, choose the additional labs. The remaining steps demonstrate structures, eight fixed-depth passive particles over six historical hours, heat evidence, El Nino and La Nina, virtual stations, evolution, blackout and a bounded wider-source region.

Reopen Guide and explicitly choose In-depth to show Understand, Method and one optional question. Open Explain, review an exact-depth query and apply it. Ask for an unsupported depth to demonstrate the honest refusal path.

Limits and next work

The tutorial uses historical prepared cases, not live conditions or equal coverage everywhere. Basic graphics disclose their 2D fallback. Automated interactions and visual inspection do not establish that unfamiliar people understand the concepts; P09.5 still requires real participants. Operator/contact identity and persistent public THREDDS remain open dependencies.

Local acceptance and retained corrections

- `npm --prefix web run build` passed TypeScript and Vite for release 0.16.0. The existing large-chunk warning remains; see `evidence/p16-build-final.log`.
- `npm --prefix web run test:science`: 60 passed, no failures or skips, including 21 new bounded-query/explanation cases. See `evidence/p16-science-final.log`. The scientific Python algorithms, source packs, tolerances and method versions were not changed; the earlier 857-test P15 backend run is historical and was not rerun for this phase.

- The final local learning/UX suite passes 78 scenarios across Chromium desktop, WebKit desktop and Chromium mobile emulation (evidence/p16-local-learning-final.json). The later final button/input layout adjustment passes a focused 12-scenario run, including all 27 demonstrations in each engine (p16-final-layout-browser.json).
- The broader initial local regression has 163 passed, four failed and one existing WebKit context-loss skip (p16-browser-third.json). Its failures exposed a resume-state race and an enlarged-text mobile overflow. Both were corrected and their focused/full learning checks now pass. The original report is retained; it was not a single green 168-test local run.
- The final local visual harness records 31 captures with no measured overflow, welcome-fit, Next-visibility, input-width or page/console-error failures (p16-visual-release-candidate/checks.json). Screenshots were inspected, not only counted. The short-screen coach may scroll internally while Next stays visible; the initial tutorial choice fits without scrolling at the checked normal-text sizes.

Earlier checks found a wider-source readiness stall, a Retry button that needed to refetch failed data, inconsistent focus after Skip, an enlarged-text welcome overflow, a small-screen Next position, and a collapsed desktop explanation input. Those are fixed. Initial browser reports remain: 21/27 passed, then 59/66. Some first-run failures were test defects: 200 m is the selected case's native index 22, not index 27 (500 m), and two controls had incorrect accessible-name expectations. The selected option text now verifies the physical depth directly. One explanation test looked for the word 'eligible' where the correct text said 'matching rules'; the assertion was corrected without changing scientific behavior.

The full initial smoke reached 25 steps before the wider readiness defect. A corrected run completed 26, and the final sequence with the save-form lesson completed all 27 without recorded page/console errors (evidence/p16-third/result.json). The browser suite adds explicit checks of native values, 100 m selection, the 20 degree isosurface threshold, eligible pairs, applied particle count/duration, ENSO season changes, actual CSV/NetCDF imports and the blackout result.

Public acceptance and handoff

Deployment `dp1_DKCJL8hUP9kNgSPePaaOHFiawV5Z` is READY on the existing Vercel project. [Public app; immutable deployment](#). The deploy command was `npx --yes vercel@59.23.2 deploy --prod --yes --scope kai-vexen-studios` and exited 0; see `evidence/p16-vercel-deploy.log`.

The broad public browser run has **162 passed, 5 failed and 1 skipped** across Chromium desktop, WebKit desktop and Chromium mobile emulation. A focused WebKit/mobile run

with unchanged assertions and deadlines had **9 passed and one initial-case loading timeout**. A further isolated full WebKit tour then **passed all 27 demonstrations**, also with unchanged assertions and deadlines. The focused runs used one browser worker to isolate the timing observations. This was not a single green 168-scenario public run. Reports: `evidence/p16-public-browser.json` and `p16-public-focused-recheck.json`, followed by `p16-public-webkit-tour-recheck.json`. The retained skip is the existing WebKit graphics-context-loss scenario.

The initial hosted failures include an approximately 19.7-second JavaScript download, requests missing the client deadline, a delayed regional module, and a regional refinement exceeding its test deadline. They did not show an incorrect numerical result. The app retained explicit loading/error states. The trace alone does not establish whether origin, network or local browser scheduling caused the delays. A later passing run does not guarantee performance on every judge's connection. A separate three-request HTTP sample returned health and case metadata in about 0.53 s each and a display subset in 1.69 s, recorded in `evidence/p16-http-timing-sample.json`; this small sample does not explain or negate the browser timing failures. Timing observations are retained in `evidence/p16-public-timing-observations.json`; broader physical-device and performance work remains in P17.

The anonymous release verifier passed **21 checks / 21 requests**, zero retries, including API version, all five original cases, all eight built JS/CSS assets against local bytes, legal pages, favicon/notices and bounded private-path checks. Command:

```
.venv/Scripts/python.exe -m science.verify_release --url
https://depth-atlas-seifuku.vercel.app --report docs/evidence/p16-public-release.json
--deployment-id dpl_DKCJL8hUP9kNgSPePaaOHFiawV5Z.
```

The final public visual harness passed **31 captures** with zero recorded page/console errors or measured layout/control failures. See `evidence/p16-visual-public/checks.json` and `p16-visual-review.md` for actual inspection, corrected defects and short-screen scrolling limits.

P16.1, P16.2 and P16.4 are delivered, with X30 and the scope of X31. P16.3 remains conditional and unconfigured. P09.5, R12 and X14 still require real participant evidence. P17 is not started. No source array, backend scientific method, numerical tolerance or runtime dependency changed. There is no Git commit or remote; a local recovery archive is recorded separately, not represented as source submission.

Follow-up: a guide beside the feature

The user requested moving guidance after trying release 0.16.0. Release 0.16.1 replaces the reserved sidebar with measured callouts. Each lesson identifies a visible control or chart. The app moves that destination into view, points to it, and keeps Back, Next and Skip with the explanation. Controls remain interactive. Manual scrolling is respected, with a return action if the target leaves the viewport. Small-screen graphics make room for the guide, and enlarged explanation text scrolls separately from navigation.

The save lesson now includes a hint directly beneath the name field in the actual form. Closing that hint resumes the tour; it does not save anything. The explanation, all 27 real demonstrations and the scientific contracts are unchanged. Placement coverage is added in tutorial-anchor.spec.ts and the complete tour also requires every lesson to find its anchor. The follow-up is deployed. Corrected local learning/placement checks pass 54/54; the final narrow-value presentation passes 15/15 placement checks. The final local visual sweep has 41 captures with no recorded layout failures. Public API/assets pass 21/21, while browser runs retain intermittent request/script delivery failures. Exact outcomes and limits are recorded in PROJECT_STATE.md.

Follow-up: teasers and direct feature tours

The additional demonstrations were available only after the core sequence or inside the lesson chooser. The new picker introduces each feature with a question, an icon, its result kind and the actual number of guided steps. It appears from the welcome teaser, the active guide's Extra features action, the core completion and the end of an individual feature tour. Two cards appear at a time over four pages. Back, More features and Close are explicit controls. Basic remains the default; In-depth can be chosen in either the welcome or picker.

Feature teaser	Demonstrations launched
Drift	Calculate eight passive particles over six hours, then play the calculated paths
El Nino & La Nina	Inspect the supported 2015 comparison, then switch to the supported 2022 season
Heat & Depth	Inspect the checked surface heat event and available depth context
Ocean structures	Inspect native connected regions satisfying the displayed rule
Virtual expedition	Build and inspect a sampling plan from the declared prior
Feature evolution	Inspect native overlap links, including splits and merges

Feature teaser	Demonstrations launched
Sensor blackout	Exclude a profile and inspect remaining and lost comparisons
Explore other regions	Load a bounded region from the wider source controls

The individual routes stop after their own steps and offer another feature or independent exploration. The complete 27-step path and complete extra-lab path remain available. The same real handlers, numerical snapshots, source limits and moving anchors are reused. The picker itself launches no lab calculations. First actions wait for the case catalogue, so selecting a teaser before the catalogue arrives cannot lose its intended destination; Skip cancels the pending tour.

Implementation: `featureTours.ts` defines routes, `FeatureTourPicker.tsx` renders teasers, and `LearningExperience.tsx` manages the bounded sequence and return path. `App.tsx` supplies catalogue readiness. `feature-tours.spec.ts` checks all direct entries, prerequisite order, bounds, mode, return/cancellation, delayed catalogue and layout.

`p16-teaser-visual.mjs` captures normal and enlarged-text layouts. Actual acceptance and release evidence are recorded in `PROJECT_STATE.md`; the earlier public transport limitations and unfamiliar-user gate remain separate.

Release 0.16.2 is deployed and verified. The final new-feature suite passes 39/39 locally across the three checked browser configurations and 13/13 on public Chromium. The 60 frontend scientific/parser/explanation checks and all 21 anonymous release/API/asset checks pass. Final visual acceptance covers 29 captures without measured failures or page errors. The broader 90/93 local run retained three picker-fit failures, which were fixed and passed the final feature suite. Screenshots subsequently caught a completion-action visibility issue, corrected and verified in the final visual sweep. No tolerance or ready-result assertion was weakened. See `PROJECT_STATE.md` and `evidence/p16-teaser-visual-review.md` for exact commands, evidence and retained limitations.

38 Review of the 31 proposed follow-ups

Historical or maintained reference: read alongside the current release overview | Source: docs/CORE_COMPLETENESS_REVIEW.md

26 September 2026. Baseline: release 0.16.4. This is a targeted source/code/document review for planning, not a fresh full acceptance run or authorization to implement all items. The official PS page returned HTTP 403 during this review; interpretation uses the preserved official statement at `reference/official-2026/project brief-statement.txt`. Do not represent this as a successful live recheck.

Update, 27 September 2026: P16A is delivered in 0.16.5. Items 1, 4, 11, 12 and 13 now have accepted import integration, co-visible inspection, preview, mapping and QC paths. Item 6 has a genuine source-checked chlorophyll profile, with no chlorophyll model field. Item 5 has verified source-type ingestion and explicit overlap limits: real glider/CTD examples still lack a matching main model case. See PHASE_16A_IMPORTS.md and PROJECT_STATE.md for actual evidence. The table below is the retained pre-implementation audit, not a claim that these fixes are still absent. P16B has now delivered a fresh historical HYCOM input, measured extension changes, repeated local protocol/scoped CF checks and a clean archived-source installation. The independent full CF checker could not install on this Windows host, public persistent THREDDS is not configured, and actual INCOIS infrastructure has not been supplied. P16C now delivers items 15-18 for bounded supported imports: numerical comparison, explicit browser saves, original-source JSON/ZIP bundles and exact replay. See PHASE_16B_FRESH_DATA.md and PHASE_16C_IMPORTED_REPLAY.md for evidence and limits. P16D implements items 19-22 and 29 through linked selections, support/sensitivity views and the Investigation Builder. Final acceptance is recorded in PROJECT_STATE.md. P16E now delivers items 14, 23, 24 and 30 within the bounded batch, manual retained-publication and offline saved-evidence scope in PHASE_16E_CONTINUITY.md. Its source, failure/recovery and public visual evidence is recorded in evidence/p16e-acceptance.json. An intermittent public HTTP/2 transfer issue remains a P17 audit follow-up. P17 has not started.

The user's distinction is useful: 1-10 concern PS coverage or evidence, while 11-31 are optional product improvements and quality/demonstration work. The list is not a verbatim official 31-item checklist. Fresh ingestion need not mean live data, and deployment outside Vercel does not establish an actual INCOIS installation. Chlorophyll is explicitly present in the saved PS background and unified-observation gap.

Effort below is relative: small means bounded presentation/verification, medium means changes across connected components with acceptance checks, and large means new data lifecycle or architectural work. These are scope judgments, not delivery-time guarantees.

#	Item	Baseline finding / work needed at 0.16.4	Effort and priority
1	Observation imports alongside model	Import state reaches InstrumentWorkspace but OceanExplorer receives only bundled catalogue profiles. Connect imports to the shared overlay.	Medium, first
2	Fresh model ingestion	Offline xarray/CF adapters exist; curated CaseStore IDs remain allowlisted. Demonstrate a new compatible file through preparation, registration, API and viewer. Not an arbitrary browser model upload.	Medium to large, core
3	Extensibility	Four trusted adapters and a product registry exist. API/frontend allowlists and metadata still need maintainer edits. Measure actual changes for the fresh-input/variable example.	Medium, core
4	Unified display	Full InstrumentWorkspace hides OceanExplorer. Add a compact linked profile inspector that retains model context on larger screens and an explicit compact layout on small screens.	Medium, first
#	Item	Baseline finding / work needed at 0.16.4	Effort and priority
5	Multi-instrument overlays	Genuine examples exist, but separate historical collections do not	Medium plus data availability, core

#	Item	Baseline finding / work needed at 0.16.4	Effort and priority
		automatically overlap the model in space/time. Demonstrate suitable overlap or clearly label unsupported comparisons.	
6	Chlorophyll	Parser already defines chlorophyll, accepted units and Argo CHLA. Current stored profiles inspected here contain temperature/salinity/oxygen, with no chlorophyll field. Demonstrate a real QC-aware chlorophyll profile; 3D model chlorophyll needs its own suitable field and integration.	Medium for profile evidence; larger for new model field
7	WMS/WCS/OPeNDAP	Recorded local THREDDS protocol evidence exists. Persistent public THREDDS is unconfigured. Reproduce and distinguish local service from hosted REST.	Verification plus hosting dependency
8	CF	Project-specific CF-1.10 exchange checker and adversarial tests exist. This is scoped validation, not universal CF certification. Add an independent checker where compatible and report its scope.	Verification
9	Portable deployment	Docker/Compose artifacts exist; earlier records state Docker unexecuted. Fresh local standards runtime was tested. Run the full app on a clean second environment.	Medium plus runtime access

#	Item	Baseline finding / work needed at 0.16.4	Effort and priority
10	Scale	P15 has bounded cache/resource tests and small four-client/browser measurements. Verify representative final workloads; do not extrapolate to large-user guarantees.	Medium verification
11	Import preview	Parsed profile metadata and feedback exist after import. A clearer pre-use summary would strengthen the import flow.	Small to medium, with core imports
12	Variable mapping	Model adapter accepts explicit maintainer mappings. A user-facing mapper must restrict units/definitions to supported conversions.	Medium; bounded mappings first
13	QC feedback	Accepted counts, excluded values, per-sample reasons, flags and raw/adjusted readings already exist. Summarize across an import and preserve distinctions.	Small to medium
14	Batch/larger imports	Current parser/browser limits protect memory and response sizes. Larger/batched uploads require resource and cancellation design.	Large, optional later
#	Item	Baseline finding / work needed at 0.16.4	Effort and priority
15	Imported comparison	Explicitly disabled for temporary imports. Server evidence currently uses the checked library. Requires validated	Large, separate optional checkpoint

#	Item	Baseline finding / work needed at 0.16.4	Effort and priority
		imported profiles through matching, source identity and result contracts.	
16	Persistent imported investigations	Explicitly disabled. Must decide local-only portable data versus server persistence and update schemas, quotas, replay and policies.	Large, separate optional checkpoint
17	Portable bundles	Existing supported investigations export settings/results/provenance. Extending this to imported payloads is additional lifecycle work.	Existing core; large import extension
18	Replay	Source fingerprints/method/settings and strict replay exist for supported saved cases. Imported data/parser identities need an extension.	Existing core; import follow-up
19	Synchronized views	Existing links can move to nearest model depth and position while preserving model time. Co-visible selections must respect incompatible timestamps/coordinates.	Medium, with unified display
20	Observation support	Evidence coverage/residual and structure-support foundations exist. Review their integration and make gaps easier to read.	Small to medium, targeted audit
21	Sensitivity	Matching assumptions and feature-threshold/evolution sensitivity exist. Verify discoverability and result interpretation.	Targeted audit/presentation

#	Item	Baseline finding / work needed at 0.16.4	Effort and priority
22	Investigation Builder	Guided cases, tutorials and saved workflows exist. A new builder is an optional connected workflow, not an unimplemented PS control.	Medium; prefer improving existing flow first
23	Dataset refresh	Maintainer acquisition tools and source journals exist. Scheduled refresh, retained-last-good publication and user-facing status form a new operational pipeline.	Large, optional later
24	Broader offline	Regional offline packs already exist. General offline investigations require explicit dependency/data bundling and limits.	Medium to large, optional
25	Independent numerical checks	Original-source/reference fixtures exist. Re-run relevant checks and add checks for new integrations; external scientific review is separate.	P17 acceptance
26	Domain-user testing	P09.5 real-participant pilot remains open. Cannot replace people with automated browser tests.	Participant dependency, arrange early
27	Stable release	Shared client/API version and latest recovery fix exist. Freeze only after final regression and clean setup. A local versioned source baseline and clean source-archive installation were established in P16B.	P17 acceptance

#	Item	Baseline finding / work needed at 0.16.4	Effort and priority
28	Documentation	Current handoffs exist but older capability notes and checkboxes need reconciliation. Preserve historical evidence while making present status explicit.	Small to medium, throughout
29	Clear workspace	Recent tutorial and navigation work exists. Use model/profile integration and participant observations for focused adjustments.	Medium with core display
30	Long operations	Cancellation, progress and retained-state patterns exist in supported tools. Audit each changed import/analysis path; do not assume universal coverage.	Targeted medium work
31	Judge demonstration	Prepare one source-backed investigation from input through visual inspection and evidence, with other labs as optional branches.	Small to medium packaging

Full checkpoint sequence

Revised after the user challenged the two-checkpoint proposal. Two checkpoints contained several tasks, but the proposal also deferred substantive items without a full delivery plan. The expanded sequence gives all 31 items a phase owner. Items 11-31 retain their status as improvements and validation suggestions rather than becoming official PS requirements. The earlier relative effort/status findings remain useful; the phase table below supersedes their vague "optional later" scheduling.

Phase	Owns review items	Scope	Relative work
P16A	1, 4, 5, 6, 11, 12, 13	Imported overlays, co-visible model/profile,	Medium to large integration

Phase	Owns review items	Scope	Relative work
		multi-instrument and chlorophyll evidence, import preview, supported variable mapping and QC	
P16B	2, 3, 7, 8, 9	Fresh model input, measured extensibility, WMS/WCS/OPeNDAP, scoped CF verification and clean portable deployment	Medium to large; data/runtime dependencies
P16C	15, 16, 17, 18	Imported comparisons, persistent imported investigations, portable bundles and reproducible replay	Large data-lifecycle change
P16D	19, 20, 21, 22, 29	Synchronized scientific views, support overlay, sensitivity display, Guided Investigation Builder and clearer workspace	Medium to large connected UI work
P16E	14, 23, 24, 30	Bounded batch/larger imports, dataset refresh, wider offline investigations and long-operation handling	Large resource/lifecycle change
P17	10, 25, 26, 27, 28, 31	Measured final scalability, independent numerical checks, actual domain-user testing, release freeze, accurate documentation and judging demonstration; plus requested security review	Broad final audit and release work

Detailed unchecked steps and acceptance conditions are in ROADMAP.md.

Documentation, numerical/interaction checks, policy review when data handling changes, and Vercel verification happen as each phase is implemented. P17 repeats the necessary final cross-feature checks after the integrated changes; it is not the first testing pass. Arrange participants early. Public standards hosting and actual institutional deployment cannot be claimed without the necessary infrastructure and evidence.

No implementation phase, test run or deployment starts through this planning clarification. First proposed work remains P16A, then stop at its verified handoff before starting the next requested checkpoint. No feature checkbox is completed by this mapping, and existing functionality should be extended or reverified rather than rebuilt merely because an item appears in the list.

Code anchors checked

- `web/src/App.tsx`: imported state, the instrument workspace, bundled-only 3D instrument prop and view hiding.
- `web/src/instruments/InstrumentWorkspace.tsx`: temporary import storage, QC presentation, imported save/compare restrictions and depth links.
- `adapters/instruments.py`: CHLA/chlorophyll definitions and units already supported; do not describe this parser as wholly missing.
- `api/case_store.py`, `adapters/cf_model.py`, `adapters/registry.py`, `science/products.py`: curated case registration, explicit mappings and extension limits.
- `science/check_cf_exchange.py`, `tests/test_standards_extensions.py`, `deploy/Dockerfile` and P04/P06/P15 evidence documents: existing bounded standards, deployment and performance claims.

39 Frontend usability rebuild

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/FRONTEND_REBUILD.md

Date: 26 September 2026. Deployed and verified release: 0.15.3.

This is a usability follow-up to P09 and P15, not completion of P16. Scientific acceptance gates are unchanged.

What changed

- A three-page illustrated introduction appears when the app is opened. Next, Back, Skip introduction and Start exploring are explicit controls. There is no automatic advance or scroll inside the introduction. Guide reopens it. Saved investigation links open their replay panel directly.

- One Tools menu replaces the long navigation rail and the duplicated row of module buttons. Tools are grouped into exploration, evidence comparison and experiments. All twelve existing destinations remain available.
- The opening workspace contains a study case, variable, three main views, an ocean scene and a colour key. The old permanent left sidebar and duplicate case summary are removed. Source dates, units, selected native values, missing-data meaning and depth exaggeration remain visible.
- Isosurfaces and current vectors open from More views. Graphics, colour controls and instrument overlays open from Colour, depth and display settings. Instrument markers start off in new explorations; saved investigations retain their original overlay choice.
- Sources open in a drawer. Case details and instrument profiles remain available. The source drawer closes before opening a dataset inspector, avoiding stacked focus traps.
- Guided case opens the existing source-checked teaching example on request. It is separate from the short introduction and retains its scientific comparison and save path.
- Compatible study-case changes keep the selected tool. A tool that requires Indian Ocean fields offers explicit Bay of Bengal or Arabian Sea choices when the current case is in the Pacific. It does not pretend that an unsupported lab works on that Pacific source.
- The Drift setup keeps a map drawing action visible and opens precise coordinates on request. Calculation, playback, numerical settings, comparisons and saving remain available.
- A late climate calculation cannot change another tool's current study case. Returning to Climate applies the completed result's context.

Verification scope

The end-to-end story is: open the introduction, enter the checked historical case, request model fields and native profiles from the existing API, inspect their rendered values, switch tools, and save or replay the same numerical result.

Checks cover keyboard focus, mobile labels, 320 px layouts, short landscape screens, 200% text, reduced motion, all tool destinations, source drawers, scientific picking, missing samples, colour controls, time changes, failed requests, retained scenes, drift calculation, regional loading and saved replay. Scientific numerical fixtures and backend contracts are not relaxed to accommodate the redesign.

Early failures are retained in the evidence directory. The first run found an ambiguous variable label, a missing mobile Back label and WebKit overflow in inline field labels.

Those were corrected. Tests were updated to use the real introduction and Tools menu rather than the removed navigation. Text comparison in a save test was corrected to use rendered text consistently. A subsequent case-switch timing check motivated the explicit inactive-climate context guard.

The [public release](#) passes 128 browser checks across Chromium desktop, WebKit desktop and Chromium mobile emulation, with one existing unsupported WebKit context-loss skip and no failures. The frontend scientific fixtures pass 39 checks. All 21 anonymous release/API/asset checks pass. The older lab regression first passed 101 of 106 scenarios; all five corrected navigation expectations pass in a focused rerun. PROJECT_STATE.md records exact reports, retained initial failures and deployment identity. Real unfamiliar-user task evidence is still required for P09.5. Browser emulation does not establish usability for every person or physical device.

Ready-to-use explanation

Ocean Navigator now opens with a short visual introduction and one ready historical ocean case. Visitors can open the cutaway, click a depth layer and read an actual model value. They choose additional tools by the question they want to answer. Detailed controls and source records appear when requested, while dates, units and the distinction between models, measurements and simulations remain clear.

Demo: open the website, advance or skip the introduction, open the cutaway, inspect a value, change a depth, then open Tools. Choose Drift Lab to show an experiment or Instruments to inspect measurements. Open Sources to check provenance. Save investigation retains the applied scientific settings and evidence.

Privacy and Terms keep the existing operator/contact placeholders. The custom favicon and required source attributions remain.

40 Globe and tutorial review

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/GLOBE_AND_TUTORIAL_REVIEW.md

27 September 2026. Authorized follow-up to P16E. This is not the P17 final audit.

What determines the order

The source is the preserved official project brief statement in `reference/official-2026/project_brief-statement.txt`, acquired from <https://www.sih.gov.in/sih2026PS> with its source manifest. The live government page returned HTTP 403 during this review. The quoted requirements and original source files remain unchanged.

The PS asks for browser-native ocean analysis: full water-column model fields, located instrument profiles, supported file ingestion, display controls and an interoperable, extensible architecture. Outreach is also explicit. It does not explicitly require a globe. The globe therefore serves a concrete task: finding the data before opening a scientific view.

Requirement and delivery map

PS priority	Existing delivery reviewed	Tutorial or interface decision	Remaining boundary
Full water column: temperature, salinity and currents	OceanExplorer, source grids, volume/cutaway, native probe, depth slice, section, isosurface and time playback	First locate a case, then demonstrate the core scientific views before optional experiments	Bounded historical cases. Adaptive 2D fallback is labelled. No live global forecast claim
Located Argo, glider, CTD and BGC profiles	Instrument contracts, sample coordinates/QC, map and co-visible model/profile workspace	Globe offers actual instrument positions; core tour adds markers and opens a timestamped source profile	Not every source has a compatible model at its location and time. Glider/CTD examples are not claimed as eligible numerical matches
NetCDF and delimited input	Source-checked preview, explicit mapping, sequential batch, parser adapters, original-byte envelopes	Core introduces both formats. Extra tour shows review and a two-file batch	Maintained supported layouts and resource limits; not arbitrary scientific files
Chlorophyll in broader scope	Genuine BGC source profile, parsed units/QC and March model context	Dedicated location-to-pro file-to-save tour	A profile is a measured vertical record. No chlorophyll model field, global concentration map or bloom extent is invented

PS priority	Existing delivery reviewed	Tutorial or interface decision	Remaining boundary
Palette/range/linear/opacity/depth stretch	Existing display controls and scientific value contracts	Core explicitly distinguishes changing appearance from changing numbers	Log scale requires positive values; exaggerated depth stays identified
REST/OPeNDA P and OGC WMS/WCS	Public REST/native exchanges; prior local January and March THREDDS protocol evidence	Sources and Data access remain available; advanced architecture does not crowd the first tutorial	Persistent public THREDDS remains unconfigured. Vercel is not an INCOIS installation
CF-compatible data	Scoped CF metadata validation, exact export/reimport and original-array evidence	File tutorial explains coordinates, units and supported adapters	Independent full compliance checker remains blocked by native UDUNITS on this host. No certification claim
Extensible/scalable browser architecture	Shared contracts, registered adapters/recipes, fresh March case, bounded operations and portable setup	Builder and import tours expose the connected workflow; source status explains published historical releases	Measured bounded workloads are not operational concurrency capacity. Actual institutional deployment and P17 performance audit remain open
Outreach and understandable results	Basic/In-depth modes, real command execution, source facts, limits and optional questions	Basic remains default, skip is immediate, coach follows targets. Extra-feature count and navigation are prominent; completion points to Tools	Automated geometry and interaction tests do not establish comprehension by unfamiliar participants

Other implemented areas retained

Area	Purpose and tutorial treatment	Scientific distinction
Comparison and observation support	Model-minus-observation, source-scoped gaps, strict matching sensitivity	Geographic overlap is not an eligible pair. Coverage is not confidence; agreement is not independent validation
Investigation Builder	Connects region, view, profile, comparison and explicit saving	Actions use the same workspaces, not a separate simulated product
Saved/imported investigations and offline viewer	Original imports, settings, hashes, numerical results and checked replay	Save/download remains explicit. Offline viewer shows supported saved values, not a new offline calculation engine
Dataset publication	Checks release, source dates and retained publication status	A newer publication does not make historical data live. Publication is a maintainer operation
Drift	Calculate then play passive virtual particles	Simulation, not a forecast or sinking model
Climate	El Nino and La Nina historical Pacific comparison	Historical anomalies and available depth context, not causal attribution or prediction
Heat & Depth	Surface event and available depth evidence	Source dates, climatology and depth coverage remain explicit
Structures and evolution	Threshold-connected water and overlap across time	Neither proves an eddy nor the travel of the same water parcel
Expedition and blackout	Virtual sampling and removal of available evidence	Reconstruction/evidence experiments do not prove operational forecast improvement
Wider regions	Bounded source exploration outside prepared cases	Coverage and tool support are not equal everywhere

Implementation decisions

- Geography is an explicit workspace state, not the temporary placeholder while model metadata loads. Late metadata cannot replace the globe. Opening a model is a deliberate action.

- Model outlines use actual declared bounds, including Pacific longitude wrapping. Each case shows its source, dates, depths and available variables.
- Instrument markers use real source positions. Moving profiles identify their representative position and direct users to full sample coordinates/QC. A marker is not a coverage surface.
- The chlorophyll example is the already maintained original source, checked against its published size and SHA-256. Previewing locates it. Opening its profile uses the normal parser, original-byte envelope and bounded import workspace.
- The core tutorial follows the PS scientific workflow. Additional tours include the newer imports, support, builder, offline and publication paths as well as all previous labs.
- Feature pagination appears above the cards with a derived total, page/range text and a high-contrast, steady-glow More features button. The count comes from the tour registry.
- The coach still follows the actual target. Actionable targets and Next receive a restrained static highlight, with no cursor effects or continual pulsing. A completion callout points to the real Tools button.
- Basic mode describes the action and output; In-depth adds concept, method and limitations. Neither saves, exports or downloads personal data automatically.

Demonstration route

1. Skip the introduction to inspect the persistent globe. Select Arabian Sea or a Pacific historical case, then explicitly open its 3D ocean.
2. Return through Find on globe. Select Chlorophyll observations and load the public BGC example. Inspect the recorded date/location, then open its measured profile beside the March model context.
3. Open Guide and start Basic. It demonstrates model depth/time/variables, instrument evidence and supported input before optional tools.
4. At the essentials boundary choose an extra feature. The picker shows the true total and a prominent route to the next pair. Select support, import review or a lab.
5. Finish a tour and choose Explore on my own. Follow the arrow to Tools, then reopen any tool without restarting the guide.

Verification reports, exact results, remaining defects and deployed release are recorded in PROJECT_STATE.md at handoff. Pending tests are not acceptance evidence.

Release verification and open transport issue

Release 0.16.11 is deployed to the existing Vercel project.

evidence/globe-tutorial-acceptance.json binds the deployed runtime, executed checks and limitations. The final local navigation suite passed on Chromium desktop/mobile and WebKit; source calculations and replay checks passed. Local and public visual captures cover five views at three screen sizes. Public assets and API matched the release on the complete independent recheck.

The normal public interaction suite passed 23 checks and failed the desktop complete tutorial during comparison. A repeat reproduced the eight-second response-header timeout. The mobile full tour and explicit failed-comparison Retry passed. HTTP/1.1 completed the desktop tour in a diagnostic configuration only. Root cause remains unconfirmed and production settings are unchanged. This existing transport defect remains assigned to P17; bounded globe/tutorial delivery does not claim uninterrupted operation on every network.

41 Hosting capacity before P16B

Historical or maintained reference: read alongside the current release overview | Source: docs/HOSTING_CAPACITY_PLAN.md

27 September 2026. Planning and read-only inspection following the user's question. P16B has not started. Release 0.16.5, production settings, storage, providers and billing are unchanged.

What failed

P16A's first deployment reported a 225.18 MB optimized Python function bundle against a 225 MB build threshold. That bundle contains server code, scientific dependencies and included source data. Excluding development-only test/benchmark/verifier files allowed the next deployment to pass. All original source packs were retained. The successful log does not give an exact final optimized size, so no exact remaining headroom is claimed.

The local cached Vercel CLI 59.23.2 includes @vercel/python 14.2.0. Its dependency externalizer defines a $225 * 1024 * 1024$ byte threshold and contains the same post-optimization failure message. It also has a VERCEL_SUPPORT_LARGE_FUNCTIONS switch and a larger-bundle fallback. This corroborates the build path observed in the logs; it does not establish the exact remote builder version or project eligibility.

Current [official function limits](#) and [Python runtime documentation](#) describe a 500 MB standard Python limit and an eligible 5 GB Large Functions public beta. Therefore 225 MB is the internal check encountered by this deployment path, not a universal current ceiling for every Python project on Vercel. Larger-function support requires Fluid compute with Active CPU, subject to documented eligibility. A supported preview must establish actual behavior before relying on it or changing production. No environment switch has been enabled during this review.

Other relevant constraints

- The normal function request/response payload limit is 4.5 MB. Our input cap is 2 MB and the import preview cap is 3.5 MB. Increasing function package capacity does not remove payload limits. [Official limits](#)
- `vercel.json` explicitly sets `maxDuration` to 30 seconds. That is our current configuration, not Vercel's universal maximum. Higher supported settings depend on the project's runtime/compute configuration and plan; browser timeouts and request/job behavior also need coordinated changes. [Duration configuration](#)
- Memory, CPU, bandwidth and storage usage are separate constraints. This review did not measure current account allowance, costs, final bundle headroom or larger-workload performance.
- Filesystem inventory, build evidence and local builder findings are recorded in `evidence/p16b-hosting-planning.json`. Remaining case-pack bytes after the existing standards-NetCDF exclusion are input-file accounting, not an exact function-size measurement.

Recommended sequence

1. Make deployment capacity the first prerequisite within P16B, not a new product phase. Check the actual project/runtime settings and test the documented larger-package option on a preview. Measure startup, imports and existing saved replay, not just whether a build succeeds. Preserve the working production release until verification passes.
2. Separate growing immutable datasets from executable code. Prefer versioned object storage with pinned manifests/checksums, bounded subset fetches and caches. Retain a small, useful demonstration fallback where practical. A Vercel-native storage option is available; select storage based on measured size/access costs and the portable-installation requirement. This is a proposal, not a provisioned service or free/unlimited-storage promise.

3. Preserve explicit upload bounds for P16C. In P16E, larger imports should use authorized direct-to-storage upload and bounded processing with progress, cancellation and retention rules. Simply raising the current input limit is insufficient. Vercel documents this architecture in [client uploads](#). Any persistent uploads require corresponding policy/security changes at implementation time.
4. Use preprocessed subsets for interaction and measured background processing for longer work. If the eventual workload needs a separate compute worker, keep the public website on Vercel and evaluate a worker only against actual requirements. No backend migration is currently established as necessary.
5. Add deployment-size and representative request/memory checks to the release gate. Do not repeatedly delete useful files to make each new case barely fit. Do not buy a plan solely on the assumption that it removes this particular packaging limit.

Phase implications

Phase	Main implication
P16B	Fresh model data can grow the package directly. Verify packaging and the data-loading strategy first, then demonstrate a genuinely new source.
P16C	Imported matching and portable records require bounded payloads and calculation budgets. Local browser saving alone does not require a server database.
P16D	UI organization has relatively little direct effect on Python package size. Keep normal browser and analysis-performance checks.
P16E	Larger/batch imports, refresh and offline packaging need deliberate storage and job design. This is the highest infrastructure concern.
P17	Verify final limits, account configuration, resource controls, reliability and security with the deployed workload.

The current app remains verified at its declared bounds. These constraints do not require dropping product features or moving the public frontend away from Vercel. Future capacity must be demonstrated, not inferred from a plan name or a larger advertised limit.

P16B execution result, 27 September 2026

The actual linked project reports Fluid Compute enabled and the deployment reports the Hobby plan. The initial 0.16.5 capacity preview built and served health. The fresh-data preview `dpl_AxWbxLXeNVLviosmx7mcb9rpJBMQ` explicitly logged that `app.py` exceeded the standard size path and enabled Large Functions beta. Its health, fresh source subset and full native chlorophyll NetCDF import passed. CLI timings include process/authentication overhead and are not presented as server-only cold starts.

Configured the non-secret project setting `VERCEL_SUPPORT_LARGE_FUNCTIONS=1` for preview and production after that proof. Subsequent deployments exercised the same path. No billing-plan purchase, provider change, database, object store or arbitrary external data fetch was added. Production remains on the existing Vercel project. Actual final bundle headroom is not inferred from the pre-optimization size or CLI compressed function size.

The March prepared pack adds 726,586 bytes; its 1,595,051-byte exchange download is served through the CDN. All old scientific pack bytes are unchanged. Runtime import bounds remain 2 MB and the function duration remains 30 seconds. Future volume growth, larger uploads, background jobs and final concurrency/security checks still require P16E/P17 work. Large Functions is currently beta and is a deployment dependency, not an unlimited-scale claim.

Evidence: `p16b-project-capacity.json`, `p16b-deploy-runtime.json`, `p16b-new-data-preview.log`, `p16b-large-runtime-preview.json` and `p16b-large-function-setting.json`. Official sources above were rechecked during P16B.

42 P05 scientific review

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: `docs/P05_SCIENCE_REVIEW.md`

Reviewed on 22 September 2026. This document records method recommendations and an independent source audit. It does not establish that the complete Phase 5 implementation has passed acceptance. The implementation and test reports must establish that separately.

Source identity and physical definitions

The pinned model is HYCOM GOFS 3.1, GLBy0.08/expt_93.0, sampled at seven historical timestamps from 7 to 10 January 2024. The provider describes the NAVO NetCDF output as in-situ temperature on 40 standard depth levels. Its output grid differs from the model's internal hybrid layers. The original `water_temp` metadata also says in-situ temperature, with `degC` units and NAVO code 15. Converting this output to potential temperature before subtracting in-situ Argo measurements would compare different quantities. See the [HYCOM analysis description](#).

The checked Argo `TEMP_ADJUSTED` attributes explicitly identify in-situ ITS-90 temperature. `PSAL_ADJUSTED` identifies practical salinity. Compare temperature and salinity independently using compatible definitions. Do not require good salinity merely to compare an already measured in-situ temperature. Practical salinity must not be relabelled as Absolute Salinity in g/kg. An unknown temperature scale, potential/conservative temperature, or other unsupported definition needs a named exclusion or a separately verified conversion. The historical CTD example already has a temperature-scale caveat and is outside this model case.

Argo R uses raw data; A/D uses adjusted data without raw fallback. BGC processing mode is parameter-specific. Preserve source flags, adjusted errors and missing values. Different flag schemes must remain distinct: Argo 2 is probably good, QARTOD 2 is not evaluated, and WOCE 2 is acceptable. See the [Argo file guide](#). A supplied adjusted error is not the uncertainty of a model residual.

Pressure-derived depth uses the existing GSW method with sea pressure in dbar and sample latitude. Depth is negative height. Dynamic height and surface geopotential are unavailable and use the documented zero defaults. This is an approximation with declared inputs, not pressure relabelled as metres. See [GSW pressure to height](#). Temperature-scale conversion, if later needed, must follow explicit source metadata; see [IPTS-68 to ITS-90](#).

Proposed matching contract

1. Use the explicitly selected bundled model snapshot. Show the observation timestamp, model timestamp and signed observation-minus-model offset. A positive offset means the observation is after the model time. The declared window permits offsets on either side of the selected snapshot, including beyond the first or last bundled timestamp. No model value at the observation timestamp is inferred. This is snapshot separation, not temporal interpolation or a claim of extra model coverage. A discovery action may suggest another actual snapshot, but switching the shared workspace must be explicit.

2. Require each observation sample to fall inside the case's actual horizontal coverage. Then choose the nearest native horizontal grid column by great-circle distance using the declared spherical radius, 6,371.0088 km. Check the user distance window. Do not search for a farther wet column when the nearest column is missing.
3. Sample the analytical float64 representation. At an exact native depth, use that finite cell. Otherwise interpolate linearly between the immediately adjacent native depth coordinates only when both cells are finite and the bracket width passes the declared maximum. Do not bridge missing cells, extrapolate above or below valid coverage, or use display-decimated values for residuals.
4. Keep observation sample indices, per-sample position/time, pressure, depth and source order. Preserve gaps and duplicate depths. The model profile is evaluated at the observation depths, not an invented regular observation grid.
5. Apply source acceptance, coordinate acceptance, physical compatibility and provenance holds before calculating a pair. A stricter QC setting can narrow acceptance. Widening windows or accepting Argo/declared flag 2 must not restore a source-metadata failure, bad coordinate, absent adjustment or raw BGC restriction.

The suggested initial settings are 6 hours, 5 km, a maximum 500 m vertical bracket and good/probably-good Argo or declared CSV values. These are transparent project settings, not universally validated oceanographic tolerances. Show them with the result and test changes to each. The original seven Bay profiles have nearest-snapshot offsets of about 2.11 to 2.32 hours and nearest-column distances of about 1.32 to 4.72 km. The additional provenance hold below removes one from quantitative eligibility.

The source QC values of the six currently unheld Bay Argo records are good for their accepted samples. Good-only versus good/probably-good can therefore yield identical real results. Demonstrate QC sensitivity using clearly identified test fixtures rather than changing real source flags. Time and distance controls provide real case sensitivity. At the first model snapshot, 7 January 00:00 UTC, a 6-hour window has no eligible Bay profile; discovery must explain the next useful snapshot instead of silently relaxing the window.

Profiles, coverage and summaries

Use residual = model minus observation, bias = arithmetic mean of accepted residuals, and RMSE = square root of their mean squared value. Count accepted source samples consistently across plots, metrics, map locations and exclusions. The summaries describe the selected, unevenly spaced, potentially correlated samples. They are not a basin-wide skill score or a count of independent evidence. Duplicate depths and dense sampling do not acquire independent statistical weight merely by being separate rows.

The coverage view should show accepted locations, actual sampled depths, timestamps and offsets under the same active rules. A depth extent is the shallowest-to-deepest span, not proof of observation at every intervening depth. Keep model context separate. Do not fill unsampled areas with observations or a confidence field. Rankings should state whether they use distance, time separation or residual, and show sample count and depth span beside results. A large residual can arise from sampling separation, representation or observation/model problems; it does not by itself identify a model failure.

HYCOM uses NCODA assimilation, including available Argo temperature and salinity profiles. The exact membership of these records in assimilation was not established. Agreement therefore cannot be described as independent validation, and disagreement is not automatically forecast error. The provider's [assimilation description](#) supports this limitation.

Independent source audit

Evidence: p05-source-audit.json. The audit fetched two original GDAC files, a bounded live HYCOM OPeNDAP ASCII response and its DAS metadata. It records exact URLs, response byte counts and SHA-256 values. Local source files were not changed.

- Argo 1902669 cycle 12: the fresh 22,408-byte file exactly matches local SHA-256 1c4592fce455fa56a92680fc9dc0ba7eec9754921953f65edf42d0104f682759. Calibration comments describe no adjustment and the OW method with reference datasets.
- Argo 5907083 cycle 12: the fresh 22,572-byte file exactly matches local SHA-256 17f3e33d95003693c24b7fc5737b1dbe9369334b6ffce679964288b5f96483dd. Its calibration text says, "Headless verification run - no calibration comments entered." The text is present in the retrieved source, not introduced by the comparison implementation. Its scientific meaning is unresolved.
- HYCOM time index 14873, depth indices 0/1, latitude index 2333 and longitude index 1085 returned packed temperatures 7999 and 8011, depths 0/2 m, time 210540 hours since 2000-01-01, latitude 13.319999694824219 and longitude 86.800048828125. The independent local raw-file probe agrees.

For Argo 1902669's first sample, depth 0.3976914255017677 m, manual float64 unpacking and two-point interpolation produce model temperature 28.00138652859846 degrees Celsius. The observed value is 28.152999877929688, giving residual -0.15161334933122816. The observation is 7,813 seconds after the model snapshot and about 1.8359222175280503 km from its native column. This reference calculation imports no production matching code.

Provenance-review hold

Recommended action for Argo 5907083: preserve its published readings, original QC and source link for inspection, but pause quantitative model comparisons pending clarification of the verification-run calibration wording. This is a provenance-review decision independent of its residual magnitude. It does not establish that the observations are fabricated or that the provider is compromised.

Key the hold to the original source SHA-256, so importing the same NetCDF under another filename cannot bypass it. Preserve calibration metadata in the observation contract, and make the hold visible in both imported and bundled profile views. CSV data have uploader-declared provenance and must not be presented as independently verified GDAC records. The audit records this hold as proposed; implementation and enforcement tests remain separate acceptance evidence.

Implementation review during P05

Read-only review covered `science/evidence.py`, `science/evidence_contracts.py` and `api/evidence_store.py`. At this point the comparison API accepts checked library records only; imported files remain inspection-only. The matcher has a SHA-based hold, so the observed 5907083 record contributes no quantitative pairs. This review does not verify a future imported-comparison route.

The reviewed implementation uses analytical arrays, a complete native-grid great-circle search, exact-depth handling and immediately adjacent finite depth brackets. It retains source sample order, calculates model-minus-observation residuals and derives metrics/exclusion counts from the same accepted rows. The independent 1902669 pair agrees exactly with the reference above. A direct local run returned 103 accepted samples for that profile. Default coverage at 7 January 12:00 UTC contained two accepted profiles and 206 pairs; widening the time window to 72 hours returned six profiles and 616 pairs. A zero-hour window returned no pairs. These are observations from a local check, not final release acceptance results.

- Snapshot endpoint behavior initially differed from this review's proposed restriction. The agreed policy now explicitly permits offsets beyond collection endpoints while retaining the selected actual model timestamp. The matcher method text reflects this choice. No temporal model extrapolation is computed.
- The initial generic temperature compatibility check would have accepted a CTD temperature with unknown scale if other gates passed. The implementation now requires `metadata.temperature_scale == ITS-90` for CTD temperature. Salinity remains independently eligible under its own rules. The pinned CTD is outside the current model

domain, so this correction prevents a latent incompatible comparison rather than changing an existing valid one.

- Coverage offset ranges and distance ranges initially included rejected samples as well as accepted samples. For a profile with changing coordinates or sample times, those ranges could describe values outside the active gates. This was corrected: accepted coverage ranges now use accepted rows; a zero-pair profile retains source time offsets for exclusion context and has no accepted distance range. A separate in-memory fixture with one accepted sample, one time-window rejection and one distance rejection passed. Its coverage range contains only the accepted sample's 2.170277777777778-hour offset and approximately 1.83592221752805-km distance. This fixture altered no source files and is not presented as an additional observation.

No speed blocker appeared in the bounded local check. Default coverage took about 0.10 seconds and a zero-hour no-match case with alternative-snapshot checks took about 0.23 seconds in one process. These incidental timings exclude network, server cold start and browser rendering; they are not a performance guarantee.

A read-only frontend pass covered the comparison charts, coverage view, workspace and runtime response readers. The chart axes use physical depth, the residual sign agrees with the backend, selected samples share a depth cursor, and curve construction stops at exclusions. Coverage draws actual accepted depth strokes rather than a filled confidence field. Metrics, source types, timestamps and assimilation limitations remain explicit. No additional urgent numerical display defect was found in this pass. The implementing team is separately changing discovery to offer a suggested snapshot before switching and clarifying that zero-pair table offsets describe the source record. Browser interaction and rendered-layout checks are still required; reading JSX is not visual acceptance.

Acceptance traps to test

- Exact-depth values beside missing neighbours, a single masked bracket endpoint, surface/bottom extrapolation, and maximum-gap equality.
- Time/distance thresholds and deterministic ties, out-of-domain points near an edge, and same-sign versus opposite-sign time offsets.
- Per-variable QC, coordinate QC, incompatible definitions, malformed imported acceptance claims and the SHA-based provenance hold.
- Source-order irregularity: Argo 2903891 begins near 1,302 m with eligible QC-1 values. Original rows 29 and 30 contain missing values with QC 4; row 31 returns near the surface. Keep this source order, exclude rows 29/30 and never mistake the first valid deep sample for a QC failure. The original-file fixture resolved an earlier review

misreading.

- No-pair results must retain explanations; charts, counts, ranking and coverage must use identical accepted pairs.
- R05, X04 and X05 can close only after their evidence exists. X06 also requires P08 replay, so P05 sensitivity work alone does not complete that entire feature row.

43 P07 component audit

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/P07_COMPONENT_AUDIT.md

Audit date: 22 September 2026. This record starts with release 0.6.0 and follows existing components and the new feature-search workspace through local and public release 0.7.1 at <https://depth-atlas-seifuku.vercel.app/>.

Scope and evidence

The reviewer first exercised the existing interface and inspected its screenshots, then reviewed the scientific code and reran numerical checks. The initial visual script completed 34 interaction/layout checks and saved 33 screenshots. Every saved screenshot was inspected. Chromium ran at 1536, 390 and 320 pixels, plus 768 pixels with 200% text. Windows WebKit ran at 1440 pixels. No page errors or horizontal page overflow occurred. This is controlled browser review, not a study with an unfamiliar judge or physical-device certification.

- Browser script: `web/benchmarks/p07-audit.cjs`.
- Initial results and screenshots: `evidence/p07-audit-initial.json` and `evidence/p07-audit-initial-*.png`.
- A WebKit image captured before its first settled compositor frame was blank apart from HTML observation markers. Waiting for the actual rendered volume and a short settling period produced the correct volume, rotated view, slice, cutaway and Basic view without changing the app. Evidence: `evidence/p07-audit-webkit.json` and corresponding images. The original blank screenshot remains available; it is not evidence of a persistent rendering failure.
- Existing scientific/API tests: 176 passed, two dependency deprecation warnings. Evidence: `evidence/p07-audit-existing-science.xml`.

- Independent source audit: all 56 analytical/display arrays across seven snapshots matched manually decoded acquired NetCDF data exactly, including every missing position. This checks 6,760,320 scalar positions across the original four variables and two representations. Evidence: `evidence/p07-audit-existing-source-arrays.json`; script: `science/audit_p07_existing_sources.py`.
- Renderer numerical tests: original 17 passed before correction; all 24 passed after adding seven independent map-vector tests.
- P07 local browser audit: all 35 checks passed across the same five configurations after the final readability fixes below. Evidence: `evidence/p07-local-071-features-visual.json`, 25 screenshots and five downloaded result files. The preceding 0.7.0 checks remain in `evidence/p07-local-final-features-visual.json`. The audit covers explicit query submission, source time changes, map drawing, keyboard coordinate entry, native-depth selection, source-value agreement, export identity, a graphics alternative and truthful empty results. This is additional coverage, not a replacement for the full release browser suite.
- The release coordinator reran 143 real local THREDDS protocol/source checks successfully during P07. Evidence: `evidence/p07-protocol-verification.json`. This remains a local service test, not public THREDDS hosting or universal standards certification.
- Final public browser audit: 35 of 35 additional feature-workspace checks passed on the public 0.7.1 alias across all five configurations, with no page errors. Evidence: `evidence/p07-public-071-features-visual.json`. Public downloaded exports identify version 0.7.1 and preserve the checked native section value. Actual full screenshots and close-ups were inspected; `evidence/p07-public-071-detail.json` records the additional WebKit, mobile, narrow and enlarged-text captures.
- After the final public visual review, the independent deployed P07 verifier passed all 99 numerical/source/API checks, and the existing public HTTP/source verifier passed 49 checks over 50 requests, including the complete native file checksum. Evidence: `evidence/p07-public-source-api.json` and `evidence/p07-public-existing-http.json`. This confirms the deployed calculations and data paths on the pinned case; the transport qualification below remains.

Concrete defect and correction

The Basic 2D current plot stretched longitude and latitude independently to fit its canvas, but used `atan2(-north, east)` for arrows. That drew a bearing as though horizontal and vertical pixels represented equal physical distances. At different window aspect ratios, an arrow could disagree with the geographic displacement it represented.

The corrected renderer projects a physical east/north displacement into the plotted longitude/latitude axes, including longitude convergence through $\cos(\text{latitude})$, then normalizes it to the declared 14-pixel glyph length. Speed values, colour values and analytical data remain unchanged. The correction is in `web/src/ocean/BasicView.tsx` and `web/src/ocean/mapVector.ts`.

Seven tests cover geographic cardinal directions, a known 3:4 vector at four latitudes, latitude convergence, unequal chart scales, desktop/mobile aspect ratios, speed-independent glyph length, and invalid/zero inputs. The browser audit intercepted the actual canvas arrow line segments and compared them with independent spherical-displacement calculations. Before correction, maximum endpoint errors were 2.629 pixels on desktop and 1.494 pixels on mobile. After correction, all 143 arrows at each size agreed within 0.00000001 CSS pixels; measured maximum error was below 0.000000000000006 pixels.

Evidence: `evidence/p07-audit-vectors-original.json`, `evidence/p07-audit-vectors-fixed.json` and associated images. The original failed comparison is retained. The corrected UI was tested through a separate Vite server at `http://127.0.0.1:5174`, proxying the 0.7.0 API at port 8004. A first dev verification attempt timed out because its proxy still pointed to the 0.6.0 API at port 8003; aligning the release versions resolved that test-environment mismatch.

P07 visual findings and corrections

The new workspace uses three visible steps: set a condition, inspect a region and draw a section. The selected amber boundary now has a clearer starting camera, a depth-exaggeration control and top/bottom depth labels. The description distinguishes the selected warm layer from the full searched depth. Plan-view positions and section clicks use the inverse SVG screen transform, which accounts for responsive letterboxing. The section names its displayed endpoints and timestamp separately from editable form inputs.

The initial P07 browser audit exposed horizontal overflow at 320 pixels. Wrapping the workspace navigation fixed it; the final measured page width is exactly 320 pixels. Initial results remain in `evidence/p07-local-features-visual.json`. Two WebKit failures in that first run were test sequencing errors: the script selected controls before React had committed the newly returned section. Waiting for the new source column before changing station/depth resolved those checks without changing numerical results.

Actual image inspection then caught a defect that the successful interaction checks did not detect: Windows WebKit drew only the upper half of native dropdown text in the query form. The frontend correction replaces the native select appearance with a simple

chevron, explicit line height and an expandable control height. The identical before/after close-ups are `evidence/p07-audit-webkit-selects-original.png` and `evidence/p07-audit-webkit-selects-fixed.png`. The final 35-check run includes this correction and has no page errors or horizontal page overflow. Representative full screenshots, the 3D boundary, section selection, empty state and close-up dropdown text were visually inspected.

The section audit independently requests the analytical source API for the selected native location. In the tested January 7 noon section, the displayed 100 m value is 22.43700011575129 degrees C at 87.52001953125 E, 13.720000267028809 N; every layout and its downloaded JSON agrees. This interface check complements the independent P07 scientific suite in `evidence/p07-independent-science.xml`; it does not substitute for its geometry and connectivity tests.

Additional element-level screenshots were taken and inspected for WebKit, 390 pixels, 320 pixels and 200% text: `evidence/p07-local-detail.json` and twelve corresponding query, inspector and section images. The closer review identified two presentation issues in that build: fixed-viewBox map/section tick labels became very small on a narrow phone, and the fixed-width depth-exaggeration dropdown truncated its selected value at 200% text. The frontend corrected both in 0.7.1. Narrow charts now retain their native 640/780-pixel drawing widths inside labelled, keyboard-focusable horizontal scrolling panels, with visible guidance and the existing keyboard coordinate/value controls. The exaggeration selector grows with its font and shows the complete 1000x; the narrow ordering control gets a full row and shorter direct wording. Scientific coordinates and values are unchanged.

The 0.7.1 browser audit scrolls the bounded chart before clicking each requested SVG position, then recomputes the screen transform. All 35 checks pass. Independent close-ups confirm readable 12-pixel mobile axis labels, access to the section's far-right end, the complete exaggeration value at 200% text and readable 390-pixel ordering. Evidence: `evidence/p07-local-071-detail.json` and corresponding images. An earlier interrupted verification overlapped the coordinator's API/frontend transition from 0.7.0 to 0.7.1; its partial evidence and reason are retained in `evidence/p07-local-responsive-transition-note.json`.

Final public release check and intermittent loading

The coordinator ran the complete 153-test browser suite against public 0.7.1. The first run had 144 passes, eight failures and one declared skip. All eight failing tests passed unchanged in a focused rerun lasting 23.4 seconds. The skip is the Windows WebKit forced WebGL context-loss test; that path is exercised in Chromium. Evidence: `evidence/p07-public-071-browser.json`, `evidence/p07-public-071-browser-recheck.json` and the preserved `evidence/p07-public-071-initial-failures/` traces. These results must not be represented as a clean 153-test first pass.

An independent offline review inspected every failed trace. Across those traces, 25 resource transfers had no completed HTTP response before the relevant deadline, and none returned an HTTP error status. Three tests stalled while retrieving the main static JavaScript bundle. Other failures involved API reads aborted around their eight-second client deadline or requests still pending when the test ended. The WebKit feature-search POST recorded the transport error `Send failed` since rewinding of the data stream failed after about 19 seconds, then began a retry. No failed trace demonstrated a returned numerical mismatch or an application exception. Evidence and reproducible offline parser: `evidence/p07-audit-public-071-failures.json`, `science/audit_p07_public_failures.py`.

The traces show intermittent public loading failures during that run. They do not identify whether local networking, browser transport, the CDN or backend latency caused them. Successful unchanged reruns and the subsequent 35-check public audit establish successful tested operation, not a measured reliability guarantee. Retain the evidence and review this behaviour if it recurs under ordinary use.

Existing component matrix

Component	Visual and interaction review	Scientific or contract check	Remaining limit
App shell, region controls and geographic preview	Real case, planned Arabian Sea context, tool labels, quick-guide keyboard return, footer and favicon reviewed	Catalogue/API tests reject pretending a planned region has model data	Only the selected Bay of Bengal model domain is connected
3D volume and optional cutaway	Closed volume and animated open cutaway inspected; water-column colour and display-cut explanations visible	Physical projection, irregular depth windows, cut-face coordinates and masks covered by renderer tests	Depth is exaggerated and volume colour blends along viewing rays

Component	Visual and interaction review	Scientific or contract check	Remaining limit
Depth slices and fixed-latitude sections	Selected-depth and east-west section views inspected, including empty 5000 m slice	Native and display arrays exactly match source decoding; missing-corner geometry tests pass	Rendering is a display approximation; native numeric inspection remains authoritative
Isosurface	Default threshold produces a visible surface; scalar and units shown	Known depth-field isosurface and missing-cell/out-of-range numerical tests pass	Linear tetrahedral approximation is explicitly documented
Current vectors	3D view inspected; Basic arrows corrected and checked at actual canvas level	Horizontal components and masks preserved; projected direction checked independently	Equal-length direction glyphs, no vertical velocity or particle transport
Kinetic energy	Derived-field notice, units, colour key, Basic section and missing-depth output reviewed	Existing independent-source derived API fixtures and registry/mask tests pass	Horizontal energy per mass only, not eddy kinetic energy
Colour, opacity, exaggeration and graphics controls	Invalid log minimum explains the problem; expanded controls reviewed; Basic mode readable	Log-domain and physical-projection tests pass; display changes do not alter source arrays; public suite checks graphics recovery	Graphics performance depends on the device; forced context-loss is tested in Chromium and explicitly skipped in WebKit
Time and selected source value	Historical dates and native source coordinates stay visible; first snapshot's 100 m temperature reads 22.303 C	Every supplied timestamp and all original fields match the acquired sources	Seven 12-hourly January 2024 snapshots, no current conditions or temporal interpolation
Argo, glider, CTD and BGC inspector	Each instrument type, measured-variable selection, timestamps, QC, map and profile chart reviewed	Existing adapter, profile-contract, pressure/depth, QC and source-provenance tests pass	Foreign examples do not extend model coverage; CTD temperature scale remains unconfirmed

Component	Visual and interaction review	Scientific or contract check	Remaining limit
Imports	Public parsing destination, temporary-memory	Existing tests cover real supported formats, malformed input, limits	Imported profiles are inspection-only; the unsupported-upload test

Component	Visual and interaction review	Scientific or contract check	Remaining limit
	handling, file-size/schema limits and example access are visible; public suite repeats import/error interactions	and source hashes	passed its unchanged public rerun after an initial loading timeout
Compare and synchronized profiles	Temperature comparison and coverage views reviewed at every viewport; observations and model times remain distinct	Existing independent-reference tests cover residual arithmetic, physical matching, exclusions, sensitivity and discovery	Agreement with assimilated Argo is not independent validation
Coverage and source holds	206 pairs from two eligible profiles shown for the selected snapshot; source-review exclusions visible in the table	Same accepted-row metrics and existing provenance-hold tests pass	Coverage is not a calibrated confidence score; one source remains held for review
Data access and extension contracts	Public-data access page, supported formats, exact-source option and unavailable public THREDDS explanation reviewed	Existing CF, adapter, registry, unit, mask and API tests pass; 143 real local protocol/source checks rerun during P07	Tested local THREDDS package; public THREDDS remains unconfigured
Privacy, terms and attribution	Static routes, readable policies, favicon and retained attribution reviewed	Text was compared with current memory-only imports, API requests and hosting behaviour	Operator/contact fields intentionally remain unpublished until final release; no legal certification claimed
Service recovery and retained frames	Public suite exercises interruption, deployment mismatch, retries and retained values	Existing contracts and API tests pass; historical recovery evidence remains archived	Initial public loading failures and successful unchanged reruns are recorded explicitly above
Component	Visual and interaction review	Scientific or contract check	Remaining limit
P07 feature search and transects	35 interaction/layout checks pass locally and publicly across five configurations; real 3D boundary, selection,	Selected section value and exports agree with the analytical source API; independent P07 numerical evidence is	Threshold regions are not eddies or water-parcel tracks; midpoint-bin volume is estimated

Component	Visual and interaction review	Scientific or contract check	Remaining limit
	form, section and empty-state images reviewed	recorded separately	

Scoped conclusions after public verification

- 1. Scientific operation:** the tested existing arrays, masks, time coordinates and corrected map-vector projection agree with their independent references. The P07 section's source value and export agree with native data. P07's additional geometry, connectivity and observation-association tests have their own evidence. These results support the implemented methods on the supplied case, not operational forecasting accuracy or unsupported ocean coverage.
- 2. Visual operation and understanding:** the exercised paths render and respond across desktop, WebKit, narrow screens and enlarged text, with overflow, WebKit dropdown clipping, tiny mobile chart ticks and enlarged-text selector truncation corrected and rechecked. Labels distinguish the searched depth, selected layer, model time and actual source value. The narrow layout remains long and scientifically detailed; charts scroll inside labelled panels to preserve readable axes. No unfamiliar-user study has yet established how quickly a judge will understand it; guided cases and real user feedback remain planned work.
- 3. Direction:** the product remains aligned with browser-native 3D ocean investigation and the accepted phased scope. P07 adds structured investigation to the core visualization without claiming later simulations, climate coverage or wider geography are finished. Public release checks now have evidence, with the initial loading failures and successful reruns preserved as part of the result.

Direction against the accepted problem statement

The current implementation addresses the requested browser-native volume, slices, isosurfaces, actual time steps, temperature/salinity/currents, geolocated profile types, bounded supported ingestion, scientific controls, REST access, CF interpretation and tested standards-service package. P07's structured threshold search and drawn section add an investigation workflow on top of those core fields. That is consistent with the accepted problem statement's aim of inspecting complex three-dimensional ocean structure.

This is not the same as saying the entire product or all official expectations are finished. R09 still needs later packaging/institutional verification, R10 retains broader integration work, and R12 still needs guided public outreach and actual user feedback. No institutional endorsement, top-five outcome, universal usability or operational forecast improvement follows from these tests.

Prepared explanation of the audit correction

We reviewed the current explorer, profiles, comparisons and data-access pages across desktop and narrow screens, then checked the scientific arrays against their original source files. The review found a direction error in the simplified current map: arrows did not account for the map's changing width and height. The corrected view projects eastward and northward velocity onto the actual chart before drawing the arrow. Numerical source values stay unchanged. A separate canvas check verifies every displayed arrow in the tested desktop and mobile views.

Reproduction commands

```
node web/benchmarks/p07-audit.cjs
node web/benchmarks/p07-audit-webkit.cjs
.venv/Scripts/python.exe -m pytest tests/test_api.py tests/test_case_pack.py
tests/test_evidence.py tests/test_evidence_provenance.py
tests/test_instruments.py tests/test_science_adapters.py
tests/test_standards_extensions.py -q --junitxml=docs/evidence/p07-audit-
existing-science.xml
.venv/Scripts/python.exe -m science.audit_p07_existing_sources
npm --prefix web run test:science
node web/benchmarks/p07-audit-dev.cjs
node web/benchmarks/p07-audit-vectors.cjs
$env:OCEAN_BENCH_NAME='p07-local-071'

node web/benchmarks/features.cjs
node web/benchmarks/p07-audit-feature-details.cjs
.venv/Scripts/python.exe -m science.audit_p07_public_failures
```

OCEAN_TEST_URL selects a different browser target. OCEAN_BENCH_NAME selects a separate evidence prefix for the general and vector audits. The dev helper uses OCEAN_API_URL, defaulting to port 8004. These scripts use unique output files and do not run or overwrite the full Playwright suite's web/test-results directory.

44 Phase 7 independent scientific review

Historical or maintained reference: read alongside the current release overview | Source: docs/P07_SCIENCE_REVIEW.md

Review date: 22 September 2026. The connected ocean case remains historical, 7 to 10 January 2024.

This review defines and independently checks a bounded query method. It does not certify HYCOM against the real ocean, identify eddies, validate an operational forecast, or establish an unfamiliar user's understanding of the interface.

Scientific interpretation

A result is a connected set of native model samples satisfying an explicit variable, inclusive threshold or closed range, selected snapshot, and depth window. Connectivity means sharing a face along one native grid axis. Edge or corner contact alone does not join regions. Missing cells never qualify and cannot bridge disconnected pieces. Flattening an array must not connect the end of one row to the beginning of another.

The source fields contain point coordinates and no supplied coordinate bounds or cell-volume measures. A region's displayed volume is therefore a **midpoint-bin spherical volume estimate**, not a provider-supplied model volume and not a measured volume of water. The estimate assumes the selected sample represents its constructed bin. It cannot account for partial wet cells, detailed bathymetry within a bin, or unresolved threshold crossings.

The outermost bin boundaries stop at the first and last sampled coordinates. Interior boundaries lie halfway between adjacent coordinates. Native depth centres are selected inclusively, and their bins are clipped to the requested depth window. A depth window between source levels can contain no selected centres and legitimately return no region. We do not extrapolate an unsampled outer half-cell or fill that window by interpolation.

For longitude bounds W and E, latitude bounds S and N, and clipped positive-down depths a and b, the estimate integrates a spherical shell:

$$V = ((R-a)^3 - (R-b)^3) / 3 * (\sin(N)-\sin(S)) * (E-W)$$

Angles are in radians and R is 6,371,008.8 metres. The independent verifier evaluates the cubic antiderivative using 70-digit Decimal arithmetic. Runtime code uses an algebraically stable expanded polynomial. This separates the numerical implementations while

checking the same declared geometry. It also preserves unequal native depth spacing and varying latitude area.

Sample mean and estimated-volume-weighted mean answer different questions. The former weights every qualifying source value equally. The latter uses the constructed bins and inherits their limitations. The interface should distinguish them. Centre extent and inferred bin extent must remain distinct as well.

Observation and section rules

An associated observation is nearby eligible evidence under the existing P05 time, distance, variable-definition, source-review and quality gates. It is not proof that the observed value passes the query threshold. Association uses the matched native horizontal column and the inferred depth bin containing the actual observation depth. Midpoint ties belong to the deeper bin, with only the final edge closed. The selected depth window and connected component still have to contain that bin and observation depth.

Default gates use a six-hour time tolerance, five-kilometre native-column tolerance, at most a 500-metre finite vertical interpolation bracket and the established good/probably-good QC rules. The held Argo calibration source remains held. Derived kinetic energy and current components do not gain false observation coverage from temperature or salinity records.

A drawn section is a straight line in longitude/latitude coordinates. It is sampled at evenly spaced requested stations, including both endpoints. Each station selects the nearest native horizontal column by full spherical great-circle distance and retains the source depth levels and missing values. Repeated native columns on short lines are expected. There is no interpolated smooth field between columns. Requested and selected coordinates, sampling offset, station count and cumulative spherical distance remain inspectable.

Section evidence lies within five kilometres of a requested station after the P05 gates. This is a **sampled-station corridor**, not the exact continuous distance to the entire segment. Fewer stations can change coverage. Model sampling and observation projection are separate operations and must not be presented as exact colocations.

Independent fixtures

`science/verify_p07_reference.py` imports no production adapter, store, feature engine, mesh geometry or observation matcher. It reads the pinned original packed NetCDF files with automatic decoding disabled and masks before applying packing scale and offset. Component labels come from SciPy's offline six-face connected-component implementation. SciPy is not required by the public runtime.

`tests/fixtures/p07-source-reference.json` contains:

- 46 source queries spanning all seven snapshots and all five supported scalar fields, including the horizontal kinetic-energy derivation from both masked velocity inputs.
- 13 synthetic test grids for face/edge/corner contact, masked bridges, flattened-row boundaries, exact threshold equality, empty and wholly missing selections, unequal spacing, clipped depth windows, means and volume ordering.
- Seven source sections, including reversed direction, a short line with repeated columns, domain endpoints, 201 stations, masked deep values and a line through Argo 1902669.
- Independent observation-association rows derived from the already pinned original-source P05 reference. Baseline accepted sample counts are checked against that reference before associating them with P07 regions or sections.

Membership hashes serialize ascending native C-order flat indices as little-endian signed 64-bit integers. Tests must check these complete memberships, not only rounded counts or a few sampled values. Floating comparisons should allow numerical rounding, while source indices, counts, masks and memberships must agree exactly.

Fixture SHA-256: `1e45922b54acffd4f707478e650a8dc7537a39f50d4dc549234455c0f808a0b0`.

The default temperature query, at least 26 degrees Celsius over 0 to 300 metres at 7 January 2024 12:00 UTC, independently selects 80,203 native samples in one component. Its estimated volume is 12,966.151584607129 cubic kilometres. There are 34 associated eligible observation samples under the stated defaults. These are regression expectations for this exact source and method, not impact metrics or a forecast result.

An exact decoded temperature value of 28.199000389431603 used as both inclusive bounds selects 74 samples in 68 separate components. This deliberately catches tolerance widening, decimal rounding, diagonal connectivity and unintended smoothing.

Acceptance status

The fixture was generated and independently regenerated byte for byte. All 83 production integration checks in `tests/test_features_reference.py` now pass. They compare complete membership hashes, counts, native source probes, extents, both means, estimated volumes, observation identities, requested versus sampled transect coordinates, original values and masks, and the exact exposed faces of synthetic and real meshes. Exact midpoint and upper-depth observation boundaries have dedicated checks.

The first test run found a small numerical defect in longitude widths: subtracting already-converted radian coordinates made two mathematically equal-width cells acquire slightly different estimated volumes and swap the intended index tie order. The runtime now subtracts the degree endpoints before converting their difference to radians. The initial failure remains in `evidence/p07-independent-science-initial.xml`; the passing result is `evidence/p07-independent-science.xml`.

The direct default native calculation took about 0.08 seconds in the first local measurement over 191,520 source samples. This excludes API overhead, observation matching, browser drawing and network transfer. It is an observed local timing, not a performance guarantee.

`science/verify_p07_public.py` independently checks a running service using only the Python standard library and the reference fixture. It checks 46 source searches, seven sections, three returned meshes with an independent boundary-volume integral, 35 original source scalar requests, seven bounded error responses and release identity. Every response is checked for its expected release and no-store API caching. Failures are retained and are not automatically retried.

The initial local service run passed 98 of 99 checks. The sole failure came from the old API process retaining the pre-fix geometry module. It is preserved in `evidence/p07-local-source-api-stale-process.json`. After restarting the API, all 99 checks passed against release 0.7.0 on `http://127.0.0.1:8004`; see `evidence/p07-local-source-api.json`.

After the final public visual review, the same independent verifier passed all 99 checks against release 0.7.1 at `https://depth-atlas-seifuku.vercel.app`. The evidence is `evidence/p07-public-source-api.json`. Every response carried the expected release and caching policy. Original-source selections, complete region membership fingerprints, extents, means, estimated volumes, observation identities, native sections and returned boundary integrals agreed with the independent references. These checks establish the implemented numerical method on the pinned historical case, not real-world forecasting skill.

Scientific interface review

The independent interface review found and reported four interpretation problems, then checked the frontend fixes:

- A section whose source values were all missing displayed a numeric 0 to 1 colour scale. It now displays a missing-source explanation and no numeric gradient.
- A section containing a single repeated native value displayed the entire colour gradient with identical endpoint labels. It now states the constant source value without suggesting a varying range.
- Clicking an exact depth-bin midpoint selected the shallower source level, whereas observation association assigns that midpoint to the deeper bin. The click rule now matches the declared half-open bin convention and includes the final inferred cell edge.
- Editable or newly drawn line endpoints could differ from the still-displayed section without an independent visible endpoint identity. The section now prints its own returned A and B coordinates, distinguishes requested stations from sampled native columns, and warns when draft coordinates differ.

All nine dedicated browser assertions pass across desktop Chromium, Windows WebKit and Chromium mobile emulation. Evidence is in `evidence/p07-science-ui-browser.json`. The initial authoring run is retained in `evidence/p07-science-ui-browser-initial.json`: it expected the wrong draft-warning wording and clicked the 337-metre query limit instead of the last selected bin's actual 325-metre boundary. Those were test mistakes, not additional application defects. The original visual defect is preserved separately in `evidence/p07-science-ui-all-missing-initial.png`; the corrected view is `evidence/p07-science-ui-all-missing-corrected.png`.

The frontend also corrected displayed precision for very short line distances and narrow depth windows. A real 300 to 301 metre section through one repeated native column now reports its nonzero 0.000155 km line length and distinct tick labels 300, 300.25, 300.5, 300.75, 301. The corrected screenshot and captured text are in `evidence/p07-science-ui-precision-corrected.png` and `.json`, with no browser page errors in that review. Regression assertions for these values are included in the final 27 passing local feature/scientific UI tests (`evidence/p07-local-accessibility-browser.json`) and the public 0.7.1 suite. All nine dedicated scientific UI cases passed on that first public run. The wider public suite and retained loading-failure rechecks are documented in `PHASE_07_FEATURES.md`.

These checks establish the stated interaction and numerical-presentation behavior on the tested browsers. They do not replace an unfamiliar-user pilot or a physical-device check.

Primary references

CF Conventions, sections 7.1 and 7.2 explain that coordinates without supplied boundaries do not establish cell extent, and distinguish cell measures from coordinates. The same document describes longitude/latitude area weights and the need to account for partial ocean cells. Our midpoint bins are an explicit project estimate, not an inference that CF guarantees such bounds.

SciPy `generate_binary_structure` specifies rank-three connectivity one without diagonal neighbours. SciPy `label` documents labeling under the supplied symmetric neighbourhood. These support the offline reference implementation; the chosen connectivity is a declared query convention rather than proof of physical coherence.

Claims that remain out of scope

No threshold-only region should be labelled an automatically detected eddy, coherent material body, transported water parcel or independently validated feature. Estimated observation coverage is not a confidence score. Changing snapshot shows a separate query result; tracking identity through time belongs to a later phase and requires its own method. Geographic coverage remains the checked Bay of Bengal subset, with the other planned regions clearly marked as planned.

45 First-time-user pilot

Historical or maintained reference: read alongside the current release overview | Source: docs/P09_USER_PILOT.md

Status: awaiting real participants. Prepared on 22 September 2026. No pilot has been run or counted yet.

Use the public Vercel app after the Phase 9 deployment is verified. Recruit one to three people who have not used it. Ask them to use their own browser/device. Do not give an app walkthrough beforehand. Tell them that the exercise tests the interface, not their ocean-science knowledge. Do not collect names, email addresses or recordings without a separate need and agreement. Anonymous labels P1, P2 and P3 are sufficient.

Read this introduction: "Please try these tasks and say what you think the app is showing. Tell me whenever something is unclear. I will record time, mistakes and any help you ask for. These are historical examples, and you do not need to know the science in advance."

Tasks

1. Open the app and identify the data year and selected sea. Explain whether this is a live forecast.
2. Start the guided investigation. Find the temperature difference at 100 m. Explain what is similar at the surface and whether the example proves that every part of the two seas differs this way.
3. Open the selected column. Explain the cube's cutaway, its colour and its vertical scale. Point out how you would inspect a different depth.
4. Compare a measurement. Identify which information comes from a sensor and which comes from the model. Find the eligible sample count. Explain whether agreement proves independent forecast accuracy.
5. Find warm water. State the temperature threshold and depth interval. Explain what a connected region means. Change the threshold and run the query, then return to the guide's setting.
6. Save the result with a short title. Reopen it from Saved and download the investigation file. Explain where the browser save lives.
7. Switch to Arabian Sea and identify its available dates. Explain whether the rectangle covers the whole sea.

For each task start timing after reading the task. Stop when they finish, abandon it, or reach five minutes. Record actual elapsed time, success/partial/failed, wrong actions, help requested and the participant's words. If you give help, record exactly what you said. Do not convert partial success into success. Ask at the end: "What was hardest to understand? What would you expect to click next?"

Results template

Participant count: pending.

App release: 0.9.1. URL: <https://depth-atlas-seifuku.vercel.app/>. Record the actual release used by each participant.

Anonymou s participant	Device / browser	Task	Seconds	Outcome	Mistakes / confusion	Help given
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Pending

Findings and fixes

No human findings recorded yet. Browser automation is separate evidence and must not be entered as a participant.

Observation	Severity and affected task	Concrete change	Retest outcome
Pending real feedback			

P09.5 can close only after actual task evidence is recorded and significant observed issues are addressed or explicitly retained. A small convenience sample cannot establish universal ease of use or a quantified productivity improvement.

46 P10 implementation contract

Historical or maintained reference: read alongside the current release overview | Source: docs/P10_API_CONTRACT.md

Backend owner: root. Frontend owner: p07_frontend. This is an implementation agreement, not test evidence.

POST /api/cases/{case_id}/expedition/plan and /experiment accept PlanQuery: {variable: 'temperature' | 'salinity', depth_index:0, budget:8, min_spacing_km:30, objective: 'gradient' | 'coverage', seed:26067}. Budget 3..16, spacing 0..150 km, seed integer 0..2147483647. Depth must be a supplied native depth at most 1000 m (index at most 32). Prior is always time index 0. Experiment evaluates indices 2,4,6, three correlated snapshots per selected case. Never uses the shared selected time for planning.

All responses have schema_version:'1', kind, method_version:'p10-sampling-v2', case_id, manifest_sha256, units, variable, depth_m, prior_time, methods:string[], limitations:string[]. All numerical values finite or null.

Plan response kind='station_plan', query, bounds:[west,south,east,north], candidate_count, evaluation_count, fulfilled:boolean, reason:string|null, stations:Station[], candidates:Candidate[], background:{longitude:number[],latitude:number[],values:(number|null)[],shape:[ny,nx]}. Candidate={id:string,longitude,latitude,x_index,y_index,prior_value,gradient_per_km:number|null}. Station adds order:number, nearest_station_km:number|null, rationale:string. Gradient is a local prior finite-difference magnitude with physical coordinate spacing.

Coverage objective is farthest-point design coverage, NOT observational coverage/confidence. No observed mismatch drives selection. No instrument placement/route safety claim.

Survey POST /api/cases/{case_id}/expedition/survey accepts

{variable,depth_index,time_index,start:[lon,lat],end:[lon,lat],stations:25}. Only endpoints inside case bounds and distinct; 2..81 stations along straight lon/lat segment. Response kind='virtual_survey', query, model_time, samples:[{order,requested_longitude,requested_latitude,longitude,latitude,x_index,y_index,distance_km,offset_km,value:number|null}], simulated:true, unique_columns:number. Nearest native column via great-circle distance, selected native depth, no spatial/time interpolation or missing filling.

Experiment response kind='sampling_experiment', query, plan:PlanResponse, evaluation_count, runs:[{strategy:'selected'|'uniform'|'random'|'persistence',seed:number|null,time_index,model_time,status:'ok'|'failed',reason:string|null,budget:number,station_ids:string[],evaluation_count:number,rmse:number|null,mae:number|null,bias:number|null}], summary:[{strategy,successful_runs,failed_runs,mean_rmse:number|null,min_rmse:number|null,max_rmse:number|null}], random_seeds:number[], held_out_times:string[], evaluation_coordinates:[{longitude,latitude}]. Three snapshots times selected + uniform + 5 deterministic random plans + persistence. Same IDW power2 residual reconstruction, all candidates excluded from evaluation. Equal exact budget, min-spacing; failures retained. No evaluation field returned or used for planning. Point-weighted error, not basin/area-weighted error. No forecast improvement/OSSE claim. Random min/max is descriptive, not uncertainty.

Recipe added to existing save/replay union: {mode:'expedition',case_id,query:PlanQuery,survey:SurveyQuery|null,experiment:boolean}. Modules named station_plan, virtual_survey, sampling_experiment, max 3. Each uses method p10-sampling-v2, random_seed=query.seed for experiment only, otherwise null. New sources.methods.expedition entry only for expedition recipe. Existing recipe method maps unchanged. Save/replay returns exact outputs, result hashes and normal export. Export CSV defaults to benchmark if experiment, otherwise survey if present, otherwise stations.

Frontend integrates shared case with one Expedition navigation button and simple task tabs/sections: Plan stations, Sample a route, Compare strategies. Keep prior timestamp visibly fixed, selected survey time editable using manifest times. Use real map coordinates, numbered stations, source units, explicit legend and responsive layout. Retain previous verified result when edits/request pending; clearly distinguish unapplied settings. Plan auto-load once, manual update thereafter. Draw route via two map clicks and numeric keyboard alternative. Save only applied results and preserve exact replay. Use accessible SVG/canvas map, no WebGL requirement for this tool. Respect reduced motion and established visual constraints.

47 P10 independent scientific method review

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/P10_INDEPENDENT_METHOD_REVIEW.md

Review date: 22 September 2026.

Status: design review plus executed independent numerical checks. The bounded scientific method passed 23 independent checks on 22 September 2026. Frontend, deployment and whole-phase acceptance remain the root task's responsibility.

The reviewer authored `tests/test_expedition_independent.py` but has not edited production implementation. The P09 unfamiliar-user pilot remains open and is separate from this review.

Scientific question and permitted claim

The defensible question is: given one frozen historical field and a fixed number of simulated measurements at a later time, how well do specified station-selection rules reconstruct withheld values of that later model field?

The resulting scores describe reconstruction of supplied HYCOM fields. They do not establish the real ocean state, forecast improvement, calibrated uncertainty, reduced operational costs, or the effect of deploying a real instrument. A scientific planning aid is useful without making those claims.

An operational observing-system simulation experiment normally includes a evaluated nature run, simulated instrument errors, assimilation and forecast components, and validation of the experimental system. NASA GMAO describes these elements in its [OSSE overview](#) and [framework method description](#). P10's bounded reconstruction exercise does not contain that system. It should not be called a validated OSSE.

Recommended frozen experiment contract

1. Choose one supported scalar variable and one actual native depth. Temperature or practical salinity has the simplest interpretation. A budget of B means B scalar values at that depth and target snapshot. It does not mean B complete profiles or a ship-time budget.

2. Freeze the prior at time index 0, 7 January 2024 at 00:00 UTC. Use only its values, masks, native coordinates and declared geographic constraints to create station candidates and selection scores.
3. Create a finite candidate lattice and a disjoint evaluation lattice deterministically from the native grid. Both definitions and index lists must be inspectable. Membership must not depend on later values or masks. Every strategy is scored at the same evaluation points.
4. Select all station locations before loading target field values. Compute gradients, normalizations, eligibility and any fallback ordering from the prior only. Store the resulting plan before reconstruction.
5. Test the predetermined later snapshots separately. The implemented contract selects indices 2, 4 and 6, which are 8, 9 and 10 January 2024 at 00:00 UTC. These are three correlated temporal holdouts within one short historical case, not three independent ocean regions. Run acceptance checks across both available regions. Do not silently average away an unsuccessful time, region or random trial.
6. At each target time, read values only at already chosen stations to create simulated measurements. Using those values for reconstruction is the measurement step, not leakage. The reconstruction function should receive the prior, station locations and measured station values, but never the full hidden target field.
7. Evaluate the resulting field against target values at the reserved evaluation locations. Finite target values may determine the scoring denominator after selection, provided that denominator is identical across strategies and missing counts are reported. They must never change the station plan.
8. Record all settings, seeds, coordinate indices, source and method identities, successful results, and failure reasons in replayable output.

The holdouts are hidden from selection and reconstruction, not secret from people. Users can inspect the historical snapshots elsewhere in Ocean Navigator. Repeatedly changing settings after seeing scores is exploratory use, so the interface must not call these results untouched external validation. Separation of information is a substantive method condition, not just hiding a chart; see the primary research discussion of leakage by [Kapoor and Narayanan](#).

Station suggestions and rationale

Use a declared objective such as covering prior spatial variation with a limited budget. A prior-gradient objective may prioritize strong local change while enforcing geographic separation. It is a heuristic, not evidence of where forecast error is greatest.

Keep these explanations separate:

Quantity	Defensible meaning	Meaning it does not establish
Prior local gradient	How quickly the earlier model scalar changes with physical distance	Future error, uncertainty or observation impact
Distance from planned stations	Geometric spread of this proposed plan	Actual observational coverage unless eligible real observations are included explicitly
Distance from real observations	Geometric separation under stated source/time/QC gates	A calibrated confidence score
Observed model mismatch	A source/QC/time/depth-gated difference at measured points	The true error at all unsampled locations

A missing gradient because a neighbor is masked should be identified as unavailable. Do not label it evidence of a uniform field. If gradients use finite differences, use actual coordinate distances, allow only stated one-sided differences, and do not interpolate through masked neighbors.

For the first release it is reasonable to omit actual-observation mismatch from the objective. Existing comparison logic allows observations within a symmetric time window, so using it without an explicit prior cutoff can expose future observations to the planner. A future mismatch-driven objective needs its own cutoff and provenance tests.

Every plan must retain requested and fulfilled budget, selected native indices, pairwise minimum separation, actual depth, bounds, objective components and reasons. Geographical separation uses a declared great-circle calculation and Earth radius. Do not use raw longitude degrees as kilometres.

An incomplete greedy plan does not prove that no feasible arrangement exists. The correct message is that the stated selection rule could not fill the budget under those constraints. A greedy maximal subset is not necessarily a maximum feasible subset.

Fair baselines and failures

Use the same finite candidate set, budget, geographic/depth limits, prior mask, spacing rule, measurement availability rule, reconstruction method and evaluation targets for every comparison.

- A uniform or space-filling baseline must have a deterministic coordinate-only definition. If it uses farthest-point selection, call it space-filling and explain that rule rather than imply a perfect regular grid.

- Random baselines need a published deterministic generator and seed sequence. Random selection followed by spacing rejection is a random feasible heuristic, not necessarily a uniform draw from all feasible station subsets.
- Use multiple predetermined random trials. Report individual trials or a full downloadable table plus a clearly defined aggregate. Do not report only the best random seed.
- Retain a no-new-measurement persistence baseline using the frozen prior. It gives the experiment an interpretable reference even if all station strategies perform similarly.
- A trial that cannot supply the exact budget must be a failed or unavailable comparison, not a successful smaller-budget score. Do not refill using target values or choose another seed because its score looks better.
- If a chosen station becomes missing at the target time, report that target/trial as unavailable under the declared strict budget rule. A later version can support a prespecified missing-measurement policy, but effective counts must then be visible and budget claims qualified.
- Missing evaluation truth is excluded identically for all strategies at that target snapshot, with the denominator retained. Zero evaluation points is a failure, never a zero-error result.

Scores need units and denominator. Unweighted pointwise RMSE and MAE on a reserved grid are acceptable when labelled pointwise. Do not describe them as basin-wide or area-weighted statistics. If area weighting is implemented, define the native cell-area approximation and verify it independently.

Report worse outcomes normally. A suggested plan is allowed to lose to persistence, uniform or random strategies. Percentage improvement relative to a zero baseline error is undefined; use a null value and explanation instead of infinity or an arbitrary percentage.

Reconstruction method

A practical fixed method is prior plus inverse-distance-squared interpolation of sampled residuals:

```
residual_i = simulated_measurement_i - prior_at_station_i
weight_i(q) = distance(q, station_i)^(-2)
reconstruction(q) = prior(q) + sum(weight_i(q) * residual_i) / sum(weight_i(q))
```

Use an explicit zero-distance branch that returns the measured station value exactly. Duplicate stations should already be forbidden. Do not introduce an undeclared distance offset or clipping. Pin whether all stations or a finite neighborhood contributes. A fixed method avoids choosing hyperparameters after inspecting the holdouts.

IDW gives nearby samples more influence, and its output depends on power and neighborhood. Power 2 is a project convention, not a theoretically optimal choice. IDW also supplies no prediction standard errors; the official [ArcGIS method documentation](#) states these limitations. Residual IDW is not a fluid model, does not conserve heat or salt, and can smooth or miss unsampled structures. It may connect nearby points across a geographical barrier unless a barrier rule is added. Restricting the model domain and showing masks does not automatically solve that issue.

Use null outputs where the frozen prior is missing. Do not fill land or below-bottom cells and present them as model support. A constant residual should be reproduced wherever prediction is supported. A linear field need not be reproduced exactly by IDW, so that is not an appropriate acceptance assertion.

Virtual survey contract

The user-drawn survey is a separate simulated sampling action. Reuse the native section conventions where possible: declared longitude/latitude route, requested stations including endpoints, actual nearest native columns by great-circle distance, exact native depths, one stated snapshot, and no interpolation unless explicitly implemented.

For each sample show requested location, selected native location, horizontal offset, actual depth, timestamp, value, units, source identity and missingness. Every sample must remain labelled simulated. Straight segments are illustrative sampling geometry, not navigable vessel or glider routes. Crossing land, weather, endurance, permissions and instrument response are outside this method.

Survey values at later times must not feed back into automatic plan selection. If the user redesigns a plan after inspecting future samples, that is an exploratory comparison and should be described as such.

Independent numerical acceptance tests

These tests should be implemented separately from the production formulas where practical. Passing a test that calls the same implementation to derive its expected result is not independent evidence.

Test	Independent expected result or invariant
IDW symmetric residuals	Station residuals 2 and 6 at equal distances give residual 4. With target prior 30, prediction is 34.
IDW unequal distances	Residuals 2 and 6 at distances 1 and 2 give $(2 + 6/4)/(1 + 1/4) = 2.8$; target prior 30 gives 32.8.

Test	Independent expected result or invariant
Coincident sample	Prediction exactly equals the simulated measurement, without NaN, infinity or smoothing.
Constant residual	All station residuals equal to 3 produce prior plus 3 at every supported target.
Zero residual	Target values identical to the prior produce the persistence field exactly.
Metric fixture	Truth [10,12], estimate [12,11]: MAE 1.5, RMSE $\sqrt{2.5}$, signed bias 0.5.
Negative result	A deliberately misleading station residual increases RMSE, and the result remains displayed.
Zero baseline error	Percentage change is undefined with a clear reason; finite numerical scores remain available.
Great-circle scale	On the equator, one longitude degree is $R \cdot \pi / 180$ km for the declared radius R; longitude convergence is checked at higher latitude.
Spacing boundary	Every selected pair meets the threshold, duplicates are rejected, and just-below/at/just-above fixtures follow the declared tolerance.
Budget failure	A tiny candidate set or excessive spacing gives an explicit incomplete plan and cannot masquerade as an equal-budget success.
Native depth	Irregular depths [0,10,30,80] retain 30 m at index 2; no equally spaced depth assumption is allowed.
Source samples	A selection of values, native indices and masks in both cases matches independently decoded original packed NetCDF, not merely the prepared array access function.
Survey sampling	Endpoints, a midpoint, a nearest-column tie and a masked column match a separate scalar reference.
Prior-only selection	Replace every future value and future mask while keeping the prior fixed: selected locations, candidate IDs, gradient scores and random choices remain unchanged.
Reconstruction isolation	Change unsampled target values only: predictions remain bitwise or tightly numerically identical; evaluation scores change.
Scoring isolation	Candidate station IDs and evaluation IDs are disjoint; all strategies and trials use the same evaluation IDs at each target.

Test	Independent expected result or invariant
Missing target station	A newly masked station is reported as a failed target/trial under the strict budget rule; no target-dependent replacement occurs.
Reproducibility	Same settings, sources, method version and seeds reproduce plan, samples, scores and fingerprints.
Replay regression	Existing P08/P09 records still replay. Add P10 method identity only to P10 recipes rather than changing all legacy identities.

Before claiming X17 complete, also inspect public outputs for at least both regions, two budgets, a non-default supported depth, an impossible or unfilled spacing request, more than one random seed, and all declared held-out times. Keep the raw tables and failure counts available. Browser interaction tests should verify that stale results retain their own settings while a new request is pending, and that saving captures the applied result rather than unfinished controls.

Acceptance recommendation

The implemented prior-only, equal-budget, disjoint-target, residual-IDW design is suitable for a bounded educational reconstruction sandbox within the checked scope below. This review alone does not close P10.4, P10.5 or X17. Replay, interface and public deployment checks remain required.

Executed independent checks

Command actually run:

```
.venv/Scripts/python.exe -m pytest tests/test_expedition_independent.py -q  
--junitxml=docs/evidence/p10-independent-method-final.xml
```

Result: **23 passed, zero skipped**, 4.31 seconds of test execution. Evidence: independent numerical test report.

The checked scope includes:

- Both Bay of Bengal and Arabian Sea cases, default surface temperature with eight stations, and 100 m salinity with five stations.
- Source candidate values decoded independently from original packed provider NetCDF with automatic scaling disabled. The acquisition-journal file hashes were checked before reading.
- Every RMSE, MAE and bias in all 24 default benchmark rows for each of the two regions. Expected values use independently decoded original source data, Cartesian unit-vector

great-circle distances, direct reciprocal-square weights and separately summed errors. These agree within an absolute and relative tolerance of $2e-12$.

- Exact scalar IDW fixtures, exact measurement reproduction at a station, zero and constant residuals, signed error, latitude-dependent distance and a physical-distance gradient with a masked-neighbor case.
- A read spy that rejects any future-field access during station planning, and a selection spy that requires all eight selection calls to finish before the first future field is loaded.
- A target mutation that changes only unmeasured future values. All 21 reconstructions remain exactly identical, while the scored errors change. This directly checks that withheld evaluation values do not enter reconstruction.
- A newly missing future station that leaves the frozen plan unchanged and produces three visible selected-strategy failures, with failed counts retained.
- A finite-grid impossible-spacing fixture, disjoint candidate/evaluation sets, requested budgets and pairwise separation.
- Six virtual surveys checked against original packed samples at surface, 100 m and 1000 m across both regions, including independent requested-to-grid offsets, cumulative distances and an exact geometric midpoint tie.

Two oracle defects were found while preparing these tests. The initial source helper read coordinates after closing its NetCDF handle; its return was moved inside the context. A scalar atan2 oracle then resolved an exactly equidistant Arabian midpoint differently because of sub-micrometre rounding. The oracle now chooses the first source-order point within $1e-10$ km of the computed minimum. Production explicitly declares and implements that same tie convention. Original failing reports are retained in `p10-independent-method-initial.xml` and `p10-independent-method-fixed-oracle.xml`. These were test-oracle construction failures, not evidence of source-value or reconstruction errors in production.

The review also caught the original contract's depth-index ceiling of 30, which admitted 800 m but excluded advertised 900 m and 1000 m levels. The implementation owner raised the ceiling to index 32. The independent 1000 m surveys now pass.

The acquired original archives are present in this workspace, so no source checks were skipped. On a fresh checkout without those archives, source-dependent tests deliberately report a skip; they do not claim independent source verification from prepared case data alone.

The run reports existing FastAPI/Starlette deprecation warnings and a NetCDF4/NumPy binary-size warning on import. The independent original-file decoding and numerical comparisons completed successfully despite those warnings. This test run does not

establish compatibility on every environment.

Observed default outcomes, without tuning

Settings: temperature, native surface depth, budget 8, minimum spacing 30 km, gradient objective, seed 26067, five predetermined random trials, and target indices 2, 4 and 6. Each case supplies 168 finite prior candidates and 4,620 reserved evaluation points. All default benchmark rows succeeded in these two cases.

Mean of per-run RMSE, degrees Celsius	Gradient suggestion	Uniform / space-filling	Random trials	Persistence
Bay of Bengal	0.162475	0.155987	0.168171	0.203111
Arabian Sea	0.311115	0.207471	0.234092	0.250760

These are arithmetic means of the three time-specific errors for deterministic strategies and the fifteen time/trial errors for random selection. They are not one pooled error over an independent validation population.

The gradient suggestion does not win universally. The space-filling baseline has a smaller mean error in both displayed cases. In the Arabian case, the gradient suggestion also performs worse than the random mean and persistence. This is a valid negative result and must remain visible. Prior gradients are useful explanations of earlier structure; they do not guarantee the most useful future measurements. No settings were changed here to manufacture a winning comparison.

Follow-up presentation and export review

A read-only frontend review was completed after the first Expedition components were written. The reviewer reported concrete issues to the implementation owner and checked the revised source. This follow-up is a source review, not a screenshot or browser-interaction pass.

The following corrections are present in the revised components:

- The displayed verified survey line and A/B endpoints come from the same `survey.query` as its plotted samples. A pending new route therefore cannot silently combine new endpoints with previous measurements. Active drawing displays draft geometry without overlaying the previous survey samples.
- Equal mean error only explains an identical space-filling design when the applied objective is actually space filling. A tie under another objective is described as equal

error, without claiming identical station choices.

- The map and legend use the same full-opacity thermal palette. Missing prior hatching, eligible candidates, suggested stations, simulated native samples and missing survey samples have explicit descriptions.
- The verified survey timestamp and route coordinates remain adjacent to the map. Source time, actual depth, nearest-column offset and simulated status are also retained in the sample inspector. Benchmark tables show point denominators and unsuccessful runs rather than concealing them in an aggregate.

The reviewer also identified that the first simulated-survey CSV lacked its actual sampling timestamp and an explicit simulation label. It retained only the prior timestamp, which could misidentify a later survey. The implementation owner added module, method version, source manifest, simulation flag and actual `sample_time_utc` columns, with dedicated export assertions. A final clarification was requested so benchmark rows containing simulated station values also carry a true simulation flag. The implementation owner's final export test report governs acceptance of that clarification.

Applied-plan, draft and replay handling was inspected in the component source. Save captures the applied plan and compatible calculated results, while pending requests disable save. Case changes remount the Expedition workspace; restored surveys set the shared time to the saved survey snapshot. Actual interaction and visual verification remain in the main task's browser evidence.

Final acceptance closure, release 0.10.1

The benchmark CSV clarification is resolved: both virtual survey and benchmark exports carry a true simulation flag, and surveys retain their actual sample timestamp. The final focused backend run passed 77 checks, including 26 independent source/method tests. Those 26 include three additional independent gradient-norm reference fixtures. See final backend evidence.

The first public exact-replay check exposed platform-dependent gradient-norm differences between Windows and Vercel, with a maximum $6.94e-18$ in the compared records. Source samples, selected stations and reconstruction scores matched. Method `p10-sampling-v2` uses explicit binary64 multiply/add/square-root operations and retains strict full-result fingerprints. Eight Windows records now replay exactly on Vercel, and eight Vercel records replay exactly on Windows. Archived `v1` records stay readable and return an explicit replay method mismatch. Existing `P08/P09` identities are preserved. Initial failures and diagnostic comparisons remain available.

Final public verification passed 116 HTTP checks and 51 browser checks. The public visual audit captured 18 views without page errors, unexpected alerts or document overflow, with representative screenshots visually inspected by the implementation owner. Full evidence and the scope of these checks are recorded in [Phase 10 acceptance](#). This follow-up does not convert automated/browser evidence into a real unfamiliar-user pilot, independent forecast validation or a proof of portability to every future platform.

48 P11 working API contract

Historical or maintained reference: read alongside the current release overview | Source: docs/P11_API_CONTRACT.md

Method p11-drift-v2. The query in science/drift_contracts.py is authoritative.

POST /api/cases/{case_id}/drift/run returns complete DriftResult JSON. POST /api/cases/{case_id}/drift/stream accepts the same query and returns newline-delimited JSON events. Progress: {type: 'progress', completed_steps, total_steps}. Final: {type: 'result', result: DriftResult}. Error: {type: 'error', code, message}. Abort cancels the browser request; the server stops after its current bounded chunk. Incomplete work is not saved or cached. This is a request-scoped calculation, not a durable cloud job.

GET /api/cases/{case_id}/drift/context?depth_index=0&time_index=0 returns DriftContext for the selected starting snapshot. Result includes the same object as background.

DriftContext: {case_id, manifest_sha256, depth_m, model_time, bounds: [west, south, east, north], longitude: number[], latitude: number[], u: (number|null)[], v: (number|null)[], shape: [ny, nx]}. Both components use m/s, flattened latitude-major. Native-coordinate preview subsampling is for display only. Background is fixed at the start snapshot, not the animated particle time.

DriftResult: {schema_version: '1', kind: 'drift_run', method_version: 'p11-drift-v2', case_id, manifest_sha256, simulated: true, query, depth_m, start_time, end_time, forcing_times: string[], bounds: [w, s, e, n], background: DriftContext, output_interval_seconds: number, completed_steps, total_steps, particles: Particle[], summary, methods: string[], limitations: string[]}.

Particle: {id: number, release_longitude, release_latitude, status: 'completed' | 'left_domain' | 'missing_velocity' | 'invalid_release', stop_elapsed_seconds, stop_reason: string|null, distance_km, arrival_elapsed_seconds: number|null, points: Point[]}. IDs start at 1. Point: {elapsed_seconds, longitude, latitude}. Every path includes its initial location and accepted final position; regular output is every lcm(1800, dt_seconds) seconds: 1800 for 300/600 steps, 3600 for 1200 steps. Stop positions are the last accepted integration position. A

rejected step does not invent a coastline impact coordinate/time.

summary: {released,valid_releases,completed,left_domain,missing_velocity,invalid_release,arrived,arrival_fraction:number|null,earliest_arrival_seconds:number|null,latest_arrival_seconds:number|null,mean_distance_km:number|null}. Arrival fraction uses ALL requested particles as denominator, including invalid releases. Mean distance uses valid releases only. Arrivals are first intersections with the optional target box along accepted integration-step segments, linearly timed within a step; initial particles inside the target arrive at zero. These are simulation counts, not real-world probabilities. Arrival need not imply a particle completes its requested duration.

Supported duration is bounded by the available source snapshots. A request beyond forcing availability is rejected, never silently shortened. All particles in a point release coincide under deterministic identical forcing; no display jitter or artificial random motion is applied. Box release is seeded uniform longitude/latitude sampling, not exact area-uniform sampling. Invalid release points remain visible in counts.

Saved recipe {mode:'drift',case_id,query,comparison:DriftQuery|null} uses modules drift_run and optionally reference_drift_run; each module has this complete result and its own seed. Only applied completed results are saveable. Replay checks source/method/results before restoration, as in earlier phases.

Archived v1 records remain readable and downloadable. Exact replay requires v2; older method identities receive an explicit method mismatch. No silent migration or approximate result hash is accepted.

49 P11 independent method review

Historical or maintained reference: read alongside the current release overview | Source: docs/P11_INDEPENDENT_METHOD_REVIEW.md

Review date: 23 September 2026. This document separates source inspection and method recommendations from implementation acceptance. A proposed check is not a passed check. The scientific cases cover 7-10 January 2024 and remain historical.

Recommendation and scope

Use the pinned OceanParcels 3.1.4 two-dimensional RK4 kernel, adapted to the application's bounded execution interface, with its MIT notice, upstream URL, source hash and changes recorded. Independently run complete trajectories through the installed upstream framework in the offline science environment. Keep the deployment description precise: an adapted OceanParcels RK4 kernel is not the entire OceanParcels runtime.

The full 3.1.4 package requires SciPy, xarray, Dask, Zarr below version 3, cgen and other dependencies. These are unnecessary runtime costs for this small fixed-depth problem. Its ScipyParticle path avoids runtime C compilation and is useful for reference testing. OpenDrift is a valid broader framework, but its recommended conda environment adds another environment and many processes outside this phase's scope. These are architectural recommendations, not measured deployment-size or timing claims. [Parcels package metadata](#), [OpenDrift installation](#).

The upstream RK4 kernel uses four interpolated vector samples for each position update. Preserve that stage ordering and coefficients. The framework has changed beyond its v3 interface; pin the source tag rather than copying code from moving latest documentation. [Pinned kernel](#), [MIT licence](#).

This phase is horizontal passive transport at a selected native depth. It has no sinking, buoyancy, windage, wave Stokes drift, turbulent diffusion, beaching/resuspension or vertical motion. A marker may be drawn as a pebble; its calculation is a passive tracer. Historical model-driven paths are simulations, not tracked objects or rescue forecasts.

Inspected forcing data

The review read each case manifest, original acquisition metadata and all seven analytical eastward/northward arrays through CaseStore. Both have shape 40 depths by 76 latitudes by 63 longitudes. Native coordinates are slightly irregular binary values; retain them rather than rebuilding axes from rounded spacing.

Property	Verified source contract
Product	HYCOM GOFS 3.1, GLBy0.08/expt_93.0
Time	Seven snapshots, 12 hours apart, from 2024-01-07 00:00 to 2024-01-10 00:00 UTC
Bay bounds	85.0400390625-90 E, 12-15 N
Arabian bounds	66-70.9599609375 E, 16-19 N

Property	Verified source contract
Components	water_u: eastward_sea_water_velocity; water_v: northward_sea_water_velocity
Units	Both m/s, source scale factor 0.0010000000474974513, offset 0
Source dimensions	Both time, depth, lat, lon on shared coordinate arrays
Depth	Positive down; fixed z-level samples, including native 15 m and 1000 m
Missing values	Original packed -30000 becomes missing, not zero flow

The released fields already provide geographic east/north components on shared points. Do not rotate them again or treat the published arrays as staggered C-grid face velocities. HYCOM's maintainer describes the released z-level velocities as being averaged from cell edges to centres before depth sampling. [HYCOM explanation](#).

Direct array inspection found no U/V mask disagreements in either case at any snapshot. Each depth plane has 4,788 locations. Bay has 4,788 finite pairs at 0, 15, 100 and 1000 m; Arabian has 4,788 at 0 and 15 m, 4,652 at 100 m and 4,513 at 1000 m. Neither has any finite values at 5000 m. These counts are identical across the seven inspected snapshots. Arabian subsurface masks provide a real missing-boundary example; a surface run alone cannot verify mask handling.

Manifest hashes inspected:

- Bay: 9598b33dc9eef89cb909f2811e1590ca580a143c1dbcc25d97e51a84faf7d28d
- Arabian: 5ddd8aaf9ef80700ad0ad7c583947bccad0b693533995f1fc923a8e33e687426

Numerical contract to verify

- 1. Coordinates and units:** interpolate U and V in m/s, then convert each RK stage at its stage latitude. Parcels v3 uses 111120 metres per latitude degree and 111120 times cosine(latitude) per longitude degree. If the app uses this convention, distances must use its consistent equivalent sphere, or explicitly document any separate distance sphere. No degrees-as-metres calculation is acceptable. [Parcels spherical conversion](#).
- 2. Interpolation:** bilinear in the actual native longitude/latitude coordinates and linear between the actual bracketing timestamps. Depth remains at the selected native plane. No time extrapolation, periodic boundary, coastal current invention or nearest-ocean jump. Exact endpoint handling and finite zero-weight corners require an explicit policy.

3. **Missing stencil:** if the declared interpolation stencil includes missing U or V, stop with a missing-forcing reason. Do not convert NaN to zero. Require this at every RK stage and at the accepted endpoint. A missing subsurface sample can indicate bathymetry or absence of data; it is not automatically an observed coastline impact.
4. **Boundary stepping:** inspect intermediate stages and the traversed segment, or impose and test a conservative per-step cell-displacement limit. Valid start/end values alone can jump across a masked strip. Stop at the last verified point and distinguish stop time from exact coastline-hit time.
5. **Time:** choose an integration step much shorter than the 12-hour forcing interval, split at forcing/output/end boundaries, and verify it by step halving. Finer numerical steps do not restore currents absent between the source snapshots.
6. **Source exhaustion:** reject impossible requested durations visibly or terminate explicitly at the last forcing time. Do not repeat the final field, silently truncate requested duration or label a partial path complete.
7. **Release sampling:** a single exact point with identical deterministic particles produces identical paths. Explain this; use a visible region for spread. For a uniform-area geographic rectangle, sample longitude uniformly and sine(latitude) uniformly. Pin the random generator, seed, particle order and retry limit. Invalid initial samples must remain counted or trigger a visible input failure; silently resampling changes the release distribution.
8. **Targets and denominators:** define the target geometry, boundary inclusion and first arrival. Count each particle once, including time zero when released inside. Report arrived / all released, plus stopped and unfinished counts. If only integration endpoints detect arrival, label that time resolution and test that narrow targets cannot be skipped unnoticed. Arrival fraction is a simulated count, not a calibrated probability.
9. **Comparison:** show each release's own start, duration, depth, source and count. A changed start date may have less available forcing. Comparison must not silently give one release a shorter run or hide its failures.
10. **Jobs and replay:** bounded deterministic segments can avoid persistent queues on Vercel, but returned state must not be accepted as trusted scientific output. Pin source hash, method, kernel, settings, particle IDs, step index and output cadence. Verify that a segmented run equals a one-shot run, that cancelled responses cannot overwrite a new job, and that Windows/public replay survives serialization. Numerical rounding for a documented output format is distinct from silently loosening saved-result integrity checks.

Required independent numerical evidence

The acceptance set should include stationary particles in zero current; analytic east/north displacement at nonzero latitude; a nonlinear or rotating manufactured field with independently calculated expected paths; varying-time forcing; native-node and interior interpolation against original packed NetCDF; both regional cases; real Arabian subsurface masks; outside-domain and end-time failures; all arrival denominators; point/region seed reproducibility; segmented/restarted equivalence; and upstream full-Parcels comparisons away from missing boundaries. The custom strict mask policy also needs separate fixtures because upstream default behavior is not our policy oracle.

For timestep convergence, retain results for at least three steps, compare against the smaller-step/reference solution and state the measured differences. Piecewise linear forcing and spatial derivative changes can reduce observed convergence order; do not promise fourth-order convergence in all real fields merely because the integrator is RK4.

Independent drifter search performed

NOAA GDP provides quality-controlled interpolated drifter positions. The six-hour product uses kriging; the hourly product uses a separate higher-frequency processing method. Their positions are observations processed onto regular times, not independent exact ground truth at every instant. Drogue status and depth, uncertainty, time overlap and possible assimilation must be reviewed before any model comparison. [NOAA GDP access description](#).

Live ERDDAP metadata was retrieved on 23 September 2026:

- `drifter_hourly_qc`: HTTP 200 metadata, time coverage ends 2022-10-31 00:00 UTC, so that endpoint cannot supply this January 2024 case.
- `drifter_6hour_qc`: HTTP 200 metadata, time coverage ends 2025-06-18 12:00 UTC, so the January 2024 query is within its advertised dates. [Six-hour metadata](#).

The following six-hour queries requested ID, WMO, time, position, velocity, position errors, drogue-loss date, drogue depth and buoy type. Both returned HTTP 404 with Your query produced no matching results. (`nRows = 0`):

- Time 2024-01-07T00:00:00Z through 2024-01-10T00:00:00Z inclusive, longitude 85.0400390625-90, latitude 12-15.
- Same time, longitude 66-70.9599609375, latitude 16-19.

Reproduce at https://erddap.aoml.noaa.gov/gdp/erddap/tabledap/drifter_6hour_qc.csv with the projected fields `ID,WMO,time,latitude,longitude,ve,vn,err_lat,err_lon,drogue_loss_date,DrogueCenterDepth,typebuoy`, the stated inclusive constraints and

```
orderBy("ID,time").
```

A wider control query for the same time, 60-95 E and 5-23 N returned HTTP 200 and real tracks. For example, GDP ID 300534062363320 starts that interval at 14.64 N, 70.777 E and lies south of the Arabian case. This confirms access and available 2024 regional data; it does not establish an overlapping validation track.

The retained HTTP search evidence records the metadata response hashes, exact URLs, status codes and responses. Its wider query returned 65 rows from five IDs. The example buoy reports drogue loss on 2023-09-27, another reason not to treat it as a matched 15 m passive tracer in January 2024.

Conclusion: no overlapping track was retrieved from the checked GDP product. No independent drifter validation has occurred. This does not prove that no suitable drifter exists in any other archive. Expanding the source/model area solely to force a validation claim is outside this bounded phase.

Acceptance status

Source semantics, actual arrays, framework source/dependencies/licence and the drifter search have been inspected. The numerical implementation was then reviewed and tested independently. Public browser behavior and cross-platform replay are separate root-task acceptance checks; this document does not pre-approve them.

Implemented numerical choices

- The supported fixed native depths are 0-1000 m. The app uses original analytical binary64 currents and coordinate values. A production-adaptor test compares every current sample and mask at 0 and 1000 m against independently decoded original files for both cases.
- A fixed-order Taylor cosine matches independent standard-library cosine within $2.3e-16$ across 12,001 latitudes spanning -30 to 30 degrees. The field rejects latitude axes beyond that domain. The calculation retains unrounded binary64 particle positions; this is not a claim of cross-platform replay before that check runs.
- The declared strict policy requires the full spatial interpolation cell, even when one corner has zero interpolation weight. Exact source-time point samples need only that snapshot; between-source samples need both. Whole traversed rectangles use all spanned temporal brackets. This conservative policy can stop before an actual boundary contact.
- Box releases are uniformly sampled in longitude and latitude, explicitly not area-uniform. This supersedes the optional area-uniform recommendation above. The

seeded generator affects release locations only.

- Output cadence is the least common multiple of the selected integration step and 1800 seconds: 1800 seconds for 300/600-second integration, 3600 seconds for 1200-second integration. The last accepted position is retained even between output frames. The initial fixed 1800-second metadata statement was identified during review and corrected in the store contract.
- Paths accumulate the stated midpoint spherical metric, using the same 111120 metres-per-degree convention. Target entry uses a closed rectangle and segment intersection, so narrow targets crossed between integration endpoints are detected. Crossing time remains approximate within that numerical step.

Independent checks actually run

Command:

```
.venv/Scripts/python.exe -m pytest tests/test_drift_independent.py -q  
--junitxml=docs/evidence/p11-independent-method.xml
```

Result: **45 passed, no skipped tests**, in 4.91 seconds for this local run. Final numerical report. A nonfatal existing NumPy/netCDF binary-size warning was emitted during provider-file import and is retained. The earlier 37-check run is retained at initial report.

The checks cover closed-form zero/constant/linearly changing velocities, cosine conversion, manufactured rotating-flow convergence, original packed-file samples and production adapter arrays, upstream framework agreement, strict masks, a real Arabian subsurface mask, outside-domain and exhausted forcing, narrow-target and time-zero arrivals, original release denominators, coincident point releases, seeds, output cadence and physical distance units.

Complete installed OceanParcels 3.1.4 ScipyParticle runs provided the separate trajectory reference. Four combinations, Bay/Arabian and 0/1000 m, each used three release positions for 24 hours at 600-second steps. All 12 final positions agreed exactly on the checked machine. The assertion tolerance is $2e-10$ degrees to allow the documented different interpolation arithmetic order, and does not weaken saved-investigation integrity rules. Source arrays and particle positions were binary64. Parcels' final accepted position was read from `lon_nextloop`, `lat_nextloop` and `time_nextloop`; reading `lon` at that point would omit the final step. A constant-flow oracle explicitly verifies this detail.

Real-case timestep checks use 1200, 600, 300 and 150 seconds. The declared numerical acceptance is less than two metres of endpoint change relative to 150 seconds for these checked paths. All combinations pass. The retained quantitative 600/300/150-second comparison records these maximum differences:

Case and depth	600 vs 150 seconds	300 vs 150 seconds
Bay, 0 m	0.001154 m	0.000343 m
Bay, 1000 m	0.0000213 m	0.00000618 m
Arabian, 0 m	0.002059 m	0.000286 m
Arabian, 1000 m	0.0000486 m	0.0000342 m

These are numerical integration differences for selected paths, not real-ocean position accuracy. Manufactured rotating flow exhibits error reduction exceeding fourteenfold at each step halving; real piecewise-interpolated fields are not assigned a universal convergence order.

No unresolved discrepancy was found in the inspected bounded numerical scope. This review is a separate implementation check within the development process, not external scientific endorsement or real drifter validation.

Frontend and export scientific presentation review

A subsequent read-only review inspected `web/src/drift/DriftMap.tsx`, `DriftWorkspace.tsx`, `contracts.ts`, the App integration and the drift branches of `api/investigation_store.py`. This is code inspection; browser interaction and screenshot results remain separately recorded by the implementation team.

The map preserves east/north orientation. Its longitude scale includes the map-centre cosine, and arrow direction compensates for each sample's latitude before projecting. Arrows are equal length and explicitly represent direction at the starting snapshot. They do not masquerade as time-varying currents during trajectory playback. Playback interpolates saved output positions for drawing only; arrival calculations use accepted integration segments. The inspector computes the displayed calendar time from each run's own historical start and the selected particle's available elapsed time. Comparing different start dates uses shared elapsed time, with an explicit explanation.

Completed numerical paths remain separate from draft controls, are hidden during geometry drawing, and are saved from applied results. Particle stop explanations refer to the last accepted numerical step, not exact contact, beaching or observed outcomes. Arrival cards use all requested particles, including invalid releases; mean distance uses valid releases. Each run declares its start/end, depth, count, integration step, target and output cadence. The readable report retains the simulation distinction and methods. CSV trajectories contain simulation status, method, manifest hash, source case, depth, seed, historical start and actual timestamp for each output position; terminal status is explicitly named `final_status`.

Two review findings were passed to the frontend owner and corrected in the inspected code:

1. **Preview masks:** the background samples every fourth native grid point and fills coarse display cells. Before the correction, this was not explicit in the map caption. A concrete Arabian 1000 m example is native cell centre 70.91998291015625 E, 17.619999885559082 N: its full interpolation stencil is invalid, while its preview patch uses a finite zero-current sample at 70.9599609375 E, 17.600000381469727 N. The caption now states that the background is subsampled, hatching describes preview samples and a coloured cell does not guarantee native support everywhere within it.
2. **Comparison background identity:** the backdrop follows the selected particle's run. The subtitle now explicitly identifies run A or B and its starting depth/time, so the background cannot silently be attributed to both experiments. Target-box legend entries also identify the corresponding run.

A small separator typo in the run card was corrected. The additional drawing-context error-state recommendation was forwarded: a failed draft-context fetch after a completed run should show its retry/error state while drawing, rather than remain labelled as loading. The final implementation corrects this behavior; the corresponding test passes in all three public browser projects in `evidence/p11-v2-public-browser.json`.

Final release follow-up recorded by the implementation lead

Initial public exact replay exposed a summary-only difference from Python 3.12's changed float summation. Retained incident comparisons show identical trajectories, per-particle distances, arrivals and statuses; the largest mean-distance difference was $1.78\text{e-}15$ km. Method `p11-drift-v2` pins sequential binary64 addition in particle order. It does not round source values, alter paths or introduce a fingerprint tolerance. D039 records the primary Python documentation and diagnostics.

The final focused backend run passes 139 tests, including the same 45 independent scientific checks and new summation/legacy-archive cases. Public HTTP verification passes 100 checks over 114 requests. Eight v2 records replay exactly from Windows to Vercel and eight in reverse; original P08/P10 records remain exact. These are deployment consistency checks, not new independent drifter validation. The final public browser suite passes 54 tests, and the 20-view public visual audit records no page errors, document overflow or unexpected alerts. See [Phase 11 acceptance](#) for evidence links and the remaining limits.

50 P12 Heat & Depth API contract

Historical or maintained reference: read alongside the current release overview | Source: docs/P12_API_CONTRACT.md

Implementation contract, 23 September 2026. Acceptance is pending. science/heat_contracts.py defines method p12-heat-v1 and HeatQuery.

Requests

GET /api/cases/{case_id}/heat/catalog returns schema_version 1, method_version, case_id, model_manifest_sha256, heat_manifest_sha256, locations, years [2023,2024], baseline_period [1982,2011], model_times, depth_limits_m [100,300,700,1000], default_query, source, methods, caveats.

Location records have id, case_id, label, latitude, longitude, requested_latitude, requested_longitude and selection_rule. IDs are bay-west, bay-center, bay-east, arabian-west, arabian-center and arabian-east. Only the selected case's locations are returned. Default query chooses the center point, year 2024, auto event, model time 0, 300 m. Auto event prefers the first event intersecting an available model day; otherwise the first event in the selected year. This deterministic selection rule is shown in the interface and is not a strongest-event search.

POST /api/cases/{case_id}/heat/analyse takes:

```
{"location_id": "bay-center", "year": 2024, "event_id": null, "model_time_index": 0, "depth_limit_m": 300}
```

Year must be 2023 or 2024. Model index is 0-6. Depth limit is one of the listed native levels. Event ID is null (automatic selection) or mhw-YYYY-MM-DD-YYYY-MM-DD, and must belong to the location/year calculation. Unsupported inputs fail rather than silently switch data.

The result is a JSON object with:

- schema_version: "1", kind: "derived", method_version, case_id.
- query: applied query, with automatic selection resolved to its actual event ID, or null when no events exist.
- model_manifest_sha256, heat_manifest_sha256, location_sha256.

- **location:** the public location record above; **source:** OISST attribution/provenance metadata.
- **baseline:** `baseline_period`, `percentile`, `window_days`, `smooth_days`, `feb29`, `missing_policy`. Full 366-day `seasonal_mean_c`, `threshold_c` and `pool_count` arrays are also included for reproducibility.
- **series:** selected year's daily records
`{date, sst_c, seasonal_mean_c, threshold_c, anomaly_c, exceeds_threshold, event_id}`.
 Missing SST/anomaly/exceedance are null; climatology remains separate. Event IDs also mark finite cool days in a merged event.
- **events:** events intersecting the selected year. Each has `id`, `start`, `end`, `peak_date`, `duration_days`, `exceedance_days`, `bridged_days`, `mean_intensity_c`, `max_intensity_c`, `cumulative_intensity_c_days`, `left_boundary_unknown`, `right_boundary_unknown`, `complete_boundaries`, `starts_before_year`, `ends_after_year`, `depth_overlap`. Full context-detected event dates are preserved.
- **selected_event:** one of these event objects or null.
- **depth_link:** `{status, message, available_time_indices, model_time}`. Status is `available`, `no_event`, `outside_model_period` or `time_outside_event`. No model time is silently replaced. An overlapping event may still require selecting one of its available model indices.
- **depth_profile:** null unless `depth_link` is available; otherwise the profile described below.
- **observations:** list of source profiles in the selected case bounds and event dates, each with `id`, `title`, `platform`, `time`, `latitude`, `longitude`, `distance_km`, `time_offset_hours`, `source_id`, `source_url`, `source_sha256`, `provenance_hold` and `note`. This is nearby event context, not a claim of collocation or independent validation. Source QC remains in the linked original inspector.
- **methods:** explicit surface and depth method strings; **caveats:** limitation strings.

Depth profile

`depth_profile` contains `latitude`, `longitude`, `requested_latitude`, `requested_longitude`, `distance_km`, `model_time`, `time_index`, `depth_limit_m`, `longitude_index`, `latitude_index` and:

- **points:** `{index, depth_m, in_situ_temperature_c, practical_salinity, pressure_dbar, absolute_salinity_g_kg, conservative_temperature_c, potential_temperature_c, sigma0_kg_m3, density_kg_m3, status}`. Missing/unavailable diagnostics are null. Status is `valid`, `missing_source`, `unsupported_source_range` or `outside_gsw_funnel`.

- **layers:** {top_depth_m, bottom_depth_m, midpoint_depth_m, temperature_gradient_c_per_m, n_squared_s2, pressure_mid_dbar, n_squared_depth_m}. Gradient uses midpoint_depth_m; N-squared uses its pressure-derived n_squared_depth_m. Preserve gaps and negative N-squared.
- **mixed_layer:** density and temperature objects, each {depth_m, status, reference_depth_m, reference_value, threshold, supported_to_m}. Status is crossing, no_crossing, missing_support or reference_unavailable. No-crossing is a lower bound within the supplied column, not an exact depth. Density uses sigma0 increase 0.03 kg/m3 from 10 m; temperature uses absolute potential-temperature difference 0.2 degrees Celsius from 10 m.
- **heat_content:** {value_j_m2, value_gj_m2, status, top_depth_m, bottom_depth_m, reference_conservative_temperature_c, cp0_j_kg_k, name}. Status available or missing_support. Label as column potential-enthalpy content relative to CT=0 degrees Celsius; do not label it heatwave excess or an anomaly.

Saved investigations

Recipe {mode: "heat", case_id, query} uses the applied resolved query. Module name heat_analysis, method p12-heat-v1, no random seed. Full result and methods are retained. Source identity adds heat_manifest_sha256 only to heat records; old record identities remain unchanged. The TypeScript source parser must accept this optional field and require it for a heat recipe. Backend replay excludes absent optional fields when comparing old source identities.

Readable report and numerical CSV/ZIP retain all daily values, detected events, profile points, adjacent-layer diagnostics, methods, source hashes and full integration/baseline definitions. Source and analysis years remain separate from the 2026 build/export date.

51 P12 independent scientific method review

Historical or maintained reference: read alongside the current release overview | Source: docs/P12_INDEPENDENT_METHOD_REVIEW.md

Review date: 23 September 2026. This document separates proposed acceptance checks from evidence actually produced. Independent numerical acceptance, local and public HTTP/source/export checks, and exact replay in both deployment directions are complete. Browser acceptance is recorded separately.

Surface event definition and reference

The surface detector follows the daily, seasonally varying threshold definition in [Hobday et al. \(2016\)](#). The project fixes the baseline at 1982-2011. It is a declared historical reference period, not a claim that every application should use that baseline. NOAA OISST at the supported source grid locations supplies the daily series; this is an analysed surface product, not an instrument profile or a measurement of the entire water column.

For each calendar month/day, collect the actual daily samples within five days of every occurrence inside the baseline. The climatology input is restricted to the baseline itself. Index windows at its first and last five days are clipped, so samples from 1981 or 2012 cannot enter. Pooling follows consecutive dates, not a circular distance between 366-day ordinal labels. The mean and linearly interpolated 90th percentile define the raw seasonal curves. February 29 is interpolated from February 28 and March 1 before both curves receive a circular 31-day moving mean.

An event requires at least five consecutive finite daily temperatures strictly above the seasonal threshold. Two already qualifying events can join across at most two observed sub-threshold days. Unknown values cannot be filled or crossed. The latter is a deliberately stricter missing-data rule than the upstream reference, which substitutes climatological means before event joining. Reference agreement is therefore checked on complete series, while missing-day behavior has separate expected fixtures. The available 2022-2025 context supplies original event boundaries for displayed 2023/2024 cases. Events reaching the context boundary must be flagged as potentially truncated.

The independent comparator is Eric C. J. Oliver's [marineHeatWaves implementation](#), commit `d7292bf08ade0af213fa760b0d7e4adfe5f52894`. Its original Python file has SHA-256 `3f47a51382f68894f34deaf6b466c944cf4f88f92a09a7f00f6d6acd609a0db8`. The code and licence are acquired under ignored `data/reference/p12/` by `python -m science.acquire_heat_reference`. They are used only for offline checks. Production does not import or vendor this GPL-3.0-or-later implementation. The test loader changes only the removed NumPy alias `np.NaN` to `np.nan` in memory; source bytes remain intact. Reference event dates must match exactly. Tiny floating-point differences in seasonal averages are assessed separately from date agreement and do not establish environmental accuracy.

Source definitions and thermodynamics

The existing checked HYCOM files declare `water_temp` as `sea_water_temperature` with units `degC`. The source adapter and case manifest define this as in-situ temperature. Salinity is the source's practical PSS-78 salinity, not Absolute Salinity. The source documentation describes the [GOFS 3.1 product](#). The checked Argo adapter also supplies in-situ temperature, practical salinity, source pressure, coordinates, per-variable quality and its provenance hold. These definitions must survive linking and exports.

The calculation sequence is pressure from positive-down depth using `p_from_z(-depth, latitude)` for model coordinates; `SA_from_SP(SP, p, longitude, latitude)`; then `CT_from_t(SA, t, p)`. Passing in-situ temperature to `CT_from_pt` would be wrong. Source instrument pressure, when supported, takes precedence over reconstructing it from rounded depth. The official [pressure conversion](#), [salinity conversion](#) and [in-situ temperature conversion](#) define the required units and signs. Unsupported definitions, invalid samples and missing pairs must remain unavailable.

Potential density anomaly uses [sigma0](#), referenced to zero sea pressure, with units `kg/m3`. It is not in-situ density minus 1000. Stratification uses [GSW Nsquared](#) at adjacent valid pressure levels, with latitude supplied. Values have units `s^-2` and belong at returned middle pressures. Negative values must not be clipped away; they describe statically unstable density structure in the sampled profile. Gaps must not be joined. The 75-term polynomial's oceanographic funnel is checked before publishing these diagnostics.

Mixed-layer estimates use a 10 m reference, with the first downward crossing of a 0.03 `kg/m3` increase in [sigma0](#) or a 0.2 C absolute change in potential temperature. These follow the threshold family assessed by [de Boyer Montegut et al. \(2004\)](#). The two criteria are distinct derived estimates, not native HYCOM MLD variables. Linear interpolation locates a crossing only inside an adjacent valid bracket. No crossing in the supported interval means a lower bound or unavailable result, not an invented bottom crossing. Actual-depth in-situ temperature gradients use adjacent differences divided by actual metre spacing and retain their sign.

The heat-content diagnostic is a declared column integral of potential enthalpy density, $\text{integral } \rho(\text{SA}, \text{CT}, p) * c_{p0} * \text{CT} \, dz$, relative to `CT = 0 C` and in `J/m2`. Here $c_{p0} = 3991.86795711963 \, \text{J}/(\text{kg K})$, as defined in the [TEOS-10 Manual](#). Numerical quadrature, integration limits and incomplete support must be explicit. This is not an anomaly from a subsurface climatology, additional heat caused by a surface event or a cyclone-energy forecast.

Acceptance plan and limits

- Compare seasonal means, percentiles and event dates with the pinned independent reference on complete synthetic series and actual source series.
- Exercise exact threshold equality, minimum event length, short gaps, missing days, leap-day treatment, context edges and baseline-only sampling.
- Check source decoding against original packed HYCOM files without production ingestion helpers.
- Check the public GSW numerical examples, plus the GSW conversion chain independently applied to original columns. Use a separate quadrature implementation and closed-form fixtures for depth spacing, gradients and integrals.
- Verify coordinate signs, pressure units, masks, temperature semantics, mixed-layer crossings and no extrapolation.
- Retain the distinction between selected surface locations, model profiles and nearby observations. A co-occurrence does not prove a subsurface marine heatwave or attribute its cause.

Independent code/reference agreement establishes the implemented numerical method for the tested inputs. It does not validate the gridded ocean analysis against withheld instruments, calibrate forecast skill, prove causality or replace the pending real-user pilot.

Executed evidence

Reference acquisition has completed; its pinned identity and acquisition helper are recorded above. The initial installed-library check found that the six-point Absolute Salinity example on the official 2015 GSW 3.05 HTML page differs from the pinned 3.6.20 release by at most 0.00007452 g/kg. We did not widen a tolerance or change production data. Instead, the test uses the upstream check cast shipped with [GSW-Python v3.6.20](#). All 98 finite Absolute Salinity values match that cast exactly. The exact cause of the older example's difference has not been established. The selected reference, checksum and original difference remain in `evidence/p12-gsw-reference-selection.json`.

The first 50 independent checks passed in `evidence/p12-independent-initial.xml`. They included complete-series synthetic comparisons with the pinned Oliver reference for ordinary and leap years, complete baseline and gap policies, eight original HYCOM columns across both cases and endpoint times, independent SciPy quadrature, a closed-form linear potential-enthalpy-density integral, signed gradients, negative stratification, first mixed-layer crossings and missing support. The initial upstream scalar-conversion deprecation warnings are retained in that report. Later runs suppress

only repetitions of that known upstream warning. At this initial stage, actual OISST point comparisons and final integrated acceptance were still pending source acquisition. The completed source-backed results are recorded below.

Static interface review identified three concrete presentation corrections: label the mixed-layer marker as a potential-temperature criterion when drawn on the in-situ-temperature curve; describe observation context as unavailable when the selected model snapshot cannot support the link, rather than claiming the library has no qualifying profiles; and select a visible event day when the full event's peak lies outside the displayed calendar year. The frontend implementation now incorporates those corrections. This is a code and scientific-copy review; browser interaction and visual acceptance are recorded separately.

Independent source-packing investigation

The retained official distributors do not always emit identical Float32 bytes. We tested the original NCEI packed daily files for 1 January 1982 and 7 January 2024 against their corresponding AOML and PSL subsets. Each comparison covers all 116 valid source cells in the retained two-latitude subset, including the six selected locations. The original source stores integer hundredths of a degree with a Float32 `scale_factor` of 0.01. That attribute's exact binary value is 0.009999999776482582.

For the checked 1982 date, legacy AOML `_T` values exactly equal the original integer divided by 100 and then converted to Float32 in all 116 cells. PSL instead exactly equals the original NetCDF Float32 scale decoding in all 116 cells. The two expressions differ in 37 cells, by at most 0.0000019073486328125 C, exactly one Float32 step at these temperatures. For the checked 2024 date, both modern AOML `_T_v1` and PSL equal the original NetCDF binary scale decoding in all 116 cells. These calculations numerically account for the observed rounding difference. They are empirical checks of the retained dates, not an assertion about the distributors' internal implementation or every historical file.

We preserved all served values and did not normalize them to a preferred rounded number. Source hashes, original packed integers, scale attributes, expected conversions and observed values are retained in `evidence/p12-source-packing-diagnostic.json`. Both exact conversion regressions pass in `evidence/p12-packing-initial.xml`. The final selected-location source tests cover all six locations on three original NCEI dates: 1 January 1982, 1 July 1996 and 7 January 2024. They require exact equality under the independently checked preferred distributor decoding, rather than a wider tolerance. The 1996 date also checks the recovered baseline archive directly. Full reference event calculations use the actual assembled source series, with exact date and duration agreement still required.

Complete-source numerical and local HTTP results

The completed pack supplies 16,071 daily values per location from 1 January 1982 to 31 December 2025. Every location has the full declared 30-year baseline and 2022-2025 event context. The preferred AOML distribution supplies 14,966 dates; original PSL responses supply 1,105 dates absent from that archive. In particular, the missing 1994-1996 baseline years were recovered from real source responses. Larger provider responses were repeatedly cut off, so the final 1996 recovery used bounded ten-day queries. Failed attempts remain in the acquisition journal. No day or temperature was interpolated to complete this archive.

All **76 independent scientific tests pass with zero skips**, recorded in `evidence/p12-independent-final.xml`. The 24 final source-backed checks add six actual complete-series comparisons with the pinned Oliver implementation and 18 exact primary NCEI comparisons. Seasonal means and thresholds agree within $1e-10$ C, event starts, ends, peaks and durations match exactly, maximum and mean intensities agree within $1e-10$ C, and cumulative intensities within $1e-8$ C-days. These are numerical implementation tolerances, not uncertainty bounds on the ocean product. The earlier 52 checks cover source semantics, published GSW references, independent native-column calculations and analytical edge cases. One installed-library binary-compatibility warning is retained in the report; the actual library/source numerical checks pass.

The local verifier also passes **88 checks across 85 HTTP requests**, with no retries and no failures, in `evidence/p12-local-http-initial.json`. It checks the six locations in both display years, source identity and exact native values, four supported Bay depth limits, eight new captures and exact replays, actual numerical CSV tables and ZIP contents, mismatch rejection, and all 20 retained P08/P10/P11 records. Eight new records are saved in `evidence/p12-local-records.json` for the separate deployment replay test. The runnable commands were:

```
.venv/Scripts/python.exe -m pytest tests/test_heat_independent.py -q
--junitxml=docs/evidence/p12-independent-final.xml
.venv/Scripts/python.exe -m science.verify_p12_public --base-url
http://127.0.0.1:8010 --expected-version 0.12.0 --output docs/evidence/p12-local-
http-initial.json --records-output docs/evidence/p12-local-records.json
```

The actual Bay events overlap the short January 2024 model case. Arabian events do not overlap it, so their unavailable depth panels and absent depth CSVs are correct scientific outputs. A Bay event beginning in 2023 can legitimately retain its January 2024 depth link when viewing the 2023 portion of that same event. For example, the Bay-center event spans 11 October 2023 to 10 February 2024. This follows the full-event boundary policy rather than restricting scientific context to the plotted calendar year.

Public deployment and exact replay in both directions

Release 0.12.0 on <https://depth-atlas-seifuku.vercel.app> passes **105 independent HTTP checks across 101 requests**, with zero failures and no automatic retries. The report is [evidence/p12-public-http-initial.json](#). It covers all six locations in both years, the four supported Bay depth limits, source URL and fingerprint identity, eight fresh captures, numerical CSV and ZIP evidence, all retained P08/P10/P11 exact replays, and all eight local P12 records replayed on Vercel with identical scientific results and exported tables. Guard checks reject altered methods, sources, recipes and result fingerprints. The same run verifies the public legal pages, favicon and source notices.

The eight fresh Vercel records in [evidence/p12-public-records.json](#) also replay exactly on the Windows server, including their numerical ZIP evidence. This reverse check passes **19 checks across 17 requests**, with zero failures or retries, in [evidence/p12-public-to-local-replay.json](#). No platform-dependent arithmetic change or relaxed comparison was required. Both directions use the accepted heat manifest SHA-256 `b2c1186439af2a66c982392961145300f0c015792327b6b96721216278d15fff` and method `p12-heat-v1`.

```
.venv/Scripts/python.exe -m science.verify_p12_public --base-url https://depth-atlas-seifuku.vercel.app --expected-version 0.12.0 --output docs/evidence/p12-public-http-initial.json --records-output docs/evidence/p12-public-records.json --replay-records docs/evidence/p12-local-records.json
.venv/Scripts/python.exe -m science.verify_p12_public --base-url http://127.0.0.1:8010 --expected-version 0.12.0 --output docs/evidence/p12-public-to-local-replay.json --replay-records docs/evidence/p12-public-records.json --replay-only
```

These results establish numerical reference agreement, faithful source transport and reproducibility for the tested bounded cases. They do not establish an operational forecast, independent validation of OISST or HYCOM against withheld ocean observations, a heatwave diagnosis throughout the water column, or a completed first-time-user pilot.

52 Phase 12 surface-temperature sources

Historical or maintained reference: read alongside the current release overview | Source: docs/P12_SURFACE_SOURCES.md

The Heat & Depth Lab uses daily NOAA Optimum Interpolation Sea Surface Temperature (OISST), distributed by NOAA AOML and NOAA Physical Sciences Laboratory (PSL), for six fixed ocean grid cells. Data acquisition is separate from event detection. The pack contains source temperatures, coordinates, dates and provenance. It does not contain a hand-picked catalogue of warm events.

Source and meaning

- Product: NOAA OISST v2.1, daily, 0.25-degree grid, degrees Celsius.
- Primary product description:
<https://www.ncei.noaa.gov/products/optimum-interpolation-sst>.
- Dataset DOI: <https://doi.org/10.25921/RE9P-PT57>.
- Main distributor: NOAA AOML ERDDAP, with [the original archive](#) through 2021 and [the separate modern archive](#) from 2022. The original archive lacks 1994-1996. The selected original and modern axes also lack nine isolated days.
- Gap recovery and cross-checks: [NOAA PSL](#), using <https://psl.noaa.gov/thredds/dodsC/Datasets/noaa.oisst.v2.highres/sst.day.mean.YEAR.nc>.

OISST blends bias-adjusted satellite and in-situ measurements and fills spatial gaps by interpolation. It is an observation-based gridded analysis. It is not an individual measured profile, the HYCOM model used elsewhere in the app, or independent proof that every ocean layer is anomalously warm. Complete gridded coverage does not mean complete direct observation coverage.

NOAA states that September 1981 to December 2015 SST and anomaly values in v2.1 are unchanged from v2. Some older annual PSL files retain legacy v2 titles and history. Those original attributes are preserved. The modern files identify Version 2.1. The source pack must document this compatible lineage rather than modifying the old source metadata.

The source product's separate published anomaly field uses a different climatology. We acquire SST only. The application calculates its declared baseline and threshold from these daily temperatures.

Fixed sample selection

Before running a heatwave detector, choose west quartile, center and east quartile longitude in each existing case rectangle, all at its center latitude. Choose the nearest OISST native cell in each coordinate independently; select the lower coordinate on an exact tie. All six locations are kept, whether or not their selected year has an event. They represent points, not regional averages or heatwave area.

Location ID	Case	Native longitude	Native latitude
bay-west	Bay of Bengal	86.375 E	13.375 N
bay-center	Bay of Bengal	87.625 E	13.375 N
bay-east	Bay of Bengal	88.875 E	13.375 N
arabian-west	Arabian Sea	67.125 E	17.375 N
arabian-center	Arabian Sea	68.375 E	17.375 N
arabian-east	Arabian Sea	69.625 E	17.375 N

The completed pack covers every day from 1 January 1982 through 31 December 2025. The fixed baseline is 1982-2011, with 2023 and 2024 analysis years. Adjacent years provide context for events crossing a year boundary. AOML is the preferred distributor on every available day. A real PSL source sample fills each absent AOML date. No date is interpolated. The detector owns the duration, threshold, leap-day, joining and missing-value rules. These source scripts do not silently invent those rules.

Time and units

The annual PSL files encode daily means at midnight UTC as days since 1800-01-01. AOML uses seconds since 1970-01-01 and noon-centered timestamps, as in the original NCEI daily product. Both original numeric times and encoded timestamps are retained. Per-day `source_file_indices` reference the units and calendar in each manifest source file. The location file has no single `time_units` value, because the source origins differ. A daily SST value is compared by its calendar day; it is not presented as an instantaneous midnight or noon instrument measurement.

Both distributors serve SST as Float32 in Celsius, with a valid range of -3 to 45 and declared missing markers. Preparation promotes those numbers exactly to JSON numbers. No temporal interpolation, smoothing, regridding or display rounding changes the analytical source pack. Missing markers become `null`. An otherwise finite out-of-range value raises an error.

The old AOML distribution can differ from PSL by one Float32 representable step. Every overlapping acquired point-day is compared. Accepted differences must remain within one Float32 ULP and the same original 0.01-degree quantization bin. Each served value is retained without a corrective rounding operation. Primary NCEI packed integers are checked with both Float32 scale multiplication and decimal hundredth-degree conversion; these are explicit numerical explanations, not a claim to have audited the distributor's internal processing code. AOML's modern distribution and the old distribution are checked separately. Exact saved-result replay remains exact; this source comparison does not loosen replay checks.

There are no per-sample instrument QC flags in the acquired SST subsets. We do not acquire the optional source error field as part of the heatwave calculation. The app must not imply that a finite interpolated SST has passed a direct instrument-quality test. Three original NCEI daily files, 1 January 1982, 1 July 1996 and 7 January 2024, support checks at all six cells. Agreement between two distributions of OISST is not independent ocean validation.

Reproduction and pack contract

From the repository root:

```
.venv/Scripts/python.exe -m science.acquire_heat --aoml
.venv/Scripts/python.exe -m science.acquire_heat --recover-gaps
.venv/Scripts/python.exe -m science.acquire_heat --primary-spots
.venv/Scripts/python.exe -m science.prepare_heat
```

data/raw/heat/ holds ignored original DAP2 and ERDDAP NetCDF responses, source metadata and checkpoint journals. Every saved response has its URL, retrieval time, size and SHA-256 fingerprint. Retries are bounded; cached checksums must match before resuming. A circuit breaker stops queued PSL work after repeated service failure. The --recover-gaps command compares the AOML missing-date list with verified PSL coverage and acquires only uncovered dates, in serial blocks of at most ten days. Six isolated recovery files from this build also retain a separate journal under recovery-gaps/. An incomplete or changed source must fail preparation rather than becoming a plausible-looking series.

casepacks/heat/manifest.json declares source, baseline period, analysis years, context years and six locations. Each location includes its requested and actual coordinates, source selection rule and prepared-file fingerprint. Each location JSON contains aligned dates, sst_c, source_timestamps, source_numeric_time and source_file_indices arrays. Original metadata and time-axis responses are copied into source-metadata/.

Access observations

CoastWatch ERDDAP access timed out in this session. The original NOAA AOML daily archive was reachable but incomplete. It was initially rejected as a sole source; later checks established a practical merge with the separate modern AOML archive and genuine PSL recovery data. The NCEI THREDDS aggregate was reachable but started in 2020. These observations do not imply that any service is permanently unavailable.

PSL annual metadata and short binary subsets were reachable. The first 2024 annual response exceeded each of three 55-second read deadlines. An annual three-cell alternative ended prematurely at about 61 seconds. Ninety-day subsets sometimes succeeded and sometimes exceeded the deadline. Thirty-day requests worked but later encountered transient 502 responses. Acquisition stopped, retained its cache, and resumed after the service recovered. Recovery work used one PSL request while AOML acquisition used one request. The final 1996 recovery used two concurrent ten-day requests with one journal writer; every ten-day request succeeded. Existing checked chunks remain available for overlap checks. No source data were fabricated to fill an access failure.

A static annual PSL download was also probed. Its 1996 file was 477,794,497 bytes. A corrected two-MiB range probe took 11.41 seconds to receive headers and 1.88 seconds to read the body, about 1.06 MiB/s. The earlier ten-second read deadline was shorter than the observed header delay, so that initial timeout did not establish slow body transfer. No full annual download was started; the existing subset recovery finished. Separate evidence preserves both probe outcomes.

Completed source checks

Preparation passed on 23 September 2026. Each of the six locations contains 16,071 daily values with no missing values, including 10,957 finite baseline days. Across the shared daily axis, 14,966 days use AOML and 1,105 use PSL. The prepared pack is 7,386,504 bytes. These counts describe source coverage at six selected points, not regional coverage or forecast skill.

All 18 comparisons against original NCEI files passed the declared source representation checks. Across acquired overlapping subsets, 973 of 6,282 point-day comparisons differed numerically. The maximum difference was 0.000003814697265625 degrees Celsius. Every difference was within one Float32 representable step and retained the same original centidegree value. The pack preserves each preferred served value; it does not round the values to make distributors agree.

Two complete preparations from the checked raw journals produced exactly the same manifest and all six location-file hashes, including source indices and metadata ordering. The accepted manifest SHA-256 is
b2c1186439af2a66c982392961145300f0c015792327b6b96721216278d15fff.

Evidence is retained in docs/evidence/p12-surface-preparation.json, p12-primary-source-comparison.json, p12-distributor-comparison.json and p12-source-rebuild.json. Individual requests, failures, successful recoveries and their elapsed times remain in the raw acquisition journals. Scientific event detection and depth diagnostics have separate acceptance checks; these source checks alone do not validate those calculations.

A further check reconstructed all 96,426 original numeric timestamps across the six locations using each referenced source file's time units and calendar. Every timestamp and calendar date matched exactly. Running gap recovery with network access deliberately blocked found the completed cache and left the acquisition journal unchanged. These checks are recorded in docs/evidence/p12-time-provenance-resume.json.

53 Phase 13 API and replay contract

Historical or maintained reference: read alongside the current release overview | Source: docs/P13_API_CONTRACT.md

Implementation contract, 23 September 2026. Source preparation and local numerical acceptance are complete; final public acceptance is recorded separately in PHASE_13_CLIMATE.md. Method p13-climate-v1.

GET /api/climate/catalog supplies three pinned historical SON events, index definitions and sources, the GODAS field source and 1991-2020 baseline, original Pacific coordinates, supported calendar windows, default query, methods and limitations. Each event maps to a real shared-workspace case: pacific-godas-2013-son, pacific-godas-2015-son OR pacific-godas-2022-son.

POST /api/climate/analyse accepts:

```
{"event_id":"son-2015","reference_event_id":"son-2013","period":"SON","longitude_index":70,"depth_index":10}
```

period is SON, 09, 10 or 11. The same calendar window applies to both historical events and the baseline. SON is an explicitly equal-weight mean of the three complete monthly means. A single-month choice remains a monthly field; event labels refer to their pinned three-month ENSO window. Longitude/depth selection uses exact native Pacific indices

and is rejected when unavailable.

The response contains `schema_version`, `kind`, `method_version`, `case_id`, `query`, `climate_manifest_sha256`, `model_manifest_sha256`, `baseline`, `field_source`, `index_definitions`, `events` (selected then reference), `period`, `panels` (selected then reference), `difference`, `scales`, `selected_point`, `observations`, `methods` and `caveats`.

Each panel has its `event`, `case_id`, `model_manifest_sha256`, `actual_period_start` and `period_end_exclusive`, `pacific`, `indian` and `profile`. A section carries `latitude`, `latitude_index`, `monotonic longitude`, `longitude_source_indices`, `depth_m`, `shape`, `dimensions`, `flat depth-major potential_temperature_c`, `baseline_c`, `anomaly_c`, `missing_value:null`, `baseline_required_year_count` and a sampling note. The profile carries the same three quantities at the selected exact Pacific longitude. No equatorial interpolation is implied.

`difference` contains matching Pacific/Indian section geometry with `values_c` equal to selected minus reference, plus `profile_c`. `scales` gives shared potential-temperature and symmetric anomaly ranges for each basin, and symmetric difference ranges. Bounds are computed from actual finite values; absent values remain null. `selected_point` retains native longitude, latitude and depth, both original/baseline/anomaly values, and their signed difference. Event panels show index labels separately from field values.

`observations` contains genuine original profile links for each event, with actual positions, dates, instrument/source identity and quality information. They are context, not pointwise validation of a monthly potential-temperature field against an instantaneous in-situ measurement.

The existing investigation service accepts mode `climate`, the selected event's real `case_id`, and the applied query. The optional `expected_climate_manifest_sha256` checks the combined field/index identity, including all three shared Pacific model manifests and original observation-library indexes. Replay includes the exact original source/method identities. New optional identities must stay absent for older recipe families. Numerical CSV exports retain sections, profiles, baseline values, index context and selected values. JSON, HTML and ZIP retain the definitions and source versions.

Main Pacific case packs use the established CaseManifest/OperationResult contracts with temperature-only availability, `sea_water_potential_temperature`, monthly support, and a declared continuous $[0,360)$ longitude axis. Other variable requests are rejected, not replaced. Shared Ocean Explorer and original instrument inspection remain available; unsupported lab paths need a direct explanation.

Exported numerical tables

File	Data rows	Meaning
climate-sections.csv	11,760	Both years, both basin sections, original potential temperature, matching baseline and anomaly
climate-difference.csv	5,880	Selected minus reference on identical native coordinates
climate-profiles.csv	56	The selected exact Pacific column in both years
climate-index-context.csv	6	Both events and their three separately labelled centered index seasons
climate-selected-point.csv	1	Exact source point values and signed difference

These counts describe the fixed supported grid. The index table retains all three context months even when the field query selects one month. Every table also carries method, source, baseline and applied-selection metadata. Missing numeric values are empty CSV cells, not zero. The standalone CSV download is the sections table; the evidence ZIP contains all five. Display rounding does not change these exported numerical values.

The Pacific 3D viewer defaults to 6000x vertical exaggeration and allows a 1-12000 range. These are retained visual recipe settings; actual depths, source values and numerical-result fingerprints do not change. Existing Indian views retain their 200x default and 1-250 slider.

54 P13 field sources and preparation

Historical or maintained reference: read alongside the current release overview | Source: docs/P13_FIELD_SOURCES.md

Prepared in September 2026. The displayed ocean periods are historical September, October and November in 2013, 2015 and 2022. Project dates do not change those source dates.

Source choice

The field source is the NOAA NCEP Global Ocean Data Assimilation System, distributed by NOAA PSL. NOAA CPC links this NetCDF distribution from its [GODAS background page](#). The [PSL dataset page](#) supplies attribution guidance, and its [THREDDS catalogue](#) lists the annual archives.

The actual source variable is `pottmp`, potential temperature in kelvin. It is a monthly mean, with the timestamp set to the first day of the averaging period. This differs from the instantaneous in-situ temperature used by the earlier HYCOM cases. The source declares COARDS metadata and does not supply a CF standard name for `pottmp`; preparation explicitly maps NOAA's potential-temperature definition to `sea_water_potential_temperature`. It does not claim that this attribute was present in the original file.

The [CF definition of sea-water potential temperature](#) references the temperature after an adiabatic change to sea-level pressure. That quantity definition supports the explicit surface-reference explanation used in the lab. It does not make potential temperature interchangeable with the original Argo in-situ readings.

Public GODAS salinity and both current components were also inspected. Current coordinates are staggered by half a degree relative to the temperature grid. P13 therefore supplies temperature fields only. The system does not fabricate missing variables or silently put those currents on the temperature grid.

Exact extraction

The common subset contains 3 months, 28 native depths, 30 native latitudes and 240 native longitudes:

- September, October and November, without a seasonal averaging step during acquisition.
- Native depths from 5 to 459 m. The original archive has 40 levels down to 4,478 m; the bounded P13 extraction stops at the last native level below 500 m.
- All native latitude centres inside 5 degrees south to 5 degrees north.
- Longitude centres from 40.5 to 279.5 degrees east, keeping the original increasing convention through the dateline.

The Pacific workspace cases retain 140.5 to 279.5 degrees east. The common climate pack also supports Indian Ocean context on the same source grid and dates. The nearest source row to the equator is 0.16612949967384338 degrees north, and must be labelled with that actual latitude. No exact-equator row is invented.

science/acquire_climate_fields.py fetches contiguous native Float32 OPeNDAP subsets and reconstructs bounded NetCDF archives. These are source subsets, not downloads of the entire original annual files. The acquisition journal retains the source URL, original source-index ranges, numeric time units, original variable metadata, checksums and any failed attempts. Each request has a 45 second timeout, with at most two attempts per year; successful years are checkpointed. Cached files must match their recorded hashes.

Baseline and transformations

The fixed field baseline is 1991 through 2020, with one separate mean for each of the three calendar months. Preparation uses the 30 actual annual fields under the declared complete-year policy below. An initial guessed `godas/pottmp.mon.1tm.1991-2020.nc` URL returned 404. Later review found the real [official climatology in the Derived directory](#), so that initial response did not establish that a provider climatology was absent.

The official product declares the same 1991-2020 period, Float32 potential temperature and a `valid_yr_count` field. Its metadata permits a mean with at least 3% of input values present, and its history records generation on 27 January 2022. P13 retains its already pinned Float64 reconstruction requiring all 30 source years, rather than silently switching to a product with a different missing-data policy and storage precision. Only the official product's DDS/DAS metadata were acquired during this later review; no numerical comparison with its climatology values has been performed. The metadata and exact source URLs are recorded in `p13-official-climatology-metadata.json`. This correction changes the source explanation, not the prepared baseline or its fingerprints.

Preparation applies this declared sequence:

1. Preserve source Float32 values and coordinate centres in the acquired archive.
2. Mask source missing values and declared valid-range failures.
3. Promote valid kelvin values to Float64, then subtract 273.15 to produce Celsius potential temperature.
4. Sum each calendar month's cell in ascending year order from 1991 through 2020 in Float64.
5. Divide by 30 only when all 30 values are finite. Otherwise keep the baseline cell missing. Preserve the contributing-year count separately.

No baseline gap is filled, no missing value is treated as zero, and no field is shifted to fit an expected ENSO appearance. The index definitions and historical labels are documented separately. The displayed GODAS anomaly baseline is a different source quantity and calculation from the surface-index baselines, even when their named year

ranges agree.

Analytical files retain Float64 values. Pacific display files choose every second latitude and every third longitude, including the final coordinate of each axis, and keep every depth. Their shape is 28 by 16 by 48. Each display file is Float32, with the largest rounding error recorded in its case manifest. This is spatial selection and numeric rounding, not a smoother or an interpolated reconstruction.

Original observations

Original Argo profiles are selected from the existing checked GDAC index before looking at their temperatures or QC. For each selected year and calendar month, candidates lie within 5S-5N and 160W-120W. Selection minimizes $\text{abs}(\text{latitude}) + \text{abs}(\text{longitude} + 140)$, then distance of the calendar day from the 15th, then source path. This is a small reproducible observation example, not an optimal observing network.

The original GDAC NetCDF files are read through the existing core Argo and instrument contracts. Adjusted-mode records never fall back to raw measurements. Per-parameter QC, original coordinates, pressure, derived depth, calibration metadata and original file hashes remain inspectable. The general source citation is [Argo GDAC, SEANOE; Argo's acknowledgement guidance](#) states that the data are freely available without restriction and requests attribution.

The observations are instantaneous in-situ temperatures. GODAS is a monthly mean of potential temperature. Co-display supplies evidence and context, but a direct temperature residual between those quantities is disabled. GODAS may have assimilated these observations, so agreement would not establish independent validation even after a justified quantity/time conversion.

Reproduction and evidence

From the repository root:

```
.venv/Scripts/python.exe -m science.acquire_climate_fields  
.venv/Scripts/python.exe -m science.prepare_climate_fields
```

The source journal is `data/raw/climate/field-acquisition.json`, with raw archives under ignored `data/raw/climate/`. Prepared source fields are in `casepacks/climate/field-manifest.json`. The three shared workspace cases are `casepacks/pacific-godas-2013-son/`, `casepacks/pacific-godas-2015-son/` and `casepacks/pacific-godas-2022-son/`. Their additional instrument libraries are under matching subdirectories of `casepacks/instruments/`.

All 31 model years and all nine original Argo files were acquired and prepared. The raw source files total 37,669,671 bytes. All 31 model acquisition attempts succeeded on their first attempt. A cached acquisition was then repeated with network calls disabled, and its acquisition journal remained byte-for-byte unchanged. Evidence: cache and resume check.

The full baseline contains 535,279 complete month/cells and 69,521 missing month/cells. Counts range from 0 to 30, and only count 30 produces a finite baseline. This count describes source availability, not calibrated confidence. Preparation evidence is p13-field-preparation.json.

The original September 2022 Argo profile has 994 accepted temperature readings but no accepted salinity readings. The older joint core contract therefore reports zero jointly eligible samples, while the per-variable instrument contract correctly retains those accepted temperature samples. The profile remains in the case, with its source flags visible. It was not replaced after seeing its QC. The other eight source profiles contain between 932 and 1,002 jointly eligible samples.

A second complete preparation reproduced all 62 checked case, instrument and field-pack files without changing a single hash. Evidence: deterministic rebuild. The frozen field manifest SHA-256 is

b5b015c3401f74e3ae9be4e92e11abf97410899be85b465a43447d3fa4b728b2.

Shared case	Manifest SHA-256	Prepared bytes
pacific-godas-201 3-son	a939fb4f64828f3d73d1bfc6fb677fbae 1b3903eb532a2aca7298edf0bbec33d	2,954,055
pacific-godas-201 5-son	dc46f59743f242e4a8dc3b2e83256ab0b b420aac3285d94c0673a9c8803ded0c	2,848,797
pacific-godas-202 2-son	5eb03d4d2f9bd7fbbbd97d19889083da6 e31d9c014a6b027a1e98a950cf5656c	2,965,850

Independent reads through NOAA's NCSS NetCDF interface matched all 1,008 compared source values exactly. These checks span 1991, 2005, 2013, 2015, 2020 and 2022, all three SON months, all 28 retained depths and the adjacent 179.5/180.5 degrees east cells on either side of the dateline. No numerical tolerance was used for the field values. Source request records and transport comparison results retain the evidence.

NCSS represents the selected latitude as 0.16612979769706726 degrees north, whereas the original OPeNDAP coordinate is 0.16612949967384338 degrees north. The difference is 0.0000002980232238769531 degrees. The source service cause was not established. Checks used the uniquely nearest native latitude and required identical time, depth and

longitude coordinates; prepared files retain the original OPeNDAP latitude. The discrepancy is recorded instead of rewriting the native grid.

These checks establish reproducibility, source transport and contract preservation. They are not independent physical validation of the GODAS analysis. Numerical climate calculations and deployed interaction checks are recorded separately by the phase acceptance work.

55 P13 independent method review

Historical or maintained reference: read alongside the current release overview | Source: [docs/P13_INDEPENDENT_METHOD_REVIEW.md](#)

Review date: 23 September 2026. These checks establish numerical/source consistency for the selected historical cases. They do not establish independent validation of the ocean state, ENSO causality, local-weather effects or forecast skill.

Index evidence

All 26 tests in `tests/test_climate_indices.py` pass. Original provider documents and independent source-format checks support the three RONI labels, their qualifying windows, the separate ONI values and the separately attributed IOD episodes. The source-method details and the 2026 operational change are in [P13_INDEX_SOURCES.md](#).

The current RONI ERSSTv6 values for SON 2013, 2015 and 2022 are -0.2, +2.0 and -1.1 C. The 2015 IOD episode is documented by BOM, while the separately sourced CPC seasonal DMI is +0.39 C. The application retains that distinction rather than forcing a threshold classification. Missing or unassigned events remain unassigned.

Field reference method

The 29 tests in `tests/test_climate_independent.py` read all 31 acquired source-year subsets directly through netCDF4. They do not import the GODAS preparation code or the production mean helper to construct expected ocean values. Original hashes, Float32 Kelvin type, monthly variable definition, coordinate axes and timestamps are checked before calculations.

Three sets of original cell values were transcribed independently at 105 m, 0.16612949967384338 N, 210.5 E. Prepared and served values must equal the original promoted Float32 Kelvin value minus 273.15 exactly. All shared Pacific analytical fields for

every selected source month are also compared exactly against the original source conversion, preserving masks and native cell order.

The climatology check uses independent scalar `math.fsum` calculations at nine locations/depths per calendar month over 1991-2020. The comparison permits at most $1e-12$ C to account for the intentionally different summation order. This is a reference arithmetic tolerance, not a source-value modification or a saved-investigation replay tolerance. Every raw cell's finite contribution count is checked across all 30 baseline years. A count below 30 must leave the baseline missing.

All twelve combinations of three historical years and four windows (September, October, November and SON) are checked. The expected point temperature, matching baseline, anomaly and selected-minus-reference difference come from original source values and independent scalar means. Closed-form fixtures separately check an equal-month seasonal mean against a day-weighted mean, missing-month propagation, difference orientation, shape errors, native depth spacing and section/profile layout.

The longitude axis is retained continuously through 179.5 and 180.5 E. Display conversion to -179.5 uses only label notation, not source-cell reordering. The source has no exact-equator row; the actual closest row is 0.16612949967384338 N. The checks preserve this position and the irregular 5-459 m depth coordinates. First-of-month source stamps are explicitly calendar-month means with start-inclusive/end-exclusive intervals.

Original observations are restricted to the selected calendar window. They remain instantaneous in-situ profiles, while GODAS contains monthly potential temperature. Source QC is preserved, including the September 2022 profile whose temperature and salinity acceptance differ. No direct residual or independent-validation claim follows from co-display.

Evidence available at this checkpoint

- p13-index-tests.xml: 26 passed.
- p13-index-audit.json: 22 source/hash/classification checks passed.
- p13-independent-formulas.xml: four closed-form checks passed before the complete source run.
- p13-independent-final.xml: all 29 source/scalar/formula checks passed on the actual completed pack.
- p13-field-transport.json: the separately owned source-preparation audit compares 1,008 values from an independent NOAA NCSS transport path with the acquired OPeNDAP fields.

- p13-field-rebuild.json: all 62 field/case/instrument files reproduce byte for byte.

science/verify_p13_public.py separately checks HTTP transport, exact saved replay, original profiles and exported numerical tables. The completed local run on <http://127.0.0.1:8012> passes 116 checks across 112 requests, with zero retries. All twelve event/window combinations, nine original profiles, native/display fields, dateline coordinates, eight fresh captures and their numerical exports, all 28 retained P08/P10/P11/P12 records, input/integrity guards and legal/static routes passed. Evidence is p13-local-http-final.json, with eight portable records in p13-local-records.json.

The initial report p13-local-http-initial.json retains 115 passes and one verifier error: an oversized analytical volume correctly returned HTTP 413 while the new verifier expected HTTP 422. Only that expected status changed before the final run. The API, scientific values and comparison precision did not change, and all eight captured result fingerprints agree between the initial and final reports. The final run rewrote the records output, so the earlier report's whole-record-file hash refers to the initial file rather than the retained final file. Both reports retain their original request and result fingerprints.

Before deployment, the combined source identity was extended to cover all three shared Pacific case manifests and their instrument-library indexes, in addition to the field and index packs. The source data and scientific method did not change. Final local acceptance was rerun against the restarted service with that complete identity:

p13-local-http-source-closure.json passes 116 checks across 112 requests, zero retries. Its eight records are p13-local-records-source-closure.json. The final combined source hash is 13b4c3aa2fc568272758f485d384044b5f98a54b87b9aa47d3d89743ebe9cd77. Earlier P13 files remain separate pre-release evidence and are not the records used for public replay.

Full production acceptance at <https://depth-atlas-seifuku.vercel1.app> on deployment dp1_4ABA1ZMJ02YqLgY6cSkugbE6yddJ passes all 133 checks across 128 requests, with zero retries. The release is 0.13.0 and the combined source identity matches the final local run. All twelve event/window results, nine original Argo profiles, shared Pacific fields, eight new saved investigations, numerical CSV/ZIP exports and all 28 retained P08/P10/P11/P12 records passed. The eight final local P13 investigations also replay and export exactly on Vercel. Evidence is p13-public-http-final.json; the eight fresh public records are p13-public-records.json.

Those public records were then replayed on the local Windows service at <http://127.0.0.1:8012>. All 19 checks across 17 requests passed, with zero retries, including exact numerical equality for each result and all five CSV tables in each ZIP. Evidence is p13-public-to-local-replay.json. Neither direction uses a numerical replay tolerance or automatic retry. These checks establish reproducibility for the selected records across this local Windows runtime and this deployed Vercel runtime; they do not establish

compatibility with every possible host or data revision.

A later UI-only correction to the sidebar's source attribution and date-range wording was deployed as `dp1_4aVLAaqS82jcbV9iJKjHD3w43UAK` at the same production alias. Scientific/API code and source packs were unchanged. A separate final smoke run passed all 19 checks across 17 requests, with zero retries: release/source identities matched, and all eight final local investigations replayed and exported all five numerical CSV tables exactly. Evidence is `p13-public-replay-sidebar-final.json`. The full 133-check production report and the earlier reverse replay report remain unchanged; they were not rerun or relabelled as checks of this later UI deployment.

Browser and visual acceptance are separate work and are not implied by the numerical results above.

56 P13 historical climate-index sources

Historical or maintained reference: read alongside the current release overview | Source: `docs/P13_INDEX_SOURCES.md`

Prepared on 23 September 2026. This document concerns the index catalogue, not acceptance of the ocean-field, browser or deployment work.

The selected seasons

The application compares September to November in three years. Each ENSO label comes from one pinned NOAA CPC RONI ERSSTv6 release. Legacy ONI values remain separately named context.

Season	RONI ERSSTv6, C	ENSO label	ONI ERSSTv5, C	ONI ERSSTv6, C	CPC DMI ERSSTv6, C	IOD episode context
SON 2013	-0.2	Neutral	-0.1	-0.1	-0.17	No episode label assigned

Season	RONI ERSSTv6, C	ENSO label	ONI ERSSTv5, C	ONI ERSSTv6, C	CPC DMI ERSSTv6, C	IOD episode context
SON 2015	+2.0	El Nino	+2.5	+2.3	+0.39	BOM documents a positive

Season	RONI ERSSTv6, C	ENSO label	ONI ERSSTv5, C	ONI ERSSTv6, C	CPC DMI ERSSTv6, C	IOD episode context
						episode from late August to mid-Novem ber
SON 2022	-1.1	La Nina	-0.9	-0.9	-0.48	BOM documents a negative episode, strong in July to September and ending by late November

These three seasonal ENSO labels agree across the separately checked definitions. Their values do not agree and are never substituted for one another. The application does not assert that every month of a year has the same ENSO or IOD phase.

RONI, ONI and the 2026 change

NOAA's [January 2026 notice](#) makes RONI the official ENSO monitoring and prediction index from 1 February 2026. The [current historical RONI page](#) identifies ERSSTv6 and a 1991-2020 baseline. It compares the Nino 3.4 anomaly with the tropical mean anomaly and rescales the difference. The application imports the provider values; it does not recalculate RONI from the bounded GODAS field.

Historical labels require five consecutive overlapping three-month windows at the threshold. The published table's classification is the label authority. A separate implementation checks inclusive $+0.5/-0.5$ C runs against those classifications. This matters because descriptive sentences on the source page alternate between threshold wording and strict greater-than/less-than wording. The actual published classifications include boundary values. All 924 seasons in each of the three pinned ENSO tables agree with the independent run check, including missing future cells.

SON 2015 falls in an 18-season RONI run from OND 2014 through MAM 2016. SON 2022 falls in a 35-season RONI run from AMJ 2020 through FMA 2023. The pack retains every value in those runs. SON 2013 has no qualifying warm or cold run.

The [ONI ERSSTv5 table](#) and [ONI ERSSTv6 table](#) are separately pinned. Their policy uses centered 30-year baselines with five-year updates. The catalogue does not invent a fixed ONI baseline or substitute RONI's fixed baseline. The earlier `ONI_v5.php` address now presents a moved-page notice, which is why actual destination tables were acquired explicitly.

Provider pages warn that recent estimates can change for two months. The selected 2013, 2015 and 2022 seasons are outside that recent provisional interval. They are described as a pinned historical snapshot, not as data guaranteed never to be revised.

IOD has separate evidence

The numerical series is the [CPC seasonal DMI text file](#). It explicitly identifies ERSSTv6 and the 1991-2020 climatology. DMI is the western tropical Indian Ocean anomaly, 50-70 E and 10 S-10 N, minus the southeastern anomaly, 90-110 E and 10 S-0 N. The [HTML table](#) matches all 920 text rows, including two missing entries. Its generic header also describes centered periods and ten-year updates; the pack preserves that inconsistency and uses the specific downloadable series definition.

The official DMI is retained as published. Subtracting the two displayed component values can differ by 0.01 C because the three text columns are separately rounded. The application does not replace DMI with that rounded subtraction.

The [Bureau of Meteorology 2015 statement](#) supports the historical positive IOD label and its approximate timing. The [2022 statement](#) supports the negative episode and its weakening through austral spring. These statements describe agency event assessments. They are not an application classification produced by thresholding a three-month CPC number. In particular, the 2015 SON DMI is +0.39 C, below +0.4 C, while the documented positive episode occupies a different interval. Both facts stay visible.

No IOD label is assigned to 2013 merely because its DMI is near zero. Exact IOD start/end dates, weekly persistence and local weather effects are not invented from annual statements. September, October and November controls expose ASO, SON and OND index windows respectively, explicitly named as centered three-month context. They do not pretend that a seasonal index is a single monthly measurement.

Other sources reviewed

The [PSL HadISST DMI description](#), raw monthly series and original NCL method were archived. Its monthly values differ from CPC ERSSTv6. The acquired method reads a separate monthly climatology but does not state its years directly. It is research evidence, not a second numerical IOD source silently blended into the application.

The [BOM IOD year list](#), 2016/2019 annual statements and [JAMSTEC's study of the 2019 event](#) were checked as contextual alternatives. They do not create additional deployed cases or demonstrate causal attribution for our three selected years.

Reproduction and checks

Original unmodified bytes and HTTP metadata are under ignored `data/raw/climate/indices/`. Seventeen source files retain URL, resolved URL, retrieval timestamp, byte count, SHA256 and available Last-Modified/ETag headers. Acquisition reuses verified cached bytes. A changed cached file stops preparation. There is no live index request in the browser or public analysis request.

Run from the project root:

```
.venv/Scripts/python.exe -m science.climate_indices --acquire --prepare  
.venv/Scripts/python.exe -m pytest tests/test_climate_indices.py -q  
--junitxml=docs/evidence/p13-index-tests.xml
```

The prepared files are `casepacks/climate/indices.json` and `casepacks/climate/index-manifest.json`, with schema `p13-indices-v1`. Two preparations reproduce identical bytes. The source snapshot identity is `a5d0378b1d738aca26556db1d3383470b2def495f0983297c921423a0f5fc26d`.

All 26 index tests pass. They cover provider classification checks, an independently downloaded two-decimal RONI series, DMI text/HTML comparison, explicit source transcriptions, five-season boundaries, cross-year windows, missing support, ordered seasons, independently sourced IOD labels, deterministic packing and altered-source rejection. The first run had 25 passes and one mistaken expected source-row count; that report remains in `evidence/p13-index-tests-initial.xml`. The corrected expectation retains both explicit missing DMI rows. No source value or tolerance was changed.

Machine-readable evidence is `p13-index-audit.json`, with final test evidence in `p13-index-tests.xml`. Index consistency does not validate GODAS subsurface fields, prove climate causality or establish operational forecast skill. Those claims are outside these checks.

57 Phase 14 observation blackout: independent code review

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/P14_BLACKOUT_INDEPENDENT_REVIEW.md

Reviewed on 24 September 2026, Asia/Calcutta. The reviewer implemented the separate feature evolution module and did not implement the blackout calculation. This is an independent review within the project, not institutional scientific endorsement.

This review inspected source code, tests and original prepared-profile metadata. It did not run application calculations, browser checks, test suites, builds or deployment. Test cases described below are inspected evidence designs, not new pass claims. A later integration closure is recorded at the end with its separate executed evidence.

Scope and conclusion

Inspected `science/blackout.py`, `science/blackout_contracts.py`, `api/blackout_store.py`, the P05 comparison rules reused by `blackout`, `api/p14_exports.py`, the relevant capture/export code in `api/investigation_store.py`, `tests/test_blackout.py`, `tests/test_p14_api.py` and the `blackout` frontend response parser.

No blocking numerical or source-QC mutation defect was found in the supported current calculation paths. Exclusions operate on evidence availability. They do not alter the ocean field, source quality flags, comparison gates or retained residuals. Two concrete integration findings were reported for correction. Their status must be closed using the subsequent implementation and verification evidence, not this review alone.

Findings reported

Source sample indices are not profile-local positions

At review time, `web/src/blackout/contracts.ts:30` required every removed source sample index to be smaller than the profile's sample count. The scientific contract does not make that assumption. `science/blackout.py:99` preserves original eligible indices, and `adapters/instruments.py:216` preserves row indices from a grouped glider source file.

Direct inspection of the checked profile JSON established these examples:

Original profile ID	Number of profile samples	Original source index interval
glider-ru29-20240 419T1430-17135378 49-0326d5ec6f	355	357 to 711
glider-ru29-20240 419T1430-17135407 42-0326d5ec6f	1,582	712 to 2,293
glider-ru29-20240 419T1430-17135425 99-0326d5ec6f	1,493	2,294 to 3,786

These records are under `casepacks/instruments/`. Their April 2024 dates do not currently overlap the supported January model window, so their blackout exclusions remove no eligible pairs. The parser error would become visible with a legitimately eligible grouped profile. The arithmetic fixture in `tests/test_blackout.py:111` already tests preservation of indices [11, 37] from a three-row input.

Recommended correction: remove the profile-count upper bound while retaining finite nonnegative integer, uniqueness and removed-pair-count checks. Add a frontend parser regression with non-contiguous source indices. Do not renumber the source samples.

Pacific blackout source exports omit the original model file record

At review time, `api/investigation_store.py:172` retained source file records only when their singular `time` field matched a selected model time. The GODAS records in `casepacks/pacific-godas-2015-son/manifest.json` use a `timestamps` list for the three monthly fields contained in the original annual subset; they have no singular `time` field.

Consequently, a Pacific blackout capture filters the model entry's `files` list to empty, dropping the original subset path and checksum from that source reference. The model manifest fingerprint remains present, and numerical eligibility remains unchanged, but the source export is incomplete. The separate Climate workflow appends its complete field-source records; blackout does not use that Climate append path.

Recommended correction: for the new blackout path, also retain model file records whose `timestamps` overlap the referenced model times. Preserve prior-phase record behavior and verify a Pacific blackout archive's source-file record explicitly.

Scientific checks from inspection

Question	Inspected implementation
Are overlapping exclusions counted once?	BlackoutQuery sorts and deduplicates selections. <code>resolve_exclusions</code> validates names against this case and builds one selected entry per profile ID. Instrument, platform, collection and individual selections form a union.
Are original QC and model values unchanged?	<code>analyze_blackout</code> reads existing comparisons and deep-copies retained coverage profiles and result objects. It does not mutate comparison rows or request a model rerun. The store uses the existing checked comparison engine.
Which indices are removed?	Only <code>row.sample_index</code> values from rows with <code>accepted == True</code> are included for an excluded profile. Rejected rows and retained-profile indices are not inserted into the lost-pair list.
Do metrics describe retained samples?	Original and remaining rows are assembled in the same library and source-row order. Existing <code>metrics</code> computes equal-sample bias, RMSE, MAE and maximum absolute residual from accepted rows. No profile weighting, interpolation or new confidence score is added.
Do counts and reasons match retained coverage?	Modified totals sum only retained profiles. Their original first-failed scientific exclusion counts are combined separately. Whole-profile user removals have dedicated counts and are not relabelled as source QC rejection.
Are cached source changes detected?	The store checks the model manifest, every source frame of the requested variable, the original observation index and all prepared profile payloads before using cached evidence. All times are checked because original coverage includes source-backed alternative-time suggestions.
Does Pacific incompatibility remain?	Blackout calls the unchanged P05 matcher. Monthly GODAS potential temperature does not become compatible with instantaneous in-situ temperature. Case-specific caveats remain attached, and unavailable Pacific salinity is rejected.
Do exports retain numerical values and identifiers?	The four blackout CSV tables copy coverage metrics, profile effects, original removed identifiers and statements. Nested identifiers/reasons remain JSON within CSV cells. Method, case, model-manifest and observation-library identities accompany each table. The model-source record omission above needs its separate correction.
Is forecast impact claimed?	Both method text and exported report wording state that the model has not been rerun. A different retained-sample error statistic is not forecast improvement, observing-system impact or calibrated confidence.

Independence of the inspected tests

The small arithmetic fixture defines residuals -1, 3 and -2 independently of the production statistics helper. It checks three eligible pairs, zero mean residual, $RMSE \sqrt{14/3}$ and MAE 2. Excluding the first two eligible rows leaves the independently specified residual -2. Other cases exercise union selections, all exclusions, an already ineligible profile, source-index preservation and output mutation without changing the original objects.

The real retained-profile statistical constants match the earlier P05 independently checked source case for platform 4903775, rather than being populated by calling blackout to obtain expected values. Some integration checks intentionally compare against the existing P05 matcher; those establish correct reuse, not a second independent ocean validation.

The archive verifier compares exported tables and replay results with the saved calculation. This checks consistency and reproducibility. Because expected CSV text uses the common formatter, that verifier alone cannot establish scientific arithmetic correctness; the independent arithmetic and source fixtures serve that separate purpose.

Limits

The review covers profile and group removal from the checked library. It does not establish optimal sensor placement, observing-system simulation experiment results, independent model accuracy, forecast benefit or performance with arbitrary new datasets. Samples within a profile are dependent, and counts are not ocean-area coverage. An unchanged error metric after removing an ineligible profile is an expected result.

Subsequent integration closure

Both findings were reproduced and corrected before deployment.

`web/science-tests/p14-contracts.test.mjs` exercises the actual TypeScript parser with a clearly synthetic fixture: source indices 11 and 37 are accepted for a three-row profile, while duplicate, negative and fractional identifiers remain rejected. Exact residual-metric changes remain distinct even when the displayed sentences round identically. All three focused tests pass; all 34 frontend numerical/geographic/parser tests pass. The initial failing fixture output remains in `evidence/p14-parser-sparse-initial.log`.

`tests/test_p14_api.py::test_monthly_blackout_save_and_export_keep_original_multitime_source` first reproduced the missing GODAS file reference. The targeted API plus existing investigation suite then passed all 40 tests after the blackout-only source filter correction.

The initial failure is `evidence/p14-monthly-source-initial.xml`; the corrected report is `evidence/p14-source-closure-tests.xml`. The final local HTTP verifier passes 49 checks, including the new Pacific record, exact exports and 36 retained earlier investigations, in `evidence/p14-local-http-final.json`. Numerical outputs, source quality rules and earlier phase reference selection are unchanged. Public and browser acceptance remain separately recorded in the Phase 14 acceptance document.

58 P14 independent method review

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: `docs/P14_INDEPENDENT_METHOD_REVIEW.md`

Review checkpoint: 24 September 2026, Asia/Calcutta. Phase 14 implementation and acceptance are still in progress at this checkpoint. This document does not establish phase completion, browser acceptance or deployment.

Scope and independence

The review covered native memberships, estimated-volume overlap, split/merge classification, missing support, explicit time gaps, depth limits, work bounds, deterministic ordering, source integrity and mutation of cached results. The associated blackout CSV tables in `api/p14_exports.py` were also reviewed for agreement with the blackout response.

Blackout tests include independent hand-small arithmetic fixtures and previously checked original-source reference values, but this is not a separate-author audit of the blackout implementation. Neither this review nor test success constitutes an exhaustive security audit or independent oceanographic validation.

Native-set verification

The reproducible checker is `science/verify_p14_links.py`. It calculates the default Arabian Sea evolution at 26 C, over the selected 0-300 m native depth centres and all seven historical model frames, with extra threshold sensitivity runs disabled for this particular arithmetic check.

For every emitted link, the checker independently forms Python sets from the complete P07 native memberships and intersects them. It sums the original midpoint-bin weights in sorted native-index order, then checks the shared-cell count, shared estimated volume,

both directional volume fractions and the overlap coefficient exactly. Expected intersections are not obtained from the P14 transition function. The checker deliberately reuses the existing geometry helper, so it does not independently verify the inferred cell-boundary geometry or the underlying ocean fields.

Command actually run:

```
.venv/Scripts/python.exe -m science.verify_p14_links --output  
docs/evidence/p14-independent-links.json
```

Result: **28 checks passed, zero failed**. These comprise all **26 emitted Arabian links** and two checks of the corrected requested-depth boundary labels. The checked evolution has 21 nodes over seven frames, six compared transitions, two split groups, two merge groups and three unmatched source nodes labelled as having no qualifying successor. Those event counts describe the declared correspondence rule. The arithmetic checker verifies the emitted links, while the dedicated evolution fixtures separately check event classification and cases with no emitted link.

The evidence report p14-independent-links.json records the model and observation identities, exact query, result fingerprint, implementation file hashes, individual expected link quantities and explicit limitations. The comparison tolerance is zero. This is a bounded local check; no network or deployment action occurs in this checker.

Findings and corrections

- 1. Requested-depth boundaries were omitted from contact labels.** The first implementation detected contacts with the full native source grid. A Bay of Bengal query with a 0 C threshold and a 100-300 m depth selection correctly limited region membership and estimated volume, but its native boundary list contained only south, north, west and east. It did not separately identify the shallow and deep selection limits. The backend now returns `selection_depth_contacts` separately from `boundary_contacts`, preserving the distinction between the requested analysis interval and the original 0-5000 m source domain. Both selected frames now return shallow and deep selection contacts, while retaining the unchanged native contacts. The new checker verifies that correction and the CSV export includes the separate field. Browser presentation remains a separate acceptance task.
- 2. A blackout CSV column used an overly narrow source-only name.** Its counts include the first failed source or matching gate, such as source review, QC, time window, domain, distance or depth support. The column is now `original_source_or_matching_exclusion_counts`. It does not mix user removals into source QC rejection counts. The coverage table already used source-or-matching wording.

No additional blocking numerical defect was found in the inspected evolution paths. This statement is limited to the inspected code and the recorded checks, not a guarantee that every request or future dataset is correct.

Blackout checks already run in this session

The explicit blackout query supports profile, instrument, platform and collection selections. Selections form a normalized union. Original source quality, quantity compatibility and matching gates remain in effect. Whole profiles are removed from the evidence view; the retained sample residuals and source flags are not rewritten. The original and modified coverage, aggregate metrics, per-profile metrics and exact lost eligible source sample indices are retained in the result.

Commands actually run before this review checkpoint:

```
.venv/Scripts/python.exe -m pytest tests/test_blackout.py -q  
.venv/Scripts/python.exe -m pytest tests/test_blackout.py tests/test_evidence.py  
-q
```

The final focused blackout run passed 46 tests. The combined blackout and existing evidence run passed 131 tests, with two existing Starlette/anyio deprecation warnings. These results were inspected in the task output; this review did not rerun broad acceptance or manufacture a JUnit report for those earlier commands.

Fixtures cover empty selections, removal of an eligible profile, removal of an already ineligible group, complete removal, overlapping group selections, unknown selections, bounded inputs, independent residual arithmetic and exact original sample indices. Tests also check unchanged native model bytes, source profiles and cached comparison results, deterministic fingerprints, and rejection of changed or missing source files after caches are warm. The real Bay reference retains 103 eligible temperature comparisons after platform 1902669 is removed from the original 206. Its remaining profile bias, RMSE and MAE agree with the earlier independently checked P05 reference values. Monthly Pacific potential temperature remains incompatible with instantaneous in-situ temperature, so those blackout requests cannot manufacture eligible residuals.

The first new index-corruption test appended whitespace to a JSON file. That does not change the canonical semantic library identity, so the expected rejection did not occur. The test was corrected to change an actual profile title in the index. No scientific gate or fingerprint tolerance was weakened to pass the test.

Scientific limits retained

- Feature correspondence describes overlap between threshold-cell sets. It does not identify the same parcel of water, prove an eddy identity or estimate transport.
- Skipped frames or unsupported elapsed intervals remain explicit gaps. Missing counterpart support remains unresolved where the rule cannot establish a qualifying match. A bounded domain, missing data and threshold changes can alter topology without proving a physical split, merge, birth or death.
- Midpoint-bin volumes are estimates because provider cell boundaries are unavailable. The checker verifies use of those weights, not their equivalence to provider cell volumes.
- Removing observations changes the displayed evidence, not the assimilative model. No model rerun, forecast improvement or operational observing-system impact follows from the blackout result.
- Samples in one profile are not independent. Coverage counts and residual summaries are not calibrated confidence or area-weighted validation. A group with no eligible comparisons can correctly be removed without changing eligible evidence.
- Earlier held-out reanalysis sampling experiments remain reconstruction experiments. They do not become operational forecast tests through association with Phase 14. Historical model-observation agreement may include assimilated observations and is not independent validation.
- Prepared case dates remain historical. The review date in 2026 does not make the January 2024 fields current observations.
- HTTP transport, cross-runtime exact replay, numerical exports, visual comprehension and real unfamiliar-user results require their own evidence. Those outcomes are not implied by this local method review.

Public verifier review

The new `science/verify_p14_public.py` was reviewed without starting its HTTP run. Its exact API-versus-local output checks, capture/replay checks and expected error statuses match the inspected routes. CSV expectations reuse the production export helper, which establishes transport consistency rather than independent CSV arithmetic.

Both changes are now present in the inspected verifier code, including rejection of empty record files and checks for the required archive files. Their HTTP outcomes should be recorded by the final acceptance run. No public verification pass or deployment is claimed here.

59 Phase 14 readiness review

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt |
Source: docs/P14_READINESS_REVIEW.md

Review date: 23 September 2026, Asia/Calcutta.

Decision

Phase 14 can be implemented from the current workspace. No dependency on the previous account's conversation history was found. Source code, prepared scientific data, contracts, decisions, acceptance evidence and phase instructions are available locally. The current environment can also inspect the existing Vercel deployment using its linked credentials.

This request authorized a readiness review. No Phase 14 implementation, phase completion, production change or deployment was performed.

Scope and fresh checks

The review covered the application architecture and module inventory, API/store composition, scientific contracts, prepared data, feature and evidence engines, frontend integration points, saved investigation replay, test coverage, deployment configuration and continuity documents. Two independent focused reviews examined frontend/replay readiness and scientific/data readiness. This was not an exhaustive line-by-line security audit or a new visual acceptance run.

Check actually performed	Result
<code>.venv/Scripts/python.exe -m pytest tests -q --junitxml=docs/evidence/p14-readiness-backend.xml</code> with output redirected to the matching log	683 passed, zero failures or skips, three existing warnings, 61.79 seconds
<code>npm --prefix web run typecheck</code>	Passed
<code>npm --prefix web run test:science</code>	31 renderer/geographic tests passed, zero failures or skips

Check actually performed	Result
Authenticated read-only npx --yes vercel@59.23.2 inspect ocean-navigator-project brief-rm8fyndyz-kai-vexe n-studios.vercel.app --scope kai-vexen-studios	Existing production deployment READY; correct public alias
Anonymous GET https://depth-atlas-seifuku.vercel.app/api/health	HTTP 200, version 0.13.0, historical_case_ready, five cases
Git history and remote inspection	Branch has no commits and no configured remote

Backend evidence: JUnit report and console log. The other command results were inspected in the current task. No deployment write permission was exercised in this review. Account-specific model entitlements, usage limits and equivalence with the previous account were not compared.

Reusable code and required additions

Area	Existing foundation	Phase 14 work
Feature evolution	science/features.py supplies native cell memberships, masks, geometry, volumes and membership hashes; api/feature_store.py retains full prepared results	Explicit correspondence across frames, split/merge/disappearance branches, ambiguous matches, time gaps and threshold sensitivity
Observation exclusion	science/evidence.py applies quantity, QC, time, distance and depth gates; instrument records include profile/platform/group identities	Explicit exclusion selections, baseline versus excluded coverage and comparison results, and affected evidence statements
User interface	FeatureWorkspace and EvidenceWorkspace already support source-aware results and task controls	Understandable evolution and exclusion views, original-versus-modified comparison, loading/cancellation and navigation integration

Area	Existing foundation	Phase 14 work
Reproducibility	Versioned investigation recipes, fingerprints, source hashes and exports exist	New recipes/method versions, exclusion identities, exact replay and frontend import/restore support while retaining older recipes
Acceptance	Native membership, source consistency, QC, replay and interface regression tests exist	Independent split/merge/gap/ambiguity fixtures, exclusion invariance tests, saved original/modified results and visual acceptance

Data readiness and scientific limits

A fresh read-only calculation covered all seven frames of both Indian cases. They are historical 7-10 January 2024 fields at 12-hour spacing, not current 2026 observations.

Case	Region counts at the default 26 C query, across the seven frames	Eligible temperature samples at zero-based frames 1 / 3 / 5 with the default six-hour matching window
Bay of Bengal	1, 1, 1, 1, 1, 1, 1	206 / 103 / 307
Arabian Sea	2, 1, 7, 1, 5, 3, 2	101 / 307 / 1052

These data support a bounded prototype. Different region counts alone do not establish splits or merges. Some frames legitimately have no eligible profiles, and excluding an already ineligible instrument can correctly produce no change. Pacific monthly potential-temperature fields retain their distinct time support and incompatibility with instantaneous in-situ temperature comparisons.

Implementation constraints:

1. Current region IDs are local to one frame/query. Matching equal IDs across frames would be wrong. Correspondence must use full native memberships and explicit rules, without asserting that the same water parcel travelled between features.
2. The presentation response is limited to 50 regions. Evolution must use the full prepared results and retain bounded computation. Existing per-frame limits do not automatically bound a multi-frame sensitivity request.
3. Exclusions must not mutate cached model results, source QC or accepted-row caches. Normalize exclusions in new cache keys and saved recipes; distinguish user exclusions from source/QC rejection.

4. Removing observations changes available evidence. It does not rerun the model or establish forecast improvement, causal impact or calibrated confidence.
5. Backend recipes, frontend mode validation, export modules and navigation restoration must change together. Preserve existing saved investigation compatibility.

Handoff

First Phase 14 action, after a separate instruction to start: define correspondence and event rules with small independent fixtures, then implement P14.1 against native memberships. Continue through evolution/sensitivity, observation exclusions, reproducible saves, numerical/interface acceptance and deployment to the existing Vercel project.

A Git baseline and backup should be established before substantial new implementation. The lack of history is a recovery risk, not an account-transfer blocker. No commit or remote was created by this review. The real unfamiliar-user pilot at P09.5 remains open and is still needed to substantiate ease-of-use claims.

Prepared explanation: Phase 14 will show how detected ocean structures change between available model frames and how the evidence changes when selected observations are excluded. It will preserve the original fields and distinguish structural change from water movement. Those features are planned; this review establishes that their required foundations are present.

60 P16D participant task script

Historical or maintained reference: read alongside the current release overview | Source: docs/P16D_USER_TASK.md

Prepared 27 September 2026. This is a test script, not completed participant evidence. P09.5, R12, X14 and the P17 domain-user gate remain open.

Recruit people unfamiliar with the app, including an ocean-data user if available. Explain that the app is a historical scientific prototype. Ask permission to record task notes. Do not collect unnecessary identity or personal data. Record browser, approximate screen size and whether assistance was needed. Let the participant think aloud without explaining where to click.

1. Open the site. Follow or skip the default Basic tutorial. Find a way to build an investigation with the January Bay case.

2. Show a depth below the surface. Explain what the colours represent and whether depth exaggeration changes the data.
3. Open an observation profile. Select a sample and identify its depth and date. Explain why its date can differ from the model date.
4. Find a useful comparison. Read one observation, interpolated model value and residual. Explain what an excluded sample means.
5. Keep a comparison reference, tighten the time window and explain the changed count. Restore the reference.
6. Save the comparison with a title. Download the JSON and reopen it. Explain where the saved copy lives.
7. Optional expert task: find a threshold structure, open observation support, inspect a gap and a supported pair, then change a matching rule. Explain why support is not confidence and model agreement is not independent validation.

For each task record completed/uncompleted, time if measured, observed wrong turns, exact confusion in the participant's own words and help supplied. A failed or assisted task remains such. Keep the participant's interpretation separate from the evaluator's inference. Summarize concrete interface changes needed, then repeat those tasks after fixes. Do not assign a usability percentage from automated checks or this unexecuted script.

61 Ocean Navigator, explained simply

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt |
Source: docs/PROJECT_EXPLAINED_SIMPLE.md

Updated: 26 September 2026. Prepared text, not a submitted report.

What we are making

Ocean Navigator is a website for looking below the ocean surface. It puts modelled fields beside real measurements, with their locations, depths and dates. Indian Ocean cases include temperature, salinity and currents. The selected Pacific cases supply monthly potential temperature for historical climate comparisons.

The model supplies calculations; instruments supply measurements. Our app helps users inspect both together. It does not create a new ocean forecast.

Why it is needed

The SIH problem describes INCOIS scientists switching between tools and files. A surface map alone cannot show changes with depth. We connect the 3D view to its numbers and measurements. Better operational advisories are the problem's intended benefit, not an improvement we have already measured.

How a user uses it

For a short introduction to an extra feature, choose Guide, then Explore extra feature tutorials. Pick a card for Drift, El Nino & La Nina, Heat & Depth, ocean structures, a virtual expedition, evolution, sensor blackout or wider regions. Its guide opens the actual tool and demonstrates it. Each topic ends with a choice to try another feature or explore independently.

Start with Basic tutorial, choose In-depth for the science behind each feature, or skip it. Both tutorials perform real app actions one step at a time. The popup moves to the relevant control or chart, points to it and brings it into view, so you do not have to search for the feature. Basic explains the output briefly; In-depth adds concepts, methods, limitations and optional questions. Pause to try the controls and resume when ready. Explain uses the current result and sources; supported queries are shown before applying them. The app opens a ready historical case. The main screen shows the ocean and its essential controls. Tools groups the other activities by exploration, comparison and experiments; source records and advanced settings open when needed. The actions below are available paths, not a checklist that must all be completed.

1. Choose a historical case: Bay of Bengal, Arabian Sea or one of three Pacific seasons.
2. Explore a variable in 3D, across depths and time.
3. Inspect an instrument profile. Indian Ocean cases also support compatible measurement comparisons.
4. Find cells meeting a condition, then draw a section through them.
5. In an Indian Ocean case, try simulated sampling, particle drift or Heat & Depth.
6. In Climate, compare the same season or month across the selected neutral, El Nino and La Nina periods.
7. In Evolution, follow overlapping structures across the supplied Indian Ocean times and inspect splits, merges, gaps and threshold sensitivity.
8. In Blackout, remove selected observation groups and inspect which evidence disappears. The model stays unchanged.

9. Save or export the analysis for someone else to recalculate.

Evolution and Blackout are in the verified Phase 14 release. Their numerical, browser, public replay and visual checks have passed; the separate real unfamiliar-user pilot remains open.

The guide uses two checked locations with similar surface temperatures but different temperatures at 100 metres. They illustrate depth differences, not whole-basin behaviour.

What project brief expects, and where we stand

What the problem asks for	What we currently deliver	What remains
3D fields, slices, isosurfaces and time animation	Two Indian Ocean cases with temperature, salinity and currents; three Pacific seasonal cases with monthly potential temperature	Broader data and scale testing
Argo, glider, CTD and BGC markers and profiles	Real examples, profile inspection and eligible comparisons	More overlapping sources
NetCDF/text ingestion and adjustable displays	Supported parsers; variable, colour, range, scale, opacity and depth-exaggeration controls	More file layouts and variables
REST/OPeNDAP, CF-NetCDF, OGC WMS/WCS	Public REST/CF downloads; separately tested Bay THREDDS services	Public standards hosting, final packaging, INCOIS installation
Scientific and public-outreach usability	Guided cases and connected workflows	P09.5 real user pilot; expanded learning mode

The deployed and verified build is 0.14.1. Phase 14 adds analyses over the existing data rather than adding coverage. The implemented build contains **five historical cases**: two Indian Ocean HYCOM cases from **January 2024**, plus three Pacific GODAS cases covering **September-November 2013, 2015 and 2022**. Pacific fields are monthly potential-temperature means at native depths from **5 to 459 m**, with a matching-calendar **1991-2020 baseline**. They supply no salinity or current fields.

Heat & Depth remains an Indian Ocean workflow. Its surface series covers **1982-2025**, with a **1982-2011 baseline** and **2023/2024 display years**. The project year is **2026**. All cases have limited historical coverage. Saves are local, comparisons are not independent model validation, and region volumes are labelled estimates.

Core requirements and our extra ideas

The viewer, instruments, ingestion, controls, standards and extensibility form the SIH core. Our extras include evidence comparisons, structure search, drawn sections, reproducible investigations and guided cases.

Phase 10 adds a **Virtual Expedition**: suggested sampling locations, simulated surveys and fair comparisons of sampling strategies. Numerical, public browser, export and replay checks have passed. These are model-based reconstruction experiments, not proven forecast improvements or safe ship routes. The suggested gradient plan loses to the simpler space-filling plan in both default examples; the app shows that result honestly.

Phase 11 adds **Drift Lab**, the particle or "pebbles" idea. Choose a release point or box, date, fixed depth and duration. The app calculates movement through the historical currents, then lets you play the paths, compare another release and count simulated arrivals in a chosen target box. The particles do not sink. Missing currents and study-area boundaries stop unsupported movement. Numerical checks and exact replay have passed, but independent validation against real drifter tracks has not been established. The current method is `p11-drift-v2`.

Phase 12 adds **Heat & Depth**. Choose one of six fixed surface-data cells, inspect daily temperature against a seasonal mean and event threshold, and select a detected historical event. If model data exist during that event, inspect the actual depth profile, two mixed-layer estimates, density, stratification and a defined column heat integral. Bay events support this depth view. The selected Arabian events fall outside the available model dates, so the app explains that depth evidence is unavailable. Surface events do not prove a heatwave at depth. Methods, sources and values can be saved, recalculated and exported.

Phase 13 adds **Climate Event Lab**. Compare the same September-November season or calendar month in the selected 2013 neutral, 2015 El Nino and 2022 La Nina cases. Inspect potential-temperature anomalies, native depth profiles and selected-minus-reference differences. ENSO labels come from a pinned official RONI release. Indian Ocean sections use matching dates and the same GODAS baseline, while DMI values and documented IOD episodes remain separately labelled. Original Argo measurements are inspectable, with their QC preserved. The app does not directly subtract instantaneous in-situ observations from monthly potential temperature or infer causality from two selected years. Its public browser, scientific replay and export checks have passed.

Phase 14 adds **Evolution** and **Blackout**. Evolution asks, "Where does a structure satisfying this condition overlap at the next supplied time?" Its graph shows candidate links, splits, merges, ambiguity and gaps. The native footprint, depth range and nearby

threshold results remain inspectable. An overlap is not proof that the same water moved there. Missing support and skipped frames do not become invented connections.

Blackout asks, "What evidence would be unavailable if these observations were removed?" Select profiles or groups, then compare original and remaining coverage, eligible comparisons and residual statistics. Original quality exclusions stay separate from the user's removals. The model does not change, and the experiment does not reverse assimilation or demonstrate a forecast impact. The original and modified results can be saved and reproduced. Final browser, visual and public acceptance has passed for the supported workflows.

The first public Evolution check found tiny differences between server and local aggregate calculations. Release 0.14.1 uses method p14-native-overlap-v2 with consistent summation. All nine new investigations now reproduce exactly in both directions between Windows and Vercel; all 36 retained older records replay. Source data and the earlier structure-search method are unchanged. Earlier Evolution files remain readable and identify their older method instead of silently becoming new results. The original failed checks remain in the record.

The P09.5 real unfamiliar-user pilot also remains open.

About 45 seconds, spoken

"Ocean Navigator lets users look below the ocean surface in a browser. It connects model fields, real measurements and saved evidence. Users can inspect depth, simulate sampling and particle drift, compare historical heat and climate cases, then reproduce the analysis. Evolution shows where defined structures overlap across time, including splits and gaps. Blackout shows which evidence disappears when observations are excluded, while keeping the model unchanged. Every result retains its source, date, units and limits. These are historical investigations, not a new live forecast."

This script describes the implemented, publicly verified Phase 14 workflows. It does not claim live forecasting, independent ocean validation or completed human usability testing.

Source: the archived [official project brief statement](reference/official-2026/project-brief-statement.txt). Feature scope and evidence: [Phase 14 methods and current acceptance](#), [Phase 13 review](#), [Phase 12 review](#), [Phase 11 review](#), [Phase 10 review](#), [Phase 9 review](#), [feature register](#) and current project state.

What Phase 15 adds

Regions lets a user choose a supported ocean rectangle beyond the original case list. It shows where data exists and what dates and depths it covers, then loads a light preview and a fuller view. A click opens an original source column. A regional pack can be saved or downloaded with an offline viewer.

62 Comparison and loading reliability

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/RELIABILITY_FOLLOWUP.md

27 September 2026. Targeted follow-up requested after release 0.16.11. This does not start the complete P17 audit.

What the evidence establishes

The previous public desktop full-tour traces contain comparison coverage/profile requests with no response headers before the client's eight-second deadline. Retrying immediately also failed. The mobile tour and an HTTP/1.1 diagnostic passed. That suggested a transport issue, without establishing its cause.

Four fresh instrumented core walkthroughs on the unchanged release passed over normal HTTP/2. Their longest observed main-thread tasks were 386-440 ms. Observed comparison responses were 257-2,376 ms. These passing runs do not erase the previous failures or prove that rendering never affects another device. The bounded Vercel request-log inspection found successful comparison responses, with no application error log in those inspected entries. Raw operational logs remain outside the tracked evidence because they can contain unrelated request metadata.

Two application weaknesses were reproduced independently:

- Valid comparison responses deliberately delayed by 9.5 seconds were cancelled by the eight-second deadline and the tutorial could not complete without Retry.
- An in-flight speculative model download continued after switching to comparison. The prior cleanup stopped scheduling more downloads but did not cancel the current one.

The initial cancellation test used an incorrect tool label. It was corrected to the actual Compare control and then reproduced the missing cancellation. Initial and corrected reports remain separate. The early TypeScript check also caught an instrumentation

variable shadowing performance and a now-unused prefetch variable; both were corrected before the build.

Changes

- JSON GET reads have a finite 25-second deadline, covering headers and body transfer. Existing caller cancellation is immediate. Existing bounded query retry, errors, schema checks and release checks remain in place.
- Preloading warms only the next time step of the selected variable. It waits until the visible field and native reading are ready, respects data-saving connections and does not load every other variable speculatively.
- Comparison and instrument tasks keep their visible model context without speculative preloading. Changing tools or selections cancels an unused prefetched query. A query adopted by the current selection is preserved, so selecting an in-flight next snapshot does not restart it.
- Foreground fetching, matching rules, data arrays, units, QC, hashes and scientific calculations are unchanged. No synthetic result, wider tolerance, new provider, persistent cache or HTTP/2-disable setting was introduced.

The distinction between stopping a scheduling loop and cancelling its request follows the installed Query implementation and its [cancellation documentation](#). The existing controller/timer approach keeps explicit cleanup, as described in the [AbortSignal documentation](#).

Verification and limits

The regression checks exercise delayed real source responses, abandoned speculation, adoption of an in-flight snapshot, normal-transport tutorial comparisons, retained scenes, exact source timestamps, malformed scientific coordinates, server-release changes and explicit Retry. Numerical replay and source checks remain required before deployment acceptance.

Final commands, results, deployed identity and any remaining issue are recorded in PROJECT_STATE.md and evidence/reliability-acceptance.json. A larger finite deadline and reduced competing downloads improve application recovery; they cannot guarantee connectivity or establish the root cause of an intermittent external transport failure. No clean public result is claimed until actually run.

The broad local command also selected `p16e-continuity.spec.ts` because Playwright treats filename filters as regular expressions. Its initial result was 76 passes and 20 failures.

Eight failures came from stale pre-globe navigation/version expectations; 12 mobile cases did not execute because the temporary test-side release import lacked a JSON import attribute. Runtime source was unchanged while correcting these test issues. The corrected recovery/publication subset passed 12/12 across Chromium desktop/mobile and WebKit, and the complete mobile import/offline file passed 12/12. These counts overlap; the original broad report is retained and is not relabelled 96/96.

Final public result

Version 0.16.12 is live at <https://depth-atlas-seifuku.vercel.app/?release=0.16.12>, runtime source 20aaa2c6c4412809d5d7262ad6c8863c8d02e2bc, deployment dpl_FcYW5c75MfC8Wryj1fCZVki43MQE. The full 39-lesson tour passed twice on desktop and once on mobile using normal browser configuration. All ten targeted public recovery checks passed, and captured Chromium responses used h2. The previous comparison timeout did not recur. The full public release check passed 24/24 with no retries; exact scientific replay passed 61 checks over 17 requests. Client scientific tests passed 82/82 locally. Evidence and overlapping counts are preserved in `evidence/reliability-acceptance.json`.

Public desktop/mobile comparison screenshots show the actual 103 eligible pairs, separate source dates, residual chart and tutorial controls. Desktop automation used the labelled Basic 2D fallback, while mobile retained the 3D context. These are checks on the stated configurations, not a universal GPU or network guarantee. The reproduced client problems are fixed; the earlier intermittent external transport cause remains unconfirmed. P17 remains next on instruction.

63 Technology stack and evaluator experience

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: `docs/STACK_AND_EXPERIENCE.md`

Updated: 2026-09-22. P01-P06 provide the React/TypeScript/Vite workspace, Three.js scientific rendering and basic graphics fallback, real HYCOM/Argo case, supported instrument ingestion, model-observation comparisons, native NetCDF download and tested local standards service. P07 adds native-grid connected-region queries and drawn sections. Exact versions are in the package/runtime/science lockfiles. Browser and numerical evidence is recorded per phase in `PROJECT_STATE.md`; it does not establish universal device support or operational scientific validation. Later labs, wider coverage and durable investigations retain their roadmap scope.

Evaluation constraint

The user clarified that national round 1 is online and the evaluator opens the app on their own device. Device, GPU, browser, screen size and connection are unknown. This supersedes the earlier answer that the demonstration would run on the user's Windows computer.

The product must open from a normal HTTPS link. Heavy scientific calculations belong on our server; browser graphics still use the evaluator's device, so moving computation to the server does not eliminate rendering limits. The user's machine is a development/test device, not the target specification.

Proposed baseline stack

Responsibility	Baseline choice	How it serves this product
Application	React + TypeScript + Vite	One browser workspace with typed state and reusable views
Interface styling and accessible controls	Tailwind CSS + Radix primitives	Custom visual identity, accessible drawers/dialogs/sliders and consistent controls
Shared interaction state	Zustand	Region, time, depth, selections and linked-view state
API data lifecycle	TanStack Query	Explicit loading/error states, cancellation where supported and controlled request caching
Scientific 3D rendering	Three.js 0.186.0 with a verified WebGL2 ray marcher and geometry pipeline	Physical irregular-depth volumes, slices, sections, isosurfaces, current vectors and native-grid picking; see D019 and PHASE_03_RENDERING.md
Basic graphics fallback	Canvas 2D	Real depth slices, sections, current vectors and native inspection without WebGL
Charts and scientific legends	D3 with React-managed chart components	Depth profiles, linked cursors, residuals, time series and honest units/scales
API and validated contracts	Python + FastAPI + Pydantic	Scientific requests, metadata, saved investigations and bounded job interfaces

Responsibility	Baseline choice	How it serves this product
Array and scientific processing	xarray + NumPy + SciPy	Supported NetCDF reading, subsetting, masks, interpolation, connected regions and diagnostics
Larger-array processing	Dask when measurements justify it	Chunked processing without assuming the whole archive fits in memory
Scientific coordinate/variable conversions	TEOS-10 / GSW; appropriate projection/geodesic utilities as needed	Pressure/depth, temperature/salinity semantics and physical distances/areas
Historical particle calculations	Pinned OceanParcels RK4 kernel adapted to a bounded NumPy runner; full Parcels offline reference	P11 fixed-depth historical trajectories with checked integration, interpolation and boundaries
Heatwave detection	Pinned marineHeatWaves reference or a tested equivalent implementation	Seasonal baseline/threshold event analysis with reference agreement
Scientific data services	THREDDS Data Server	Tested WMS, WCS and OPeNDAP paths for supported datasets
Long-running calculations	Separate Python worker process; RQ + Redis is the initial queue candidate	Jobs with status, limits and cancellation rather than blocking API requests or browser interaction
Scientific storage	Versioned NetCDF originals and bounded NetCDF/Zarr derived caches	Traceable input and result storage; binary display assets derived separately
Investigation metadata	SQLite initially; PostgreSQL only when the deployment/concurrency requires it	Source/catalogue records, manifests, saved cases and job references, not huge array blobs
Packaging and hosting	Vercel for the public application, plus a portable Linux container / Docker Compose recipe	P01 frontend/CDN and FastAPI are deployed. Long-running science workers, THREDDS and durable storage need a separate measured hosting design; no additional service is selected
Verification	pytest, TypeScript checks and Playwright	Numerical and API checks plus browser interaction/replay verification

Do not install every optional component at P01. Add each when its feature requires it. Choose one verified frontend/runtime dependency set rather than chasing latest versions during the sprint.

Feature-to-engine mapping

Feature	Main implementation
3D Ocean Explorer	Three.js receives bounded real scalar fields from the backend; analytical inspection remains separate
Instrument overlays	Python source adapters/QC produce consistent records; Three.js and chart components will link positions and profiles
Evidence Inspector	Python conversions/interpolation/matching; linked profile and residual charts
Feature search	NumPy/SciPy masks and connected regions; verified cell geometry; Three.js surfaces and picking
Particle/"pebbles" simulation	P11 request-scoped server calculations stream progress; browser animates the resulting positions and trails
Heat & Depth Lab	xarray/NumPy + reference heatwave method + GSW diagnostics; common 3D/profile/time-series views
El Nino, La Nina and IOD	Versioned historical datasets/indices, xarray anomaly/season alignment and synchronized regional views; no new climate-model training
Virtual Expedition	Explicit numerical candidate rules and constraints, simulated transect sampling and separated benchmark evaluation
Feature evolution	Declared numerical correspondence across region masks with split/merge handling
Observation blackout	Recalculate coverage and comparison eligibility from filters; do not silently rerun or change the underlying ocean model
Saved investigations	Versioned JSON manifest + metadata store + source/result references; CSV and readable reports
Interoperability	Real THREDDS protocol requests plus documented app REST endpoints and adapter interfaces
Global expansion	Data-service subsetting, bounded caches, progressive display detail and geographic edge-case checks
Public/teacher mode	Same verified data and calculations with simplified controls and guided explanations

Visual direction

The user explicitly prioritizes very impressive visuals/simulations and ease of use. Treat these as acceptance criteria throughout the phases, not optional decoration at the end.

Use flat navy/neutral surfaces, rectangular controls, direct labels and sourced scientific graphics. Scientific playback and necessary loading feedback remain functional interactions. Keep the custom SVG favicon, public legal-page links and required attribution.

Build a polished ocean observatory with a dominant interactive scene, readable typography, restrained controls, depth cues, smooth transitions and excellent picking/inspection. Desired interactions include:

- A regional ocean volume opening to expose internal layers.
- A cutting plane that reveals an underwater section and stays linked to a profile cursor.
- Selected structures with clean surfaces, purposeful lighting and clear dimensional labels.
- Particle trails showing direction, speed and elapsed simulation time, with controlled visual density.
- Synchronized historical El Nino/La Nina comparisons with identical scales when comparing the same variable.
- A real same-surface/different-depth pair that reveals its difference through one interaction.
- Bathymetry/coastline context where a suitable sourced dataset is available.

P03 evaluated the proposed vtk.js path and selected Three.js with a custom irregular-axis ray marcher, recorded in D019. The current renderer uses scalar-dependent opacity and simple shading. Costly cinematic lighting remains deferred because the measured software-rendering path already needs adaptive quality. Do not assume medical-volume or dedicated-GPU demonstrations establish ocean performance on evaluator devices.

Scientific fields use explicit false-color scales; they are not underwater photographs. Surface reflections/waves, if used for scene context, are illustrative unless driven by a supplied wave product. Smooth display interpolation does not add new temporal/spatial information. If playback time or vertical scale is exaggerated, label it.

Particle paths must come from the actual supported solver. Do not add random motion to make them look turbulent and then describe that motion as a physical result. The "pebbles" are passive tracers unless additional object physics is explicitly implemented and validated.

Quality adapts to the evaluator

Default to Auto with a useful conservative first frame. Avoid a device questionnaire or a graphics menu before the user can investigate.

Mode	Rendering behavior	Scientific behavior
Enhanced	Richer shading, higher display resolution, more visible trajectories and detail where the device sustains it	Same declared analytical sources/methods; any distinct calculation setting remains explicit
Balanced	Responsive volume/slice views with controlled shading, resolution and trajectory density	Same analytical values on inspection; display reductions labelled
Basic	2D sections, maps/charts and simplified geometry when 3D is unavailable or unstable	Retains useful data inspection; clearly explains that full 3D is unavailable on this device

Auto adapts using capability checks and measured runtime behavior. WebGPU is not a prerequisite. A WebGL2 failure must lead to a useful explanation/fallback, not an endless loading screen. High-DPI displays must not silently multiply rendering work without bounds.

Display fewer trajectories without silently changing a simulation's calculation sample. Preserve full declared outputs or explain result decimation. Changing rendering resolution must not change a reported residual, volume or diagnostic.

Large calculations run on the server with explicit jobs, limits, cached results and cancellation. New requests must not silently substitute old/precomputed output. A precomputed or cached case is legitimate when labelled with its sources, dates and settings.

Usability rules

- Open directly into a useful real example with an obvious next action.
- Offer task-based entry points such as explore depth, release particles and compare climate events; do not expose all expert settings at once.
- Keep selected source/time/variable/units visible and consistent across panels.
- Keep advanced scientific settings expandable, with sensible documented defaults.
- Use familiar words, contextual explanations, reset/undo and clear disabled reasons.

- Provide read-only exploration without an account where deployment allows it; reviewers should not need development credentials.
- Preserve accessible contrast, keyboard interaction, reduced-motion preferences and responsive layouts.
- A progress state must show what is happening; errors must explain recovery.
- Tests with unfamiliar users determine whether the flow is easy. Attractive screenshots alone do not establish usability.

Proposed performance targets, not measured claims

These targets guide P03/P09/P15/P17. Revise them transparently after the first real-data benchmark.

- A useful first case within roughly 5 seconds on a declared ordinary broadband/representative laptop test; initial payload budget must be measured.
- Aim for at least 30 fps during typical 3D interaction on the declared balanced reference setup. Adapt display quality when performance drops; do not promise this on every device.
- Immediate visual acknowledgement of controls; longer calculations show progress and remain cancellable.
- Test Chromium, Firefox and WebKit engines, small/large viewports and throttled CPU/network behavior. Engine automation is not a substitute for all real-device GPU testing.
- Obtain real integrated-GPU and at least one different-device/browser check when available. Low-GPU behavior cannot be established solely by CPU throttling.
- Verify GPU context loss, failed requests, slow data, blocked 3D, no-data states and the Basic fallback.
- Test concurrent evaluator-like sessions and server memory/job limits before publishing unrestricted operations.

P01 browser interaction, mobile layout and 200% text-size overflow checks have been run on the geographic shell. No scientific renderer/frame-rate benchmark has been run. We cannot guarantee identical visual quality or universal device compatibility. The release goal is the best stable experience the device supports, with a truthful useful fallback.

References checked for this recommendation

- [vtk.js scientific visualization](#)

- [Kitware cinematic volume rendering and its tradeoffs](#)
- [Three.js renderer documentation](#)
- [THREDDS available services](#)
- [FastAPI background-task guidance](#) distinguishes heavier computation that benefits from external job tooling.
- [Marine heatwave reference implementation](#)

These documents establish component capabilities. They do not demonstrate that our proposed integration or visual targets have already been achieved.

Implemented P03 baseline

The stack table originally proposed [vtk.js](#). P03 verified a single [Three.js](#) scientific engine and a Canvas fallback; no [vtk.js](#) dependency was installed. Other rows still describe future phases where those components have not been added. See [rendering methods and limitations](#) and the actual phase-03 evidence. Enhanced graphics is still a future option; the implemented choices are Auto, Balanced 3D and Basic 2D.

P11 bounded execution decision, 23 September 2026

The checked two-case particle scope uses an attributed [OceanParcels 3.1.4](#) RK4 kernel adapted to our existing NumPy runtime. The complete framework remains an offline reference dependency, pinned separately. Real progress and cancellation use request-scoped streamed calculations with bounded steps, particles and warm caches. No Redis instance or persistent worker is required or claimed for this release. Larger durable jobs still need an actual deployed service before such capability can be advertised. See D037-D038 and PHASE_11_DRIFT.md.

64 project brief submission and supporting-material checklist

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/SUBMISSION_CHECKLIST.md

Rechecked: 2026-09-27. Scope: national idea shortlisting to qualify for the grand finale, which is the goal described in this conversation. The grand-finale event has a separate later handover process; do not assume the same upload rules apply.

Source prepared: The user selected public visibility on 27 September.

<https://github.com/GauravR17A/depth-atlas> is published and verified without login. Use <https://depth-atlas-seifuku.vercel.app/> for the app. This satisfies preparation of the requested links, not proof of portal submission or organizer acceptance. The private planning archive and competition identifier are excluded from that repository.

What was actually verified

The SIH homepage linked the 2026 institute guidelines and 2026 idea-presentation template. Both were downloaded directly from www.sih.gov.in, opened/extracted locally and preserved unchanged. On 27 September, direct HTTPS downloads again returned 200; the guideline and template SHA-256 hashes match the 21 September originals exactly. The current project brief page was also read and saved; its functional scope and 30 September 2026 deadline remain present. The page file itself changed, so it is not described as byte-identical. Evidence: [p17-official-source-check.json](#). Old 2024/2025 results are not treated as current rules.

Official sources:

- [2026 guidelines](#), printed pages 10-13 / physical PDF pages 16-18 and 20 for submissions and selection.
- [2026 presentation template](#), especially the final instruction slide.
- [Official project brief](#).
- Local copies and text extracts: [reference/official-2026/](#).

The signed-in team dashboard and any team-specific SPOC/nodal-center instructions have not been inspected. Additional fields or requirements there remain to be checked. This document distinguishes published requirements from our preparation recommendations.

User-reported repository requirement, 27 September 2026: The user states that a repository link is requested for their presentation/portal. Treat that link as required for this team's preparation. The exact field, notice and public-versus-private access instruction have not yet been supplied or inspected. Do not use the absence of that field from general public documents to dismiss the user's portal requirement. A link to a private repository grants no permissions by itself.

Focused recheck: Fresh direct downloads of the official 2026 guidelines and presentation template returned 200 and matched the earlier source hashes. The guideline's submission list on physical PDF page 17 (printed page 11) ends with the idea presentation PDF. No [GitHub/repository/source-code](#) requirement was found in the extracted guideline text or

the template's slide and notes text. The project brief entry likewise did not supply a repository-visibility rule. These public documents do not establish what the user's authenticated submission form requires. Evidence:

evidence/github-requirement-check.json. No private dashboard, repository or collaborator access was changed.

The user clarified that the repository request appeared on the portal while registering the problem statement, with repository links also appearing in some presentations. No URL, exact field wording, screenshot or reviewer invitation procedure was supplied. Use the PS-registration form as the applicable user-reported source. A public review repository is the recommendation for anonymous judge access unless the organizer supplies an accepted private-access process; this recommendation is not a verified public-visibility mandate or authorization to publish.

Public identity clarified by the user on 26 September 2026: Team Seifuku, ID 150123; leader Adwita Kurle; website creator and operator Gaurav Ranade; Pune. The project has no university affiliation or university involvement in development. All six names and the public contact are recorded in [TEAM.md](#). This supplies the public-page and draft-deck identity. Portal registration, required private contact fields, nomination and institutional authorization remain unverified, so their checklist entries stay open.

A. Published shortlisting requirements

The guideline's submission list calls for verification of team/contact details and the college authorization PDF, plus the chosen PS, idea title/description and a presentation PDF. Its current deadline is 30 September 2026; a PS can close earlier at its 500-idea limit. It says roughly four or five teams per PS may be shortlisted, subject to the sponsor's decision. [Official guidelines](#)

Ready	Item	What we need to prepare or verify	Owner
[]	Portal eligibility/access	Confirm SPOC nomination and team-leader access are in place	User / college SPOC
[]	Team information	Verify registered team name/ID, leader and five other members, required contact/profile details	User / college SPOC

Ready	Item	What we need to prepare or verify	Owner
[]	College authorization PDF	Obtain/verify the current prescribed institutional letter and required signatures/seal	User / college SPOC
[]	PS selection	Select project brief and verify the current title/category/theme	User, with prepared project metadata
[]	Submission confirmation	Verify uploaded content, final submitted state and any receipt/acknowledgment	User / authorized submitting workflow

Do not put private team phone numbers, authorization letters or identity documents in a public repository by default. They belong in the organizer's required submission channel.

Presentation format

The downloaded PPTX contains seven slides because the last one holds instructions; remove that instruction slide from the submission. Keep the editable PPTX for our own preparation. [Official template](#)

Slide	Content to prepare for our project
1	PS/team identifiers and required title-page metadata
2	Problem, proposed solution and distinctive connected workflow; prominently include original lab ideas using honest working/planned labels
3	Technical design, data sources and implementation flow; small real prototype illustration when available
4	Feasibility, data/rendering/scientific risks and concrete mitigations
5	Intended users, practical benefits and measured evidence where it exists
6	Credible research references and accessible supporting resources within the prescribed layout

Our 44-page build guide is internal planning material, not the six-slide submission deck. A demo link or QR can be supporting material if the layout/portal permits it; the deck must remain understandable without clicking it.

The deck identifies Seifuku, Adwita Kurle and Gaurav Ranade accurately, uses a real earlier prototype capture, and separates working behavior from external acceptance gaps. The PDF was exported through installed PowerPoint and its six rendered pages were inspected. Its 546 KB size is an observed file size, not confirmation of the unknown portal limit. Source and review: `submission/build_deck.py` and `docs/evidence/p17-deck-review`. No upload was made.

B. Supporting engineering package we should prepare

Prepared wording is maintained in `PHASE_WRITEUPS.md` as each phase is built. It includes report passages, presentation text, demonstration steps and evidence-backed limits. This is reusable source text, not a finished deck, exported report or submitted document.

The published 2026 submission list and project brief entry do **not list these as mandatory separate shortlisting uploads**. Prepare them for credibility, review and later handover. If the actual dashboard/SPOC requests one, update its status to required with the source of that instruction.

Ready	Asset	Minimum useful content	Build phase
[]	Reviewer-accessible source repository	Required for this team's preparation per the user's portal report. Actual front end/backend/science/adapters, clear setup and submitted release identity. Confirm visibility and reviewer permissions from the applicable field/notice.	P01 onward; P17 freeze
[]	Runnable release/archive	Production build or source archive plus pinned dependencies, environment example and repeatable run command; no credentials	P01, P06, P17
[]	Deployed demo link	Stable browser URL, clear data dates, a ready real case and reviewer access instructions if needed	P01 onward; verify P17

Ready	Asset	Minimum useful content	Build phase
[]	Demo video and offline copy	Actual app behavior: inspect depth, select a profile, run a supported query/lab, save/reopen; no invented functionality	P09, P17
[x]	README / quick start	README rebuilt around the current workflow, exact setup, source dates, team roles and limits; DEVELOPMENT keeps detailed commands	P17
[x]	Technical handover notes	P17_DELIVERY, SCIENTIFIC_CONTRACTS, DEVELOPMENT and phase method reports supply architecture, APIs, adapters and deployment boundaries	P17
[x]	Scientific-method and data notes	Existing phase reports, case manifests and source notes retain coordinate conversions, QC, matching, lab methods and limitations	P02-P17
[]	Requirement coverage matrix	R01-R12 mapped to actual implementation, demonstration steps, evidence and outstanding gaps	P06, P17
[]	Test and benchmark report	Real numerical checks, protocol requests, fresh setup, measured loading/memory and small usability results	Throughout; P17 package
Ready	Asset	Minimum useful content	Build phase
[x]	User guide	Anchored Basic/In-depth tutorials, the public import guide and P17_DEMO cover the current first-investigation workflow; actual	P09, P16, P17

Ready	Asset	Minimum useful content	Build phase
		participant testing stays open	
[]	Reproducible sample investigation	A small licensed sample or retrieval recipe, settings/results, checksums and replay instructions	P02, P08
[]	Attribution and contribution record	Data/software citations and licences; original contribution vs reused libraries; truthful assistance disclosures if requested	Throughout
[x]	Presentation support	Six-slide draft PPTX/PDF, real source screenshot, architecture/data flow, PORTAL_TEXT and P17_DEMO judge questions are prepared for review	P17

A public GitHub repository is not established as a mandatory rule by these sources. Use whatever visibility the organizer permits, but ensure reviewers can actually access the material provided. Repository hosting and any publication remain future actions; no remote repository has been created in this task. Prepared package status, clean-install evidence, the real-behavior recording and remaining licence/hosting decisions are indexed in [P17_DELIVERY.md](#). Unchecked rows can contain partial evidence; they are not silently marked complete while a named component remains open.

C. What the PS specifically expects us to demonstrate

The PS is a functional specification for browser-based 3D model/observation visualization, supported input formats, interaction controls, extension and open-standard interoperability. It does not contain a separate required archive/repo/video/report upload list. [project brief](#)

Use [FEATURE_REGISTER.md](#) for detailed coverage. Prepare evidence of a real depth-resolved field, a genuine instrument profile, working controls, supported imports, actual standards requests, and reproducible server setup. Our extra labs strengthen the proposal but do not substitute for missing required behavior.

There is no APK requirement in this browser-based PS or the inspected general shortlisting checklist. A research paper, model-training notebook or complete ocean-data archive is also not established as a required upload.

D. Items to confirm in the actual dashboard or organizer notice

- Exact PDF size limit and portal text limits.
- Whether repository, website, video or additional-document links have dedicated fields, and which are required.
- Any specific repository visibility, video length, report template, filename or hosting instructions.
- Current deadline/capacity and whether an upload is draft or final.
- Any additional PS-provider instructions received after nomination.
- Grand-finale deliverable/handover requirements once shortlisted.

The general guidelines require college identification and a consent-letter copy for attendance at the finale. Those are later administrative requirements, not public engineering assets. Obtain the current applicable instructions before travel. [Official guidelines](#)

E. Packaging and release discipline

Current review materials and their locations:

```
submission/  
  project_brief_OceanNavigator_Idea_DRAFT.pdf  
  project_brief_OceanNavigator_Idea_DRAFT.pptx  
  build_deck.py  
  LINKS.md  
  PORTAL_TEXT.md  
docs/  
  P17_DEMO.md  
  P17_DELIVERY.md  
  P17_CAPABILITY_AUDIT.md  
  SCIENTIFIC_CONTRACTS.md  
  DEVELOPMENT.md  
  
  PHASE_WRITEUPS.md  
deploy/  
  deployment configuration and run instructions
```

The deck's claims must reflect the actual implementation status on the submission date. Keep all provided links consistent with one tested release, check them without developer-only permissions, and retain the actual submission acknowledgment. A

prepared draft, successful deploy or local source archive is not a submitted idea.

65 Running and verifying Depth Atlas

Historical or maintained reference: read alongside the current release overview | Source: docs/DEVELOPMENT.md

The workspace includes scientific 3D exploration, instrument inspection/imports, eligible model comparisons, native-grid structure search, saved investigations, guided cases and Virtual Expedition. Checked historical model cases cover bounded Bay of Bengal and Arabian Sea grids. Separate glider, CTD and BGC examples retain their own dates and locations. See the phase method documents and PROJECT_STATE.md for supported inputs, exact verification results and deployment status.

Requirements

Release identity lives in `api/release.json`, shared by the browser bundle and Python API. Bump it when deploying changes to client/API behavior and align root/web package versions. The API sends `x-ocean-app-version`; an older open tab offers a reload instead of parsing an incompatible response. Health/catalogue recover from transient failures on their 30-second foreground refresh or when a stale tab becomes visible. Pre-0.2.1 tabs need one manual reload to acquire this behavior. See D017 and `web/e2e/service-recovery.spec.ts`.

- Node.js 22.12+ in the 22.x line (local verification used 22.17.1).
- Python 3.10-3.12 locally; Vercel and the container target Python 3.12.
- npm, Git, and an authenticated Vercel CLI for publication.
- Docker only if using the optional portable server package.

No data-provider credentials are needed for the checked examples or temporary file imports. Never put tokens in frontend code, tracked files or public environment variables.

First setup on Windows

From the repository root:

```
python -m venv .venv
.venv/Scripts/python.exe -m pip install -r requirements-dev.txt
npm --prefix web ci
```

Runtime-only installations can use `requirements.txt`. The exact frontend dependency tree is in `web/package-lock.json`. The `.in` files declare intentional direct requirements; the `.txt` files pin resolved dependencies.

To run all scientific adapter tests or reacquire/rebuild model data, install `requirements-science.txt` instead of `requirements-dev.txt`. The checked packs are included under `casepacks/`, so startup needs no source download. P04 observation imports use `netCDF4`, `NumPy` and `GSW` in the runtime; `xarray` and `pydap` remain offline preparation dependencies.

Development with live frontend updates

Terminal 1, at the repository root:

```
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8000  
--reload
```

Terminal 2, at the repository root:

```
npm --prefix web run dev
```

Open `http://127.0.0.1:5173`. Vite proxies `/api` to port 8000, so local development follows the same-origin production contract. No permissive CORS policy is needed.

Production build locally

```
npm --prefix web run build  
.venv/Scripts/python.exe -m science.publication --candidate casepacks --manifest  
api/publication.json  
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8000
```

Open `http://127.0.0.1:8000`. Build and validate the matching publication before starting the API. A release change requires the frontend, `api/release.json` and publication to agree; startup rejects a stale publication. Stop a server with `Ctrl+C` before rebuilding its served files or starting another on the same port. The container recipe now runs the same publication validation during its image build. Docker execution remains unverified on this host. See [P17 delivery](#) for the isolated setup record.

Service contracts

Endpoint	Meaning
/api/health	API liveness, app version, explicit data readiness and case count
/api/catalog	Six bounded historical cases: January Indian Ocean, March Bay of Bengal and three Pacific seasons
/api/cases/bay-bengal-2024-01	Scientific coverage, sources, variables, profiles, hashes and methods
/api/cases/{case_id}/subset	Bounded analytical slices or display volumes; specify variable, time and depth/operation
/api/cases/{case_id}/profiles/{profile_id}	Original/adjusted readings, QC and derived depth for a genuine profile
/api/instruments	Version 2 observation catalogue, supported examples and limitations
/api/instruments/profiles/{profile_id}	Checked common Argo, glider, CTD or BGC observation profile
/api/instruments/examples/{filename}	One allowlisted original/example file, verified by SHA-256
POST /api/instruments/import?filename=...	Raw bounded CSV/NetCDF bytes; decoded profiles returned without application persistence
/observation-import-guide.txt	Exact supported file layouts, QC policy and import limits
/api/docs	Generated API documentation
/api/openapi.json	Machine-readable API schemas
/third-party-notices.txt	Bundled software/data attribution
/privacy	Static Privacy Policy for the current prototype
/terms	Static Terms & Conditions for the current prototype

Errors have { "error": { "code", "message", "request_id" } }. Request IDs also appear in X-Request-ID. API HTTP responses use no-store; immutable case arrays have a bounded process-local LRU. Health separates service liveness from data_status and case_count. Historical data readiness does not establish live data or complete scientific feature implementation.

Verification

With the production server running:

```
.venv/Scripts/python.exe -m pip install -r requirements-science.txt
.venv/Scripts/python.exe -m pytest tests -q
npm --prefix web run build
cd web
npx playwright install chromium firefox webkit
npm run test:e2e
```

OCEAN_TEST_URL overrides the browser test base URL. The suite checks actual frontend/backend interaction, empty-data truthfulness, region/view controls, modal keyboard focus, error recovery and malformed API responses. It uses request interception for explicit failure scenarios, not fabricated science.

The first Windows run passed Chromium, WebKit and Chromium mobile emulation. Firefox could not start because Windows reported a side-by-side configuration error, before any page loaded. Its four cases remain unverified; do not report them as passing. Reports are under docs/evidence/.

P02 adds historical sample/profile and missing-value/failure tests. Its initial run caught a WebKit dialog-focus issue, fixed by explicitly retaining the clicked trigger. The subsequent supported-engine run passed all 18 tests; six Firefox cases still cannot launch in this environment. To target working engines directly on this PowerShell host without npm argument forwarding ambiguity:

```
node web/node_modules/@playwright/test/cli.js test
--config=web/playwright.config.ts --project=chromium-desktop --project=webkit-
desktop --project=chromium-mobile
```

Playwright's WebKit engine on Windows and mobile emulation do not establish coverage on physical Apple/mobile hardware. Those P02 checks did not exercise WebGL volume rendering. P03 adds that coverage, described below.

Publishing to Vercel

The user explicitly selected Vercel and authorized ongoing deployment. The existing project is linked through the ignored .vercel/project.json; use it instead of creating another project.

```
npx --yes vercel@59.23.2 deploy --prod --yes
```

Public production URL: <https://depth-atlas-seifuku.vercel.app>

The Depth Atlas domain is a verified production domain on the same ocean-navigator-project brief Vercel project, not a duplicate project. The earlier address remains active. Future production deployments should retain the configured production

domains. Browser storage is origin-specific; moving saved investigations requires explicit export/import.

`vercel.json` selects FastAPI and builds the React client. `deploy/prepare_vercel.py` copies only compiled frontend assets into root `public/` for the Vercel CDN. `app.py` exports the same API used locally. Python dependencies are pinned in `requirements.txt`; `.python-version` selects Python 3.12. No Git remote is required for CLI deployment.

The function explicitly includes `casepacks/**`. It does not install `requirements-science.txt` or perform source downloads during deployment. Source acquisition and numerical preparation are separate offline steps; checked read-only assets are deployed with the app. Keep raw archives out of deployment.

Vite also builds `web/about.html`, `web/privacy.html` and `web/terms.html` as standalone static pages. `cleanUrls` exposes `/about`, `/privacy` and `/terms` on Vercel; explicit FastAPI routes provide the same paths locally. The pages work without JavaScript. The user confirmed the public identity on 26 September 2026; see `TEAM.md` and keep all three pages synchronized. Privacy distinguishes IndexedDB investigations, the `localStorage` regional pack, temporary imports and API replay. Review the policies whenever data handling changes. P17 now owns the authorized security audit; consult its security report and `PROJECT_STATE` for verified scope instead of assuming these pages establish security.

On a fresh machine, run `npx --yes vercel@59.23.2 login`, then link the existing `ocean-navigator-project` brief project in the user's intended account/team. Do not overwrite another project or purchase a plan without authorization. The project root is this repository, not `web/` or `deploy/`.

Check the public URL in a clean browser session after every deployment, including `/api/health`, `catalogue`, `scripts/styles`, `favicon` and `attribution`. A READY deployment alone does not prove correct routes or anonymous access. Do not enable Vercel's `public` `source/log` visibility setting to make the website accessible.

Portable infrastructure package

```
docker compose -f deploy/compose.yaml up --build
```

This builds the frontend and serves it with FastAPI as a non-root process, on loopback port 8000. The image has a liveness check. Docker is not installed on the current Windows machine, so this container recipe has not yet been executed. Actual INCOIS installation remains a separate verification task.

Vercel is the public application host. Future long scientific jobs, THREDDs and persistent scientific storage need a measured deployment design: serverless request limits and ephemeral local files cannot be treated as an always-running worker or durable database.

Retain the portable service boundary and decide supported jobs/storage when those phases are implemented; no additional paid service has been selected.

Phase 3 renderer checks

The implemented renderer uses Three.js 0.186.0 and a Canvas 2D fallback. See [PHASE_03_RENDERING.md](#) for physical-coordinate, mask, picking and quality contracts. No extra scientific runtime dependency is required in the deployed API. Eligible API responses are gzip-compressed.

Run `npm --prefix web run test:science` for the 17 independent grid/geometry and display-motion checks. Node 22 runs these TypeScript helpers with `--experimental-strip-types`; the resulting experimental warning is expected. The normal build performs TypeScript checks for the full frontend and browser tests.

The browser suite now includes scientific views, independent NetCDF value fixtures, actual screen picking, playback, invalid colour ranges, missing depths, missing WebGL, Chromium context loss and Auto degradation using an explicitly scaled frame clock. One WebKit context-loss case is deliberately skipped because that injection runs in Chromium. This is separate from the unavailable Firefox binary.

Run `node web/benchmarks/phase-03.cjs` with the production server running for screenshots and measured frame/resource observations. Use `OCEAN_TEST_URL` for another target and `OCEAN_BENCH_NAME` for a distinct evidence filename. Run benchmarks without the browser suite concurrently. Headless software rendering and mobile emulation are not real-device coverage.

Use `node web/benchmarks/clarity.cjs` for the current cutaway layout and all-view review, including 320-pixel width and enlarged text. The original `phase-03.cjs` captures the earlier release design.

Release 0.3.2 adds `web/e2e/continuity.spec.ts` for retained-frame identity, depth/cache requests, cut transition/resource behavior, request ordering, context preservation and native-coordinate rejection. Run `node web/benchmarks/continuity.cjs` for the current all-view/responsive audit and `node web/benchmarks/interaction.cjs` for depth requests and transition frames. Benchmark separately from the browser suite. Historical `phase-03.cjs` and `clarity.cjs` describe earlier interface releases.

Phase 4 instruments and deployment

`science/instruments.py` defines the version 2 contract. `adapters/instruments.py` dispatches supported layouts and validates units, coordinate metadata and source QC.

`science/acquire_instruments.py` downloads bounded originals;

`science/prepare_instruments.py` produces casepacks/`instruments/`. Regenerate the library only as a controlled preparation step, then restart the API because its immutable index is cached. Never rebuild deployed packs inside a request.

The public parser receives file bytes, never a URL or filesystem path. Limits are 2 MB, 24 profiles and 5,000 source samples per file, with a 3.5 MB decoded JSON ceiling. Native NetCDF operations run under a process lock. Browser memory holds imported profiles until cleared or reloaded, capped at 24 profiles/12,000 samples. Policies describe the upload and filename/log handling. There is no durable upload service.

`pyproject.toml` mirrors the exact runtime pins in `requirements.txt` for Vercel's uv resolver. Supported server resolution targets are Linux and Windows AMD64, Python 3.10-3.12. `netCDF4` 1.7.4 requires a different NumPy version for Windows ARM64, which has not been verified as a server target. This does not restrict client browsers or require users to install Python. Keep runtime dependency lists and release versions synchronized. See the [uv resolution settings](#) and [Vercel Python runtime](#).

The backend suite includes original-file observation checks. `web/e2e/instruments.spec.ts` covers marker selection, charts, every genuine example import, malformed input and keyboard controls. `node web/benchmarks/instruments.cjs` records desktop, WebKit, mobile/narrow and enlarged-text screenshots. `OCEAN_TEST_URL` and `OCEAN_BENCH_NAME` select the target and evidence prefix. Run browser suites and visual reviews sequentially; concurrent Playwright runs share output paths.

`.venv/Scripts/python.exe -m science.verify_p04_public` verifies the Vercel release, all 13 profiles against local assets, five downloadable/importable examples, error responses, static assets and three independently sourced model probes. It writes `docs/evidence/p04-public-http.json`, including failures. Preserve failed reports under distinct filenames before repeating. These checks do not establish scientific independence from assimilated observations or physical-device coverage.

Phase 5 evidence comparisons

The existing FastAPI service now exposes comparison and coverage GET routes beneath `/api/cases/{case_id}/evidence/`. No extra runtime dependency, remote per-request source download, database or background worker is required. Both routes share bounded matching settings and immutable checked assets. See `PHASE_05_EVIDENCE.md` for the full method and scope.

Run `.venv/Scripts/python.exe -m science.verify_p05_reference --check` to confirm the frozen independent reference from the original files under `ignored data/raw/`. This verifier does not import the application matcher. Run `.venv/Scripts/python.exe -m pytest tests -q` for science/API checks, including deliberate mismatches, source holds, QC narrowing and all 576 frozen settings scenarios.

Browser acceptance includes `web/e2e/evidence.spec.ts`. Use `node web/benchmarks/evidence.cjs` for comparison/coverage/rules screenshots at desktop, WebKit, mobile/narrow and enlarged-text sizes. Set `OCEAN_TEST_URL` and `OCEAN_BENCH_NAME` for the target and a unique report prefix. Run browser suites and visual reviews sequentially. The P05 local verification service uses port 8002 so it cannot reuse an earlier process's cached library index.

The final P05 public audit command is `.venv/Scripts/python.exe -m science.verify_p05_public`. `OCEAN_TEST_URL` chooses the base URL and `OCEAN_HTTP_REPORT` chooses a unique report path. It verifies scientific outputs strictly and records any failures without automatic retries. Public/local asset checks account only for demonstrated generated filename references, HTML line endings and the exact unused test-derived `.table` utility documented in the asset audit.

P06 scientific exchange and extensions

The public release includes Use the data, the native CF NetCDF download, product metadata and the derived horizontal-energy field. Build commands are unchanged. `deploy/prepare_vercel.py` verifies the exchange checksum and copies it to the public CDN directory; the large NetCDF is excluded from the Python function bundle.

Rebuild the exchange deterministically with `python -m science.prepare_standards`. Check scientific boundaries with `python -m pytest tests -q`. Offline requirements now include pinned OWSLib and Pillow for independent protocol verification.

Run `.venv/Scripts/python.exe deploy/run_standards.py --install` from the root on Windows to install vendor-checksummed portable Java/Tomcat/TDS into `.runtime/p06` and serve loopback port 8096. In a second terminal run `.venv/Scripts/python.exe -m science.verify_p06_standards`. That verifier requires the original acquired HYCOM files;

serving the bundled exchange does not. Detailed Linux/institutional and optional Compose instructions are in `deploy/standards/README.md`. The Windows launcher and fresh-base setup were executed; Docker was not.

Use `python -m science.verify_p06_public` for anonymous HTTP and complete public download verification. P06 browser checks are `web/e2e/standards.spec.ts`; the responsive screenshot runner is `web/benchmarks/standards.cjs`. They use the existing `OCEAN_TEST_URL` and `OCEAN_TEST_REPORT` or `OCEAN_BENCH_NAME` environment variables. Do not run multiple Playwright test suites against the same test-results directory concurrently.

P07 connected regions and arbitrary sections

`science/features.py` operates on the full native float64 case. `api/feature_store.py` associates eligible observations and exposes bounded search, selected-region and section POST routes. `web/src/features/` contains the linked query, map, inferred boundary, section and numerical export interface. See [PHASE_07_FEATURES.md](#) for the exact method and limits.

Run the full backend suite and all 24 renderer checks:

```
.venv/Scripts/python.exe -m pytest tests -q
npm --prefix web run test:science
```

The independent fixture generator is `science/verify_p07_reference.py`. It reads original acquired files and uses offline SciPy labeling and high-precision volume arithmetic, without importing the application's feature engine. Keep the frozen fixture and original source hashes when changing numerical code. Restart the API after changes to cached scientific modules or source packs.

For deployed numerical acceptance, use a distinct evidence path so earlier failures remain available:

```
.venv/Scripts/python.exe science/verify_p07_public.py --base https://depth-atlas-seifuku.vercel.app --output docs/evidence/p07-public-source-api.json
.venv/Scripts/python.exe -m science.verify_p06_public --base https://depth-atlas-seifuku.vercel.app --output docs/evidence/p07-public-existing-http.json
```

`web/e2e/features.spec.ts` and `features-science.spec.ts` cover query identity, source values, masks, section coordinates, selection, exports and failure recovery.

`web/benchmarks/features.cjs` and `p07-audit-feature-details.cjs` capture responsive visual evidence, including scrollable chart panels and 200% text. Run these separately from the full browser suite and inspect the images as well as the assertions. The P07 local acceptance server uses port 8004; the normal documented port remains 8000.

P08 saved investigations

`science/investigations.py` defines the typed recipes and cross-language fingerprints. `api/investigation_store.py` recalculates and packages records using the existing scientific stores. The three POST routes under `/api/investigations/` capture, replay and export. They do not write durable records to a server filesystem. Browser storage uses IndexedDB; portable files and bounded fragment links support another browser or device. See [PHASE_08_INVESTIGATIONS.md](#) for compatibility, limits and source handling.

Run the standard production build, full backend suite and renderer checks. `web/e2e/investigations.spec.ts` adds saved-state, reference-policy, file/link replay, mismatch, storage-denial, export and keyboard checks. Run the screenshot/export workflow separately with `node web/benchmarks/investigations.cjs`, setting `OCEAN_TEST_URL` and `OCEAN_BENCH_NAME` to the intended target and unique evidence prefix. P08 local acceptance uses a freshly started API on port 8005.

Fresh-process and public replay commands:

```
.venv/Scripts/python.exe science/verify_p08_replay.py --capture
tests/fixtures/p08-replay-cases.json --output docs/evidence/p08-local-
records.json
.venv/Scripts/python.exe science/verify_p08_replay.py --replay
docs/evidence/p08-local-records.json --output docs/evidence/p08-fresh-process-
replay.json
.venv/Scripts/python.exe science/verify_p08_public.py --base https://depth-atlas-
seifuku.vercel.app --records docs/evidence/p08-local-records.json --output
docs/evidence/p08-public-replay.json
```

The first two commands are separate processes. The third checks complete deployed result identity and actual archive/CSV contents against those records. It does not import the application, retry failures or replace the existing independent source fixtures. Use new output names when repeating a failed run. Browser, report and portable-file timestamps are capture dates in 2026; scientific timestamps remain those of the historical sources.

Phase 9 case preparation

Both registered cases use `science.case_recipes.RECIPES`. From the workspace root:

```
.venv/Scripts/python.exe -m science.acquire_case --case arabian-sea-2024-01
.venv/Scripts/python.exe -m science.prepare_case --case arabian-sea-2024-01
.venv/Scripts/python.exe -m science.prepare_case_instruments --case arabian-
sea-2024-01
.venv/Scripts/python.exe -m science.prepare_standards --case arabian-sea-2024-01
.venv/Scripts/python.exe -m science.prepare_guides
.venv/Scripts/python.exe -m science.verify_p09_sources
.venv/Scripts/python.exe -m science.verify_p09_observations
```

Acquisition is serial and resumes checksum-verified source checkpoints. Original raw files stay under ignored `data/raw/arabian-sea/`. The prepared model pack, separate instrument index, native exchange manifest and guided recipe are versioned assets. Do not overwrite the original Bay source library to append a case: that would invalidate existing investigation source fingerprints. Rebuild or change sources deliberately and retain original hashes.

`/api/instruments?case_id=...` selects the case library. The default remains the original Bay library. `/api/data-access?case_id=...` selects the native download and declares the actual protocol scope. `/api/guided-cases` returns the source-checked teaching pair. New registered cases require a recipe, packed assets, an allowlisted ID and tests, not arbitrary client-supplied paths.

Latest P09 development API: `http://127.0.0.1:8007`. Start a fresh process after changing Python modules; older ports can retain in-memory modules and caches. `npm --prefix web run build` refreshes its served frontend. Run browser suites sequentially because they share Playwright's result directory. On this Windows shell, use the direct Node Playwright CLI when passing project arguments, as described in earlier phase notes.

Phase 10 Virtual Expedition

The checked P10 local service uses `http://127.0.0.1:8008`. The normal startup command remains `.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8008`. Rebuild the frontend with `npm --prefix web run build` after browser changes.

`science/expedition.py` contains prior-only selection and fixed residual reconstruction. `api/expedition_store.py` freezes all designs before loading target arrays. `science/expedition_contracts.py` bounds the supported requests. New POST endpoints are `/api/cases/{case_id}/expedition/plan`, `/survey` and `/experiment`. The browser tool lives in `web/src/expedition/` and uses existing shared cases and investigation capture/replay. No additional runtime package or persistent store was introduced.

Run `.venv/Scripts/python.exe -m pytest tests/test_expedition.py tests/test_expedition_independent.py -q` for focused API/replay and independent scientific checks. Original-source tests require the acquired packed files under ignored `data/raw/`; without them, those tests explicitly skip. Do not treat a skip as independent source verification.

The HTTP deployment verifier compares all supported responses with local calculations and checks portable records and exports. This is deployment consistency evidence; the independent source tests establish the numerical reference. Use fresh evidence filenames for rechecks:

```
.venv/Scripts/python.exe science/verify_p10_public.py --base-url https://depth-atlas-seifuku.vercel.app --expected-version 0.10.1 --output docs/evidence/p10-public-new-check.json --replay-records docs/evidence/p10-v2-local-records.json
```

From web, run `node benchmarks/p10-visual.mjs <base-url> <evidence-directory>` for the screenshot audit. Run Playwright suites sequentially and preserve any failed traces before rerunning. Source/method identities, exact source timestamps, simulation labels and seeds are part of exported investigations. App patch identity is separate from scientific method identity.

Phase 11 Drift Lab

Local verification uses `http://127.0.0.1:8009`. Start a fresh API after changing Python code:

```
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8009  
npm --prefix web run build
```

The Vercel runtime requirements are unchanged. `science/vendor/parcels_rk4.py` adapts the pinned MIT-licensed integration kernel; `science/drift.py` supplies checked interpolation, masks, release sampling, integration and accounting. `api/drift_store.py` bounds current caches and completed results. The `/api/cases/{case_id}/drift/context`, `/run` and `/stream` endpoints share native case data. Stream responses are newline-delimited JSON with genuine step progress, explicit no-transform/no-store caching and no application gzip buffering. The calculation is request-scoped; no durable job queue is configured.

Install the separate offline reference environment to reproduce the full OceanParcels checks:

```
.venv/Scripts/python.exe -m pip install -r requirements-drift-reference.txt  
.venv/Scripts/python.exe -m pytest tests/test_drift_independent.py  
tests/test_drift_api.py -q
```

Full-framework reference tests require Parcels 3.1.4. Original-source tests require the acquired NetCDF archives. Missing dependencies/archives produce explicit skips, which do not count as source or framework verification. Runtime deployment does not install this reference-only environment.

From web, run the direct Node Playwright CLI with `e2e/drift.spec.ts`, the selected supported projects and `OCEAN_TEST_URL`. Run suites sequentially to preserve their shared output directory. `node benchmarks/p11-visual.mjs <base-url> <evidence-directory>` captures the scientific and responsive views. Keep original failed reports before a recheck. Verify final public replay in both directions using saved records from the exact final scientific method, in addition to local numerical tests.

The deployed method is `p11-drift-v2` in release 0.11.1. It fixes the cross-version summary summation policy; original v1 archives remain readable but intentionally report a replay method mismatch. To repeat the final public HTTP check with new output names:

```
.venv/Scripts/python.exe science/verify_p11_public.py --base-url https://depth-atlas-seifuku.vercel.app --expected-version 0.11.1 --output docs/evidence/p11-public-new-check.json --records-output docs/evidence/p11-public-new-records.json --replay-records docs/evidence/p11-v2-local-records.json
```

The verifier performs no automatic retries and checks actual streams, request limits, source context, old investigations, record fingerprints and exported archive contents. Deployment consistency does not replace the independent source and numerical reference tests. The final recorded run passed 100 checks across 114 requests; all eight resulting public records also replayed exactly through the Windows API. See `evidence/p11-v2-public-to-local-replay.json`.

Phase 12 Heat & Depth Lab

P12 verification uses a fresh local API at `http://127.0.0.1:8010`. Rebuild the frontend and start the API from the workspace root:

```
npm --prefix web run build
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8010
```

`science/heat.py` implements the declared surface baseline, event detection and native-column thermodynamic diagnostics. `api/heat_store.py` checks the separate SST source pack and links a selected event to actual model dates. The two heat endpoints are `GET /api/cases/{case_id}/heat/catalog` and `POST /api/cases/{case_id}/heat/analyse`. Missing baseline data, unsupported selections and unavailable depth dates remain explicit. The browser module is under `web/src/heat/`.

Source acquisition and preparation are offline operations. See `P12_SURFACE_SOURCES.md` for the exact distributor recipes, raw archives and checksums. The deployed API reads the prepared pack; visitors do not trigger NOAA downloads. Runtime dependencies remain unchanged. The independently pinned Oliver reference is acquired into ignored `data/reference/p12/` for tests and is not distributed in the app.

```
.venv/Scripts/python.exe -m science.acquire_heat_reference
.venv/Scripts/python.exe -m pytest tests/test_heat_independent.py
tests/test_heat_api.py tests/test_investigations.py -q
.venv/Scripts/python.exe -m science.verify_p12_public --base-url http://127.0.0.1:8010 --output docs/evidence/p12-local-http.json --records-output docs/evidence/p12-local-records.json
```

Original source and reference checks require their recorded offline archives. A skipped source test is not scientific verification. The API acceptance suite requires the real

prepared SST pack, without a synthetic fallback.

Run `web/e2e/heat.spec.ts` in the three supported browser projects using the direct Node Playwright CLI. Run browser suites and `node benchmarks/p12-visual.mjs <base-url> <evidence-directory>` sequentially from `web` because browser reports share an output directory. Inspect retained screenshots as well as automated checks.

Use new evidence paths for rechecks. After deployment, run the HTTP verifier with the public URL and `--replay-records docs/evidence/p12-local-records.json`, saving the resulting public records separately. Replay those records through the local API as a separate reverse-direction check. Exact source/method/result identities remain required; numerical tolerances in independent physical-method comparisons do not relax saved-record integrity.

Phase 13 Climate Event Lab

Current P13 API verification uses a fresh process at `http://127.0.0.1:8012`. Port 8011 was the initial browser process and can retain earlier Python modules. Build and start from the workspace root:

```
npm --prefix web run build
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8012
```

Runtime dependencies are unchanged. Production reads the immutable climate field and official-index packs; visitors do not trigger external source downloads. Offline acquisition and preparation are documented in `P13_FIELD_SOURCES.md` and

`P13_INDEX_SOURCES.md`. After those independent packs are ready,

`.venv/Scripts/python.exe -m science.prepare_climate` freezes their hashes in the combined manifest. The public API is `GET /api/climate/catalog` and `POST /api/climate/analyse`; see `P13_API_CONTRACT.md`.

```
.venv/Scripts/python.exe -m pytest tests/test_climate_indices.py
tests/test_climate_independent.py tests/test_climate_api.py -q
.venv/Scripts/python.exe -m science.verify_p13_public --base-url
http://127.0.0.1:8012 --expected-version 0.13.0 --output docs/evidence/p13-local-
new-check.json --records-output docs/evidence/p13-local-new-records.json
```

Use new evidence filenames for rechecks. Independent source tests require the actual archived original files; missing archives do not establish scientific verification. Public acceptance repeats the verifier against Vercel with `--replay-records` pointing to eight local records and saves eight public records separately. A separate local `--replay-only` run checks the reverse direction. Strict source/method/result identities remain required. Run browser suites and visual capture scripts sequentially because Playwright shares its results directory.

Pacific cases retain source longitudes on a monotonic 0-360 axis, monthly support intervals and potential-temperature metadata. The fixed earlier Indian Ocean packs and existing replay identities remain unchanged. New source products need explicit quantity/time compatibility before reusing another scientific lab.

Phase 14 evolution and observation blackout

Current P14 acceptance uses a fresh API at `http://127.0.0.1:8014`. Earlier phase processes can retain earlier Python modules and should not be used for final P14 evidence. Build and start from the workspace root:

```
npm --prefix web run build
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8014
```

Runtime dependencies are unchanged. The implementation reads the existing immutable model and instrument packs. Evolution uses full native memberships and version `p14-native-overlap-v2`; observation exclusion uses `p14-blackout-v1`. The endpoints are `POST /api/cases/{case_id}/evolution/run`, `POST /api/cases/{case_id}/blackout/run` and `GET /api/cases/{case_id}/blackout/catalog`. Their bounded queries, source hashes, exported tables and scientific limits are in [Phase 14 methods](#).

```
.venv/Scripts/python.exe -m pytest tests/test_evolution.py
tests/test_evolution_reference.py tests/test_blackout.py tests/test_p14_api.py -q
.venv/Scripts/python.exe -m science.verify_p14_links --output
docs/evidence/p14-independent-new-check.json
.venv/Scripts/python.exe -m science.verify_p14_public --base-url
http://127.0.0.1:8014 --output docs/evidence/p14-local-new-check.json --records-
output docs/evidence/p14-local-new-records.json
```

The original-source evolution oracle needs the archived packed HYCOM files. A missing source test does not count as verification. Use new evidence filenames for rechecks. The public verifier supports repeated `--replay-records` inputs for older investigations. After deployment, verify the local records on Vercel, save separate public records, then run locally with `--replay-only --replay-records` pointing to those public records. Source, method and numerical replay fingerprints must remain exact in both directions.

Run `evolution.spec.ts` and `blackout.spec.ts` through the direct Node Playwright CLI in the three supported browser projects. Set `OCEAN_TEST_URL` and an absolute `OCEAN_TEST_REPORT` path. From `web`, `node benchmarks/p14-visual.mjs <base-url> <evidence-directory>` captures actual graph, extent, gap, exclusion, saved-record and responsive views. Browser suites and visual capture runs must be sequential to avoid contention and shared output-directory changes. Inspect screenshots in addition to checking the JSON report. Do not rebuild the frontend during a browser run.

Release 0.14.1 versions ordered evolution aggregates and adds an eight-result, per-instance cache. Source checks precede every lookup; duplicate requests share calculation and receive separate copies. Earlier v1 investigation JSON remains readable, with an explicit method mismatch on replay. Include `evolution-legacy.spec.ts` when checking this behavior. Final v2 replay inputs are `evidence/p14-v2-local-records.json`; the original `p14-local-records-final.json` deliberately remains a v1 archive fixture. Never overwrite it or relabel its method. The parser test loader uses built-in Node module hooks, tested on Node 22.17.1, without an added runtime dependency.

P15 wider-region preparation and verification

The prepared `casepacks/wider/` sources ship with deployment, so regional requests need no provider credentials or live provider connection. The public API accepts allowlisted dataset IDs only. `science.acquire_wider` and `science.acquire_gobai` are offline maintainer commands requiring the science environment. Acquisition journals and original slabs/ranges remain under ignored `data/raw/wider/`; they are needed for original-source reconstruction tests, not for serving the app. A fresh acquisition must review any changed provider metadata/hash rather than overwriting a published source identity silently.

```
.venv/Scripts/python.exe -m science.acquire_wider
.venv/Scripts/python.exe -m science.acquire_gobai
npm --prefix web run build
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8015
```

Regional method: `p15-native-subset-v1`. Contracts and limits are in `science/wider_contracts.py`; server cache/source integrity in `api/wider_store.py`; UI in `web/src/wider/`. `science.verify_p15_public` checks the served catalogue, selections, profiles, exact pack replay and CSV. `science.verify_p15_performance` records fresh-store/cache measurements and bounded four-client HTTP samples. `web/benchmarks/p15-visual.mjs` checks real views, a fully offline exported file and throttled regional requests. `web/benchmarks/p15-evaluators.mjs` records four isolated concurrent Chromium workflows. Use explicit new report paths to preserve earlier outcomes. See `PHASE_15_WIDER_COVERAGE.md` for precise invocation and final evidence.

The Vercel build retained all checked sources and automatically optimized its initially 279.34 MB function bundle. Runtime compatibility is established by public endpoint, replay and browser checks, not by the build message alone. No new runtime package or external storage account was added. The original linked hosting project remains in use.

P16 learning and explanations

The tutorial bridge and result contracts are in `web/src/learning/bridge.ts`. `lessons.ts` gives the core-first order, Basic wording, in-depth concepts/methods and optional questions. Each existing workspace registers its actual action handler and a structured current-result snapshot. `explain.ts` implements reviewed terminology, result-kind explanations and a bounded command grammar. No external model/provider secret or runtime dependency is required.

Build with `npm --prefix web run build`; `RUN npm --prefix web run test:science` for the renderer/parser and learning logic checks. The learning browser suite is `web/e2e/learning.spec.ts`; it covers full real demonstrations, interruption/retry, supported and unsupported queries, missing output and accessible controls. Use the existing Playwright config and unique `OCEAN_TEST_REPORT/output` paths. `web/benchmarks/p16-visual.mjs` captures Basic/In-depth/explanation layouts across five viewport sizes and enlarged text; `p16-smoke.mjs` records the complete 27-step path. Both accept `OCEAN_TEST_URL` and `OCEAN_VISUAL_DIR`.

The current local P16 service is port 8019. Start the existing Python environment with `-m uvicorn api.app:app --host 127.0.0.1 --port 8019` after building the frontend. Existing packed data and source methods are unchanged. On Windows, launch background helpers hidden. The public service remains the linked Vercel project; actual deployment and verification evidence is in `PROJECT_STATE.md`. P16 adds no standalone model server, analytics or persistent tutorial-answer store.

P16 anchored-tour follow-up

The active local service for release 0.16.1 is `http://127.0.0.1:8020`. The placement hook is `web/src/learning/useTourAnchor.ts`; lesson targets stay in `lessons.ts`. Retained hidden workspaces must not be used as anchors. Preserve the real-action and result-identity bridge when changing tutorial presentation. Navigation remains outside the scrollable explanation; selecting a lesson closes its menu.

`web/e2e/tutorial-anchor.spec.ts` checks positions, pointer/target separation, responsive resizing, usable controls, manual scroll recovery, modal save guidance, reduced motion and enlarged text. `web/benchmarks/p16-anchor-visual.mjs` captures the resulting layouts with `OCEAN_TEST_URL` and `OCEAN_VISUAL_DIR`. Run browser suites and screenshot harnesses sequentially. Keep unique evidence paths.

When bumping the app release, edit only the top-level version fields and the root package-lock entry. Do not replace all matching version strings: `h11==0.16.0` is a dependency pin, not the app version. The initial 0.16.1 deployment caught this mistake before promotion; dependency pins were restored and structurally compared to the prior

checkpoint. The final deployment uses the unchanged dependencies.

P16 feature-teaser follow-up

Release 0.16.2 uses local port 8021, launched hidden with `.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8021`. Build first with `npm --prefix web run build`. `featureTours.ts` maps each teaser to existing lesson IDs. Each route must begin with an independently runnable preparation action; Drift playback must follow its calculation. The lesson command waits for the case catalogue, and browsing the picker must not dispatch a lab action.

Add `feature-tours.spec.ts` to the existing learning/anchor browser suite. The `p16-teaser-visual.mjs` harness accepts `OCEAN_TEST_URL` and `OCEAN_VISUAL_DIR`, records viewport fit and captures the welcome, four feature pages, Drift, its completion and enlarged text. As before, run interaction and visual suites sequentially, with unique report paths. Public status is recorded in `PROJECT_STATE.md`.

Site-information follow-up, release 0.16.3

Local server: `http://127.0.0.1:8022`, started hidden using `.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8022`. Rebuild with `npm --prefix web run build` after editing static pages or styles. The server serves the compiled pages. `web/benchmarks/site-pages.mjs` checks no-JavaScript layouts, native anchors, footer/contact navigation, enlarged text, print colours and return to a real model value. Set `OCEAN_TEST_URL` and `OCEAN_PAGE_EVIDENCE` for another host/report directory; run `node web/benchmarks/site-pages.mjs` from the repository root. This Windows WebKit build does not tab to links by default, so its check records explicit link focus; Chromium checks sequential Tab entry. No new dependency or environment secret is required.

`science.verify_release` now includes `/about` among the anonymous static-page byte checks. `TEAM.md` is the source for confirmed public names and contact details. The site-page release itself did not complete security work. P17 is the separately authorized audit and its results are recorded in `PROJECT_STATE`.

Startup recovery, release 0.16.4

Use `http://127.0.0.1:8023` for this release. Build first, then start a new hidden uvicorn process so both the Python release constant and compiled client agree. The old 8022 process retains its earlier release constant. The new `startup-recovery.spec.ts` covers missing entry/lazy modules, one-time stale-page recovery, preserved state and independent-project identity.

The inline guard in `web/index.html` deliberately has no bundle dependency. `StartupReady` signals a completed mount. Successful mount removes only the temporary `_startup_recovered` query key; shared fragments and unrelated parameters remain. No browser storage is used. A scoped Vercel rewrite and equivalent local API route map only the known retired `/assets/workspace-fwVeVJXC.js` entry to `startup-recovery.js`. Do not introduce a wildcard missing-module rewrite. The entry and bridge use no-store; normal hashed assets retain long caching. CSP changes must preserve this guard and be verified against startup recovery; P17 records the actual selected policy.

P16A observation review, release 0.16.5

Use `http://127.0.0.1:8025` for the matching release process. Build first, then launch `.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8025` in a hidden background process. Earlier ports retain their own release constants. No new runtime package, provider account or environment secret is required.

`POST /api/instruments/inspect?filename=...` accepts the same bounded raw file body as the original import endpoint. Optional `X-Ocean-Column-Mapping` is URI-encoded JSON mapping documented target names to `{source, units}` or null. Units are specified for measurements only. The request is read incrementally up to 2 MB.

`adapters/import_preview.py` validates a temporary CSV representation, returns the original file identity and preview, and writes no upload. Existing `/api/instruments/import` remains available. The frontend uses `ImportReview.tsx` for explicit review/acceptance, cancellation and source-example preview. `context.ts` locates eligible samples without altering the original profile.

`science.prepare_p16a_chlorophyll` pins the genuine source file, checks its expected hash and writes the additional public example catalogue and source evidence. A changed provider file requires source review; do not replace the expected hash just to make preparation pass. Keep `casepacks/instruments/index.json` unchanged: existing saved investigations depend on its exact source identity. Supplemental examples are listed in `import-examples.json`; they enter scientific views only through the import path.

Useful checks:

```
.venv/Scripts/python.exe -m pytest tests/test_import_preview.py
tests/test_instruments.py tests/test_evidence.py
tests/test_evidence_provenance.py tests/test_investigations.py
tests/test_guided_cases.py -q
.venv/Scripts/python.exe -m science.verify_p16a --report
docs/evidence/p16a-example-overlap.json
$env:OCEAN_TEST_URL='http://127.0.0.1:8025'
$env:OCEAN_TEST_REPORT='../docs/evidence/p16a-browser-release.json'
node web/node_modules/@playwright/test/cli.js test -c web/playwright.config.ts
```

```
import-review.spec.ts instruments.spec.ts instrument-context.spec.ts
--project=chromium-desktop --project=chromium-mobile --project=webkit-desktop
--workers=2

node web/benchmarks/p16a-visual.mjs
```

Use unique evidence paths on repeat runs. Call Playwright directly on this Windows setup; npm argument pass-through stripped selection flags in an initial attempt. Run visual captures after interaction suites. `science.verify_p16a --url <base-url> --report <path>` checks served example bytes and interpretation against local references. Standard `science.verify_release` still verifies version/asset alignment after Vercel deployment. Scope, source gaps and limits are in `PHASE_16A_IMPORTS.md`.

The first P16A deploy exceeded Vercel's 225 MB optimized function limit by 0.18 MB. The function excludes `web/benchmarks/**`, `web/science-tests/**` and offline `science/verify_*.py`; the files remain available locally for verification and portable source distribution. Runtime imports do not reference these verifier commands. Existing scientific packs and runtime dependencies are preserved. Budget source-pack growth before adding fresh model data in P16B; this deployment has limited remaining function size.

P16B: clean portable installation and fresh input

Release 0.16.7 was installed from the local versioned source archive at `.runtime/p16b-portable-final`, with fresh runtime-only Python dependencies and `npm ci`. The API, source-value, model replay/export/download checks and 24 browser checks passed against port 8028. This is an isolated Windows installation on the same host, not an INCOIS deployment or a tested Docker container.

To reproduce from a source checkout or archive:

```
python -m venv .venv
.venv/Scripts/python.exe -m pip install -r requirements.txt
npm --prefix web ci --no-fund
npm --prefix web run build
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8000
```

Wait for Application startup complete before checking `/api/health`. The app does not require raw source downloads, xarray, provider keys or the developer's existing virtual environment for normal use. Full scientific preparation still uses `requirements-science.txt`. Start servers hidden when launching them as background helpers on Windows.

`casepacks/** -text` in `.gitattributes` prevents Git from changing scientific JSON line endings and breaking the recorded byte hashes. All 304 staged case-pack files were checked byte-for-byte against the working files; all 260 files present before P16B retain their exact bytes. Do not renormalize pinned source packs.

Fresh-input acquisition and the loopback standards commands are documented in PHASE_16B_FRESH_DATA.md. The existing Vercel project now has the non-secret VERCEL_SUPPORT_LARGE_FUNCTIONS=1 setting for preview and production, after an actual larger-function preview passed. No plan/provider change was made. This is a packaging option, not an increase to the app's 2 MB import limit or 30-second function duration. A separately linked installation should inspect its own eligibility before using it. Keep deployment and dataset-size checks at each release.

P16C imported comparison, capture and replay

science/imported.py defines the 2 MB original-file envelope (base64, SHA-256, filename, parser version, optional CSV mapping). api/imported_store.py reparses it with the existing bounded preview adapter and gives EvidenceStore a request-only observation collection. The imported subclass bypasses every process-wide library/profile/result cache. Do not feed client-normalized profiles directly to scientific methods or add imported data to casepacks/instruments/index.json.

POST /api/cases/{case_id}/evidence/imported/comparison takes {import_source, profile_id, settings}; /coverage takes {import_source, settings}. Coverage and discovery include the selected original file only. Existing capture/replay/export accept an optional import_source for instrument/comparison recipes. Bounded JSON dependencies stop input over 3 MB. The envelope checks original bytes and parser version before parsing; source identities bind the context and all parsed profiles. An imported response over 4.2 MB fails explicitly. New source/model/parser versions do not silently migrate old records.

Source bytes live in app tab state until explicit save. IndexedDB uses the existing 25-record, 32 MB and 8 MB limits. Local files are cleared from the imported comparison query cache with the workspace; deleting saved records uses the existing separate control. The portable JSON retains original bytes and parsed source profiles; the evidence ZIP additionally exposes the original file. Imported share URLs are disabled. Reopen replaces the active imported set only after exact recalculation. The optional instrument model_time_index field is omitted when absent to keep legacy fingerprints stable.

Run `python -m pytest tests/test_imported_investigations.py tests/test_investigations.py tests/test_import_preview.py tests/test_evidence.py tests/test_evidence_provenance.py tests/test_instruments.py tests/test_fresh_model.py tests/test_api.py -q`, then the direct Playwright `imported-investigations/import-review/investigations suites. python -m science.verify_p16c --url http://127.0.0.1:8031 --report docs/evidence/p16c-local-http.json` verifies exact served/local results, raw-file exports, rejection paths and four legacy P08 archives. `web/benchmarks/p16c-visual.mjs` captures

three screen sizes. This does not replace P17 or actual domain-user testing. PHASE_16C_IMPORTED_REPLAY.md describes scientific/storage boundaries.

P16D support and connected workflow

POST /api/cases/{case_id}/features/support takes {query, region_id, settings, import_source?}. Query is the existing P07 threshold contract, settings the P05 comparison contract; their variable/time must agree. ImportedSource is the bounded P16C original-byte envelope. The separate p16d-native-support-v1 response provides accepted rows, distinct native cells/columns, gaps, rules and provenance. No frozen P05/P07 response changes. Body limit 3 MB, decoded output budget 4.2 MB; no automatic truncation. Imported collections are request-scoped and uncached.

UI entry: Build investigation in current-study controls. Structure support appears on demand inside Find structures. Comparison and source profile use shared source sample IDs with separate model time; unsupported picks never become residuals. Support exports are snapshots, with structure/comparison saves retaining the existing replay path. See PHASE_16D_CONNECTED_WORKFLOW.md and P16D_USER_TASK.md.

Verification: `python -m pytest tests/test_structure_support.py -q; npm --prefix web run test:science`; direct Playwright CLI with `connected-investigation.spec.ts`; `python -m science.verify_p16d --url URL --report REPORT.json`. Use the same release version in backend, web build and test fixtures. Do not rebuild a served directory or bump the release file while browser acceptance is running; mock version headers also read that file. Public checks must follow a completed deployment.

P16E continuity

See PHASE_16E_CONTINUITY.md for enforced budgets, scientific/offline boundaries and the publication runbook. `science/publication.py` validates prepared candidates, stages checked bytes under a bounded maintainer store and atomically selects a retained release. `OCEAN_PUBLICATION_ROOT` pins a checked store release at startup for local/institutional installations. Do not configure this writable-store path on Vercel. `deploy/prepare_vercel.py` builds the bundled publication inventory; `/api/publication` returns bounded public status without internal paths.

The observation queue reads sequentially and reuses the existing original-byte parser. Source/decoded limits apply independently to queue and workspace. Imported scientific requests still carry one original file and use the existing replay identities. `investigations/offline.ts` builds a self-contained saved-evidence viewer. Keep scientific series exact, source text inert, CSP network-denying, and original JSON available. Explicit

save/replay/export cancellation uses generation checks and stops before the local atomic commit stage.

Checks: `pytest tests/test_publication.py tests/test_api.py`
`tests/test_import_preview.py tests/test_imported_investigations.py`
`tests/test_investigations.py -q; npm --prefix web run test:science; Playwright`
`p16e-continuity.spec.ts`, import review and saved/imported investigation suites;
`web/benchmarks/p16e-visual.mjs`; existing release and P16C replay verifiers. Tests
generating 20,000 levels label them resource fixtures, not observations.

Globe and tutorial follow-up (0.16.11)

The default workspace is a deliberate data globe. Keep its state independent of metadata loading; only an explicit user or tutorial command opens the selected ocean model. Discovery uses source bounds/profile coordinates, while the learning registry determines tutorial order and feature count. More features stays above cards, and the completed-tour handoff targets the real Tools button. Tutorials reuse the real scientific command path.

After a version change, build both frontend and matching publication before starting: `npm --prefix web run build`, then `.venv/Scripts/python.exe deploy/prepare_vercel.py`. This is also the existing Vercel preparation path. The final local server used `.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8036 --no-access-log`. A mismatched old publication fails startup intentionally. Keep scientific case-pack bytes unchanged.

Run client scientific checks and direct Playwright suites `data-globe.spec.ts`, `tutorial-discovery.spec.ts`, `learning.spec.ts` and `connected-investigation.spec.ts` as appropriate. The visual harness is `web/benchmarks/globe-tutorial-visual.mjs`, with `OCEAN_TEST_URL`, `OCEAN_TUTORIAL_VISUAL=1` and `OCEAN_VISUAL_DIR`. Final evidence is in `evidence/globe-tutorial-acceptance.json`.

`web/playwright-transport.config.ts` is a diagnostic-only alternative disabling HTTP/2. Normal acceptance uses `web/playwright.config.ts`. A diagnostic pass does not fix the repeated public comparison timeout or justify changing production transport without further evidence. The normal suite failure, Retry success and diagnostic pass remain separately recorded for P17.

Targeted reliability follow-up (0.16.12)

JSON GET requests now allow 25 seconds for headers and body, with immediate caller cancellation and existing bounded transient retry. OceanExplorer warms only the next selected-variable frame, once the current field/native reading are ready. It cancels unused speculative queries on context changes and preserves a request adopted by an active observer. The parent enables speculative work only for the primary ocean task. Source validation, response identities and scientific methods remain unchanged.

`comparison-reliability.spec.ts` includes real-source delayed-response, abandoned-prefetch, adoption and normal-transport core-tour regressions. Its diagnostic attachment records API protocol/timing and browser long tasks, without disabling HTTP/2. Use the normal Playwright config for acceptance. Existing continuity, service-recovery and full learning suites cover surrounding behavior. On this session the final local service is port 8037; build the matching publication before startup as above. `RELIABILITY_FOLLOWUP.md` records the diagnosis and limits.

P17 release and Firefox verification

Release 0.17.0 is deployed on the existing Vercel project. [P17_DELIVERY.md](#) records its immutable deployment, runtime/packaging commits, clean installation and public checks. Earlier port numbers and release counts above describe their dated runs; use `PROJECT_STATE` for the currently running service. The final public release passed 24 release checks, 61 exact replay checks and both complete desktop/mobile tutorials on normal transport. These are separate evidence scopes.

The previously installed global Firefox binary still failed before page startup. P17 installed a fresh Playwright-managed Firefox into the isolated `.runtime/p17-browsers` directory, leaving the global binary unchanged. Its six-test launch/navigation probe passed. The broader Firefox suite then recorded **43 passes, one picking-test failure and one intentional context-loss skip**. The retained report is `p17-firefox-release.json`, not a relabelled clean run.

The independent pointer/source oracle showed that Firefox delivered a different subpixel pointer location from the requested one, crossing a native-cell midpoint. The application returned the correct source cell/value for the delivered event. The test now aims at the pixel centre for Firefox; production picking code, source indices and numerical tolerances are unchanged. The corrected picking check passed **three repeated runs**, with no skips or retries. Evidence: independent diagnosis and repeated acceptance. Do not add repeats to the broader suite's unique-test count or claim all physical Firefox/GPU configurations were tested.

To use that isolated browser install, set `PLAYWRIGHT_BROWSERS_PATH` to the absolute `.runtime/p17-browsers` path and run the normal Playwright configuration with `--project=firefox-desktop`. Retain source-reference checks for picking rather than weakening tolerances. Chromium/WebKit, mobile emulation, actual hardware, external usability, independent oceanography review and institutional installation are separate evidence categories. Docker execution, full independent CF certification, actual INCOIS deployment and private portal verification remain unestablished.

66 Product decisions and boundaries

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt |
Source: docs/DECISIONS.md

D001 : Full scope is retained

On 2026-09-21 the user requested that everything be built in phases so work can be directed and resumed reliably. The earlier recommendation to omit several labs from a ten-day prototype is not authority to remove those labs. They remain planned, with their own prerequisites and release checks.

D002 : Target and audience

The user reports a national-round target of top five among 100 teams. No placement probability or judging rubric has been established.

Primary workflow: explore model data, discover a structure, inspect observational evidence, run a supported investigation and save a reproducible record. Primary users are ocean analysts/research students; forecasters, educators and public audiences receive appropriate views. Preserve one connected workspace rather than unrelated mini-apps.

D003 : Geography

Start with a bounded Bay of Bengal case, then an Arabian Sea case. Select boundaries and periods by actual data availability; they are not yet locked. Add selected tropical Pacific cases with the Climate Event Lab. Wider global access follows explicit compatibility, coverage and performance checks. A world map does not establish worldwide analytical support.

D004 : Technology direction, not yet a locked dependency list

The proposed feature-by-feature baseline is in `STACK_AND_EXPERIENCE.md`: React + TypeScript + Vite, Tailwind/Radix, Zustand/TanStack Query, vtk.js for scientific 3D, D3 charts, Python + FastAPI/Pydantic + xarray/NumPy/SciPy, and Dask only when workload measurements justify it. Consider NetCDF/Zarr cache files and SQLite metadata initially. A separate Python worker with RQ/Redis is the queue candidate for longer calculations; Three.js is an optional presentation-only addition subject to a focused visual/performance check. These are proposed defaults, not installed or benchmarked dependencies.

Evaluate THREDDS for real WMS/WCS/OPeNDAP paths. Evaluate Parcels for bounded passive drift, TEOS-10/GSW for seawater conversions, and a documented marine-heatwave reference implementation. Pin versions after small compatibility checks. Reuse components; do not build a new rendering engine or scientific forecast model.

Deploy a thin application early. The user subsequently selected Vercel (D013); no paid service purchase is authorized. Prepare a portable server package. A public prototype is distinct from installation on INCOIS infrastructure, which needs their environment and access.

D005 : Data and analytical integrity

- Candidate sources: Copernicus GLORYS, INCOIS services, Argo, a suitable glider archive, CCHDO, NOAA OISST and documented ENSO/IOD indices.
- Access to a catalogue is not proof of a successful subset download or a usable matched case.
- Preserve source versions, licences, checksums, timestamps, variable meanings, units, masks, QC and processing steps.
- Temperature/salinity definitions and pressure/depth conversions must be explicit. Preserve irregular depths and distinguish display resampling from analytical data.
- Physics-only products do not provide invented biogeochemical fields. Missing variables remain unavailable until sourced.
- Synthetic fixtures are useful for tests and must be labelled. Real demonstration records must be genuine.

D006 : Feature-search semantics

The initial search finds connected regions satisfying explicit scalar and depth conditions. It does not discover a new physical phenomenon simply by giving a region a name. Query controls are the reliable first interface; language assistance can translate into a visible structured query later. Physical volumes use real cell geometry. Tracking threshold regions does not establish that the same water moved between them.

D007 : Expedition and sensitivity limits

Candidate sampling locations have explicit objectives, constraints and reasons. Coverage, gradients and measured mismatches must remain distinguishable. Do not turn an unexplained weighted score into calibrated uncertainty.

A reconstruction sandbox gives competing strategies the same prior information, budget and evaluation protocol. Hidden evaluation values must not leak into selection. A reanalysis-based sandbox measures its declared reconstruction task; it does not establish forecast improvement, operational mission safety or real-world savings.

D008 : Labs and explanations

- Drift begins as historical, horizontal passive transport at selected depth, with no invented vertical motion, windage or pollutant chemistry.
- Surface marine heatwaves need a declared seasonal baseline and event method. Surface detection does not establish a subsurface event.
- Climate comparisons are historical and use compatible season/baseline/index definitions; they do not establish causal attribution or new ENSO forecasts.

D009 : Ten-day checkpoint and full roadmap

The initial public prototype target is ten days. P01-P09 describe the first competition-focused candidate, but phase duration has not been measured. Check progress daily; reduce the scope claimed by a release if needed without deleting future features from the roadmap. Do not compress away validation to claim all 17 phases fit in ten days.

The current user clarification that this is a national-round effort governs planning. Subsequent direct reading of the 2026 guidelines confirms 30 September 2026 for national idea submission/shortlisting and a proposed December 2026 finale. The precise finalist-event date, team-specific instructions and any weighted rubric remain unknown. See D011 and `SUBMISSION_CHECKLIST.md`.

D010 : Boundaries of "everything"

Everything means the software capabilities in the reference guide and the subsequent discussion, including the later/conditional software features listed in the feature register. It does not mean inventing an original coupled ocean/climate forecast model, manufacturing sensors, delivering operational rescue/oil-spill forecasts, or predicting local weather from an ENSO slider. The guide itself excluded those claims. Such work would need a separately scoped scientific project.

Reference provenance

- User source: D:\Downloads\project
brief_Ocean_Project_Introduction_and_Build_Guide.pdf, planning edition dated 19 September 2026, 44 pages.
- Preserved unchanged at reference/project
brief_Ocean_Project_Introduction_and_Build_Guide.pdf.
- Archive integrity is recorded in reference/SOURCE.txt after copying.
- The PDF is reference material and proposed design, not evidence that any capability exists.
- Official PS: [project brief on the SIH portal](#), reviewed during the planning discussion. Recheck operative requirements before final submission.
- Source foundations: [GLORYS](#), [Argo guidance](#), [TEOS-10](#), [CF](#), [THREDDS](#), [vtk.js](#), [Parcels](#), [NOAA experiment-design guidance](#).

D011 : Verified submission format and original USP visibility

On 2026-09-21, the user requested an official submission checklist and asked where El Nino/La Nina and the particle/"pebbles" simulation were covered. Those capabilities already existed in P13 and P11 respectively.

The current official homepage linked a 2026 guidelines PDF and 2026 idea-template PPTX. Direct HTTP downloads succeeded and both were read locally. They establish the six-slide maximum PDF shortlisting deck and listed portal fields. The template's seventh slide is an instruction slide for removal. The published list and this PS do not establish mandatory repo/video/report uploads. Those are recommended supporting assets unless current dashboard or organizer instructions add requirements.

General selection criteria include novelty, complexity, clarity, feasibility, practicality, sustainability, impact, usability and future development. No numerical weighting was

verified. Prepare the submission alongside development; do not wait for every roadmap phase to finish before producing the required deck, and clearly label what is working versus proposed.

Official documents are archived in `reference/official-2026/`, with retrieval URLs and hashes in `source-manifest.json`. The detailed submission checklist records what is verified, recommended and still unknown.

D012 : Unknown evaluator hardware; visual quality and usability

On 2026-09-21 the user corrected the target-device assumption: national round 1 is online and the evaluator opens the app on their own unknown device. The earlier selection of the user's Windows computer must not drive the release specification. The user explicitly wants ease of use and exceptionally impressive visuals/simulations.

Use server-side scientific computations, bounded browser graphics, adaptive quality, a useful first case and a truthful 2D/basic fallback. Browser rendering still depends on client hardware. No required WebGPU, discrete GPU, installation or developer account. Full 3D support is capability-dependent; never promise identical graphics on every device. Preserve scientific values when reducing display quality. Add visual/UX checks throughout P03, P09, P15 and P17.

The stack/experience document records visual intent, feature-to-engine mapping and proposed measurable targets. Implementation and benchmarking have not started.

D013 : Phase 1 foundation and Vercel hosting

Use the already authenticated Vercel account/team and the newly linked `ocean-navigator-project` brief project for future updates. The public production URL is `https://depth-atlas-seifuku.vercel.app`; anonymous HTTP and browser checks verified it. No plan/domain purchase or new GitHub repository was required.

P01 uses React/TypeScript/Vite, Tailwind, Radix Dialog, TanStack Query, Zustand, D3 geographic projections and FastAPI/Pydantic. Its coastline preview comes from Natural Earth via `world-atlas`, with attribution and licences served alongside the app. Planned region boxes are navigation extents, not confirmed dataset bounds. No scientific data is available; health/catalogue responses and disabled controls say so explicitly.

Build dependencies and local test dependencies are pinned separately. Vercel uses Python 3.12, while the local Windows interpreter is Python 3.10.11. The API and static frontend share an origin. Vercel builds the frontend and serves compiled assets from its CDN, with FastAPI for the API. Keep the Python API portable for future institutional

deployment.

Vercel's serverless request lifecycle and ephemeral filesystem cannot stand in for a permanent RQ worker, THREDDS server or durable SQLite data. Before those capabilities are implemented, benchmark and choose an appropriate supported service/storage arrangement; keep the public Vercel app. The Docker/Compose recipe is unexecuted because Docker is not installed. Firefox browser tests are blocked by the local browser binary's side-by-side configuration; Chromium, WebKit and mobile-emulated interactions passed. Do not call these scientific-volume or universal-device benchmarks.

D014 - Restrained product design and prototype legal pages

On 2026-09-21, the shell was simplified accordingly: rectangular controls, flat surrounding surfaces, direct task copy, explicit missing-data labels and no unused profile count badge. Existing source-based geography, required attribution and functional loading feedback are retained.

The custom SVG favicon is retained with square corners. `/privacy` and `/terms` are static HTML pages linked from the workspace footer, with dedicated titles, mobile styles, direct-load routes and no JavaScript requirement. They describe the current prototype and its actual code/hosting behavior; they do not claim legal certification. Vercel's current Privacy Notice and runtime log documentation were reviewed; the app has no analytics SDK, account/upload workflow or persistent visitor-state storage.

The user explicitly instructed that public team/operator identity and contact email remain blank until final phase completion. Do not invent contact details. Both pages explain that those details are not yet published, and P17.3 now requires their addition and a final data-flow/policy review.

D015 - Historical case selection and 2026 date discipline

P02 was authorized on 21 September 2026. The user explicitly emphasized that the current year is 2026. Project/retrieval dates therefore remain 2026; January 2024 is deliberately selected historical scientific coverage, not a claim of current conditions. UI and provenance display this distinction.

Use the publicly accessible HYCOM GOFS 3.1 + NCODA GLBy0.08/expt_93.0 archive for the first bounded model case and seven genuine INCOIS-DAC Argo delayed-mode profiles from Ifremer GDAC. Actual overlap established the 7-10 January 2024 window and approximately 85.04-90 E, 12-15 N box. The exact first source longitude is 85.0400390625 E; rounded display labels must not replace native coordinates. Source output contains 40 fixed depths, separate from the 41 hybrid model layers.

Copernicus metadata and credential requirements were inspected, but an authenticated subset download was not established. INCOIS ESSDP was reachable, but no INCOIS numerical-model download was verified. These remain source/institutional integration options; the HYCOM case does not claim to be an INCOIS operational model or endorsed product. See `PHASE_02_DATA.md` for the audit, licences and reproducible acquisition.

D016 - Immutable data pack and lightweight Vercel requests

P02 scientific preparation runs offline with pinned xarray/NumPy/netCDF4/pydap/GSW dependencies. It preserves packed source subsets and original Argo records under ignored `data/raw/` and emits a versioned, hash-checked `casepacks/` artifact for the repository and existing Vercel deployment. Normal website use requires no external marine-data account or live connection to those archives.

The public API reads preprocessed arrays with the Python standard library and Pydantic. It bounds each response to 100,000 scalar values and caches at most three decompressed arrays. Analytical float64/native-coordinate arrays are separate from float32/decimated display arrays, with explicit transformation/error metadata. Original missing values remain missing. There is no persistent user database, worker process or arbitrary file-upload capability in this phase.

The historical dataset inspector provides actual model samples and profile tables as P02 acceptance evidence. 3D rendering, profile map markers, comparison methods, additional sensors and simulations stay assigned to their existing phases. No later phase is marked complete by this inspector.

D017 - Open-tab compatibility and service recovery

On 2026-09-21 the user reported "Service unavailable" after P02 deployment. Anonymous API probes returned 200 and a fresh browser loaded the real dataset. Replaying the unmodified P01 JavaScript bundle against the current API reproduced the error: its validators only accepted the empty P01 catalogue and health data state. Reloading loaded P02 and restored the dataset. The user's exact tab was not instrumented, so this is a reproduced cause consistent with the report, not proof that every reported failure had this cause.

Release 0.2.1 introduces a shared version in `api/release.json`, imported by the frontend and API. Every API response includes `X-Ocean-App-Version`. The browser checks that header before parsing and offers **Reload workspace** for a different release instead of misreporting a connection failure. Response-validation failures also have a distinct message and reload action; incompatible responses never invent scientific data. Bump the

shared release version for future deployments that change API/client behavior and keep package metadata aligned. Do not silently reload an active investigation.

Health and catalogue queries refresh every 30 seconds while active and on return to a stale tab. Temporary failures recover automatically; incompatible/update responses avoid immediate retry loops. The existing 8-second request bound and one transient retry remain. P01/P02 bundles already open cannot acquire the new recovery code without a reload; the user must refresh once. Scientific assets and P03 scope are unchanged.

Evidence: `evidence/service-incident-initial.json`,
`evidence/service-incident-fresh-browser.json`,
`evidence/service-incident-legacy-reproduction.json`, and the service-incident API/browser reports. Firefox and Docker verification remain open as previously recorded.

D019 - One scientific renderer with explicit irregular-depth sampling

On 22 September 2026 P03 selected Three.js 0.186.0 after rendering the real case in Chromium and WebKit. The initial stack had proposed vtk.js. Its image-data path uses origin and spacing, so that route would require resampling our unequal depth intervals onto a uniform image grid. This does not mean vtk.js cannot handle other dataset types. For this bounded case, a single Three.js engine with explicit physical-axis lookup keeps all 40 original depths and also supports the required slices, tetrahedral isosurfaces and vectors. No vtk.js dependency was installed. References: [vtk.js ImageData](#), [Three.js Data3DTexture](#), [Three.js ShaderMaterial](#).

The ray marcher, projection, transfer function, mask policy, source picking and graphics thresholds are recorded in PHASE_03_RENDERING.md. CPU numerical checks and independent NetCDF/browser picking checks are required. This choice does not authorize scientific analysis on the decimated display field. The analytical endpoint remains the source of numerical inspection.

Headless software rendering is slow on the tested machine. Auto therefore measures actual post-draw frame intervals, reduces render resolution and offers a data-backed Basic fallback. A CPU draw-call timer alone underestimated the GPU cost and was replaced. Additional cinematic lighting and unbounded global volume loads remain deferred to measured later work. The renderer is dynamically loaded; its roughly 576 kB minified chunk exceeds Vite's advisory, recorded with its compressed size and observed load time rather than hidden. Existing policy pages, favicon and real source/software credits are retained; the Terms description now reflects actual 3D functionality.

Remaining implementation choices

- Later regional datasets, versions, bounds, periods and additional supported input schemas; P02's first case is now pinned.
- Resource budget and service/storage arrangements for future scientific jobs and standards services; public application hosting is Vercel.
- Further matching methods and source-specific policies beyond the bounded P05 native-column method in D024.
- Fixed chart/rendering/runtime versions after compatibility checks.
- Team dashboard limits/additional fields and any team-specific national-round or finalist-event instructions.
- Public team/operator identity and contact email, intentionally deferred to P17 by the user.

D020 - Expose depth structure and explain display choices

On 22 September 2026, user feedback showed that v0.3.0's valid deeper water was too faint to distinguish from missing data. The P03 clarity revision uses scalar-independent volume opacity, brighter cold colours, a source-aligned cutaway and an explicit initial 0-1,000 m window. Full-column and whole-volume views remain available. Gray hatching identifies cells with missing corners and does not describe seafloor geometry. No scientific values or source masks are filled or altered.

The scene appears before its detailed controls. A colour key, native value and short view-specific instruction explain the display. Evidence opens on demand. Equal-length current arrows show direction while colour and native values show speed. Basic volume and isosurface modes show an explicitly labelled vertical section. Text-relative responsive breakpoints prevent a narrow side panel when text is enlarged.

This improves presentation and interaction based on the user's feedback and browser checks. It does not complete the unfamiliar-user research gates in R12 or X14. P04 and all later labs retain their existing scope and status.

D021 - Retain verified views and animate display clipping only

On 22 September 2026, repeated blank loading interrupted exploration. Version 0.3.2 retains the previous verified view during field changes, along with its original source date, variable, colour key and isosurface threshold. A pending or error notice names what remains displayed. Late responses cannot change a newer query selection. Exact native profiles are loaded as complete source columns, reducing repeated network requests when only depth changes.

The cube starts whole. A visible Open cutaway / Close cutaway button beside the camera controls replaces the less discoverable Volume style dropdown. A short clipping reveal preserves source coordinates and scalar values. The moving preview temporarily reduces ray work and restores full settings when it settles. No temporal/scalar interpolation is added. Reduced-motion preferences disable decorative and cutaway transitions.

Geographic context no longer unmounts the explorer. Hidden rendering and playback pause; the camera and controls remain available on return. The ten-minute inactive cache and limited prefetch apply only to this pinned historical case. Future mutable/live sources must define an appropriate freshness and dataset-version policy.

D022 - Preserve observation provenance and make eligibility explicit

P04 uses a separate version 2 instrument contract so the checked P02 model pack remains unchanged. Per-variable raw/adjusted choice, QC scheme and flags survive alongside per-sample coordinates, timestamps, pressure and GSW-derived depth. Core Argo R uses raw values; A/D uses adjustments. BGC uses each parameter's mode. Missing adjustments never substitute raw data. Raw BGC remains inspect-only. Argo, QARTOD, WOCE and declared CSV quality categories are distinct.

Genuine glider, CTD and BGC examples can come from separate historical collections. Their location/time remains explicit and they do not imply extra model coverage. The selected glider source has reversed salinity range metadata. Preserve finite readings for inspection, explain the defect and exclude that variable from the accepted curve. Missing ship-CTD oxygen remains missing. Charts break at excluded levels, and isolated eligible samples are visible dots rather than disappearing between gaps.

The 3D model displays historical profile locations with their own timestamps. Selecting one opens the instrument workspace; depth linking only navigates to the nearest model grid point without changing its timestamp. P05 will implement eligible physical/time/space matching before any residual or agreement claim.

D023 - Bound observation uploads and retain the Vercel deployment path

Supported CSV and NetCDF files are parsed in memory under explicit byte/profile/sample/decoded-response limits. Application code does not persist uploads or fetch supplied URLs. Profiles stay in browser memory until cleared or reloaded; filenames can occur in hosting logs. Policies and the import guide describe this boundary. Arbitrary NetCDF/Excel and durable upload storage remain unsupported.

The deployed runtime now includes netCDF4, NumPy and GSW. xarray remains an offline model-preparation dependency. `pyproject.toml` mirrors runtime pins and constrains uv resolution to Linux and Windows AMD64. The initial universal dependency resolution failed on Windows ARM64; a subsequent attempt put the project-only environment setting in `uv.toml` and was rejected. The final setting belongs in `pyproject.toml`, preserving the tested NumPy version. These server targets do not limit browser users. Full license texts are distributed and exposed under `/licenses/`.

D024 - Match original samples against an explicit native model column

P05 compares one selected actual model snapshot with each original observation sample. It requires compatible definitions, source QC, coordinates inside the native model subset, a bounded great-circle separation and finite adjacent depth levels. Temperature remains in-situ Celsius and salinity remains practical salinity. Only physical-depth interpolation is used; model values are never taken from the display grid, extrapolated beyond the column or filled through masked gaps. Residual sign is model minus observed. The 6-hour, 5-km and 500-m bracket defaults are explicit project settings.

Coverage derives from the same accepted rows as the charts and metrics. It shows actual sample depths and accepted time/distance ranges, without a confidence field. Empty profiles keep source time offsets as diagnostic context. Discovery uses explicit nearest eligible or largest-absolute-residual ranking; a different suggested model timestamp requires the user's in-app selection. Snapshot suggestions rank actual eligible sample offsets and distances, including moving profiles.

The comparison and ocean renderer share timestamp state. Matched-column navigation explains the difference between an interpolated comparison value and the nearest native-level probe. A temporary reference allows controlled changes to matching/QC settings and restoration in the open workspace. Durable replay/export stays in P08. Original-file numerical fixtures, deliberate mismatch tests, contract checks and browser interaction checks are required for acceptance. See `PHASE_05_EVIDENCE.md`.

D025 - Hold a source for unresolved calibration provenance

The 5907083 original Argo file contains a calibration comment mentioning a verification run. A fresh GDAC download has the same SHA-256 as the cached source. The meaning of this wording is unresolved. Keep the original file, values and provider QC unchanged and inspectable, but exclude that source fingerprint from quantitative comparison pending clarification. This project hold is based on provenance, not residual magnitude, and is not a claim of fabrication or provider rejection. Preserve calibration text, equations and coefficients with their spaces and separate entries in the observation metadata.

The current ship CTD temperature lacks confirmed scale metadata and is also unavailable for quantitative temperature comparison. Practical salinity uses its declared PSS-78 definition. Other temporal/geographic/QC gates still apply. Imports remain an inspection capability in P05; this release does not silently grant quantitative eligibility to arbitrary uploaded files.

D026 - Separate persistent standards service from the Vercel viewer

P06 uses actual THREDDS 5.9 on a pinned Java 17/Tomcat 10.1 runtime to serve the supported historical grid through WMS, WCS and DAP2. The Windows launcher and a fresh service base were executed and independently checked. The optional Docker stack is unexecuted because Docker is absent. Keep the public app and REST API on Vercel and serve the checked native NetCDF as a CDN asset. Do not present localhost, a temporary tunnel or the Vercel Python function as a publicly deployed persistent THREDDS server.

The exchange file preserves source-coordinate values and float64 decoded fields. Explicit CF metadata normalization keeps the original salinity units/name attached and represents the documented practical-salinity interpretation as dimensionless. THREDDS WMS/WCS regularizes horizontal coordinates slightly; the tested tolerance is 0.0001 degree with measured WCS longitude change 0.0000826928 degree. This service behavior is declared, and exact-coordinate consumers should use the native file or DAP. It does not justify altering source coordinates or analytical values.

D027 - Demonstrate extensions through declared adapters and native-grid products

The trusted adapter registry contains HYCOM and CF exchange entry points with a shared ModelGrid output. The product registry contains horizontal kinetic energy, calculated on native float64 components before display selection and rounding. The existing renderer handles this scalar without geometry changes.

Additional products still require public allowlist/schema and presentation metadata updates. R10 remains partly open for P15 integrations.

D028 - Query native cells and declare inferred geometry

P07 uses inclusive numerical thresholds, native depth-centre selection and six-face adjacency without interpolation or diagonal joining. Source coordinates do not include cell bounds. Midpoint bins clipped to the sampled domain therefore define an explicitly estimated spherical volume. The result retains centre and cell extents, arithmetic and weighted means, a complete membership fingerprint, source identity and limitations. This implements the roadmap's physical-geometry intent without claiming missing provider geometry exists. Search results are neither automatically identified eddies nor tracked water parcels.

Selected meshes merge only coplanar exposed cell faces. The complete calculation uses the native grid, separately from rendering. Drawn sections use nearest native columns along declared longitude/latitude stations and expose sampling offsets. Eligible observation context reuses P05 rules and source holds. Midpoint depth ties use the deeper bin, with the last boundary closed. The section's 5-km observation rule measures distance to sampled stations and is not described as an exact continuous corridor.

D029 - Audit current components before making broader claims

The user requested a full visual and scientific audit after P07. Keep original failing evidence and focused rechecks, and fix demonstrated defects. The audit identified a direction error in the basic current map caused by ignoring unequal plot-axis scale and longitude convergence. The fix projects physical east/north velocity before normalizing arrow length; source speed and values remain unchanged. Independent P07 geometry checks also identified subtraction after radian conversion as a source of unequal volumes for equal-width cells. Longitude widths are now subtracted before conversion, restoring deterministic equal-volume ordering.

Passing source, numerical and browser checks establishes the documented tested scope. It does not certify the model against the real ocean, prove unfamiliar-user comprehension, guarantee every evaluator device, or complete later product phases. Keep deployment limitations and remaining official requirement gates visible.

D030 - Keep portable investigations with the user and replay statelessly

P08 uses explicit IndexedDB saves, portable JSON files and bounded self-contained replay links. No persistent server store is configured or purchased, and the Vercel function filesystem is not a database. The interface distinguishes a browser-local save from a cloud backup and explains storage eviction, deletion and sharing. Source imports remain temporary and are excluded from saved investigations. Privacy and Terms pages describe the actual new storage and replay requests.

Capture retains one applied analysis with source/method identities, settings and complete numerical result modules. Reopening recalculates in the current service and verifies fingerprints before changing the workspace. Missing sources, method changes and result mismatches stop restoration. Numeric hashing uses explicit binary64 tags to survive Python/JavaScript JSON round trips. Patch versions are recorded without invalidating unchanged scientific outputs; scientific method changes must be versioned. Hashes are consistency checks, not signatures or validation against the real ocean.

Report, settings and numerical exports retain source attribution, historical dates, QC/exclusions and method limits. HTML is printable to PDF; CSV keeps full numerical precision with explicit missingness. P08 defines a common module/parameters/seed/output envelope for subsequent experiments without enabling unsupported labs or arbitrary code execution. See PHASE_08_INVESTIGATIONS.md for the declared supported scope and evidence.

D031 - Extend cases without invalidating original investigations

P09 adds an explicit second case recipe and case-scoped observation identities. The original Bay model manifest and observation index remain unchanged, so prior investigations still replay. Arabian Sea adds a separate six-profile library from five original files. Scientific parsing and matching rules remain shared. Native exchange and public REST support both cases; the verified THREDDS scope remains Bay only.

D032 - Disclose selection in the guided contrast

The two-column teaching example uses a frozen, reproducible search over real source values. Its surface-tolerance and maximum-100-m-contrast rule are shown to the user and checked independently from original packed NetCDF. It illustrates why surface similarity can conceal a depth difference. It is not advertised as a basin average, independent model validation or causal explanation. The guide's separate Argo comparison explicitly identifies its different location.

D033 - Separate automated checks from real user understanding

P09 implements a public/student guide with navigation, explanations and recovery controls. Browser and numerical checks can verify supported interactions and source consistency. They cannot supply real first-time-user task times or comprehension findings. P09.5 and X14 remain open until actual pilot evidence exists. A deployment can ship the tested technical work while this gate remains pending.

D034 - Freeze sampling designs before evaluating later fields

P10 uses one scalar variable at a native depth from 0 to 1000 m. The first historical snapshot supplies the candidate wet mask, local gradients and design coordinates. Every sixth native row/column plus final indices defines the finite candidate lattice. All other finite prior locations form a disjoint evaluation set, shared across every strategy. Later snapshots at indices 2, 4 and 6 are loaded only after the selected, space-filling and five seeded random designs are frozen. The same target-finite evaluation subset and fixed inverse-distance-squared residual reconstruction apply to all strategies. A zero-new-measurement persistence baseline remains visible.

Budgets count scalar measurements, not profiles or vessel costs. Incomplete greedy designs and missing target station values remain failed comparisons. No future-driven replacement, favourable-seed selection or suppressed negative score is allowed. The initial default gradient design loses to the space-filling baseline in both regions and to persistence in the Arabian case; these results remain part of the experiment. Random ranges are descriptive, not confidence intervals. Repeated tuning is exploration, not external validation. No forecast gains or operational OSSE claim follows from this model reconstruction.

D035 - Keep virtual sampling reproducible and separate from navigation

The virtual survey samples a straight longitude/latitude segment at declared positions and one exact native depth/time. Great-circle nearest-column selection reports requested coordinates, source coordinates and offset; ties within 1e-10 km select the first latitude-major source index. Source masks stay null. No interpolation or route-safety model is implied. The sampling method has its own version and is added to saved source identities only for expedition recipes, preserving prior investigations. Plans, simulated surveys, benchmark trials, seeds, failures and source references use the existing stateless replay and local-save system.

D036 - Fix the demonstrated arithmetic difference without weakening replay

The initial 0.10.0 deployment passed local and public browser workflows, but eight Windows-generated P10 records failed exact replay on Vercel. Recursive comparison isolated the differences to the reported gradient magnitude, at most $6.94e-18$ in the captured records. Native values, station indices, distances and reconstruction errors were identical. Strict fingerprints correctly stopped replay.

Method `p10-sampling-v2` replaces platform-dependent `np.hypot` with separate binary64 multiply, add and square-root operations for the bounded gradient components. No output rounding, fuzzy fingerprint comparison, source-value changes or ranking-policy change is introduced. Release 0.10.1 uses this method and passed exact replay in both directions between Windows and Vercel, with eight records per direction. This is verified platform compatibility, not a proof for every possible future processor or dataset.

Initial `p10-sampling-v1` records remain readable and downloadable, but replay reports an explicit method mismatch. They are not silently migrated or labelled altered. Existing P08/P09 recipe identities remain unchanged. Retain the first failing verification report and both sets of diagnostic records as evidence of the correction.

D037 - Adapt the pinned particle kernel to a bounded request-scoped runner

P11 adapts the MIT-licensed OceanParcels 3.1.4 AdvectionRK4 kernel, with source hash and attribution, to the existing NumPy server runtime. The complete framework is installed separately for independent trajectory comparisons. Production is described as an adapted kernel with our own data adapter and bounded runner, not the full framework. This avoids adding its offline xarray/SciPy/Dask/Zarr dependencies to the Vercel function. It also removes the need for an undeployed Redis/worker service for these small historical cases.

Jobs live for one request. Real step-count progress is streamed; cancellation stops after the current bounded chunk, and incomplete results never enter the completed-result cache. Warm caches have explicit entry limits. Durable queued work is not provided or implied. Independent source, analytical, full-framework, boundary and convergence checks govern the custom adapter rather than treating attribution alone as validation.

D038 - Stop at missing support and keep simulation accounting explicit

Drift uses native east/north m/s at one fixed native depth, bilinear spatial and linear temporal interpolation, and the Parcels 111120-metre-per-degree sphere convention. A tested fixed-order cosine calculation preserves checked cross-platform arithmetic without rounding source values. Every RK4 stage, endpoint and the rectangle of traversed interpolation cells must be supported. No coast bounce, nearest-wet relocation or hidden final-field extrapolation occurs. A failed step keeps the last accepted position and time; this is not an observed beaching event.

Box release sampling is seeded uniform longitude/latitude, with no claim of exact area uniformity. Point releases coincide under the deterministic forcing. Target entry uses first accepted-step segment intersection, timed linearly within the step, and arrival fractions include every requested release. Mean path distance includes valid releases only and uses a declared midpoint-metric approximation. All starting times, durations, masks, failures and seeds remain visible in comparisons and saved results.

The full-resolution numerical field remains separate from the subsampled current preview. A preview cell can be coloured even when a native interpolation stencil inside it is missing; the interface states this. It also identifies which run supplies the static background. No matching NOAA GDP tracks were retrieved within either checked case/date rectangle. This leaves independent drifter validation open and does not negate the separate numerical/reference checks.

D039 - Pin summary arithmetic across Python versions

The initial public P11 check found eight failed replay/export checks across four of eight Windows records. Recursive comparison showed identical trajectories, individual path distances, source values, times, statuses and arrivals. Only mean path distance differed, by at most $1.78\text{e-}15$ km. Windows Python 3.10 accumulated the values sequentially while Vercel Python 3.12 used the changed compensated `sum()` behavior. The [Python 3.12 release notes](#) document this change; retained incident values reproduce the difference.

Method `p11-drift-v2` computes the summary with explicit left-to-right binary64 additions in particle order. No source value, path, timestep, fingerprint tolerance or output precision is changed. Release 0.11.1 records the method revision. Initial v1 archives remain readable/downloadable, but replay reports an explicit method mismatch. They are not silently migrated or labelled altered. Older P08/P09/P10 methods remain unchanged. Keep the original 90-pass/8-failure public report and diagnostic comparisons.

Final acceptance passed independently: eight Windows-created v2 records replay exactly on Vercel Python 3.12.14, and eight Vercel-created records replay exactly on Windows Python 3.10.11. Public verification passes 100 checks, including all original P08/P10 archive replays, and final browser checks pass 54 cases. Evidence: `evidence/p11-v2-public-http.json`, `evidence/p11-v2-public-to-local-replay.json` and `evidence/p11-v2-public-browser.json`. This establishes checked-platform consistency, not a guarantee for every future environment.

D040 - Declare the surface baseline and keep fixed locations independent of events

P12 uses NOAA OISST daily analysed SST at fixed west-quarter, centre and east-quarter locations in each existing case rectangle, snapped to native OISST coordinates before detecting events. These six points are not a regional average or heatwave-area map. The fixed 1982-2011 baseline, 11-day sample pools, linear 90th percentile, interpolated leap day and 31-day circular smoothing are explicit. Baseline windows cannot use values outside the baseline years. A complete daily baseline is required; missing analysis days are not filled or bridged. Qualifying five-day events may merge across at most two observed intervening days. Events are calculated over 2022-2025 context before selecting the 2023/2024 display year.

The detector is independent application code, tested against the separately acquired pinned Oliver reference on complete inputs. The upstream reference's different missing-data policy is not silently inherited. The original GPL-licensed comparator stays in ignored offline reference files and is not shipped in the runtime. Ordered arithmetic is used where aggregation would otherwise depend on Python version. Initial synthetic/reference checks passed; final real-source and deployed acceptance must be recorded separately.

D041 - Keep thermodynamic definitions and missing support explicit

P12 interprets the supplied HYCOM temperature as in-situ and salinity as practical PSS-78, matching original metadata. GSW converts depth to pressure, SP to SA and in-situ temperature to CT; potential temperature is separately referenced to zero pressure. Density and N-squared use the documented GSW validity funnel. Negative N-squared is preserved. Adjacent-layer gradients and stratification never bridge a missing source level.

Mixed-layer depth is a derived threshold estimate, not an unavailable provider MLD variable. Both criteria use the 10 m reference: sigma0 increase 0.03 kg/m³ or absolute potential-temperature change 0.2 degrees Celsius. The first supported crossing is interpolated in its native bracket; no crossing is a bound within the selected profile

window. The column integral is explicitly potential-enthalpy content relative to $CT=0$, integrating in-situ density times cp_0 times CT over fully supported native depths. It is neither a subsurface anomaly nor heatwave excess. A separate long-term subsurface baseline would be required for those claims.

D042 - Link events to actual dates without fabricating depth coverage

Heat & Depth resolves automatic selection to the first event intersecting the supplied model dates, or otherwise the first event in the selected year. This choice is disclosed and does not alter the fixed point catalogue. Original context-detected start/end dates remain intact when an event crosses the display year. The selected model timestamp is never silently replaced; only a model day inside the event enables event-linked diagnostics. Other dates return an explicit unavailable or choose-an-overlapping-snapshot state. Nearby observation links retain source QC, coordinates, date offsets and the existing provenance hold; they do not imply collocation or independent validation.

Heat investigations add their source manifest identity only to the new recipe family. Existing source identities and method versions stay intact. The saved heat recipe must contain the applied resolved event selection, not an unsubmitted automatic draft. Exports retain the baseline curves, daily values, full event boundaries, diagnostic definitions and source references.

D043 - Merge documented distributions of the same SST product without filling gaps

The initial PSL-only acquisition encountered slow subsets, interrupted responses and temporary server errors. NOAA AOML supplied most historical years much faster, but its actual coordinate axes omitted 1994-1996 and isolated later dates. Dataset titles alone did not establish coverage. Preparation therefore prefers the recorded AOML values and uses recorded PSL data only where AOML has no day. Every required calendar day and fixed coordinate must be present before publishing the source manifest. This is a merge of actual NOAA OISST distributions, not interpolation or a substitute product. Original source files, retrieval URLs, checksums, numeric time values and per-file units remain traceable for every day.

The original packed NCEI files explain a small representation difference in the checked legacy sample. AOML's older distribution exactly matches Float32 conversion after dividing the packed integer by 100. PSL matches the original Float32 scale-factor multiplication. On the checked 1982 subset, those operations can differ by one Float32 step, $1.9073486328125e-6$ degrees Celsius. The checked modern AOML and PSL values

both match NCEI Float32 decoding exactly. See `evidence/p12-source-packing-diagnostic.json` for original integers, attributes, hashes, exact operations and 232 valid-cell comparisons. This establishes the checked decoding behavior, not an assumption that every future distribution has identical packing.

The prepared analytical series preserves each selected distributor's exact served value. It does not round values to force cross-distributor identity. Reference event calculations use that same declared series, and saved investigations still require exact fingerprints. Independent link-day tests require exact primary-source equality; separate legacy packing tests verify the demonstrated alternate operation. Numerical event comparison tolerances and replay integrity policies are unchanged.

D044 - Pin the actual 2026 ENSO definition and keep IOD assessments separate

P13 uses the official RONI v6 snapshot retrieved on 23 September 2026 as its primary historical ENSO index. NOAA's 2026 adoption and the ERSSTv6 source are documented in `P13_INDEX_SOURCES.md`. Historical ONI v5 and ONI v6 remain separately named reference values, without mixing classifications, releases or baselines. The selected SON 2013, 2015 and 2022 seasons are respectively neutral, El Nino and La Nina under all three checked definitions. Warm/cold classifications must satisfy the declared five-consecutive-season rule and match provider labels.

CPC ERSSTv6 DMI numerical values and BOM historical IOD episode assessments remain different evidence. In particular, SON 2015 DMI is +0.39 degrees Celsius, while BOM documents a positive episode with different temporal support. No arbitrary rounding or revised threshold is used to make those statements identical. The specific downloaded DMI series identifies 1991-2020, while the broader HTML header uses different baseline wording; the discrepancy is recorded and the exact series definition is retained. Monthly field controls show explicitly named centered three-month index context.

D045 - Add genuine monthly potential-temperature cases without changing quantity semantics

P13 prepares GODAS source data for SON 2013, 2015 and 2022 into the existing CaseManifest and native/display array contracts. Only potential temperature is supplied. The source's original kelvin values are converted to Celsius in float64; its original variable description establishes the explicit potential-temperature mapping. It is not relabelled as HYCOM in-situ temperature. Native monthly timestamps and full support intervals remain visible. Source longitudes stay monotonic on a declared 0-360 degree axis; east/west display labels do not reorder the data.

The shared explorer uses supported-variable controls and source-specific defaults. Original Argo profiles remain inspectable with their own dates and parameter QC, but monthly potential temperature is not directly compared to instantaneous in-situ temperature under the existing evidence method. Missing salinity/current fields are not fabricated. The existing Drift, Heat and Expedition methods retain their bounded Indian Ocean support and explain unsupported Pacific inputs. These new climate cases do not silently imply that every earlier scientific module supports a different forcing product.

D046 - Use matched calendar windows and complete-source baselines

Each GODAS monthly baseline uses all 30 actual 1991-2020 fields at the same native cell. An incomplete cell stays missing. SON is the equal-weight mean of complete September, October and November monthly means, with explicit ordered arithmetic. The event and baseline always use the same calendar window, geometry, units and potential-temperature definition. The nearest native latitude to the equator is sampled without interpolation. Between-event differences are selected minus reference and do not establish causal attribution.

The combined climate manifest fingerprints the independently prepared field and official-index snapshots. Climate investigations retain this identity, both event selections, original source references, baseline values, profiles and numerical differences. Older recipe families omit the new optional identity to preserve their existing fingerprints. Reproduction still requires exact results on the checked platforms; independent scientific comparison tolerances never relax replay integrity. Missing cells, unavailable variables and original observation QC remain visible rather than being adjusted to strengthen the demonstration.

The final P13 combined manifest also pins all three shared Pacific case manifests and their original observation-library indexes. This closes source identity for both compared years and each supported event. Earlier pre-release P13 records retain their original identity and evidence; public acceptance uses new records from the final closure. The P08-P12 source identities and replay outputs remain unchanged.

D047 - Correspondence is a native overlap graph, not water identity

P14 connects full native threshold memberships across adjacent 12-hour frames of the two supported Indian cases. A link passes when shared estimated volume divided by the smaller region volume meets the selected minimum; both directional fractions remain visible. Connected bipartite groups retain one-to-one, one-to-many, many-to-one and many-to-many structure. Frame-local IDs never become persistent parcel identities. Unmatched membership with missing counterpart support is unresolved. Skipped frames are explicit gaps and have no interpolated links. Nearby threshold runs repeat the calculation; their spread is not probability. Resource bounds reject excessive complexity rather than silently truncating to the P07 first-50 presentation list.

D048 - Exclude evidence without rewriting observations or rerunning the ocean

P14 blackout resolves the normalized union of profile, instrument, platform and collection selections within the source library. Original QC, quantity, temporal, spatial and vertical matching gates stay unchanged. The baseline is retained, and eligible pairs and sample-weighted residual summaries are recomputed for remaining whole profiles. User exclusions are separately counted from source or matching rejection. Exact source sample indices identify the eligible comparisons lost. Excluding already ineligible observations can correctly have no effect on eligible evidence. Saved original and modified investigations retain selection identities, source hashes and versioned methods. No assimilation rerun, forecast benefit or calibrated confidence is inferred.

Original sample indices can be sparse identifiers from a larger source file; they are not required to fall below a grouped profile's row count. The client validates integer, nonnegative, unique identities and counts without imposing that false positional rule. Blackout exports also retain source-file records listing several matching timestamps, such as a GODAS seasonal subset, alongside the original checksums. This extension is scoped to blackout so earlier investigation reference selection and replay identities remain unchanged.

D049 - Version reproducible evolution aggregates and bound repeated work

The first public P14 check found last-bit differences between Windows and Vercel in copied P07 per-region volumes and weighted means. Complete response comparison found six different leaves for default Bay temperature and four for Bay salinity.

Memberships, geometry-dependent frame totals, overlap links, fractions and sensitivity summaries agreed exactly in those checked responses. P14 now calculates its own region volume, mean and weighted mean using explicitly ordered `math.fsum`, under `p14-native-overlap-v2`. Original fields, memberships, geometry, P07 calculations and independent scientific tolerances remain unchanged. Exact cross-runtime replay is still required; rounding results or weakening fingerprints is not an accepted fix.

The original v1 archives remain readable and their original JSON can be downloaded with its original hash. Recalculation requires the matching scientific method, so a v1 replay against v2 reports an explicit method mismatch instead of silently replacing its result. Release 0.14.1 carries this correction and passed final public acceptance on 24 September 2026, including exact numerical replay in both directions between Windows and Vercel. Initial failures remain recorded separately. See [Phase 14 acceptance](#).

Concurrent hosted save/replay verification also exposed repeated expensive calculation and a 30-second function timeout. An eight-result per-instance cache reuses identical completed P14 calculations, with duplicate concurrent requests sharing one calculation. Source integrity is checked before every lookup, returned results are copied, exceptions remain failures and cached results are bounded. This is a correction to the supported P14 request path; it does not complete P15 scaling, offline or broader resource benchmarks.

D050 - Wider source access preserves native definitions

D051 - Portable regional evidence has explicit resource bounds

Progressive display and bounded caches are separate from native export precision. Four operations per server instance, bounded arrays, request/pack sizes and clear capacity errors prevent unbounded work; these are not distributed account quotas. Local pack checksums establish integrity, while online exact replay compares the current pinned sources. An exported standalone HTML viewer supplies a real offline path without claiming that the entire app is an offline application. Details and evidence are recorded in `PHASE_15_WIDER_COVERAGE.md`.

D052 - Organize the existing frontend before P16

On 26 September 2026 the user prioritized a frontend rebuild because the initial screen exposed too many controls and long explanations. The app now uses one task-grouped Tools menu, a compact case/variable context, a scene-first explorer and optional source/settings panels. The short visual introduction is manually paged with Next, Back and Skip. It opens on normal page loads without storing a new preference; replay links open their existing replay panel directly.

New explorations start with instrument markers off so the first click can inspect model water. The explicit overlay control and instrument workspace remain available, and replay retains the original saved overlay flag. Units, dates, missing data and source identity are preserved. Compatible case changes keep the current laboratory; incompatible Indian Ocean tools offer a deliberate compatible-case choice. Background climate results update shared case context only while Climate is active. See `FRONTEND_REBUILD.md` for scope and `PROJECT_STATE.md` for verified evidence.

D053 - Teach by operating the real workspace, core functions first

The project brief statement was rechecked on the official 2026 page on 26 September 2026. Model fields throughout the water column, co-displayed instrument profiles, ingestion and configurable source-aware views are the core requirements. Outreach is explicit; the statement does not publish numerical feature weights. We therefore demonstrate model depth and sensor comparison before extra experiments. This ordering is a product interpretation, not an official judging rubric.

The default welcome selects Basic. In-depth is an explicit choice explaining that it adds concepts, methods and limits. Both use the same real actions and result snapshots. Next advances one action; there is no forced continuous tour. Back, replay, practice, Skip and a lesson chooser retain visitor control. The core has 17 steps, followed by 10 optional lab steps. Source examples and simulations retain their existing parsing, scientific gates and dates. A save lesson opens the real form without automatically saving or sharing.

D054 - Ground explanations in structured output with a bounded path

P16 explanations read current parsed values, units, parameters, provenance kind, references and limits. Reviewed terminology is distinct from calculated evidence. A strict request grammar exposes a query for review before application; requested depths must exist in the selected source coordinates. Unsupported questions, forged values or attribution, and arbitrary external retrieval are refused. Loading, failed and missing output cannot become a completed explanation.

The roadmap explicitly permits the checked learning and explanation path. This does not claim open-ended conversation, additional-language validation, operational advice or demonstrated beginner comprehension. The real P09.5 participant gate remains separate.

D055 - Anchor each tutorial step to its actual control or result

On 26 September 2026 the user rejected the fixed tutorial column because it left visitors searching for the relevant feature. The P16 follow-up places a compact callout beside a measured, visible target, with a pointer and highlight. It brings that destination into view once per step, follows scroll and resize, and offers an explicit return action after manual scrolling. The guide does not continually pull the user back. Hidden retained workspaces are excluded from target selection.

Basic and In-depth retain their real actions, evidence snapshots, Next/Back, practice, replay and Skip. Short screens adapt the highlighted visual and allow explanatory content to scroll while navigation stays available. The real save dialog contains its own adjacent guidance and an explicit close-and-continue action. The tutorial still does not save or share automatically. No scientific algorithm, source array or method gate changes. Automated placement checks do not replace the open unfamiliar-user pilot.

D056 - Make additional lab tutorials discoverable and directly selectable

On 26 September 2026 the user accepted the moving guide but requested teasers linking to additional features. A two-card, four-page picker presents all eight supported extra topics with a short question, a provenance label and the actual guided-step count. It is reachable from the welcome, active tutorial and core completion. Each topic launches only its own existing real demonstrations and ends with an explicit choice to try another feature or explore independently. A separate action retains the complete extra-lab sequence.

Basic remains the default; the picker also exposes the optional In-depth style with its concepts and methods clearly labelled. Drift always calculates before playback, and Climate includes both supported El Nino and La Nina steps. Browsing teasers does not issue lab commands. These are discovery and navigation changes over the existing ten lab demonstrations, not eight new scientific implementations or a claim of additional data coverage. No new dependency, provider, stored progress, analytics or personal-data handling is added.

The lesson runner waits for the case catalogue before issuing the first action. A visitor can therefore choose a teaser while metadata is loading and still reach the correct workspace when it arrives. Skip removes that pending tutorial. Completion cards may use the full available viewport height because they no longer point at a target; this keeps the next-feature and independent-exploration choices visible on small phones.

D057 - Publish the confirmed team identity and keep the security audit in P17

On 26 September 2026 the user supplied Team Seifuku's six names, team ID 150123, leader Adwita Kurle, website operator Gaurav Ranade, public email, Pune location and university affiliation. TEAM.md preserves the exact supplied names and roles. About, Privacy and Terms use these details. No additional member roles, mentor details, social links, endorsements or institutional operation are inferred.

About introduces the working app and its historical coverage. The standalone pages remain readable without JavaScript and share workspace, data-access and attribution navigation. Privacy describes actual IndexedDB investigation storage separately from the localStorage regional pack, temporary uploads, replay requests, hosting logs and email. Provider retention is not assigned an invented fixed period. The pages describe the prototype's practical terms and data handling; they are not a claim of legal certification.

The early firewall overview and npm audit were read-only. This follow-up changes no secrets, rate limits, firewall rules, authentication or scientific methods. P17.3 stays unchecked and covers the later security review, including rate limiting and API-key handling. The team identity task is completed early without silently starting the whole final-release phase.

D058 - Distinguish the team leader from the website creator and operator

On 26 September 2026 Gaurav clarified that he created the website as well as operating it. Credit Gaurav Ranade as Website creator & operator on About and in the Privacy/Terms identity details. Adwita Kurle remains Team leader. TEAM.md, the README and prepared text use the same distinction. This is the user's confirmed role, not an inferred role for other members. It changes no data handling, scientific behavior or P17 scope.

The same follow-up clarified that the university did not participate in development and that the project has no university affiliation. Remove the university from public project identity and current reusable text. This clarification supersedes D057's affiliation interpretation. Historical records retain their dated context; source-provider and software-licence university attributions remain intact. SIH submission requirements for institutional nomination/authorization remain separate, unverified requirements rather than proof of project affiliation.

D059 - Recover stale entry documents without discarding active work

The reported blank page requested `/assets/workspace-fwVeVJXC.js`, the entry recorded for release 0.14.1. The current 0.16.3 document instead referenced `workspace-DEfSpNRW.js`; the retired file returned 404 in both the user's screenshot and anonymous HTTP checks. Vercel logs confirmed that missing route. The exact reason the user's browser retained or received the old document cannot be established from the screenshot alone. Current HTML already had `no-cache`; a query-string change alone is not treated as the underlying fix.

Release 0.16.4 serves an explicitly scoped recovery bridge for that retired entry path. It reloads the entry document once with a fresh request key, preserving other query context and the fragment, then stops with a usable link if the same stale page recurs. A successful mount removes the temporary recovery key. This is not a substitute bundle or a wildcard rewrite for arbitrary missing modules.

The entry document now uses `no-store` and contains inline startup content and failure handling, independent of the main bundle. A missing entry or 15-second startup delay produces a visible retry choice. A Vite lazy-import failure displays a reload notice warning about unsaved work; it never forces an active tab to reload. The regional lazy tool also has a local error boundary so other tools remain usable. No cookies, `localStorage`, `sessionStorage`, analytics or new service are used for recovery. P17 must account for the inline guard when reviewing a future Content Security Policy.

Relevant platform guidance: [Vite load-error handling](#) and [Vercel cache-control headers](#). The implementation is checked by failed-download, stale-document, bounded-recovery and preserved-workspace browser scenarios. This targeted availability fix does not begin the deferred P17 security audit.

D060 - Give all 31 follow-ups an explicit proposed phase

On 26 September 2026 the user objected that the proposed P16A/P16B sequence appeared to cover only two features. The checkpoints each contained several tasks, but imported comparisons, persistence, larger imports, refresh and offline expansion had also been left as unscheduled choices. Replace that incomplete delivery plan with P16A-P16E followed by the existing P17. The numbered coverage table in `CORE_COMPLETENESS_REVIEW.md` assigns every item 1-31 a phase owner; `ROADMAP.md` supplies unchecked steps and acceptance conditions.

P16A covers observation integration and the complete supported import flow. P16B proves fresh model input and architecture. P16C extends comparisons, persistence and replay to imported investigations. P16D connects scientific views and guided investigation. P16E handles larger workloads, refresh and offline continuity. P17 retains final science, security, usability, performance and release evidence. Preserve existing implementation and original phase history; an already working capability needs extension or verification rather than an unnecessary rebuild.

This is a proposed full-scope plan, not a blanket start instruction or a claim that all items are mandatory PS requirements. Continue phase by phase only when requested. Real participants, unavailable source data, public standards hosting and institutional infrastructure remain external dependencies. No completion percentage, delivery-time guarantee or new feature evidence follows from planning.

D062 - Preview imports, preserve source meaning and retain the frozen replay library

On 26 September 2026 the user authorized P16A with "lets start". Implement the shared imported-marker and co-visible model/profile workflow, explicit preview/QC and bounded CSV mapping. Reuse native source adapters, retain separate observation/model dates and require Add after manual review. The tutorial may auto-add its own bundled public example. Files remain request-only at the server and temporary in the tab. Imported numerical comparison and persistent lifecycle remain P16C.

Choose only conversions with declared, compatible physical definitions and no missing ancillary inputs. Keep original file identity separate from normalized CSV and mapping

hashes. Marker placement uses an actual coordinate-eligible sample inside the current model rectangle; a track-only summary never assigns a first-sample timestamp to a later track point.

Use genuine Argo GDAC BGC 1902594 cycle 34 for chlorophyll. It is geographically inside the Bay case but dated March rather than January 2024. Display that mismatch and the absent chlorophyll model field. Preserve the distinction between CHLA_ADJUSTED and CHLA_FLUORESCENCE. Source acquisition is pinned and rejects changed upstream bytes.

The first library insertion broke saved-investigation source fingerprints, as caught by regression. Restore the original immutable index and publish the new file through a separate import-example catalogue. This supplies the real data demonstration while retaining the exact source identities required by existing replay. A broader source-library migration is unnecessary for this checkpoint. PHASE_16A_IMPORTS.md records scope, evidence and remaining matched glider/CTD data dependencies. No later phase is started by this decision.

The first P16A Vercel build exceeded the optimized function-size limit. Exclude non-runtime benchmark/test files and offline verifier commands from the function rather than removing data or runtime features. These sources remain in the project. Treat deployment-size budgeting as a concrete prerequisite for P16B source expansion. Public runtime/asset checks remain required after the packaging change.

D063 - Verify the Vercel packaging path before selecting additional infrastructure

27 September 2026: the user requested an explanation and recommendation about hosting limits. The observed 225 MB failure is an internal check in the deployment path; the local Python builder contains the same threshold and error. Current official docs describe 500 MB standard Python packages and eligible 5 GB Large Functions beta. This does not automatically establish project eligibility or available headroom. HOSTING_CAPACITY_PLAN.md records the evidence and the distinction.

Recommend a capacity/configuration preview as the first P16B prerequisite, followed by versioned external data storage/subsets where growth warrants it. Retain the public Vercel frontend. Do not assume buying Pro resolves packaging, or prematurely migrate the backend. Payload, duration, memory, storage and cost need separate measurements. Larger-upload/job architecture belongs to P16E; imported persistence must retain P16C's explicit storage/policy contract. No infrastructure purchase, switch or new phase is authorized merely by this advice request.

D064 - Prove a fresh compatible model with explicit limits and a portable baseline

On 27 September 2026 the user authorized P16B with "lets ace it . Start". Reuse HYCOM's registered scientific decoder with newly acquired 28-30 March source snapshots. Make dates and observation requirements recipe-driven; retain existing source and replay bytes. This is a new historical input from the same model family, not a novel provider or arbitrary-model upload. The March chlorophyll observation overlaps this model in area/time, but no model chlorophyll or imported numerical comparison is inferred. Surface heatwave data remain case-specific.

Extend the local THREDDS launcher to checksum-verified exchanges and a configurable loopback port. Both source-to-exchange and actual protocol responses are checked. The IOOS independent checker could not install because Windows native UDUNITS support is missing; retain that finding rather than claiming universal CF compliance.

An actual Vercel preview exceeded the standard packaging path and successfully enabled Large Functions beta. Its scientific subset and native NetCDF import passed. Set `VERCEL_SUPPORT_LARGE_FUNCTIONS=1` for preview and production in the existing project, leaving plan, provider, upload bounds and function duration unchanged. Public production checks are still required. A larger package allowance is not evidence of concurrency capacity.

Establish a local versioned source baseline for a clean archive-based runtime installation. Preserve scientific case-pack bytes with `.gitattributes -text`, including JSON; ordinary Git text normalization would invalidate existing integrity/replay identities. Do not publish a remote repository or claim actual institutional/container deployment from this local proof.

D065 - Reparse embedded originals and isolate imported investigations

27 September 2026: the user authorized P16C with "lessgo start". Use a bounded original-file envelope with explicit mapping/parser identity, rather than trusting normalized profiles supplied by the browser or inserting private uploads into the frozen public library. Reuse existing scientific eligibility and native-grid calculations through request-scoped stores that bypass global profile/result caches. Preserve old source and recipe fingerprints.

Use explicit IndexedDB saves and portable JSON/ZIP files. Original bytes travel back to the service for comparison/replay/export but do not become server records. Disable imported replay URLs so observation content never enters a share fragment. State the

storage, quota, source-rights and network/model dependencies in the UI and policies. Changing files clears a comparison reference; coverage is explicitly limited to the selected file. Preserve profile model-time context through an optional recipe field omitted from old serialized recipes.

Keep the 2 MB parser limit and 3 MB JSON input envelope. Bound imported investigation responses to 4.2 MB under the hosted payload ceiling; larger workload changes remain P16E. No provider, account, paid plan or persistent server storage is introduced. No P17 security completion is claimed. Scientific acceptance requires independent raw-observation/native-model checks and exact portable replay, followed by deployed API/browser/visual checks.

D066 - Link source identities and expose bounded support without rewriting existing science

27 September 2026: the user authorized P16D with "lessgo". Retain source sample IDs across profile/comparison rather than guessing from list offsets, similar depths or source dates. Shared model time remains distinct from observation time. An eligible comparison positions the model probe at the nearest native depth, while its residual retains P05 interpolation. Independent model selections and invalidated pairs clear the observation link instead of fabricating a new match.

Add the support API as a separate P16D contract over existing P05/P07 results. Associate samples with native threshold membership using the existing midpoint-depth-bin rule. Count unique cells/columns and show gaps, accepted residuals and matching sensitivity; never call this confidence, independent validation or sampled volume. Imported support reparses one original file in a request-scoped store. Preserve frozen scientific pack bytes and old replay contracts.

Use an optional five-step Investigation Builder and co-visible model/comparison panels. Keep existing moving tutorials, prepared cases and direct lab access. Saving remains explicit and only a successful browser write marks the applicable builder result saved. The support JSON is a derived snapshot with source identities, not a new replay format. Existing structure/comparison saves supply checked replay. P16E handles broader workload/offline lifecycle; P17 security remains deferred.

D067 - Bound aggregate imports and retain explicit offline/publication snapshots

27 September 2026: the user authorized P16E with "go all out lessgo". Preserve original 2 MB file, parser identity and 3 MB replay-envelope limits. Measured decoded expansion supports sequential batch review instead of larger single uploads. Bound the queue separately from the 32-profile/20,000-sample workspace, retain per-file failures and ignore late cancelled responses. Existing parser and numerical methods remain unchanged.

Use self-contained downloaded HTML for saved profile/comparison/native-column inspection. Keep original investigation bytes, QC, source dates and attribution. Do not claim offline recomputation or a fully cached app. No service worker or new automatic browser persistence is introduced. Hash-check the payload and prohibit network access in the viewer.

Use maintainer-only versioned publication: prepared candidate, validation, bounded isolated staging, atomic pointer and retained old releases. Readers pin a release at startup. Vercel continues to publish bundled data through a compatible application deployment; it does not become a writable persistent refresh server. No external scheduler is invented. Source refresh/acquisition uses supported trusted recipes, not arbitrary URLs or public mutations. P17's broad security audit remains deferred.

Keep observation review mounted while model metadata loads or the model context changes. Ingestion does not depend on a model response. Only explicit investigation restoration resets this panel's local lifecycle; ordinary context updates must not discard a pending queue. A public slow-response failure exposed the old metadata-dependent key and prompted a deterministic regression.

D068 - Give geography a data-discovery task and teach the PS workflow first

27 September 2026: the user requested a purposeful globe before tutorial updates, no automatic replacement by the cube, visible extra-feature navigation, glowing action controls and a Tools handoff. Treat this as a bounded P16E follow-up, not authorization to mark the P17 audit complete.

The preserved official PS prioritizes depth-resolved model fields, located measurements, supported ingestion, display controls and interoperability. It does not explicitly mandate a globe. Geography now locates actual model bounds and measured profiles, preserving source dates and variable definitions. Chlorophyll is an original BGC profile, never a fabricated surface map. Source examples use the existing parser, checksums and import

budgets. Location/time overlap does not establish numerical eligibility.

The app starts on a persistent data globe behind the default Basic welcome. Users deliberately open a model or profile. The core tour locates the case, explores its full water column and compares a sensor before demonstrating file ingestion. Additional tours include the newer connected workflow without dropping the original labs. Keep Basic actions/results distinct from In-depth concepts/methods. Save and download remain explicit.

Move extra-tour pagination above its cards, state the registry-derived feature total and highlight More features and actionable targets with a steady mint outline. End completed tours with an anchored arrow to Tools. Preserve keyboard focus, small-screen fit and reduced-motion behavior. Automated checks establish interaction and geometry only; actual user comprehension and independent scientific review remain separate evidence.

D069 - Prioritize selected data and bound transport recovery

27 September 2026: the user authorized fixing the remaining comparison errors after 0.16.11. Treat this as targeted reliability work, without starting the broader P17 audit. Reproduce application weaknesses separately from the unconfirmed intermittent HTTP/2 failure. Keep normal browser transport in acceptance checks.

Allow JSON GET reads 25 seconds instead of eight, retaining immediate caller cancellation, bounded query retry and source/release validation. Preload only the selected variable's next snapshot once the visible field and native reading are ready. Cancel unused speculative requests when selection/tool changes, but preserve an in-flight request adopted by a visible query. Disable speculative work while other tools use the co-visible model. Do not change scientific methods, original bytes, quality gates or the user's saved results. Record failures and successful repeats separately; a diagnostic pass does not prove an external transport cause was fixed.

D070 - Freeze a bounded release and separate engineering evidence from external acceptance

Date: 27 September 2026.

P17 audits the current PS/USP implementation before declaring a release. Source code, scientific fixtures, browser checks, clean installation and public deployment provide internal evidence. P17_CAPABILITY_AUDIT records every row; the human acceptance gates remain open until actual findings exist.

The API rejects excessive request bytes before parsing, bounds uploads by a real absolute deadline, applies per-worker token buckets and finite concurrency, and sends constrained browser policies. Production limits stay active in browser acceptance. Independent parameterized backend tests use separate simulated peers rather than increasing production budgets. A process-local limiter is not a distributed firewall or native-parser sandbox. P17_SECURITY records the implementation and remaining infrastructure responsibilities.

The exact Sources control, rather than its position in a toolbar, receives focus after closing its drawer or nested inspector. Historical browser tests now follow the explicit globe entry and expanded-location controls; numerical assertions are preserved. A fresh isolated Firefox test binary replaces a broken global installation for verification only.

The final deployment reuses the linked Vercel project. Prepared submission documents stay outside its upload/function package. The six-slide official-template draft, actual browser recording and local source package are review artifacts; no portal submission, public repository, institutional affiliation or scientific endorsement is implied.

P17 final evidence clarification: Firefox rounded one intended canvas pointer across a native-cell boundary. The actual delivered coordinate selected the scientifically correct cell, independently verified against the original file. The test now aims at a pixel centre and retains exact source-value expectations; production picking and numerical tolerances did not change. Repeated targeted checks pass. Performance reports retain three samples per condition and a bounded four-client request sample, with no sustained-load or real-person speedup claim. Private portal instructions, participant/domain review and actual institutional installation stay open after publication of 0.17.0.

P17 evidence and downloaded reference bytes are exempt from Git line-ending normalization. Their reported SHA-256 identities must survive checkout and source archiving. Original provider HTML and diagnostic whitespace are retained; authored source/document whitespace is checked separately.

D071 - Team domain and consolidated documentation

27 September 2026: the user requested Seifuku in the website address and a main documentation PDF. Add `seifuku-ocean-navigator.vercel.app` as a verified production domain on the existing Vercel project. Preserve the original address and deployment. Domain configuration does not change runtime or source science; browser-local storage remains bound to each origin.

Create a current handbook followed by dated scientific, engineering, phase, methods, visual-review and public-policy records, with linked contents and PDF bookmarks. Earlier

milestone statements are explicitly historical. Keep the maintained source records intact and provide an editable consolidated Markdown, reproducible renderer and source inventory. The document is a prepared artifact, not a portal submission. Keep it in the existing deployment-excluded submission folder.

Cover follow-up: use the existing Seifuku artwork from Downloads/Logo.png without alteration. Compose a navy atlas cover with a vector orthographic map from the same Natural Earth geometry used in the app. Mark only the demonstrated Bay of Bengal location; the globe is geographic context, not a claim of worldwide model coverage. Keep the body, source attribution and scientific limitations readable in the existing light document layout.

D072 - Depth Atlas and a neutral public source release

27 September 2026: the user selected Depth Atlas, requested the website address and PDF title be updated, authorized a public GitHub repository for judge access, and required omission of the competition identifier. Publish a fresh source snapshot rather than uploading private development history. Preserve that history locally. Keep scientific arrays, case identities, methods and replay calculations unchanged; update only public download addresses in exported compatibility fixtures and verify the replay/result payloads remain identical. Preserve source and third-party attribution. No blanket open-source licence is selected for original code.

Use depth-atlas-seifuku.vercel.app on the existing Vercel project. Preserve old production origins so earlier links and browser-local saves remain accessible. Export/import is needed to move saves between origins. Keep the browser storage schema and database name compatible. The main handbook receives the new cover, title, headers and address, retaining the existing large type, spacing and original team logo. The exported handbook omits the problem identifier while original source records remain local.

67 Depth Atlas: prepared phase text

Historical or maintained reference: read alongside the current release overview | Scope-edited excerpt | Source: docs/PHASE_WRITEUPS.md

These passages are ready to adapt for the project report, presentation and demo. They describe the implementation at the stated milestone. They are not a final submission. Keep earlier phase records intact and update the current introduction as the app develops. The user-confirmed public identity is recorded in TEAM.md. Historical phase sections retain the facts known at their respective milestones.

Current project introduction

Ocean Navigator is a browser application for project brief that connects ocean model fields, original instrument observations and reproducible investigations. Two bounded Indian Ocean cases supply January 2024 temperature, salinity and currents for 3D exploration, structure search, simulated sampling and passive particle drift. Heat & Depth links checked historical surface events to depth evidence only where their dates overlap. Three Pacific cases supply monthly potential temperature for selected 2013, 2015 and 2022 climate comparisons. Supported CSV and NetCDF imports, original profile inspection, quality gates, sources, dates and units remain visible. Phase 14 adds tested native-region evolution and explicit observation exclusions, with reproducible original and modified evidence. These are historical investigations, not a new live forecasting service.

Learning mode now demonstrates real features through default Basic and optional In-depth paths.

Hosting explanation prepared on 27 September 2026

The latest release encountered a server-package size check during deployment. Removing development-only files from the deployed function allowed the release to pass while retaining its scientific data and behavior. The public website remains on Vercel. Before adding larger datasets, the next phase will need to verify the project's supported package configuration and separate growing data storage from server code where necessary. Package size, file-upload limits and calculation duration are different constraints. No plan upgrade, new storage service or backend migration has been completed. HOSTING_CAPACITY_PLAN.md records the current evidence and proposed approach.

P16A: review observations while keeping the model visible

Reusable explanation

Ocean Navigator now keeps the model and instrument profile together. A visitor can preview a supported CSV or NetCDF file, check its variables, units and quality flags, and add it to the current tab. Eligible imported positions appear in the instrument layer. Selecting a reading shows its original value, depth, location and time while the model keeps its own date.

The import review supports explicit CSV column mapping and a small set of physically defined unit conversions. Unsupported quantities and conversions produce an explanation. A genuine Argo BGC example demonstrates chlorophyll with raw values, source adjustments and excluded readings preserved. Its March 2024 observation date differs from the January model, and the screen says so.

Why the change matters

Previously, imports could be inspected but did not join the shared model instrument layer, and opening the full profile replaced the model view. The connected view removes that context switch. Review before Add makes the file interpretation visible before it affects the workspace. This is an interaction improvement; no measured user-task speedup is claimed.

Demonstration and evidence

Open Tools, Instruments, Import observations, then Formats and genuine example files. Preview the Argo example and Add. Close the profile, select its imported marker, choose a sample and View nearest model depth. Then preview SR1902594_034.nc to show genuine chlorophyll and its different date. The model has no chlorophyll volume, and the imported profile is not offered as an eligible numerical comparison.

PHASE_16A_IMPORTS.md and PROJECT_STATE.md record the actual parser, source-value, mapping, browser, visual and deployment checks. The 165-check scientific regression preserves older saved-investigation replay. Additional import examples have their own catalogue so they cannot silently change the frozen library used by saved results. Intermediate failures and their fixes remain in the evidence folder.

Release acceptance: 165 scientific/backend checks, five API checks and 60 client logic checks passed. The 45 instrument/location checks, 72 renderer/tutorial/save regression checks and 15 final import checks passed locally in their recorded configurations. Production passed 24 anonymous release checks, 36 desktop/mobile browser checks and exact agreement for six public example files and their parsed results. Visual captures and manual findings are retained. These are implementation checks, not an independent scientific endorsement or a real-user study.

Limits and next work

Uploads remain temporary and bounded. Imported comparison, saving and portable replay belong to P16C. The available glider and CTD examples have no overlapping main model case; they remain honest source examples until compatible data are supplied. This phase does not establish independent scientific validation, live data coverage or actual user-study results. P16B is the next checkpoint for fresh model ingestion, measured extensibility, standards and portable installation. P17 security work remains separate.

Site information: Seifuku identity, Privacy and Terms

Reusable explanation

Ocean Navigator is an independent project by Team Seifuku, team ID 150123, based in Pune. Adwita Kurle leads the team. Gaurav Ranade created and operates the public website. About Us introduces all six members, the app's purpose and its supported historical datasets. Privacy explains what the website processes, what stays on the visitor's device and how to remove local copies. Terms explains appropriate use, source attribution and the limits of scientific results.

What changed and why

The user supplied the missing public identity on 26 September 2026. We replaced the blank operator/contact wording, added the About page and connected all three pages to the workspace footer. Privacy now distinguishes saved investigations in IndexedDB from the separate regional pack in localStorage. It also covers temporary imports, filenames in request URLs, replay links, hosting logs and contact emails. The pages remain readable without JavaScript and have mobile and print layouts.

Real demonstration

Open About Us from the workspace footer. Read the team and contact details, then open Privacy Policy. Use Your device to find the two kinds of local storage and their deletion controls. Open Terms & Conditions to read the historical-data and simulation limits. Back to workspace returns to the live application and its optional tutorial.

Verification, limits and next work

The build and five API checks pass. The final local browser run passes eleven grouped page/navigation checks, covering desktop Chromium, 320-pixel Chromium and 390-pixel WebKit layouts, no-JavaScript reading, native anchors, contact links, enlarged text and print colours. Chromium checks sequential keyboard entry. This Windows WebKit build requires explicit link focus for the keyboard activation check; it does not establish default Safari keyboard navigation or real-device support. The public deployment passes 22 anonymous release checks and the same eleven browser groups with zero page errors. Final evidence is recorded in PROJECT_STATE.md.

The earlier test harness stalled while inserting styles in a no-JavaScript context, and an attempted default WebKit link-tabbing check failed. These reports are retained. The accepted run uses direct font-size adjustment and explicitly records WebKit's focus limitation. Page publication is not legal certification, independent scientific review or a security audit. P09.5 and optional P16.3 remain open.

Tutorial discovery follow-up: choose what to try next

Reusable explanation

The guide now previews the extra features instead of expecting visitors to find them after the main walkthrough. Choose a teaser to open that feature and follow its moving guide. Drift demonstrates virtual particle paths. Climate compares the supported El Nino and La Nina seasons. Other choices cover heat and depth, connected ocean structures, a virtual expedition, feature evolution, missing-sensor evidence and wider source regions.

What changed and why

Eight topic cards link directly to the existing ten real lab demonstrations. Two cards appear at a time, with Back and More features to browse. Each card states what it shows, whether it is a simulation or analysis, and how many guided steps it contains. Basic remains the default. In-depth adds concepts, methods and limits. Each short tour ends with a choice to try another feature or use the controls independently. The complete extra-lab tour remains available.

Demonstration path

Open Guide and choose Explore extra feature tutorials. Select Drift, read the eight-particle, six-hour calculation and advance to its playback. Choose Try another feature, then El Nino & La Nina. Compare the 2015 and 2022 examples. Use More features to find the other six topics. The main tutorial also offers Choose an extra feature after its essential steps.

Evidence and limits

Release 0.16.2 is deployed on the existing Vercel project. The final feature suite passes 39 checks across Chromium desktop, WebKit desktop and Chromium mobile emulation. The final visual review covers 29 captures, including narrow portrait, short landscape and enlarged text. The 60 frontend scientific/parser/explanation checks pass. Exact public results, earlier failures and corrections are in PROJECT_STATE.md. These routes use the existing scientific actions, source values and failure states. They do not add data coverage, turn model experiments into forecasts or replace the real unfamiliar-user pilot. No account, external model, analytics or automatic save was added. The next implementation phase remains P17, on a separate instruction.

Frontend rebuild: one clear starting point

Report and presentation text

The application now begins with three short illustrated steps and a ready historical ocean case. Visitors can open a cutaway and click the water to inspect a model value. A single Tools menu groups the remaining work into exploration, comparison and experiments. Detailed settings, source records and the longer guided investigation open on request. Dates, units and the distinction between source measurements, model fields and simulations remain part of the workspace.

Demonstration

Advance or skip the introduction. Open the cutaway and inspect a value, then change the depth or time. Open Tools to choose Instruments or Drift Lab. Show Sources, and use Save investigation to keep the applied settings and numerical evidence. The earlier labs remain available; the interface gives each task its own space.

Evidence, limits and next work

See FRONTEND_REBUILD.md and the final handoff in PROJECT_STATE.md for actual test counts, failures corrected, screenshots and the verified Vercel release. Technical browser checks do not replace an unfamiliar-user task pilot.

P00: planning and scope

Report text

We reviewed the problem statement, the project guide and existing ocean data tools, then divided the work into 17 phases with defined dependencies and acceptance checks. The plan covers the required model and observation viewer before adding investigation tools and the proposed labs. Each feature has a status and an evidence requirement so that an interface mockup cannot be mistaken for a working scientific capability. Submission requirements, source references, technical choices and unresolved questions are recorded separately.

Presentation text

The build is organized into testable phases. We track the core requirements and proposed innovations separately, and mark a feature complete only when its behavior has been checked.

Evidence and limits

See [the roadmap](#), [feature register](#), [USP index](#) and [submission checklist](#). Planning is complete; the full product is still under development. The signed-in submission portal has not been inspected.

P01: application foundation and deployment

Completed: 21 September 2026.

Report text

We built the first browser workspace using React and TypeScript with a FastAPI backend. It provides region and variable controls, a geographic preview, an evidence panel and a quick guide. The coastline geometry comes from Natural Earth and includes source attribution. The app is deployed on Vercel and can be opened without an account or local installation. The foundation also includes a favicon, Privacy Policy and Terms pages. At this milestone, scientific datasets were not connected, and the interface stated that limitation.

The interface uses restrained colours, rectangular controls and direct labels. Loading, error and unavailable states explain what the user can do. The layout and keyboard interactions were checked on desktop and mobile browser emulation.

Presentation text

The first phase established a publicly accessible workspace and deployment process. It lets users choose a region, inspect geographic context and find source information. The scientific data layer was added in the next phase.

Demonstration path

Open the workspace, switch between Globe and Map, change the region, open the evidence panel and use the quick guide. The current deployment includes subsequent P02 work, so it no longer represents the original empty P01 dataset state.

Evidence and limits

See build and deployment evidence and [development instructions](#). This phase established the interface and service connection. It did not implement scientific 3D rendering. Firefox automation could not launch on the Windows test machine, and the Docker recipe has not been executed.

P02: real data and scientific contracts

Completed: 21 September 2026. Scientific coverage: 7 to 10 January 2024. These are historical conditions, not live 2026 observations.

Report text

We connected a bounded Bay of Bengal case using HYCOM GOFS 3.1 model analysis and seven Argo profiles from the INCOIS data assembly centre, distributed through the Ifremer global archive. The case contains seven model snapshots, 40 irregular depth levels and four variables: temperature, salinity, eastward current and northward current. The model domain is approximately 85.04 to 90 degrees east and 12 to 15 degrees north. Values below the local seafloor or otherwise unavailable remain missing.

The dataset inspector lets users select a model variable, timestamp and depth, or open an Argo record to inspect its samples and quality flags. Source dates, units, provenance and processing limits accompany the data. The preparation pipeline preserves source records and separates analytical values from a smaller display grid. The deployed API serves a checked data package, so normal use does not depend on downloading files from an external ocean archive.

This phase establishes the data foundation for the 3D explorer. It does not yet provide volume rendering, instrument markers or model and observation comparison charts.

Presentation text

The prototype now uses a real historical Bay of Bengal case. Users can inspect seven model snapshots across 40 depths and open seven Argo profiles with their quality flags and source information. We preserve missing cells and distinguish model analysis from observations. The next phase will render these fields in 3D.

Demonstration path

1. Open the app and select Bay of Bengal.
2. Select **Inspect dataset** and point out the historical dates and source details.
3. Change the sample variable, timestamp and depth. At a depth with no valid value, explain the missing-data message.
4. Open an Argo record and inspect its sample table and quality flags.
5. Open the source details and explain the case's geographic and scientific limits.

Evidence and limits

The phase passed 21 API and scientific tests, 60 independent source-value comparisons and a repeatable data-package rebuild. The initial 18 browser checks passed in Chromium desktop, Chromium mobile emulation and WebKit after a focus issue was corrected. The later recovery release expanded the browser suite to 27 checks, all passing locally and on the public deployment.

See [data preparation and provenance](#), [scientific contracts](#), source-value checks, API and scientific results and public browser results.

Coverage is one historical regional case. Global coverage, live feeds, glider/CTD/BGC support, user uploads and standards services remain unfinished. Agreement between Argo and HYCOM would not by itself establish independent validation because the model assimilates observations. Firefox and Docker verification remain open. Test results do not establish compatibility with every evaluator device.

Maintenance: service status and updates

Deployed: 21 September 2026 UTC. Follow-up verified: 22 September 2026, Asia/Calcutta.

Report text

An older browser tab could reject the new dataset format after deployment and incorrectly display a service outage. We reproduced this using the original frontend against the updated API. A fresh page loaded the dataset correctly. Release 0.2.1 adds a shared application version and a clear reload action when the browser and server releases differ. It also distinguishes unreadable responses from connection failures and retries temporary service failures. Tabs opened before the fix require one reload to receive it.

Demonstration text

The app now explains when an open workspace needs an update. Temporary connection failures have a retry path, and an unreadable response cannot be presented as scientific data.

Evidence

See the reproduced failure, deployment checks, public browser checks and the fresh-page check. These verify the recovery behavior; they do not claim that the service can never fail.

Text to prepare with each next phase

Add each write-up as the implementation develops, with its status and date clearly stated. P01-P08, P10 and P11 now have completed bounded implementation records; P09 still needs its real-user pilot, and P12-P17 remain planned. Do not write future capabilities as completed work.

For each phase, keep a short report passage and a shorter presentation or spoken-demo passage. Include what the user can do, what changed and why, the actual data or method, a demonstration path, test evidence and remaining limits. Link detailed engineering records instead of filling the presentation text with implementation details. Record meaningful fixes alongside the phase they affect.

Use plain language, concrete verbs and supported claims. Avoid em dashes, stock promotional phrases, fabricated metrics and claims about selection or scientific endorsement. Keep a distinction between measured results, intended benefits and future plans.

P03: 3D Ocean Explorer

Initial visual review, before the 0.3.1 correction below: user feedback identified a clarity problem in the default volume view. Warm upper water dominates while valid colder water is too faint against the background. Some deeper levels are genuinely missing. Numerical checks passed, but the presentation needs to distinguish these cases more clearly. The diagnostic review did not change the application.

Implementation date: 22 September 2026. Source period: 7 to 10 January 2024. Release: 0.3.0. The deployment and final verification record are in PROJECT_STATE.md.

Report text

We added a scientific 3D explorer for the verified Bay of Bengal case. Users can view temperature and salinity through the water column, select a depth slice, inspect an east-west section, extract an isosurface and display horizontal current vectors. Time controls use the seven supplied model snapshots. Colour ranges, palettes, logarithmic mapping, opacity and vertical exaggeration can be adjusted without changing numerical source values.

The renderer retains the model's unequal depth spacing and missing-value masks. A smaller horizontal grid keeps the browser workload bounded, while numerical inspection reads the native analytical grid. Clicking a view identifies the nearest source point and shows its value, units and depth. A volume click uses the selected probe depth. Current speed is derived from the eastward and northward components, which are also shown.

Auto graphics starts with 3D, reduces rendering resolution when frames are slow and can switch to a real 2D slice or section. The Basic view remains useful when WebGL is unavailable or its graphics context is lost. Reduced-motion settings disable automatic playback but retain manual time selection. This makes the same source values accessible across different browser capabilities.

Presentation text

The 3D explorer makes the model's vertical structure visible. We can move from a full volume to a depth slice or section, follow actual timestamps and inspect source values at a selected location. Missing water-column data remain empty. A simpler 2D view provides the same numerical inspection when a device cannot draw the full scene smoothly.

Spoken demonstration

This is a historical Bay of Bengal case from January 2024. The colours show model temperature, and the depth scale is exaggerated to make the water column easier to see. I can rotate the volume, open a section and select a depth. Clicking the view reads a point from the original model grid. I can also switch to salinity or horizontal currents, then move to the next recorded timestamp. The source dates and units stay visible. On a slower device, Basic keeps the slice, section and data inspection available.

Demonstration path

1. Open the current release and confirm Service connected and the historical case dates.
2. Orbit Volume, then choose Depth slice. Select 100 metres and click the field to inspect a native point.
3. Choose Section to see the actual unequal depth structure. Click within the section and read the selected depth and source value.
4. Choose Isosurface. Change the threshold from 20 degrees Celsius to 99 to show that an absent surface produces an empty-state explanation.
5. Select salinity, then Current vectors. Explain the component values and speed; these arrows are not particle paths.
6. Change Snapshot or start playback. Dates advance through the supplied 12-hourly records, not current 2026 conditions.
7. Expand the colour settings, adjust a palette or depth exaggeration, and show that the numerical probe stays unchanged.
8. Choose Basic 2D and inspect the same data. Return to Auto or Balanced 3D when appropriate.

Evidence and limits

Rendering methods, numerical checks, source-picking fixtures, response sizes and device measurements are recorded in [PHASE_03_RENDERING.md](#). The 12 numerical grid tests and 21 backend tests cover geometry, masks, variable semantics and source handling. Browser checks cover the views, analytical picking, colour/time controls and failure recovery. The final suite passed 50 checks locally and 50 on Vercel, with one deliberate WebKit context-loss skip per run.

This is a small historical model case, not live conditions or equal ocean-wide coverage. Volume colours are composited display values, so exact measurements come from the native probe. Thin structures can be missed by finite ray sampling. The section is an east-west plane; arbitrary transects belong to later work. Software rendering was slow in the measured Chromium setup and triggered the Basic fallback. Physical mobile devices, Safari on Apple hardware and Firefox remain separate verification needs. Instrument markers and profile charts are next in P04. Drift, heat/depth and climate labs remain planned in their original phases.

P03 revision: making the water column readable

Implementation: 22 September 2026. Release 0.3.1. Deployment verification is recorded in [PROJECT_STATE.md](#).

Report text

Feedback on the first 3D release showed that the coloured surface was clear but the deeper water looked empty. We revised the display so that a cutaway exposes the model's vertical layers. The initial view covers the upper 1,000 metres; users can select the full 5,000-metre grid. These windows use the original depth coordinates. Cells with missing samples receive gray hatching on cross-sections, which is distinct from both cold-water colour and the opening cut for viewing.

The interface now gives each view a short explanation and keeps the colour key and selected source value beside the scene on larger screens. Detailed controls follow the scene, with additional settings collapsed until needed. The same layout stacks on smaller screens and with enlarged text. Current arrows show direction at equal lengths, while colour and inspected values show speed. Basic mode opens a vertical section so the depth structure remains visible without 3D graphics.

Presentation text

The opening in this volume is a viewing cut, so we can see the layers inside. Colours represent the selected model variable. Gray hatching marks missing data. Clicking a coloured cut face reads the nearest original model point, with its depth and units. We can compare the surface with 1,000 metres below, expand to the full column, and inspect the same source values in Basic mode.

Short judge demonstration

1. Open the current release. Identify Bay of Bengal, the January 2024 historical dates and the temperature colour key.
2. Explain that the open corner exposes depth layers. Click a coloured cut face and read the selected value and depth.
3. Use **At the surface**, then **1,000 m below** to compare actual model layers. Numerical values come from the original grid.
4. Return to **3D volume**, choose **Full column**, and explain the hatched deep samples. A hatched-face click returns no value.
5. Select **Current vectors**. Equal-length arrows indicate direction; the colour key and numeric components show speed.
6. Open **Evidence** to inspect the seven existing Argo records. Map overlays and profile charts are the next phase, not part of this release.
7. Show **Basic 2D** if needed, then **Reset exploration** to restore the initial data/view settings. Graphics preference is retained.

Verification and limits

The 16 grid tests and 21 backend tests pass. Browser acceptance checks include a cut-face value independently read from the original NetCDF file, a missing-depth click, native picking after zooming, every view, source timestamps, failure recovery and context loss. The public suite passed 53 checks with one intentional WebKit context-loss skip. The last full local suite passed 52; its one outdated mobile-hint assertion was corrected and the affected check passed in all three supported projects. Evidence and screenshot reviews are linked in PROJECT_STATE.md.

The source case remains unchanged. Hatching is not measured bathymetry, and volume colour is composited display information. Native grid rows can be smaller than a screen pixel in the overview; zoom or use the coordinate selectors for exact row selection. The visual review uses Windows Chromium, Windows WebKit and mobile viewport emulation. It does not establish compatibility with every judge's device or replace testing with

unfamiliar users.

P03 revision: uninterrupted exploration and an optional cutaway

Implementation: 22 September 2026. Release: 0.3.2. Confirm the final deployment and acceptance record in PROJECT_STATE.md.

Report text

We revised the loading behavior so that users can keep their visual context while changing ocean fields. The previous verified field stays visible until the next one is ready, and retains its original variable, timestamp and colour key. A compact notice explains the pending selection. Cached fields can be revisited, and changing depth at the same location reads directly from a previously verified native profile. Nearby snapshots and alternative variables are prepared in the background when the connection permits it.

The cube now starts whole. A button beside it opens or closes the cutaway with a short reveal. This animation changes the display cut without changing scientific values, depth coordinates or missing-data masks. Reduced-motion preferences apply the change immediately. The explorer also retains its camera and selected depth when users visit geographic context.

Presentation and demonstration text

The ocean stays in view as I move between fields. If new data are still loading, the app tells us exactly which field and date we are seeing. I can change depth without downloading another value, then open the cutaway to expose the layers inside. Closing the cut restores the whole volume. These transitions help preserve context; the numerical measurements still come from the original model grid.

Start with the whole temperature volume, select Open cutaway, inspect a source value, and close it again. Change depth from 100 to 500 to 1,000 metres. Switch to salinity and back, then visit Geographic context and return to Ocean data. The camera and depth selection remain. A slower network may still need time for an uncached field; its pending state retains the labelled previous view.

Evidence and limits

The acceptance record covers source values, masks, native-coordinate validation, retained labels/thresholds, late-response handling, failure recovery, reduced motion, resource reuse and responsive layouts. See `PROJECT_STATE.md` and `PHASE_03_RENDERING.md` for actual counts and reports. All P04-P17 capabilities remain at their previous status. This work does not claim every future feature is complete, every device is equally fast or human usability testing is finished.

Final 0.3.2 verification: 17 numerical checks, 21 backend checks and 74 local browser checks passed, with one deliberate WebKit context-loss skip. The public suite initially passed 71 checks and hit three request timeouts; all three passed unchanged on focused recheck. Anonymous HTTP/source checks also passed after an initial static-asset read timeout. These observations confirm the tested paths but do not guarantee network availability or identical graphics on every device.

The final public interaction check confirmed zero additional data requests for the tested depth sequence in Chromium and Windows WebKit. Fresh desktop and mobile-sized screenshots show the whole-volume default, visible cutaway button and retained field during loading. A screenshot run initially timed out waiting for salinity; its instrumented recheck completed successfully, with the previous temperature still correctly labelled while waiting. The initial timeout remains recorded in the evidence. Mobile layouts were emulated, not tested on physical phones.

P04: genuine instruments and supported imports

Completed: 22 September 2026. Release 0.4.0 on Vercel. Data dates are historical and remain separate from this build date.

Report text

Ocean Navigator now presents real Argo, glider, ship CTD and BGC observations through one instrument workspace. The checked library contains 13 profiles across four historical collections. Users can select locations on a map, follow a glider track and inspect measurements against physical depth. A selected sample retains its own position, timestamp, source variable and quality flag. Argo raw and adjusted values remain available for inspection, with the processing mode applied separately to each parameter.

The interface separates usable readings from missing or excluded samples. Chart lines stop at gaps, and isolated eligible samples remain visible. The included glider file has contradictory salinity range metadata, so the app preserves its readings for inspection while excluding them from the accepted curve. Missing oxygen in the selected CTD cast

produces an empty-state explanation. These checks keep a complete-looking chart from concealing source problems.

Users can also import supported observation NetCDF files, CTD Exchange casts and a documented CSV layout. The app checks metadata, units, quality flags and file limits before displaying the result. Five genuine example files are available to download and import. Uploads are processed in memory, and imported profiles stay in the open browser workspace until cleared or reloaded. The Privacy Policy and Terms describe that handling.

Presentation and spoken demonstration

The numbered locations on the ocean model open real historical Argo profiles. Here the chart shows temperature against depth, while the selected reading gives us its source date, location and quality flag. I can inspect raw and adjusted values or move the model view to the nearest depth. That movement is navigation only; a scientific comparison needs matching rules, which are the next phase.

The collection selector also opens genuine glider, ship CTD and BGC records. The glider retains a different position and time for each sample. The BGC example includes adjusted dissolved oxygen. These examples use their own regions and dates, so they do not imply that the Bay of Bengal model covers those locations.

Judge demonstration path

1. Open the current release and select a numbered Argo location on the volume, or select Instruments.
2. Choose Argo 2903891, inspect a sample near 100 metres and expand Raw, adjusted and coordinate details.
3. Select View nearest model depth. Read the observation/model timestamp notice and confirm the model keeps its original timestamp.
4. Return to Instruments. Choose Tropical Atlantic, September 2023, to inspect BGC oxygen and its A processing mode; switch to temperature to see its separate R mode.
5. Choose Caribbean Sea, April 2024, then the third glider profile. Open Track detail and change the sample to see its position/time. Select salinity and Show excluded values to read the source-metadata reason.
6. Choose Southwest Indian Ocean, February 1996. Temperature is present; oxygen is missing and has no accepted curve.
7. Open Import observations and Formats and genuine example files. Download the documented Argo CSV example, import it, inspect the readings, then clear the imported

profiles.

Evidence, limits and next work

The final backend suite passes 51 checks, including 30 observation/import checks. The existing renderer has 17 passing numerical checks. Local browser acceptance passed 98 checks with one intentional WebKit context-loss skip; the isolated-sample chart addition then passed its three focused checks. The public suite passed 94 checks and hit four page/data-loading timeouts; all four passed an unchanged recheck. One WebKit context-loss check is intentionally skipped. Public HTTP checks verify all 13 profiles, all five original/example imports, rejected inputs, static routes and independent model values. Desktop, WebKit, 390/320-pixel layouts and enlarged text were reviewed. See PHASE_04_INSTRUMENTS.md and PROJECT_STATE.md for reports and earlier failures.

This release supports documented input layouts, not arbitrary NetCDF or spreadsheets. Foreign observation examples do not extend model coverage. Glider salinity remains excluded because its provider metadata needs correction. Quantitative collocation, residuals, coverage analysis, live feeds, durable uploads and institutional installation are not implemented. Physical devices, Firefox and unfamiliar-user review remain open.

P05: model comparisons, coverage and matching assumptions

Completed: 22 September 2026. Release 0.5.0 is deployed and verified on Vercel.

Report text

Ocean Navigator now compares real observations with a selected historical model snapshot. The matching method checks physical definitions, quality flags, position, time and depth before calculating a residual. Each sample shows its observation, the interpolated native model value and the difference between them. The two profile charts share a depth cursor, and users can open the matched model column in the ocean explorer.

The coverage view shows where accepted observations exist and which depths they sample. It uses the same rules as the comparison charts. Changing a time window, distance limit, depth bracket or quality category updates the accepted counts and explains exclusions. Users can keep one result temporarily, change assumptions and restore the original settings. Discovery can find the nearest eligible match or the largest absolute residual among accepted pairs. A suggestion involving another model timestamp requires an explicit selection.

Source review remains part of the workflow. One Argo file contains unresolved calibration-verification wording, confirmed in a fresh source download. Its original values and quality flags remain available for inspection, while quantitative use is held pending clarification. This prevents a complete-looking chart from concealing a provenance concern.

Presentation and spoken demonstration

I can now ask whether a measurement and model value form a valid pair. This view keeps their timestamps separate and shows the distance to the chosen model column. The model is interpolated only between valid neighbouring depths. Here the residual means model minus observation, so its sign has a clear interpretation.

The map and depth marks show the observations that actually passed these rules. If I narrow the time window too far, the comparison becomes empty and explains why. That empty result matters: the app does not invent a measurement or relax the rules silently. I can restore the earlier settings and inspect the same result again.

Judge demonstration path

1. Open Compare. The initial January 7 midnight snapshot has no eligible pairs under the default window. Select Find a useful comparison, then explicitly use the suggested snapshot.
2. For the numerical reference demonstration, select January 7 at 12:00 UTC and Argo 1902669. Confirm 103 eligible pairs. Source sample 0 shows approximately 28.153 degrees Celsius observed, 28.001 model and -0.152 residual.
3. Use the sample selector or click a chart depth. Both charts move together. Open the native-bracket details to inspect the interpolation, then View matched model column to visit its position and shared timestamp in 3D.
4. Return to Compare and select Evidence coverage. At January 7 noon the defaults give 206 pairs across two profiles. Each depth stroke is an accepted observation sample.
5. Keep the profile result as a reference, open Matching rules and apply a 2-hour window. The result has zero pairs. Restore reference settings to recover the original selection.
6. Inspect Argo 5907083 to see the source-review hold. Open its original readings to inspect the unchanged QC and calibration note. Show an out-of-domain collection to explain why an observation library is different from model coverage.
7. Switch to practical salinity or rank by largest absolute residual. Read the sample count and assimilation caveat before interpreting the numbers.

Evidence and limits

The original-file reference generator is separate from the application matcher. It checks seven profiles, four snapshots, 576 matching scenarios and labelled synthetic boundary probes. Final science/API, browser, layout and public-release results are recorded in PHASE_05_EVIDENCE.md and PROJECT_STATE.md.

The HYCOM analysis assimilates observations including Argo, so agreement is not independent validation or a forecast improvement claim. Metrics describe the selected samples, without ocean-area weighting or independence between neighbouring depths. The source-review hold, unknown CTD temperature scale and foreign observation domains remain visible limitations. Imported files are inspectable but are not part of the checked comparison library. Durable replay/export belongs to P08.

Final P05 acceptance: 137 science/API checks and 17 renderer checks passed. Local browser acceptance passed 116 checks with one deliberate skip. The public run passed 113 and flagged two loading timeouts plus an overly broad test locator; all three passed a focused recheck, with the original reports retained. All 18 new comparison browser checks passed. The final deployment passes 66 HTTP checks and the desktop, WebKit, narrow-screen and enlarged-text visual audit. A CSS footnote-width conflict found in screenshot review was fixed and redeployed. These results establish the tested paths, not universal device support or uninterrupted network service.

Phase 6: data access and extensions

Report text

Ocean Navigator now offers the historical Bay of Bengal case as a native-grid NetCDF download, alongside its bounded REST API. The file retains the source coordinates, irregular depth levels, decoded numerical values and missing-data masks. Its metadata declares the variable definitions, units, time calendar, processing history and original case fingerprint.

A separate THREDDS service was configured and tested through real WMS map requests, WCS coverage downloads and OPeNDAP subsets. An independent checker compared the responses with the original packed source files. The checks identified slight coordinate regularization in WMS/WCS, which is documented. The native download and OPeNDAP retain exact source coordinates. The Vercel website provides REST and downloads; the THREDDS package has been verified locally and still needs a persistent public or institutional host.

The phase also demonstrates extensions. A second model adapter reads the CF exchange format into the same grid contract. Horizontal kinetic energy is registered as a derived

scalar and works in the existing viewer. It is calculated from the two horizontal currents before display decimation, with both missing-value masks preserved.

Presentation text

- The same historical case can be inspected in our viewer, downloaded as NetCDF or requested through tested scientific protocols.
- A registered adapter and a new derived variable use the existing grid and renderer. Source definitions and missing values remain explicit.
- Public REST and downloads are deployed on Vercel. The separately tested THREDDS service requires a persistent host for public use.

Spoken demonstration

Here I select Kinetic energy. The scene uses the same model grid, but the quantity is derived from horizontal current components. The note explains the formula and what it excludes. I can inspect a native point, change depth or open an equal-value surface without changing the scientific definition.

The Use the data page provides the native NetCDF and its checksum. It also links an actual 100-metre temperature request. A researcher can read the numerical data outside this browser. We tested the separate standards server against the original source values, and we make its current hosting availability clear.

Demonstration path and limits

1. Select Kinetic energy from Variable. Read the formula and horizontal-only limitation.
2. Open the cutaway, switch to Depth slice and inspect 100 m. The selected point displays a derived value in square metres per square second.
3. Select 5000 m. Missing model currents produce a missing derived value, not zero energy.
4. Open Use the data in the footer or Sources panel. Download the file and inspect its checksum, source identity and bounded API example.
5. For the developer/institutional demonstration, start the packaged THREDDS service and run the documented independent protocol verifier.

The initial implementation checks passed 176 science/API tests and 17 renderer checks. The protocol/source verifier passed 143 checks, and rebuilding the exchange file produced an identical checksum. Browser testing caught a unit-label encoding defect, which was corrected before publication; initial evidence is retained. Final public and

responsive results are recorded in PROJECT_STATE.md. Physical-device, Firefox and Docker verification remain open.

Final P06 public acceptance: all nine new browser checks passed. The wider suite passed 117 checks, failed eight and deliberately skipped one unavailable context-loss check. All eight failures passed a focused recheck without code or timeout changes. The initial loading/connection failures remain documented. The release also passed 49 HTTP checks, a complete download checksum check and fifteen responsive visual checks. Public THREDDS hosting remains unconfigured; REST and the native download are deployed on Vercel.

Phase 7: search structures and draw sections

Report text

Ocean Navigator now searches the native model grid for connected samples that meet an explicit threshold and depth range. Users can inspect a region's source values, extent and estimated volume, compare eligible observation coverage, and draw a vertical section through the selected study area. The search operates on original-resolution numerical arrays, separately from the display grid.

A connected region is a mathematical query result. It is not automatically an eddy or a parcel of water. The source does not provide cell boundaries, so the application labels volume as an estimate from midpoint coordinate bins. The method retains unequal depth spacing, geographic area, source masks and the requested depth limits. Those assumptions travel with numerical exports.

Drawn sections show native column values at declared sampling stations. Users can inspect both the requested line and the actual sampled coordinates. Observation links use the established time, distance, quality and source-provenance gates. Nearby observations provide context rather than a confidence score or proof that the model threshold is correct.

Presentation text

- Ask a visible numerical question using a variable, threshold, historical snapshot and depth range.
- Inspect connected native-grid regions with explicit adjacency, estimated volume and eligible observation links.
- Draw a line and inspect the source values below it, including missing samples and actual sampling offsets.

- Export the result and its method. Saved investigations and replay are the next phase.

Spoken demonstration

Here I open Find structures and ask for water at least 26 degrees Celsius between the surface and 300 metres. The date is a historical model snapshot, not today's ocean. Amber shows the selected connected region. The volume is labelled estimated because the source gives sample coordinates without cell boundaries.

I can change the threshold and see which native samples still qualify. The result also tells me which observation samples are eligible nearby under the stated matching rules. Now I draw from A to B and inspect the vertical section. The values remain on native source levels. Clicking a sample shows its value, depth and the model column used.

Demonstration path and limits

1. Open Find structures. Choose 7 January 2024 12:00 UTC, Temperature, At least 26 degrees Celsius and 0 to 300 m. Run the search.
2. Inspect the selected region, its estimated volume and eligible observation samples. Read the method note before interpreting the shape.
3. Raise the threshold or choose a closed range. Check the new region count and the source-specific result date.
4. Draw two points on the plan view, or enter endpoint coordinates. Inspect a section value and its source column.
5. Export the numerical result. Explain that saved investigations, durable links and fresh-session replay belong to P08.

At this exact noon snapshot, the default query selects 80,203 native samples in one connected region, with an estimated volume of 12,966.152 cubic kilometres and 34 eligible samples from two profiles. Changing the time or query can change these values. They are source-specific analysis results, not performance or impact metrics.

Review, fixes and release evidence

We reviewed the existing explorer, instrument profiles, comparisons, imports, data access and new structure search visually, then checked the numerical results against independent references. The review corrected current-arrow directions on differently sized maps, equal-volume ordering, exact depth-boundary association, section picking and legends for missing or constant data. Narrow charts now keep readable labels inside scrollable panels. WebKit controls and enlarged-text labels were also corrected.

Release 0.7.1 is deployed on Vercel. The full backend suite passed 281 tests, the renderer suite passed 24, and all 35 public visual checks passed. After the visual review, all 99 independent P07 public API checks and all 49 existing public HTTP/source checks passed. The broader public browser suite passed 144 tests initially; eight resource-loading failures passed unchanged rechecks, and one unsupported WebKit context-loss injection was deliberately skipped. Initial traces and rechecks remain part of the evidence. The exact cause of the intermittent transfers is unresolved.

The implemented calculations match their declared methods on the supplied historical case. Reviewed views are usable in the tested browser configurations, while actual unfamiliar-user comprehension and physical devices still need their planned checks. The product remains aligned with the problem statement; broader coverage, public persistent standards hosting and the later labs are not claimed as complete. P08 adds saved investigations, replay and report/CSV exports.

This is prepared report and demonstration text. It has not been submitted as a presentation or evaluated by the national-round judges.

Phase 8: save, replay and export an investigation

Report text

Ocean Navigator now preserves an applied investigation together with its source versions, scientific methods and calculated results. A record can describe an ocean view, a checked observation profile, a model comparison or a structure search with a drawn section. Comparison records can also retain a reference result, allowing the same change in matching assumptions to be repeated later.

Reopening sends the saved recipe to the calculation service. The service checks source and method versions, recalculates the outputs and compares their numerical fingerprints before restoring the workspace. A missing source, changed method or different result stops the replay with an explanation. It does not silently replace the dataset. Fingerprints check consistency; they do not establish author identity or prove a scientific conclusion.

Users can keep records in their current browser, download a portable investigation file or copy a replay link for another device. The readable report, numerical CSV and complete evidence ZIP retain the relevant settings, methods, sources and limitations. Missing numerical values remain missing. Reports use the historical source dates separately from the date the investigation was recorded.

Presentation text

- Save the applied question, source versions, settings and calculated evidence together.

- Recalculate before reopening, with explicit checks for changed sources, methods and results.
- Repeat a comparison under its original and modified matching policies.
- Share a replay link or download the investigation, readable report and numerical evidence.

Spoken demonstration

I have selected a model value below the surface. I choose Save investigation and give it a name. The record keeps the depth, date, variable and display settings with the source values and method versions.

Now I can reopen it from Saved, or open the downloaded file in another browser. Recalculate and reopen checks that the same recipe still produces the same results before returning to the investigation. The report explains what I examined, and the CSV contains the numerical evidence. For a comparison, the reference and changed matching rules are both preserved.

Demonstration path and limits

1. In Explore, select a source depth and a view. Choose Save investigation, name it and select Save on this browser.
2. Download Investigation JSON and the Complete evidence ZIP. Open the HTML report and inspect its source date, units and source references.
3. Reload the app, open Saved, select the record and choose Recalculate and reopen. Confirm the selected depth and value return.
4. Copy the replay link and open it in a fresh browser session, or use Open investigation file. Recalculate before interpreting the restored view.
5. In Compare, keep a reference, change a matching rule and apply it. Save and reopen the comparison, then restore the reference settings to repeat the sensitivity check.

Records are stored locally in the current browser, without accounts or a server database. Browser data can be cleared or evicted, so a downloaded investigation file is the portable backup. A replay link contains its title and settings and still needs compatible source data and methods on the service. Temporary observation uploads, unsent form drafts, camera orbit and running animations are not saved. Current model coverage remains the bounded historical Bay of Bengal case.

Verification and next work

The final backend suite passed 307 tests and the renderer suite passed 24. All 27 final public P08 browser checks passed, including deleting a browser record and reopening its downloaded backup. Four investigation types reproduced their complete outputs in a second fresh Python process. All 19 public replay/export checks and all 49 existing HTTP/source checks passed on release 0.8.1. The earlier full 0.8.0 browser suite passed 172 checks initially, with four loading failures passing unchanged rechecks and one deliberate WebKit skip. All 30 separate public visual checks passed, including 320 px width and enlarged dialog text. Initial failures and traces remain available.

The review fixed WebKit keyboard-focus return and corrected a deployment packaging issue that omitted attribution text from evidence ZIPs. ZIP creation now stops explicitly if required source credits are absent. The final downloaded archives contain the records, settings, reports, numerical files and attribution. Source arrays and scientific methods were not changed by this phase.

Browser tests and checked calculations establish the tested workflows, not universal judge-device performance, independent model validation or unfamiliar-user comprehension. The cause of intermittent public loading failures remains unresolved. P09 is the next planned phase: an Arabian Sea case, a guided investigation, a public/student path and an unfamiliar-user task pilot. Later simulation and climate labs retain their existing phases.

This is prepared report, slide and demonstration text, not a submitted competition document or a claim of judge approval.

Phase 9: two Indian Ocean cases and a guided investigation

Status: technical implementation deployed as 0.9.1; first-time-user pilot pending. Project date: 22 September 2026. Source dates: January 2024.

Report text

Ocean Navigator now supports bounded Bay of Bengal and Arabian Sea cases through the same tools. Users can inspect depth structure, compare eligible Argo measurements with nearby model columns, search for connected regions and save the evidence. The guide starts with two real model columns whose surface temperatures differ by 0.007 degrees Celsius but whose temperatures at 100 m differ by 4.239 degrees Celsius. The selection rule is shown because this pair was chosen to illustrate contrast. It is not a basin average or an explanation of the cause.

The Arabian Sea case adds seven historical HYCOM snapshots and six Argo profile records from five original files. Its data retain native coordinates, missing values and observation quality information. Numerical verification independently checked 10,725,120 model values across the two cases. The original Bay case identity is preserved, allowing previously saved investigations to replay.

Slide text

- Two bounded Indian Ocean cases use one analysis workspace.
- A source-verified surface and depth example introduces the investigation.
- Four guided steps connect the view, a measurement, a structure query and saved evidence.
- Source dates, quality gates and selection limits remain visible.
- First-time-user pilot results are pending and are not replaced by browser automation.

Spoken demonstration

"These are January 2024 cases, not live conditions. I will start the guided investigation. The two columns are almost the same temperature at the surface, but they differ by about 4.24 degrees at 100 metres. We disclose how the example was chosen. I can open one column, compare a separate real Argo measurement with the nearby model, and search for water at or above 26 degrees in the upper 300 metres. Finally I save the applied query and its evidence. Reopening it recalculates the result against the same sources."

Demonstration path

Open the public workspace. Start guided investigation. Change Compare at depth between Surface and 100 m. Open this column. Continue to Compare a measurement, then Find warm water. Inspect the 26-degree and 0-to-300-m settings. Continue to Save the evidence. Use Save investigation, choose a title and Save on this browser. Recalculate and reopen from Saved. Change Study case to Arabian Sea and repeat. The guide's float and its two-column example have different locations and are labelled accordingly.

Reality check and next work

The source pair illustrates a depth contrast; it does not prove every part of either sea behaves this way. Assimilative model agreement is not independent forecast validation. Both cases expose REST and native CF downloads, while the separate verified THREDDS package still covers Bay only and lacks a persistent public host. Real unfamiliar-user understanding remains an open P09.5 gate. Expedition, particle simulation, Heat & Depth, climate events and the remaining labs keep their later phases. P10 is not started by this release.

Phase 9 release checks and corrections

Final release 0.9.1 passes 69 public browser checks for the guide, instruments and investigation workflows, 20 public case/source/isolation checks and 19 replay/export checks against original P08 records. Local evidence includes 317 backend tests, 36 focused save-isolation checks, 26 renderer/transport tests and 23 visual captures. The initial full public run had eight loading failures, all of which passed unchanged rechecks. The failures remain documented, and uninterrupted availability is not claimed.

Corrections include the Arabian salinity display range, case-neutral chart captions, accessible depth-selection labels, larger mobile chart labels, cancelled-calculation feedback, correct network/JSON error classification, a duplicate startup observation request and a save-API case-library validation gap. No fake observations or metrics were added. The real first-time-user pilot is still required to close Phase 9. Documentation and this text are prepared for later use; no final report or submission was filed.

The deployed screenshot audit covers 23 distinct views. Its first run stopped at Arabian current rendering; unchanged focused rechecks and direct current-component requests passed. The initial failure is retained, and its precise cause is not claimed to be solved. Human task feedback remains the next action.

Phase 10: Virtual Expedition and sampling experiments

Status: P10.1-P10.5 complete, deployed and verified as release 0.10.1. Project date: 22 September 2026. Source dates: January 2024.

Report text

Virtual Expedition lets a user choose a measurement budget, suggest sampling locations, sample a virtual line and compare sampling strategies. It supports temperature and salinity at supplied depths from the surface to 1000 metres in the two checked Indian Ocean cases. Suggested stations explain the prior gradient and geographic spread used to select them. These are separate from observed mismatch and observational confidence.

The comparison fixes every plan before reading the later model snapshots. Each strategy uses the same candidate pool, exact measurement budget, minimum spacing and reconstruction method. The evaluation points are separate from all possible station locations. A space-filling baseline, five fixed random plans and the unchanged prior provide context. Impossible-to-fill greedy designs and missing measurements remain visible failures. The gradient plan loses to the space-filling baseline in both default examples, demonstrating why a suggestion should be tested rather than called optimal.

All survey readings are explicitly simulated. The app preserves source depth, time, coordinates, missing values and sampling offsets. Plans and experiments can be saved, recalculated and exported with their settings, seeds, source references and numerical results. This phase adds a repeatable model reconstruction experiment, not an operational ship route, an observing-system impact study or proof of better forecasts.

Slide text

- Choose a station budget, spacing and planning objective.
- Inspect why each candidate was selected.
- Draw a line and sample actual historical model values.
- Compare equal-budget designs on the same withheld locations.
- Keep unsuccessful plans and worse reconstruction scores visible.
- Save the settings, sources, simulated samples and every benchmark trial.

Spoken demonstration

"Suppose we can take eight temperature measurements. This tool suggests locations using the first historical snapshot. We can inspect why each was selected, draw a separate survey line, and sample the model along it. The test compares our plan with a space-filling plan and five fixed random plans. Every strategy is scored on the same locations that were excluded from sampling. Here, the gradient plan performs worse than the simpler space-filling plan. We keep that result visible. This helps explain sampling tradeoffs; it does not claim that we have improved an ocean forecast."

Verification and next work

The full backend suite passed 364 tests before the final portability correction. After that correction, 77 focused backend checks passed, including 26 independent scientific checks. They decode the original packed source files, verify numerical samples and benchmark errors, and check that hidden evaluation values cannot influence planning or reconstruction. The renderer and transport suite passed 26 checks. The full local browser run passed 218 tests with one unsupported WebKit context-loss injection skipped, followed by 24 passing final Expedition checks.

On the final public release, 116 HTTP checks and 51 browser checks passed. An 18-view public screenshot audit found no page errors, unexpected alerts or document overflow; representative desktop and narrow layouts were visually inspected. Eight saved records replayed exactly from Windows to Vercel and eight from Vercel to Windows. See [Phase 10 acceptance](#) for the reports and their scope.

Verification found and fixed a tiny platform-dependent gradient calculation that initially prevented exact replay. Method version v2 now passes both checked platforms, while original v1 records remain readable and report a clear method mismatch. Further corrections separated draft routes from verified samples, included actual survey timestamps and simulation labels in CSV, and made map legends and comparison explanations consistent. Initial failures and diagnostic evidence remain available. No numerical scores were tuned to make the suggested plan win.

P09.5 real unfamiliar-user feedback remains open. The next planned implementation is P11 Drift Lab, the particle or pebbles simulation. No next phase is started automatically. This text is prepared for reports, slides and demos, not submitted material.

Phase 11: Drift Lab

Status: P11.1-P11.5 complete, deployed and verified as release 0.11.1. Project date: 23 September 2026. Source dates: 7-10 January 2024.

Report text

Drift Lab lets users release passive tracers at a point or within a small area and calculate their movement through supplied historical currents. The experiment stays at one native depth, keeps its original source dates and stops where the current data no longer support a trajectory. Users can compare two releases, inspect individual paths, count simulated target entries and save the complete settings and results. Playback displays the calculated positions with a clear separation between visual interpolation and numerical integration.

The calculation adapts a pinned OceanParcels RK4 kernel to the application's bounded server runner. A separate full-framework reference and independently decoded source files check the implementation. All requested particles remain in the arrival denominator, including invalid releases. The results are historical simulations, not observed tracks, safe travel routes or operational rescue probabilities.

Slide text

- Release particles from a point or a seeded area at a native depth.
- Calculate movement using time-varying historical eastward and northward currents.
- Play or scrub the paths and inspect each stopping reason.
- Compare declared releases and approximate target-entry times.
- Save, recalculate and export settings, trajectories and source identity.

Spoken demonstration

"I will release 24 particles inside this box at the surface. The app calculates their movement using January 2024 currents, then lets us play the paths. The background arrows show the starting snapshot; the paths use the changing currents. I can select a particle, see its elapsed time and inspect why it stopped. A second release lets us compare a different starting area or date. If I choose a target box, arrivals are simulated counts out of every requested particle. Saving retains the settings and source data identity so the calculation can be repeated."

Verification and limitations

Independent local verification passes 45 scientific checks, including original source values, analytical motion, masks, boundaries, target crossing, numerical convergence and comparison with the complete pinned particle framework. Twelve checked paths match the framework endpoints locally. Real drifter validation remains unavailable within the queried study bounds and dates.

The final release passes 139 focused backend checks, 54 public browser checks and 100 public API checks. Eight saved runs replay exactly in each direction between Windows and Vercel. Twenty public visual captures show the setup, paths, comparisons, supported failure states and responsive layouts without document overflow or unexpected page errors. Representative desktop and narrow views were also inspected. These checks do not replace real first-time-user feedback or physical-device testing. See [Phase 11 acceptance](#) for the full evidence and earlier regression results.

Testing found a tiny difference in summary mean-distance arithmetic between Python versions that stopped exact replay. The correction explicitly adds individual distances in a fixed order and records a new method version. Paths and source values were unchanged. Old method records remain readable and explain why they cannot replay under the new method. Other fixes preserved completed results after cancelled or failed drafts, clarified which run supplies the static current background, retained full arrival denominators and included the correct numerical trajectories in exports. Initial failures remain documented.

P09.5 real-user feedback remains open. No later phase starts automatically. This is prepared report and demonstration text, not a submitted report.

Phase 12: Heat & Depth Lab

Status: P12.1-P12.5 complete, deployed and verified on Vercel as release 0.12.0. Project date: 23 September 2026. The surface baseline is 1982-2011; display years are 2023 and 2024; available model depth snapshots remain 7-10 January 2024.

Report text

Heat & Depth detects historical marine heatwaves at six fixed NOAA OISST grid cells in the Bay of Bengal and Arabian Sea study areas. The graph compares daily analysed surface temperature with a seasonal mean and 90th-percentile threshold calculated from the complete 1982-2011 baseline. An event needs at least five consecutive days above the threshold. The app preserves its full dates, including year crossings, and explains its duration and intensity relative to the seasonal mean.

Surface temperature alone does not describe the water column. When an event overlaps the supplied January 2024 HYCOM case, the user can inspect a separate native depth profile and available observation context. The calculations retain original temperature, salinity, coordinates and depth spacing. They provide potential-density anomaly, signed temperature gradients, stratification and two stated mixed-layer estimates. Column potential-enthalpy content has an explicit reference and integration limit. Missing support produces an unavailable diagnostic or a lower-bound statement.

The current Bay events support this depth link. Arabian events at the selected cells do not overlap the short model case, so the app explains that depth evidence is unavailable. It does not substitute a different date or classify a surface event as a heatwave throughout the water column. Saving and exporting retain the applied settings, full event boundaries, seasonal curves, numerical values and source fingerprints.

Slide text

- Inspect daily analysed SST against a declared 30-year seasonal baseline.
- Select events calculated from the threshold and duration rules.
- Link an event only to depth snapshots actually available on its dates.
- Inspect separate temperature, salinity, density and stratification evidence.
- Compare the two stated mixed-layer criteria and inspect integration limits.
- Retain methods, source identities, baseline curves and numerical values in exports.

Spoken demonstration

"This is a historical Bay of Bengal surface record. The solid line shows analysed SST; the other lines show its seasonal mean and event threshold. The selected event began in October 2023 and ended in February 2024, although this graph displays 2024. A January model snapshot lets us inspect a separate depth profile during that event. The mixed-layer estimates use different criteria, so they can disagree. The column integral has a stated reference and does not measure additional heat caused by this event. Switching to the Arabian case shows real surface events but no overlapping depth data. I can save this analysis and export the actual values and methods."

Demonstration path

Open the public workspace and choose the Bay of Bengal case. Select Heat & Depth, the centre surface location and event year 2024. Use Analyse surface record if settings have changed. Read the full event dates, 11 October 2023 to 10 February 2024, then inspect a visible daily sample using the chart or daily sample control. Keep an overlapping January model snapshot. Change Depth variable and Column and integration limit; compare the two mixed-layer estimates. Open Methods, baseline and source identities to inspect the definitions. Use Save heat investigation, then recalculate the saved record and download its evidence ZIP. Switch to Arabian Sea to demonstrate the genuine unavailable-depth state. Displayed numbers are rounded for readability; the numerical exports preserve the calculated values.

Verification and limits

The final backend suite passes 590 tests with zero skips, including 76 independent scientific checks. Those checks compare all six actual surface series with a pinned reference implementation, check 18 original NCEI source values exactly under their verified packing rules, and independently verify source definitions, calendar rules, native HYCOM profiles, TEOS-10 quantities and column integrals. The complete archive has 16,071 daily values per location. Missing distributor years were recovered from original NOAA records without filling temperatures or selecting locations to produce events.

Local HTTP acceptance passes 88 checks. Public acceptance passes 105 checks, including eight exact Windows-to-Vercel replays and their numerical exports. The reverse run passes 19 checks: all eight Vercel-produced records replay and export exactly on Windows. These verifier runs have no failures or retries. The 76 independent tests are included in the 590 backend total, and the replay checks are included in their respective HTTP totals.

All 54 public Heat and saved-investigation browser scenarios pass across Chromium desktop, WebKit desktop and Chromium mobile emulation, with no failures, skips or flaky tests. Twenty public visual captures have no page errors, document overflow or unexpected alerts. Representative public surface, depth, gradient, unavailable-data and 320 px views were manually inspected. The curves, rounded display values, unavailable states and mobile scrolling guidance remain readable in those checked layouts. See [Phase 12 acceptance](#) for evidence filenames, earlier results and corrections.

Testing corrected a mixed-layer crossing case where potential-temperature differences changed sign, clarified the MLD marker and unavailable observation context, and fixed a narrow WebKit select overflow. Three initial browser failures came from an incorrect test assumption that a 2023 event could not extend into the January 2024 depth case; the tests now select an actual non-overlapping event. Initial failures remain retained. The tiny distributor packing differences were traced to checked decoding alternatives and preserved, rather than hidden by rounding source data or loosening exact reference comparisons.

These are six point samples and a short historical model case, not basin-wide heatwave coverage or an operational forecast. Numerical agreement and repeatable exports do not establish independent ocean validation or unfamiliar-user understanding. P09.5 actual user feedback, physical-device coverage and the documented external hosting/runtime limits remain open. This is prepared report, slide and demonstration text; it has not been submitted.

Phase 13: Climate Event Lab

Status: P13.1-P13.5 and X23-X24 complete. Release 0.13.0 is deployed and verified on Vercel. Project date: 23 September 2026. Historical cases: SON 2013, 2015 and 2022; GODAS anomaly baseline: 1991-2020.

Report text

Climate Event Lab compares real historical Pacific depth structure during selected neutral, El Nino and La Nina seasons. Both views use the same calendar window, source coordinates and anomaly baseline. Users can inspect a native water column from 5 to 459 m, read its exact values, compare the reference year and follow the original source records. The same dates also supply separately labelled Indian Ocean context and IOD index information.

The Pacific fields are monthly GODAS potential-temperature analyses. Their first-of-month timestamps represent whole averaging months. September-November is an explicitly equal-weight mean of those three complete monthly fields, and the baseline requires all 30 corresponding years from 1991 to 2020. Missing support stays missing. The fields remain distinct from the instantaneous in-situ measurements in the original Argo profiles.

The baseline is a reproducible calculation from the original annual source fields. NOAA also supplies a climatology for those years under a different missing-data policy. The project retains its declared complete-year calculation, and does not claim numerical equivalence with that separately available provider product.

The app uses the official RONI definition adopted in 2026 for its pinned historical ENSO labels and keeps ONI references separate. Seventeen source documents support four named index definitions and three event records. IOD episode assessments come from their named agency sources; the displayed seasonal DMI value does not automatically determine that episode label. Choosing an individual month also shows the separately named centered three-month index window.

Slide text

- Compare selected 2013 neutral, 2015 El Nino and 2022 La Nina seasons.
- Use the same September-November window or one matching calendar month.
- Inspect potential-temperature anomalies, profiles and exact between-year differences.
- Open the real Pacific case in the existing 3D explorer and inspect original Argo profiles.

- Use Find structures to inspect connected potential-temperature threshold cells within the supplied depth range.
- Keep Indian Ocean depth context, DMI values and documented IOD episodes separate.
- Save and reproduce the applied settings, sources, baselines and numerical evidence.

Demonstration path

"These panels compare the same season in 2015 and 2013. The label comes from the official ENSO index; the colours below show a different quantity, the model's potential-temperature anomaly at each depth. Both plots use the same scale. I can select a water column and compare its actual source values and the reference baseline. October changes both fields to the same month. The Indian Ocean section uses those dates too, but it does not prove that one climate pattern caused another. These Argo links open the original measurements with their quality flags. Saving retains the exact selections and source versions for replay."

Verification and remaining limits

Source preparation reproduces all 62 generated files exactly. Independent source-service requests match 1,008 native field values. All 26 index tests, 22 source audit checks and 29 independent field/method tests pass. The final backend regression passes 683 tests with zero skips; this total includes the 38 climate API checks and the new independent tests.

The final local HTTP verifier passes 116 checks; production passes 133. Eight local investigations and all their numerical exports reproduce exactly on Vercel, and eight public records reproduce exactly on Windows in the separate 19-check reverse run. All 28 retained earlier investigations still replay. There were no verification retries or relaxed numerical fingerprints.

The broad public run passes all 111 browser scenarios across Chromium desktop, WebKit desktop and Chromium mobile emulation. A final source-copy correction then passes nine focused public scenarios. The final public visual audit has 24 captures with no page errors, unexpected alerts or document overflow; earlier 23-view audits are retained. Actual screenshots were inspected, including historical anomaly sections, constant zero differences, native profiles, Pacific 3D/cutaway and narrow layouts. Earlier broad browser failures are retained and resolved by documented focused rechecks, rather than described as initially passing. Evidence and exact counts are in [Phase 13 acceptance](#).

The first browser run found a real legacy parser assumption about instantaneous profile overlap; monthly source support is now accepted through its explicit contract. Shared API review corrected a Pacific HYCOM label, replaced unrelated Bay-specific caveats and added monthly potential-temperature context to standalone results. Earlier test

corrections fixed two endpoint paths, an unsupported 0 m request and an expected HTTP status for a correctly rejected oversized response. Original failures remain recorded. No source values, event classifications or exact replay comparisons were relaxed. Visual review also found and corrected a longitude-boundary rounding error that zoomed nearby Pacific floats out to a global map. The corrected map uses their original coordinates. The Pacific volume now uses visibly labelled depth exaggeration, and the climate table explains display rounding. A last public sidebar audit caught a remaining hardcoded HYCOM label. The corrected sidebar uses GODAS metadata, full monthly dates and actual coordinate/depth bounds. A final replay/export smoke passes 19 checks after that UI-only redeploy; scientific values and methods did not change.

Original observation QC stays visible. The September 2022 profile has 994 accepted temperature readings and no accepted salinity readings. Its joint eligibility remains zero, while the valid temperature readings remain inspectable. The app does not replace that profile to improve its appearance or calculate a direct residual against the monthly potential-temperature analysis.

These selected cases do not establish complete Pacific coverage, causal attribution or a new climate forecast. Missing source cells and original parameter-specific observation QC remain visible. P09.5 actual first-time-user testing and the documented physical-device and external validation limits remain open. This is prepared report and demo text, not a submitted report.

Phase 14: feature evolution and observation blackout

Status: P14.1-P14.4 and X25-X26 complete. Release 0.14.1 is deployed and verified on Vercel. Project date: 24 September 2026; supported evolution frames: 7-10 January 2024.

Reusable explanation

Feature evolution follows defined threshold regions through the available historical model frames. The interface shows the overlap rule, branching relationships, nearby threshold choices and missing time support. Users can select a region, inspect its native extent and see why it connects to another region or stops. A connection between regions does not mean the same water travelled between them.

Observation blackout lets a user exclude selected profiles or instrument groups and compare the available evidence before and after. Source observations, quality flags and the ocean model stay unchanged. The result identifies lost eligible comparisons and recalculates residual summaries for the remaining samples. This does not measure the impact of removing instruments from an operational forecasting system.

Demonstration

Open Evolution in the Arabian Sea case and keep the default 26 C, 0-300 m condition. Select a branching region, inspect its overlap percentages and compare the nearby threshold rows. Change the selected frame spacing to every second frame to see an explicit gap. Switch to Bay of Bengal and open Blackout. Select one contributing profile, then apply the exclusion: the checked default changes from 206 eligible pairs to 103. Inspect the lost source sample indices. Save original evidence and modified evidence separately, then recalculate and export each record.

Slide text

- Follow native threshold regions across actual model timestamps.
- Inspect split, merge and ambiguous links under a stated overlap rule.
- Compare nearby thresholds and keep missing intervals visible.
- Exclude a profile or group and see exactly which eligible comparisons disappear.
- Keep source quality flags, historical fields and original evidence unchanged.
- Save and reproduce both the original and modified investigations.

Spoken explanation

"Here we define a region by temperature and depth. Each column is an actual model snapshot. The lines show where those threshold regions overlap, including branches. They do not trace the same water. In Blackout, I remove a contributing profile from the evidence view. The ocean model stays the same, but fewer measurements remain for comparison. I can inspect exactly what was lost and save both versions for someone else to reproduce."

Acceptance and limitations

The final backend passes 807 tests, and all 34 frontend numerical/geographic/parser tests pass. Independent evolution checks use original packed HYCOM values and separate segmentation/geometry arithmetic; 28 additional correspondence/depth-contact checks pass. The final public browser run passes 87 scenarios across Chromium desktop, WebKit desktop and Chromium mobile emulation. The 43-view public visual audit reports no page/console errors, unexpected alerts or document overflow; nine actual captures were opened and inspected. This does not replace testing with unfamiliar users.

Public HTTP acceptance passes 67 checks across 94 requests with no retries. All nine new investigations and numerical exports reproduce exactly between Windows and Vercel in both directions; all 36 retained earlier records replay. The separate reverse check passes

19, and the anonymous release-content check passes 19. Focused scientific checks are included in the full backend total, not added to it. Evidence and original failed runs are retained in [Phase 14 methods and acceptance](#).

Review fixed sparse source-ID handling, monthly source-file attribution, selected-depth labels and late responses overwriting new input. The initial public run also found last-bit differences in region summaries and expensive repeated work. Ordered summation under evolution method v2 and a bounded source-validated cache corrected those paths. Exact fingerprints were not relaxed. Old v1 archives remain readable and their original JSON can be downloaded; recalculation explicitly requires the matching method.

The implementation uses bounded January 2024 Indian Ocean cases. Monthly Pacific fields are not treated as 12-hour feature evolution, and monthly potential temperature is not compared directly with instantaneous in-situ temperature. Region overlap does not establish movement of the same water. Blackout recalculates available evidence without rerunning or unassimilating the model. The existing P09.5 unfamiliar-user pilot remains open. This text is ready for reports, slides and demonstrations; it has not been submitted.

P15: wider regional access, completed 26 September 2026

Reusable explanation

Ocean Navigator now lets a user explore supported areas beyond the original five cases. The user sees the available source, months, variables and vertical range before loading. A light preview appears first, followed by a fuller display. Native profiles and downloaded values preserve the original coordinates and numerical precision.

What changed and why

The new Regions workspace accepts bounded areas across wider source coverage, including the Atlantic, southern Indian Ocean and date line. Progressive loading, bounded caches, cancellation and adaptive graphics keep requests manageable. Native regional packs can be saved, replayed online or opened through a standalone offline viewer. The original five cases and their scientific labs remain available.

Demonstration path

Verification and limits

The backend suite passes 857 checks and frontend numerical/parser tests pass 39. All eight source arrays match 28,964,160 original values, including missing values. Seven new regional packs replay exactly between local and hosted environments, and 45 older investigations still reproduce. The final visual patch passes 45 public browser scenarios; the earlier broader release run passes 113 with one existing WebKit skip. Twenty final views were captured and nine inspected, including a corrected pressure-axis label. Four simultaneous browser workflows passed. These are implementation and source-consistency checks, not independent ocean validation or proof that 100 users can run concurrently.

The supplied wider data covers September and October 2022. It is historical, with explicit source, geographic and vertical limits. Earlier scientific labs are not enabled for every new area. Real unfamiliar-user comprehension remains untested at P09.5. Final methods, exact commands, initial failures and evidence are in PHASE_15_WIDER_COVERAGE.md. This text is ready for reports and demos; it has not been submitted.

Next planned work

P16 adds learning mode and grounded explanations over verified evidence. It has not started.

P16: learn by using the real application

Report and presentation text

Ocean Navigator now demonstrates its features through two tutorials. Basic is selected at startup and shows how each feature works and how to read its output. In-depth is explicitly chosen and adds the concepts, method, limitations and optional questions. Both operate the real application one step at a time. Visitors decide when to continue, replay, skip or try the controls themselves.

The order follows the central project brief tasks: depth-resolved model fields, instrument profiles and eligible comparisons first, then variable/time controls, supported files and reproducible evidence. Additional experiments follow as an optional second part. The official statement does not provide numerical judging weights; this sequence is our interpretation of its priorities.

Explain reads current values, units, source references and applied settings. It can also define reviewed concepts or prepare a supported query such as salinity at an exact native

depth. The user sees the query before applying it. Unknown questions and missing evidence stay explicit.

Short spoken explanation

Start with Basic and watch the app open real views for you. Next takes one action at a time and explains the result beside it. Choose In-depth if you want the science behind the controls. You can pause, explore and resume. Explain keeps the current numbers and sources together, and it says when the data cannot answer a question.

Real demonstration

Start Basic, open the cutaway and inspect 100 metres. Continue to the sensor overlay, original Argo profile and eligible model comparison. Read the residual as model minus observation. Continue through time, variables, genuine imports and the save form. At the end of the essentials, either explore freely or try the additional labs. Show the eight-particle historical drift calculation and the separate El Nino/La Nina comparisons. In In-depth mode, show a concept, its method and one optional question. In Explain, review an exact-depth query and contrast it with an unsupported request.

Verification and limitations

Final counts, report links and retained initial failures are recorded in PHASE_16_LEARNING.md and PROJECT_STATE.md. Tests check real actions and outputs, source links, interruption/recovery, refusal of invented measurements, viewport layouts, keyboard access and reduced motion. The source methods and scientific backend were not changed. Automated completion is not a real unfamiliar-user study; P09.5 remains open. This is prepared report and demo text, not a submitted deck or scientific endorsement.

Next planned work

P17 is the separate final audit, reproducibility and submission phase. Operator/contact details and participant evidence still need real input. P17 has not started.

P16 follow-up: guidance at the point of use

Reusable explanation

The tutorial takes you to the feature it is explaining. A small popup points to the current button, control or chart, performs the demonstration and shows its actual output. Next moves to the next feature. You can go back, try the control yourself or leave the tour at any point.

What changed and why

The original tutorial stayed in a side column. Visitors still had to find the relevant feature elsewhere. The revised guide moves between visible targets, highlights the current one and brings it into view. It follows scrolling and resizing. Guidance for saving appears inside the real save form, beside the name field.

Demonstration

Open the app and start Basic. Watch the ocean appear. Select Next to move the popup to the cutaway button, then Next again to reach the inspection-depth control. Continue to the sensor profile chart. Use Try it myself to pause, or Skip to explore. In-depth uses the same guided actions and adds concepts, methods and limits.

Scope and limits

This changes the presentation of working features. It does not change the ocean calculations, source dates or quality rules. Automated interaction and layout checks support technical behavior; actual beginner comprehension still needs the planned unfamiliar-user pilot. P17 remains a separate phase. Final verification evidence is recorded in PROJECT_STATE.md.

Follow-up verification

Release 0.16.1 is deployed to the existing Vercel project. Local tutorial checks pass 54/54 and the final layout checks pass 15/15; the visual review covers 41 captures. Public release checks pass 21/21 and all 27 steps completed in the tested browser profiles. Public runs also encountered intermittent data and script delivery failures, retained in the evidence rather than described as solved. This supports the moving-guide implementation, not a claim of flawless availability or proven first-time comprehension.

Identity correction: website creator and operator

Gaurav Ranade created and operates the Ocean Navigator website. Adwita Kurle leads Team Seifuku. Release 0.16.4 makes this distinction explicit on About, Privacy and Terms and preserves the other confirmed team details. Demonstrate it by opening About Us and reading the two role cards. Build, deployment and public-page evidence are recorded in PROJECT_STATE.md. The user also clarified that the project has no university affiliation or university involvement in development. The university name and affiliation have been removed from all three pages and current project identity records. No scientific calculation or personal-data handling changed. P17 remains the next phase only on instruction.

Startup recovery: an old app link should not leave a blank screen

An older page was requesting a JavaScript file retired after release 0.14.1. Without that file, the application could not start. The corrected release recognizes that specific old entry URL and requests the current page once. It preserves shared-link context and stops if the problem repeats. The latest entry page also contains its own loading and recovery message, so a failed main download has an explanation and a reload option.

If a regional tool fails after the app has opened, the rest of the workspace stays usable. A visible notice offers a reload and explains that unsaved work will be lost. It does not refresh an active investigation automatically. No scientific values, methods or source dates change.

Demonstration: open the updated public link and skip the tutorial to reach the real model. The acceptance checks also deliberately block the entry script, replay a stale entry document and block the regional tool module. Local checks pass fifteen scenarios across Chromium desktop/mobile and WebKit. Exact deployment and public results are recorded in PROJECT_STATE.md. These controlled tests establish recovery behavior; they do not establish universal availability or identify every cause of earlier network failures. The next planned phase remains P17 on instruction.

Planned completion sequence after the 31-item review

Planning update, 26 September 2026. This supersedes earlier next-phase wording; it describes planned work rather than delivered features.

The follow-up plan connects imports to the model, proves the application with a fresh compatible model dataset, extends comparison and saving to imported investigations, simplifies the complete investigation journey, and improves larger imports, refresh and offline use. These are P16A-P16E. P17 then checks the integrated release, its security,

scientific results, measured performance, documentation and judging demonstration. All 31 review items are mapped in CORE_COMPLETENESS_REVIEW.md.

The intended demonstration is one continuous task: import supported observations, inspect them alongside the model, compare eligible measurements, inspect support and sensitivity, then save and replay the investigation. This complete extended path is not yet an accepted demo. Release 0.16.4 remains the current app. The planning update ran no new scientific or browser acceptance checks. Source compatibility, real-user participation and infrastructure availability remain practical dependencies. First implementation work is P16A only when instructed.

P16B: fresh model data and portable setup, 27 September 2026

Prepared wording for reports, slides and a spoken demo. Final public evidence is recorded in PROJECT_STATE.md; this text has not been submitted anywhere.

Short explanation: Ocean Navigator can load a new compatible historical model case through the same acquisition, preparation, API and viewer. The added Bay of Bengal case covers 28-30 March 2024. It preserves the source grid, dates, depths, units and missing values. Its location and time range include our real chlorophyll observation, which can now be inspected beside the model. Chlorophyll is still an observed quantity; the model does not supply a chlorophyll field.

What changed and why: case dates and observation requirements are configurable, and the API registers maintained recipes. The service catalogue now tells the browser which tools have the required case support. This avoids offering the March case to methods checked only for January. The native file is downloadable, and its WMS/WCS/OPeNDAP paths have been exercised on a local standards server. A clean source archive runs with fresh dependencies outside Vercel. Scientific pack files retain exact byte identities in Git.

Demo: select the March Bay case, choose 29 March 00:00 UTC, and inspect temperature at 100 m. Open Instruments, preview SR1902594_034.nc, then add it. Show the chlorophyll profile next to the dated model. Open Use the data for the native NetCDF and provenance. If demonstrating Expedition, Evolution or Heat & Depth, choose one of their checked January cases when prompted.

Verification: independent packed-source decoding checks every cell of 25 native arrays, including the derived current-energy field. Scientific, source-integrity, browser, saved-model replay, protocol, clean-installation and public checks are linked from the phase record. Initial failures and the corrections remain documented. Desktop, tablet and mobile captures retain separate model/observation dates and no horizontal page overflow.

Limits: this is a new historical input from the existing HYCOM provider, not a new model family or live stream. Imported numerical comparison and persistence are next. Public persistent THREDDS, actual INCOIS installation, Docker execution and independent scientific review remain open. The external IOOS CF checker could not be installed on this Windows host because native UDUNITS support is missing; scoped CF checks are not universal certification.

Next: P16C adds eligible comparisons and durable save/export/replay for imported observations. P16C requires a separate start instruction.

P16C: imported data that can be checked again

Ocean Navigator can compare an imported observation with a compatible historical model and keep enough information to repeat that calculation. The app saves the original file, its column mapping, the matching rules and the results together when the user chooses Save on this browser. Reopening reparses the file and recalculates the evidence before displaying it.

The March Bay of Bengal example now connects a real BGC profile to the March model. At the selected 29 March snapshot, the default rules yield 245 eligible temperature pairs and 243 salinity pairs. Excluded readings keep their QC reasons. Observed chlorophyll is available for inspection, but this model has no chlorophyll field. Model agreement does not establish independent validation because HYCOM assimilates observations.

Demo: select the March case and 29 March 00:00 UTC, preview and add SR1902594_034.nc, choose Compare with model, inspect the profile and exclusions, save, download JSON and reopen it in another browser. For sensitivity, retain the six-hour comparison, narrow the window to one hour and observe that no pairs qualify without changing the source data.

Local saves use browser storage, with 25-record/32 MB limits. JSON and ZIP include the original observation file, so sharing depends on source permission. The model pack and matching parser/method versions must still be available online. No uploaded file is placed in a replay URL or saved to a server database. Numerical, browser and visual verification is recorded in PHASE_16C_IMPORTED_REPLAY.md; release 0.16.8 passed 203 backend tests, 60 client calculation checks, 61 public scientific/replay checks and 40 public desktop/mobile browser checks. The deployed scientific charts and save disclosures were reviewed at desktop, tablet and mobile sizes. Counts from different suites overlap and are not a total of unique features. P16D will connect these pieces into a guided investigation workflow. This is prepared report/demo text, not a submitted presentation.

P16D: one investigation across connected views

Ocean Navigator now keeps a source reading connected as you move between its profile, the model comparison and the model scene. Each dataset keeps its original date. The model probe shows a native depth, while a compared value can be interpolated to the observation depth. If the matching rules fail, the app explains the exclusion and does not invent a residual.

Build investigation offers five steps: choose data, explore depth, inspect a reading, compare and save. The model can stay beside the comparison, or the comparison can expand for more space. The existing tutorials and scientific labs remain available. Saving only happens when the user requests it. A replay recalculates the evidence before restoring the view.

Find structures now offers an observation-support view. It shows eligible pairs in the selected region, their model-minus-observation residuals and the native cells with no eligible sample in the selected source. Tightening time, distance or quality rules can change support without changing the model. The January demonstration has 34 eligible pairs touching 23 of 80,203 cells in the selected region under default rules. These counts describe sparse sample support, not confidence or independent validation.

Demo: open Build investigation, follow the five steps and save a comparison. Then choose Find structures at 7 January 2024, 12:00 UTC. Open observation support for the default warm region, keep a reference and set the time window to zero. Restore the reference and open a pair in comparison. For the imported path, use the March Bay case and original BGC file SR1902594_034.nc; its measured chlorophyll remains separate from model temperature.

Verification covers irregular depth-bin boundaries, original source dates, accepted P05 rows, duplicate-cell counts, stricter-rule empty states, imported identities, browser navigation and replay. Final counts and deployment evidence are in PROJECT_STATE.md and PHASE_16D_CONNECTED_WORKFLOW.md. The support export is a result snapshot, not a replay bundle; save the structure or comparison separately for recalculated replay. Actual unfamiliar-user and oceanographer testing remains open. This is prepared wording, not a submitted report or presentation.

P16D release verification, 27 September 2026: version 0.16.9 is live on the existing Vercel project. The declared phase passed 163 backend tests, 73 client checks, 24 public support checks, 61 imported/legacy replay checks, 34 public desktop/mobile browser checks and 12 responsive visual captures. These overlapping counts are not percentages of PS completion. Initial test and network failures are preserved with their diagnosis and repeat evidence in PROJECT_STATE.md. Real-user comprehension and the P17 audit remain open.

P16E prepared text: larger investigations and continuity

Short explanation: Ocean Navigator can review several observation files together, keep successful imports when one fails and carry a saved scientific view into an offline file. Data publication is checked before activation, so a failed candidate leaves the previous release available.

What changed and why: Imported observations now have an eight-file review queue and explicit per-file states. The workspace accepts up to 20,000 samples under source and decoded-data limits. Requests remain small enough for the existing hosted comparison and replay path. A source-status view explains historical coverage and publication dates. Offline HTML supports saved observation profiles, model comparisons and native columns with the same source values and QC. Cancellable operations retain earlier usable results.

Demonstration path: Import genuine Argo and BGC files, inspect each preview, add ready files, save a chlorophyll profile and open its Offline viewer with the network unavailable. Select a sample and show its source timestamp and quality flag. Return to Sources to show publication identity and historical coverage. A maintainer example rejects a corrupted dataset update while the prior release remains available.

Evidence and limitations: The parser workload measured six genuine formats and a separately labelled 20,000-sample resource fixture. Backend publication tests check failure, cancellation, quotas, atomic selection and process restart. Browser/source-value, offline tamper and hosted release results are recorded in the final acceptance report. Offline charts display saved evidence; they do not fetch new ocean conditions or recompute analyses. No automatic external scheduler or new persistent server is configured. This text is prepared for reports and demonstrations, not a submitted final report.

Accepted release: 0.16.10 is deployed on the existing Vercel project. The relevant checks include 66 backend tests, 79 client checks, 21 public publication/import checks, 61 exact imported/legacy replay checks and 12 final public visual captures. These counts overlap and do not represent PS completion percentages. The latest full public phase suite had 20 passes and two failures. Six targeted desktop/mobile recovery checks subsequently passed, including an interrupted large-file request and explicit Retry. An intermittent Chromium HTTP/2 transfer failure remains recorded for P17. We fixed a real import-reset race caused by late model metadata, portable publication ordering and unclear source dates in offline charts. Actual user comprehension and independent scientific review still require participants.

Globe and tutorial follow-up: prepared explanation

Short explanation: Start by finding a dataset on Earth. Open its ocean model to look through depth and time, or choose a sensor to inspect what it measured. The guide demonstrates the same controls the user can operate, then points to Tools for the other analyses.

What changed and why: The globe now shows real model areas and source-profile locations. It remains on screen until the user opens another view. A chlorophyll example locates a real BGC float and opens its depth profile beside the appropriate historical model context. The tutorials follow the SIH scientific workflow first and include the newer import, observation-support, saved-evidence and publication features. The extra-feature picker states how many tours exist, keeps navigation above the cards and highlights the next action. A completion arrow makes Tools easier to find.

Demonstration: Select a region on the globe and open its 3D ocean. Inspect temperature at depth, switch to salinity or currents, then compare a real profile with the model. Return to the globe for the chlorophyll example. Finish an extra-feature tour and use the highlighted Tools button to continue independently.

Evidence and limitations: GLOBE_AND_TUTORIAL_REVIEW.md maps PS priorities to actual delivery and remaining boundaries. Final executed checks and the deployed link are recorded in PROJECT_STATE.md. The globe shows bounded source coverage, not a global live ocean. A chlorophyll dot is a measured profile location, not a mapped bloom. Tutorials do not automatically save personal data or download files. Browser tests are not evidence of comprehension by actual judges or oceanographers. This is reusable report/demo text, not a submitted report.

Release and checks: 0.16.11 is live at <https://depth-atlas-seifuku.vercel.app/?release=0.16.11>. The follow-up passed 153 relevant backend tests, 82 client scientific checks, 36 final local navigation regressions across Chromium desktop/mobile and WebKit, 61 exact local replay checks and 24 public source-support checks. Final public release verification passed 24/24 on an independent repeat. Fifteen public screenshots across three sizes confirm layout and target visibility. These counts overlap and are not a percentage of PS completion.

Remaining issue: the normal public interaction suite passed 23 checks and failed the desktop full tutorial at an intermittent comparison-request timeout. The mobile full tour and explicit comparison Retry passed. A diagnostic desktop run over HTTP/1.1 also passed, but this does not fix or certify normal transport. P17 should investigate that remaining issue alongside the planned audit. The source error is shown honestly and never replaced with a fabricated comparison.

Targeted loading reliability follow-up

Short explanation: Ocean Navigator gives the selected scientific result priority. It stops an unused background download when the user moves to another task, and allows a slow connection more time to deliver a valid comparison before reporting an error.

What changed: the app warms only the next model time step, after the current reading is ready. Switching tools cancels unused speculative work. Choosing a snapshot already loading keeps the same request. JSON reads retain a finite deadline, explicit error states and the same source/version checks. Scientific calculations and matching rules are unchanged.

Demonstration: follow the Basic tour from the globe through the model and sensor comparison. The comparison still returns the real eligible pairs. Change snapshots and switch to another tool while data load; unused background work stops and the current selection remains authoritative. A controlled delayed-response test checks that a valid slower comparison finishes without a manual Retry.

Evidence and limits: RELIABILITY_FOLLOWUP.md separates reproduced application defects from the intermittent transport cause that remains unconfirmed. Final executed local/public tests and deployment are recorded in PROJECT_STATE.md. No uninterrupted-network guarantee or P17 completion is claimed. This is prepared report/demo wording, not a submitted report.

Verified release: 0.16.12 is deployed. The complete public tutorial passed twice on desktop and once on mobile with normal browser transport, and ten targeted public recovery checks passed. Exact scientific replay passed 61 checks, release/API/assets passed 24 checks, and local client science passed 82 checks. The earlier comparison timeout did not recur in these final runs. Initial failures and corrected test expectations remain in the evidence record; counts overlap and are not percentages of PS completion.

P17: checked release and demonstration package

Short explanation: Ocean Navigator brings a historical ocean model and real sensor measurements into one browser workspace. A user chooses a supported dataset on the globe, explores its depth and time, checks a profile against compatible model values, and saves an investigation that can be recalculated. The current release supports scientific exploration and teaching. It does not issue operational advisories.

What changed and why: The final audit checked every tracked PS requirement and additional feature against code and evidence. It added independent numerical checks, bounded public requests before parsing, upload deadlines, per-worker rate and concurrency limits, and restrictive browser resource policies. A keyboard-focus defect in

Sources was fixed. Dependency and secret-signature checks found no listed advisories or candidate credentials in their recorded scope. This is targeted hardening, not a guarantee of complete security.

Demonstration: Open the Basic guide or skip to the globe. Open the January Bay case, rotate the water column, choose the optional cutaway and inspect a depth slice. Compare Argo profile 1902669 against the compatible 7 January 2024, 12:00 UTC model snapshot. Show eligible pairs, original dates and residuals, then save, export and recalculate the investigation. Open Tools for the other labs. The short backup video records these actual actions from the public release. The longer live script includes instrument inspection and structure support.

Verification: The complete integrated backend suite passed 963 tests and client calculations passed 82. Public release/API/assets passed 24 checks, exact replay passed 61 checks, and the complete 39-lesson guide passed on both desktop and mobile emulation. Chromium, Firefox and WebKit checks retain their original failures and subsequent targeted corrections in the acceptance index; overlapping runs are not summed into a feature-completion score. The isolated portable installation passed source, replay and browser checks on the same Windows computer.

Measured behavior: In three fresh browser contexts per condition, the median automated route from navigation through a checked comparison took 6.837 seconds on desktop and 9.333 seconds under the stated mobile network/CPU emulation. Each retained Interactive 3D and exact source results. These timings exclude human reading time and do not prove an evaluator's device performance. All eight requests in two four-client regional-data rounds succeeded. P17_PERFORMANCE records the conditions and resource limits.

Prepared materials: A six-slide PDF/PPTX draft uses the official idea template. README, setup, methods, attribution, capability coverage, demo narration and judge questions are ready. The 43.12-second silent backup is a real browser recording, not a narrated pitch or a hosted submission. No idea, source repository or video was submitted to the organizer by this work.

Limits and next work: Actual unfamiliar-user testing, oceanographer review, an unrelated physical-device check and institutional acceptance remain open. Persistent public THREDDS, full independent CF checking and compatible glider/CTD matchup cases also remain explicit dependencies. The mobile tutorial review does not establish comprehension. The next step is to observe real people using the prepared task script, correct demonstrated problems, rehearse and verify the private submission instructions. This is prepared report and spoken-demo text, not a submitted final report.

Team address and main documentation

Ocean Navigator is now available at <https://seifuku-ocean-navigator.vercel.app/>. The address belongs to the existing production project and serves the same verified release 0.17.0. The earlier address remains available. Browser saves remain on the origin where they were created; use a portable investigation to move work between addresses.

The main documentation combines a current project explanation and requirement-delivery matrix with the phase history, scientific methods, source references, setup, security, verification, visual reviews and public policies. Linked contents and PDF bookmarks make the longer reference chapters navigable. Current status is separated from older milestone records. The PDF, editable Markdown and rebuild script are in submission. This is a documentation artifact, not a completed organizer submission.

The new address passed all 24 release/API/asset checks without retries and an actual browser journey covering the globe, 3D cutaway, original model reading, eligible comparison, explicit browser save and exact recalculation. The document is checked for excluded editorial topics, page bounds, text extraction and navigation. The application scientific methods and runtime files are unchanged. Existing external acceptance gates remain open.

The redesigned cover uses the original Seifuku logo, a source-backed ocean map and three short actions: locate, explore and compare. The cover links directly to the team address. Its geography provides context; it does not represent an additional dataset or worldwide coverage. The source inventory records visual-asset hashes as well as document sources.

Readability revision: body text is now 13 pt, table text 11.5 pt, and captions 11.5 pt. Wider text frames, compact contents rows and continuous chapters reduce unnecessary blank areas. Short code blocks can flow across pages. Cover labels and team details are larger. A further spacing pass separates major chapters with a rule and 26/19 pt gaps, gives section headings 23 pt before and 11 pt after, and uses 18.5 pt body line spacing. The full source material and linked navigation remain included.

Repository preparation requirement check

The team reports that its presentation/portal requests a repository link. That link is now tracked as required preparation. A fresh check of the public 2026 guideline, presentation template and project problem statement did not establish whether that repository must be public. The exact authenticated form or team-specific notice still needs inspection. Reviewers must be able to access the submitted code: a private URL alone does not grant permission. Neutral naming can reduce obvious competition keywords, but cannot make public code confidential. No repository was created or uploaded during this check. The next step is to resolve reviewer access, prepare the required source package, and verify it before submission.

Depth Atlas public source and identity update, 27 September 2026

Reusable explanation: Depth Atlas is Team Seifuku's browser workspace for exploring ocean model fields, inspecting real instrument profiles and saving reproducible investigations. The new name and public source address make the current prototype easier to identify and review.

What changed and why: The name, team domain, app labels and documentation cover now agree. The mobile header retains the full name without horizontal overflow. A public repository contains the runnable code, checked datasets, tests, methods and source credits. Private planning and earlier development history remain local. Scientific arrays and calculations are unchanged.

Demonstration: Open depth-atlas-seifuku.vercel.app, skip or follow the guide, select the Bay dataset, open the 3D field and inspect a native value. Open Compare, inspect the supported profile and save the result. Recalculate it from Saved. Review the source and setup at github.com/GauravR17A/depth-atlas.

Verification: The public release passed 24 release checks and 61 exact replay checks. Desktop, phone and narrow browser layouts passed the branding journey. The source package passed 854 backend tests with 109 explicitly skipped raw-archive checks, plus 82 client checks. All 145 affected checks passed in the original archive-equipped workspace. Test counts overlap. The public repository was verified without login, with no competition identifier or configured credential signature found in its final upload candidate.

Limits and next work: These are technical checks, not participant or oceanographer acceptance. Browser storage is tied to the website origin. Transfer saved work explicitly when moving to the new address. The current draft presentation/old backup retain their earlier branding; refresh those selected assets before a final submission. Gather real user/domain feedback and confirm the applicable portal instructions.

68 Visual and interaction review records

Historical or maintained reference: read alongside the current release overview | Source: Maintained visual review collection

The following dated reviews preserve observed layouts, corrected issues and unverified usability claims. The final release overview describes the current accepted scope.

Globe and tutorial visual review

27 September 2026. Release 0.16.11. Runtime source
6f7339d0842feae5b70e016b1746e62092ff6701.

Local final captures: `globe-tutorial-visual-final/`, generated by `web/benchmarks/globe-tutorial-visual.mjs` against port 8036. Fifteen screenshots cover globe/model discovery, chlorophyll location, extra-feature picker, Tools handoff and a glowing cutaway control at 1536x960, 768x1024 and 393x851. The recorded checks show no page errors or horizontal overflow, and the feature picker and Tools callout fit all three viewports.

Manually inspected the desktop globe, desktop/mobile feature picker, desktop control highlight, mobile Tools arrow and tablet chlorophyll view. The globe and actual source dates remain readable, the case-opening action is explicit, the feature total and More features action are conspicuous, and the popup arrow points to the real Tools control. The model and profile remain available behind the learning callout. The chlorophyll screen describes a source-profile location and model-context overlap, not a surface concentration field.

The first mobile Tools capture showed the button's glow too close to the paragraph. Explicit flex spacing separates text and button. The final mobile capture confirms the correction, and a geometry assertion checks a gap between the paragraph and the button. The first 320x568 picker overflowed; the compact layout now passes every page at 320x568, 390x844 and 844x390 without scrolling. Pointer/target, keyboard, enlarged text and reduced-motion checks remain in the interaction suites.

These are rendered-browser and internal interaction checks. They do not constitute actual unfamiliar-user testing, an oceanographer review or measurements on every physical GPU/device. Final public captures and results are listed separately in `PROJECT_STATE.md`.

Final public captures: `globe-tutorial-visual-public/` contains the same fifteen views from the deployed canonical URL. All three viewport checks show no page errors/overflow and fitting picker/Tools callouts. Manually inspected the public desktop feature picker, mobile Tools arrow and mobile globe. The displayed release is 0.16.11; the next-page action remains above the cards with steady mint emphasis, and the completion arrow points to the actual Tools button. This capture run does not override the separate repeated full-desktop-tutorial transport timeout documented in the acceptance record.

Source record: `docs/evidence/globe-tutorial-visual-review.md`.

P13 frontend checkpoint

Updated 23 September 2026. Frontend public acceptance is complete. The root task maintains overall phase closure.

The Climate Event Lab compares September to November in 2013, 2015 and 2022 using the same GODAS native coordinates and fixed 1991-2020 calendar-window baseline. Users can inspect potential temperature, a baseline anomaly or selected-minus-reference difference, select one exact column/depth in both panels, and save/export/replay the applied comparison. Draft form changes do not relabel completed charts or saves. ENSO uses pinned RONI v6; historical ONI versions and IOD episode labels remain separate. Indian Ocean sections describe the same dates and field source without causal attribution.

Pacific cases also open in the shared 3D/slice/section workspace, bounded structure search and original instrument inspector. Only potential temperature is available. Native depth begins at 5 m and ends at 459 m. Monthly model fields are not compared directly with instantaneous in-situ Argo temperatures. Source profile coordinates, time and QC remain original. The broad Pacific view defaults to explicitly labelled 6000x vertical exaggeration, adjustable through 12000x. Existing Indian defaults and ranges are preserved. Dateline labels and instrument map placement use continuous source coordinates with east/west display labels.

Initial P13 browser run: 24 pass and 3 failures exposed a legacy parser assumption that every model/profile overlap had an instantaneous nearest timestamp. The parser now separately accepts the new explicitly non-comparable calendar-month support contract. The second run passes 27 across Chromium desktop, WebKit desktop and Chromium mobile.

Initial existing-browser regression: 260 pass, 15 failures, one skip. All failures were five cases repeated across three projects: two Indian continuity assertions affected by an unnecessary quantity-label change, corrected by restoring their original labels, and three

obsolete expectations that the Pacific region had no data. The latter now verify actual available Pacific data and unavailable variables/labs. Initial reports and failure traces are retained.

The initial 30-case final P13 run passes, including exact Pacific 6000x save/replay and unchanged native values when adjusting the view to 10000x. The second full existing-browser run records 268 passes, seven failures and one skip. Six failures are the same two Indian structure-result wording changes across three browsers, now restored to their prior wording. The seventh is a revised test using the desktop-only sources button on mobile; the shared evidence-panel path is used instead. The second report and traces remain archived. The full suite is not described as an all-green run; a separate corrected focused recheck covers the affected modules.

TypeScript/Vite build and all 30 renderer/geographic numerical checks pass. Initial screenshots include the original thin Pacific 3D view and the real climate comparison. The final caption build also clarifies matching single-month baselines, IOD agency-statement labels, missing-baseline hatching and public metadata. The corrected 69-case focused recheck passes across Chromium desktop, WebKit desktop and Chromium mobile, resolving all seven failures in the second broad run. It includes all climate, feature, feature-section science and workspace tests. A subsequent 54-case climate/instrument recheck also passes after the exact-coordinate map correction below.

A short demonstration: open Climate, compare 2015 with 2013, switch from baseline anomaly to actual potential temperature, select a native depth/longitude, inspect the separate IOD episode and numerical DMI context, then save the applied result. Open a Pacific event in the shared explorer and inspect its source profile. The three historical years and bounded latitude/depth coverage are explicit limits; this is not a climate forecast or globally equivalent data coverage.

Final local visual correction and freeze

The first 23-capture visual audit exposed a real map defect that automated overflow checks could not detect. Computing the circular-longitude cut with modulo after adding 360 introduced a floating-point boundary error for the actual 2015 Argo longitudes. A nearby point could be assigned to the next world, forcing an incorrect global map scale. The helper now uses the exact next sorted source coordinate. A new regression uses the real longitudes -139.687, -140.093 and -140.031, plus repeated positions. All 31 renderer/geographic checks pass. The 54-case climate/instrument browser recheck covers corrected Pacific centring and existing Indian glider, CTD, BGC, upload and map-keyboard paths.

Final 23 screenshots are in `p13-visual-local-final`, with zero page errors, alerts or horizontal page overflow. Manual review covered their charts, values, geographic placement, original profiles, 3D/cutaway, narrow layouts and enlarged text. The earlier map capture remains in `p13-visual-local`. The corrected small equatorial Pacific map contains no coastline at this zoom because its real position is in open ocean; original coordinates remain visible. The final point note explains that visible values are rounded while exports retain calculation precision. Zero-difference legends use a constant neutral colour and display zero without a negative-zero sign.

Frozen build: `workspace-CC4CTtTg.js`, `scene-CKByfc7t.js`, `workspace-0oAPrcy8.css`. Build log: `p13-frontend-build-frozen.log`. Final renderer log: `p13-renderer-tests-map-final.log`. Corrected browser reports: `p13-browser-corrected-final.json` (69 passes) and `p13-browser-map-final.json` (54 passes). These are separate runs, not a claim that the earlier 276-case broad run was entirely green. Public deployment/browser/replay acceptance is still pending at this checkpoint.

First public acceptance and final sidebar correction

The first production browser run passes 111 of 111 checks with no skips: 30 climate, 27 saved-investigation, 18 feature, 24 instrument and 12 workspace checks across three browser projects. Report/log: `p13-browser-public.json` and `.log`. The first public visual audit has 23 captures, zero errors/alerts/page overflow, preserved in `p13-visual-public`. Public cutaway, climate sections, exact profiles and corrected geographic placement were manually reviewed.

A final production-sidebar audit then found an attribution defect outside those cropped workspace captures: the shared About this case paragraph still hardcoded HYCOM for the new GODAS cases. The retained screenshot is `p13-public-sidebar-attribution-initial.png`. Sidebar text now uses the real source label, E/W/N/S coordinates and actual depth limits. Its Pacific time-period field uses full calendar-month support, while the existing Indian snapshot-date display is preserved. Nine focused local browser checks pass, including both regions. The old fixed seven-snapshot playback tooltip was also replaced by Play available times; three additional Pacific browser assertions pass. No numerical fields or scientific methods changed.

Corrective build: `workspace-C3Ymu47W.js`, `scene-BgH9yDKL.js`, `workspace-0oAPrcy8.css`, logged in `p13-frontend-build-corrective-final.log`. Sidebar checks are `p13-sidebar-local-final.json` and `p13-sidebar-local-title-final.json`. The earlier 111-check public report and 23 captures remain unchanged. A corrective deployment, focused final public recheck and final 24-capture audit including the sidebar are pending at this checkpoint.

Final public closure

The corrective production deployment `dp1_4aVLAaqS82jcbV9iJKjHD3w43UAK` is ready at <https://depth-atlas-seifuku.vercel.app>. All nine focused final public checks pass across Chromium desktop, WebKit desktop and Chromium mobile. They verify the corrected Pacific source, full monthly support, available-times playback title and preserved Indian date/source paths. Report and log: `p13-sidebar-public-final.json` and `p13-sidebar-public-final.log`. The frozen asset filenames above remain the final build.

The final public visual audit contains 24 captures in `p13-visual-public-final`, with zero page errors, alerts or horizontal page overflow in its `checks.json`. The new `pacific-case-sidebar.png` capture includes the complete source and date sidebar. Manual review confirmed NOAA GODAS attribution, September 1 to November 30 support for the 2015 case, native 5 to 459 m depth limits, and correctly formatted east/west bounds. The paired El Nino anomaly sections, explicitly labelled 6000x Pacific 3D view and contained 320px section layout were also inspected. No remaining frontend acceptance blocker was found.

The earlier 111-check public report and 23 public captures remain unchanged. This final nine-check run verifies the small metadata correction after that broad acceptance run; it is not presented as another full regression. Initial failed reports, traces and visual evidence remain available alongside the focused corrected runs.

Source record: `docs/evidence/p13-frontend-checkpoint.md`.

Phase 13 visual review

Review date: 23 September 2026. Screenshot inspection is separate from automated browser checks and the still-open P09.5 unfamiliar-user pilot.

Initial final-build captures

The root reviewer opened actual images in `p13-visual-local/`: `pacific-shared-cutaway.png`, `el-nino-neutral-anomaly.png`, `pacific-original-instrument.png`, `profiles-390.png`, `pacific-320.png` and `pacific-feature-section.png`.

The Pacific cutaway visibly exposes the supplied water column. Its 6000x display exaggeration is labelled, the first level is 5 m, and actual depth/value readouts remain separate from display geometry. The climate panels share a signed scale and native longitude/depth alignment. Missing source or baseline cells use visible hatching. Selected profiles and the value table remain readable at 390 px. At 320 px the scientific sections scroll internally, with explicit guidance and native-coordinate controls; the document itself does not overflow. The feature section preserves actual native samples, distance,

requested versus sampled locations and source dates.

The original instrument screenshot exposed an unnecessarily wide and off-centre geographic view of three nearby Pacific floats. Frontend review traced this to rounding at a longitude cut boundary. Deployment was held while the map helper was corrected and an actual-coordinate regression was added. These initial captures are retained; they are not evidence that the original map presentation passed.

Corrected local captures

The final map build passes 31 renderer/geographic checks, including the actual 2015 source longitudes and repeated station case. Its climate/instrument browser run passes 54 checks across the three supported projects. The corrected capture directory `p13-visual-local-final/` contains 23 views, with no page errors, alerts or document overflow.

The root reviewer opened the corrected `pacific-original-instrument.png` and `synchronized-native-profiles.png`. The three nearby actual float markers now sit together in the central open-ocean window. The absence of land in this regional window is consistent with its location; no decorative coastline is invented. Profiles, source sample counts, QC and original positions remain unchanged. The climate value table now explains display rounding and retained export precision, so a subtraction of rounded display numbers is not mistaken for the exact calculation.

Public captures

The production audit in `p13-visual-public/` completes 23 captures with no page errors, alerts or document overflow. The root reviewer opened `la-nina-neutral-anomaly.png`, `same-event-zero-difference.png`, `pacific-shared-cutaway.png` and `profiles-390.png`. The 2022 and neutral fields share their actual signed scale; a same-event comparison correctly displays a constant zero scale. The cutaway keeps the explicit depth exaggeration and native depths. The 390 px profile and rounded-value explanation remain legible. Frontend review additionally covers the public source map, Indian context, feature section, 320 px and enlarged-text captures. No further visual defect was found in these inspected views. This is checked-layout evidence, not a completed real-user comprehension study.

Final public sidebar correction

The retained `p13-public-sidebar-attribution-initial.png` showed the remaining hardcoded HYCOM statement on a Pacific sidebar. The corrected release has a separate24-view audit in `p13-visual-public-final/`, with zero errors, alerts or document overflow. The root reviewer opened `pacific-case-sidebar.png`: it states NOAA GODAS monthly potential temperature, three genuine Argo records, the full 2015-09-01 to 2015-11-30 monthly interval, actual east/west and north/south coverage, and 28 native levels from 5 to 459 m. The final public interaction recheck passes all nine source/date/title cases. Earlier public captures and broad111-case evidence remain unmodified. No further visual issue was found in the checked final views.

Source record: `docs/evidence/p13-visual-review.md`.

Phase 14 visual review

Review date: 24 September 2026, Asia/Calcutta. Local and public outcomes are recorded separately. Screenshots use real prepared data and actual API responses, not substituted display fields.

Local review

`p14-visual-local-initial/checks.json` records 43 captures with zero page errors, console errors, visible alerts or document overflow. The harness completed real original/modified saves and downloaded an evidence ZIP. Its checks include 320 px and 390 px viewports and 200% root text size; these are browser emulations, not physical-device tests.

The desktop graph shows all 21 default Arabian region instances and their branches. Snapshot columns, actual dates, threshold, minimum overlap and selected node are readable. The extent view labels native longitude/latitude and actual selected depths, with source-domain and selected-depth contacts kept separate. The blackout chart distinguishes removed and retained sample depths without changing the source positions. Its original/modified counts agree with the checked API output. The Pacific card shows the actual monthly interval and potential-temperature quantity, with zero eligible residuals. Narrow cards and enlarged controls remain usable within the checked viewport.

Two small copy refinements followed inspection: singular wording for one cell/profile, and a visible horizontal-scroll hint above the graph on small screens. The original captures are preserved. Both refinements are included in the final public review below.

Public review

Release 0.14.1 passes the final public harness: 43 captures, zero page errors, console errors, visible alerts or document overflow. Original and modified blackout saves work, and the downloaded evidence ZIP is 168,638 bytes. Results are in public visual checks.

The branch graph and native-coordinate extent remain readable. Selected shallow/deep contacts are distinguished from source-domain contacts. Exact removed source IDs are visible, and empty residuals say unavailable rather than displaying zero. The narrow graph has its scroll hint; singular counts use singular wording. Monthly potential-temperature limitations and the full averaging interval remain explicit. The inspected 320 px and enlarged-text controls wrap without hiding their labels or actions. These technical checks do not establish comprehension by unfamiliar users. P09.5 remains open, and no physical-device or Firefox test is claimed.

Source record: `docs/evidence/p14-visual-review.md`.

P15 final visual inspection

Inspected on 26 September 2026, release 0.15.2. `p15-v2-visual-public/checks.json` records twenty actual captures, eighteen layout checks and the offline/throttled paths. Nine final images were opened: 01-source-overview, 06-volume-cutaway, 09-dateline, 11-oxygen-pressure, 12-oxygen-native-uncertainty, 14-standalone-offline, 18-narrow-result, 19-enlarged-text and 20-slow-basic.

Source availability and the first-load action are readable. The 3D cutaway is optional and its depth exaggeration is labelled. Date-line coordinates retain east/west labels; salinity g/kg, oxygen micromoles/kg and pressure dbar are distinct. Native profiles retain irregular vertical levels and exact tables. Missing uncertainty support is hatched. The 320 px controls/actions fit without horizontal document overflow, and the full 1481.9 dbar label is visible after the final margin correction. Enlarged-root-text and Save-Data/basic views remain usable in the captured paths.

The first public visual pass exposed the clipped pressure-axis leading digit despite a green document-overflow check. The initial captures remain under `p15-v1-visual-public`. The fix has a browser check against actual text width plus independent geographic click verification. The final inspected screenshots confirm it. No generated imagery, decorative animation or invented measurements were used.

The standalone HTML was opened from disk with browser networking disabled and made zero HTTP requests. It displays provider uncertainty, native cells and exact source indices. Final throttled regional loading used 400 ms latency and 50,000 bytes/second; preview response 0.92 seconds and completed view 3.44 seconds. This starts after shell load and is

not an initial-page-load guarantee.

Technical layout/interaction checks do not establish actual first-time-user comprehension. P09.5 remains open. WebKit/Chromium desktop and Chromium mobile emulation were tested; this is not physical-device or Firefox certification.

Source record: docs/evidence/p15-visual-review.md.

Anchored tutorial visual review

26 September 2026. Final release 0.16.1; local assets match production in the 21-check release verifier.

Inspected actual screenshots, including desktop cutaway/profile/save, narrow volume/depth/Method, mobile profile, landscape volume/profile, and the enlarged-text Method view. The popup points to the selected button or chart. The exact control remains usable. The full cube fits alongside the landscape callout; narrow volume/depth cards show the real source number before secondary explanation. Sensor axes and source dates remain unchanged. Save guidance appears directly under the name field in the real modal.

The final local harness has 41 captures with no recorded overlap/viewport/Next failures and no console/page errors. The first placement draft had a landscape overlap, fixed by adapting the highlighted view. Position easing was removed because it visually detached the arrow during scrolling. The original initial screenshots captured the middle of automatic scrolling; the dedicated harness waits for scrolling to settle.

On 320 by 568 screens and at 200% text, the explanation body requires scrolling. Back, Next, Skip and the lesson chooser remain available. Large figures can exceed one screen; the highlight points to the visible part rather than representing it as a complete chart. This is not a physical-device or unfamiliar-user comprehension study.

The public visual attempt captured five desktop views before a profile load failed, and did not finish its sweep. Its log and partial captures are retained separately. Public browser/release results and unresolved transport timeouts are in PROJECT_STATE.md.

Source record: docs/evidence/p16-anchor-visual-review.md.

Extra-feature tutorial visual review

Date: 26 September 2026. Local release 0.16.2 at http://127.0.0.1:8021.

The final p16-teaser-visual-accepted/checks.json records 29 captures with no measured viewport, horizontal overflow or completion-control visibility failures and no page errors.

The harness covers 1536 x 960 desktop, 390 x 844 mobile, 320 x 568 narrow portrait and 844 x 390 landscape. Each has a welcome, all four teaser pages, the real Drift calculation and its completion. A 768 x 900 capture adds 200% text and reduced motion.

Actual screenshots were inspected for the desktop feature cards, mobile welcome, narrow final feature page, narrow completion, landscape picker and completion, and enlarged picker. The eight topics are presented in pairs with direct Start actions, short questions, provenance labels and actual step counts. Cards and their next-page controls fit the checked normal-text sizes. Code-native icons identify the topics.

Visual inspection found that the first completion layout required scrolling to reach the final actions on small screens. Completion now uses available viewport height, with closer spacing and no decorative check icon on short screens. The final landscape capture shows all three actions. The preceding `p16-teaser-visual-final` report retains the one completion-control failure. The earlier `p16-teaser-visual-release` sweep did not check every completion action, so its zero failures did not establish that those buttons fit. The harness now measures all completion buttons.

The first browser layout checks also found clipped picker footers in narrow portrait and short landscape. Compact card spacing and a shorter landscape header corrected those failures. `p16-teaser-feature-acceptance.json` passes all 39 feature-tour scenarios over Chromium desktop, WebKit desktop and Chromium mobile emulation. Existing anchor tests and all 27 actions passed in the broader learning run; its three picker-fit failures and their corrected checks remain documented separately.

At enlarged text sizes, the picker and long lesson content may scroll inside their panels; keyboard access is preserved by the dialog and existing guide controls. At 320 x 568 the welcome diagram shrinks to keep both tutorial styles, Skip, the feature teaser and Start visible. These are automated/emulated layouts and manual screenshot inspection, not a physical-device certification or a substitute for the open unfamiliar-user pilot. Public transport reliability is verified and reported separately.

Source record: `docs/evidence/p16-teaser-visual-review.md`.

P16 visual inspection

Reviewed 26 September 2026. Final hosted release: 0.16.0.

The public harness `p16-visual-public/checks.json` records 31 captures at 1536x960, 768x900, 390x844, 320x568 and 844x390, plus a 768x900 welcome with 200% root text. It checks document width against the layout client width, initial welcome fit, Next visibility and usable request-input width. All measured checks pass and no page or console errors were recorded.

Actual screenshots opened during review include the desktop welcome, Basic ocean/cutaway and explanation panel; mobile Basic and In-depth methods; the narrow welcome and Basic coach; landscape welcome and concepts; and the enlarged-text welcome. The real source field, dates, depth stretch and selected values remain visible. Basic has short action/output copy with two current facts. In-depth has separate conceptual/method pages. Rectangular controls, the original favicon and source styling are retained.

Visual inspection caught issues beyond the first overflow check: a narrow-screen Next button below the visible coach and a desktop request input squeezed by the generic full-width primary-button style. Next is now sticky inside the coach and the request input has usable width. A Save button at enlarged mobile text caused the browser's layout viewport to widen; its text now wraps within the header. The visual harness now compares against `clientWidth` rather than the expanded `innerWidth`. Two animation frames are awaited before capture to allow completed dialog content to paint.

The initial choice does not scroll at the checked normal-text sizes. The coach and explanation panel can scroll on short screens or when more detail is opened, with Next retained in view. Landscape concept content can require that internal scroll. This is not a claim of no scrolling throughout the app. Keyboard/reduced-motion/fallback interactions are checked separately in the browser reports.

Automation and screenshot review do not prove comprehension by real beginners, performance on every physical device or accessibility certification. The P09.5 participant pilot and broader P17 audit remain separate.

Source record: `docs/evidence/p16-visual-review.md`.

P16A visual review

Reviewed 26-27 September 2026 using the compiled 0.16.5 app at `localhost:8025` and the public Vercel release. Captures are in `p16a-visual-release/`, `p16a-visual-final/` and `p16a-visual-public/`. The same build's public bytes are checked by `p16a-public-release.json`.

The harness captures connected model/profile, import preview and chlorophyll at 1536x960, 768x1024 and 393x851. All nine states have no document-level horizontal overflow and no uncaught page errors. This is a bounded layout check, not a usability study.

Manually inspected the connected desktop/mobile, chlorophyll desktop/tablet and mobile preview. The desktop model and profile are visible together. Source dates remain distinct; colour keys and numerical inspection sit below the model instead of squeezing

its canvas. Mobile stacks the views and bounds the model panel; a visible hint explains how to reach depth/time controls within it. Preview counts fit the narrow screen. Location/track/provenance and CSV mapping expand on demand.

The first chlorophyll capture exposed a cramped disabled Save control with its hint wrapping to the right. The final layout places the hint below the disabled control with a readable width. The example list collapses after a preview is ready. Two literal question-mark separators were replaced with middle dots before deployment; public captures verify the final build. Original source gaps in chlorophyll curves remain visible rather than smoothed over.

The 45-check final-release instrument set and 72-check existing renderer/tutorial/saved-investigation regression passed. After the last layout changes, the 15 import checks passed again across Chromium desktop/mobile and WebKit desktop. Reduced motion, enlarged text and keyboard interaction are included in the existing tutorial regression. Firefox did not launch in the earlier run (spawn UNKNOWN), so no Firefox rendering acceptance is claimed.

On small screens the model and full profile chart cannot both occupy the initial viewport. They remain in one scrollable workspace. Automated checks and this manual review do not establish comprehension for a first-time oceanography user; P09.5 remains open.

Public acceptance: all nine captured states have no document overflow or page errors. Manually inspected the deployed chlorophyll desktop, chlorophyll mobile and preview mobile views. The corrected Save hint, source-quality table, distinct dates and middle-dot separator render as intended. The initial source sample in the BGC example has missing pressure/chlorophyll; it is explicitly marked missing/excluded, while the accepted curve remains available for selecting another source sample. It is not replaced by a fabricated reading. The production import/instrument/startup suite passed 36/36 on Chromium desktop/mobile.

Source record: <docs/evidence/p16a-visual-review.md>.

P16B visual review, 27 September 2026

Inspected local desktop March volume and mobile chlorophyll screenshots, the deployed desktop side-by-side model/chlorophyll view, and the final 0.16.7 tablet volume. Automated captures cover desktop 1536x960, tablet 768x1024 and mobile 393x851 at two views each. No horizontal document overflow or uncaught page errors was recorded.

The March date, source, temperature units, native depth and exaggeration are visible in the Explorer. The color key explains values rather than natural water color. Instrument inspection retains model context on desktop and stacks it on smaller screens. Separate

observation and model timestamps remain visible. The chlorophyll annotation explicitly says no model chlorophyll is supplied; overlap is not a numerical-match claim. Accepted/excluded counts and physical axes remain readable. Imports are correctly described as temporary and unsaveable until P16C.

The additional-tool checks found that Expedition and Evolution were visible for an unsupported March case. That navigation defect is fixed in 0.16.7 using the actual backend method registries. All 24 fresh-case and capability checks pass in the final clean installation across desktop Chromium, mobile emulation and WebKit. These checks are not unfamiliar-user testing or physical-device certification.

Source record: <docs/evidence/p16b-visual-review.md>.

P16C visual review

Reviewed 27 September 2026. This is an engineering visual review, not participant or oceanographer approval.

The local and deployed harness captured comparison controls, matched profile/residual charts, save disclosure and saved-record actions at 1536x960, 768x1024 and 393x851. Each run produced 12 images. Both checks.json reports show no document or dialog horizontal overflow and no uncaught page errors. The real source-backed example shows 245 eligible temperature pairs from 1,475 source samples and 1,230 exclusions.

Manually inspected initial local desktop/mobile charts and mobile save disclosure. That exposed an unnecessary reference-cleared notice on first opening and a question-mark separator in the imported selector. The notice now appears only when an actual reference was cleared; the selector uses a plain colon. The imported-file scope notice was shortened. Final local desktop/tablet charts and mobile saved actions, and public desktop charts, mobile controls and tablet save disclosure were inspected after those corrections.

The curves use distinct solid/dashed observation/model styling, physical depth increasing downward, explicit temperature/residual units and the same eligible pairs. No extrapolated segment or visualized chlorophyll model was added. Observation and model timestamps remain separate. Tablet charts fit side by side, while mobile charts stack. Long saved-record content scrolls inside the existing dialog; its actions remain keyboard/browser tested. Imported saves clearly disclose inclusion of original bytes and local storage. The disabled replay-link action has an adjacent explanation directing the user to files.

Existing scientific controls remain available, with advanced matching rules expandable and a Go to results link for long/narrow layouts. P16D will improve the connected workflow; no claim is made that these screenshots prove novice comprehension. Browser

viewport emulation does not prove performance on a physical phone or every evaluator GPU. Local Chromium desktop/mobile and WebKit are tested; final public coverage is recorded separately in `p16c-public-browser.json`.

Evidence folders: `p16c-visual-local` (initial), `p16c-visual-final-local` (corrected) and `p16c-visual-public` (deployed). Each includes its check report. Source data dates are March 2024; the implementation/review date is September 2026.

Source record: `docs/evidence/p16c-visual-review.md`.

P16D visual review

Reviewed 27 September 2026. This is an engineering review, not participant or oceanographer evidence.

The harness captures the optional Investigation Builder, co-visible model/comparison, sensitivity table and structure-support footprint at 1536x960, 768x1024 and 393x851. The final local run produced 12 captures with no document horizontal overflow or uncaught page errors. Initial mobile review prompted a compact five-number step navigation with full accessible names. The active step title and Next action remain explicit. Support-column rings make sparse samples visible without enlarging their actual native-cell coverage.

Manually inspected the local mobile builder, desktop/mobile comparison, desktop/mobile support and tablet sensitivity. In the public candidate, reviewed desktop/mobile builder, desktop/tablet comparison, desktop support and tablet sensitivity. The native model probe and the interpolated comparison retain separate values and an explanation. Depth increases down the scientific plots. Solid and dashed curves distinguish observation/model, and the residual has its own physical scale and units. The matching table compares the applied reference and current result, with a note that each uses its own eligible samples.

The support footprint uses hatching for gaps, mint cells/rings for columns with an eligible pair and amber for the selected pair. It labels coordinate bounds and states that collapsing depth does not establish support at every depth. Real sparse coverage stays sparse; counts are not presented as confidence. Large and small screens retain the selected reading's observed value, interpolated model value, residual and original timestamp.

The first public visual run stopped after nine captures because its mobile comparison did not arrive within 30 seconds. The overlapping browser run recorded an HTTP/2 transport failure and metadata timeouts. This incomplete visual run is retained at `p16d-visual-public` and is not accepted as a complete set. The Vercel log and

interrupted-browser report preserve that event. The capture harness now waits for a real native value before the builder screenshot and closes the browser even if capture fails. The sequential repeat completed all 12 captures at three sizes, with no document horizontal overflow or uncaught page errors (p16d-visual-public-final/checks.json). Final desktop builder and mobile support footprint, linked charts and sensitivity table were manually reviewed. The source value and model scene were present in the final builder capture. The unchanged runtime also passed all 34 public browser checks.

Browser viewport emulation does not establish physical-device GPU performance or novice comprehension. The participant task script remains unexecuted. No universal device, reliability or scientific endorsement claim follows from these screenshots.

Source record: docs/evidence/p16d-visual-review.md.

P16E visual and interaction review

Reviewed 27 September 2026. Local automated captures: p16e-visual-local-final/checks.json, 12 screens across 1536x960, 768x1024 and 393x851. No document, saved-dialog or offline-page horizontal overflow and no uncaught page errors. Final public captures are a separate acceptance gate.

Manually viewed the desktop batch queue and saved panel, mobile batch queue and desktop/mobile offline chlorophyll chart. The first review exposed excessive limit text in the import panel and scaled-down chart labels on mobile. Detailed limits now expand on demand; the chart draws at its actual CSS width with readable text, explicit positive-down depth and five depth ticks. The revised mobile chart was inspected. Missing samples remain absent, negative source chlorophyll values are retained under the source QC instead of silently clamped, and excluded points remain optional.

An additional scientific-label review found that the offline header highlighted the January model context beside a March observation. Observation dates are now visible above the chart, separately from model timestamps and record age. The imported offline test checks 29 March 2024 explicitly. The correction passed Chromium desktop/mobile and WebKit. Earlier screenshots remain historical evidence; the public final screenshots must include the correction.

Initial browser test issues were in fixture/selectors: a malformed CSV correctly needed mapping rather than a hard failure, imported IDs include the instrument kind, the profile selector is scoped to a collection, and closed details expose no innerText. These were corrected without weakening the intended checks. Windows Playwright WebKit also blocks any file URL in its forced offline mode, reproduced with a plain local HTML file. WebKit offline-viewer checks therefore open the real file while aborting all HTTP/HTTPS

routes. They observe zero network requests and exercise reload, exact values and tamper rejection. Chromium uses the actual offline context flag. This does not constitute physical iOS/Safari testing.

The main local regression suite passed 60 tests on Chromium desktop/mobile. Four WebKit offline modes and a subsequent three-browser imported-date test passed. Original fingerprints and exact source values were compared rather than judging plots by appearance alone. Actual unfamiliar-user/domain-user testing remains open.

The first complete public phase suite had 18 passes and two mobile assertions that ended before the app request deadlines. The batch test waited ten seconds for a control while the final original was still being read; the save test waited thirty seconds while the bounded request allows forty-five. Test waits now follow actual queue-ready and saved states within those deadlines. This does not change app timeouts or hide failed imports. A repeated targeted public run and the final visual capture determine acceptance. The earlier interrupted runner had no final report and is not counted as passed.

A repeated public mobile test then exposed a genuine lifecycle defect. The instrument panel used the asynchronously arriving model manifest ID as a React key, so a late response recreated the panel and erased a pending observation review. Its key now changes only when explicitly restoring an investigation. Import review is independent of model availability and stays usable across model context updates. A delayed-metadata regression checks that the pending review remains mounted, then accepts the original file after the model becomes available. Final acceptance uses the deployment containing this fix.

The public old-build delayed-metadata regression failed exactly after metadata delivery removed Cancel review. The fixed local sequence passed in Chromium desktop/mobile and WebKit. The expanded local suite had 31 passes and two premature reloads after clicking Remove local copy. Both affected configurations passed twice after the test waited for the actual transaction-completion notice. These test corrections keep the same storage outcome requirement. Runtime app checks and final hosted evidence use source 6b8409a.

The latest full public phase run, p16e-public-browser-accepted.json, actually reports 20 passes and two failures. Its filename is not a claim of a clean run. One original-file transfer received HTTP 200 headers but Chromium then reported net::ERR_HTTP2_PING_FAILED while receiving its body; the UI retained the existing profile and offered Retry. The other test ended after ten seconds while the comparison was still checking samples. The trace shows successful catalogue/coverage requests and a pending comparison request. That assertion now allows the existing query timeout/retry sequence to finish; no scientific threshold or runtime timeout changed.

The targeted public recovery run, `p16e-public-recovery.json`, passed all six checks on desktop/mobile without Playwright retries. It repeats the affected comparison/imported offline scenarios and deliberately interrupts a large original-file request. The explicit Retry path then uses the real public parser, keeps prior evidence, saves the BGC profile and downloads its offline viewer. A successful repeat does not resolve the cause of the intermittent HTTP/2 transfer failure. Broader public network/performance investigation remains a P17 follow-up. Original failed reports and traces are retained.

Final public visual run: `p16e-visual-public/checks.json`, twelve captures across desktop 1536x960, tablet 768x1024 and mobile 393x851. All three configurations reported no document/dialog/offline horizontal overflow and no uncaught page errors. Manually inspected final desktop offline chlorophyll and publication status, tablet import queue, and mobile offline chlorophyll and saved-investigation panel. Observation dates are visibly separate from model context and save age. Mobile controls and chart labels fit their containers; content remains scrollable where needed. This is a bounded visual/interaction review, not evidence that unfamiliar users understood the application or that every physical device works.

Source record: `docs/evidence/p16e-visual-review.md`.

P17 visual review

Reviewed on 27 September 2026 against release 0.17.0. It is not a participant usability study or an oceanographer's acceptance review.

Scope and evidence

All 28 original images were opened and inspected: the first result of each of the 14 extra-feature tutorials in Playwright WebKit desktop and Chromium Pixel 7 emulation. The original source paths, copied paths, dimensions and SHA-256 hashes are in `capture-manifest.json`. Original files were copied without editing. Capture names are `p17-visual/{webkit-desktop|chromium-mobile}-{feature}.png`.

The capture test is `web/e2e/p17-visual.spec.ts`. It waits for the tutorial result and an anchored coach, records the full page, and checks the coach fits the viewport, document width does not overflow and no page errors occurred. It does not wait for every secondary lazy-loaded chart, exercise every control, or establish ease of understanding. The parent release run supplies the test pass/fail record separately.

The configured desktop viewport is 1440 by 900 CSS pixels. Pixel 7 emulation uses 412 by 839 CSS pixels. The PNGs have higher physical pixel dimensions and include scrollable document content. Full-page capture can affect layout and scroll position, so a

screenshot's apparent spotlight alignment is not sufficient evidence of a live anchoring defect.

Per-feature observations

Feature	Desktop image inspection	Mobile image inspection	Scope or limitation
Builder	The five-stage data, depth, reading, comparison and save strip has a clear current step and action. The ocean scene remains present.	The strip becomes numbered steps, with the current action and tutorial controls visible.	A first step is shown, not a completed saved investigation.
Chlorophyll	Globe marker and source card identify a genuine BGC profile, date, position and measured variables. The guide points at its details.	Globe and profile details stack; the guide occupies part of the lower card while showing the location result.	One observed location is explicitly distinguished from a concentration surface or bloom extent. The profile-opening action is outside this first-result capture's visible guide focus.
Imports	The model context remains beside the source-profile panel. The CSV preview lists accepted, excluded and missing samples by variable.	The model precedes the profile panel. The preview table and coach are readable without document-wide horizontal overflow.	This is a preview before adding the file. The coach covers lower preview actions while explaining the result; Try it myself is visible.
Publication	Historical source dates, release identity and publication status are separate. The page explains its maintainer-driven update process.	Source and publication records stack. Long timestamps wrap while remaining visible.	This does not demonstrate an automatic live scheduler.
Offline	The actual save dialog and explicit browser-save action are visible. Storage and offline limitations are explained.	The dialog uses internal vertical scrolling; close and continue controls remain visible.	This capture proves the save form opens, not that an offline file was saved or reopened. Those are separate interaction tests.

Feature	Desktop image inspection	Mobile image inspection	Scope or limitation
Drift	Calculated particle paths, coordinates, playback controls and the simulation explanation are present. The guide points beside the result.	The path chart has internal horizontal scrolling and explicit instructions to scroll or use coordinate and particle controls.	The capture does not measure animation smoothness or validate real drifter trajectories.
Climate	Two historical sections share the stated baseline and colour scale. Profiles, selected depths and anomaly values remain labelled.	The card and controls fit, but the coach covers much of the selected section in this captured state. Other section and profile content remains in the document.	Narrow-screen teaching limitation: pause with Try it myself to inspect the plot unobstructed. The source sections themselves require sideways scrolling. Do not claim that both complete sections fit on a phone.
Heat	Surface event chart and separate subsurface evidence have distinct source labels. Event duration, date boundaries, mixed-layer definitions and units are visible.	Controls and diagnostics stack. A visible part of the event chart sits above the coach; lower context remains available by scrolling.	The page is long on a phone. A surface heat event is not presented as proof of a subsurface heatwave.
Structures	The threshold query, plan map, native-cell count and units are visible. The original capture caught the secondary 3D mesh preparing.	The 3D boundary has loaded, with selected top/bottom depths and vertical exaggeration explained. The plan map has internal horizontal scrolling.	A settled desktop probe below confirms the actual mesh rendered. The original loading capture is retained rather than replaced.
Expedition	Numbered stations, a sampling budget, model map and station context are visible.	Form and station details stack; the map shows a horizontally scrollable subset.	Not every station is simultaneously visible at this width. The station controls are an alternative to map interaction. These are planning/sampling simulations, not an operational cruise plan.

Feature	Desktop image inspection	Mobile image inspection	Scope or limitation
Evolution	Seven dated source frames, overlap links, threshold sensitivity and boundary warnings are visible.	Two frames fit at once; remaining frames and table columns require internal scrolling. The frame and region selectors remain available.	The interface explicitly distinguishes shape correspondence from transport of the same water. This Bay case has one continuing region; it does not visibly demonstrate a split in this screenshot.
Blackout	Original and remaining pair counts, retained and excluded depth marks, residuals and source identities are visible.	The coach and Next fit, but the kept-profile column is outside the initial horizontal chart view. Tables also need internal scrolling.	Narrow-screen teaching limitation: the two chart columns cannot be compared simultaneously in this captured view. The original/remaining counts and explanatory text remain available. No model rerun or observing-system improvement is claimed.
Wider regions	A bounded source region and its model field retain source, month, variable, units and geographic bounds.	Region controls precede the field and guide. Labels and source context remain in the document.	A nearly uniform field colour is not itself evidence of a rendering error. The view does not establish equal support across all oceans.
Support	Region boundary, eligible-pair counts, unsupported-cell hatching, selected observation and residual context are visible.	The original full-page capture appeared to highlight a thin strip below the map. A focused viewport probe confirmed the live target is the correct support map.	See geometry probe below. Coverage remains a count of eligible source support, not confidence or validation.

Focused follow-up checks

Executed `node docs/evidence/p17-visual-settled-probe.cjs` against the already running release at <http://127.0.0.1:8038/>. The command exited 0 in 8.43 seconds. No application runtime files were changed and neither browser recorded a page error. Geometry and exact times are preserved in `p17-visual-settled-probe.json`. The five supplementary images are listed with hashes in `supplemental-capture-manifest.json`.

- Desktop Structures: waited for `.feature-mesh-canvas` canvas to become visible and captured the settled result. Viewport and full page show the real surface alongside the selected plan region. The coach's highlighted target exactly matches the plan element's rectangle. The earlier preparing state was transient, not a persistent missing surface.
- Mobile Support: after settling, the target was `.support-map`, 316 by 384.31 CSS pixels at $x=48$, $y=537.11$. The spotlight covered its visible portion down to the viewport margin. Settled viewport shows the arrow pointing to the map. Full-page capture changed the scroll position and layout geometry. After using Find this view, the target and spotlight both measured $x=48$, $y=12.11$, $width=316$, $height=384.31$, with the coach below it. Relocated viewport shows both clearly. Try it myself removed the coach while the support map stayed visible. This probe did not reproduce a persistent wrong target or inaccessible map.

Interpretation and remaining work

No persistent blank app, missing tutorial navigation, or wrong scientific target was established by this review. The first-result layouts consistently expose actual outputs and source context. Rectangular controls, restrained navy and mint surfaces, plotted source data and clear guide actions match the project's visual direction.

Several original full-page images have a long empty tail after the footer.

`LearningExperience.tsx` deliberately inserts `tour-scroll-space` only while an active, unpaused tutorial is displayed; `useTourAnchor.ts` uses that space to bring lower results into view. The empty tail is not unrendered data. It remains visible in full-page evidence and should not be mistaken for the normal unguided workspace's content area.

Mobile analytical pages remain long. Some plots and tables need their own horizontal scrolling, and tutorial cards can cover part of a result while presenting its explanation. Climate and Blackout are specific candidates for participant feedback. Their captured Next and Try it myself controls remain visible, but this review does not establish that an unfamiliar person will discover every horizontal chart area or understand every scientific term.

This review covers first extra-feature result states and the two focused follow-ups only. It does not establish every lesson's visual state, keyboard or screen-reader usability, colour-vision accessibility, real-device GPU performance, touch accuracy on physical phones, motion comfort, expert scientific endorsement, or novice task completion. The independent numerical suite, full tutorial interaction runs, accessibility checks and device-performance evidence must be read separately. The planned unfamiliar-user and oceanography-user acceptance gates remain open.

Source record: docs/evidence/p17-visual-review.md.

Site pages: visual and interaction review

26 September 2026. Release 0.16.3. This checks the content follow-up, not P17's full-app audit.

Final evidence

site-pages-local-accepted/checks.json and site-pages-public/checks.json each record eleven passing grouped checks and zero page errors. Nine checks cover the three pages in desktop Chromium, 320 by 568 Chromium and 390 by 844 WebKit. Two check About -> workspace -> About -> Privacy -> Terms, including the ready native model value 22.303 in the workspace. Each report directory contains 24 screenshots: top/footer views, team layouts and three enlarged-text captures.

Pages render with JavaScript disabled, loaded CSS and a favicon. Local and public checks cover contact mail links, current-page footer links, section anchors, print colours and horizontal overflow. Chromium's Tab/Enter sequence reaches Skip to content and focuses the document. This Windows WebKit build did not move focus through native links with Tab or Alt+Tab; its accepted checks explicitly focus the link before testing Enter activation. They do not establish default Safari keyboard navigation or physical-device accessibility.

Manual inspection included the initial desktop About view, desktop team cards, 320-pixel Privacy and Terms/footer views, the 390-pixel WebKit About view, enlarged Privacy text and the public desktop About page. Names and the team ID are readable; card columns collapse with width, policy jumps wrap, email text fits and footer links remain visible. Keyboard outlines around the focused document are intentional. The site uses its existing navy/neutral palette, rectangular controls and local SVG favicon.

Earlier failures and tooling limits

A later unquoted PowerShell element reference was interpreted as a splat, so that click did not run; the attempted Privacy screenshot at that point showed About and is not evidence of Privacy. A corrected direct Privacy open succeeded, but the CLI viewport/screenshot commands stalled. Their helper processes were stopped. The completed Playwright runs above provide the actual final navigation and mobile evidence.

The first batch browser script stalled while adding a style tag in a no-JavaScript context. It was stopped, and direct font-size adjustment replaced that test operation. Two intermediate runs then exposed the Windows WebKit tabbing behavior described above. Those reports and the explicit diagnostic remain retained; neither is described as a passing suite. The final browser source is web/benchmarks/site-pages.mjs.

Print styling was updated to give the new definition lists, team cards and contact details black text on a transparent background. No scientific method or source-data change was required. Human usability feedback, physical devices, broader delivery reliability and security hardening remain separate work.

Source record: docs/evidence/site-pages-visual-review.md.

Frontend rebuild: visual review

Reviewed on 26 September 2026 against public release 0.15.3.

The first public harness captured 32 screenshots and 44 layout/intro-fit checks with no document overflow, intro clipping or recorded console/page errors. Manual inspection found that mobile-explorer.png captured the initial loading state before the field arrived. The harness was corrected to wait for the actual native value, 22.303 degrees Celsius, and disappearance of the field loader before taking mobile, narrow and enlarged-layout ocean screenshots. Initial evidence remains in ux-visual-public.

The corrected public run in ux-visual-public-final contains 32 captures, 44 checks, zero reported fit/overflow failures and zero recorded page/console errors. This harness adjustment did not change deployed product code.

Images opened and inspected from the first public run:

- desktop-explorer.png: prominent ocean scene, date/source strip, three primary view tabs, optional cutaway and labelled source value/colour key.
- desktop-tools.png: all twelve destinations grouped into three task categories with a visible close button.
- desktop-drift.png: numbered release/calculate/compare flow, map and primary calculation action; detailed coordinates collapsed; model limitations visible.
- desktop-step-1.png: code-authored depth illustration explicitly labelled as an illustration, short explanation, Skip and Next.
- narrow-step-2.png: all content and controls fit a 320 by 568 viewport without scrolling.
- landscape-step-3.png: diagram, short first-action instructions, Back and Start exploring fit an 844 by 390 viewport.
- mobile-tools.png: readable rectangular tool rows. This requested tool catalogue can scroll on mobile; the no-scroll requirement applies to the introduction.

Images opened and inspected from the corrected final public run:

- mobile-explorer.png: actual rendered ocean is visible at 390 by 844, with accessible view and cutaway controls and no horizontal clipping.
- narrow-explorer.png: controls fit at 320 by 568. The normal workspace scrolls vertically to the scene and lower controls, unlike the fixed introductory pages.
- zoom-equivalent-explorer.png: a narrower 640 by 720 layout retains the source date, task controls and actual field. Real 200% text and reduced-motion checks are separately covered in the browser suite.

Keyboard focus rings are intentionally visible in these screenshots. The final public browser suite passed 128 checks with one inherited WebKit context-loss skip and no failures. Screenshots and automated checks do not certify human comprehension or physical-device compatibility. The P09.5 unfamiliar-user pilot remains open.

Source record: <docs/evidence/ux-visual-review.md>.

69 Depth Atlas

Current release reference | Scope-edited excerpt | Source: README.md

A browser workspace for inspecting ocean model fields beside real instrument profiles. Built by **Team Seifuku**.

[Open the public app](#) · [About](#) · [Privacy](#) · [Terms](#)

The app brings location, depth, time, observations and saved evidence into one connected workspace. It uses historical source data and declared calculations. It is a research and demonstration prototype, not an operational forecast or advisory service. The current release, exact checks and remaining dependencies are recorded in PROJECT_STATE.md.

Main project documentation PDF consolidates the current guide, scientific methods, phase records, setup, security and verification. The editable Markdown and rebuild script are in [submission/](#).

Earlier application addresses remain available. Browser-local saves are origin-specific; export/import an investigation when moving between addresses.

Try one investigation

1. Use the default **Basic tutorial**, or skip it to explore. The globe shows the actual area and dates of each available case.
2. Open a Bay of Bengal case. Rotate the 3D field, open the optional cutaway, change depth and inspect a native reading.
3. Open **Instruments**, select a real profile, and inspect its units, time and quality flags beside the model.
4. Use **Compare** for supported temperature/salinity samples. Read exclusions and matching rules along with residuals.
5. Explicitly save the result. Reopen it to recalculate and verify source, method and result identity, or export its evidence.

The guide has 18 core steps. Fourteen additional feature tours are available through **Explore extra feature tutorials**; the completed guide points to **Tools**. In-depth mode adds scientific concepts and limits. [Demo script and judge questions](#).

The 43-second silent technical backup shows actual public release 0.17.0 behavior: globe, depth, comparison, save, replay and Tools. Capture metadata and separate narration cues identify the tested source and deployment. It is a local recording, not a narrated pitch or uploaded submission.

What is available

Area	Working scope and boundary
3D exploration	Depth-resolved temperature, salinity and horizontal currents; slices, isosurfaces, time steps, sections, color controls and native-value inspection. Source masks and irregular depths are preserved.
Data and profiles	Supported CSV/NetCDF import preview, explicit compatible mappings, QC feedback and batches of up to eight bounded files. Argo, BGC, glider and CTD source examples retain their own positions and dates.
Connected evidence	Model/profile co-display, eligible numerical comparison, source-sample selection, structure support, sensitivity and an Investigation Builder. Coverage is not confidence.
Reproducible work	Browser-local investigations, original imported bytes when explicitly saved, portable JSON, checked replay, readable reports, CSV, evidence ZIPs and bounded offline viewers. No account or cloud database.

Area	Working scope and boundary
Scientific labs	Structures, Expedition, fixed-depth passive Drift, Heat & Depth, historical ENSO/IOD Climate, Evolution and observation Blackout. Each has its own tested case scope and stated limits.
Publication and deployment	Read-only publication status, checked case-pack identities, retained last usable publication, same-origin API and a portable source/server setup. A maintainer initiates updates; no live-feed scheduler is supplied.

Coverage consists of selected Bay of Bengal and Arabian Sea grids, including January and March 2024 cases, plus selected Pacific seasons in 2013, 2015 and 2022. These are not basin-wide measurements. Chlorophyll is an observed BGC profile variable; the bundled model cases do not contain a chlorophyll field. Genuine glider/CTD examples can be imported and inspected, but their dates/locations lack a matched main model case.

The Indian model cases expose REST and native CF NetCDF exchange. A separate local THREDDS package has verified January/March WMS/WCS/OPeNDAP routes. Persistent public THREDDS, an actual INCOIS installation and independent full CF conformance certification remain unestablished. [Capability audit](#).

Run locally

Use Node.js 22.12+ in the 22.x line, Python 3.10-3.12 and npm. Checked data packs are included; no data-provider token is needed for startup. From the project root in PowerShell:

```
python -m venv .venv
.venv/Scripts/python.exe -m pip install -r requirements.txt
npm --prefix web ci
npm --prefix web run build
.venv/Scripts/python.exe -m science.publication --candidate casepacks --manifest api/publication.json
.venv/Scripts/python.exe -m uvicorn api.app:app --host 127.0.0.1 --port 8000
```

Open <http://127.0.0.1:8000>. The frontend, API and publication must use the same release identity. Rebuild and regenerate the manifest before restarting after a release change. Use `requirements-dev.txt` for API tests or `requirements-science.txt` for the full scientific/reference environment. Large original archives are separate from the runnable checked packs; a skipped source test is not a passed scientific check.

The frontend uses React, TypeScript, Vite, Three.js and D3. FastAPI serves bounded scientific operations using NumPy, netCDF4 and GSW; xarray is used in offline model preparation. [Setup, contracts and verification commands](#) · [Portable delivery and architecture](#).

For development, run Uvicorn on port 8000 and `npm --prefix web run dev` in another terminal. Vite serves port 5173 and proxies the API. Deployment uses the existing Vercel project. Never put secrets in frontend code or public environment variables. Institutional deployment must configure its own HTTPS, access rules, resource limits and persistent standards service.

Evidence and limitations

The final audit is P17. Its evidence distinguishes implemented behavior, internal numerical/source checks, deployed verification and external review. Automated layouts do not establish human comprehension. An actual unfamiliar-user pilot and independent oceanography review remain open until results are recorded. No national selection, endorsement, forecast improvement or universal device-performance claim is made.

Earlier phase reports retain their original failures, rechecks and dated counts. Do not add overlapping counts into one test total or use an old phase's result as proof of the current release. The source/version and evidence index in PROJECT_STATE are authoritative for the latest verified deployment.

Record	Purpose
PROJECT_STATE.md	Current release, active work, actual commands, evidence and next action
P17_CAPABILITY_AUDIT.md	Requirement and original-feature audit
P17_DELIVERY.md	Architecture, clean setup, package inventory and institutional limits
P17_DEMO.md	Focused live demonstration, backup route and judge questions
SUBMISSION_CHECKLIST.md	Official shortlisting requirements versus supporting assets
SCIENTIFIC_CONTRACTS.md	Quantities, units, QC, provenance and result contracts
ROADMAP.md / FEATURE_REGISTER.md	P01-P17 and evidence-backed capability status
USP_INDEX.md	Original ideas, including Drift, Heat & Depth and ENSO
PHASE_WRITEUPS.md	Plain report, slide and spoken-demo text
DEVELOPMENT.md	Setup, tests, APIs and deployment
TEAM.md	Confirmed public identity and roles

Team Seifuku is led by **Adwita Kurle**. **Gaurav Ranade** created and operates the website. Contact: **ranadegaurav30@gmail.com**, Pune, Maharashtra, India. The project has no university affiliation or university involvement in development. Other member names are recorded in TEAM.md without invented roles.

P17 is the last planned implementation phase; external acceptance gaps are not closed by a release label.

70 Standards service setup

Historical or maintained reference: read alongside the current release overview | Source: [deploy/standards/README.md](#)

The public app stays on Vercel. THREDDS is a separate persistent Java service. A browser user needs no Java, Python or local setup.

Tested Windows setup

From the repository root, with the documented Python science environment installed:

```
.venv/Scripts/python.exe deploy/run_standards.py --install
```

This downloads vendor-checksummed Java 17.0.20.1, Tomcat 10.1.60 and THREDDS 5.9 into ignored `.runtime/p06`. The pinned ZIPs are Windows x64; the launcher does not change machine-wide Java or PATH. An incomplete download fails its checksum rather than being executed. If that occurs, remove only that named archive and repeat the command.

The launcher checks the bundled exchange file, prepares a dedicated Tomcat base and runs in the terminal. Open <http://127.0.0.1:8096/thredds/catalog.html>. Stop with Ctrl+C. Do not start a second instance on the same port. To check a fresh runtime base after stopping the first:

```
.venv/Scripts/python.exe deploy/run_standards.py --base .runtime/p06/fresh-server  
.venv/Scripts/python.exe -m science.verify_p06_standards
```

The second command runs in another terminal while the server is ready. It needs the original acquired model files for independent reference values. To obtain those when absent, follow [docs/PHASE_02_DATA.md](#). Normal service startup uses the bundled checked exchange file and needs no source-server connection.

Requests

```
http://127.0.0.1:8096/thredds/wms/ocean/bay-
bengal-2024-01.nc?service=WMS&version=1.3.0&request=GetCapabilities
http://127.0.0.1:8096/thredds/wcs/ocean/bay-
bengal-2024-01.nc?service=WCS&version=1.0.0&request=GetCapabilities
http://127.0.0.1:8096/thredds/wcs/ocean/bay-bengal-2024-
01.nc?service=WCS&version=1.0.0&request=GetCoverage&coverage=temperature&format=NetCDF3&time=2024-01-07T12:00:00Z&vertical=100&bbox=86,12.5,87,13.5
http://127.0.0.1:8096/thredds/dodsC/ocean/bay-
bengal-2024-01.nc.dods?temperature[1:1:1][19:1:19][10:1:12][10:1:12]
```

DAP endpoints are datasets; use a DAP client such as pydap or xarray to decode binary responses. The verifier uses OWSLib to construct WMS/WCS requests, reads NetCDF coverage files, and compares them with source values. WMS 1.3.0 uses longitude/latitude for CRS:84 and latitude/longitude for EPSG:4326. WCS 1.0.0 uses west/south/east/north in the tested native geographic grid. Depth selections are positive down in metres.

The native NetCDF and DAP coordinates are exact. THREDDS can regularize horizontal axes for WMS/WCS. The tested WCS longitude change is at most 0.0000826928 degree; the explicit test tolerance is 0.0001 degree. Numerical values and masks passed at 1e-12 absolute tolerance. No interpolation or universal reprojection support is claimed.

Linux / institutional server

Use maintained Java 17 and Tomcat 10.1 releases. The same Python launcher accepts `--java /path/to/java --tomcat-home /path/to/tomcat --war /path/to/thredds.war --base /path/to/ocean-runtime`. Use the WAR checksum from the lock file. Do not use `--install` for Linux because its JRE archive is Windows-specific. That Linux path is documented but has not been executed here.

An optional Compose profile is also supplied:

```
docker compose -f deploy/compose.yaml --profile standards up --build
```

Docker is unavailable on the development machine. This recipe is not recorded as tested. The application Dockerfile now includes `adapters/`, required by supported instrument imports. The separate TDS image verifies its WAR, runs as a non-root user and mounts data read-only. Host ports bind only to loopback. Tomcat cache, generated files and logs are writable within its container; they are not user investigation storage.

Resources and public deployment

The tested launcher uses 128 MB initial and 768 MB maximum Java heap, 12 connector threads and 1024-by-1024 maximum WMS images. DAP ASCII and binary limits are configured at 5 MB and 64 MB. WCS has no claimed equivalent response-size cap in this configuration. Its exposure is bounded by the single published case, but concurrent requests can still increase memory/CPU use. One post-test process sample used approximately 357 MB working set. This is a measurement from one local run, not a capacity guarantee.

The optional container has a 1.5 GB memory limit, two-CPU limit and 128-process limit. Reserve additional memory for the Python app, host and filesystem cache. Disk must hold the source artifact, approximately 100 MB WAR, Java/Tomcat files, expanded libraries, caches and rotated logs. Benchmark the actual institutional machine and expected concurrency before selecting production resources.

To expose TDS publicly, the institution must provide a persistent host, HTTPS reverse proxy, DNS, patching and resource/log management. Keep manager/admin applications disabled and remote-dataset fetching disabled. Configure the proxy's request/rate/time limits and re-run the protocol verifier through its public URL. This project has no verified INCOIS server access, no public TDS hostname and no authorization to buy an extra hosting service. Public Vercel REST and native downloads are independent of that deployment.

Configuration templates use `__OCEAN_DATA__`, substituted by the launcher or Docker build. It is not a THREDDS environment-variable placeholder. Original prototype attempts using `${tds.content.root.path}` in the catalogue failed to resolve as a dataset path; the explicit absolute directory fixed that startup error.

71 Portal text for review

Historical or maintained reference: read alongside the current release overview | Source: submission/PORTAL_TEXT.md

Prepared 27 September 2026. Not submitted. The signed-in dashboard's exact character limits, attachment size, nomination status and required fields remain unverified. The current published shortlisting deadline is 30 September 2026; verify availability and any round-specific notice before submission.

Suggested idea title: Ocean Navigator: 3D Ocean Models and Observation Evidence

Short description: Ocean Navigator is a browser workspace for exploring depth-resolved ocean models beside real instrument profiles. Users select a historical case, inspect 3D fields, review quality-controlled observations, compare eligible samples and save reproducible evidence. Guided actions explain the results and their limits.

Extended description: Ocean Navigator addresses project brief through one connected browser workflow for numerical model fields and in-situ observations. It provides depth-resolved 3D rendering, slices, isosurfaces, time controls, colour settings and native source-value inspection. Supported NetCDF and delimited observation imports expose detected variables, units, coordinates and QC before users add them. Genuine Argo, BGC, glider and CTD examples retain their source dates and locations. Model comparisons apply explicit quantity, time, distance, depth and QC rules.

Saved investigations preserve source and method identities, settings and results for checked replay and portable export. Additional tools include structure support and sensitivity, fixed-depth particle drift, heat/depth analysis, historical ENSO/IOD comparison, sampling experiments, feature correspondence and observation exclusions. A geographic data globe and Basic/In-depth tutorials guide users through real actions. These are historical research demonstrations with stated limits, not operational forecasts.

The frontend uses React, TypeScript and Three.js with a bounded FastAPI backend and modular scientific adapters. Public REST/native NetCDF and separately verified local THREDDS WMS/WCS/OPeNDAP paths support interoperability. Actual INCOIS deployment, persistent public THREDDS, full independent CF certification and external user/domain acceptance remain to be established.

Team metadata to verify against registration: Seifuku; team ID 150123; leader Adwita Kurle; PS project brief; Software; Disaster Management. Website creator/operator: Gaurav Ranade. Public contact: ranadegaurav30@gmail.com. The project has no university affiliation or university development involvement. This does not establish portal nomination or eligibility.

Members: Adwita Kurle, Gaurav Ranade, Anushree Dixit, Spruha Kurle, Hrishikesh Tarade, Tanisha Natrajan.

Demo: <https://depth-atlas-seifuku.vercel.app/>

Deck for review: project brief_OceanNavigator_Idea_DRAFT.pdf, six slides using the supplied official template. Keep its PPTX as editable source. Confirm portal metadata and current limits before approving a submission copy. No authorization letter, private contact record, repository or submission receipt is fabricated here.

72 Submission preparation links

Current release reference | Source: submission/LINKS.md

Prepared 27 September 2026. This folder is a review package, not an uploaded submission.

Resource	Location
Public browser app	https://depth-atlas-seifuku.vercel.app/
Release 0.17.1 entry	https://depth-atlas-seifuku.vercel.app/?release=0.17.1
Current production release	https://depth-atlas-seifuku.vercel.app/
Source/data access	https://depth-atlas-seifuku.vercel.app/data-access
Public identity and contact	https://depth-atlas-seifuku.vercel.app/about
Privacy and Terms	https://depth-atlas-seifuku.vercel.app/privacy and https://depth-atlas-seifuku.vercel.app/terms
Official project brief	https://www.sih.gov.in/sih2026PS
Current published guidelines	https://www.sih.gov.in/letters/2026/SIH%202026%20Guidelines.pdf
Official presentation template	https://www.sih.gov.in/letters/2026/SIH2026-IDEA-Presentation-Format.pptx
Main project documentation	Complete PDF, editable Markdown
Local draft deck	[Six-slide PDF](project brief_OceanNavigator_Idea_DRAFT.pdf), [editable PPTX](project brief_OceanNavigator_Idea_DRAFT.pptx)
Draft portal wording	PORTAL_TEXT.md
Demo script and judge questions	P17_DEMO.md
Local silent technical backup	Actual public-browser recording, metadata and narration cues
Demonstration investigation	Portable saved evidence
Package and technical handover	P17_DELIVERY.md

No public repository, video-hosting URL or team social link has been created or supplied. Do not invent one. Public source/repository access requires a separate authorized publication decision and a licence review. The release record supplies the exact current immutable deployment and source commit; the canonical link alone is not a release freeze.

Release 0.17.0 deployment ID: `dp1_6REGcXUreo8pXinNuhQp331W2ANe`, Ready. Runtime source: `5a5e6160a7165427d6f5a174b58083f5a2db7750`; packaging source: `0f5c417f607f04aa586ae94f16fbe3aec5b35da7`. The packaging-only change excludes submission files from Vercel. Public evidence is linked in [P17_DELIVERY.md](#): 24 release checks, 61 exact replay checks and two complete desktop/mobile tutorial runs passed. Counts have different scopes and are not one unique-test total.

The local actual-behavior recording is complete: 43.12 seconds, 1440 by 900 pixels, 3,707,695 bytes, nine steps without page errors. Video seeking/decoding and selected decoded frames were checked; the exported investigation passed exact replay. It is a silent technical backup with separate narration cues, not a narrated pitch. No external hosting, video upload or portal submission was performed. Portal metadata, nomination, size/character limits and submission status remain unverified.

The Seifuku address was added on 27 September 2026 as a production domain of the existing project. The earlier ocean-navigator-project `brief.vercel.app` address remains active. No new application deployment was needed; release 0.17.0 is unchanged. Both the 24-check public release verifier and the actual save/replay browser journey passed on the new address.

Current Depth Atlas follow-up: release 0.17.1, deployment `dpl_3zagWeChqNm3PvVsop2Na1cFhqoY`, runtime source `c623165`. All 24 public release checks, 61 exact replay checks and the desktop/mobile/narrow branding journey passed. The earlier domain-only record above is historical.

Public source: <https://github.com/GauravR17A/depth-atlas>, verified without login on 27 September 2026. Public snapshot commit `4af2a933feb90a012406008e220638a336427c6b`. Main handbook: `Depth_Atlas_Main_Documentation.pdf`. The earlier draft deck and backup video retain their historical branding.

73 Privacy Policy

Current release reference | Scope-edited excerpt | Source: web/privacy.html

SITE INFORMATION

Last updated 27 September 2026

This policy explains how the public Depth Atlas prototype handles information when you browse, import observations, save work or contact us.

Who operates this website

Depth Atlas is an independent project by Team Seifuku. Gaurav Ranade created and operates this public website.

- Team: Seifuku
- Website creator & operator: Gaurav Ranade
- Location: Pune, Maharashtra, India
- Contact: ranadegaurav30@gmail.com

Information used to serve the site

Vercel hosts the site and API. Requests can include your IP address, browser information, requested URL, request time, response status and diagnostic identifiers. Hosting logs and application request IDs help with delivery, troubleshooting and security.

To limit abusive traffic, each API worker keeps short-lived request counters keyed by a salted digest of the client network address. These counters are held in memory, expire after ten minutes without a request and reset when the worker restarts. They do not contain uploaded files or calculation settings. Rate limits can be shared by people using the same network.

The app sends your selected dataset, dates, variables, study coordinates and calculation settings to its API. This includes particle releases, virtual stations, climate comparisons, feature thresholds and observation exclusions. These are study locations, not your device location. Calculations use packaged public sources; the app does not send your selections to their original data providers.

This release has no accounts, payments, app-added advertising, analytics, session replay or tracking pixels. It does not request device geolocation or set application cookies. Following an external source link opens a website with its own privacy practices.

What stays on your device

- **Unsaved work:** View selections, loaded fields and temporary imported profiles stay in tab memory. Clearing imports or reloading removes the temporary profiles. Nearby fields may be prefetched through the same API to make exploration smoother.
- **Saved investigations:** Save investigation and Keep local copy use IndexedDB in this browser for titles, settings, source references and calculated results. When you save an imported observation or comparison, the record also includes the original file, column mapping and parsed profiles. The limit is 25 records and 32 MB. Remove records from Saved or clear this site's browser data.
- **Regional pack:** Save regional pack on device uses localStorage for one pack, up to 4 MB. Saving another replaces it. Remove local pack deletes it. This is separate from the Saved investigation list.
- **Downloaded files:** Reports, CSV, JSON, ZIP and offline HTML stay wherever you save or share them. Browser storage is not a cloud backup and can be removed by browser cleanup or private-browsing rules.

The Offline viewer download contains saved results, replay settings, source credits and any original imported file already embedded in the investigation. It makes no network requests and creates no additional browser storage. Delete the downloaded HTML to remove that copy. Its charts do not fetch new data or recalculate results.

Local copies remain until removed, replaced or cleared by you or your browser. The app does not use sessionStorage. Your browser may also cache site files.

Observation imports

Choosing a supported CSV or NetCDF observation file sends its contents to the API for parsing. The application processes it in memory and does not write it to a database or saved upload directory. Original bytes, mapping choices and parsed profiles remain in the current tab until cleared or reloaded unless you explicitly save an investigation.

Comparing an import or checking its support for a structure sends the original file and settings to the API again; the app reparses it in memory and does not cache uploaded files on the server.

A batch sends up to eight files one at a time. Cancelling stops pending client requests; a bounded server request already received may finish. Completed previews and earlier workspace profiles remain until discarded, cleared or reloaded.

The filename appears in the request URL and may appear in hosting logs. Application code does not log the file contents. Only upload data you are authorized to send to this public service; do not include confidential or personal information. The observation-file limit is 2 MB, with additional schema and sample limits.

Saved files and replay links

Support snapshots contain selected observation readings, calculated residuals, matching settings and source hashes, without the original upload. They are not replay bundles. Imported-investigation JSON and ZIP downloads include the original uploaded file. They are not anonymized. Imported files are never placed in replay URLs; use a downloaded file only when you are authorized to share its data.

An investigation JSON is checked in the browser. Recalculation sends its title, settings, source fingerprints and expected result fingerprints to the API. It does not create a server-side saved record. For an imported investigation, recalculation and verified exports also send the embedded original file and mapping to the API. Reopening replaces the active imported set with the verified file. Other temporary imports are not included in a saved record.

A replay link includes its title and settings in the URL fragment. The fragment is not sent in the initial HTTP request, but the app sends the recipe when you choose Recalculate and reopen. Anyone receiving the link can read those settings. Avoid personal or confidential information in titles or shared files.

Wider regional requests and offline packs

Regional requests send the source, month, bounds, vertical limits and variable to the API. A bounded server memory cache can reuse identical public-data calculations; it is not an account or a durable history database.

Importing regional JSON checks and displays the pack locally. Verify replay online sends the pack to the API to compare it with current source data. Downloaded offline HTML makes no network requests. Local regional storage follows the rules above.

Tutorials and explanations

Tutorials call the same APIs as the visible controls. Example imports use bundled public files, not files from your device. Applying a suggested view makes the usual API requests. Saving, downloading or sharing still requires your action.

Hosting, email and retention

Vercel may process hosting information outside your country. Hosting log retention depends on the provider and project configuration. See Vercel's Privacy Notice and runtime log documentation. We do not promise a fixed provider-wide deletion period.

If you email the operator, your email address, message and any attachments are processed through the email service to answer your request. They are separate from the app's browser storage. Avoid sending sensitive files. The app has no server database of visitor profiles or saved investigations.

Your choices and contact

You can use the viewer without uploading a file or saving work. Clear temporary imports in the workspace, delete investigations in Saved, remove a regional pack from Regions, or clear this site's data in your browser. Clearing local storage does not delete files you downloaded or messages you sent by email.

For privacy questions or requests concerning information you sent us, contact Gaurav Ranade at ranadegaurav30@gmail.com. Explain the request without including passwords or unnecessary personal information. We cannot remotely access or erase copies stored only on your device.

We will revise this page when data handling changes. The date at the top identifies the current published wording.

74 Terms and Conditions

Current release reference | Scope-edited excerpt | Source: <web/terms.html>

SITE INFORMATION

Last updated 27 September 2026

These terms describe how you may use the public Depth Atlas prototype and how to interpret and share its results.

Project and operator

Depth Atlas is an independent project by Team Seifuku. Gaurav Ranade created the website and operates it from Pune, Maharashtra, India. Adwita Kurle is the team leader. See [About Us](#) for the team.

This is a development prototype. It is not an official INCOIS advisory service. The problem statement and source citations do not imply institutional approval, deployment or scientific endorsement.

Using the app

You may explore the public prototype for learning, research exploration, demonstration and evaluation. Use it lawfully and respect the published resource limits. Do not disrupt or overload the service, bypass access controls, exploit vulnerabilities or present its results as official advisories.

Only import files you are authorized to send to the service. Do not upload confidential or personal information. Supported observation files are parsed temporarily; unsupported schemas and oversized files are rejected. An accepted import is not independent scientific validation. Imported investigations embed the original file in local records and JSON/ZIP and offline HTML downloads. Only share these copies where you have permission under the original data terms; checksums do not establish ownership or authenticity.

Read results with their limits

The app provides historical model fields, original instrument profiles, derived analyses and simulations. It does not show live ocean conditions. Read each result with its source, date, coordinates, units, quality flags and method limitations. Display colours and exaggerated depth help make data visible; they are not the ocean's natural appearance.

- **Models and measurements:** Model output is not a direct measurement. Agreement with observations that may have been assimilated does not establish independent validation. Imported temperature and salinity can be compared after explicit matching checks. Observed chlorophyll has no matching model field in this release. Observation support describes this collection and these rules, not calibrated confidence.
- **Drift and virtual expeditions:** Particles are fixed-depth passive tracers in historical currents, not sinking pebbles or measured drifters. Simulated stations, routes and reconstruction experiments do not establish safe navigation, forecast improvement or operational observing-system impact. Arrival fractions are not real-world probabilities.
- **Heat and climate:** A surface event does not prove unusual heat throughout the water column. Climate comparisons cover selected historical seasons against a declared baseline; they do not establish causation or predict future events.
- **Evolution and blackout:** Overlapping shapes do not prove that the same water travelled between locations. Excluding observations changes the displayed comparison evidence, not the underlying model or its assimilation.

Do not use the prototype for navigation, search and rescue, hazard warnings, fisheries operations or other safety-critical decisions. Use the responsible official services for operational information.

Saves, downloads and sharing

Saved investigations and regional packs are local browser copies, not cloud backups. Download a portable copy if you need to keep it or move it to another device. Browser cleanup can remove local records. Imported replay reparses the embedded original file with the recorded parser and column mapping. Model data remains a separate pinned dependency. Replay depends on the source and method versions remaining available; a mismatch prevents reopening with silently substituted results.

Offline viewers display stored snapshots, not updated conditions or freshly recalculated results. They support observation profiles, model comparisons and saved native model columns. Full model grids and other laboratory computations require the online service. Deleting a browser record does not delete downloaded copies.

Checksums and matching replay establish file consistency or reproducibility, not author identity or independent scientific accuracy. Shared links reveal their titles and settings to recipients. Keep source attribution, historical dates, quality flags and limitations with any shared report or numerical subset.

Sources and attribution

Third-party datasets and software retain their own licences and citation requirements. Access to the site does not change those rights. See Source & software credits and the source labels beside each result.

When sharing a view, credit Depth Atlas by Team Seifuku as the interface and retain the original dataset and method references. Do not imply that the team, INCOIS or data providers endorse your interpretation.

Learning and explanation tools

Basic and In-depth tutorials demonstrate supported functions over prepared historical cases. The explanation panel reads current structured results and reviewed definitions. Suggested queries are shown before being applied. These aids do not certify a result, provide missing measurements, answer arbitrary questions or predict hazards.

Availability, privacy and changes

The prototype may change or be unavailable during development. Uninterrupted access, complete datasets and error-free results are not guaranteed. Nothing in these terms removes rights that applicable law does not permit us to limit.

The Privacy Policy explains requests, temporary imports, browser storage and hosting. Material changes to these terms will appear here with an updated date.

Questions and issue reports

Contact Gaurav Ranade for website questions, corrections or issue reports. Describe the problem and the affected page or result; do not send credentials or confidential data.

- Team: Seifuku
- Website creator & operator: Gaurav Ranade
- Location: Pune, Maharashtra, India
- Contact: ranadegaurav30@gmail.com

75 Source inventory and editorial scope

This edition includes the maintained scientific, engineering, phase and public-policy records listed below. Repetitive internal handoff notes are consolidated in the opening handbook. Development-session instructions and requested out-of-scope topics are not included. Original files remain unchanged. Raw logs and third-party documents are referenced rather than reproduced.

Source	Included record
docs/TEAM.md	Public project identity
docs/SCIENTIFIC_CONTRACT S.md	Scientific contracts
docs/P17_CAPABILITY_AUDI T.md	P17 capability and scientific acceptance audit
docs/P17_SECURITY.md	P17 security review and request safeguards
docs/P17_PERFORMANCE.md	P17 measured performance
docs/P17_DELIVERY.md	P17 delivery and technical handover
docs/P17_DEMO.md	P17 demonstration and judge questions
docs/P17_USER_VALIDATION .md	Final participant and scientific review protocol
docs/FEATURE_REGISTER.md	Complete feature register
docs/USP_INDEX.md	Original USP features: where each one is built
docs/ROADMAP.md	Full phased build plan
docs/PHASE_02_DATA.md	Phase 2: historical Bay of Bengal case
docs/PHASE_03_RENDERING. md	Phase 3: scientific ocean rendering
docs/PHASE_04_INSTRUMENT S.md	Phase 4: observations and supported imports
docs/PHASE_05_EVIDENCE.m d	Phase 5: model comparisons and observation coverage

Source	Included record
docs/PHASE_06_STANDARDS.md	P06: standards and extension contract
docs/PHASE_07_FEATURES.md	Phase 7: connected structures and drawn sections
docs/PHASE_08_INVESTIGATIONS.md	Phase 8: saved investigations and numerical replay
docs/PHASE_09_GUIDED_CASES.md	Phase 9: guided Indian Ocean cases
docs/PHASE_10_EXPEDITION.md	Phase 10: Virtual Expedition
docs/PHASE_11_DRIFT.md	Phase 11: Drift Lab
docs/PHASE_12_HEAT_DEPTH.md	Phase 12: Heat & Depth Lab
docs/PHASE_13_CLIMATE.md	Phase 13: Climate Event Lab and selected tropical Pacific cases
docs/PHASE_14_EVOLUTION_BLACKOUT.md	Phase 14: feature evolution and observation blackout
docs/PHASE_15_WIDER_COVERAGE.md	Phase 15: wider coverage and scale
docs/PHASE_16A_IMPORTS.md	P16A: imported observations in a connected model view
docs/PHASE_16B_FRESH_DATA.md	P16B: fresh model input and portable setup
docs/PHASE_16C_IMPORTED_REPLAY.md	P16C: imported comparison and reproducible saving
docs/PHASE_16D_CONNECTED_WORKFLOW.md	P16D: connected scientific investigation
docs/PHASE_16E_CONTINUITY.md	P16E: import workload, publication and offline continuity
docs/PHASE_16_LEARNING.md	Phase 16: learning through real actions
docs/CORE_COMPLETENESS_REVIEW.md	Review of the 31 proposed follow-ups

Source	Included record
docs/FRONTEND_REBUILD.md	Frontend usability rebuild
docs/GLOBE_AND_TUTORIAL_REVIEW.md	Globe and tutorial review
docs/HOSTING_CAPACITY_PLAN.md	Hosting capacity before P16B
docs/P05_SCIENCE_REVIEW.md	P05 scientific review
docs/P07_COMPONENT_AUDIT.md	P07 component audit
docs/P07_SCIENCE_REVIEW.md	Phase 7 independent scientific review
docs/P09_USER_PILOT.md	First-time-user pilot
docs/P10_API_CONTRACT.md	P10 implementation contract
docs/P10_INDEPENDENT_METHOD_REVIEW.md	P10 independent scientific method review
docs/P11_API_CONTRACT.md	P11 working API contract
docs/P11_INDEPENDENT_METHOD_REVIEW.md	P11 independent method review
docs/P12_API_CONTRACT.md	P12 Heat & Depth API contract
docs/P12_INDEPENDENT_METHOD_REVIEW.md	P12 independent scientific method review
docs/P12_SURFACE_SOURCES.md	Phase 12 surface-temperature sources
docs/P13_API_CONTRACT.md	Phase 13 API and replay contract
docs/P13_FIELD_SOURCES.md	P13 field sources and preparation
docs/P13_INDEPENDENT_METHOD_REVIEW.md	P13 independent method review
docs/P13_INDEX_SOURCES.md	P13 historical climate-index sources
docs/P14_BLACKOUT_INDEPENDENT_REVIEW.md	Phase 14 observation blackout: independent code review

Source	Included record
docs/P14_INDEPENDENT_MET HOD_REVIEW.md	P14 independent method review
docs/P14_READINESS_REVIEW. md	Phase 14 readiness review
docs/P16D_USER_TASK.md	P16D participant task script
docs/PROJECT_EXPLAINED_S IMPLY.md	Ocean Navigator, explained simply
docs/RELIABILITY_FOLLOWU P.md	Comparison and loading reliability
docs/STACK_AND_EXPERIENC E.md	Technology stack and evaluator experience
docs/SUBMISSION_CHECKLIS T.md	project brief submission and supporting-material checklist
docs/DEVELOPMENT.md	Running and verifying Depth Atlas
docs/DECISIONS.md	Product decisions and boundaries
docs/PHASE_WRITEUPS.md	Depth Atlas: prepared phase text
Maintained visual review collection	Visual and interaction review records
README.md	Depth Atlas
deploy/standards/README. md	Standards service setup
submission/PORTAL_TEXT.m d	Portal text for review
submission/LINKS.md	Submission preparation links
web/privacy.html	Privacy Policy
web/terms.html	Terms and Conditions
docs/evidence/globe-tuto rial-visual-review.md	Included in Visual and interaction review records
docs/evidence/p13-fronte nd-checkpoint.md	Included in Visual and interaction review records

Source	Included record
docs/evidence/p13-visual-review.md	Included in Visual and interaction review records
docs/evidence/p14-visual-review.md	Included in Visual and interaction review records
docs/evidence/p15-visual-review.md	Included in Visual and interaction review records
docs/evidence/p16-anchor-visual-review.md	Included in Visual and interaction review records
docs/evidence/p16-teaser-visual-review.md	Included in Visual and interaction review records
docs/evidence/p16-visual-review.md	Included in Visual and interaction review records
docs/evidence/p16a-visual-review.md	Included in Visual and interaction review records
docs/evidence/p16b-visual-review.md	Included in Visual and interaction review records
docs/evidence/p16c-visual-review.md	Included in Visual and interaction review records
docs/evidence/p16d-visual-review.md	Included in Visual and interaction review records
docs/evidence/p16e-visual-review.md	Included in Visual and interaction review records
docs/evidence/p17-visual-review.md	Included in Visual and interaction review records
docs/evidence/site-pages-visual-review.md	Included in Visual and interaction review records
docs/evidence/ux-visual-review.md	Included in Visual and interaction review records